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Arthur Cayley

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THE COLLECTED MATHEMATICAL PAPERS

OF

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IN THE UNIVERSITY OF CAMBRIDGE.

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A. E. Forsyth.
17 December 1895.

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CLASSIFICATION.

Analysis:

556.

ON STEINER'S SURFACE.

[From the Proceedings of the London Mathematical Society, vol. v. (1873–1874), pp. 14–25. Read December 11, 1873.]

I have constructed a model and drawings of the symmetrical form of Steiner's Surface, viz. that wherein the four singular tangent planes form a regular tetrahedron, and consequently the three nodal lines (being the lines joining the mid-points of opposite edges) a system of rectangular axes at the centre of the tetrahedron.

Before going into the analytical theory, I describe as follows the general form of the surface.

Take the tetrahedron, and inscribe in each face a circle (there will be, of course, two circles on each face touching at the mid-point of each edge, and having their centres on the lines joining the mid-point to the opposite vertex). The three circles on any face contain, the central circle within each and the three marginal circles outside each. Now truncate the tetrahedron, reducing the altitudes to three-fourths their original values; each circle will then be on a new face. Produce the three lines joining the mid-points of the edges beyond the centre, so as each to meet two of the new faces; and from the centre of each new face excavate downwards by planes passing through these lines, each circle will be scooped out so as to become the bounding curve of an excavated cavity. The surface will thus consist of four excavations bounded each of them by a trinodal quartic, and united together so as to form a continuous surface.

We may imagine the tetrahedron placed in two different positions, (1) resting with one of its faces on the horizontal plane, (2) with two opposite edges horizontal, or say with the horizontal plane passing through the centre of the tetrahedron and being parallel to two opposite edges.

1. In the first position, the horizontal plane meets the surface in a trinodal quartic.

This is at once seen to be the inverse of a circular cubic with three real asymptotes. In fact, the general equation of a circular cubic is

$y^2+(x-\alpha)(x-\beta)(x-\gamma)=0$; and if the origin is on the curve, then taking $\rho^2=x^2+y^2$ and inverting in regard to the origin, we obtain the trinodal quartic $\rho^2y^2+\rho^2(x-\alpha)(x-\beta)(x-\gamma)=0$, viz. $(x^2+y^2)^2+(x-\alpha)(x-\beta)(x-\gamma)=0$, or $x^2+y^2+\sqrt{(x-\alpha)(x-\beta)(x-\gamma)}=0$, a trinodal quartic.

But the section is also obtained directly. The equation of the surface is $\sqrt{x}+\sqrt{y}+\sqrt{z}+\sqrt{w}=0$, where $x+y+z+w=0$ if $x=0$, $y=0$, $z=0$, $w=0$ are the equations of the four faces of the tetrahedron. The section by the plane $w=\lambda$ is given by these two equations; and writing X, Y, Z for the coordinates referred to the triangle which is the section of the tetrahedron, the equations of the section become $\sqrt{X}+\sqrt{Y}+\sqrt{Z}+\sqrt{\lambda(X+Y+Z)}=0$.

To simplify, I write $q=\dfrac{\lambda}{2\lambda-1}$, that is $\lambda=\dfrac{q}{2q+1}$, the equation then is

$$\sqrt{X}+\sqrt{Y}+\sqrt{Z}+\sqrt{(2q+1)(X+Y+Z)}=0;$$

or, proceeding to rationalize, we have first

$$q(X+Y+Z)=\sqrt{YZ}+\sqrt{ZX}+\sqrt{XY},$$

and thence

$$q^2(X+Y+Z)^2-(YZ+ZX+XY)=2\sqrt{XYZ}(\sqrt{X}+\sqrt{Y}+\sqrt{Z})$$

and finally

$$\left[q^2(X+Y+Z)^2-YZ-ZX-XY\right]^2=4(2q+1)XYZ(X+Y+Z).$$

This is a quartic curve, having for double tangents the four lines $X=0$, $Y=0$, $Z=0$, $X+Y+Z=0$, the last of these being the line infinity touching the curve in two imaginary points, since obviously the whole real curve lies within the triangle.

This is as it should be: the double tangents are the intersections of the plane $w=\lambda$ by the singular planes of the surface. To find the points of contact, writing for instance $Z=0$, the equation becomes $q^2(X+Y)^2-XY=0$, that is

$$\frac{Y}{X}=\frac{-1+2q\pm\sqrt{4q^2-3}}{2},$$

giving the two points of contact equi-distant from the centre; these are imaginary if $q > \frac{1}{2}$, but otherwise real, which agrees with what follows. (See the Table afterwards referred to.)

The nodal lines of the surface are $(x-y=0,\ z-w=0)$, $(y-z=0,\ x-w=0)$, $(z-x=0,\ y-w=0)$.

We may imagine the tetrahedron placed in two different positions, (1) resting with one of its faces on the horizontal plane, (2) with two opposite edges horizontal, or say with the horizontal plane passing through the centre of the tetrahedron and being parallel to two opposite edges; or, what is the same thing, the nodal lines form a system of rectangular axes, one of them, say that of z, being vertical.

And I proceed to consider, in the two cases respectively, the form of the curve of intersection by a plane parallel to the horizontal plane.

For the first position: taking X, Y, Z as the coordinates referred to the triangle which is the section of the tetrahedron, we have the quartic equation in the form

$$\left[q^2(X+Y+Z)^2-YZ-ZX-XY\right]^2=4(2q+1)XYZ(X+Y+Z).$$

For the point in question we have $X:Y:Z=-q:-q:2q+1$; and taking the perpendicular from the vertex on a side as unity, the values $-q$, $-q$, $2q+1$ will be absolute magnitudes. We thus see that the curve must have the three nodes $(2q+1,-q,-q)$, $(-q,2q+1,-q)$, $(-q,-q,2q+1)$; and it is easy to verify that this is so.

The curve will pass through the centre $X=Y=Z$, if $(\sqrt{3}-3)^2=12(2q+1)$, that is, if $3(3q-1)^2-4(2q+1)=0$, or if $(3q+1)^2(q-1)=0$.

If $q=1$, that is, $\lambda=\frac{3}{4}\Lambda$, the equation is

$$(X^2+Y^2+Z^2+YZ+ZX+XY)^2-16XYZ(X+Y+Z)=0,$$

where the curve is, in fact, a pair of imaginary conics meeting in the four real points $(3,-1,-1)$, $(-1,3,-1)$, $(-1,-1,3)$, $(1,1,1)$.

To verify this, observe that, writing

$$A=(Y-Z)(2X+Y+Z),\quad B=(Z-X)(X+2Y+Z),\quad C=(X-Y)(X+Y+2Z),$$

and therefore $A+B+C=0$, the function in (X,Y,Z) is $=\frac{1}{2}(A^2+B^2+C^2)$, and thus the equation may be written in the equivalent forms each of which shows that the curve breaks up into two imaginary conics.

The foregoing value $q=1$, or $\lambda=\frac{3}{4}\Lambda$, belongs to the summit or highest real point of the surface.

In the case $3q+1=0$, that is, $\lambda=\frac{3}{2}\lambda$, the equation is

$$[(X+Y+Z)^2-4(YZ+ZX+XY)]^2=108XYZ(X+Y+Z),$$

which is, in fact, the equation of a curve having the centre, or point $X=Y=Z$, for a triple point.

To verify this, write $X=\beta-\gamma+u$, $Y=\gamma-\alpha+u$, $Z=\alpha-\beta+u$; also

$$2\Delta=(\beta-\gamma)^2+(\gamma-\alpha)^2+(\alpha-\beta)^2, \quad n=(\beta-\gamma)(\gamma-\alpha)(\alpha-\beta).$$

Then we have $X+Y+Z=3u$, $YZ+ZX+XY=3u^2-\Delta$, $XYZ=u^3-u\Delta+n$, and the equation is

$$\begin{aligned}[9u^2-9(3u^2-\Delta)]^2-324u(u^3-u\Delta+n)&=0,\\ (-2u^2+\Delta)^2-4u(u^3-u\Delta+n)&=0,\\ \Delta^2-4un&=0,\end{aligned}$$

where the lowest terms in $\beta-\gamma$, $\gamma-\alpha$, $\alpha-\beta$ are of the order 3, and the theorem is thus proved.

The case in question, $q=-\frac{1}{3}$ or $\lambda=\frac{1}{4}\Lambda$, is where the plane passes through the centre of the tetrahedron.

When $q=\frac{1}{2}$, or $\lambda=\frac{2}{3}\Lambda$, the equation is

$$(X^2+Y^2+Z^2-2YZ-2ZX-2XY)^2=128XYZ(X+Y+Z).$$

Here each of the lines $X=0$, $Y=0$, $Z=0$ is an osculating tangent having with the curve a 4-pointic intersection.

When $q=0$, or $\lambda=\frac{1}{2}\Lambda$, the equation is

$$(YZ+ZX+XY)^2-4XYZ(X+Y+Z)=0,$$

that is, $Y^2Z^2+Z^2X^2+X^2Y^2-2XYZ(X+Y+Z)=0$; viz. each angle of the triangle is here a cusp.

When $q=-\frac{1}{2}$, or $\lambda=\frac{1}{3}\Lambda$, the curve is

$$[X^2+Y^2+Z^2-2(YZ+ZX+XY)]^2=0,$$

viz. the plane is here the base of the tetrahedron, and the section is the inscribed circle taken twice.

For tracing the curves, it is convenient to find the intersections with the lines $Y-Z=0$, $Z-X=0$, $X-Y=0$ drawn from the centre of the triangle to the vertices; each of these lines passes through a node, and therefore besides meets the curve in two points.

Writing, for instance, $Y = X$, the equation becomes

$$[q^2(2X+Z)^2 - 2XZ - X^2]^2 - 4(2q+1)X^2Z(2X+Z) = 0;$$

viz. this is

$$[qZ + (2q+1)X]^2[q^2Z^2 + (2q^2-2q-1)XZ + (4q^2-4q+1)X^2] = 0,$$

where the first factor gives the node. Equating to zero the second factor, we have

$$[2q^2 + (2q-1-q)Z]^2 = Z^2[(2q-1-q)^2 - 4q^2 + 4(2q-1)] = Z^2(1-q)(1+2q);$$

or, finally,

$$q^2Z = [-2q+1 \pm 1 \pm q\sqrt{(1-q)(1+2q)}]X,$$

giving two real values for all values of q from $q = 1$ to $q = -\frac{1}{2}$. (See the Table afterwards referred to.)

We may recapitulate as follows:

$q > 1$, or $\lambda > \frac{3}{4}\Lambda$; the curve is imaginary, but with three real acnodes, answering to the acnodal parts of the nodal lines:

$q = 1$, or $\lambda = \frac{3}{4}\Lambda$; the summit appears as a fourth acnode:

$q < 1 > \frac{1}{2}$, or $\frac{3}{4}\Lambda > \lambda > \frac{2}{3}\Lambda$; the curve consists of three acnodes and a trigonoid lying within the triangle and having the sides of the triangle for bitangents of imaginary contact:

$q = \frac{1}{2}$, or $\lambda = \frac{2}{3}\Lambda$; the curve consists of three acnodes and a trigonoid having the sides of the triangle for osculating tangents:

$q < \frac{1}{2} > 0$, or $\frac{2}{3}\Lambda > \lambda > \frac{1}{2}\Lambda$; the curve consists of three conjugate points and an indented trigonoid having the sides of the triangle for bitangents of real contact:

$q = 0$, or $\lambda = \frac{1}{2}\Lambda$; curve has the summits of the triangle for cusps:

$q < 0 > -\frac{1}{2}$, or $\frac{1}{2}\Lambda > \lambda > \frac{1}{3}\Lambda$; curve has three crunodes, or say it is a cis-centric trifolium:

$q = -\frac{1}{2}$, or $\lambda = \frac{1}{3}\Lambda$; curve has a triple point, or say it is a centric trifolium:

$q < -\frac{1}{2} > -\frac{1}{3}$, or $\frac{1}{3}\Lambda > \lambda > \frac{1}{4}\Lambda$; curve has three crunodes, or say it is a trans-centric trifolium:

$q = -\frac{1}{3}$, or $\lambda = \frac{1}{4}\Lambda$; curve is a two-fold circle:

$q < -\frac{1}{3}$, or $\lambda < \frac{1}{4}\Lambda$; the curve becomes again imaginary, consisting of three acnodes answering to the acnodal parts of the nodal lines.

Table of Values

The Table shows the critical values. The columns are q, q^2, q^3, λ, and the apsidal distances and contacts for the radius vector $X = Y$, the value of $Z : X$ being calculated from the foregoing formula.

q	q^2	q^3	λ	Contact, $Z = 0$, $X : Y$	Apses, $Y = X$, $X : Z =$
.75	1.00	imposs.	1	imposs.	.032 or 7.968
.70	.666	.6666	.5	imposs.	.006 or 22.44
.65	.4285	.320	or 3.124	.006	22.44
.6	.25	.0721	13.9279	.059	67.941
.55	.1111	.011	78.988	.15	337.85
.5	0	or ∞	.25	∞	
.45	.0909	.005	118.99	.39	457.61
.4	.1666	.029	33971	.496	127.504
.35	.2308	.060	1672	.648	61.79
.3	.2857	.099	10.151	.812	37.187
.25	.3333	.141	6.854	1.25	25
.2	.375	.207	4.904	1.218	17.89
.15	.4118	.276	3.622	1.48	13.25
.10	.4444	.372	2.690	1.813	9.937
.05	.4737	.515	1.941	3.15	6.46
0	.5	1	4		

where the asterisks show the critical values of $\lambda : \Lambda$.

It is worth while to transform the equation to new coordinates X', Y', Z' such that $X' = 0$, $Y' = 0$, $Z' = 0$ represent the sides of the triangle formed by the three nodes.

Writing for shortness $X + Y + Z = P$, $YZ + ZX + XY = Q$, $XYZ = R$, the equation is $(q^2 P - Q)^2 = 4(2q + 1)PR$.

The expressions of X, Y, Z in terms of the new coordinates are of the form $X' + \theta P'$, $Y' + \theta P'$, $Z' + \theta P'$, where $P' = X' + Y' + Z'$; writing also $Q' = Y'Z' + Z'X' + X'Y'$, $R' = X'Y'Z'$, then the values of P, Q, R are $(1+3\theta)P'$, $Q'+(2\theta+3\theta^2)P'^2$, $R'+\theta Q'P'+(\theta^2+\theta^3)P'^3$.

And the transformed equation is

$$\begin{aligned}\big[q^2(1+3\theta)P'^2 - Q' - (2\theta+3\theta^2)P'^2\big]^2 \\ - 4(2q+1)(1+3\theta)P'\big[R' + \theta Q'P' + (\theta^2+\theta^3)P'^3\big] = 0,\end{aligned}$$

which is satisfied by $Q' = 0$, $R' = 0$, if only

$$\big[q^2(1+3\theta) - 2\theta - 3\theta^2\big]^2 = 4(2q+1)(1+3\theta)(\theta^2+\theta^3),$$

or, if for a moment $q(1 + 3\theta) = \mu$, the equation is

$$(\mu^2 - 2\theta - 3\theta^2)^2 = 4(\theta^2+\theta^3)(2\mu + 1 + 3\theta),$$

that is,

$$\mu^4 + \mu^2(-6\theta^2 - 4\theta) + \mu(-8\theta^3 - 8\theta^2) - 3\theta^4 - 4\theta^3 = 0,$$

that is,

$$(\mu+\theta)^2(\mu^2 - 2\theta\mu - 3\theta^2 - 4\theta) = 0.$$

If the new axes pass through the nodes, then $3\theta + 1 = 0$; that is, $q^2(1 + 3\theta) + \theta = 0$, which equation gives the value of θ for which the new axes have the position in question; substituting in the first instance for q the value $\dfrac{q}{3q+1}$, the equation becomes

$$[2\theta(1+\theta)P'^2 + Q']^2 = 4(1+\theta)P'[\theta^2(1+\theta)P'^2 + \theta Q'P' + R'],$$

that is, $Q'^2 = 4(1 + \theta)P'R'$; or, finally, substituting for θ its value in terms of q, the required equation is

$$(Y'Z' + Z'X' + X'Y')^2 = 4\frac{q+1}{3q+1}X'Y'Z'(X' + Y' + Z').$$

In particular, for $q = 0$ the equation is

$$(Y'Z' + Z'X' + X'Y')^2 - 4X'Y'Z'(X' + Y' + Z') = 0,$$

which is right, since, in the case in question (the tricuspidal curve), we have X, Y, $Z = X'$, Y', Z'.

I remark, in passing, that, taking the equation to be

$$(Y'Z' + Z'X' + X'Y')^2 = mX'Y'Z'(X' + Y' + Z'),$$

we may write herein $Z' = 1$, and then the equation takes the form $(Y' + X' + X'Y')^2 = mX'Y'(1 + X' + Y')$; or, what is the same thing, $\sqrt{mX'} + \sqrt{mY'} + \sqrt{X'Y'} + 1 = 0$.

The curve may be represented as a unicursal quartic; in fact, taking α, β, γ as the roots of

$$(\alpha-\theta)(\beta-\theta)(\gamma-\theta)=\theta^3-u\theta^2+v\theta-w=0,$$

we may write

$$x=\frac{\sqrt{m}\sqrt{(m-3)^3}}{2\sqrt{m(m-3)}}\cos\theta+\frac{2(m-3)}{m}\cos 2\theta,$$
$$y=\frac{\sqrt{m}\sqrt{(m-3)^3}}{2\sqrt{m(m-3)}}\sin\theta+\frac{2(m-3)}{m}\sin 2\theta,$$

which are the formulae for the description of the trinodal quartic as a unicursal curve.

I consider now the second position; viz. the horizontal plane now passes through the centre of the tetrahedron, and is parallel to two opposite edges. The equations of the nodal lines are here $(y = 0,\ z = 0)$, $(x = 0,\ z = 0)$, $(x = 0,\ y = 0)$; and if for convenience we assume the distance of the mid-points of opposite edges to be $= 2$, or the half of this $= 1$, then the equations of the faces are

$$\begin{aligned} X &= x + y + z - 1 = 0, \\ Y &= -x - y + z - 1 = 0, \\ Z &= x - y - z - 1 = 0, \\ W &= -x + y - z - 1 = 0, \end{aligned}$$

and the equation of the surface is $\sqrt{X} + \sqrt{Y} + \sqrt{Z} + \sqrt{W} = 0$.

Proceeding to rationalise, this is $X + Y + 2\sqrt{XY} = Z + W + 2\sqrt{ZW}$, viz.

$$2z + 2\sqrt{XY} = 2\sqrt{ZW};$$

we thence have $4z^2 + 4z\sqrt{XY} + XY = ZW$; or, since $ZW - XY = 4z + 4xy$, this is $4z^2 + 4z\sqrt{XY} + XY = 4z + 4xy + XY$, whence

$$z + \sqrt{XY} - z = \sqrt{ZW};$$

$$\begin{aligned} (z + xy - z^2)^2 &= z^2 - \tfrac{1}{4}(z-1)^2 - (x^2 - y^2)^2; \\ 2xyz + \tfrac{1}{2}z^2 &+ \tfrac{1}{2}z(x^2 + y^2) = 0, \end{aligned}$$

a form which puts in evidence the nodal lines.

Considering z as constant, we have the equation of the section; this is a quartic having the node $(x = 0,\ y = 0)$, and two other nodes at infinity on the two axes respectively; moreover, the curve has for bitangents the intersections of its plane with the faces of the tetrahedron; or what is the same thing, attributing to z its constant value, the equations of the bitangents are

$$x+y+z-1 = 0, \quad -x-y+z-1 = 0, \quad x-y-z-1 = 0, \quad -x+y-z-1 = 0.$$

Observe that when $c = 1$, we have so that the only real point is $x = 0$, $y = 0$; viz. this is a tacnode, having the real tangent $x + y = 0$.

For $c = 0$ (central section) the equation becomes $x^2y^2 = 0$; viz. the curve is here the two nodal lines each twice.

It is now easy to trace the changes of form.

$c = 1$; curve is a tacnode, as just mentioned, tangent the dexter diagonal.

$c < 1 > 0$; the tacnode breaks up into two conjugate points on the dexter diagonal, and the curve consists of a loop (enclosed oval) at the origin, and two branches each extending to infinity.

$c = 0$; the loop is reduced to zero, and the two branches are each the axis of x or y taken twice; the only real points of the curve are on the axes.

$c < 0 > -1$; the curve consists of four infinite branches, two on each side of each axis.

$c = -1$; each of the four branches has a cusp at infinity, viz. the line infinity is an osculating tangent.

$c < -1 > -3$; the infinite branches have each two (real or imaginary) inflexions; there is a crunode at infinity on each axis.

$c = -3$; the crunodes unite in a triple point at infinity.

$c < -3$; the crunodes disappear (are acnodes), and each of the four branches has again two (real or imaginary) inflexions.

The nodal lines of the surface are given by the equations $(y = 0,\ z = 0)$, $(x = 0,\ z = 0)$, $(x = 0,\ y = 0)$. To find the tangent cone at a point on a nodal line, we may proceed as follows.

The general equation of the tangent cone at any point (x_0, y_0, z_0) of a surface $F(x, y, z) = 0$ is obtained by writing $x_0 + \frac{X}{W}$, $y_0 + \frac{Y}{W}$, $z_0 + \frac{Z}{W}$ for x, y, z, and then equating to zero the terms of the lowest order in X, Y, Z, W.

For the surface in question, the equation is

$$F = (z + xy - z^2)^2 - z^2 + \tfrac{1}{4}(z - 1)^2(x^2 - y^2)^2 = 0.$$

The tangent cone at any point of the nodal line $y = 0$, $z = 0$ (so that x_0 is arbitrary) is obtained by writing $x_0 + X$, Y, Z for x, y, z; the equation becomes

$$(Z + x_0Y + XY - Z^2)^2 - Z^2 + \tfrac{1}{4}(Z + x_0^2 - 1)^2(X^2 - Y^2)^2 = 0,$$

and taking the terms of the lowest order, we have

$$(Z + x_0Y)^2 - Z^2 + \tfrac{1}{4}(x_0^2 - 1)^2(X^2 - Y^2)^2 = 0.$$

Equating to zero the discriminant in regard to Z, we have

$$4x_0^2(X^2 - Y^2)^2[(x_0^2 + 1)Y^2 + X^2] - (x_0^2 - 1)^2(X^2 - Y^2)^4 = 0.$$

That is,

$$4x_0^2[(x_0^2 + 1)Y^2 + X^2] - (x_0^2 - 1)^2(X^2 - Y^2)^2 = 0;$$

and substituting herein the values $X = \dfrac{-x_0}{x_0^2 - 1}$ and $Y = \dfrac{1}{x_0^2 - 1}$, we have the equation of the cone, viz. this is

$$\gamma(x^2 + 2y^2 - 1) + 2\gamma(xy + y^2) - \tfrac{1}{4} = 0;$$

or, what is the same thing,

$$\gamma(x^2 + 2y^2 - 1) + 2\gamma(xy + y^2) - \tfrac{1}{4} = 0.$$

This is a quadric cone having for its principal planes $y = 0$ (plane through the nodal line and the centre) and $x + y = 0$, $x - y = 0$ (planes through the nodal line and the vertices of the tetrahedron).

The section by the plane $x = 0$ is the conic $2\gamma y^2 + 2\gamma y^2 - \frac{1}{4} = 0$, that is $4\gamma y^2 = \frac{1}{4}$, or $y = \pm\frac{1}{4\sqrt{\gamma}}$; and similarly for the other principal sections. The cone touches the plane $z = 0$ along the line $x + y = 0$.

The analytical theory is thus completed for the two positions of the tetrahedron; and we have a complete discussion of the form of the sections by planes parallel to the horizontal plane. The results are in accordance with the model and drawings which I have constructed.

The general equation of Steiner's surface may be written in the form

$$\alpha^2+\beta^2+\gamma^2+\delta^2-2\beta\gamma-2\gamma\alpha-2\alpha\beta-2\alpha\delta-2\beta\delta-2\gamma\delta=0,$$

where α, β, γ, δ are linear functions of the coordinates. The four singular tangent planes are $\alpha=0$, $\beta=0$, $\gamma=0$, $\delta=0$; the three nodal lines are given by the equations $\alpha=\beta=\gamma+\delta$, $\beta=\gamma=\alpha+\delta$, $\gamma=\alpha=\beta+\delta$; and the surface has the property that any plane through a nodal line meets the surface in two conics.

557.

ON CERTAIN CONSTRUCTIONS FOR BICIRCULAR QUARTICS.

[From the Proceedings of the London Mathematical Society, vol. v. (1873–1874), pp. 29–31. Read March 12, 1874.]

I call to mind that if F, G are any two points and F', G' their antipoints; then the circle on the diameter FG and that on the diameter $F'G'$ are concentric orthotomics, viz. they have the same centre, and the sum of the squared radii is $= 0$.

Moreover, if the circles B, B' are concentric orthotomics, and the circle A is orthotomic to B, then it is a bisector of B', viz. it cuts B' at the extremities of a diameter of B'; and B is then said to be a bifid circle in regard to A.

Given two real circles, these have an axial orthotomic, the circle, centre on the line of centres at its intersection with the radical axis, which cuts at right angles the given circles; viz. this axial orthotomic is real if the circles have no real intersection; but if the intersections are real, then the axial orthotomic is a pure imaginary.

The above remarks have an obvious application to the theory of bicircular quartics; viz. a bicircular quartic is the envelope of a variable circle, having its centre on a conic, and orthotomic to a circle: it may be that this circle is a pure imaginary. We then replace it by the concentric orthotomic, and say that the curve is the envelope of a variable circle having its centre on a conic and bisecting a circle. We have thus a real form for cases which originally present themselves under an imaginary form.

The Bicircular Quartic with given vertices.

First, if the vertices are all real, say f, g, h, k (these being the foci or double foci of the curve); then taking F, G, H, K to be the points whose coordinates are f, g, h, k respectively, and F', G' the antipoints of F, G, and H', K' the antipoints of H, K; we have the two concentric orthotomic circles $F'G'$ and $H'K'$.

The axial orthotomic of these two circles is real (since the circles have no real intersection), and calling this the circle Ω, the curve is the envelope of a variable circle having its centre on a conic and orthotomic to Ω.

To obtain the conic, we remark that the vertices f, g belong to the circle $F'G'$, and h, k to the circle $H'K'$; and if on the line of centres (which is also the radical axis of the two circles) we describe a conic having $F'G'$ and $H'K'$ as its director circles, then the variable circle is orthotomic to Ω, and its centre describes the conic. The conic is in fact a central conic, the axes of which are the line of centres and the perpendicular thereto through the centre.

If the conic is an ellipse, the curve is a simple oval; if it is a hyperbola, the curve consists of two ovals.

Secondly, if the vertices are two real, two imaginary, say f, $g = \alpha \pm \beta i$; h, k, we modify the first or third construction; viz. if F', G' are the antipoints of F, G; then on $F'G'$ as diameter describe a circle, and on HK as diameter a circle. On the line terminated by the two centres (as transverse or conjugate axis) describe a conic Θ_1, and describe the axial bisector-orthotomic circle Ω_1 of the two circles; viz. this is the circle (centre on the axis of symmetry) which bisects the circle $F'G'$, and cuts at right angles the circle HK; then the curve is the envelope of the variable circle having its centre on Θ_1 and orthotomic to Ω_1.

Thirdly, if the vertices are all imaginary, say f, $g = \alpha \pm \beta i$; h, $k = \gamma \pm \delta i$, we modify the first or third construction in like manner.

The curve is in this case quadrantal: having, besides the original axis of symmetry, another axis of symmetry through O, at right angles thereto.

558.

A GEOMETRICAL INTERPRETATION OF THE EQUATIONS OBTAINED BY EQUATING TO ZERO THE RESULTANT AND THE DISCRIMINANTS OF TWO BINARY QUANTICS.

[From the Proceedings of the London Mathematical Society, vol. v. (1873–1874), pp. 31–33. Read March 12, 1874.]

Consider the equations $U = (a, b, \ldots \wr x, 1)^{\lambda} = 0$, $U' = (a', b', \ldots \wr x, 1)^{\lambda'} = 0$; and equating to zero the discriminants of the two functions respectively, and also the resultant of the two functions, let the equations thus obtained be

$$\Delta = (a, b, \ldots)^{\lambda-2} = 0, \quad \Delta' = (a', b', \ldots)^{\lambda'-2} = 0, \quad R = (a, b, \ldots)^{\lambda}(a', b', \ldots)^{\lambda'} = 0.$$

I take $(a, b, \ldots)$, $(a', b', \ldots)$ to be linear functions of the coordinates (x, y, z); and t to be an indeterminate parameter. Hence $U = 0$ represents a line the envelope whereof is the curve $\Delta = 0$, or, what is the same thing, the equation $U = 0$ represents any tangent of the curve $\Delta = 0$; this is a unicursal curve of the order $2\lambda - 2$ and class λ, with $3(\lambda - 2)$ cusps and $\frac{1}{2}(\lambda - 2)(\lambda - 3)$ nodes.

Similarly $U' = 0$ represents a line the envelope whereof is the curve $\Delta' = 0$, a unicursal curve of the order $2\lambda' - 2$ and class λ', with $3(\lambda' - 2)$ cusps and $\frac{1}{2}(\lambda' - 2)(\lambda' - 3)$ nodes.

The envelope of the line which is the join of the corresponding points of contact of the two curves is the curve $R = 0$, a unicursal curve of the order $\lambda + \lambda'$, with $\frac{1}{2}(\lambda + \lambda' - 1)(\lambda + \lambda' - 2)$ nodes and no cusps; consequently of the class $2(\lambda + \lambda' - 1)$.

It is to be shown that the curve $R = 0$ touches the curve $\Delta = 0$ in $\lambda' + 2\lambda - 2$ points, and similarly the curve $\Delta' = 0$ in $2\lambda' + \lambda - 2$ points.

In fact, consider any tangent T' of the curve Δ'; let this meet the curve Δ in a point A, and let Q be the tangent at A to the curve Δ; suppose, moreover, that T is the tangent of Δ corresponding to the tangent T' of Δ'. Then if Q and T coincide, the corresponding tangent of T' will be Q, and the curve R will pass through A. It is easy to see that in this case the curves R, Δ will touch at A.

Again, if P be a tangent from A to the curve Δ, then, if P and T coincide, the corresponding tangent of T' will be P, and the curve R will pass through A; but in this case the point A will be a mere intersection, not a point of contact, of the two curves.

Hence the points of contact of R with Δ correspond to the tangents T' which touch Δ, that is, to the $2(\lambda - 1)$ tangents from the cusps of Δ' to the curve Δ, together with the λ' common tangents of Δ and Δ'; in all $2(\lambda - 1) + \lambda' = \lambda' + 2\lambda - 2$ points of contact. And similarly the number of points of contact of R with Δ' is $2\lambda' + \lambda - 2$.

559.

[NOTE ON INVERSION.]

[From the Proceedings of the London Mathematical Society, vol. v. (1873–1874), p. 112.]

The inverse of the anchor ring (in the foregoing paper called the cyclide) is in fact the general binodal cyclide or binodal bicircular quartic; viz. assuming it to be a cyclide (bicircular quartic), to see that it is binodal, it need only be observed that the anchor ring is binodal (has two real or imaginary conic points, viz. these are the intersections of the circles in the several axial planes); and to see that it is the general binodal cyclide, we have only to count the constants; viz. the general cyclide or surface

$$(x^2+y^2+z^2)^2+(x^2+y^2+z^2)(\alpha x+\beta y+\gamma z) \\ +(a,b,c,d,f,g,h,l,m,n)(x,y,z,1)^2=0$$

contains 13 constants, and therefore the binodal cyclide $13-2=11$ constants. But the anchor ring, irrespective of position, contains 2 constants; centre of inversion, taken in given axial plane, has 2 constants; radius of inversion, 1 constant; in all $2+2+1=5$ constants for the form, and 6 for the position, in all 11 constants, the same number as for the general binodal cyclide.

560.

[ADDITION TO LORD RAYLEIGH'S PAPER "ON THE NUMERICAL CALCULATION OF THE ROOTS OF FLUCTUATING FUNCTIONS."]

[From the Proceedings of the London Mathematical Society, vol. v. (1873–1874), pp. 123, 124. November 22.]

Prof. Cayley, to whom Lord Rayleigh's paper was referred, pointed out that a similar result may be attained by a method given in a paper by Encke, "Allgemeine Aufloesung der numerischen Gleichungen," Crelle, t. xxii. (1841), pp. 193–248, as follows:

Taking the equation

$$0 = 1 - ax + bx^2 - cx^3 + dx^4 - ex^5 + fx^6 - gx^7 + hx^8 - \dots;$$

if the equation whose roots are the squares of these is

$$0 = 1 - a_1x + b_1x^2 - c_1x^3 + \dots,$$

then

$$\begin{aligned} a_1 &= a^2 - 2b, \\ b_1 &= b^2 - 2ac + 2d, \\ c_1 &= c^2 - 2bd + 2ae - 2f, \\ d_1 &= d^2 - 2ce + 2bf - 2ag + 2h, \end{aligned}$$

&c.; and we may in the same way derive a_2, b_2, c_2, &c. from a_1, b_1, c_1, &c., and so on.

As regards the function

$$\phi^n(x) = \frac{2^n \cdot n!}{x^n} \{J_n(x)\} = 1 - \frac{x^2}{2 \cdot 2n+2} + \frac{x^4}{2 \cdot 4 \cdot 2n+2 \cdot 2n+4} - \dots,$$

we have as follows:

$$
\begin{aligned}
a^{-1} &= 2^1 \cdot n+1, \\
b^{-1} &= 2^2 \cdot n+1 \cdot n+2, \\
c^{-1} &= 2^3 \cdot 3 \cdot n+1 \ldots n+3, \\
d^{-1} &= 2^4 \cdot 3 \cdot n+1 \ldots n+4, \\
e^{-1} &= 2^5 \cdot 3 \cdot 5 \cdot n+1 \ldots n+5, \\
f^{-1} &= 2^6 \cdot 3^2 \cdot 5 \cdot n+1 \ldots n+6, \\
g^{-1} &= 2^7 \cdot 3^2 \cdot 5 \cdot 7 \cdot n+1 \ldots n+7, \\
h^{-1} &= 2^8 \cdot 3^2 \cdot 5 \cdot 7 \cdot n+1 \ldots n+8,
\end{aligned}
$$

and

$$
\begin{aligned}
a_1^{-1} &= 2^4 \cdot (n+1)^2 \cdot n+2, \\
b_1^{-1} &= 2^7 \cdot (n+1 \cdot n+2)^2 \cdot n+3 \cdot n+4, \\
c_1^{-1} &= 2^{10} \cdot 3 \cdot (n+1 \ldots n+3)^2 \cdot n+4 \ldots n+6, \\
d_1^{-1} &= 2^{13} \cdot 3 \cdot (n+1 \ldots n+4)^2 \cdot n+5 \ldots n+8, \\
a_2 &= \frac{5n^2+23n+42}{2^8 \cdot (n+1)^2 \cdot (n+2)^2 \cdot n+3 \cdot n+4}, \\
b_2 &= \frac{429n^4+7640n^3+53702n^2+185430n+311387n+202738}{2^{16} \cdot (n+1)^8 \cdot (n+2)^4 \cdot (n+3 \cdot n+4)^2 \cdot n+5 \cdot n+6 \cdot n+7 \cdot n+8}.
\end{aligned}
$$

If $n=0$, whence

$$
p_1 = a_2 = \frac{101369}{2^{15} \cdot 3^2 \cdot 5 \cdot 7} = p_1 \text{ suppose;}
$$

$$
p_1 = 2.404825.
$$

[The quantities p_1, p_2, $\ldots$ are the roots of the function $J_n(x)$ in increasing order of magnitude, so that, as these roots are all real, it follows that for $J_0(x)$,

$$
a = 2p_1^{-2}, \quad a_1 = 2p_1^{-4}, \quad a_2 = 2p_1^{-6}, \quad a_3 = 2p_1^{-8}, \quad \ldots]
$$

[561
]

561.

ON THE GEOMETRICAL REPRESENTATION OF CAUCHY'S THEOREMS OF ROOT-LIMITATION.

[From the Transactions of the Cambridge Philosophical Society, vol. xii. Part ii. (1877), pp. 395–413. Read February 16, 1874.]

There is contained in Cauchy's Memoir "Calcul des Indices des Fonctions," Journ. de l'Ecole Polytech. t. xv. (1837) a general theorem, which, though including a well-known theorem in regard to the imaginary roots of a numerical equation, seems itself to have been almost lost sight of.

In the general theorem (say Cauchy's two-curve theorem) we have in a plane two curves $P=0$, $Q=0$, and the real intersections of these two curves, or say the "roots," are divided into two sets according as the Jacobian $\partial_x P \cdot \partial_y Q - \partial_x Q \cdot \partial_y P$ is positive or negative, say these are the Jacobian-positive and the Jacobian-negative roots: and the question is to determine for the roots within a given contour or circuit, the difference of the numbers of the roots belonging to the two sets respectively.

I reproduce the whole theory, presenting it under a completely geometrical form, viz. I establish between the two sets of roots the distinction of right- and left-handed: and (availing myself of a notion due to Prof. Sylvester) I give a geometrical form to the theoretic rule, making it depend on the "intercalation" of the intersections of the two curves with the circuit: I also complete the Sturmian process in regard to the sides of the rectangle: the memoir contains further researches in regard to the theory.

1. Right- and Left-handed Roots.

Consider any point of intersection of the two curves. There will be about this point four regions, sable and argent being opposite to each other, as also gules and azure; whence selecting an order sable, gules, argent, azure; if to have the colours in this order we have to go about the point, or root, right-handedly, the root is right-handed:

but if to have the colours in this order we have to go about the root left-handedly, the root is left-handed.

It is easy to see that if the Jacobian is positive, the root is right-handed; and if the Jacobian is negative, the root is left-handed.

2. The Circuit.

I consider in general a circuit not containing within it any root; as a simple example let the circuit lie altogether in a sable region: the intercalation is here $= 0$, and the circuit is said to be neutral.

If the circuit contains within it a single root, and if this is right-handed, the intercalation is $+P - Q$; and the Index of the circuit, $=$ number of right-handed roots minus number of left-handed roots, is $= +1$.

3. Intercalation-theory for a right line.

Considering then the case where the trajectory is a line parallel to the axis of X, P and Q will denote given rational functions of x; the curves $P = 0$, $Q = 0$ being of course each of them a set of right lines parallel to the axis of y: the regions will be bands each of them included between two consecutive lines $P = 0$, or between two consecutive lines $Q = 0$.

A band included between two consecutive lines $P = 0$ is a P-band; and so a band included between two consecutive lines $Q = 0$ is a Q-band. In regard to a P-band, the signs of P on the two sides thereof are $+$ and -, or - and $+$; we may distinguish these as the $+P$ side and the -P side respectively: and the like as regards a Q-band.

4. Bicolumns.

We have a bicolumn: that is, a double column of signs; say the first or P-column, consisting of the signs of P at the several roots of $Q = 0$; and the second or Q-column, consisting of the signs of Q at the several roots of $P = 0$; the roots are here taken in order of increasing magnitude, and the bicolumn is read downwards.

The intercalation of the bicolumn is derived from the two columns taken separately; the P-column contributes only P's, the Q-column contributes only Q's: the two letters are written down in the order in which they present themselves; the signs $+$ and - are attributed as above: and we have thus the intercalation.

5. Gain of a bicolumn.

The gain of a bicolumn is the number of the letters of the intercalation taken with the proper signs. The rule for the signs is: for the first letter the sign is +, and the signs are alternately + and -: but the signs of the P's and the signs of the Q's are each of them independently + and - according to the sign of the letter in the intercalation.

As regards the final result, a bicolumn may be replaced by its gain, or single letter with its sign.

6. Decomposition of a bicolumn.

A bicolumn may be divided in any manner into parts, taking always the last row of any part as being also the first row of the next succeeding part. This being so, the gain of the whole bicolumn is equal to the sum of the gains of its parts.

In a bicolumn of two rows, if we reverse either row (that is, write therein $-$ for $+$ and $+$ for $-$), we reverse the signs of the letters contributed by the reversed column, and leave the gain unaltered—of course the intercalation (considered irrespectively of sign) is in each case unaltered.

It is convenient to take the signs in such manner that for the initial value of x, the signs of P, Q shall be each positive: or, what is the same thing, taking P, Q with their proper signs, we may in the bicolumn, by reversing if necessary, take the signs of the first row each positive.

7. Application to Sturm's theorem.

For a rectangle the sides of which are parallel to the axes, the index is equal to the gain of the bicolumn formed by the signs at the corners: viz. reading the corners in order, beginning with the left-hand corner of the under side, and going round right-handedly, we have a bicolumn of four rows: and the index is equal to the gain of this bicolumn.

The index is thus equal to the sum of the gains of the two side-bicolumns and the two terminal columns; and we have thence the expression for the index in the form required for the application of Sturm's theorem.

8.

Consider a circuit not containing within it any root; as a simple example let the circuit lie altogether in a sable region: the intercalation is here $= 0$, and the circuit is said to be neutral.

If the circuit contains within it a single root, and if this is right-handed, the intercalation is $+P - Q$; and the Index of the circuit, $=$ number of right-handed roots minus number of left-handed roots, is $= +1$.

If the circuit contains within it any number of right-handed roots, and any number of left-handed roots, then the Index is equal to the number of right-handed roots less the number of left-handed roots.

9.

In the case of a circuit not containing any root, the intercalation is either $(+P-Q)$, say this is a positive circuit, or else $(-P+Q)$, say this is a negative circuit. There is of course the neutral circuit (PQ) for which the intercalation vanishes.

10.

Consider a circuit not containing within it any root; as a simple example let the circuit lie altogether in a sable region: the intercalation is here $= 0$, and the circuit is said to be neutral.

If the circuit contains within it a single root, and if this is right-handed, the intercalation is $+P - Q$; and the Index of the circuit, $=$ number of right-handed roots minus number of left-handed roots, is $= +1$.

11.

Consider a circuit not containing within it any root; as a simple example let the circuit lie altogether in a sable region: the intercalation is here $= 0$, and the circuit is said to be neutral.

If the circuit contains within it a single root, and if this is right-handed, the intercalation is $+P - Q$; and the Index of the circuit, $=$ number of right-handed roots minus number of left-handed roots, is $= +1$.

12.

In the second case, that is, when the intercalations are contrary, they counteract each other in forming the intercalation of the circuit: it is the difference of the numbers of letters, or twice the difference of the indices, which is evenly even, and the half of this, or the difference of the indices, which is the index of the circuit: one intercalation contains a number of letters greater by an odd number than the number of letters in the other intercalation, and the index of the circuit is the half of an odd number, that is, it is a fraction with denominator 2, or say it is a fractional index: this can only happen when the circuit passes through a root.

13.

Suppose CA similar to ABC, and AG similar to CDA; then $ABCA$ is also similar to ABC, and $AGDA$ similar to CDA; viz. ABC, GA and $ABCA$ are similar to each other, and contrary to AC, CDA, $ACDA$ which are also similar to each other.

Also

$$\text{Ind. } ABCDA = \text{Ind. } ABC + \text{Ind. } CDA,$$
$$\text{Ind. } ABCA = \text{Ind. } ABC + \text{Ind. } CA,$$
$$\text{Ind. } ACDA = \text{Ind. } CDA + \text{Ind. } AC,$$

and thence

$$\text{Ind. } ABCDA = \text{Ind. } ABCA + \text{Ind. } ACDA.$$

14.

The theory of intercalations as thus applied to a polygon gives a complete geometrical representation of Cauchy's theorem for any contour whatever. For a rectangle the sides of which are parallel to the axes, the index is equal to the gain of the bicolumn formed by the signs at the corners: viz. reading the corners in order, beginning with the left-hand corner of the under side, and going round right-handedly, we have a bicolumn of four rows: and the index is equal to the gain of this bicolumn.

The index is thus equal to the sum of the gains of the two side-bicolumns and the two terminal columns; and we have thence the expression for the index in the form required for the application of Sturm's theorem.

15.

Consider a circuit not containing within it any root; as a simple example let the circuit lie altogether in a sable region: the intercalation is here $= 0$, and the circuit is said to be neutral.

If the circuit contains within it a single root, and if this is right-handed, the intercalation is $+P - Q$; and the Index of the circuit, $=$ number of right-handed roots minus number of left-handed roots, is $= +1$.

16.

Suppose CA similar to ABC, and AG similar to CDA; then $ABCA$ is also similar to ABC, and $AGDA$ similar to CDA; viz. ABC, GA and $ABCA$ are similar to each other, and contrary to AC, CDA, $ACDA$ which are also similar to each other.

Also

$$\begin{aligned} \text{Ind. } ABCDA &= \text{Ind. } ABC + \text{Ind. } CDA, \\ \text{Ind. } ABCA &= \text{Ind. } ABC + \text{Ind. } CA, \\ \text{Ind. } ACDA &= \text{Ind. } CDA + \text{Ind. } AC, \end{aligned}$$

and thence

$$\text{Ind. } ABCDA = \text{Ind. } ABCA + \text{Ind. } ACDA.$$

17.

In the case where the circuit is a polygon, and most easily when it is a rectangle the sides of which are parallel to the two axes respectively, the excess in question can be actually determined by means of an application of Sturm's theorem successively to each side of the polygon, or rectangle.

18. Intercalation-theory for a right line.

Considering then the case where the trajectory is a line parallel to the axis of X, P and Q will denote given rational functions of x; the curves $P = 0$, $Q = 0$ being of course each of them a set of right lines parallel to the axis of y: the regions will be bands each of them included between two consecutive lines $P = 0$, or between two consecutive lines $Q = 0$.

A band included between two consecutive lines $P = 0$ is a P-band; and so a band included between two consecutive lines $Q = 0$ is a Q-band. In regard to a P-band, the signs of P on the two sides thereof are + and -, or - and +; we may distinguish these as the $+P$ side and the -P side respectively: and the like as regards a Q-band.

19.

We have a bicolumn: that is, a double column of signs; say the first or P-column, consisting of the signs of P at the several roots of $Q = 0$; and the second or Q-column, consisting of the signs of Q at the several roots of $P = 0$; the roots are here taken in order of increasing magnitude, and the bicolumn is read downwards.

20.

The intercalation of the bicolumn is derived from the two columns taken separately; the P-column contributes only P's, the Q-column contributes only Q's: the two letters are written down in the order in which they present themselves; the signs + and - are attributed as above: and we have thus the intercalation.

21. A bicolumn may be divided in any manner.

A bicolumn may be divided in any manner into parts, taking always the last row of any part as being also the first row of the next succeeding part. This being so, the gain of the whole bicolumn is equal to the sum of the gains of its parts.

In a bicolumn of two rows, if we reverse either row (that is, write therein $-$ for $+$ and $+$ for $-$), we reverse the signs of the letters contributed by the reversed column, and leave the gain unaltered—of course the intercalation (considered irrespectively of sign) is in each case unaltered.

It is convenient to take the signs in such manner that for the initial value of x, the signs of P, Q shall be each positive: or, what is the same thing, taking P, Q with their proper signs, we may in the bicolumn, by reversing if necessary, take the signs of the first row each positive.

22.

The gain of a bicolumn is the number of the letters of the intercalation taken with the proper signs. The rule for the signs is: for the first letter the sign is $+$, and the signs are alternately $+$ and -: but the signs of the P's and the signs of the Q's are each of them independently $+$ and - according to the sign of the letter in the intercalation.

As regards the final result, a bicolumn may be replaced by its gain, or single letter with its sign.

23.

We may, in regard to the final result, replace any three consecutive rows by a single row containing the product of the signs: the bicolumn is thus reduced to a bicolumn of two rows, the gain of which is equal to the gain of the original bicolumn.

24.

The application to Sturm's theorem is immediate. For a rectangle the sides of which are parallel to the axes, the index is equal to the gain of the bicolumn formed by the signs at the corners: viz. reading the corners in order, beginning with the left-hand corner of the under side, and going round right-handedly, we have a bicolumn of four rows: and the index is equal to the gain of this bicolumn.

The index is thus equal to the sum of the gains of the two side-bicolumns and the two terminal columns; and we have thence the expression for the index in the form required for the application of Sturm's theorem.

[End of Papers 556–561, covering pages 1–30 of Volume IX, corresponding to pages 1–50 of the PDF source.]

Transcription note: The source PDF contains significant OCR degradation. Mathematical formulae have been reconstructed from context. The original papers continue beyond this point in the source volume.

The Collected Mathematical Papers of Arthur Cayley, Vol. IX (1874–1877)

Pages 51–100

566. ON THE TRANSFORMATION OF THE EQUATION OF A SURFACE TO A SET OF CHIEF AXES.

[From the Quarterly Journal of Pure and Applied Mathematics, vol. xii. (1873), pp. 34–38.]

. . .

Moreover, p_1, p_2 being thus determined, we have a_1, β_1, γ_1, d_1 proportional to the determinants formed with the matrix

$$\begin{array}{ccccc} 1 & , & h & , & g \\ A & , & h & , & f \\ B & , & g & , & f \\ C & , & 1 & , & c \end{array}$$

say $a_1 = k\mathfrak{A}_1$, $\beta_1 = k\mathfrak{B}_1$, $\gamma_1 = k\mathfrak{C}_1$, $d_1 = k\mathfrak{D}_1$, where $\mathfrak{A}_1$, $\mathfrak{B}_1$, $\mathfrak{C}_1$, $\mathfrak{D}_1$ are the determinants in question; and then $1 = k^2(\mathfrak{A}_1^2 + \mathfrak{B}_1^2 + \mathfrak{C}_1^2)$, or we have $k^2 = 1/(\mathfrak{A}_1^2 + \mathfrak{B}_1^2 + \mathfrak{C}_1^2)$.

But we find at once that

$$(a, \ldots)(a_1, \beta_1, \gamma_1)^2 = \frac{\mathfrak{V}_1}{\sqrt{(2\mathfrak{U}_1^2 + 3\mathfrak{B}_1^2 + \mathfrak{C}_1^2)}},$$

and in the same manner

$$(a, \ldots)(a_1, \beta_1, \gamma_1)(a_2, \beta_2, \gamma_2) = \frac{\mathfrak{V}_2}{\sqrt{(3\mathfrak{U}_2^2 + 3\mathfrak{B}_2^2 + \mathfrak{C}_2^2)}}.$$

Hence the transformed equation is

$$\frac{\mathfrak{V}_1^2}{2\mathfrak{U}_1^2 + 3\mathfrak{B}_1^2 + \mathfrak{C}_1^2} X^2 + \frac{\mathfrak{V}_2^2}{3\mathfrak{U}_2^2 + 3\mathfrak{B}_2^2 + \mathfrak{C}_2^2} Y^2 + Z^2$$
$$+ 2YZ \frac{\mathfrak{V}_3}{\sqrt{4}} + \text{\&c.} = 0,$$

where it will be recollected that $\mathfrak{V} = \sqrt{(A^2 + B^2 + C^2)}$. The &c. refers as before to the terms $(X, Y, Z)^3$ and higher powers, which are obtained from the corresponding terms in hx, ky, lz, by substituting for these their values $hx = aL_1X + aL_2Y + aL_3Z$, &c., where the coefficients have the values above obtained for them.

It will be observed, that the radii of curvature are $\sqrt{p_1}$, $\sqrt{p_2}$, and that the process includes an investigation of the values of these radii of curvature similar to the ordinary one; the novelty is in the terms in XZ, YZ, and Z^2.

But regarding X, Y as small quantities of the first order, Z is of the second order, and the terms in XZ, YZ are of the third order, and that in Z^2 of the fourth order.

567. ON AN IDENTICAL EQUATION CONNECTED WITH THE THEORY OF INVARIANTS.

[From the Quarterly Journal of Pure and Applied Mathematics, vol. xii. (1873), pp. 115–118.]

Write

$$a = g - h, \quad b = h - f, \quad c = f - g,$$

equations implying a fourth equation forming with them the system

$$\begin{aligned} -h + g - a &= 0, \\ h - f - b &= 0, \\ -g + f - c &= 0, \\ a + b + c \quad &= 0, \end{aligned}$$

and also

$$af + bg + ch = 0.$$

Then, putting for shortness

$$\begin{aligned} P &= (bg - ch)(ch - af)(af - bg), \\ Q &= a^2g^2h^2 + b^2h^2f^2 + c^2f^2g^2 + a^2b^2c^2, \\ R &= a^2f^2(a^2 + f^2) + b^2g^2(b^2 + g^2) + c^2h^2(c^2 + h^2), \end{aligned}$$

we have

$$2P + Q - R = 0,$$

viz. substituting for a, b, c their values $g - h$, $h - f$, $f - g$, this is an identical equation.

The direct verification is however somewhat tedious, and the equation may be proved more easily as follows:

In the terms $a^2 + f^2$, $b^2 + g^2$, $c^2 + h^2$ of R, substituting for a, b, c their values, we find

$$R = (f^2 + g^2 + h^2)(a^2f^2 + b^2g^2 + c^2h^2) - 2fgh(a^2f + b^2g + c^2h),$$

which may be written

$$R = -2(f^2 + g^2 + h^2)(bcgh + cahf + abfg) - 2fgh(a^2f + b^2g + c^2h),$$

since $af + bg + ch = 0$ gives $a^2f^2 + b^2g^2 + c^2h^2 = -2(bcgh + cahf + abfg)$.

We have then

$$2P = -2bcgh(bg - ch) - 2cahf(ch - af) - 2abfg(af - bg),$$

and thence

$$\begin{aligned} 2P + Q = 2bcgh(f^2 + g^2 + h^2 - bg + ch) \\ + 2cahf(f^2 + g^2 + h^2 - ch + af) \\ + 2abfg(f^2 + g^2 + h^2 - af + bg) \\ + 2fgh(a^2 f + b^2 g + c^2 h), \end{aligned}$$

which is at once converted into $2P + Q = R$, viz. we thus obtain the equation in question.

Instead of a, b, c, f, g, h, I write $a = W - YZ$, $b = W - ZX$, $c = W - XY$, $f = X$, $g = Y$, $h = Z$: we have therefore

$$\begin{aligned}
-hY + gZ - aW &= 0,\\
hX \quad - fZ - bW &= 0,\\
-gX + fY \quad - cW &= 0,\\
aX + bY + cZ \quad &= 0,
\end{aligned}$$

and as before $af + bg + ch = 0$.

Moreover, omitting a common factor, the new values of P, Q, R are

$$\begin{aligned}
&P = XYZW(bg - ch)(ch - af)(af - bg),\\
&Q = a^2g^2h^2X^4 + b^2h^2f^2Y^4 + c^2f^2g^2Z^4 + a^2b^2c^2W^4,\\
&R = a^2f^2(a^2 + f^2)X^4 + b^2g^2(b^2 + g^2)Y^4 + c^2h^2(c^2 + h^2)Z^4,
\end{aligned}$$

and the equation $2P + Q - R = 0$ subsists.

But we have identically

$$\begin{aligned}
&(bg - ch)X + (ch - af)Y + (af - bg)Z\\
&\quad = -(af + bg + ch)W + (bg - ch + ch - af + af - bg)W = 0,
\end{aligned}$$

since $af + bg + ch = 0$.

Whence, writing for shortness

$$\lambda = bg - ch, \quad \mu = ch - af, \quad \nu = af - bg,$$

we have $\lambda X + \mu Y + \nu Z = 0$, giving $X : Y : Z = \mu\nu : \nu\lambda : \lambda\mu$, or say X, Y, $Z = \mu\nu t$, $\nu\lambda t$, $\lambda\mu t$; and thence $XYZW = \lambda^2\mu^2\nu^2t^3W$; and if for a moment $t^3W = \Omega$, then

$$\begin{aligned}
&P = \lambda^3\mu^3\nu^3 \cdot \Omega,\\
&Q = \lambda^2\mu^2\nu^2(a^2g^2h^2\mu^2\nu^2 + \&\text{c.}),\\
&R = \lambda^2\mu^2\nu^2(a^2f^2(a^2 + f^2)\mu^2\nu^2 + \&\text{c.}),
\end{aligned}$$

where the terms of Q and R are completed by the cyclic interchange of the letters; viz. in Q the remaining terms are $b^2h^2f^2\nu^2\lambda^2 + c^2f^2g^2\lambda^2\mu^2 + a^2b^2c^2\lambda^2\mu^2\nu^2$, and in R they are $b^2g^2(b^2 + g^2)\nu^2\lambda^2 + c^2h^2(c^2 + h^2)\lambda^2\mu^2$.

Writing $\mu\nu$, $\nu\lambda$, $\lambda\mu = X$, Y, Z, the equation $2P + Q - R = 0$ becomes

$$2a^2b^2c^2XYZ + a^2g^2h^2X^2(b^2h^2f^2Y^2 + c^2f^2g^2Z^2 + a^2b^2c^2XYZ) \\ + \&\text{c.} - a^2f^2(a^2 + f^2)X^2(b^2g^2(b^2 + g^2)Y^2 + \&\text{c.}) - \&\text{c.} = 0,$$

where the full expression of the equation would be obtained by the cyclic interchange of the capital and also of the small letters.

But by what precedes, each of the non-cyclic terms contains the factor XYZ, and omitting this, the whole equation divides by $\lambda\mu\nu$; we have thus an equation of the form $2a^2b^2c^2 + \Pi = 0$, where Π is a sum of terms each divisible by λ, μ, or ν; and since the whole is symmetric, we must have $\Pi = -2a^2b^2c^2$.

The original equation $2P + Q - R = 0$ is thus proved.

568. NOTE ON THE INTEGRALS $\int \cos x^2\,dx$ AND $\int \sin x^2\,dx$.

[From the Quarterly Journal of Pure and Applied Mathematics, vol. xii. (1873), pp. 118–126.]

Mr Walton has raised, in relation to these integrals, a question which it is very interesting to discuss.

Taking for greater convenience the limits to be $-\infty$, $+\infty$, and writing

$$2u = \int_{-\infty}^{+\infty} \cos x^2\,dx, \qquad 2v = \int_{-\infty}^{+\infty} \sin x^2\,dx,$$

he finds, by a process which he states to be that of Ch. xxv. of the *Intégrales Eulériennes*—a process in fact equivalent to that of the memoir, "Sur les intégrales définies, &c.," *Crelle*, t. viii. (1831)—the values $u = \sqrt{\tfrac{1}{8}\pi}$, $v = \sqrt{\tfrac{1}{8}\pi}$.

I remark that there is no foundation for the process in question; the error is in the assumption as to the limits of r, θ; viz. in the original expressions for $4(u^2 - v^2)$, $8uv$, we integrate over the area of an indefinitely large square (or rectangle); and the assumption is that we are at liberty, instead of this, to integrate over the area of an indefinitely large circle.

Consider in general in the plane the positive and negative quadrants of the coordinates (x, y); viz. the four quadrants are

Positive quadrants	Negative quadrants
$(+,+)$	$(+,-)$
$(-,-)$	$(-,+)$

The areas of these quadrants for the circle and for the square are

	Circle	Square
$(+,+)$	$\frac{1}{4}\pi r^2$	r^2
$(-,-)$	$\frac{1}{4}\pi r^2$	r^2
$(+,-)$	$-\frac{1}{4}\pi r^2$	0
$(-,+)$	$-\frac{1}{4}\pi r^2$	0

where the areas of the negative quadrants are taken to be negative.

Hence the integrals for the square and for the circle respectively are

$$\begin{aligned} 4(u^2 - v^2) &= r^2(\cos r^2 + \sin r^2), \qquad\qquad 0, \\ 8uv &= r^2(\cos r^2 - \sin r^2), \qquad\qquad 0, \end{aligned}$$

and the assumed equations $u^2 - v^2 = 0$, $uv = 0$, giving each of them $u = 0$, $v = 0$, are thus proved to be erroneous.

To show that the integrals have determinate values, observe that we have

$$\int_0^\infty \cos x^2\,dx = \int_0^\infty \frac{\cos t\,dt}{2\sqrt{t}},$$

and writing for the successive positive values $t = \frac{1}{2}\pi$, $\frac{3}{2}\pi$, $\frac{5}{2}\pi$, &c., we have an infinite series of terms alternately positive and negative, and of the form

$$\int_{(2n-1)\frac{\pi}{2}}^{(2n+1)\frac{\pi}{2}} \frac{\cos t\,dt}{2\sqrt{t}},$$

where the numerators are alternately $+$ and $-$, and each of them is $= 2$; and the denominators increase, so that the series is convergent, and the integral has a determinate value.

Similarly,

$$\int_0^\infty \sin x^2\,dx = \int_0^\infty \frac{\sin t\,dt}{2\sqrt{t}},$$

and the expression on the right-hand side is a convergent series of positive terms; so that this integral also has a determinate value.

But in consequence of the alternately positive and negative values of $\cos x^2$ and $\sin x^2$, we cannot infer that the like process is applicable to the integrals of these functions.

To show that it is in fact inapplicable, it will be sufficient to prove that the integrals in question have determinate values; for this being so, the double integrals over the square and over the circle must each of them have the same determinate value.

Writing $r^2 = \rho$, the integral over the circle is

$$\int_0^\infty d\rho \int_0^{2\pi} (\cos\rho\cos^2\theta - \sin\rho\sin^2\theta)\,d\theta = \pi\int_0^\infty \cos\rho\,d\rho,$$

which is not a determinate value. But this is

$$\int_0^\infty dx \int_0^\infty \cos(x^2+y^2)\,dy,$$

and if we write $y = x\tan\theta$, $dy = x\sec^2\theta\,d\theta$, the limits being $\theta = 0$ to $\theta = \frac{\pi}{2}$, we have

$$x\int_0^{\frac{\pi}{2}} \cos(x^2\sec^2\theta)\sec^2\theta\,d\theta,$$

and integrating from $x = 0$ to $x = \infty$, we obtain a determinate value.

Consider the sector included between the radii vectores θ and $\theta + d\theta$, and describe about the origin an indefinitely small square having its sides parallel to the axes; the sides are each $= \delta$, and the area is δ^2. We have for this area, integrating over it the element $\cos(x^2 + y^2)\,dx\,dy$:

The integral over the square is $\cos(x^2 + y^2)\delta^2$, where (x, y) are the coordinates of the centre; or, if the origin is at a corner, it is $\frac{1}{4}\delta^2$. The integral over the sector is in the limit $\frac{1}{2}d\theta \cdot r^2 \cos r^2$.

We have thus for the element of the integral over the square, extending to the several portions of the square which correspond to the sector $d\theta$:

$$\frac{1}{4}\delta^2 \cos(x^2 + y^2) \cdot \frac{r^2\,d\theta}{\frac{1}{4}\delta^2} = r^2 \cos r^2\,d\theta,$$

and hence for the whole integral over the square

$$\int_0^{2\pi} r^2 \cos r^2\,d\theta = 2\pi r^2 \cos r^2,$$

which, for r large, may be taken to be $-\dfrac{1}{2}\sin r^2$; viz. r being large, we have A_r differing from the above value $-\dfrac{1}{2}\sin r^2$ by a quantity of the order $\dfrac{1}{r^4}$.

It is obviously immaterial whether we integrate from $x^2 = n\pi$ to $(n+1)\pi$ or to $(n+1)\pi + \epsilon$, where ϵ has any value less than π; for by so doing, we alter the value of the integral by a quantity of the order $\dfrac{1}{r^4}$.

The values of u, v being shown to be determinate, I see no ground for doubting that these are the values of the more general integrals

$$\int_0^\infty e^{-ax^2} \cos x^2 \, dx, \qquad \int_0^\infty e^{-ax^2} \sin x^2 \, dx,$$

(a real and positive) when a is supposed to continually diminish and ultimately become $= 0$.

We have, in fact, (u as above)

$$\int_0^\infty e^{-2a\rho} \cos \rho \frac{d\rho}{2\sqrt{\rho}} = u\sqrt{\frac{a}{a^2 + b^2}},$$

where b is any real quantity; and similarly for the integral with $\sin \rho$.

We have

$$\int_0^\infty e^{-ax^2} \cos bx^2 \, dx = \frac{1}{2}\sqrt{\pi}\sqrt{\frac{a}{a^2+b^2}},$$

and writing herein $a = 1$, $b = 1$, we have

$$\int_0^\infty e^{-x^2} \cos x^2 \, dx = \frac{1}{2}\sqrt{\pi}\sqrt{\frac{1}{2}} = \sqrt{\frac{\pi}{8}}.$$

Also

$$\int_0^\infty e^{-x^2} \sin x^2 \, dx = \frac{1}{2}\sqrt{\pi}\sqrt{\frac{1}{2}} = \sqrt{\frac{\pi}{8}}.$$

But the question is whether, when a is diminished to 0, these values remain the same; and this depends on the question whether the integrals $\int_0^\infty \cos x^2 \, dx$, $\int_0^\infty \sin x^2 \, dx$ have determinate values.

I have, in the foregoing investigation, sought to prove that they have determinate values; and if this is so, then the values are $\sqrt{\frac{\pi}{8}}$ and $\sqrt{\frac{\pi}{8}}$ respectively.

569. ON THE CYCLIDE.

[From the Quarterly Journal of Pure and Applied Mathematics, vol. xii. (1873), pp. 148–163.]

The cyclide may be considered as the envelope of a variable sphere having its centre on a given conic and cutting at right angles a given sphere; or, what is the same thing, touching at right angles a given sphere.

The equation of a sphere having its centre at (α, β, γ) and radius $= \delta$ is

$$(x-\alpha)^2 + (y-\beta)^2 + (z-\gamma)^2 = \delta^2,$$

or say

$$x^2 + y^2 + z^2 - 2\alpha x - 2\beta y - 2\gamma z + \kappa = 0,$$

where $\kappa = \alpha^2 + \beta^2 + \gamma^2 - \delta^2$.

The condition in order that this may cut at right angles the fixed sphere

$$x^2 + y^2 + z^2 - 2lx + d = 0,$$

is $d + \kappa - 2l\alpha = 0$, or say $\kappa = 2l\alpha - d$; and the equation of the variable sphere thus is

$$x^2 + y^2 + z^2 - 2\alpha x - 2\beta y - 2\gamma z + 2l\alpha - d = 0,$$

where α, β, γ are connected by the equation of the conic, which may be taken to be

$$(a, b, c, f, g, h)(\alpha, \beta, \gamma)^2 = 0.$$

Two conics having the same foci may be represented by means of a parameter θ in the form

$$\frac{x^2}{a+\theta} + \frac{y^2}{b+\theta} = 1,$$

where $a > b$, and the foci are given by $\theta = -a$, $\theta = -b$.

The cyclide is thus the envelope of the sphere

$$x^2 + y^2 + z^2 - 2\alpha x - 2\beta y - 2\gamma z + 2l\alpha - d = 0,$$

where

$$\frac{\alpha^2}{a+\theta} + \frac{\beta^2}{b+\theta} = 1,$$

and

$$\gamma = \text{arbitrary}.$$

The equation of the cyclide is obtained by eliminating θ between the two equations; and the result is

$$\left\{(x-l)^2 + y^2 + z^2 + ld - d\right\}^2 = 4(a+\theta)(b+\theta)\gamma^2,$$

or, as it may be written,

$$\begin{aligned}&\left\{(x-l)^2 + y^2 + z^2 + ld - d\right\}^2 \\ &\qquad = 4(a+\theta)(b+\theta)\left\{(x-l)^2 + y^2 + z^2\right\}.\end{aligned}$$

Taking the centre of the fixed sphere as origin, the equations may be written

$$X = \frac{x}{r^2}, \qquad Y = \frac{y}{r^2}, \qquad Z = \frac{z}{r^2},$$
$$R^2 = X^2 + Y^2 + Z^2 = \frac{1}{r^2},$$

and the inverse of the cyclide is a surface of the fourth order.

If in the equation of the cyclide we write $z = 0$, we obtain the equation of the focal conic in the plane of xy; viz. this is

$$\left\{(x-l)^2 + y^2 + ld - d\right\}^2 = 4(a+\theta)(b+\theta)(x-l)^2.$$

But we have
$$K\alpha^2 + L\beta^2 + M\gamma^2 = 0,$$

where K, L, M are linear functions of θ; and the equation of the cyclide may be written

$$K\left\{(x-l)^2 + y^2 + z^2 + ld - d\right\}^2 + L\cdot 4(a+\theta)(x-l)^2 + M\cdot 4(a+\theta)(b+\theta) = 0.$$

Reverting to the equation

$$(x^2+y^2+z^2-2l\alpha-2m\beta-2n\gamma+d)(\alpha^2+\beta^2+\gamma^2-\delta^2)-(\alpha x+\beta y+\gamma z-\delta^2)^2 = 0,$$

this may be written

$$\{x^2+y^2+z^2+d-2(l\alpha+m\beta+n\gamma)\}(\alpha^2+\beta^2+\gamma^2-\delta^2)-(\alpha x+\beta y+\gamma z-\delta^2)^2 = 0,$$

and if we write

$$\alpha x + \beta y + \gamma z = \delta^2,$$

the equation becomes

$$(x^2 + y^2 + z^2 + d - 2\delta^2)(\alpha^2 + \beta^2 + \gamma^2 - \delta^2) = 0.$$

The second factor gives $\delta = 0$, and then $\alpha x + \beta y + \gamma z = 0$; but the first factor gives

$$x^2 + y^2 + z^2 + d - 2\delta^2 = 0,$$

which is a sphere; and the locus of the centres is the conic

$$(a, b, c, f, g, h)(\alpha, \beta, \gamma)^2 = 0.$$

If in the equation of the cyclide we write $x = l$, we obtain the section by the plane through the centre of the fixed sphere perpendicular to the axis; and the points X_1, L_2 by which the several spheres are touched are obtained by taking the section of the cyclide by the plane $x = l$.

We have to show that these points lie on a conic. The equation of the cyclide is

$$\left\{(x-l)^2 + y^2 + z^2 + ld - d\right\}^2 = 4(a+\theta)(b+\theta)\left\{\frac{(x-l)^2}{a+\theta} + \frac{y^2}{b+\theta}\right\},$$

or say

$$\left\{y^2 + z^2 + ld - d\right\}^2 = 4(a+\theta)(b+\theta)\left\{\frac{y^2}{b+\theta}\right\},$$

that is,

$$\left\{y^2 + z^2 + (l-d)(l-x)\right\}^2 = 4(a+\theta)(y^2+z^2).$$

And writing $x = l$, this becomes

$$(y^2+z^2)^2 = 4(a+\theta)y^2 + 4(b+\theta)z^2,$$

or say

$$(y^2+z^2)^2 - 4ay^2 - 4bz^2 = 4\theta(y^2+z^2).$$

Hence eliminating θ, we have

$$\{(y^2+z^2)^2 - 4ay^2 - 4bz^2\}^2 = 16(a+\theta)(b+\theta)(y^2+z^2)^2,$$

which is a curve of the eighth order; but this breaks up into the curve of the fourth order, which is the section of the cyclide, and into the conic $y^2 + z^2 = 0$.

If in the general equation we write

$$K = 2\alpha x + 2\beta y + 2\gamma z - 2l\alpha - d,$$

the equation may be written

$$K^2 - 4(a+\theta)(b+\theta)(x^2+y^2+z^2 - 2l\alpha - 2m\beta - 2n\gamma + d) = 0,$$

where θ is determined by

$$\frac{\alpha^2}{a+\theta} + \frac{\beta^2}{b+\theta} = 1.$$

It may be remarked that the equation of the cyclide may be written in the form

$$\begin{vmatrix} 0 & x^2+y^2+z^2 & x & y & z \\ x^2+y^2+z^2 & 0 & \alpha & \beta & \gamma \\ x & \alpha & a+\theta & 0 & 0 \\ y & \beta & 0 & b+\theta & 0 \\ z & \gamma & 0 & 0 & c+\theta \end{vmatrix} = 0,$$

where θ is a parameter.

The nodal curve consists of the conics which are the sections of the cyclide by the planes $x = 0$, $y = 0$, $z = 0$; and the nodes are the intersections of these conics.

In fact, representing the equation of the cyclide in the form

$$U = (x^2+y^2+z^2)^2-2(a+d)x^2-2(b+d)y^2-2(c+d)z^2+4lxyz+4mzx^2+\&c. = 0,$$

we have

$$\frac{dU}{dx} = 4x(x^2+y^2+z^2-a-d)+4lyz+8mzx+\&c.,$$

and similarly for $\dfrac{dU}{dy}$ and $\dfrac{dU}{dz}$.

The nodes are given by $\dfrac{dU}{dx} = 0$, $\dfrac{dU}{dy} = 0$, $\dfrac{dU}{dz} = 0$; and these equations give $x = 0$, $y = 0$, $z = 0$, or the intersection of the three planes.

The parabolic cyclide corresponds to the case where one of the focal conics is a parabola; the form of the equation is then

$$(x^2+y^2+z^2)^2-2ax^2-2by^2-2cz^2+4lxyz = 0,$$

where the coefficients are subject to certain relations.

It may be shown that the cyclide has four conical points, which are the intersections of the four pairs of planes

$$\begin{aligned} x &= 0, & y &= 0; \\ x &= 0, & z &= 0; \\ y &= 0, & z &= 0; \\ lx + my + nz &= 0, & \alpha x + \beta y + \gamma z &= 0. \end{aligned}$$

The section of the cyclide by the plane $z = 0$ is

$$(x^2 + y^2)^2 - 2ax^2 - 2by^2 = 0,$$

which is a bicircular quartic; and the four foci of this quartic are the points where the plane $z = 0$ meets the four focal conics of the cyclide.

The centres of inversion of the cyclide are the points from which the surface can be inverted into itself; they are the points where the tangent cones are circular, and they lie on the normal lines through the nodes.

Consider the fixed sphere $x^2+y^2+z^2 = a^2$, and the circle $x^2+y^2 = b^2$, $z = 0$, where $b < a$. The envelope of a variable sphere having its centre on the circle and touching the sphere is a cyclide; and the equation may be written

$$\{x^2 + y^2 + z^2 + a^2 - b^2\}^2 = 4a^2(x^2 + y^2).$$

This is a particular form of the cyclide; the focal conics are the circle and its axis.

The equation may be written

$$(z^2 + a^2 - b^2)^2 + 2(x^2 + y^2)(z^2 + a^2 - b^2) + (x^2 + y^2)^2 = 4a^2(x^2 + y^2),$$

or say

$$(z^2 + a^2 - b^2)^2 = (x^2 + y^2)\{4a^2 - 2(z^2 + a^2 - b^2) - (x^2 + y^2)\}.$$

viz. this is

$$(z^2 + a^2 - b^2)^2 = (x^2 + y^2)(2a^2 + 2b^2 - z^2 - x^2 - y^2).$$

Or writing $r^2 = x^2 + y^2$, the equation is

$$(z^2 + a^2 - b^2)^2 = r^2(2a^2 + 2b^2 - z^2 - r^2).$$

The section by a plane through the axis of z is a curve of the fourth order; and writing $z = 0$, the section by the plane of xy is

$$(a^2 - b^2)^2 = r^2(2a^2 + 2b^2 - r^2),$$

that is,

$$r^4 - 2(a^2 + b^2)r^2 + (a^2 - b^2)^2 = 0,$$

or

$$\{r^2 - (a+b)^2\}\{r^2 - (a-b)^2\} = 0.$$

The section thus consists of the two concentric circles $r = a + b$ and $r = a - b$, which are the circles of the system.

Consider the section by a plane through the axis of z, and write the equation in the form

$$(z^2 + a^2 - b^2)^2 + r^2(z^2 + r^2 - 2a^2 - 2b^2) = 0.$$

Writing herein $r = 0$, we have $z^2 + a^2 - b^2 = 0$, that is $z = \pm\sqrt{b^2 - a^2}$, which are imaginary if $b < a$. But if $b > a$, we have real values, and the surface has real conical points on the axis of z.

The general form of the cyclide may be exhibited by means of elliptic coordinates. Let λ, μ, ν be the roots of the equation

$$\frac{x^2}{a + \theta} + \frac{y^2}{b + \theta} + \frac{z^2}{c + \theta} = 1,$$

then the equation of the cyclide may be written in the form

$$F(\lambda, \mu, \nu) = 0,$$

where F is a symmetric function of the coordinates.

The cyclide is the envelope of a sphere touching three fixed spheres; and the equation may be written in the form

$$\sqrt{R_1} \pm \sqrt{R_2} \pm \sqrt{R_3} = 0,$$

where R_1, R_2, R_3 are the squares of the distances from the centres of the three fixed spheres.

If we write

$$R_1 = (x - \alpha_1)^2 + (y - \beta_1)^2 + (z - \gamma_1)^2 - \delta_1^2,$$

and similarly for R_2, R_3, the equation of the envelope is obtained by eliminating α_1, β_1, γ_1, δ_1 between the four equations

$$R_1 = 0, \quad R_2 = 0, \quad R_3 = 0, \quad \text{and} \quad \frac{dR}{d\theta} = 0.$$

The result is a surface of the fourth order, which is the cyclide; and it has the property that it is its own inverse with respect to any of the three fixed spheres.

570.ON THE SUPERLINES OF A QUADRIC SURFACE IN FIVE-DIMENSIONAL SPACE.

[From the Quarterly Journal of Pure and Applied Mathematics, vol. xii. (1873), pp. 222–225.]

The theory of the lines on a quadric surface in ordinary space is well known; but the corresponding theory in five-dimensional space presents some novel features.

Consider the quadric surface in five-dimensional space

$$X_1^2 + X_2^2 + X_3^2 + X_4^2 + X_5^2 + X_6^2 = 0,$$

where the coordinates are connected by the linear relation

$$a_1X_1 + a_2X_2 + a_3X_3 + a_4X_4 + a_5X_5 + a_6X_6 = 0.$$

But in five-dimensional space, a quadric surface does not contain lines in the ordinary sense; but it contains "superlines," which are three-dimensional spaces lying entirely on the quadric.

Thus considering any two lines, the equations may be written

$$X_1 = \lambda_1 x_1 + \mu_1 x_2 + \nu_1 x_3,$$
$$X_2 = \lambda_2 x_1 + \mu_2 x_2 + \nu_2 x_3,$$
$$X_3 = \lambda_3 x_1 + \mu_3 x_2 + \nu_3 x_3,$$
$$X_4 = \lambda_4 x_1 + \mu_4 x_2 + \nu_4 x_3,$$

where (x_1, x_2, x_3) are the coordinates of a point on the line, and the coefficients are connected by certain relations.

The condition that the line should lie entirely on the quadric is that the equation

$$X_1^2 + X_2^2 + X_3^2 + X_4^2 + X_5^2 + X_6^2 = 0$$

should be satisfied identically for all values of x_1, x_2, x_3.

Substituting the values of X_1, X_2, &c., and equating to zero the coefficients of x_1^2, x_2^2, x_3^2, x_1x_2, x_1x_3, x_2x_3, we obtain six equations, which are the conditions that the line should lie on the quadric.

theory of the superlines is connected with the theory of the complex of lines in four-dimensional space. The superlines are the loci of points which satisfy the equation of the quadric and also a certain system of linear equations.

The number of superlines on a quadric in five-dimensional space is infinite; and they form a system which may be investigated by means of the theory of matrices.

If we represent the coordinates of a point on the quadric by means of a matrix of two rows and four columns, the equation of the quadric is equivalent to the condition that the determinant of the matrix should vanish.

571. A DEMONSTRATION OF DUPIN'S THEOREM.

[From the Quarterly Journal of Pure and Applied Mathematics, vol. xii. (1873), pp. 225–228.]

The theorem of Dupin, in regard to the orthogonal intersection of surfaces, may be demonstrated as follows:

Let $U = 0$, $V = 0$, $W = 0$ be three surfaces which intersect at right angles; then at any point of intersection the normals are at right angles to each other.

Let (x, y, z) be the coordinates of a point on the curve of intersection of the surfaces $U = 0$ and $V = 0$; then the direction-cosines of the normal to $U = 0$ are proportional to

$$\frac{dU}{dx}, \quad \frac{dU}{dy}, \quad \frac{dU}{dz},$$

and similarly for the other surfaces.

The condition that the normals to $U = 0$ and $V = 0$ should be at right angles is

$$\frac{dU}{dx}\frac{dV}{dx} + \frac{dU}{dy}\frac{dV}{dy} + \frac{dU}{dz}\frac{dV}{dz} = 0.$$

that is,

$$\frac{dP}{dx} = \frac{dQ}{dy}, \quad \frac{dP}{dy} = -\frac{dQ}{dx},$$

and passing thence to the second differential coefficients, we may write

$$\frac{d^2P}{dx^2} = \frac{d^2Q}{dxdy}, \quad \frac{d^2P}{dxdy} = \frac{d^2Q}{dy^2},$$

so that we have

$$\begin{aligned} \delta P &= L\delta x + M\delta y, \\ \delta Q &= -M\delta x + L\delta y, \\ \delta^2 P &= (b, a, -b \between \delta x, \delta y)^2, \\ \delta^2 Q &= (-a, b, a \between \delta x, \delta y)^2. \end{aligned}$$

Hence, for the maximum or minimum elevation of the path, we have $\delta P = 0$, where $\delta Q = 0$; that is, $L\delta x + M\delta y = 0$, where $-M\delta x + L\delta y = 0$; whence $L^2 + M^2 = 0$, giving $L = 0$, $M = 0$.

that is,

$$\frac{dP}{dx} = 0, \quad \frac{dP}{dy} = 0;$$

and if the two normals intersect in the point (X, Y, Z), we have

$$X - x = P\frac{dP}{dx},$$
$$Y - y = P\frac{dP}{dy},$$
$$Z - z = P\frac{dP}{dz},$$

where P is the perpendicular from (x, y, z) on the tangent plane.

viz. by what precedes, this may be written under either of the forms

$$\frac{X - x}{\frac{dP}{dx}} = \frac{Y - y}{\frac{dP}{dy}} = \frac{Z - z}{\frac{dP}{dz}} = P.$$

Let the surfaces be $U = 0$, $V = 0$, $W = 0$, intersecting orthogonally; then we have

$$\begin{gathered}
\frac{dU}{dx}\frac{dV}{dx} + \frac{dU}{dy}\frac{dV}{dy} + \frac{dU}{dz}\frac{dV}{dz} = 0, \\
\frac{dV}{dx}\frac{dW}{dx} + \frac{dV}{dy}\frac{dW}{dy} + \frac{dV}{dz}\frac{dW}{dz} = 0, \\
\frac{dW}{dx}\frac{dU}{dx} + \frac{dW}{dy}\frac{dU}{dy} + \frac{dW}{dz}\frac{dU}{dz} = 0.
\end{gathered}$$

Differentiating the first equation with respect to x, we have

$$\frac{d^2U}{dx^2}\frac{dV}{dx} + \frac{dU}{dx}\frac{d^2V}{dx^2} + \&\text{c.} = 0,$$

and similarly for the other equations.

Adding these equations, we obtain

$$\frac{d^2U}{dx^2} + \frac{d^2U}{dy^2} + \frac{d^2U}{dz^2} = 0,$$

and similarly for V and W; that is, the sum of the principal curvatures at any point of the surface of intersection is zero.

Hence the surfaces cut each other in lines of curvature; and the theorem is proved.

572. THEOREM IN REGARD TO THE HESSIAN OF A QUATERNARY FUNCTION.

[From the Quarterly Journal of Pure and Applied Mathematics, vol. xii. (1873), pp. 229–232.]

Let $U = (a, b, c, d, f, g, h, l, m, n)(x, y, z, w)^2 = 0$ be a quaternary quadric function; then the Hessian is the determinant

$$H = \begin{vmatrix} a & h & g & l \\ h & b & f & m \\ g & f & c & n \\ l & m & n & d \end{vmatrix}.$$

And from it we derive the equation

$$H = \frac{d^2U}{dx^2}\frac{d^2U}{dy^2}\frac{d^2U}{dz^2}\frac{d^2U}{dw^2} - \&\text{c.} = 0.$$

Hence, writing $P' = P$, which implies $k' = k$ and $l' = l$, we have

$$\frac{d^2U}{dx^2} = k\frac{d^2U}{dx^2} + l\frac{d^2U}{dxdy} + \&c.,$$

and the equation of the Hessian may be written in the form

$$\begin{vmatrix} a & h & g & l \\ h & b & f & m \\ g & f & c & n \\ l & m & n & d \end{vmatrix} = 0.$$

Attending only to the leading terms, we have

$$abcd - af^2 - bg^2 - ch^2 - dl^2 + \&c. = 0,$$

and the discriminant of the quaternary function is

$$\Delta = abcd - af^2 - bg^2 - ch^2 - dl^2 + 2afgl + 2bhlm + 2cghn + \&c.$$

The relation between the Hessian and the discriminant is that the former is the second differential coefficient of the latter; and we have

$$H = \frac{d^2\Delta}{da\,db\,dc\,dd}.$$

The theorem is that if the Hessian of a quaternary function vanishes, then the function is expressible as the sum of three cubes; and conversely, if the function is the sum of three cubes, then the Hessian vanishes.

Let $U = x^3 + y^3 + z^3 + w^3 + 6mxyzw$; then the Hessian is

$$H = \begin{vmatrix} 6x & 6mz & 6my & 6mw \\ 6mz & 6y & 6mx & 6mv \\ 6my & 6mx & 6z & 6mu \\ 6mw & 6mv & 6mu & 6w \end{vmatrix} = 0,$$

where (u, v, w) are the new coordinates.

The condition that this should vanish identically is that $m = 0$, and then $U = x^3 + y^3 + z^3 + w^3$, which is the sum of four cubes.

But if we write $w = -(x+y+z)$, then $U = x^3+y^3+z^3-(x+y+z)^3$, which is the sum of three cubes; and the Hessian vanishes.

573.NOTE ON THE $(2, 2)$ CORRESPONDENCE OF TWO VARIABLES.

[From the Quarterly Journal of Pure and Applied Mathematics, vol. xii. (1873), pp. 232–234.]

The $(2, 2)$ correspondence of two variables is a particular case of the general (m, n) correspondence; and it presents some features which are worthy of special attention.

Let x and y be two variables connected by the equation

$$(a, b, c, d)(x, 1)^2(y, 1)^2 = 0,$$

which may be written in the form

$$(a_0x^2 + 2a_1x + a_2)y^2 + 2(b_0x^2 + 2b_1x + b_2)y + (c_0x^2 + 2c_1x + c_2) = 0.$$

To each value of x correspond two values of y, and to each value of y correspond two values of x; and the correspondence is thus a $(2, 2)$ correspondence.

The condition that the two values of y should be equal is that the discriminant of the quadratic in y should vanish; and this gives an equation of the fourth order in x.

Similarly, the condition that the two values of x should be equal gives an equation of the fourth order in y.

The $(2, 2)$ correspondence is thus represented by a curve of the fourth order; and the genus of this curve is 1, so that the correspondence is elliptic.

The general theory of the $(2, 2)$ correspondence may be investigated by means of the theory of elliptic functions; and the results are analogous to those which have been obtained for the $(2, 2)$ correspondence on a curve of the third order.

574.ON WRONSKI'S THEOREM.

[From the Quarterly Journal of Pure and Applied Mathematics, vol. xii. (1873), pp. 234–240.]

Wronski's theorem is a theorem in elimination, which gives the resultant of a system of equations in the form of a determinant.

Let ϕ_1, ϕ_2, …, ϕ_n be n functions of n variables x_1, x_2, …, x_n; then the theorem asserts that the resultant of the equations $\phi_1 = 0$, $\phi_2 = 0$, …, $\phi_n = 0$ is equal to a determinant formed from the coefficients of the functions.

The proof of the theorem depends on the properties of the functional determinant, or Jacobian, of the system of functions.

Let the functions be

$$
\begin{aligned}
\phi_1 &= a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n,\\
\phi_2 &= a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n,\\
&\cdots\\
\phi_n &= a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n,
\end{aligned}
$$

then the resultant is the determinant

$$
R = \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix}.
$$

The series of equalities may be presented as follows, writing as above A to denote the function

$$
A = \frac{1}{1} - \frac{1}{1 \cdot 2} + \frac{1}{1 \cdot 2 \cdot 3} - \cdots ,
$$

and similarly for the other functions.

viz. it is

$$\frac{1}{1}A^0 - \frac{1}{1}A^1 + \frac{1}{1\cdot 2}A^2 - \frac{1}{1\cdot 2\cdot 3}A^3 + \cdots,$$

where in each case the function on the left hand is to be expanded only as far as the power which is contained in the determinant.

The numerical coefficients in the top-lines of the several determinants are the reciprocals of

$$n(n-1)\cdots 2\cdot 1, \quad n(n-1)\cdots 2, \quad n(n-1), \quad n, \quad 1,$$

where n is the index of the highest power of θ.

The demonstration of Wronski's theorem therefore ultimately depends on the establishment of the formula

$$\frac{d}{dx}\begin{vmatrix} \phi_1 & \phi_2 & \cdots & \phi_n \\ \phi_1' & \phi_2' & \cdots & \phi_n' \\ \vdots & \vdots & \ddots & \vdots \\ \phi_1^{(n-1)} & \phi_2^{(n-1)} & \cdots & \phi_n^{(n-1)} \end{vmatrix} = \begin{vmatrix} \phi_1 & \phi_2 & \cdots & \phi_n \\ \phi_1' & \phi_2' & \cdots & \phi_n' \\ \vdots & \vdots & \ddots & \vdots \\ \phi_1^{(n)} & \phi_2^{(n)} & \cdots & \phi_n^{(n)} \end{vmatrix}.$$

This is the fundamental formula of the theory; and from it all the other results follow by integration.

The theorem may be extended to the case where the functions are not linear, but are any functions of the variables; and the result is that the resultant is equal to the functional determinant, divided by a certain factor which depends on the degrees of the functions.

The demonstration of Wronski's theorem therefore ultimately depends on the establishment of the formula for the differentiation of a determinant; and when this is established, the theorem follows as a corollary.

[574. On Wronski's Theorem *(continued)*]

101. and expanding the left hand as far as θ^4, this is

$$\begin{aligned} &= 1 \\ &= 1 - 4(\tfrac{1}{2}\theta + \tfrac{1}{6}\theta^3 + \tfrac{1}{24}\theta^5) \\ &\quad + 10(\tfrac{1}{4}\theta^2 + \tfrac{1}{6}\theta^4) \\ &\quad - 20(\tfrac{1}{8}\theta^3 + \tfrac{1}{24}\theta^5) \\ &\quad + \tfrac{35}{4}\theta^4 + \tfrac{63}{24}\theta^6 \\ &\quad - \tfrac{1}{16}\theta^5 + \tfrac{1}{32}\theta^6 - \theta^4, \end{aligned}$$

which agrees.

Reverting to the above equations, and expanding the several terms

$$(\phi\theta)' = 2\phi'\theta', \quad (\phi^2\theta)'' = 2\phi\phi'' + 2\phi'^2, \quad \&\text{c.},$$

then, since in each case the left-hand side contains ϕ', ϕ'', ϕ''', &c. but not ϕ, it is clear that on the right-hand side the terms involving ϕ must disappear of themselves; and assuming that this is so, the equality takes the more simple form obtained by writing in the foregoing expressions $\phi = 0$, viz. we thus have

$$(\phi\theta)' = 0, \quad (\phi^2\theta)'' = 2\phi'\theta', \quad \&\text{c.}$$

In order to simplify the formulae, I replace the series ϕ', $\frac{1}{2}\phi''$, $\frac{1}{6}\phi'''$, $\frac{1}{24}\phi^{iv}$, &c. by b, c, d, e, &c., and θ', $\frac{1}{2}\theta''$, $\frac{1}{6}\theta'''$, &c. by b', c', d', &c.; then the formulae assume the following simple form, viz. writing

$$\Theta = b + c\theta + d\theta^2 + e\theta^3 + \&\text{c.},$$

then we have

	1	θ	θ^2	θ^3	θ^4
Θ	b	c	d	e	
Θ^2	b^2	$2bc$	$2bd + c^2$	$2be + 2cd$	
Θ^3	b^3	$3b^2c$	$3b^2d + 3bc^2$	$3b^2e + 6bcd + c^3$	
Θ^4	b^4	$4b^3c$	$4b^3d + 6b^2c^2$	$\cdots$	

viz. for Θ^n the right-hand gives the development as far as θ^{n-1}.

It will be observed, that in the determinants the several lines are the coefficients in the expansions of Θ, Θ^2, Θ^3, &c. respectively. The demonstration is very easy; it will be sufficient to take the equation for Θ^2.

Assume

$$\Theta = \cdots + r\theta^7 + q\theta^6 + p\theta^5 + n\theta^4 + m\theta^3 + l\theta^2 + k\theta,$$

where clearly $k = b'$, and write also

$$\begin{aligned} \Theta &= B_1 + C_1\theta + D_1\theta^2 + E_1\theta^3 + \cdots\,, \\ \Theta^2 &= B_2 + C_2\theta + D_2\theta^2 + \cdots\,, \\ \Theta^3 &= B_3 + C_3\theta + \cdots\,, \end{aligned}$$

102. where $B_1 = b$, $B_2 = b'$, $B_3 = b''$; we wish to show that

$$\begin{aligned} \beta B_3 + \gamma C_3 + \delta D_3 + \epsilon E_3 &= 0, \\ \gamma B_3 + \delta C_3 + \epsilon D_3 &= 0, \\ \delta B_3 + \epsilon C_3 &= 0, \end{aligned}$$

for this being the case, neglecting the terms in θ^5, θ^6, &c., and writing

$$\frac{B_3,\ C_3,\ D_3,\ E_3}{B_2,\ C_2,\ D_2,\ E_2} = 0,$$

then eliminating β, γ, δ, ϵ, we have

$$\begin{vmatrix} B_3 & C_3 & D_3 & E_3 \\ B_2 & C_2 & D_2 & E_2 \\ B_1 & C_1 & D_1 & E_1 \\ A_1 & C_1 & D_1 & E_1 \end{vmatrix} = 0,$$

in which equation the term which contains B_3 is B_3 multiplied by the determinant without the term in question (that is, with A_1 for its corner term).

To prove the subsidiary theorems, multiply the expression of Θ by $e^{-\phi}$, and differentiate in regard to θ, we have

$$(\theta\Theta)' - \frac{\phi'}{\phi}\theta\Theta = \theta\phi' - \phi''\theta^2 + \phi'''\theta^3 - \&c.$$

Multiplying by $e^{\Theta} = B_1\theta + C_1\theta^2 + D_1\theta^3 + E_1\theta^4$, we see that

$$A\beta + C_1\gamma + D_1\delta + E_1\epsilon$$

is the coefficient of θ in $A\frac{(\theta\Theta)'}{e^{\Theta}}$; and similarly

$$\frac{1}{4}(\theta\Theta)' \qquad \frac{1}{4}(\theta\Theta)'$$

$A\gamma + C_1\delta + D_1\epsilon$ is the coefficient of θ^2 in $A\frac{(\theta\Theta)'}{e^{\Theta}}$, and $A\delta + C_1\epsilon$ that of θ^3 in $\frac{1}{7}\frac{(\theta\Theta)'}{e^{\Theta}}$.

Now, m being any positive integer, $\frac{1}{e^{m\Theta}}$ expanded in ascending powers of θ contains negative and positive powers of θ, but of course no logarithmic term; hence differentiating in regard to θ, $\frac{1}{e^{(m+1)\Theta}}$ contains no term in θ^0; and the expressions in question are thus each $= 0$; which completes the demonstration.

The foregoing formulae giving the expansion of D_m up to θ^{m-1} in terms of the coefficients in the expansions of Θ, Θ^2, ..., Θ^{m-1} are I think interesting.

[This is a well-known method made use of by Jacobi and Murphy.]

575. On a Special Quartic Transformation of an Elliptic Function

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xii. (1873), pp. 266–269.]

103. It is remarked by Jacobi that a transformation of the order $n'n''$ may lead to a modular equation

$$\frac{\Lambda'}{\Lambda} = \frac{n'}{n''}\frac{\mathrm{K}'}{\mathrm{K}},$$

and in particular when $n' = n''$, or the order is square, then the equation may be

$$\frac{\Lambda'}{\Lambda} = \frac{\mathrm{K}'}{\mathrm{K}};$$

viz. that instead of a transformation we may have a multiplication.

A quartic transformation of the kind in question may be obtained as follows: writing

$$X = (a, b, c, d, e \between x, 1)^4 = a(x-\alpha)(x-\beta)(x-\gamma)(x-\delta),$$

H the Hessian, Φ the cubi-covariant, I and J the two invariants, then there is a well known quartic transformation leading to

$$\frac{dz}{\sqrt{(Z)}} = \frac{2\sqrt{(-2)}\,dx}{\sqrt{(X)}},$$

where $Z = 2\Phi^3 - I\Phi H^2 + 2JH^3$.

In fact we have

$$Z = \frac{1}{f}\{(4H^3 - IH)X + JX^2\}^2 = \frac{1}{f}\Phi^6,$$

that is,

$$\sqrt{Z} = \frac{1}{\sqrt{f}}\Phi^3\sqrt{X},$$

104. so that, by Jacobi's general principle, it at once appears that we have a transformation of the form in question.

Now we may establish a linear transformation such that to the roots z_1, z_2, z_3 of the equation $z^3 - Iz + 2J = 0$ correspond the values α, β, γ of y; and this being so, we have between y, z the relation

$$\frac{dz}{\sqrt{(Z)}} = \frac{\sqrt{(-2)}\,dy}{\sqrt{(Y)}},$$

where $Y = a(y-\alpha)(y-\beta)(y-\gamma)(y-\delta) = (a, b, c, d, e \between y, 1)^4$; that is, we have

$$\frac{py+q}{y-\delta} = X',$$

such that

$$\frac{dy}{\sqrt{(Y)}} = \frac{2\,dx}{\sqrt{(X)}},$$

which is a quartic transformation giving a duplication of the integral.

The foundation of the theorem is that we can determine p, q in such wise that the functions

$$\frac{p\alpha+q}{\alpha-\delta}, \quad \frac{p\beta+q}{\beta-\delta}, \quad \frac{p\gamma+q}{\gamma-\delta}$$

shall be the roots z_1, z_2, z_3 of the equation $z^3 - Iz + 2J = 0$.

For writing

$$\begin{aligned}
A &= (\beta-\gamma)(\alpha-\delta),\\
B &= (\gamma-\alpha)(\beta-\delta),\\
C &= (\alpha-\beta)(\gamma-\delta),
\end{aligned}$$

and observing the equations

$$\begin{aligned}
I &= -\tfrac{1}{4}(A^2+B^2+C^2),\\
J &= -\tfrac{1}{4}(BC+CA+AB),\\
\sqrt[3]{J^2} &= -\tfrac{1}{12}(B-C)(C-A)(A-B),
\end{aligned}$$

(since $A+B+C=0$) and the equation in z is

$$\{z-\tfrac{1}{2}a(B-C)\}\{z-\tfrac{1}{2}a(C-A)\}\{z-\tfrac{1}{2}a(A-B)\},$$

and the equations for the determination of p, q thus are

$$\begin{aligned}
p\alpha+q &= \tfrac{1}{2}a(\alpha-\delta)(B-C) = \tfrac{1}{2}a(\alpha-\delta)\{2(\beta\delta+\gamma\delta)-(\alpha+\delta)(\beta+\gamma)\},\\
p\beta+q &= \tfrac{1}{2}a(\beta-\delta)(C-A) = \tfrac{1}{2}a(\beta-\delta)\{2(\gamma\delta+\alpha\delta)-(\beta+\delta)(\gamma+\alpha)\},\\
p\gamma+q &= \tfrac{1}{2}a(\gamma-\delta)(A-B) = \tfrac{1}{2}a(\gamma-\delta)\{2(\alpha\delta+\beta\delta)-(\gamma+\delta)(\alpha+\beta)\},
\end{aligned}$$

105. giving

$$\begin{aligned}
p &= \tfrac{1}{2}a\{-3\delta^2+2\delta(\alpha+\beta+\gamma)-\beta\gamma-\gamma\alpha-\alpha\beta\},\\
q &= \tfrac{1}{2}a\{\delta^2(\alpha+\beta+\gamma)-2\delta(\beta\gamma+\gamma\alpha+\alpha\beta)+3\alpha\beta\gamma\},
\end{aligned}$$

or, as these may also be written

$$\begin{aligned}
p &= \tfrac{1}{2}a\{(\beta-\delta)(\gamma-\delta)+(\gamma-\delta)(\alpha-\delta)+(\alpha-\delta)(\beta-\delta)\},\\
q &= \tfrac{1}{2}a\{\alpha(\beta-\delta)(\gamma-\delta)+\beta(\gamma-\delta)(\alpha-\delta)+\gamma(\alpha-\delta)(\beta-\delta)\};
\end{aligned}$$

and observe also

$$p\delta+q=\tfrac{1}{2}a(\alpha-\delta)(\beta-\delta)(\gamma-\delta).$$

Taking X in the standard form $=(1-x^2)(1-k^2x^2)$, and writing

$$\gamma=-1,\quad \delta=1,\quad \alpha=+\tfrac{1}{l},\quad \beta=-\tfrac{1}{l},$$

we have

$$\frac{py+q}{y-1}=\frac{-\tfrac{1}{2}\{2k^2l+(1+k^2l^2)(1+k^2x^2)+(1-10k^2l^2+k^4l^4)x^2\}}{l},$$

$$A=-\frac{1}{l^2}(1+l)^2,\quad B=\frac{1}{l^2}(1-l)^2,\quad C=\frac{4}{l^2},$$

$$\begin{gathered}
z_1=\tfrac{1}{4}(1+6k+k^2),\quad z_2=\tfrac{1}{4}(1-6k+k^2),\quad z_3=-\tfrac{1}{4}(1+k^2);\\
Z=z^3-\tfrac{1}{4}z(1+14k^2+k^4)+\tfrac{1}{27}(1+k^2)(1-34k^2+k^4)\\
=(Z-Z_1)(Z-Z_2)(Z-Z_3),\\
p=\tfrac{1}{4}l(1-5k),\quad q=\tfrac{1}{4}l(5-k);
\end{gathered}$$

giving as they should do

$$\frac{p+q}{l-1}=\frac{1-k}{-1}.$$

Write for shortness

$$-\frac{1}{2}\{2k^2l(1+k^2)(1+l^2)+(1-10k^2l^2+l^4)k^2x^2\}=Q,$$

so that

$$\frac{py+q}{y-1}=\frac{Q}{l}.$$

106. then

$$\begin{aligned}
X&=\frac{1}{k-1}\cdot\frac{y-1}{y+1},\quad & Q&=\frac{P+1}{k+1},\\
X&=\frac{1}{k+1}\cdot\frac{y+1}{y-1},\quad & Q&=\frac{P-1}{k-1},\\
X&=\frac{2}{k-1}\cdot\frac{1}{y-1},\quad & Q&=\frac{P+1}{y+1}.
\end{aligned}$$

The last of these is

$$\frac{1-k}{1+k} = \frac{P+1}{y+1} \cdot \frac{y-1}{2},$$

that is,

$$\frac{1-k}{1+k} \cdot \frac{y+1}{y-1} = \frac{(1-ky)}{(1-y)} \cdot \frac{1}{x^2} = \frac{(1-ky)(1+y)}{(1-y)(1+ky)},$$

$$\frac{y+1}{y-1} = \frac{(1+x^2)(1-k^2x^2)}{(1-x^2)(1+k^2x^2)},$$

from which the foregoing equation

$$\frac{dy}{\sqrt{(Y)}} = \frac{2\,dx}{\sqrt{(X)}}$$

may be at once verified.

C. IX. 14

576. Addition to Mr Walton's Paper "On the Ray-Planes in Biaxal Crystals"

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xii. (1873), pp. 273–275.]

107. Instead of Mr Walton's a', b', c' write a, b, c, and assume

$$\alpha, \beta, \gamma = b - c, \quad c - a, \quad a - b;$$

$$\delta, \epsilon, \zeta = b + c, \quad c + a, \quad a + b.$$

Also instead of his x^2, y^2, z^2 write x, y, z.

Then instead of the octic cone we have the quartic cone, or say the quartic curve

$$\begin{aligned} &a^2b^2c^2 \sec^2\theta \cdot xyz(x+y+z) \\ &= a^2z\{(bc-a^2)x + a(\beta y - \gamma z)\}^2 \\ &\quad + b^2zx\{(ca-b^2)y + b(\gamma z - \alpha x)\}^2 \\ &\quad + c^2xy\{(ab-c^2)z + c(\alpha x - \beta y)\}^2; \end{aligned}$$

viz. we may herein consider x, y, z as trilinear coordinates, the ratios $x : y : z$ being positive for a point within the fundamental triangle.

The curve passes through the angles of the triangle, and it touches the sides in the points

$$(x = 0, \quad \beta y - \gamma z = 0), \quad (y = 0, \quad \gamma z - \alpha x = 0), \quad (z = 0, \quad \alpha x - \beta y = 0)$$

respectively.

Moreover, the tangents at the angles of the triangle lie each of them outside the triangle.

Hence, supposing a, b, c each positive, and $a > b > c$, we have α and γ each positive, β negative, and the form of the curve is as shown in the figure, or else the like form with the oval lying outside the triangle.

And it is hence clear that, if the side AC ($y = 0$) instead of touching the curve meets it in a node, this is a conjugate point arising from the evanescence of the oval; and in this case no part of the curve lies within the triangle.

Now considering any point $x : y : z = l : m : n$, we obtain a tetrad of points $x : y : z = \pm\sqrt{l} : \pm\sqrt{m} : \pm\sqrt{n}$ on the octic cone or curve;

and in order that the point on the octic curve may be real, we must have l, m, n all of the
108. same sign; that is, the point on the quartic curve must lie within the triangle.

Hence, when in the quartic curve the oval becomes a conjugate point, the octic curve has no real branch, but it consists wholly of conjugate points; viz. it consists of the points A, B, C as conjugate points; two imaginary conjugate points answering to the point α of the figure, two other imaginary conjugate points answering to the point γ; and two conjugate points answering to the point β, these last being not ordinary conjugate points, but conjugate tacnodal points, or points of contact of two imaginary branches of the curve.

The case in question, β a conjugate point on the quartic curve, answers to Mr Walton's critical value of $\sec^2\theta$, viz. in the present notation

$$\sec^2\theta = \frac{1}{\alpha\gamma}.$$

To show this I consider the intersection of the curve by the line $\gamma z - \alpha x = 0$; and I write for convenience $\gamma z = \alpha x = \gamma\alpha u$, that is, $x = \gamma u$, $z = \alpha u$.

Substituting these values, the equation divides by $y\gamma u$, or omitting this factor it is

$$\begin{aligned} &a^2b^2c^2\sec^2\theta\cdot\gamma u(u+\alpha u+\gamma u)\\ &= a^2\alpha u\{(bc-a^2)\gamma u + a(\beta y - \gamma\alpha u)\}^2\\ &\quad + b^2\gamma\alpha u^2\{(ca-b^2)y + b(\gamma\alpha u - \alpha\gamma u)\}^2\\ &\quad + c^2\gamma u y\{\alpha\gamma u(ab-c^2) + c(\alpha\beta y - \beta\gamma\alpha u)\}^2, \end{aligned}$$

or observing that we have $\alpha+\gamma = -\beta$,
109. The required condition is that the coefficient of u^2 shall vanish; viz. we then have

$$\begin{aligned} -\alpha\beta\gamma\sec^2\theta &= a^2\beta^2 + \gamma^2 b^2\\ &= (b-c)(a+b)^2 + (a-b)(b+c)^2\\ &= (a-c)\{3b^2 + b(a+c) - ac\}, \end{aligned}$$

that is,

$$\alpha\gamma\sec^2\theta = 4b^2 + ac,$$

agreeing with Mr Walton's value.

Giving $\sec^2\theta$ this value, and throwing out the factor u, the equation becomes

$$u\{2a^2\alpha\gamma^2+2\gamma ca\beta+\alpha\gamma(ca-b^2)-\alpha\gamma^2(4b^2+\alpha\gamma)\}+\gamma(\alpha^2\alpha^2+c^2\gamma^2)=0;$$

or, what is the same thing,

$$\alpha\gamma u\{2a(a+b)(b-c)^2+2c(c+b)(b-a)^2+(ca-b^2)(b-c)(a-b)-4b^2(b-c)(b-a)\} \\ +\gamma(\alpha^2\alpha^2+c^2\gamma^2)=0.$$

say

$$\alpha\gamma Ku+(\alpha^2\alpha^2+c^2\gamma^2)\gamma=0,$$

viz. this equation determines the remaining intersection of the curve by the line $\gamma z-\alpha x=0$; the point in question lies outside the triangle, that is, $u:y$ should be negative; or α, γ, $\alpha^2\alpha^2+c^2\gamma^2$ being each positive, we should have K positive; we in fact find

$$\begin{aligned} K &= 4b^4+b^2(a^2+c^2-6ac)+4a^2c^2 \\ &= 4(b^2-ac)^2+b^2(a+c)^2, \end{aligned}$$

which is as it should be.

577. Note in Illustration of Certain General Theorems obtained by Dr Lipschitz

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xii. (1873), pp. 346–349.]

110. The paper by Dr Lipschitz, which follows the present Note [in the Quarterly Journal, l.c.] is supplemental to Memoirs by him in Crelle, vols. LXX., LXXII., and LXXIV.; and he makes use of certain theorems obtained by him in these memoirs; these theorems may be illustrated by the consideration of a particular example.

Imagine a particle not acted on by any forces, moving in a given surface; and let its position on the surface at the time t be determined by means of the general coordinates x, y.

We have then the vis-viva function T, a given function of x, y, $\dot{x}$, $\dot{y}$; and the equations of motion are

$$\frac{d}{dt}\frac{\partial T}{\partial \dot{x}} - \frac{\partial T}{\partial x} = 0, \qquad \frac{d}{dt}\frac{\partial T}{\partial \dot{y}} - \frac{\partial T}{\partial y} = 0,$$

which equations serve to determine x, y in terms of t, and of four arbitrary constants; these are taken to be the initial values (or values corresponding to the time $t = t_0$) of x, y, $\dot{x}$, $\dot{y}$; say the values are α, β, α', β'.

We have the theorem that x, y are functions of α, β, $\alpha'(t-t_0)$, $\beta'(t-t_0)$.

Suppose for example that x, y, z denote ordinary rectangular coordinates, and that the particle moves on the sphere $x^2+y^2+z^2 = c^2$; to fix the ideas, suppose that the coordinates z are measured vertically upwards, and that the particle is on the upper hemisphere; that is, take $z = +\sqrt{(c^2-x^2-y^2)}$, we have

$$T = \tfrac{1}{2}(\dot{x}^2+\dot{y}^2+\dot{z}^2),$$

where $\dot{z}$ denotes its value in terms of x, y, $\dot{x}$, $\dot{y}$; viz. we have $x\dot{x}+y\dot{y}+z\dot{z} = 0$, or

$$\dot{z} = -\frac{x\dot{x}+y\dot{y}}{z} = -\frac{x\dot{x}+y\dot{y}}{\sqrt{(c^2-x^2-y^2)}}.$$

111. the proper value of T is thus

$$T = \tfrac{1}{2}\left\{\dot{x}^2+\dot{y}^2+\frac{(x\dot{x}+y\dot{y})^2}{c^2-x^2-y^2}\right\};$$

but it is convenient to retain z, $\dot{z}$, taking these to signify throughout their foregoing values in terms of x, y, $\dot{x}$, $\dot{y}$.

The constants of integration are, as before, α, β, α', β'; but we use also γ, γ' considered as signifying given functions of these constants, viz. we have

$$\gamma = \sqrt{(c^2 - \alpha^2 - \beta^2)} \quad \text{and} \quad \gamma' = -\frac{\alpha\alpha' + \beta\beta'}{\gamma}$$

(in fact, $\alpha^2 + \beta^2 + \gamma^2 = c^2$ and $\alpha\alpha' + \beta\beta' + \gamma\gamma' = 0$; γ, γ' being thus the initial values of z, $\dot{z}$).

Now, writing

$$\sigma = \frac{\sqrt{(\alpha'^2 + \beta'^2 + \gamma'^2)}}{c}(t - t_0),$$

the required values of x, y and the corresponding value of z are

$$x = \alpha\cos\sigma + \frac{c\alpha'}{\sqrt{(\alpha'^2 + \beta'^2 + \gamma'^2)}}\sin\sigma,$$
$$y = \beta\cos\sigma + \frac{c\beta'}{\sqrt{(\alpha'^2 + \beta'^2 + \gamma'^2)}}\sin\sigma.$$

To verify that these are functions of α, β, $\alpha'(t - t_0)$, $\beta'(t - t_0)$, write

$$\alpha(t - t_0) = u, \quad \beta'(t - t_0) = v; \quad \text{and take also } \gamma'(t - t_0) = w;$$

we have $\alpha u + \beta v + \gamma w = 0$, viz. $w = -\frac{1}{\gamma}(\alpha u + \beta v)$, is a function of α, β, u, v; and then

$$\sigma = \frac{\sqrt{(u^2 + v^2 + w^2)}}{c},$$

$$x = \alpha\cos\sigma + \frac{u}{\sigma}\sin\sigma, \quad y = \beta\cos\sigma + \frac{v}{\sigma}\sin\sigma, \quad z = \gamma\cos\sigma + \frac{w}{\sigma}\sin\sigma;$$

so that x, y (and also z) are each of them a function of α, β, u, v, that is α, β, $\alpha'(t - t_0)$, $\beta'(t - t_0)$, which is the theorem in question.

The original variables are x, y; the quantities $\alpha'(t-t_0)$, $\beta'(t-t_0)$, or u, v are Dr Lipschitz' "Normal-Variables," and the theorem is that the original variables are functions of their initial values, and of the normal-variables.

112. The vis-viva function T may be expressed in terms of the normal-variables and their derived functions; viz, it is easy to verify that we have

$$T = \tfrac{1}{2}\frac{(\alpha u' + \beta v' + \gamma w')^2}{u^2 + v^2 + w^2} + \tfrac{1}{2}\frac{c^2 \sin^2 \sigma}{\sigma^2}(u'^2 + v'^2 + w'^2),$$

where w denotes $-\frac{1}{\gamma}(\alpha u + \beta v)$ and consequently w' denotes $-\frac{1}{\gamma}(\alpha u' + \beta v')$; introducing herein differentials instead of derived functions, or writing

$$\phi(du) = \tfrac{1}{2}\left(1 - \frac{\sin^2 \sigma}{\sigma^2}\right)(u\,du + v\,dv + w\,dw)^2 \\ + \tfrac{1}{2}\frac{c^2 \sin^2 \sigma}{\sigma^2}(du^2 + dv^2 + dw^2),$$

where w, dw denote $-\frac{1}{\gamma}(\alpha u + \beta v)$, $-\frac{1}{\gamma}(\alpha\,du + \beta\,dv)$ respectively; then $\phi(du)$ is the function thus denoted by Dr Lipschitz: and writing herein $t - t_0 = 0$, and thence $u = 0$, $v = 0$, $w = 0$, $\sigma = 0$, the resulting value of $\phi(du)$ is

$$f_0(du) = \tfrac{1}{2}(du^2 + dv^2 + dw^2),$$

where $f_0(du)$ is the function thus denoted by him; the corresponding value of $f_0(u)$ is

$$= \tfrac{1}{2}(u^2 + v^2 + w^2).$$

We have thus an illustration of his theorem that the function $\phi(du)$ is such that we have identically

$$\phi(du) - \{d\sqrt{f_0(u)}\}^2 = \frac{2\sqrt{f_0(u)}}{m}\left[f_0(du) - \{d\sqrt{f_0(u)}\}^2\right],$$

where m is a function of u, v independent of the differentials du, dv; the value in the present example is in fact $m^2 = c^2 \sin^2 \sigma$; or the identity is

$$\phi(du) - \{d\sqrt{f_0(u)}\}^2 = \frac{c^2 \sin^2 \sigma}{f_0(u)}\left[f_0(du) - \{d\sqrt{f_0(u)}\}^2\right],$$

in verification whereof observe that we have

$$d\sqrt{f_0(u)} = \frac{u\,du + v\,dv + w\,dw}{2\sqrt{f_0(u)}} = \frac{u\,du + v\,dv + w\,dw}{\sqrt{(u^2 + v^2 + w^2)}}.$$

The value of the left-hand side is thus

$$= \tfrac{1}{2}\left(1 - \frac{\sin^2\sigma}{\sigma^2}\right)(u\,du + v\,dv + w\,dw)^2 + \tfrac{1}{2}\frac{c^2\sin^2\sigma}{\sigma^2}(du^2 + dv^2 + dw^2)$$

$$- \frac{(u\,du + v\,dv + w\,dw)^2}{u^2 + v^2 + w^2},$$

viz. this is

$$\frac{c^2\sin^2\sigma}{u^2 + v^2 + w^2}\left\{\tfrac{1}{2}(du^2 + dv^2 + dw^2) - \frac{(u\,du + v\,dv + w\,dw)^2}{2(u^2 + v^2 + w^2)}\right\};$$

or, finally, it is

$$\frac{c^2\sin^2\sigma}{f_0(u)}\left[f_0(du) - \{d\sqrt{f_0(u)}\}^2\right],$$

which is right.

578. A Memoir on the Transformation of Elliptic Functions

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113. The theory of Transformation in Elliptic Functions was established by Jacobi in the *Fundamenta Nova* (1829); and he has there developed, transcendentally, with an approach to completeness, the general case, n an odd number, but algebraically only the cases $n = 3$ and $n = 5$; viz. in the general case the formulae are expressed in terms of the elliptic functions of the nth part of the complete integrals, but in the cases $n = 3$ and $n = 5$ they are expressed rationally in terms of u and v (the fourth roots of the original and the transformed moduli respectively), these quantities being connected by an equation of the order 4 or 6, the modular equation.

The extension of this algebraical theory to any value whatever of n is a problem of great interest and difficulty: such theory should admit of being treated in a purely algebraical manner; but the difficulties are so great that it was found necessary to discuss it by means of the formulae of the transcendental theory, in particular by means of the expressions the exponential of $\frac{\pi i}{\mathrm{K}} \int \frac{d\Phi}{\sqrt{(1-k^2 \sin^2 \Phi)}}$, or say by means of the q-transcendents.

Several important contributions to the theory have since been made:—Sohnke, "Equationes modulares pro transformatione functionum ellipticarum," Crelle, t. xvi. (1836), pp. 97–130, (where the modular equations are found for the cases $n = 3$, 5, 7, 11, 13, 17, and 19); Joubert, "Sur divers Equations analogues aux equations modulaires dans la theorie des fonctions elliptiques," Comptes Rendus, t. XLVII. (1858), pp. 337–345, (relating among other things to the multiplier equation for the determination of Jacobi's M); and Königsberger, "Algebraische Untersuchungen aus der Theorie der elliptischen Functionen," Crelle, t. LXXII. (1870), pp. 176–275; together with other papers by Joubert and by Hermite in later volumes of the Comptes Rendus, which need not be more particularly referred to.

In the present Memoir I carry on the theory, algebraically, as far as I am able; and I have, it appears to me, put the purely algebraical question in a clearer light than has hitherto been done; but I still find it necessary to resort to the transcendental theory.

I remark that the case $n=7$ (next succeeding those of the *Fundamenta Nova*), on account of the peculiarly simple form of the modular equation $(1-u^8)(1-v^8)=(1-u^2v^2)^4$, presents but little difficulty; and I give the complete formulae for this case, obtaining them as well algebraically as transcendentally; I also to a considerable extent discuss algebraically the case of the next succeeding prime value $n=11$.

For the sake of completeness I reproduce Sohnke's modular equations, exhibiting them for greater clearness in a square form, and adding to them those for the non-prime cases $n=9$ and $n=15$; also a valuable table given by him for the powers of $f(q)$; and I give other tabular results which are of assistance in the theory.

The General Problem.
Article Nos. 1 to 6.

1. Taking n a given odd number, I write

$$\frac{1-y}{1+y}=\frac{1-x}{1+x}\left(\frac{P-Qx}{P+Qx}\right)^2,$$

where P, Q are rational and integral functions of x^2, $P\pm Qx$ being each of them of the order $\frac{1}{2}(n-1)$, or, what is the same thing, $(1\pm x)(P\pm Qx)^2$ being each of them of the order n; that is,

	$n=4p+1,$	$n=4p+3,$
Order of P in x^2 is	$p,$	$p;$
Order of Q in x^2 is	$p-1;$	$p;$

whence in the first case the number of coefficients in P and Q is $(p+1)+p=\frac{1}{2}(n+1)$, and in the second case the number is $(p+1)+(p+1)=\frac{1}{2}(n+1)$, as before.

Taking $P=\alpha+\beta x^2+\gamma x^4+\cdots$, $Q=\beta'+\delta' x^2+\epsilon' x^4+\cdots$, the formula is

$$\frac{1-y}{1+y}=\frac{1-x}{1+x}\left(\frac{\alpha-\beta x+\gamma x^2-\cdots}{\alpha+\beta x+\gamma x^2+\cdots}\right)^2,$$

the number of coefficients being as just explained.

Starting herefrom I reproduce in a somewhat altered form the investigation in the *Fundamenta Nova*, as follows.

2. If the coefficients are such that the equation remains true when we therein change simultaneously x into $\frac{1}{kx}$ and y into $\frac{1}{\lambda y}$, then the variables x, y will satisfy the differential equation

$$M\frac{dy}{\sqrt{(1-y^2.1-\lambda^2y^2)}} = \frac{dx}{\sqrt{(1-x^2.1-k^2x^2)}}$$

(M a constant, the value of which, as will appear, is given by $\frac{1}{M} = 1 + 2\beta$); and the problem of transformation is thus to find the coefficients so that the equation may remain true on the above simultaneous change of the values of x, y.

In fact, observing that the original equation and therefore the new equation are each satisfied on changing therein simultaneously x, y into $-x$, $-y$, it follows that the equation may be written in the four forms

$$\begin{aligned}
1-y &= (1-x)A^2(-x),\\
1+y &= (1+x)B^2(-x),\\
1-\lambda y &= (1-kx)C^2(-x),\\
1+\lambda y &= (1+kx)D^2(-x),
\end{aligned}$$

the common denominator being, say E, where A, B, C, D, E are all of them rational and integral functions of x; and this being so, the differential equation will be satisfied.

3. To develop the condition, observe that the assumed equation gives

$$x(P^2+2PQ+Q^2x^2) - y(P^2+2PQx^2+Q^2x^4) = 0,$$

where $\mathfrak{A}$, $\mathfrak{B}$ are functions each of them of the degree $\frac{1}{2}(n-1)$ in x^2.

(Hence, if with Jacobi $\frac{1}{M}$ denotes the value $\left(\frac{y}{x}\right)_{x=0}$, we have $\frac{1}{M} = \left(1+2\frac{\beta}{\alpha}\right)_{x=1} = 1+2\beta$, as mentioned.)

Suppose in general that U being any integral function $(1, x^2)^p$, we have

$$U^* = (1-kx^2)^pU\left(\frac{1}{kx}\right),$$

viz. let U^* be what U becomes when x is changed into $\frac{1}{kx}$ and the whole multiplied by $(1-kx^2)^p$.

Let y^* be the value of y obtained by writing $\frac{1}{kx}$ for x; then, observing that in the expression for y the degree of the numerator exceeds by unity that of the denominator, we have

$$yy^* = \frac{1}{k}\frac{\mathfrak{A}\mathfrak{A}^*}{\mathfrak{B}\mathfrak{B}^*},$$

and the functions $\mathfrak{A}$, $\mathfrak{B}$ may be such that this shall be a constant value, $= \frac{1}{\lambda}$; viz. this will be the case if $\frac{1}{\lambda} = \frac{1}{k}\frac{\mathfrak{A}\mathfrak{A}^*}{\mathfrak{B}\mathfrak{B}^*}$, which being so, the required condition is satisfied.

4. I shall ultimately, instead of k, λ introduce Jacobi's u, v ($u = \sqrt[4]{k}$, $v = \sqrt[4]{\lambda}$); but it is for the present convenient to retain k, and instead of λ to introduce the quantity Ω connected with it by the equation $\lambda = k\Omega^2$; or say the value of Ω is $= \sqrt{\frac{\lambda}{k}}$.

The modular equation in its standard form is an equation between u, v, which, as will appear, gives rise to an equation of the same order between u^2, v^2; and writing herein $\Omega = \frac{v^2}{u^2}$, the resulting equation contains only integer powers of u^4, that is, of k, and we have an Ω-form of the modular equation, or say an Ω-modular equation, of the same order in Ω as the standard form is in v; these Ω-forms for $n = 3$, 5, 7, 11 will be given presently.

5. Suppose then, Ω being a constant, that we have identically

$$\frac{1}{\Omega}k^{\frac{1}{2}(n-1)}\mathfrak{A}^2 = \mathfrak{B}\mathfrak{B}^*;$$

this implies

$$\frac{1}{\Omega}k^{\frac{1}{2}(n-1)}\mathfrak{A}\mathfrak{A}^* = \mathfrak{B}^2,$$

(In fact, if

$$\begin{aligned}
\mathfrak{A} &= a + ax^2 + \cdots + qx^{n-3} + sx^{n-1},\\
\mathfrak{B} &= b + cx^2 + \cdots + rx^{n-3} + tx^{n-1},\\
\mathfrak{A}^* &= s + qk^1x^2 + \cdots + ck^{\frac{1}{2}(n-1)}x^{n-1} + ak^{\frac{1}{2}(n-1)}x^{n-1},\\
\mathfrak{B}^* &= t + rk^1x^2 + \cdots + dk^{\frac{1}{2}(n-1)}x^{n-1} + bk^{\frac{1}{2}(n-1)}x^{n-1},
\end{aligned}$$

and the assumed equation gives

$$\frac{1}{\Omega}k^{\frac{1}{2}(n-1)}\mathfrak{A}^2 = \mathfrak{B}\mathfrak{B}^*, \qquad \frac{1}{\Omega}k^{\frac{1}{2}(n-1)}\mathfrak{A}\mathfrak{A}^* = \mathfrak{B}^2,$$

that is,

$$\mathfrak{B} = \frac{1}{\Omega}k^{\frac{1}{2}(n-1)}\mathfrak{A}^*.)$$

From these equations $\mathfrak{A}\mathfrak{B} = \Omega^n$, that is, $= \frac{v^2}{u^2}$, as it should be; so that Ω signifying as above, the required condition will be satisfied if only $\mathfrak{A} = \Omega k^{\frac{1}{2}(n-1)}\mathfrak{B}^*$; or substituting for $\mathfrak{A}$, $\mathfrak{B}$ their values, if only

$$(P^2 + 2PQx^2 + Q^2x^4)^2 = \Omega k^{\frac{1}{2}(n-1)}(P^2 + 2PQ + Q^2x^2),$$

where each side is a function of x^2 of the order $\frac{1}{2}(n-1)$, or the number of terms is $\frac{1}{2}(n+1)$, the several coefficients being obviously homogeneous quadric functions of the $\frac{1}{2}(n+1)$ coefficients of P, Q.

We have thus $\frac{1}{2}(n+1)$ equations, each of the form $U = \Omega V$, where U, V are given quadric functions of the coefficients of P, Q, say of the $\frac{1}{2}(n+1)$ coefficients α, β, γ, δ, &c., and where Ω is indeterminate.

6. We may from the $\frac{1}{2}(n+1)$ equations eliminate the $\frac{1}{2}(n-1)$ ratios $\alpha : \beta : \gamma : \cdots$, thus obtaining an equation in Ω (involving of course the parameter k) which is the Ω-modular equation above referred to; and then Ω denoting any root of this equation, the $\frac{1}{2}(n+1)$ equations give a single value for the set of ratios $\alpha : \beta : \gamma : \delta : \cdots$, so that the ratio of the functions P, Q is determined, and consequently the value of y as given by the equation

$$\frac{1-y}{1+y} = \frac{(1-x)(P-Qx)^2}{(1+x)(P+Qx)^2} = \frac{x(P^2 + 2PQ + Q^2x^2)}{P^2 + 2PQx^2 + Q^2x^4}.$$

The entire problem thus depends on the solution of the system of $\frac{1}{2}(n+1)$ equations,

$$(P^2 + 2PQx^2 + Q^2x^4)^2 = \Omega k^{\frac{1}{2}(n-1)}(P^2 + 2PQ + Q^2x^2).$$

The Ωk-Modular Equations, $n = 3$, 5, 7, 11.
Article No. 7.

7. For convenience of reference, and to fix the ideas, I give these results, calculated, as above explained, from the standard or uv-forms.

For $n = 3$:

	k^0	k^1	k^2	k^3	k^4
Ω^0	1	-4	$+6$	-4	1
Ω^1					
	$= 0$,	$n = 3$;	$n = 1$, we have $\frac{1}{4}(k-1) = 0$.		

For $n = 5$:

	k^0	k^1	k^2	k^3	k^4	k^5	k^6
Ω^0	1	-16	$+100$	-16	1		
Ω^1		$+10$	-20	$+15$	-20	$+10$	
Ω^2							
	$= 0, \quad n = 5; \quad n = 1,$ we have $-\frac{1}{16}(k^2 - 1)^2 = 0.$						

116. For $n = 7$:

	k^0	k^1	k^2	k^3	k^4	k^5	k^6	k^7	k^8
Ω^0	1	-64	$+924$		$+70$		-64	1	
Ω^1		$+56$	-112	$+140$	-112	$+56$			
Ω^2									
	$= 0, \quad n = 7; \quad n = 1,$ we have $\frac{1}{64}(k - 1)^2(k^2 + 3k + 1)(k^2 + 3k + 1) = 0.$								

117. For $n = 11$:

	k^0	k^1	k^2	k^3	k^4	k^5	k^6	k^7	k^8	$\cdots$
Ω^0	1	-1024								
Ω^1		$+\cdots$								

$= 0, \quad n = 11.$

Equation-systems for the cases $n = 3$, 5, 7, 9, 11.
Art. Nos. 8 to 10.

8. $n = 3$, cubic transformation.

$k = u^4$, $\Omega = \dfrac{v^2}{u^2}$ (here and in the other cases).

$P = \alpha$, $Q = \beta$.

The condition here is

$$k^2\alpha^2\beta^2 + (2\alpha\beta + \beta^2)^2 = \Omega k\{(\alpha^2 + 2\alpha\beta) + \beta^2\alpha^2\},$$

and the system of equations thus is

$$2\alpha\beta + \beta^2 = \Omega k(\alpha^2 + 2\alpha\beta),$$

and similarly in the other cases; for these it will be enough to write down the equation-systems.

$n = 5$, quintic transformation.

$$2\alpha\gamma + 2\alpha\beta + \beta^2 = \Omega(2\alpha\gamma + 2\beta\gamma + \beta^2),$$
$$\gamma^2 + 2\beta\gamma = \Omega k^2(\alpha^2 + 2\alpha\beta).$$

$n = 7$, septic transformation.

$$P = \alpha + \gamma x^2, \quad Q = \beta + \delta x^2.$$

$$k^2(2\alpha\gamma + 2\alpha\beta + \beta^2) = \Omega(\gamma^2 + 2\beta\delta + 2\beta\gamma),$$

$$\gamma^2 + 2\beta\gamma\delta + 2\alpha\delta + 2\beta\delta = \Omega k^2(2\alpha\gamma + 2\beta\gamma + 2\alpha\delta + \beta^2),$$

$$\delta^2 + 2\gamma\delta = \Omega k^4(\alpha^2 + 2\alpha\beta).$$

$n = 9$, enneadic transformation.

$$P = \alpha + \gamma x^2 + \epsilon x^4, \quad Q = \beta + \delta x^2.$$

$$k^4\alpha^2 = \Omega\epsilon^2,$$

$$k^2(2\alpha\gamma + 2\alpha\beta + \beta^2) = \Omega(2\gamma\epsilon + 2\epsilon\delta + \delta^2),$$

$$2\alpha\epsilon + \gamma^2 + 2\alpha\delta + 2\gamma\beta + 2\beta\delta = \Omega(2\alpha\epsilon + \gamma^2 + 2\gamma\delta + 2\epsilon\beta + 2\beta\delta),$$

$$2\gamma\epsilon + 2\gamma\delta + 2\epsilon\beta + \delta^2 = \Omega k^2(2\beta\gamma + 2\alpha\delta + 2\gamma\beta + \beta^2),$$

$$\epsilon^2 + 2\gamma\delta = \Omega k^4(\alpha^2 + 2\alpha\beta).$$

$n = 11$, endecadic transformation.

$$P = \alpha + \gamma x^2 + \epsilon x^4, \quad Q = \beta + \delta x^2 + \zeta x^4.$$

$$k^4(2\alpha\gamma + 2\alpha\delta + \beta^2) = \Omega(\epsilon^2 + 2\epsilon\zeta + 2\delta\zeta),$$

$$k^2(2\alpha\epsilon + \gamma^2 + 2\alpha\delta + 2\gamma\beta + 2\beta\delta) = \Omega(2\gamma\epsilon + 2\gamma\zeta + 2\epsilon\delta + 2\beta\delta + \delta^2),$$

$$2\gamma\epsilon + 2\alpha\zeta + 2\gamma\delta + 2\epsilon\beta + 2\beta\zeta + \delta^2 = \Omega k^2(2\alpha\epsilon + \gamma^2 + 2\alpha\zeta + 2\gamma\delta + 2\epsilon\beta + 2\beta\delta),$$

$$\gamma^2 + 2\gamma\zeta + 2\epsilon\delta + 2\delta\zeta = \Omega k^4(2\alpha\gamma + 2\alpha\delta + 2\gamma\beta + \beta^2),$$

$$2\epsilon\zeta + \zeta^2 = \Omega k^6(\alpha^2 + 2\alpha\beta).$$

And so on.

120. 9. It will be noticed that if the coefficients of $P + Qx$ taken in order are α, β, ..., ρ, σ, then in every case the first and last equations are

$$\Omega^{\frac{1}{2}(n-1)}\alpha\sigma = \Omega^n\alpha\sigma, \qquad 2\rho\sigma + \sigma^2 = \Omega k^{\frac{1}{2}(n-1)}(\alpha^2 + 2\alpha\beta).$$

Putting in the first of these $k = u^4$, $\Omega = \dfrac{v^2}{u^2}$, the equation becomes

$$v^{n-1}\alpha\sigma = u^{n-1}v\alpha\sigma,$$

where each side is a perfect square; and in extracting the square root we may without loss of generality take the roots positive, and write

$$v^{\frac{1}{2}(n-1)}\alpha\sigma = u^{\frac{1}{2}(n-1)}v\alpha\sigma.$$

This speciality, although it renders it proper to employ ultimately u, v in place of k, Ω, produces really no depression of order (viz. the Ωk-form of the modular equation is found to be of the same order in Ω that the standard or uv-form is in v), and is in another point of view a disadvantage, as destroying the uniformity of the several equations: in the discussion of order I consequently retain Ω, k.

Ultimately these are to be replaced by u, v; the change in the equation-systems is so easily made that it is not necessary here to write them down in the new form in u, v.

10. The case $\alpha = 0$ has to be considered in the discussion of order, but we have thus only solutions which are to be rejected; in the proper solutions α is not $= 0$, and it may therefore for convenience be taken to be $= 1$.

We have then $\sigma = \Omega^{\frac{1}{2}n} \div v$.

The last equation becomes therefore

$$2\rho\Omega^{\frac{1}{2}n}v^{-1} + \Omega^n v^{-2} = \Omega k^{\frac{1}{2}(n-1)}(1+2\beta),$$

or recollecting that β is connected with the multiplier M by the relation

$$v^2 = 1 + 2\beta,$$

that is,

$$\frac{1}{M} = 1 + 2\beta,$$

and substituting for $1 + 2\beta$ its value, the equation becomes

$$\frac{2\rho\Omega^{\frac{1}{2}n}v + \Omega^n}{v^2} = \frac{\Omega k^{\frac{1}{2}(n-1)}}{M},$$

that is, the first and the last coefficients are $1, -\dfrac{\Omega^{\frac{1}{2}n}}{v}$, and the second and the penultimate coefficients are each expressed in terms of v, M.

The cases $n = 3$, $n = 5$ are so far peculiar, that the only coefficients are α, β, or α, β, γ; in the next case $n = 7$, the only coefficients are α, β, γ, δ, and we have in this case all the coefficients expressed as above.

121. The Ωk-form—Order of the Systems. Art. Nos. 11 to 22.

11. In the general case, n an odd number, we have Ω, and $\frac{1}{2}(n+1)$ coefficients connected by a system of $\frac{1}{2}(n+1)$ equations of the form

$$\frac{U}{U'} = \frac{V}{V'} = \cdots = \Omega.$$

Omitting the $\frac{U}{U'}(=\Omega)$, there remains a system of $\frac{1}{2}(n-1)$ equations of the form $\frac{U}{U'} = \frac{V}{V'} = \cdots$, or say

$$\begin{vmatrix} U, & V, & W, & \ldots \\ U', & V', & W', & \ldots \end{vmatrix} = 0,$$

which determine the ratios $\alpha : \beta : \gamma : \cdots$ of the coefficients; and to each set of ratios there corresponds a single value of Ω.

The order of the system, or number of sets of ratios, is $= \frac{1}{2}(n+1) \cdot 2^{\frac{1}{2}(n-1)} = (n+1) \cdot 2^{\frac{1}{2}(n-3)}$; and this is consequently the number of values of Ω, or the order of the equation for the determination of Ω; viz. but for reduction, the order in Ω of the Ωk-modular equation would be $= (n+1) \cdot 2^{\frac{1}{2}(n-3)}$.

In the case $n = 3$, this is $= 4$, which is right, but for any larger value of n the order is far too high; in fact, assuming (as the case is) that the order is equal to the order in v of the uv-form, the order should for a prime value of n be $= n+1$, and for a composite value not containing any square factor be $=$ the sum of the divisors of n.

I do not attempt a general investigation, but confine myself to showing in what manner the reductions arise.

12. I will first consider the cubic transformation; here, writing for convenience $\Omega^{\frac{1}{2}} = q$, the equations give

$$kq\beta^2 + 2\beta - q\alpha^2 = 0 \quad \text{and} \quad kq^2\alpha^2 = \Omega\beta^2,$$

that is,

$$k^2q^2\alpha^2(\beta+2) - (2\beta+1) = 0,$$

The equation in q gives

$$(k^2q^2-1)^2 - 4q^2(kq^2-1)^2 = 0,$$

and we have thence

$$kq^4 - 4k^2q^3 + 6kq^2 - 4q + k = 0,$$

the modular equation; and then

$$kq^2(\beta^2 - 1 + 2\beta(kq^2 - 1)) = 0, \quad \text{that is, } \beta = -1 + 2\beta(kq^2 - 1) = 0,$$

or $2\beta = -\dfrac{1}{kq^2 - 1}$; say we have $\alpha = \Omega = -1$, $\beta = 2(1 - k\Omega)$; consequently

$$\frac{1-y}{1+y} = \frac{1-x}{1+x}\left\{\frac{1 + 2(\Omega - 1)x}{1 - 2(\Omega - 1)x}\right\}^2,$$

which completes the theory.

C. IX. 16

122. 13. Reproducing for this case the general theory, it appears *a priori* that Ω is determined by a quartic equation; in fact, from the original equations eliminating Ω, we have an equation

$$\begin{vmatrix} U, & V \\ U', & V' \end{vmatrix} = 0,$$

where U, U', V, V' are quartic functions of α, β; that is, the ratio $\alpha : \beta$ has four values, and to each of these there corresponds a single value of Ω; viz. Ω is determined by a quartic equation.

14. Considering next the case $n = 5$, the quintic transformation; the elimination of Ω gives the equations

$$\frac{U}{U'} = \frac{V}{V'} = \frac{W}{W''},$$

where U, U', &c. are all quadric functions of α, β, γ.

We have thence $4 \cdot 4 - 2 \cdot 2 = 12$ sets of values of $\alpha : \beta : \gamma$; viz. considering α, β, γ as coordinates in piano, the curves $UV - U'V = 0$, $UW - U'W = 0$ are quartic curves intersecting in 16 points; but among these are included the four points $U = 0$, $U' = 0$ (in fact, the point $\alpha = 0$, $\gamma = 0$ four times), which are not points of the curve $VW - V'W = 0$; there remain therefore $16 - 4 = 12$ intersections, agreeing with the general value $(n+1) \cdot 2^{\frac{1}{2}(n-3)}$.

Hence Ω is in the first instance determined by an equation of the order 12; but the proper order being $= 6$, there must be a factor of the order 6 to be rejected.

To explain this and to determine the factor, observe that the equations in question are

$$k^2\alpha^2(2\alpha\gamma + 2\beta\gamma + \beta^2) - \gamma^2(2\alpha\gamma + 2\alpha\beta + \beta^2) = 0,$$

$$k^2\alpha^2(\alpha + 2\beta) - \gamma(\gamma + 2\beta) = 0;$$

at the point $\alpha = 0$, $\gamma = 0$, the first of these has a double point, the second a triple point; or there are at the point in question 6 intersections; but 4 of these are the points which give the foregoing reduction $16 - 4 = 12$; we have thus the point $\alpha = 0$, $\gamma = 0$, counting twice among the twelve points.

Writing in the two equations $\beta = 0$, the equations become

$$k^2\alpha^2\gamma^2 - \gamma^4 = 0, \qquad k^2\alpha^4 - \gamma^4 = 0,$$

viz. these will be satisfied if $k^2\alpha^2 - \gamma^2 = 0$, that is, the curves pass through each of the two points $(\beta = 0, \gamma = \pm k\alpha)$, and these values satisfy (as in fact they should) the third equation

$$k^2(2\alpha\gamma + 2\alpha\beta + \beta^2)\alpha(\alpha + 2\beta) - \gamma(\gamma + 2\beta)(2\alpha\gamma + 2\beta + \beta^2) = 0.$$

It is moreover easily shown that the three curves have at each of the points in question a common tangent; viz. taking A, B, C as current coordinates, the tangent at the point (α, β, γ) of the second curve has for its equation

$$A(2\alpha^2 + 3\alpha\beta)k^2 + B(k^2\alpha^2 - \gamma^2) - C(2\gamma^2 + 3\gamma\beta) = 0;$$

and for $\beta = 0$, $\gamma = \pm k\alpha$, this becomes

$$2k^2A + k^2(k+1)B + 2C = 0,$$

viz. this is the line from the point $(\beta = 0, \gamma = \pm k\alpha)$ to the point $(1, -2, 1)$. And similarly for the other two curves we find the same equation for the tangent.

123. Hence among the 12 points are included the point $(\gamma = 0, \alpha = 0)$ twice, and the points $(\beta = 0, \gamma = \pm k\alpha)$ each twice: we have thus a reduction $= 6$.

15. Writing in the equations $\gamma = 0$, $\alpha = 0$, the first and third are satisfied identically, and the second becomes $\beta^2 = \Omega k^2$, that is, the equations give $\Omega = 1$; writing $\beta = 0$, they become

$$k^2\alpha^2\gamma^2 = \Omega\gamma^2\beta^2, \quad \alpha\gamma = \Omega\alpha\gamma, \quad \gamma^4 = \Omega\gamma^2 k^2\alpha^2,$$

viz. putting herein $\gamma^2 = k^2\alpha^2$, the equations again give $\Omega = 1$; hence the factor of the order 6 is $(\Omega - 1)^6$, and the equation of the twelfth order for the determination of Ω, is

$$(\Omega - 1)^6\{(\Omega, 1)^6\} = 0,$$

where $(\Omega, 1)^6 = 0$ is the Ωk-modular equation above written down.

16. Reverting to the equation

$$\frac{1-y}{1+y} = \frac{(1-x)(P-Qx)^2}{(1+x)(P+Qx)^2},$$

it is to be observed that for $\alpha = 0$, $\gamma = 0$, that is, $P = 0$, this becomes simply $y = x$, which is the transformation of the order 1; the corresponding value of the modulus λ is $\lambda = k$, and the equation $\lambda = \Omega^2 k$ then gives $\Omega^2 = 1$, which is replaced by $\Omega - 1 = 0$.

If in the same equation we write $\beta = 0$, that is, $Q = 0$, then (without any use of the equation $\gamma^2 = k^2\alpha^2$) we have $y = x$, the transformation of the order 1; but although this is so, the fundamental equation

$$(P^2 + 2PQx^2 + Q^2x^4)^2 = \Omega k^{\frac{1}{2}(n-1)}(P^2 + 2PQ + Q^2x^2),$$

which, putting therein $Q = 0$, becomes

$$(P^2)^2 = \Omega k^{\frac{1}{2}(n-1)}P^2, \quad \text{that is, } (k^2\alpha + \gamma x^2)^2 = \Omega k^{\frac{1}{2}(n-1)}(\alpha + \gamma x^2),$$

is not satisfied by the single relation $\Omega - 1 = 0$, but necessitates the further relation $k^2\alpha = \gamma^2$.

The thing to be observed is that the extraneous factor $(\Omega - 1)^6$, equated to zero, gives for Ω the value $\Omega = 1$; but that this does not in any case (except $n = 3$) lead to the transformation of order 1.

124. 17. For the case $n = 7$, we have

$$P = \alpha + \gamma x^2, \quad Q = \beta + \delta x^2,$$

and the equation-system is

$$\begin{aligned} k^2(2\alpha\gamma + 2\alpha\beta + \beta^2) &= \Omega(\gamma^2 + 2\beta\delta + 2\beta\gamma), \\ \gamma^2 + 2\beta\gamma\delta + 2\alpha\delta + 2\beta\delta &= \Omega k^2(2\alpha\gamma + 2\beta\gamma + 2\alpha\delta + \beta^2), \\ \delta^2 + 2\gamma\delta &= \Omega k^4(\alpha^2 + 2\alpha\beta). \end{aligned}$$

Eliminating Ω, we have two equations, which being quadrics in $(\alpha, \beta, \gamma, \delta)$, intersect in $2 \cdot 2 \cdot 2 \cdot 2 = 16$ points; but we have the reductions arising from the points $\alpha = 0$, $\gamma = 0$ (4 times) and $\beta = 0$, $\delta = 0$ (4 times); there remain thus $16 - 8 = 8$ points, agreeing with the general formula $(n+1) \cdot 2^{\frac{1}{2}(n-3)} = 8 \cdot 2^2 = 32$, which is too high by the factor $2^{\frac{1}{2}(n-3)} = 4$. The proper order in Ω being $= n + 1 = 8$, the extraneous factor is of the order $32 - 8 = 24$.

But without going into the complete theory, I observe that the points $\alpha = 0$, $\gamma = 0$ give $\Omega = 1$ twice over; and the points $\beta = 0$, $\delta = \pm k\alpha$, $\gamma = \pm k^2\alpha$ give the factor $(\Omega - 1)^4$, and so on.

The complete modular equation for $n = 7$ is $(1 - u^8)(1 - v^8) = (1 - u^2v^2)^4$, as will be shown.

18. One remark may be added as to the case $\alpha = 0$, which gives the rejected solutions. For $n = 3$, $\alpha = 0$ gives $\beta = 0$, and the equation $y = x$; for $n = 5$, $\alpha = 0$ gives the equations $\beta^2 = \Omega\gamma^2$, $\gamma^2 = \Omega k^2\beta^2$, viz. $\Omega^2k^2 = 1$, and then $y = x$; for $n = 7$, $\alpha = 0$ gives a nontrivial transformation. But these are all to be rejected.

125. 19. It is easy to see that the condition for $y = x$, or transformation of order 1, is that P and Q shall be proportional, that is $\alpha : \beta = \gamma : \delta = \epsilon : \zeta = \cdots$, which implies $\Omega = 1$; and conversely $\Omega = 1$ implies this proportionality. The factor $(\Omega - 1)$ thus corresponds to the transformation of order 1.

20. The factor of the equation in Ω which arises from the points $\alpha = 0$, $\gamma = 0$, &c. is in every case a power of $(\Omega - 1)$, but for the larger values of n the theory of the complete reduction is very complicated.

21. As regards the uv-form of the modular equation, this can be obtained from the Ωk-form by writing therein $\Omega = \dfrac{v^2}{u^2}$, $k = u^4$, and clearing. It was in this way that Sohnke calculated the modular equations for the prime values $n = 3$, 5, 7, 11, 13, 17, 19.

22. The formulae may be presented in a more complete form by writing a, b, c, &c. for the coefficients, and determining the ratios

$\dfrac{b}{a}$, $\dfrac{c}{a}$, &c. as rational functions of Ω and k; but the formulae thus become very complicated.

The Modular Equations, $n = 3$, 5, 7, 9, 11, 13, 15, 19. Art. Nos. 23 to 28.

23. I reproduce Sohnke's modular equations in the square form, viz. writing them as equations between u and v, and arranging in a square. I add the equations for $n = 9$ and $n = 15$, which he does not give.

$n = 3$:

	v^0	v^2	v^4	v^6
u^0	1	−2	−2	1
u^2	−2			
u^4	−2			
u^6	1			

$= 0.$

$n = 5$:

	v^0	v^2	v^4	v^6	v^8	v^{10}
u^0	1		−4		1	
u^2						
u^4	−4					
u^6						
u^8	1					
u^{10}						

$= 0.$

126. 24. For $n = 7$:

	v^0	v^2	v^4	v^6	v^8	v^{10}	v^{12}	v^{14}
u^0	1				−2			1
u^2								
u^4								
u^6								
u^8	−2							
u^{10}								
u^{12}								
u^{14}	1							

$= 0.$

In fact the equation is $(1-u^8)(1-v^8) = (1-u^2v^2)^4$, which expanded gives the form above.

25. For $n = 11$:

	v^0	v^2	v^4	v^6	v^8	v^{10}	v^{12}	v^{14}	v^{16}	v^{18}	v^{20}	v^{22}
ν^0	1											1

$$= 0.$$

26. For $n = 13$:

	v^0	v^2	v^4	v^6	v^8	v^{10}	v^{12}	v^{14}	v^{16}	v^{18}	v^{20}	v^{22}	v^{24}	v^{26}
ν^0	-1													1

$$= 0.$$

27. For $n = 15$:

	v^0	v^2	v^4	v^6	v^8	v^{10}	v^{12}	v^{14}	v^{16}	v^{18}	v^{20}	v^{22}	v^{24}	v^{26}	v^{28}	v^{30}
ν^0	1															1

$$= \{(v-1)^6(v+1)\}^2.$$

28. For $n = 19$:

	v^0	v^2	v^4	v^6	v^8	v^{10}	v^{12}	v^{14}	v^{16}	v^{18}	v^{20}	v^{22}	v^{24}
ν^0	-1		$+114$		-114		-2584		-6859		-2584		-11

$$= \{(v+1)^{10}(v-1)\}.$$

C. IX. 17

127. The Tables for $f(q)$.

26. Sohnke gives a valuable table of the powers of $f(q)$, where

$$f(q) = q^{\frac{1}{24}}(1-q)(1-q^2)(1-q^3)\cdots = q^{\frac{1}{24}}\prod_{n=1}^{\infty}(1-q^n).$$

The table gives the expansions of $f(q)$, $f(q^2)$, $f(q^3)$, &c., and also the products which occur in the theory of transformation.

The products required for the multiplier equation are of the form

$$A_n(q) = \frac{(1+q)(1+q^2)(1+q^3)\cdots(1+q^n)}{(1-q)(1-q^2)(1-q^3)\cdots(1-q^n)} \cdot \frac{(1+q^n)(1+q^{2n})\cdots}{(1-q^n)(1-q^{2n})\cdots},$$

and Sohnke's table facilitates the calculation of these products.

27. I remark that if n is prime, then putting in the modular equation $u = 1$, the equation in the case $n \equiv 1$ or $7 \pmod 8$ becomes $(v -$

$1)^{n+1}=0$, but in the case $n\equiv 3$ or $5 \pmod 8$ it becomes $(v+1)^n(v-1)=0$.

And there is in every case a pair of terms $v^n u^n$ and vu, having coefficients equal in absolute magnitude, but of opposite signs, or of the same sign, in the two cases respectively.

Each Table is symmetrical in regard to its two diagonals respectively, so that every non-diagonal coefficient occurs (with or without reversal of sign) 4 times; viz. in the case $n\equiv 1$ or $7 \pmod 8$ this is a perfect symmetry, without reversal of sign; but in the case $n\equiv 3$ or $5 \pmod 8$ it is, as regards the lines parallel to either diagonal, and in regard to the other diagonal, alternately a perfect symmetry without reversal of sign and a skew symmetry with reversal.

Thus in the case $n=19$, the lines parallel to the dexter diagonal are -1 (symmetrical), $+114$, -114 (skew), 0, -2584, -6859, -2584, (symmetrical), and so on.

The same relation of symmetry is seen in the composite cases $n=9$ and $n=15$, both belonging to $n\equiv 1$ or 7, mod. 8.

28. If, as before, n is prime, then putting in the modular equation $u=1$, the equation in the case $n\equiv 1$ or $7 \pmod 8$ becomes $(v-1)^{n+1}=0$, but in the case $n\equiv 3$ or $5 \pmod 8$ it becomes $(v+1)^n(v-1)=0$.
128. 29. In the case n a composite number not containing any square factor, then dividing n in every possible way into two factors $n=ab$ (including the divisions $n\cdot 1$ and $1\cdot n$), and denoting by β an imaginary bth root of unity, a value of v is $\pm\sqrt[4]{2}\beta^{\frac{1}{4}}\sqrt[4]{a}\cdot f(q^{\frac{1}{a}})\div f(q^a)$; so that the whole number of roots (or order of the modular equation) is $=\nu$, if ν be the sum of the divisors of n.

Thus $n=15$, the values are

$$\sqrt[4]{2}q^{\frac{1}{4}}f(q^{15})\div f(q^{\frac{1}{15}}),\quad \sqrt[4]{2}q^{\frac{3}{4}}f(q^5)\div f(q^3),\quad \sqrt[4]{2}q^{\frac{5}{4}}f(q^3)\div f(q^5),\quad \sqrt[4]{2}q^{\frac{15}{4}}f(q)\div f(q^{15}$$

1, 3, 5, 15 roots; and the order of the modular equation is $=24$.

The modular equation might thus be obtained as for a prime number; but it is easier to decompose n into its prime factors, and consider the transformation as compounded of transformations of these prime orders.

Thus $n=15$, the transformation is compounded of a cubic and a quintic one.

If the v of the cubic transformation be denoted by θ, then we have

$$\theta^4+2\theta^3u^2-2\theta u-u^4=0;$$

and to each of the four values of θ corresponds the six values of v belonging to the quintic transformation given by

$$v^4 + 2v^3\theta^2 - 2v\theta - \theta^4 = 0.$$

The equation in v is thus

$$\prod_{r=1}^{4}(v^4 + 2v^3\theta_r^2 - 2v\theta_r - \theta_r^4) = 0,$$

where θ_1, θ_2, θ_3, θ_4 are the roots of the equation in θ, viz. we have

$$\theta^4 + 2\theta^3u^2 - 2\theta u - u^4 = (\theta - \theta_1)(\theta - \theta_2)(\theta - \theta_3)(\theta - \theta_4);$$

and it was in this way that the equation for the case $n = 15$ was calculated.

Observe that writing $u = 1$, we have

$$(\theta + 1)^2(\theta - 1) = 0, \quad \text{or say } \theta_1 = \theta_2 = -1, \quad \theta_3 = \theta_4 = +1.$$

The equation in v thus becomes $\{(v-1)^2(v+1)\}^2(v+1)^4(v-1) = 0$, that is, $(v-1)^6(v+1)^8 = 0$.

129. 30. The case where n has a square factor is a little different; thus $n = 9$, the values are 1, 3, 9 roots; but here ω being an imaginary cube root of unity, the second term denotes the three values, the first of which is $= u$, and is to be rejected; there remain $1 + 2 + 9 = 12$ roots, or the equation is of the order 12.

Considering the equation as compounded of two cubic transformations, if the value of v for the first of these be θ, then we have

$$\theta^4 + 2\theta^3u^2 - 2\theta u - u^4 = 0;$$

and to the four values of θ correspond severally the four values of v given by the equation

$$v^4 + 2v^3\theta^2 - 2v\theta - \theta^4 = 0.$$

One of these values is however $v = -u$, since the uv-equation is satisfied on writing therein $v = -u$; hence, writing

$$\theta^4 + 2\theta^3u^2 - 2\theta u - u^4 = (\theta - \theta_1)(\theta - \theta_2)(\theta - \theta_3)(\theta - \theta_4),$$

we have an equation

$$(v^4 + 2v^3\theta_1^2 - 2v\theta_1 - \theta_1^4)(v^4 + \cdots - \theta_2^4)(v^4 + \cdots - \theta_3^4)(v^4 \cdots - \theta_4^4) = 0,$$

which contains the factor $(v+u)^4$ and, divested hereof, gives the required modular equation of the order 12; it was in fact obtained in this manner.

Observe that writing $u=1$, we have $(\theta+1)^2(\theta-1)=0$, or say $\theta_1=\theta_2=-1$, $\theta_3=\theta_4=+1$; the modular equation then becomes

$$\{(v-1)^2(v+1)\}^2(v+1)^4(v-1)^4(v+1)^2=0,$$

that is, $(v-1)^4(v+1)^8=0$.

C. IX. 18

130. The Multiplier Equation.
Art. No. 29.

29. The theory is in many respects analogous to that of the modular equation.

To each value of v there corresponds a single value of M; hence M, or what is the same thing $\dfrac{1}{M}$, is determined by an equation of the same order as v, viz. n being prime, the order is $=n+1$.

The last term of the equation is constant, and the other coefficients are rational and integral functions of u^4, of a degree not exceeding $\frac{1}{2}(n-1)$; and not only so, but they are, $n\equiv 1 \pmod 4$, rational and integral functions of $u^4(1-u^4)$, and $n\equiv 3 \pmod 4$, alternately of this form and of the same form multiplied by the factor $(1-2u^4)$.

The values are in fact given as transcendental functions of q; viz. denoting by M_0, M_1, M_2, …, M_n the values corresponding to v_0, v_1, v_2, …, v_n respectively, and writing

$$A_n(q)=\frac{(1+q)(1+q^2)(1+q^3)\cdots}{(1-q)(1-q^2)(1-q^3)\cdots}\cdot\frac{(1+q^n)(1+q^{2n})\cdots}{(1-q^n)(1-q^{2n})\cdots}$$

$$=\frac{(1-q^2)(1-q^4)(1-q^6)\cdots(1-q^{2n})\cdots}{(1-q)(1-q^2)(1-q^3)\cdots(1+q)(1+q^2)(1+q^3)\cdots}$$

$$=1+2q+2q^4+2q^9+2q^{16}+\cdots,$$

then we have

$$\frac{1}{M_0}=(-)^n\frac{A_n(q)}{A_n(q^n)},\quad M_1=\frac{A_n(\omega q)}{A_n(q^n)},\quad \ldots,\quad M_n=\frac{A_n(\omega^n q)}{A_n(q^n)},$$

(ω an imaginary nth root of unity).

Hence, the form of the equation being known, the values of the numerical coefficients may be calculated; and it was in this way that Joubert obtained the following results.

I have in some cases changed the sign of Joubert's multiplier, so that in every case the value corresponding to $u = 0$ shall be $M = 1$.

The equations are:

$n = 3$, $M = 0$, this is $M^2(m^{-1} + m - 2) - 3 = 0$.

$n = 5$,

$$\left(\frac{1}{M}\right)^6 + \frac{1}{M^4}(-10) + \frac{1}{M^2}(-26 + 256(1 - 2u^4)) + 5 = 0.$$

$n = 7$,

$$\left(\frac{1}{M}\right)^8 + \frac{1}{M^6}(-14) + \frac{1}{M^4}(-70 + 224(1 - 2u^4))$$

$$+\frac{1}{M^2}(-140 + 21 \cdot 25(1 - 2u^4)) + \frac{1}{M}(48 + 2048u^4(1 - u^4)) + 7 = 0.$$

$n = 11$,

$$\left(\frac{1}{M}\right)^{12} + \frac{1}{M^{10}}(-66) + \frac{1}{M^8}(- \cdots)$$

$$+\frac{1}{M^6}(\cdots) + \frac{1}{M^4}(\cdots) + \frac{1}{M^2}(\cdots) + 11 = 0.$$

131. The modular equations may be written in square form as follows:

$n = 9$:

	v^0	v^2	v^4	v^6	v^8	v^{10}	v^{12}	v^{14}	v^{16}
u^0	1	-16	$+8$	$+16$	$+10$	-16	-24	$+15$	1

$$= (v-1)^6(v+1)^3.$$

$n = 11$:

	v^0	v^2	v^4	v^6	v^8	v^{10}	v^{12}	v^{14}	v^{16}	v^{18}	v^{20}
ν^0	1	-32	-22	$+44$	$+88$	$+22$	-165	$+132$	$+44$	-44	1

$$= (v+1)^6(v-1).$$

$n = 13$:

	v^0	v^2	v^4	v^6	v^8	v^{10}	v^{12}	v^{14}	v^{16}	v^{18}	v^{20}	v^{22}
ν^0	-1	$+64$	-52	-65	$+208$	-429	$+520$	$+52$	-429	$+208$	-65	$+52$

$$+1 = (v+1)^{12}(v-1).$$

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132. For $n = 13$ (continued):

	v^0	v^2	v^4	v^6	v^8	v^{10}	v^{12}	v^{14}	v^{16}	v^{18}	v^{20}	v^{22}
ν^0	-1	$+64$	-52	-65	$+208$	-429	$+520$	$+52$	-429	$+208$	-65	$+52$
$+1$												

$$= (v+1)^{12}(v-1).$$

133–135. [Tabular data for higher order modular equations.]

The tables continue with the coefficients for $n = 13$, 15, 17, 19 in square form. The coefficients exhibit the symmetrical properties described in Art. 27.

136. Various remarks arise on the Tables.

27. Attending first to the cases n a prime number; the only terms of the order $n+1$ in v or u are $u^{n+1} \pm v^{n+1}$; viz. $n = 3$ or $5 \pmod 8$ the sign is $-$, but $n = 1$ or $7 \pmod 8$ the sign is $+$.

And there is in every case a pair of terms $v^n u^n$ and vu, having coefficients equal in absolute magnitude, but of opposite signs, or of the same sign, in the two cases respectively.

Each Table is symmetrical in regard to its two diagonals respectively, so that every non-diagonal coefficient occurs (with or without reversal of sign) 4 times; viz. in the case $n \equiv 1$ or $7 \pmod 8$ this is a perfect symmetry, without reversal of sign; but in the case $n \equiv 3$ or $5 \pmod 8$ it is, as regards the lines parallel to either diagonal, and in regard to the other diagonal, alternately a perfect symmetry without reversal of sign and a skew symmetry with reversal.

Thus in the case $n = 19$, the lines parallel to the dexter diagonal are -1 (symmetrical), $+114$, -114 (skew), 0, -2584, -6859, -2584 (symmetrical), and so on.

The same relation of symmetry is seen in the composite cases $n = 9$ and $n = 15$, both belonging to $n \equiv 1$ or 7, mod. 8.

If, as before, n is prime, then putting in the modular equation $u = 1$, the equation in the case $n \equiv 1$ or $7 \pmod 8$ becomes $(v-1)^{n+1} = 0$, but in the case $n \equiv 3$ or $5 \pmod 8$ it becomes $(v+1)^n(v-1) = 0$.

27. In the case n a composite number not containing any square factor, then dividing n in every possible way into two factors $n = ab$ (including the divisions $n \cdot 1$ and $1 \cdot n$), and denoting by β an imaginary bth root of unity, a value of v is $\pm\sqrt[4]{2}\beta^{\frac{1}{4}}\sqrt[4]{a} \cdot f(q^{\frac{1}{a}}) \div f(q^a)$; so that the whole number of roots (or order of the modular equation) is $= \nu$, if ν be the sum of the divisors of n.

Thus $n = 15$, the values are 1, 3, 5, 15 roots; and the order of the modular equation is $= 24$.

The modular equation might thus be obtained as for a prime number; but it is easier to decompose n into its prime factors, and consider the transformation as compounded of transformations of these prime orders.

Thus $n = 15$, the transformation is compounded of a cubic and a quintic one.

If the v of the cubic transformation be denoted by θ, then we have

$$\theta^4 + 2\theta^3u^2 - 2\theta u - u^4 = 0;$$

and to each of the four values of θ corresponds the six values of v belonging to the quintic transformation given by

$$v^4 + 2v^3\theta^2 - 2v\theta - \theta^4 = 0.$$

The equation in v is thus

$$\prod_{r=1}^{4}(v^4 + 2v^3\theta_r^2 - 2v\theta_r - \theta_r^4) = 0,$$

where θ_1, θ_2, θ_3, θ_4 are the roots of the equation in θ, viz. we have

$$\theta^4 + 2\theta^3 u^2 - 2\theta u - u^4 = (\theta - \theta_1)(\theta - \theta_2)(\theta - \theta_3)(\theta - \theta_4);$$

and it was in this way that the equation for the case $n = 15$ was calculated.

Observe that writing $u = 1$, we have

$$(\theta + 1)^2(\theta - 1) = 0, \quad \text{or say } \theta_1 = \theta_2 = -1, \quad \theta_3 = \theta_4 = +1.$$

The equation in v thus becomes $\{(v-1)^2(v+1)\}^2(v+1)^4(v-1) = 0$, that is, $(v-1)^6(v+1)^8 = 0$.

137. where θ_1, θ_2, θ_3, θ_4 are the roots of the equation in θ, viz. we have

$$\theta^4 + 2\theta^3 u^2 - 2\theta u - u^4 = (\theta - \theta_1)(\theta - \theta_2)(\theta - \theta_3)(\theta - \theta_4);$$

and it was in this way that the equation for the case $n = 15$ was calculated.

Observe that writing $u = 1$, we have $(\theta + 1)^2(\theta - 1) = 0$, or say $\theta_1 = \theta_2 = -1$, $\theta_3 = +1$.

The equation in v thus becomes $\{(v-1)^2(v+1)\}^2(v+1)^4(v-1) = 0$, that is, $(v-1)^6(v+1)^8 = 0$.

28. The case where n has a square factor is a little different; thus $n = 9$, the values are 1, 3, 9 roots; but here ω being an imaginary cube root of unity, the second term denotes the three values, the first of which is $= u$, and is to be rejected; there remain $1 + 2 + 9 = 12$ roots, or the equation is of the order 12.

Considering the equation as compounded of two cubic transformations, if the value of v for the first of these be θ, then we have

$$\theta^4 + 2\theta^3 u^2 - 2\theta u - u^4 = 0;$$

and to the four values of θ correspond severally the four values of v given by the equation

$$v^4 + 2v^3\theta^2 - 2v\theta - \theta^4 = 0.$$

One of these values is however $v = -u$, since the uv-equation is satisfied on writing therein $v = -u$; hence, writing

$$\theta^4 + 2\theta^3 u^2 - 2\theta u - u^4 = (\theta - \theta_1)(\theta - \theta_2)(\theta - \theta_3)(\theta - \theta_4),$$

we have an equation

$$(v^4+2v^3\theta_1^2-2v\theta_1-\theta_1^4)(v^4+\cdots-\theta_2^4)(v^4+\cdots-\theta_3^4)(v^4\cdots-\theta_4^4)=0,$$

which contains the factor $(v+u)^4$ and, divested hereof, gives the required modular equation of the order 12; it was in fact obtained in this manner.

Observe that writing $u=1$, we have $(\theta+1)^2(\theta-1)=0$, or say $\theta_1=\theta_2=-1$, $\theta_3=\theta_4=+1$; the modular equation then becomes

$$\{(v-1)^2(v+1)\}^2(v+1)^4(v-1)^4(v+1)^2=0,$$

that is, $(v-1)^4(v+1)^8=0$.

C. IX. 18

138. The Multiplier Equation.
Art. No. 29.

29. The theory is in many respects analogous to that of the modular equation.

To each value of v there corresponds a single value of M; hence M, or what is the same thing $\dfrac{1}{M}$, is determined by an equation of the same order as v, viz. n being prime, the order is $=n+1$.

The last term of the equation is constant, and the other coefficients are rational and integral functions of u^4, of a degree not exceeding $\frac{1}{2}(n-1)$; and not only so, but they are, $n\equiv 1 \pmod 4$, rational and integral functions of $u^4(1-u^4)$, and $n\equiv 3 \pmod 4$, alternately of this form and of the same form multiplied by the factor $(1-2u^4)$.

The values are in fact given as transcendental functions of q; viz. denoting by M_0, M_1, M_2, …, M_n the values corresponding to v_0, v_1, v_2, …, v_n respectively, and writing

$$\begin{aligned} A_n(q) &= \frac{(1+q)(1+q^2)(1+q^3)\cdots(1+q^n)}{(1-q)(1-q^2)(1-q^3)\cdots(1-q^n)}\cdot\frac{(1+q^n)(1+q^{2n})\cdots}{(1-q^n)(1-q^{2n})\cdots} \\ &= \frac{(1-q^2)(1-q^4)(1-q^6)\cdots(1-q^{2n})\cdots}{(1-q)(1-q^2)(1-q^3)\cdots(1+q)(1+q^2)(1+q^3)\cdots} \\ &= 1+2q+2q^4+2q^9+2q^{16}+\cdots, \end{aligned}$$

then we have

$$\frac{1}{M_0}=(-)^n\frac{A_n(q)}{A_n(q^n)},\quad M_1=\frac{A_n(\omega q)}{A_n(q^n)},\quad \ldots,\quad M_n=\frac{A_n(\omega^n q)}{A_n(q^n)},$$

(ω an imaginary nth root of unity).

Hence, the form of the equation being known, the values of the numerical coefficients may be calculated; and it was in this way that Joubert obtained the following results.

I have in some cases changed the sign of Joubert's multiplier, so that in every case the value corresponding to $u = 0$ shall be $M = 1$.

The equations are:

$n = 3$, $\frac{1}{M} = 0$, this is $\frac{1}{M^2}(m^{-1} + m - 2) - 3 = 0$.

$n = 5$,

$$\left(\frac{1}{M}\right)^6 + \frac{1}{M^4}(-10) + \frac{1}{M^2}(-26 + 256(1 - 2u^4)) + 5 = 0.$$

$n = 7$,

$$\left(\frac{1}{M}\right)^8 + \frac{1}{M^6}(-14) + \frac{1}{M^4}(-70 + 224(1 - 2u^4))$$

$$+\frac{1}{M^2}(-140 + 21 \cdot 25(1 - 2u^4)) + \frac{1}{M}(48 + 2048u^4(1 - u^4)) + 7 = 0.$$

139. $n = 5$ (continued):

$$\left(\frac{1}{M}\right)^6 + \frac{1}{M^4} \cdot (-10) + \frac{1}{M^2} \cdot 4 + 35(1 - 2u^4)$$

$$+\frac{1}{M} \cdot (-80) + 5 = 0.$$

$n = 7$:

$$\left(\frac{1}{M}\right)^8 + \frac{1}{M^6} \cdot (-14) + \frac{1}{M^4}(-70 + 224(1 - 2u^4))$$

$$+\frac{1}{M^2}(-140 + 21 \cdot 25(1 - 2u^4)) + \frac{1}{M}(48 + 2048u^4(1 - u^4)) + 7 = 0.$$

$n = 11$,

$$\left(\frac{1}{M}\right)^{12} + \frac{1}{M^{10}} \cdot (-66) + \frac{1}{M^8} \cdot 112(1 - 2u^4)$$

$$+\frac{1}{M^6}(\cdots) + \frac{1}{M^4}(\cdots) + \frac{1}{M^2}(440(1 - 2u^4)) + 11 = 0.$$

$n=0$, this is $(\cdots)=0$; $u=1$, it is $(\cdots)=0$.

140. $n=11$ (continued):

$$
\begin{gathered}
+\frac{1}{M^4}\cdot 3168(1-2u^4)\\
+\frac{1}{M^2}\cdot\{-4620-3\cdot 11\cdot 256u^4(1-u^4)\}\\
+\frac{1}{M}\{+4752+11\cdot 4096u^4(1-u^4)\}(1-2u^4)\\
+\frac{1}{M^2}\{-3465-3\cdot 7\cdot 11\cdot 512u^4(1-u^4)\}\\
+\frac{1}{M}\{+1760+11\cdot 83\cdot 2048u^4(1-u^4)\}(1-2u^4)\\
+\{-594-9\cdot 11\cdot 37\cdot 256u^4(1-u^4)-3\cdot 11\cdot 131072[u^4(1-u^4)]^2\}\\
+\frac{1}{M}\{120+15\cdot 4096u^4(1-u^4)-524288[u^4(1-u^4)]^2\}(1-2u^4)\\
-11=0.
\end{gathered}
$$

The Multiplier as a rational function of u, v.
Art. Nos. 30 to 36.

30. The multiplier M, as having a single value corresponding to each value of v, is necessarily a rational function of u, v; and such an expression of M can, as remarked by Königsberger, be deduced from the multiplier equation by means of Jacobi's theorem,

$$\frac{1}{M}=\frac{n}{\lambda}\frac{k(1-k^2)}{\frac{d\lambda}{dk}}\frac{d\lambda}{dk},$$

viz. substituting for k, λ their values u^4, v^4, and observing that if the modular equation be $F(u,v)=0$ so that the value of $\dfrac{d\lambda}{dk}$ is $=-F'(v)\div F'(u)$, this is

$$\frac{1}{M}=\frac{n(1-u^4)u}{v(1-v^4)}\frac{F'(v)}{F'(u)};$$

and then in the multiplier equation separating the terms which contain the odd and even powers, and writing it in the form $\Phi(M^2)+\dfrac{1}{M}\Psi(M^2)=0$, this equation, substituting therein for M^2 its value, gives the value of M rationally.

The rational expression of M in terms of u, v is of course indeterminate, since its form may be modified in any manner by means of the equation $F(u, v) = 0$; and in the expression obtained as above, the orders of u and v in the numerator and denominator may be very high.

141. 31. Denoting, as above, by M_0, M_1, ..., M_n the values which correspond to v_0, v_1, ..., v_n respectively, and writing

$$S\frac{1}{M} = \frac{1}{M_0} + \frac{1}{M_1} + \cdots + \frac{1}{M_n}, \quad \text{\&c.},$$

we have $S\frac{1}{M}$, $S\frac{1}{M^2}$, &c., all of them expressible as determinate functions of u; and we have moreover the theorem that each of these is a rational and integral function of u: we have thus the series of equations

$$\begin{gathered} S\frac{1}{M} = A, \\ S\frac{1}{M^2} = B, \\ \vdots \\ S\frac{1}{M^{n+1}} = H, \end{gathered}$$

where A, B, ..., H are rational and integral functions of u.

These give linearly the different values of $\frac{1}{M}$; in fact, we have

$$(v_0 - v_1) \cdots (v_0 - v_n)\frac{1}{M_0} = H - GSv_1 + FSv_1v_2 - \cdots \pm ASv_1v_2 \cdots v_n,$$

where Sv_1, Sv_1v_2, &c. denote the combinations formed with the roots v_1, v_2, ..., v_n (these can be expressed in terms of the single root v_0); and we have also

$$(v_0 - v_1) \cdots (v_0 - v_n) = F'(v_0):$$

the resulting equation is consequently $F'v_0 \cdot \frac{1}{M_0} = R(u, v_0)$, R a determinate rational and integral function of (u, v_0); but as the same formula exists for each root of the modular equation, we may herein write M, v in place of M_0, v_0; and the formula thus is

$$F'v \cdot \frac{1}{M} = R(u, v),$$

viz. we thus obtain the required value of $\frac{1}{M}$ as a rational fraction, the denominator being the determinate function $F'v$, and the numerator being, as is easy to see, a determinate function of the order n as regards v.

32. The method is applicable when M is only known by its expression in terms of q; but if we know for M an expression in terms of v, u, then the method transforms this into a standard form as above.

By way of illustration I will consider the case $n = 3$, where the modular equation is

$$v^4 + 2vu^2 - 2vu - u^4 = 0,$$

and where a known expression of M is $\dfrac{1}{M} = 1 + \dfrac{2u^2}{v}$.

Here writing S_{-1}, S_0 $(= 4)$, S_1, &c. to denote the sum of the powers -1, 0, 1, &c. of the roots of the equation, we have

$$\begin{aligned} S_{-1} &= S_1 + 2u^2 S_{-1} = 0, \\ S_0 &= S_2 + 2u^2 S_0 = 0, \\ S_1 &= S_3 + 2u^2 S_1 = 6u; \end{aligned}$$

142. and observing that v_0, being ultimately replaced by v, we have

$$\begin{aligned} Sv_1 &= Sv_0 - v, \\ Sv_1v_2 &= Sv_0v_1 - vSv_0 + v^2, \\ Sv_1v_2v_3 &= Sv_0v_1v_2 - vSv_0v_1 + v^2Sv_0 - v^3, \end{aligned}$$

that is,

$$\begin{aligned} Sv_1 &= -2u^2 - v, \\ Sv_1v_2 &= 2u^2v + v^2, \\ Sv_1v_2v_3 &= 2u - 2u^2v^2 - v^3, \end{aligned}$$

we have

$$F'v \cdot \frac{1}{M} = (S_1 + 2u^2 S_{-1}) \frac{1}{M_0}$$

viz. this is

$$\begin{aligned} + (2u^2 + v)(S_2 + 2u^2S_0) + (2u^2v + v^2)(S_1 + 2u^2S_0) \\ + (-2u + 2u^2v^2 + v^3)(S_0 + 2u^2S_{-1}), \end{aligned}$$

But we have

$$S_{-1} = -\frac{3}{u}, \quad S_0 = 4, \quad S_1 = -2u^2, \quad S_2 = u^4, \quad S_3 = 6u - 8u^5;$$

and the equation thus is

$$(2v^2 + 8vu^2 - 4)\frac{1}{M} = Svu \cdot (vu^2 + 2u^2v + 1) \cdot u;$$

to verify which observe that, substituting herein for $\frac{1}{M}$ its value $1 + \dfrac{2u^2}{v}$, the equation becomes

$$(2v^2 + 8vu^2 - 4u)(v + 2u^2) - Svu \cdot (vu^2 + 2u^2v + 1) = 0;$$

that is,

$$2v^4 + 4u^2v^2 - 4vu - 2u^4 = 0,$$

as it should do.

33. Any expression whatever of M in terms of u, v is in fact one of a system of four expressions; viz. we may simultaneously change

$$u \text{ into } \frac{1}{v}, \quad M \text{ into } (-)^{\frac{n-1}{2}}\frac{1}{nM}, \quad u \text{ into } \frac{1}{u};$$

that is, signs are

	u^{n-1}	v^{n-1}	$(-)^{\frac{n-1}{2}}$	$(-)^{\frac{n-1}{2}}$
$n=1$	$+$	$+$	$+$	$+$
$n=3$	$+$	$+$	$+$	$+$
$n=5$	$+$	$+$	$+$	$-$
$n=7$	$+$	$+$	$-$	$+$

143. Thus $n = 3$, starting from $\dfrac{1}{M} = 1 + \dfrac{2u^2}{v}$, we have

	M	v	u	u^2M
(1)	$\frac{1}{M}$	v	u	u^2M
(2)	Mu^2	$\frac{1}{v}$	u	$\frac{1}{u^2M}$
(3)	$\frac{v^2}{M}$	v	$\frac{1}{u}$	$\frac{v^2}{u^2M}$
(4)	$\frac{1}{Mu^2v^2}$	$\frac{1}{v}$	$\frac{1}{u}$	$\frac{1}{Mu^2v^2}$

and of course if from any two of these we eliminate M, we have either an identity or the modular equation; thus we have the modular equation under the six different forms:

$$\begin{array}{ll}(1,2) & (v+2u^2)(u-2v^2)+3uv=0,\\ (1,3) & v^2(v+2u^2)-u(u^2+2v)=0,\\ (1,4) & (v+2u^2)(v^2-2u)+3u^2=0,\\ (2,3) & (u-2v^2)(u^2+2v)+3v^2=0,\\ (2,4) & v(v^2-2u)-u^2(u-2v^2)=0,\\ (3,4) & (u^2+2v)(v^2-2u)+3u^2v^2=0.\end{array}$$

34. Next $n=5$.

Here, starting from $\frac{1}{M}=\dfrac{v-v^4}{u+u^4}$, the changes give

$$\frac{1}{M}=\frac{v-v^4}{u+u^4},\quad Mu^2=\frac{u^2(u-u^4)}{v(1-uv^3)},\quad \frac{v^2}{M}=\frac{v^2(v^2-u^3)}{u(1+u^3v)},\quad \frac{1}{Mu^2v^2}=\frac{1-u^3v}{u^2v^2(v-u^3)}$$

viz. the third and the fourth forms agree with the first and the second forms respectively; that is, there are only two independent forms, and the elimination of M from these gives

$$5uv(1-uv^3)(1+u^3v)-(v-v^4)(u+u^4)=0,$$

which is a form of the modular equation.

144. 35. In the case $n=7$ (post No. 43), the forms are

$$\frac{1}{M}=\frac{1-7u(1-uv)(1-uv+u^2v^2)}{v-u^7},$$

starting from

$$\frac{1}{M}=\frac{1-7u(1-uv)(1-uv+u^2v^2)}{v-u^7}$$

as to this see (3),

$$Mu^2=\frac{u^2-7v(1-uv)(1-uv+v^2u^2)}{u^7-v},$$

$$\frac{v^2}{M}=\frac{v^2-7u^2(1-uv)(1-uv+u^2v^2)}{u-v^7},$$

$$\frac{1}{Mu^2v^2}=\frac{1-7v^2(1-uv)(1-uv+u^2v^2)}{u^2(v-u^7)},$$

so that here again the third and the fourth forms are identical with the second and the third forms respectively; there are thus only two forms, and the elimination of M gives

$$(u - v^7)(v - u^7) + 49uv(1 - uv)^2(1 - uv + u^2v^2)^2 = 0,$$

which is a form of the modular equation.

36. If in the foregoing equation we make the change u, v, M into v, $\pm u$, $\pm nM$, it becomes

$$\frac{1}{\pm nM} = \frac{R(v, \pm u)}{\pm F'v \cdot nM},$$

combining these equations, we have

$$\frac{F'v}{v(1 - v^4)} \cdot \frac{1}{M} = \frac{R(u, v)}{u(1 - u^4)} \cdot \frac{1}{nM},$$

or, substituting herein the foregoing value M,

$$\frac{1}{M} = \frac{v(1 - v^4)}{n(1 - u^4)u} \cdot \frac{R(v, \pm u)}{R(u, v)},$$

for $n = 3$ or $5 \pmod 8$,

$$\frac{1}{M} = \frac{v(1 - v^4)}{u(1 - u^4)} \cdot \frac{R(v, -u)}{R(u, v)},$$

for $n = 1$ or $7 \pmod 8$, which must agree with the modular equation.

Thus in the last-mentioned case $n = 3$, we have

$$\frac{1}{M}F'v - 4 = 3(v^2u^2 + 2u^2v + 1)u, \quad R(u, v) = (v - u^2 + 2u^2v + 1)u,$$

$$R(v, -u) = (vu^2 - 2uv^2 + 1)v;$$

$$\frac{v(1 - v^4)}{u(1 - u^4)} = \frac{(vu^2 - 2uv^2 + 1)v}{(v - u^2 + 2u^2v + 1)u},$$

or, say

$$\frac{v(1 - v^4)}{u(1 - u^4)} = \frac{R(v, -u)}{R(u, v)},$$

and therefore the equation is

$$v(1 - v^4)R(u, v) - u(1 - u^4)R(v, -u) = 0,$$

which is right; because Jacobi, p. 82, [Ges. Werke, t. I., p. 137], for the modular equation, gives

$$1 - v^4 = (1 - u^2v^2)(v^2u^2 + 2u^2v + 1),$$

$$1 - u^4 = (1 - u^2v^2)(u^2v^2 - 2uv^2 + 1).$$

Observe that the general equation

$$\frac{1}{M} = \frac{v(1-v^4)}{u(1-u^4)} \cdot \frac{R(v, \pm u)}{R(u, v)}$$

no longer contains the functions $F'v$, $F'u$, which enter into Jacobi's expression of M.

145. Theorem in connexion with the multiplication of Elliptic Functions.

Art. Nos. 37 to 40.

37. The theory of multiplication gives an important theorem in regard to transformation.

Starting with the $n$thic transformation

$$\frac{1-y}{1+y} = \frac{1-x}{1+x}\left(\frac{\alpha - \beta x + \gamma x^2 - \cdots}{\alpha + \beta x + \gamma x^2 + \cdots}\right)^2 = \frac{1-x}{1+x}\left(\frac{P - Qx}{P + Qx}\right)^2,$$

we may form a like transformation,

$$\frac{1-z}{1+z} = \frac{1-y}{1+y}\left(\frac{\alpha' - \beta' y + \gamma' y^2 - \cdots}{\alpha' + \beta' y + \gamma' y^2 + \cdots}\right)^2 = \frac{1-y}{1+y}\left(\frac{P' - Q'y}{P' + Q'y}\right)^2,$$

such that the combination of the two gives a multiplication, viz. for the relation between y, z, deriving w from v as v from u, we have $w = u$; and instead of M we have $M_1 = \pm\frac{1}{nM}$; that is, we have

$$\frac{dx}{\sqrt{1 - x^2 \cdot 1 - u^4x^2}} = \frac{M\,dy}{\sqrt{1 - y^2 \cdot 1 - v^4y^2}}$$

and thence

$$\frac{dy}{\sqrt{1 - y^2 \cdot 1 - v^4y^2}} = \frac{M'\,dz}{\sqrt{1 - z^2 \cdot 1 - w^4z^2}},$$

$$\pm\frac{dz}{\sqrt{1 - z^2 \cdot 1 - u^4z^2}} = \frac{n\,dx}{\sqrt{1 - x^2 \cdot 1 - u^4x^2}},$$

or, writing $x = \sin\phi$, we have $z = \pm\sin n\phi$; $\pm$ is here $(-)^{\frac{n-1}{2}}$, viz. it is $-$ for $n = 3$ or $7 \pmod 8$, and $+$ for $n = 1$ or $5 \pmod 8$.

Now in part effecting the substitution, we have

$$\frac{1-z}{1+z} = \frac{1-x}{1+x}\left(\frac{P-Qx}{P+Qx}\right)^2\left(\frac{P'-Q'y}{P'+Q'y}\right)^2,$$

where y denotes its value in terms of x.

And from the theory of elliptic functions, replacing $\sin n\phi$, $\sin\phi$ by their values $\pm z$, x, we have an equation

$$\frac{1-z}{1+z} = \frac{1-x}{1+x}\left(\frac{A-Bx+Cx^2-\cdots}{A+Bx+Cx^2+\cdots}\right)^2,$$

where $A-Bx+Cx^2-\cdots$, $A+Bx+Cx^2+\cdots$ are given functions each of the order $\frac{1}{2}(n^2-1)$; viz. the coefficients are given functions of k, or, what is the same thing, of u^4.

Comparing the two results, we see that in the $n$thic transformation the sought-for function, $\alpha+\beta x+\gamma x^2+\cdots$ of the order $\frac{1}{2}(n-1)$, is a factor of a given function $A+Bx+Cx^2+\cdots$ of the order $\frac{1}{2}(n^2-1)$.

38. Considering the modular equation as known, then by what precedes we have

$$\alpha+\beta x+\gamma x^2+\cdots = \alpha'\left(1+\frac{\beta}{\alpha}x+\cdots+\frac{m^n}{\alpha}x^{\frac{1}{2}(n-1)}\right),$$

that is, the given function $A+Bx+Cx^2+\cdots$ has a factor $1+\dfrac{\beta}{\alpha}x+\cdots+\dfrac{m^n}{\alpha}x^{\frac{1}{2}(n-1)}$, of which one (the last) coefficient $\dfrac{m^n}{\alpha}$ is known, and we are hence able theoretically to

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146. determine all the other coefficients rationally in terms of u, v; that is, the modular equation being known, we can theoretically complete the solution of the transformation problem.

I do not, however, see the way to obtaining a convenient solution in this manner.

39. The formula in question for $n = 3$ is

$$\frac{1+\sin 3\phi}{1-\sin 3\phi} = \frac{1-\sin\phi}{1+\sin\phi}\left\{\frac{1+2\sin\phi-2k^2\sin^3\phi-k^2\sin^4\phi}{1-2\sin\phi+2k^2\sin^3\phi-k^2\sin^4\phi}\right\}^2,$$

which, putting therein $x = \sin\phi$, $z = -\sin 3\phi$, and replacing k by u^4, may be written

$$\frac{1+z}{1-z} = \left(\frac{1+x}{1-x}\right)\left(\frac{1-2x+2u^4x^2-u^4x^4}{1+2x+2u^4x^2-u^4x^4}\right)^2,$$

where the signs $(\mp)$ indicate denominators which are obtained from the numerators by changing the signs of z, x respectively.

The theorem in regard to $n = 3$ thus is, $1 + \frac{u^2}{v}x$ is a factor of $1 - 2x + 2u^4x^2 - u^4x^4$; viz. writing in the last-mentioned function $x = -\frac{v}{u^2}$, we ought to have

$$= 1 + 2\frac{v}{u^2} - 2\frac{v^2}{u^4} - \frac{v^4}{u^8},$$

that is,

$$u^4 + 2uv^3 - 2u^3v - u^8 = 0,$$

which is in fact the modular equation.

40. And so for $n = 5$, if $x = \sin\phi$, $z = \sin 5\phi$; and for $n = 7$, if $x = \sin\phi$, $z = -\sin 7\phi$; the formulae are:

$n = 5$,

$$\frac{1+z}{1-z} = (1+x)\left\{\frac{1+2x-4x^2-4x^3-10u^4x^4+4(2+7u^4)x^5+\cdots}{(-)}\right\}^2.$$

$n = 7$,

$$\begin{aligned}\frac{1+z}{1-z} &= (1+x)\{1+2x-4x^2-4x^3-4x^4-14u^4x^5+4(2+7u^4)x^6+6u^4x^7\\ &- 14u^4x^8 + 4u^4(3+2u^4)x^9 - 28u^4(3+2u^4)x^{10} + 4u^4(1-u^4)x^{11}\\ &+ 28u^4(4+u^4)x^{12} - 4u^4(2+3u^4)x^{13} + 4u^4(16+51u^4+8u^8)x^{14}\\ &- 5u^8x^{15} - u^4(144+305u^4+16u^8)x^{16} + 10u^8x^{17}\\ &- 8u^8(4+25u^4+16u^8)x^{18} + 4u^8x^{19} + 8u^8(8+57u^4+46u^8)x^{20}\\ &- 2u^8x^{21} + 56u^8(2+u^4)x^{22} - u^{12}x^{23}\\ &- 4u^8(56+161u^4+56u^8)x^{24}\}^{(-)}\end{aligned}$$

147. Term in $\{\}$ has factor

$$1 + \frac{\alpha}{v}x + \frac{\beta}{v}x^2 + \cdots;$$

$u = 1$, term in $\{\}$ is $= (1+x)^3(1-x)^3$.

The transformations $n = 3, 5, 7, 11$.

Art. Nos. 41 to 51.

41. The cubic transformation, $n = 3$.

I reproduce the results already obtained; since there are only two coefficients α, β, these are also the last but one and last coefficients ρ, σ.

Hence, from the values of α, β, ρ, σ, we have

$$\alpha = 1, \quad 2\alpha = -\frac{1}{\Omega} \cdot \frac{2u^2}{v},$$

$$\frac{2u^2}{v} = \frac{1}{M} - 1, \quad \frac{2u^2}{v} = 1 - \frac{1}{M},$$

the two values of $\frac{2u^2}{v}$ are thus $-\frac{1}{M}+1$ and $1-\frac{1}{M}$, giving the modular equation

$$v^4 + 2vu^2 - 2vu - u^4 = 0;$$

and we then have

$$\frac{1-y}{1+y} = \frac{1-x}{1+x}\left(\frac{v-u^2x}{v+u^2x}\right)^2.$$

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148. 42. The quintic transformation, $n = 5$.

Here there are the three coefficients α, β, γ, or β, γ are the last but one and last coefficients ρ, σ; we have $\alpha = 1$.

$$2\beta = \frac{v}{u}\left(\frac{1}{M} - 2\right), \quad 2\beta = 1 - \frac{1}{M}, \quad \gamma = -\frac{v^2}{u^2}.$$

Comparing the two values of β, we have $\dfrac{1}{M} = \dfrac{v-u^4v}{u-u^4v}$, so that only the modular equation remains to be determined.

The unused equation is

$$2\alpha\gamma + 2\alpha\beta + \beta^2 = \frac{v^2}{u^4}(2\alpha\gamma + 2\beta\gamma + \beta^2),$$

which, putting therein $\alpha = 1$, may be written

$$(2\gamma + \beta^2)(u^4 - v^4) = 2\beta(v^2u^2 - u^4);$$

attending to the value of β, this divides by u^2-v^2; in fact the equation may be written

$$\beta vu^2(1-u^2v^2)+(v^4-u^8)\gamma=0,$$

and then completing the substitution, and integralizing, this becomes

$$\beta vu^2(1-u^2v^2)+(v^4-u^8)\gamma=\beta uv(u^2+v^2)(1-u^2v)(1-uv^3),$$

viz. this is

$$4(1-u^2v^2)uv\{2u^2(1-u^2v^2)-(u^2+v^2)(1-vu^2)\}+(v^4-u^8)\gamma=0;$$

and the term in $\{\}$ being $=-(u^2-v^2)(1+vu^2)$, the whole again divides by u^2-v^2, and the equation thus becomes

$$(v^4+u^4)(u^2-v^2)-4uv(1-u^2v)(1+u^4)=0,$$

which is the modular equation.

43. The septic transformation, $n=7$.

I do not propose to complete the solution directly from the fundamental equations for α, β, γ, δ, but resort to the known modular equation, and to an expression of M which I obtain by means thereof.

The modular equation is

$$(1-u^8)(1-v^8)-(1-u^2v^2)^4=0,$$

149. which may also be written

$$(u^4-v^4)(u-v^4)+14uv(1-uv)^2(1-uv+u^2v^2)^2=0,$$

as can be at once verified; but it also follows from Cauchy's identity

$$(x^3+y^3+z^3-3xyz)^2=(x+y+z)^2(x^2+y^2+z^2-xy-yz-zx)^2.$$

We then have

$$\frac{1}{M}=\frac{1}{n}\frac{v(1-v^4)F'v}{u(1-u^4)F'u},$$

Moreover

$$uF'u=-2u^8(1-v^8)+uv(1-u^2v^2)^3,$$

and similarly

$$uF'v=-2v^8(1-u^8)+uv(1-u^2v^2)^3,$$

whence

$$F'v = \frac{1}{v}(u - v^8), \quad F'u = \frac{1}{u}(v - u^8);$$

$$\frac{1}{M} = \frac{-7u(v - u^8)}{v(u - v^8)}.$$

Writing this under the form

$$\frac{1}{M} = \frac{-7uv(v - v^8)(u - v^8)}{v^2(u - v^8)^2} = \frac{49u^2(1 - uv)^2(1 - uv + u^2v^2)^2}{(u - v^8)^2},$$

I find, as will appear, that the root must be taken with the sign $-$, and that we thus have $\dfrac{1}{M} = \dfrac{-7u(1 - uv)(1 - uv + u^2v^2)}{u - v^8}$, whence also $M = \dfrac{v(1 - uv)(1 - uv + u^2v^2)}{v - u^8}$.

44. Recurring now to the fundamental equations for the septic transformation, the coefficients are α, β, γ, δ, and we have

$$\alpha = 1, \quad 2\beta = \frac{v}{u}\left(\frac{1}{M} - 2\right),$$

so that the coefficients are all given in terms of v, M.

The unused equations are

$$u^2(2\alpha\gamma + 2\alpha\beta + \beta^2) = v^2(2\gamma\delta + 2\beta\delta + \beta^2),$$
$$u^4(\gamma + 2\beta\gamma + 2\alpha\delta + 2\beta\delta) = v^4(2\alpha\gamma + 2\beta\gamma + 2\alpha\delta + \beta^2),$$

which, substituting therein for α, β, γ, δ the foregoing values, give two equations; from these, eliminating M, we should obtain the modular equation, and then M in terms of u, v.

150. Substituting in the first instance for α, β their values, the equations are

$$u^2(2\beta + 2\gamma + \beta^2) = v^2\left\{-\frac{v}{u} + 2 - (\beta + \gamma)\right\},$$

$$\gamma^2 + 2\beta\gamma + (2 + 2\beta)\delta = u^4v^4\left\{\gamma + 2\beta\gamma + 2\frac{v}{u}\delta + M\right\}.$$

The first of these is

$$4(1 - uv)(2\beta + 2\gamma) + 4\beta^2 - 4\frac{v^2}{u^2}\gamma = 0,$$

viz. this is

[equation continues]

or observing that in this equation the coefficient of $-v^2$ is

$$(1-uv)^2\{2-2uv+2u^2v^2-1-u^2v^2\} = (1-uv)^2(1-uv)^2 = (1-uv)^3(1+uv),$$

the equation becomes

$$(1-v^8)\beta + \frac{v}{u}(1-uv)^3(1+uv) + 1 - uv - 4(1-uv)(1+u^4) = 0.$$

45. This should be satisfied identically by the foregoing value of $\frac{1}{M}$; viz. it should be satisfied on writing therein

$$\frac{1}{M} = \frac{-7uv - v^8}{u - v^8}, \quad \frac{1}{M} = \frac{-7u(1-uv)(1-uv+u^2v^2)}{u-v^8},$$

that is, we should have

$$\beta\gamma - 7(v-u^8)(1-uv) - 14u(1-uv)^2(1+u^3v^3) \\ + (u-v^8)\{1-uv-4(1-uv)(1+u^4)\} = 0,$$

where observe that the $-$ sign of the second term is the sign of the foregoing value of $\frac{1}{M}$; so that the identity being verified, it follows that the correct sign has been attributed to the value of $\frac{1}{M}$.

46. Multiplying by v, the equation is

$$-7(1-uv-\tfrac{1}{2}uv)(1-v^8) - 14uv(1-uv)^2(1+u^3v^3) \\ +\{1-v^8-1-uv\}\{-8(1-uv)+1-uv\}+4(1-uv)(v-u^8)(u-v^8) = 0,$$

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[Paper 578]

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Remarks on the Tables, Art. No. 27

27. The following Tables I. to VI. show for $n = 9$, 11, 13, 15, 17, 19, the values of the several coefficients in the expansions

$$\frac{1-y}{1+y} = \frac{1-x}{1+x}\left(\frac{1-\beta x+\gamma x^2-\cdots\pm\sigma x^{\frac{1}{2}(n-1)}}{1+\beta x+\gamma x^2+\cdots+\sigma x^{\frac{1}{2}(n-1)}}\right)^2,$$

as rational functions of u, v; and also the values of $\frac{1}{M}$, and of certain other functions which present themselves in the investigation.

Table I. $n = 9$.

The table gives the coefficients of the expansion of $(v-1)^8(v+1)$, arranged by powers of v. The coefficients are read off from the several columns.

	v^9	v^8	v^7	v^6	v^5	v^4	v^3	v^2	v	1
$(v-1)^8(v+1)$	1	-7	20	-28	14	14	-28	20	-7	-1

Table II. $n = 11$.

The table gives the coefficients of the expansion of $(v-1)^{10}(v+1)^2$.

	v^{11}	v^{10}	v^9	v^8	v^7	v^6	v^5	v^4	v^3	v^2	v	1
$(v-1)^{10}(v+1)^2$	1	-9	34	-64	55	11	-55	64	-34	9	1	-1

Table III. $n = 13$.

The table gives the coefficients of the expansion of $(v+1)^{13}(v-1)$.

	v^{13}	v^{12}	v^{11}	v^{10}	v^9	v^8	v^7	v^6	v^5	v^4	v^3
$(v+1)^{13}(v-1)$	1	12	55	120	125	-48	-161	-88	88	161	48

Table IV. $n = 15$.

The table gives the coefficients of the expansion of $(v-1)^{14}(v+1)$.

	v^{15}	v^{14}	v^{13}	v^{12}	v^{11}	v^{10}	v^9	v^8	v^7	v^6
$(v-1)^{14}(v+1)$	1	-13	64	-182	273	-156	-130	260	-182	0

Table V. $n = 17$.

The table gives the coefficients of the expansion of $(v-1)^{16}(v+1)$.

	v^{17}	v^{16}	v^{15}	v^{14}	v^{13}	v^{12}	v^{11}	v^{10}	v^9	v^8
$(v-1)^{16}(v+1)$	1	-15	90	-320	650	-728	119	560	-455	-18

Table VI. $n = 19$.

The table gives the coefficients of the expansion of $(v+1)^{19}(v-1)$.

	v^{19}	v^{18}	v^{17}	v^{16}	v^{15}	v^{14}	v^{13}	v^{12}	v^{11}	v^{10}
$(v+1)^{19}(v-1)$	1	18	120	460	969	798	-714	-1748	-605	137

The Multiplier Equation, Art. Nos. 28 to 51

28. The multiplier equation is the equation connecting the multiplier M with the moduli u, v; or, what is the same thing, the equation connecting $\frac{1}{M}$, u, v.

I take first the case $n = 7$; here we have

$$\alpha = 1, \quad \beta = \tfrac{1}{2}\left(\frac{1}{M} - 1\right), \quad \gamma = \tfrac{1}{2}u^3v^3\left(\frac{1}{M} - \frac{u^4}{v^4}\right), \quad \delta = \frac{u^7}{v},$$

where

$$\frac{1}{M} = \frac{-7u(1-uv)(1-uv+u^2v^2)}{u - v^7}.$$

29. The conclusion is

$$\alpha = 1, \quad \beta = \tfrac{1}{2}\left(\frac{1}{M} - 1\right), \quad \gamma = \tfrac{1}{2}u^3v^3\left(\frac{1}{M} - \frac{u^4}{v^4}\right), \quad \delta = \frac{u^7}{v},$$

where

$$\frac{1}{M} = \frac{-7u(1-uv)(1-uv+u^2v^2)}{u-v^7},$$

and of course

$$\frac{1-y}{1+y} = \frac{1-x}{1+x}\left(\frac{1-\beta x+\gamma x^2-\delta x^3}{1+\beta x+\gamma x^2+\delta x^3}\right)^2;$$

but the resulting form may admit of simplification.

46. For the verification of these values, observe that the first equation is

$$\begin{aligned}&\alpha\gamma - \beta^2 + 2(\alpha\delta - \beta\gamma) - \gamma^2 + 2\beta\delta - 2\gamma\delta\\ &\qquad = u^{12}(2\alpha\gamma + 2\alpha\delta + 2\beta\delta - \beta^2) - u^6v^6(2\beta\gamma + 2\gamma + 2\delta + \gamma^2).\end{aligned}$$

47. The conclusion is

$$\alpha = 1, \quad \beta = \tfrac{1}{2}\left(\frac{1}{M} - 1\right), \quad \gamma = \tfrac{1}{2}u^3v^3\left(\frac{1}{M} - \frac{u^4}{v^4}\right), \quad \delta = \frac{u^7}{v},$$

where

$$\frac{1}{M} = \frac{-7u(1-uv)(1-uv+u^2v^2)}{u-v^7},$$

and of course

$$\frac{1-y}{1+y} = \frac{1-x}{1+x}\left(\frac{1-\beta x+\gamma x^2-\delta x^3}{1+\beta x+\gamma x^2+\delta x^3}\right)^2;$$

but the resulting form may admit of simplification.

48. The endecadic transformation, $n = 11$.

I have not completed the solution, but the results, so far as I have obtained them, are interesting. The coefficients are α, β, γ, δ, ϵ, ζ; and we have, as in general,

$$\alpha = 1, \qquad 2\epsilon = u^7v^3\left(\frac{1}{M} - \frac{u^4}{v^4}\right),$$

$$2\beta = \frac{1}{M} - 1, \qquad \zeta = \frac{u^{11}}{v}.$$

The unused equations then are

$$\begin{aligned}
u^{14}(2\alpha\gamma+2\alpha\beta+\beta^2) &= v^2(\epsilon^2+2\epsilon\zeta+2\delta\zeta),\\
u^{8}(\gamma^2+2\alpha\epsilon+2\alpha\delta+2\beta\gamma+2\beta\delta) &= v^2(2\gamma\epsilon+2\gamma\zeta+2\delta\epsilon+2\beta\zeta+\delta^2),\\
u^{-2}(2\gamma\epsilon+2\alpha\zeta+2\gamma\delta+2\beta\epsilon+2\beta\zeta+\delta^2) &= v^2(\gamma^2+2\alpha\epsilon+2\alpha\zeta+2\gamma\delta+2\beta\epsilon+2\beta\delta),\\
u^{-10}(\epsilon^2+2\gamma\zeta+2\delta\epsilon+2\delta\zeta) &= v^2(2\alpha\gamma+2\alpha\delta+2\beta\gamma+\beta^2);
\end{aligned}$$

but I attend only to the first and the last, which, it will be observed, contain γ, δ linearly. If in the first instance we substitute only for α, ζ their values, the equations become

$$\begin{aligned}
u^{12}\beta(2+\beta)-\frac{v^2}{u^2}\epsilon\left(\epsilon+2\frac{u^{11}}{v}\right)+u^{12}\cdot 2\gamma - vu^9\cdot 2\delta &= 0,\\
u^{-12}\epsilon^2-\frac{v^2}{u^2}\beta^2+\left\{\frac{1}{uv}-\frac{v^2}{u^2}(1+\beta)\right\}\cdot 2\gamma+\left\{\frac{1}{uv}-\frac{v^2}{u^2}+u^{-12}\epsilon\right\}\cdot 2\delta &= 0;
\end{aligned}$$

say, for a moment, these are

$$\begin{aligned}
A+P\cdot 2\gamma+Q\cdot 2\delta &= 0,\\
B+R\cdot 2\gamma+S\cdot 2\delta &= 0,
\end{aligned}$$

giving

$$1:2\gamma:2\delta = PS-QR:QB-SA:RA-PB.$$

Here

$$\begin{aligned}
PS-QR &= \frac{u^{11}}{v}+\epsilon-u^{10}v^2+u^8-u^7v^3(1+\beta)\\
&= \tfrac{1}{2}\left\{\frac{2u^{11}}{v}+\left(u^7v^3\frac{1}{M}-\frac{u^{11}}{v}\right)-2u^{10}v^2+2u^8-2u^7v^3-\left(u^7v^3\frac{1}{M}-u^7v\right.\right.
\end{aligned}$$

where the terms containing $\frac{1}{M}$ disappear of themselves, viz. this is

$$\begin{aligned}
&= \tfrac{1}{2}\left(\frac{u^{11}}{v}-2u^{10}v^2+2u^8-u^7v^3\right)\\
&= -\tfrac{1}{2}\frac{u^7}{v}(v^4+2v^3u^3-2vu-u^4);
\end{aligned}$$

observe that the term in $\{\ \}$, equated to zero, gives the modular equation for the case $n=3$. It thus appears that γ and δ are given as fractions, having in their denominator this function $u^4+2uv-2u^3v^3-v^4$.

49. To complete the calculation, we have

$$
\begin{aligned}
QB - AS = -vu^9 \left(\frac{\epsilon^2}{u^{12}} - \frac{v^2}{u^2}\beta^2\right) \\
- \left\{u^{12}\beta(2+\beta) - \frac{v^2}{u^2}\epsilon\left(\epsilon + 2\frac{u^{11}}{v}\right)\right\}\left\{\frac{1}{uv} - \frac{v^2}{u^2} + \frac{\epsilon}{u^{12}}\right\};
\end{aligned}
$$

viz. multiplying by 8, and substituting for 2β, 2ϵ their values, this is

$$
\begin{aligned}
8(QB - AS) = -2vu^9 \left\{u^2v^6\left(\frac{1}{M} - \frac{u^4}{v^4}\right)^2 - \frac{v^2}{u^2}\left(\frac{1}{M} - 1\right)^2\right\} \\
-\left\{u^{12}\left(\frac{1}{M} - 1\right)\left(\frac{1}{M} + 3\right) - u^{12}v^3\left(\frac{1}{M} - \frac{u^4}{v^4}\right)\left(\frac{1}{M} + \frac{3u^4}{v^4}\right)\right\}\left(\frac{1}{uv} - \frac{2v^2}{u^2} + \frac{v^3}{u^3}\frac{1}{M}
\end{aligned}
$$

or, what is the same thing,

$$
\begin{aligned}
-\frac{8v}{u^7}(QB - AS) = 2\left\{u^4v^3\left(\frac{1}{M} - \frac{u^4}{v^4}\right)^2 - v^4\left(\frac{1}{M} - 1\right)^2\right\} \\
+\left\{\left(\frac{1}{M} - 1\right)\left(\frac{1}{M} + 3\right) - v^3\left(\frac{1}{M} - \frac{u^4}{v^4}\right)\left(\frac{1}{M} + \frac{3u^4}{v^4}\right)\right\}\left(u^4 - 2v^3u^3 + \frac{v^4}{M}\right),
\end{aligned}
$$

viz. the left-hand side is

$$
\begin{aligned}
= 2\left\{-v^4(1 - u^4v^4)\frac{1}{M^2} + 2v^4(1 - u^8)\frac{1}{M} + u^{12} - v^4\right\} \\
+\left\{\frac{1}{M^2}(1 - v^8) + \frac{2}{M}(1 - u^4v^4) - 3(1 - u^8)\right\}\left\{\frac{1}{M} + u^3(u - 2v^3)\right\};
\end{aligned}
$$

or, say we have $-\frac{8v}{u^7}(QB - SA) = \Pi$, where

$$
\begin{aligned}
\Pi = {} & \frac{1}{M^3} \cdot v^4(1 - v^8) \\
& + \frac{1}{M^2} \cdot v(v^3 - 2u)(1 - v^8) \\
& + \frac{1}{M} \cdot 4u^5v^5 + v^4(3 - u^8) - 4u^3v^3 + 2u^4 \\
& + \quad \cdot - 2v^4 + 6v^3u^3(1 - u^8) + u^4(-3 + 5u^8);
\end{aligned}
$$

wherefore the value of 2γ is $= \frac{1}{4}\Pi \div (v^4 + 2v^3u^3 - 2vu - u^4)$. Similarly, writing

$$\begin{aligned}\Pi' = {} & \frac{1}{M^3}\cdot v^4(1-v^8)\\ & +\frac{1}{M^2}\cdot v(v^3-2u)(1-v^8)\\ & +\frac{1}{M}\cdot 4u^5v^5+v^4(3-u^8)+4uv-2u^4v^3\\ & + \quad \cdot v^4(-5+3u^8)+6vu(1-u^8)+2u^{12},\end{aligned}$$

we find

$$2\delta = \tfrac{1}{4}\frac{u^3}{v}\Pi' \div (v^4+2v^3u^3-2vu-u^4);$$

in verification whereof observe that this being so, the first equation gives the identity

$$\left\{\left(\frac{1}{M}-1\right)\left(\frac{1}{M}+3\right)-v^3\left(\frac{1}{M}-\frac{u^4}{v^4}\right)\left(\frac{1}{M}+\frac{3u^4}{v^4}\right)\right\}(v^4+2v^3u^3-2vu-u^4)+\Pi-\Pi$$

50. The result is that, writing for the moment $v^4+2v^3u^3-2vu-u^4=\Delta$, the values of the coefficients are

$$\begin{array}{cccccc} \alpha, & \beta, & \gamma, & \delta, & \epsilon, & \zeta,\\ =1, & \tfrac{1}{2}\left(\tfrac{1}{M}-1\right), & \tfrac{1}{8}\tfrac{\Pi}{\Delta}, & \tfrac{1}{8}\tfrac{u^3}{v}\tfrac{\Pi'}{\Delta}, & \tfrac{1}{2}u^7v^3\left(\tfrac{1}{M}-\tfrac{u^4}{v^4}\right), & \tfrac{u^{11}}{v},\end{array}$$

and

$$\frac{1-y}{1+y}=\frac{1-x}{1+x}\left(\frac{1-\beta x+\gamma x^2-\delta x^3+\epsilon x^4-\zeta x^5}{1+\beta x+\gamma x^2+\delta x^3+\epsilon x^4+\zeta x^5}\right)^2;$$

the modular equation is known, and to complete the solution we require only an expression for M in terms of u, v.

51. We may herein illustrate the following theorem, viz. we may simultaneously change u, v, $\frac{1}{M}$, $\alpha:\beta:\gamma:\delta:\epsilon:\zeta$; into $\frac{1}{u}$, $\frac{1}{v}$, $\frac{v^4}{u^4}\frac{1}{M}$, $\zeta:\epsilon:\delta:\gamma:\beta:\alpha$.

Thus making the change in the equation

$$\frac{\beta}{\alpha}=\tfrac{1}{2}\left(\frac{1}{M}-1\right),$$

we have

$$\frac{\epsilon}{\zeta}=\tfrac{1}{2}\left(\frac{v^4}{u^4}\frac{1}{M}-1\right);\ \text{that is, } \tfrac{1}{2}\frac{v^4}{u^4}\left(\frac{1}{M}-\frac{u^4}{v^4}\right)=\tfrac{1}{2}\left(\frac{v^4}{u^4}\frac{1}{M}-1\right),$$

which is right.

So in the equation $\frac{\gamma}{\alpha} = \frac{1}{8}\frac{\Pi}{\Delta}$, if for a moment (Π), (Δ) are what Π, Δ become, the equation is $\frac{\delta}{\zeta} = \frac{1}{8}\frac{(\Pi)}{(\Delta)}$, that is, $\frac{1}{u^8}\frac{\Pi'}{\Delta} = \frac{(\Pi)}{(\Delta)}$, or $(\Pi) = \frac{1}{u^8}\frac{(\Delta)}{\Delta}\Pi'$; but obviously $\frac{(\Delta)}{\Delta} = -\frac{1}{u^4v^4}$; and the equation thus is $(\Pi) = -\frac{1}{u^{12}v^4}\Pi'$, or say $u^{12}v^4(\Pi) = -\Pi'$; that is,

$$\begin{aligned}
-\Pi' = u^{12}v^4 \Bigg\{ & \frac{v^{12}}{u^{12}} \cdot \frac{1}{M^3} \cdot \frac{1}{v^4}\left(1 - \frac{1}{v^8}\right) \\
& + \frac{v^8}{u^8} \cdot \frac{1}{M^2} \cdot \frac{1}{u^3}\left(\frac{1}{u} - \frac{2}{v^3}\right)\left(1 - \frac{1}{v^8}\right) \\
& + \frac{v^4}{u^4} \cdot \frac{1}{M} \cdot \frac{4}{u^5v^5} + \frac{1}{v^4}\left(1 - \frac{3}{u^8}\right) - \frac{4}{u^3v^3} + \frac{2}{u^4} \\
& - \frac{2}{v^4} + \frac{6}{v^3u^3}\left(1 - \frac{1}{u^8}\right) + \frac{1}{u^4}\left(-3 + \frac{5}{u^8}\right)\Bigg\},
\end{aligned}$$

which is right.

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52. I recur to the formula

$$\frac{1-y}{1+y} = \frac{1-x}{1+x}\left(\frac{\alpha - \beta x + \gamma x^2 + \cdots \pm \sigma x^{\frac{1}{2}(n-1)}}{\alpha + \beta x + \gamma x^2 + \cdots + \sigma x^{\frac{1}{2}(n-1)}}\right)^2,$$

and seek to express the ratios $\alpha : \beta : \cdots : \sigma$ in terms of q. Writing with Jacobi

$$\omega = \frac{mK + m'iK'}{n},$$

we have in general

$$\alpha + \beta x + \gamma x^2 + \cdots + \sigma x^{\frac{1}{2}(n-1)} = \alpha\left(1 + \frac{x}{2\omega}\right)\left(1 + \frac{x}{4\omega}\right)\cdots\left(1 + \frac{x}{(n-1)\omega}\right),$$

(snc = sin co am; viz. $2\omega = \mathrm{sn}(K - 2\omega)$, &c.);

and the values of α, β, …, θ which correspond to the moduli v_0, v_1, … v_n, or say the values $(\alpha_0, \beta_0, \ldots, \theta_0)$, $(\alpha_1, \beta_1, \ldots, \theta_1)$, …, $(\alpha_n, \beta_n, \ldots, \theta_n)$, are obtained by giving to ω the values

$$\begin{array}{ccccc}
\omega_0, & \omega_1, & \omega_2, & \ldots, & \omega_n, \\
= \dfrac{2K}{n}, & \dfrac{2K + iK'}{n}, & \dfrac{4K + iK'}{n}, & \ldots, & \dfrac{iK'}{n},
\end{array}$$

viz. the cases ω_0, ω_n correspond to Jacobi's first and second real transformations, and the others to the imaginary transformations.

I remark that $\omega = \omega_0$ gives for $2g\omega$ an expression which is rational as regards q, but $\omega = \omega_n$ gives an expression involving $q^{\frac{1}{n}}$, the real nth root of q; the other values $\omega_1, \omega_2, \ldots$ give the like expressions, involving $\alpha q^{\frac{1}{n}}, \alpha^2 q^{\frac{1}{n}}, \ldots$ (α an imaginary nth root of unity), the imaginary nth roots of q.

53. I consider first the expression

$$\frac{1}{2g\omega_0}, = \frac{1}{\mathrm{sn}(K - 2g\omega_0)}, = \frac{\mathrm{dn} 2g\omega_0}{\mathrm{cn} 2g\omega_0}.$$

Here, writing $2g\omega_0 = \dfrac{2K\xi}{\pi}$ (ξ for Jacobi's x, as x is being used in a different sense), that is,

$$\xi = \frac{\pi}{2K} \cdot 2g \cdot \frac{2K}{n}, = \frac{2g\pi}{n},$$

(and hence $e^{i\xi} = e^{g\frac{2\pi i}{n}} = \alpha^g$, $e^{2i\xi} = \alpha^{2g}$, if $\alpha = e^{\frac{2\pi i}{n}}$, an imaginary nth root of unity), we have (Jacobi, p. 86, [*Ges. Werke*, t. I., p. 143])

$$\begin{aligned}\frac{1}{2g\omega_0} &= \mathrm{dn}\frac{2K\xi}{\pi} \div \mathrm{cn}\frac{2K\xi}{\pi} \\ &= \frac{C}{B} \cdot \frac{2e^{i\xi}}{1 + e^{2i\xi}} \cdot \frac{(1 + qe^{2i\xi}) \cdots (1 + qe^{-2i\xi}) \cdots}{(1 + q^2 e^{2i\xi}) \cdots (1 + q^2 e^{-2i\xi}) \cdots},\end{aligned}$$

where

$$\frac{C}{B} = \frac{\{(1 + q^2) \cdots\}^2}{\{(1 + q) \cdots\}^2} = f^2(q);$$

that is,

$$\frac{1}{2g\omega_0} = \frac{2\alpha^g}{1 + \alpha^{2g}} \cdot f^2(q) \cdot \frac{(1 + \alpha^{2g} q) \cdots (1 + \alpha^{n-2g} q) \cdots}{(1 + \alpha^{2g} q^2) \cdots (1 + \alpha^{n-2g} q^2) \cdots},$$

where, for shortness, I write $(1 + qe^{2i\xi}) \cdots$ to denote the infinite product

$$(1 + qe^{2i\xi})(1 + q^3 e^{2i\xi})(1 + q^5 e^{2i\xi}) \cdots,$$

and similarly $(1 + q^2 e^{2i\xi}) \cdots$ to denote the infinite product $(1 + q^2 e^{2i\xi})(1 + q^4 e^{2i\xi})(1 + q^6 e^{2i\xi}) \cdots$, and the like for the terms in $e^{-2i\xi}$; the notation, accompanied by its explanation, is quite intelligible, and it would be difficult to make one which would be at the same time complete and not cumbrous. Then attributing to g the values 1, 2, $\dots$, $\frac{1}{2}(n-1)$, and forming the symmetric functions of these expressions, we have the values of $\frac{\beta}{\alpha}$, $\frac{\gamma}{\alpha}$, &c., or α being put $=1$, say the values of $\beta, \gamma, \dots, \sigma$.

54. I stop to notice a verification afforded by the value of β_0. Putting $u=0$, that is, $q=0$, we have

$$\frac{1}{2g\omega_0} = \frac{2\alpha^g}{1+\alpha^{2g}},$$

and hence

$$\beta_0 = 2\left\{\frac{\alpha}{1+\alpha^2} + \frac{\alpha^2}{1+\alpha^4} + \frac{\alpha^3}{1+\alpha^6} + \dots + \frac{\alpha^{\frac{1}{2}(n-1)}}{1+\alpha^{n-1}}\right\},$$

we have $2\beta_0 = \frac{1}{M_0} - 1$; and putting as above $u=0$, the value of $\frac{1}{M_0}$ is $= (-)^{\frac{n-1}{2}}n$; whence

$$(-)^{\frac{1}{2}(n-1)}n - 1 = 4\left\{\frac{\alpha}{1+\alpha^2} + \frac{\alpha^2}{1+\alpha^4} + \frac{\alpha^3}{1+\alpha^6} + \dots + \frac{\alpha^{\frac{1}{2}(n-1)}}{1+\alpha^{n-1}}\right\},$$

a theorem relating to the imaginary nth roots of unity, n an odd prime. In particular,

$$n=3, \quad -4 = 4\left\{\frac{\alpha}{1+\alpha^2}\right\}, \text{ at once verified by } \alpha^2+\alpha+1=0;$$

$$n=5, \quad 4 = 4\left\{\frac{\alpha}{1+\alpha^2} + \frac{\alpha^2}{1+\alpha^4}\right\}, \text{ verified by } \alpha^5-1=0,$$

viz. the theorem is also true for the real root $\alpha=1$; in fact, the term in { } is

$$\{\alpha(1+\alpha^4)+\alpha^2(1+\alpha^2)\}\div(1+\alpha^2)(1+\alpha^4), \text{ that is, } (\alpha+1+\alpha^2+\alpha^4)\div(1+\alpha^2+\alpha^4+\alpha), = 1$$

$$n=7, \quad -8 = 4\left\{\frac{\alpha}{1+\alpha^2} + \frac{\alpha^2}{1+\alpha^4} + \frac{\alpha^3}{1+\alpha^6}\right\},$$

which may be verified by means of $\alpha^6+\alpha^5+\alpha^4+\alpha^3+\alpha^2+\alpha+1=0$; and so on.

55. I further remark that we have

$$\frac{1}{M_0} = (-)^{\frac{1}{2}(n-1)}\left\{\frac{\operatorname{sn}2\omega_0 \cdot \operatorname{sn}4\omega_0 \cdots \operatorname{sn}(n-1)\omega_0}{2\omega_0 \cdot 4\omega_0 \cdots (n-1)\omega_0}\right\}^2.$$

But Jacobi (p. 86, [$l.c.$]),

$$\operatorname{sn} 2g\omega_0 = \operatorname{sn}\frac{2K\xi}{\pi},$$

$$= \frac{AK}{\pi i}\cdot\frac{e^{2i\xi}-1}{e^{i\xi}}\cdot\frac{(1-q^2e^{2i\xi})\cdots(1-q^2e^{-2i\xi})\cdots}{(1-qe^{2i\xi})\cdots(1-qe^{-2i\xi})\cdots},$$

where (p. 89, [$l.c.$, p. 146])

$$\frac{AK}{\pi} = \frac{\sqrt[4]{q}}{\sqrt{k}}, = \tfrac{1}{2}\left\{\frac{(1+q)(1+q^3)\cdots}{(1+q^2)(1+q^4)\cdots}\right\}^2 = \tfrac{1}{2}f^{-2}(q),$$

that is,

$$\operatorname{sn} 2g\omega = f^{-2}q\cdot\frac{\alpha^{2g}-1}{2i\alpha^g}\cdot\frac{(1-\alpha^{2g}q^2)\cdots(1-\alpha^{n-2g}q^2)\cdots}{(1-\alpha^{2g}q)\cdots(1-\alpha^{n-2g}q)\cdots}.$$

Hence

$$\frac{\operatorname{sn} 2g\omega_0}{2g\omega_0} = \frac{\alpha^{2g}-1}{i(\alpha^{2g}+1)}\cdot\frac{1-\alpha^{2g}q^2\cdots}{1+\alpha^{2g}q^2\cdots}\cdot\frac{1+\alpha^{2g}q\cdots}{1-\alpha^{2g}q\cdots}\cdot\frac{1-\alpha^{n-2g}q^2\cdots}{1+\alpha^{n-2g}q^2\cdots}\cdot\frac{1+\alpha^{n-2g}q\cdots}{1-\alpha^{n-2g}q\cdots};$$

and giving to g the values 1, 2, …, $\frac{1}{2}(n-1)$, and multiplying the several expressions, we have the value of $\frac{1}{M_0}$, viz. this is

$$\frac{1}{M_0} = (-)^{\frac{1}{2}(n-1)}\Pi\left\{\frac{(\alpha^{2g}+1)^2}{i^2(\alpha^{2g}+1)^2}\right\}R(q),$$

where $R(q)$ denotes the product of the several factors which contain q.

56. The $(i^2)^{\frac{1}{2}(n-1)}$ of the denominator gives a factor i^{n-1}, $=(-)^{\frac{n-1}{2}}$, which destroys the factor $(-)^{\frac{n-1}{2}}$. We have then a factor

$$\Pi\left(\frac{\alpha^{2g}-1}{\alpha^{2g}+1}\right)^2, \text{ which is } = (-)^{\frac{1}{2}(n-1)}n.$$

In fact, $n=3$, this is

$$\left(\frac{\alpha^2-1}{\alpha^2+1}\right)^2 = -3,$$

viz. the numerator is $\alpha - 2\alpha^2 + 1$, $= -3\alpha^2$, and the denominator is $(-\alpha)^2$, $= \alpha^2$.

So $n = 5$, the formula is

$$\left(\frac{\alpha^2-1}{\alpha^2+1}\cdot\frac{\alpha^4-1}{\alpha^4+1}\right)^2 = 5,\ \text{that is,}\ \frac{(\alpha^2-1)^4}{(\alpha^4+1)^2} = 5;$$

or

$$\frac{\alpha^8 - 4\alpha^2 + 6\alpha^4 - 4\alpha + 1}{\alpha^8 + 2\alpha^4 + 1} = 5,$$

viz. this is $5(1 + \alpha^8 + 2\alpha^4) - (1 - 4\alpha - 4\alpha^2 + \alpha^8 + 6\alpha^4) = 0$, which is right; and so in other cases.

We thus have

$$\frac{1}{M_0} = (-)^{\frac{1}{2}(n-1)} n \cdot R(q),$$

which, on putting therein $u = 0$, that is, $q = 0$, gives, as it should do, $\frac{1}{M_0} = (-)^{\frac{1}{2}(n-1)} n$.

57. As regards the expression of $R(q)$, observe that, giving to g its different values, the factors $1 - \alpha^{2g}q^2$ and $1 - \alpha^{n-2g}q^2$ are all the factors other than $1 - q^2$ of $1 - q^{2n}$, and so as to the other pairs of factors; viz. we have

$$R(q) = \left(\frac{1-q^{2n}\cdots}{1-q^2\cdots}\frac{1+q^n\cdots}{1+q\cdots}\frac{1+q^2\cdots}{1+q^{2n}\cdots}\frac{1-q\cdots}{1-q^n\cdots}\right)^2,$$

viz. this is

$$= \left(\frac{1-q^{2n}\cdots}{1+q^{2n}\cdots}\frac{1+q^n\cdots}{1-q^n\cdots}\right)^2 \div \left(\frac{1-q^2\cdots}{1+q^2\cdots}\frac{1+q\cdots}{1-q\cdots}\right)^2,$$

that is,

$$\frac{1}{M_0} = (-)^{\frac{1}{2}(n-1)} n \frac{\phi^2(q^n)}{\phi^2(q)},$$

agreeing with a former result.

58. We have of course the identity $2\beta_0 = \frac{1}{M_0} - 1$; that is,

$$4S\frac{\alpha^g}{1+\alpha^{2g}} f^2(q)\cdot\frac{(1+\alpha^{2g}q)\cdots(1+\alpha^{n-2g}q)\cdots}{(1+\alpha^{2g}q^2)\cdots(1+\alpha^{n-2g}q^2)\cdots} = (-)^{\frac{1}{2}(n-1)} n \frac{\phi^2(q^n)}{\phi^2(q)} - 1,$$

$(g = 1,\ 2,\ \ldots \frac{1}{2}(n-1))$, which, putting therein $q = 0$, is an identity before referred to; a form perhaps more convenient is obtained by dividing each side by $f^2(q)$.

59. I notice further that we have

$$v_0 = u^n\{2\omega_0 4\omega_0 \cdots (n-1)\omega_0\};$$

the term in { } is

$$\Pi \frac{1+\alpha^{2g}}{2\alpha^g} f^{-2}(q) \frac{(1+\alpha^{2g}q^2)\cdots(1+\alpha^{n-2g}q^2)\cdots}{(1+\alpha^{2g}q)\cdots(1+\alpha^{n-2g}q)\cdots},$$

where we have $\Pi \dfrac{1+\alpha^{2g}}{\alpha} = (-)^{\frac{1}{2}(n^2-1)}$. For example, $n=3$, the term is $\dfrac{1+\alpha^2}{\alpha} = -1$; $n=5$, it is

$$\frac{(1+\alpha^2)(1+\alpha^4)}{\alpha\cdot\alpha^2}, = \frac{1+\alpha^2+\alpha^4+\alpha}{\alpha^3}, = -1;$$

$n=7$, it is

$$\frac{(1+\alpha^2)(1+\alpha^4)(1+\alpha^6)}{\alpha\cdot\alpha^2\cdot\alpha^4}, = \frac{1+\alpha+\alpha^2+\alpha^3+\alpha^4+\alpha^5+2\alpha^6}{\alpha^6} = 1;$$

and so on. The term in question thus is

$$(-)^{\frac{1}{2}(n^2-1)} \cdot \frac{1}{(\sqrt{2})^{n-1}} f^{-n+1}(q) \frac{1+q^{2n}\cdots 1+q\cdots}{1+q^n\cdots 1+q^2\cdots},$$

that is,

$$(-)^{\frac{1}{2}(n^2-1)} \frac{1}{(\sqrt{2})^{n-1}} f^{-n}(q) f(q^n).$$

This has to be multiplied by $u^n, = (\sqrt{2})^n q^{\frac{n}{8}} f^n(q)$, and we thus obtain

$$v_0 = (-)^{\frac{1}{2}(n^2-1)} \sqrt{2} q^{\frac{n}{8}} f(q^n),$$

agreeing with a former result.

We have in what precedes a complete q-transcendental solution for the *transformatio prima*; viz. the original modulus $k^2 (= u^8)$ being given as a function of q, then, as well the new modulus $\lambda_0^2 (= v_0^8)$ and the multiplier M_0, as also the several functions which enter into the expression

$$\frac{1-y}{1+y} = \frac{1-x}{1+x} \left\{ \frac{\left(1-\dfrac{x}{2\omega_0}\right)\cdots\left(1-\dfrac{x}{(n-1)\omega_0}\right)}{\left(1+\dfrac{x}{2\omega_0}\right)\cdots\left(1+\dfrac{x}{(n-1)\omega_0}\right)} \right\}^2,$$

are all of them expressed as functions of q.

60. I consider in like manner the expression

$$\frac{1}{2g\omega_n}\cdot\frac{1}{\operatorname{sn}(K-2g\omega_n)}, = \frac{\operatorname{dn}2g\omega_n}{\operatorname{cn}2g\omega_n}.$$

Here, writing $2g\omega_n = \dfrac{2K\xi}{\pi}$ (ξ instead of Jacobi's x as before), that is,

$$\xi = \frac{\pi}{2K}\cdot 2g\cdot\frac{iK'}{n} = \frac{g\pi iK'}{nK},$$

and thence

$$e^{i\xi} = e^{-\frac{g}{n}\frac{\pi K'}{K}}, = q^{\frac{g}{n}},$$

we have

$$\begin{aligned}\frac{1}{2g\omega_n} &= \operatorname{dn}\frac{2K\xi}{\pi}\div\operatorname{cn}\frac{2K\xi}{\pi}\\ &= f^2(q)\cdot\frac{2q^{\frac{g}{n}}}{1+q^{\frac{2g}{n}}}\cdot\frac{(1+q^{1+\frac{2g}{n}})\cdots(1+q^{1-\frac{2g}{n}})\cdots}{(1+q^{2+\frac{2g}{n}})\cdots(1+q^{2-\frac{2g}{n}})\cdots},\end{aligned}$$

where the notations are as follows:

$(1+q^{1+\frac{2g}{n}})\cdots$ is the infinite product $(1+q^{1+\frac{2g}{n}})(1+q^{3+\frac{2g}{n}})(1+q^{5+\frac{2g}{n}})\cdots$,

and

$(1+q^{2+\frac{2g}{n}})\cdots$ is the infinite product $(1+q^{2+\frac{2g}{n}})(1+q^{4+\frac{2g}{n}})(1+q^{6+\frac{2g}{n}})\cdots$;

and the like as to the expressions with exponents containing $-\dfrac{2g}{n}$.

And then attributing to g the values $1, 2, \ldots, \frac{1}{2}(n-1)$, and forming the symmetric functions of these expressions, we have the values of $\frac{\beta}{\alpha}, \frac{\gamma}{\alpha}, \ldots, \frac{\sigma}{\alpha}$; or α being put $=1$, say the values of $\beta, \gamma, \ldots, \sigma$.

It is easy to see, and I do not stop to prove that, if instead of $\omega=\omega_n$ we have $\omega=\omega_1, \omega_2, \ldots$, or ω_{n-1}, we simply multiply $q^{\frac{1}{n}}$ by an imaginary nth root of unity; that is, we replace the real nth root $q^{\frac{1}{n}}$ by an imaginary nth root of q.

In the case $u=0$, that is, $q=0$, we have $\dfrac{1}{2g\omega_n}=0$, and hence $\beta=0$; and the like for the values $\omega_1, \omega_2, \ldots, \omega_{n-1}$: the equation $2\beta = \frac{1}{M}-1$ gives consequently for $\frac{1}{M}$, n values each $=1$, agreeing with the multiplier equation.

61. We have for M_n the formula

$$\frac{1}{M_n} = (-)^{\frac{1}{2}(n-1)} \left\{ \frac{\text{sn}2\omega_n \text{sn}4\omega_n \cdots \text{sn}(n-1)\omega_n}{2\omega_n 4\omega_n \cdots (n-1)\omega_n} \right\}^2,$$

and, as before,

$$\text{sn}2g\omega_n = f^{-2}(q) \cdot \frac{q^{\frac{2g}{n}} - 1}{2iq^{\frac{g}{n}}} \cdot \frac{(1-q^{2+\frac{2g}{n}}) \cdots (1-q^{2-\frac{2g}{n}}) \cdots}{(1-q^{1+\frac{2g}{n}}) \cdots (1-q^{1-\frac{2g}{n}}) \cdots};$$

hence

$$\frac{\text{sn}2g\omega_n}{2g\omega_n} = \frac{q^{\frac{2g}{n}} - 1}{i(q^{\frac{2g}{n}}+1)} \cdot \frac{(1-q^{2+\frac{2g}{n}}) \cdots}{(1+q^{2+\frac{2g}{n}}) \cdots} \cdot \frac{(1+q^{1+\frac{2g}{n}}) \cdots}{(1-q^{1+\frac{2g}{n}}) \cdots} \cdot \frac{(1-q^{2-\frac{2g}{n}}) \cdots}{(1+q^{2-\frac{2g}{n}}) \cdots} \cdot \frac{(1+q^{1-\frac{2g}{n}}) \cdots}{(1-q^{1-\frac{2g}{n}}) \cdots};$$

and we thence derive the value of $\frac{1}{M_n}$; viz. observing that we have in the denominator $(i^2)^{\frac{1}{2}(n-1)}, = (-)^{\frac{1}{2}(n-1)}$ which destroys this factor in the expression of $\frac{1}{M_n}$, this is

$$\frac{1}{M_n} = \Pi \left\{ \frac{1-q^{\frac{2g}{n}}}{1+q^{\frac{2g}{n}}} \cdot \frac{(1-q^{2+\frac{2g}{n}}) \cdots (1+q^{1+\frac{2g}{n}}) \cdots (1-q^{2-\frac{2g}{n}}) \cdots (1+q^{1-\frac{2g}{n}}) \cdots}{(1+q^{2+\frac{2g}{n}}) \cdots (1-q^{1+\frac{2g}{n}}) \cdots (1+q^{2-\frac{2g}{n}}) \cdots (1-q^{1-\frac{2g}{n}}) \cdots} \right\}^2.$$

62. We have also

$$v_n = u^n \{2\omega_n 4\omega_n \cdots (n-1)\omega_n\},$$

where the term in { } is

$$= \Pi f^{-2}(q) \frac{(1+q^{\frac{2g}{n}})}{(1+q^{\frac{2g}{n}})} \cdot \frac{(1+q^{2+\frac{2g}{n}}) \cdots (1+q^{2-\frac{2g}{n}}) \cdots}{2q^{\frac{g}{n}}(1+q^{1+\frac{2g}{n}}) \cdots (1+q^{1-\frac{2g}{n}}) \cdots};$$

or, observing that the sum of the exponents $\frac{g}{n}$ is $\frac{1}{n}\{1+2\cdots+\frac{1}{2}(n-1)\} = \frac{n^2-1}{8n}$, this is

$$= f^{-n+1}(q) \cdot \frac{1}{(\sqrt{2})^{n-1} q^{\frac{n^2-1}{8n}}} \frac{(1+q^{\frac{2}{n}}) \cdots (1+q) \cdots}{(1+q^2) \cdots (1+q^{\frac{1}{n}}) \cdots};$$

or, the last factor being $f(q^{\frac{1}{n}}) \div f(q)$, the expression is

$$f^{-n}(q) \frac{1}{(\sqrt{2})^{n-1}} q^{-\frac{n}{8}+\frac{1}{8n}} f(q^{\frac{1}{n}});$$

or, multiplying by u^n, $= (\sqrt{2})^n q^{\frac{n}{8}} f^n(q)$, we have

$$v_n = \sqrt{2} q^{\frac{1}{8n}} f(q^{\frac{1}{n}}),$$

agreeing with a former result.

We have in what precedes the complete q-transcendental solution for the *transformatio secunda*; viz. the original modulus $k(= u^4)$ being given as a function of q, then, as well the new modulus $\lambda_n (= v_n^4)$ and the multiplier M_n, as also the several functions which enter into the formula

$$\frac{1-y}{1+y} = \frac{1-x}{1+x} \left\{ \frac{\left(1 - \dfrac{x}{2\omega_n}\right) \cdots \left(1 - \dfrac{x}{(n-1)\omega_n}\right)}{\left(1 + \dfrac{x}{2\omega_n}\right) \cdots \left(1 + \dfrac{x}{(n-1)\omega_n}\right)} \right\}^2,$$

are all expressed in terms of q. The expressions all contain $q^{\frac{1}{n}}$, and by substituting for this an imaginary nth root of q, we have the formulae belonging to the several $(n-1)$ imaginary transformations.

63. As an illustration of the formulae for the *transformatio secunda* I write $n = 7$; and putting for greater convenience $q = r^7$, that is, $r = q^{\frac{1}{7}}$, then we have

$$v_7 = \sqrt{2} r^{\frac{1}{8}} f(r), \quad \frac{1}{M_7} = \frac{\phi^2(r)}{\phi^2(r^7)},$$

$$\frac{1}{2\omega_7} = 2f^2(r^7)A, \quad \frac{1}{4\omega_7} = 2f^2(r^7)B, \quad \frac{1}{6\omega_7} = 2f^2(r^7)C,$$

where

$$A = r \cdot \frac{5 \cdot 19 \cdots \quad 9 \cdot 23 \cdots}{2 \cdot 16 \cdots \quad 12 \cdot 26 \cdots},$$
$$B = r^2 \cdot \frac{3 \cdot 17 \cdots \quad 11 \cdot 25 \cdots}{4 \cdot 18 \cdots \quad 10 \cdot 24 \cdots},$$
$$C = r^3 \cdot \frac{1 \cdot 15 \cdots \quad 13 \cdot 27 \cdots}{6 \cdot 20 \cdots \quad 8 \cdot 22 \cdots},$$

where the numerator of A denotes $(1 + r^5)(1 + r^{19}) \cdots (1 + r^9)(1 + r^{23}) \cdots$, and so in other cases, the difference of the exponents being always $= 14$. And we have, as mentioned, the identical equation

$$f^2(r^7)(A + B + C) = \tfrac{1}{4} \left\{ \frac{\phi^2 r}{\phi^2 r^7} - 1 \right\}.$$

The values of the several expressions up to r^{50} are as follows: Mr J. W. L. Glaisher kindly performed for me the greater part of the calculation.

Numerical Table for $n = 7$, $q = r^7$.

Ind. of r	A	B	C	Sum	Mult. by $f^2(r^7)$	$\phi^2(r) \div \phi^2(r^7)$
0				0	0	+ 1
1	+ 1			+ 1	+ 1	+ 4
2		+ 1		+ 1	+ 1	+ 4
3	− 1		+ 1	0	0	0
4			+ 1	+ 1	+ 1	+ 4
5	+ 1	+ 1		+ 2	+ 2	+ 8
6	+ 1	− 1		0	0	0
7	− 1			− 1	− 1	− 4
8	− 1			− 1	− 3	− 12
9	+ 1	− 1	− 1	− 1	− 3	− 12
10	+ 2	+ 1	− 1	+ 2	+ 2	+ 8
11	− 1		− 1	− 2	− 4	− 16
12	− 2	− 1	− 1	− 4	− 8	− 32
13		− 2		− 2	− 2	− 8
14	+ 2	− 1		+ 1	+ 3	+ 12
15	+ 1	− 1	+ 1	+ 1	+ 8	+ 32
16	− 2	+ 2	+ 2	+ 2	+ 9	+ 36
17	− 2	− 2	+ 2	− 2	− 6	− 24
18	+ 1	+ 1	+ 2	+ 4	+ 13	+ 52
19	+ 2	+ 2	+ 2	+ 6	+ 24	+ 96
20		− 3	+ 1	− 2	− 6	− 24
21	− 2	+ 2	− 1	− 1	− 8	− 32
22	− 2	+ 1	− 2	− 3	− 20	− 80
23	+ 2	− 4	− 3	− 5	− 24	− 96
24	+ 3	+ 3	− 4	+ 2	+ 16	+ 64
25		− 1	− 4	− 5	− 33	− 132
26	− 4	− 3	− 3	− 10	− 62	− 248
27	− 2	+ 5	− 1	+ 2	+ 16	+ 64
28	+ 4	− 3	+ 1	+ 2	+ 19	+ 76
29	+ 5	− 1	+ 3	+ 7	+ 46	+ 184
30	− 3	+ 6	+ 5	+ 8	+ 56	+ 224
31	− 7	− 6	+ 7	− 6	− 40	− 160
32	+ 1	+ 1	+ 7	+ 9	+ 77	+ 308
33	+ 9	+ 5	+ 4	+ 18	+ 144	+ 576
34	+ 3	− 8	+ 1	− 4	− 38	− 152
35	− 9	+ 5	− 1	− 5	− 42	− 168
36	− 7	+ 2	− 5	− 10	− 99	− 396
37	+ 7	− 9	− 9	− 11	− 122	− 488
38	+ 11	+ 10	− 11	+ 10	+ 88	+ 352
39	− 4	− 3	− 10	− 17	− 168	− 672
40	− 13	− 8	− 7	− 28	− 310	− 1240
41	− 2	+ 13	− 3	+ 8	+ 82	+ 328
42	+ 13	− 8	+ 3	+ 8	+ 88	+ 352
43	+ 8	− 3	+ 9	+ 14	+ 204	+ 816

64. As already mentioned, the foregoing expressions of the coefficients in terms of q may be applied to the determination of the coefficients as rational functions of u, v.

Representing by θ any one of the coefficients $\alpha, \beta, \gamma, \ldots, \sigma$, consider the sum

$$Sv^{\gamma f}\frac{\theta}{\alpha},$$

f a positive integer, and the summation extending as before to the $n+1$ values of v, and corresponding values of $\frac{\theta}{\alpha}$. This is a rational function of u, and it is also integral. As to this observe that the function, if not integral, must become infinite either for $u = 0$ (this would mean that the expression contained a term or terms $Au^{-\bullet}$) or for some finite value of u. But the function can only become infinite by reason of some term or terms of $Sv^f\frac{\theta}{\alpha}$ becoming infinite; viz. some term $\frac{1}{2g\omega}$ must become infinite; or attending to the equation

$$v = u^n\{2\omega 4\omega \cdots (n-1)\omega\},$$

it can only happen if $u = 0$, or if $v = \infty$; and from the modular equation it appears that if $v = \infty$, then also $u = \infty$: the expression in question can therefore only become infinite if $u = 0$, or if $u = \infty$. Now $u = 0$ gives the ratios $\frac{\beta}{\alpha}, \frac{\gamma}{\alpha}, \ldots$, each of them a determinate function of n, that is finite; and gives also $v = 0$, so that the expression does not become infinite for $u = 0$; hence it does not become infinite either for $u = 0$ or for any finite value of u; wherefore it is integral. The like reasoning applies to the sum $Sv^{-f}\frac{\theta}{\alpha}$; viz. this is a rational function of u; and it is quasi-integral, viz. there are no terms having a denominator other than a power of u, the highest denominator being u^{nf}; viz. the expression contains negative and positive integer powers of u, the lowest power (highest negative power) being $\frac{1}{u^{nf}}$.

65. It is to be observed, further, that writing the expression in the form

$$v_0^f\frac{\theta_0}{\alpha_0} + S'v^f\frac{\theta}{\alpha},$$

(where S' refers to the values $v_1, v_2, \ldots, v_n$ of the modulus), and considering the several quantities as expressed in terms of q, then in the sum S' every term involving a fractional power $q^{\frac{h}{n}}$ acquires by the summation the coefficient $(1 + \alpha + \alpha^2 + \cdots + \alpha^{n-1})$, and therefore disappears; there remains only the radicality $q^{\frac{1}{8}}$ occurring in the expressions of the v's; and if $nf \equiv \mu$ (mod. 8), $\mu = 0$, or a positive integer less than 8, then the form of the expression is $q^{\frac{\mu}{8}}$ into a rational function of q. Hence this, being a rational and integral function of u, must be of the form

$$Au^{\mu} + Bu^{\mu+8} + Cu^{\mu+16} + \&\text{c.}$$

66. We have thus in general

$$Sv^f \frac{\theta}{\alpha} = Au^{\mu} + Bu^{\mu+8} + \&\text{c.};$$

and in like manner

$$Sv^{-f} \frac{\theta}{\alpha} = A'u^{-nf} + B'u^{-nf+8} + \&\text{c.}$$

We may in these expressions find a limit to the number of terms, by means of the before-mentioned theorem that we may simultaneously interchange u, v; α, β, $\ldots$, ρ, σ into $\frac{1}{u}$, $\frac{1}{v}$; σ, ρ, $\ldots$, β, α. Starting from the expression of $Sv^f \frac{\theta}{\alpha}$, let ϕ be the corresponding coefficient to θ; viz. in the series α, β, $\ldots$, θ, $\ldots$, ϕ, $\ldots$, ρ, σ, let ϕ be as removed from σ as θ is from α; then the equation becomes

$$Sv^{-f} \frac{\phi}{\sigma} = Au^{-\mu} + Bu^{-\mu-8} + \&\text{c.},$$

where $\frac{\phi}{\sigma} = \frac{\phi}{\alpha}\frac{\alpha}{\sigma} = \frac{v}{u^n}\frac{\phi}{\alpha}$; the equation thus is

$$Sv^{n-f} \frac{\phi}{\alpha} = Au^{n-\mu} + Bu^{n-\mu-8} + \&\text{c.};$$

and by what precedes the series on the right-hand side can contain no negative power higher than $\frac{1}{u^{n(n-f)}}$; that is, the series of coefficients A, B, C, $\ldots$ goes on to a certain point only, the subsequent coefficients all of them vanishing.

In like manner from the equation for $Sv^{-f} \frac{\theta}{\alpha}$ we have

$$Sv^{f+1} \frac{\phi}{\alpha} = A'u^{(n+1)f} + B'u^{(n+1)f-8} + \&\text{c.},$$

where the indices must be positive; viz. the series of coefficients A', B', $\ldots$ goes on to a certain point only, the subsequent coefficients all of them vanishing.

67. The like theory applies to the expression $\frac{1}{M}$. We have, putting as before $nf \equiv \mu$ (mod. 8),

$$Sv^f \frac{1}{M} = Au^\mu + Bu^{\mu+8} + \cdots ,$$
$$Sv^{-f} \frac{1}{M} = A'u^{-nf} + B'u^{-nf+8} + \cdots$$

and we find a limit to the number of terms by the consideration that we may simultaneously change u, v, $\frac{1}{M}$ into $\frac{1}{u}$, $\frac{1}{v}$, $\frac{v^4}{u^4}\frac{1}{M}$; the equations thus become

$$Sv^{n-f} \frac{1}{M} = Au^{4-\mu} + Bu^{4-\mu} + \cdots$$

(where, if $f =$ or < 4, there must be on the right-hand side no negative power of u; but if $f > 4$, then the highest negative power must be $\frac{1}{u^{(f-4)n}}$), and

$$Sv^{n+f} \frac{1}{M} = A'u^{nf+4} + B'u^{nf-4} + \cdots ,$$

where on the right-hand side there must be no negative power of u.

68. It is to be remarked that β, ρ being always given linearly in terms of $\frac{1}{M}$ it is the same thing whether we seek in this manner for the values of β, ρ or for that of $\frac{1}{M}$; but the latter course is practically more convenient. Thus in the cases $n = 5$, $n = 7$ we require only the value of $\frac{1}{M}$.

In the case $n = 11$, where the coefficients are α, β, γ, δ, ϵ, ζ, it has been seen that γ, δ are given as cubic functions of $\frac{1}{M}$: seeking for them directly, their values would (if the process be practicable) be obtained in a better form, viz. instead of the denominator $(F'v)^3$ there would be only the denominator $F'(v)$.

69. I consider for $\frac{1}{M}$ the cases $n = 3$ and 5:

$$n = 3, \quad f = 0,\ 1,\ 2,\ 3,\ \text{then}\ \mu = 0,\ 3,\ 6,\ 1;$$

and we write down the equations

$$\begin{aligned}
&S\frac{1}{M} = A, &&\text{giving } S\frac{v^4}{M} = Au^4,\\
&S\frac{v}{M} = A'u^3, &&\quad ,, \qquad S\frac{v^5}{M} = A'u,\\
&S\frac{v^2}{M} = 0, &&\quad ,, \qquad S\frac{v^6}{M} = 0;
\end{aligned}$$

viz. if we had in the first instance assumed $S\frac{1}{M} = A + Bu^8 + \cdots$, this would have given $S\frac{v^4}{M} = Au^4 + Bu^{-4} + \cdots$, whence B and the succeeding coefficients all vanish; and so in other cases. We have here only the coefficients A, A'; and these can be obtained without the aid of the q-formulae by the consideration that for $u = 1$ the corresponding values of v, $\frac{1}{M}$ are

$$v = 1,\ -1,\ -1,\ -1,$$
$$\frac{1}{M} = 3,\ -1,\ -1,\ -1,$$

Algebraic theory of the modular equation. Art. Nos. 70 to 79.

70. The modular equation gives rise to a series of curves of the orders $n+1$, $2(n+1)$, $4(n+1)$, ... successively derived from each other; viz. starting from the curve $U = V$ of the order $n+1$ (I use U, V to denote the two functions of u, v which, equated to zero, give the modular equation), this has a $(n+1)$-tuple point at the origin, and its deficiency is therefore $\frac{1}{2}n(n-1)$. The next curve is derived from this by writing u^2, v^2 for u, v; it has thus a $2(n+1)$-tuple point at the origin, and its deficiency is therefore $= n(n-1)$. But the curve has also $2(n+1)$ dps on the axes $u = 0$, $v = 0$, viz. these arise from the $(n+1)$ values $u^{n+1} = 1$ and $v^{n+1} = 1$ respectively; and there are besides $2(n+1)$ dps; for the total number is

$$\tfrac{1}{2}\{2(n+1)-1\}\{2(n+1)-2\} - 2(n+1)\{2(n+1)-1\} + 1,$$

$= (2n+1)(2n) - 2(n+1)(2n+1) + 1, = -2n^2 - 2n - 1$, that is, $2n^2 + 2n$ dps, $= 2(n+1) + 2(n^2-1)$; so that there are $2(n^2-1)$ remaining dps, and the deficiency is thus $n(n-1) - 2(n+1) - (n^2-1), = -3n-1$, viz. there is a negative deficiency, and we thus see *a priori* that the curve is reducible. In fact it breaks up into the two curves $U \pm V = 0$ of the same order $n+1$.

71. The third curve is derived from the second by writing u^2, v^2 for u, v; viz. this is the curve $U^2 - V^2 = 0$ of the order $4(n+1)$; it has at the origin a $4(n+1)$-tuple point, and there are on the axes $4(n+1)$ $+\ 4(n+1)$ dps from the values $u^{2(n+1)} = 1$, $v^{2(n+1)} = 1$ respectively; and there are besides $4(n^2-1)$ dps from the $2(n^2-1)$ dps of the second curve; so that the deficiency is $2n(n-1) - 4(n+1) - 2(n^2-1), = -6n$, a negative deficiency; and the curve is in fact $U+V\cdot U-V\cdot=0$, viz. it is composed of the 4 curves $U\pm V$, $U\pm iV=0$.

72. But to the curve $U^2-V^2=0$ we may apply a different process; viz. we may consider it as derived from the curve $U-V=0$ of the order $n+1$ by writing therein u^4, v^4 for u, v; that is, from the curve of the order $4(n+1)$ which has at the origin a $4(n+1)$-tuple point, and on the axes $4(n+1)+4(n+1)$ dps from $u^{4(n+1)}=1$, $v^{4(n+1)}=1$ respectively; the deficiency would be $=2n(n-1)-4(n+1)$, $=2n^2-6n-4$, and there remain therefore this number of dps.

73. The modular equation

$$v^8=u^8+2uv(1-u^4v^4)+u^2v^2(u^4+v^4-u^4v^4)=0$$

gives for the values of u^8 corresponding to a given value of v^8 the equation

$$(1-u^8)^2(v^8-u^8)=4u^2v^2(1-u^4)^2,$$

viz. this is

$$v^8(1-u^8)^2=u^8(1-u^8)^2+4u^2v^2(1-u^4)^2,$$

which is of the degree 8 in regard to u^8; that is, corresponding to a given value of v^8 we have 8ν values of u^8, giving the same number, 8ν, of v^8; that is, the values of v^8 corresponding to a given value of u^8 would group themselves in eights corresponding to the 8 values of u. There is, in fact, no such grouping; the equations are $(u,v)^p=0$, $(u^8,v^8)^p=0$; to a given value of u^8 correspond 8 values of u, and therefore 8ν values of v, but these give in eights the same value of v^8, so that the number of values of v^8 is $=\nu$.

74. I consider the case $n=3$; here, writing x, y for u, v, we have here the sextic curve

I. $y^4-x^4+2xy(x^2y^2-1)=0$;

and it is easy to see that the remaining forms wherein x, y denote u^2, v^2; u^4, v^4; and u^8, v^8 respectively, are derived herefrom as follows; viz.

II. $(y^2-x^2)^2-4xy(xy-1)^2=0$, that is,

$$y^4+6x^2y^2+x^4-4xy(x^2y^2+1)=0;$$

III. $(y^2+6xy+x^2)^2-16xy(xy+1)^2=0$, that is,

$$y^4+6x^2y^2+x^4-4xy(4x^2y^2-3x^2-3y^2+4)=0;$$

IV. $(y^2+6xy+x^2)^2-16xy(4xy-3x-3y+4)^2=0$, that is,

$$
\begin{aligned}
&y^4-762x^2y^2+x^4\\
&-4xy\{64x^2y^2-96x^2y-96xy^2+33x^2+33y^2-96x-96y+64\}=0,
\end{aligned}
$$

where it may be noticed that the process is not again repeatable so as to obtain a sextic equation between x, y standing for u^{16}, v^{16} respectively.

The curve I. has a dp (fleflecnode) at the origin, viz. the branches are given by $y^2-2x=0$, $-x^2-2y=0$; and it has 2 cusps at infinity, on the axes $x=0$, $y=0$ respectively; viz. the infinite branches are given by $y+2x^2=0$, $-x+2y^2=0$ respectively. These same singularities present themselves in the other curves.

The curve II. has the four dps $(x^2-y^2=0,\ xy-1=0)$, that is,

$$(x=y=1),\ (x=y=-1),\ (x=i,\ y=-i),\ (x=-i,\ y=i).$$

Corresponding hereto we have in the curve III. the 2 dps $(x=y=1,\ x=y=-1)$, and in the curve IV. the dp $(x=y=1)$.

The curve III. has besides the 4 dps $y^2+6xy+x^2=0$, $xy+1=0$, that is,

$$(1+\sqrt{2},\ 1-\sqrt{2}),\ (1-\sqrt{2},\ 1+\sqrt{2}),\ (-1-\sqrt{2},\ -1+\sqrt{2}),\ (-1+\sqrt{2},\ -1-\sqrt{2});$$

and corresponding hereto in the curve IV. we have the 2 dps

$$(3+2\sqrt{2},\ 3-2\sqrt{2}),\ (3-2\sqrt{2},\ 3+2\sqrt{2}).$$

The curve IV. has besides the 4 dps $(y^2+6xy+x^2=0,\ 4xy-3x-3y+4=0)$, or say $(2x-\frac{3}{2})(2y-\frac{3}{2})+\frac{7}{4}=0$, $2(x+\frac{9}{4})^2+2(y+\frac{9}{4})^2-\frac{171}{8}=0$. Hence the 4 curves have respectively the dps and deficiency following:—

	dps.	Def.
2, 1	= 3,	7,
2, 1, 4	= 7,	3,
2, 1, 2, 4	= 9,	1,
2, 1, 1, 2, 4	= 10,	0;

viz. the curve IV. representing the equation between u^8 and v^8 is a unicursal sextic.

Arranging the results in a tabular form and adding the values of the deficiency, we have

		dps.	Def.
I.	$1+12+8$	$=21,$	$=15,$
II.	$1+12+4+12$	$29,$	$7,$
III.	$1+12+2+\ \ 6+12$	$33,$	$3,$
IV.	$1+12+1+\ \ 3+\ \ 6+12$	$35,$	$1,$

so that the curve IV. is a curve of deficiency 1, or bicursal curve. It appears by Jacobi's investigation for the quintic transformation (*Fund. Nov.* pp. 26—28, [*Ges. Werke*, t. I., pp. 77—79]) that we can in fact express x, y, that is, u^8, v^8, rationally in terms of the parameters α, β connected by the equation

$$\alpha^3 = 2\beta(1+\alpha+\beta),$$

which is that of a general cubic (deficiency $=1$); in fact, we have

$$\frac{2-\alpha}{\alpha-2\beta} = \frac{v^4}{u^4}, \quad \beta = \frac{u^8}{v},$$

that is,

$$u^8(=x) = \beta^2\left(\frac{2-\alpha}{\alpha-2\beta}\right), \quad v^8(=y) = \beta^2\left(\frac{2-\alpha}{\alpha-2\beta}\right)^5,$$

where α, β satisfy the relation just referred to. The actual verification of the equation IV. by means of these values would be a work of some labour.

79. In the general case p an odd prime, then in I. we have at the origin one dp (in the nature of a fleflecnode) and at infinity two singular points each $=\frac{1}{2}(p-1)(p-2)$ dps. I infer, from a result obtained by Professor Smith, that there are besides $(p-1)(p-3)$ dps; but I have not investigated the nature of these. And the Table of dps and deficiency then is

I.	$1+(p-1)(p-2)+\ \ (p-1)(p-3)$	$=$
II.	$1+(p-1)(p-2)+\frac{1}{2}(p-1)(p-3)+\frac{1}{2}(p^2-1)$	
III.	$1+(p-1)(p-2)+\frac{1}{4}(p-1)(p-3)+\frac{1}{4}(p^2-1)+\frac{1}{2}(p^2-1)$	
IV.	$1+(p-1)(p-2)+\frac{1}{8}(p-1)(p-3)+\frac{1}{8}(p^2-1)+\frac{1}{4}(p^2-1)+\frac{1}{2}(p^2-1)$	2

viz. his values of the deficiencies being as in the last column, the total number of dps must be as in the last but one column.

[End of Paper 578]

Address delivered by the President on presenting the Medals

[Paper 579]

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...with these twelve elements, the expressions for the variations assume the canonical form

$$\frac{dk_i}{dt} = \frac{dR}{de_0}, \quad \frac{de_0}{dt} = -\frac{dR}{dk_i}, \ \&\text{c.}$$

The concluding part of the Memoir contains approximate calculations which seems to show that the whole process is a very practicable one: but the author remarks that it is only doing justice to Delaunay to say that, starting from his (Delaunay's) final differential equations, and regarding the planet as adding new terms to the disturbing function, there would be obtained equations of the same degree of rigour as those of his own Memoir.

Everything in the Lunar Theory is laborious, and it is impossible to form an opinion as to the comparative facility of methods; but irrespectively of the possible applications of the method, the Memoir is, from the boldness of the conception and beauty of the results, a very remarkable one, and constitutes an important addition to Theoretical Dynamics*.

I come now to the planets *Neptune* and *Uranus*: it is well-known how, historically, the two are connected. The increasing and systematic inaccuracies of Bouvard's Tables of *Uranus* were found to be such as could be accounted for by the existence of an exterior disturbing planet; and it was thus that the planet *Neptune* was discovered by Adams and Le Verrier before it was seen in the telescope, in September 1846. It was afterwards ascertained that the planet had been seen twice by Lalande, in May 1795. The theory of *Neptune* was investigated by Peirce and Walker: viz. Walker, by means of the observations of 1795, and those of 1846—47, and using Peirce's formulae for the perturbations produced by *Jupiter*, *Saturn*, and *Uranus*, determined successfully two sets of elliptic elements of the planet. The values first obtained showed that it was necessary to revise the perturbation-theory, which Peirce accordingly did, and with the new perturbations and revised normal places, the second set of elements

(*Walker's Elliptic Elements II.*) was computed. With these elements and perturbations there was obtained for the planet from the time of its discovery a continuous ephemeris, published in the *Smithsonian Contributions*, *Gould's Astronomical Journal*, and since 1852 in the *American Ephemeris* and the *Nautical Almanac*. The theory was next considered by Kowalski in a work published at Kasan in the year 1855. The long period inequalities are dealt with by him in a manner different from that adopted by Peirce, so that the two theories are not directly comparable, but Professor Newcomb, by a comparison of the ephemerides with observation, arrives at the conclusion that the theory of Kowalski (although derived from observations up to 1853, when the planet had moved through an arc of 16°) was on the whole no nearer the truth than that of Walker;

*Since the above was written, Professor Newcomb has communicated to me some very interesting details as to the extent to which he has carried his computations, and in particular he mentions that, considering the action of each planet from *Mercury* to *Saturn*, he has (in regard to the terms the coefficients of which might become large by integration) estimated the probable limiting value of more than fifty such terms of period from a few years to several thousands without finding any which could become sensible, except the term leading to Hansen's first inequality produced by *Venus*.

[Transcription ends at page 200]

585. A New Theorem on the Equilibrium of Four Forces Acting on a Solid Body.

[From the Philosophical Magazine, vol. xxxi. (1866), pp. 78, 79;
Camb. Phil. Soc. Proc. vol. I. (1866), p. 235.]

Defining the "moment of two lines" as the product of the shortest distance of the two lines into the sine of their inclination, then, if four forces acting along the lines 1, 2, 3, 4 respectively are in equilibrium, the lines must, as is known (Möbius), be four generating lines of an hyperboloid; and if 12 denote the moment of the lines 1 and 2, and similarly 13 the moment of the lines 1 and 3, &c., the forces are as

$$\sqrt{(23 \cdot 34 \cdot 42)} : \sqrt{(34 \cdot 41 \cdot 13)} : \sqrt{(41 \cdot 12 \cdot 24)} : \sqrt{(12 \cdot 23 \cdot 31)}.$$

Calling the four forces P_1, P_2, P_3, P_4, it follows as a corollary that we have

$$P_1 P_2 \cdot 12 = 12 \cdot 34 \sqrt{(13 \cdot 42 \cdot 14 \cdot 23)} = P_3 P_4 \cdot 34;$$

viz. the product of any two of the forces into the moment of the lines along which they act is equal to the product of the other two forces into the moment of the lines along which they act,—which is equivalent to Chasles's theorem, that, representing a force by a finite line of proportional magnitude, then in whatever way a system of forces is resolved into two forces, the volume of the tetrahedron formed by joining the extremities of the two representative lines is constant.

586. On the Mathematical Theory of Isomers.

[From the Philosophical Magazine, vol. xlvii. (1874), pp. 444—446.]

I consider a "diagram," viz. a set of points H, O, N, C, &c. (any number of each), connected by links into a single assemblage under the condition that through each H there passes not more than one link, through each O not more than two links, through each N not more than three links, through each C not more than four links. Of course through every point there passes at least one link, or the points would not be connected into a single assemblage.

In such a diagram each point having its full number of links is saturate, or nilvalent: in particular, each point H is saturate. A point not having its full number of links is univalent, bivalent, or trivalent, according as it wants one, two, or three of its full number of links.

If every point is saturate the diagram is saturate, or nilvalent; or, say, it is a "plerogram"; but if the diagram is susceptible of n more links, then it is n-valent; viz. the valency of the diagram is the sum of the valencies of the component points.

Since each H is connected by a single link (and therefore to a point O, C, &c. as the case may be, but not to another point H), we may without breaking up the diagram remove all the points H with the links belonging to them, and thus obtain a diagram without any points H: such a diagram may be termed a "kenogram": the valency is obviously that of the original diagram plus the number of removed H's.

If from a kenogram, we remove every point O, C, &c. connected with the rest of the diagram by a single link only (each with the link belonging to it), and so on indefinitely as long as the process is practicable, we arrive at last at a diagram in which every point O, C, &c. is connected with the rest of the diagram by two links at least: this may be called a "mere kenogram."

Each or any point of a mere kenogram may be made the origin of a "ramification"; viz. we have here links branching out from the original point, and then again from the derived points, and so on any number of times, and never again uniting. We can thus from the mere kenogram obtain (in an infinite variety of ways) a diagram. The diagram completely determines the mere kenogram; and consequently two diagrams cannot be identical unless they have the same mere kenogram.

Observe that the mere kenogram may evanesce altogether; viz. this will be the case if the diagram or kenogram is a simple ramification.

A ramification of n points C is $(2n+2)$-valent: in fact, this is so in the most simple case $n = 1$; and admitting it to be true for any value of n, it is at once seen to be true for the next succeeding value. But no kenogram of points C is so much as $(2n + 2)$-valent; for instance, 3 points C linked into a triangle, instead of being 8-valent are only 6-valent.

We have therefore plerograms of n points C and $2n + 2$ points H, say plerograms C_nH_{2n+2}; and in any such plerogram the kenogram is of necessity a ramification of n points C; viz. the different cases of such ramifications are:

$n = 1$,	$n = 2$,	$n = 3$,	$n = 4$,	$n = 5$,	$n = 6$,
$*$	$*$	$*$	$*$	$*$	$*$
		$*$	$*$	$*$	$*$
			$*$	$*$	$*$
					$*$

where the mathematical question of the determination of such forms belongs to the class of questions considered in my paper "On the Theory of the Analytical Forms called Trees."

The different forms of univalent diagrams C_nH_{2n+1} are obtained from the same ramifications by adding to each of them all but one of the $2n + 2$ points H; that is, by adding to each point C except one its full number of points H, and to the excepted point one less than the full number of points H. The excepted point C must therefore be univalent at least; viz. it cannot be a saturate point, which presents itself for example in the diagrams $n = 5$ (7) and $n = 6$ (8).

And in order to count the number of distinct forms (for the diagrams C_nH_{2n+1}), we must in each of the above ramifications consider what is the number of distinct classes into which the points group themselves, or, say, the number of "allotrious" points. For instance, in the ramification $n = 3$ there are two classes only; viz. a point is either terminal or medial; or, say, the number of allotrious points is $= 2$: this is shown in the diagrams by means of the asterisks; so that in each case the points which may be considered allotrious are represented by asterisks, and the number of asterisks is equal to the number of allotrious points.

Thus, number of univalent diagrams C_nH_{2n+1}:

$$\begin{array}{ll}
n=1, & 1 \\
n=2, & 1 \\
n=3, & 2 \\
n=4, & (\alpha)\,2;\ (\beta)\,2;\ \text{together } 4 \\
n=5, & (\alpha)\,3;\ (\beta)\,4;\ (\gamma)\,1;\ \text{together } 8 \\
n=6, & (\alpha)\,3;\ (\beta)\,5;\ (\gamma)\,2;\ (\delta)\,3;\ \text{together } 13
\end{array}$$

where it will be observed that, $n = 5$ (7), and $n = 6$ (8), the numbers of allotrious points are 2 and 4 respectively; but since in each of these cases one point is saturate, they give only the numbers 1 and 3 respectively.

It might be mathematically possible to obtain a general solution; but there would be little use in this; and for even the next succeeding case, No. of bivalent diagrams C_nH_{2n}; the extreme complexity of the question would, it is probable, prevent the attainment of a general solution.

Passing to the chemical signification of the formulæ, and instead of the radicals C_nH_{2n+1} considering the corresponding alcohols $C_nH_{2n+1}\cdot$OH, then, $n = 1, 2, 3, 4$, the numbers of known alcohols are 1, 1, 2, 4, agreeing with the foregoing theoretic number (see Schorlemmer's Carbon Compounds, 1874); but $n = 5$, the number of known alcohols is $= 2$, instead of the foregoing theoretic number 8.

It is, of course, no objection to the theory that the number of theoretic forms should exceed the number of known compounds; the missing ones may be simply unknown; or they may be only capable of existing under conceivable, but unattained, physical conditions (for instance, of temperature); and if defect from the theoretic number of compounds can be thus accounted for, the theory holds good without modification.

But it is also possible that the diagrams, in order that they may represent chemical compounds, may be subject to some as yet undetermined conditions; viz. in this case the theory would stand good as far as it goes, but would require modification.

587. A Smith's Prize Dissertation.

[FROM THE MESSENGER OF MATHEMATICS, VOL. III. (1874), PP. 1—4.]

Write a dissertation:

On the general equation of virtual velocities.

Discuss the principles of Lagrange's proof of it and employ it [the general equation] to demonstrate the Parallelogram of Forces.

Imagine a system of particles connected with each other in any manner and subject to any geometrical conditions, for instance, two particles may be such that their distance is invariable, a particle may be restricted to move on a given surface, &c. And let each particle be acted upon by a force [this includes the case of several forces acting on the same particle, since we have only to imagine coincident particles each acted upon by a single force].

Imagine that the system has given to it any indefinitely small displacement consistent with the mutual connexions and geometrical conditions; and suppose that for any particular particle the force acting on it is P, and the displacement in the direction of the force (that is, the actual displacement multiplied into the cosine of the angle included between its direction and that of the force P) is $= \delta p$. Then δp is called the virtual velocity of the particle, and the principle of virtual velocities asserts that the sum of the products $P\,\delta p$, taken for all the particles of the system, and for any displacement consistent as above, is $= 0$; say that we have $\Sigma P\,\delta p = 0$.

This is also the general equation of virtual velocities: as to the mode of using it, observe that the displacements δp are not arbitrary quantities, but are in virtue of the mutual connexions and other geometrical conditions connected together by certain linear relations; or, what is the same thing, they are linear functions of certain independent arbitrary quantities δu. Substituting for δp their expressions in terms of δu, we have $\Sigma P\,\delta p = \Sigma U\,\delta u$, where the several expressions U are each of them a linear function of the forces P, and where on the right hand Σ refers to the several quantities δu; and the resulting equation is $\Sigma U\,\delta u = 0$; viz. since the quantities δu are independent, the equation divides itself into a set of equations $U_1 = 0$, $U_2 = 0, \ldots$ which are the equations of equilibrium of the system.

Lagrange imagines the forces produced by means of a weight W at the extremity of a string passing over a set of pulleys, as shown in the figure, viz. assuming the forces commensurable and equal to mW, nW, &c., we must have m strings at A, A'; n strings at B, and so on.

Suppose any indefinitely small displacement given to the system; each string at A is shortened by ...

[The dissertation continues with a discussion of Lagrange's proof and its application to the parallelogram of forces, including consideration of cases with one-sided constraints such as a particle resting on a plane but free to move off in one direction.]

The principle as not brought out with sufficient clearness by Lagrange himself.

Parallelogram of forces. Let P, Q, R be the forces, α, β, γ their inclinations to any line; then taking δs the displacement in the direction of this line, the displacements in the directions of the forces are $\delta s\cos\alpha$, $\delta s\cos\beta$, $\delta s\cos\gamma$, and the equation $\Sigma P\,\delta p = 0$ assumes the form

$$(P\cos\alpha + Q\cos\beta + R\cos\gamma)\,\delta s = 0,$$

that is, we have

$$P\cos\alpha + Q\cos\beta + R\cos\gamma = 0,$$

viz. this equation holds whatever be the fixed line to which the forces are referred. It is easy to see that, supposing it to hold in regard to any two lines, it will hold generally, and that the relation in question is thus equivalent to two independent conditions; and forming these we may obtain from them the theorem of the parallelogram of forces.

588. Problem.

[From the Messenger of Mathematics, vol. iii. (1874), pp. 50—52.]

It is required to place two given tetrahedra in perspective; or, what is the same thing, the tetrahedra being $ABCD$, $A'B'C'D'$ respectively, to place these so that the lines AA', BB', CC', DD' may meet in a point O.

The following considerations present themselves in regard to the solution of this problem. Take the tetrahedron $ABCD$ to be given in position, and the point O at pleasure; then drawing the lines OA, OB, OC, OD, we may in a determinate number of ways (viz. in 16 different ways) place the tetrahedron $A'B'C'D'$ in perspective with $ABCD$, so that the corresponding vertices may lie on these lines respectively; and the like as regards the accented letters.

Then A, B, C, F, G, H are the cosines of the angles which the edges of the tetrahedron $ABCD$ subtend at O; they are consequently the cosines of the six sides of the spherical quadrangle obtained by the projection of $ABCD$ on a sphere centre O; and they are therefore not independent, but are connected by a single equation; substituting for A, B, C, F, G, H their values, we have a relation between a, b, c, f, g, h, x, y, z, w; viz. this is the relation which connects the ten distances of the five points in space O, A, B, C, D.

589. On Residuation in Regard to a Cubic Curve.

[From the Messenger of Mathematics, vol. iii. (1874), pp. 62—65.]

The following investigation of Prof. Sylvester's theory of Residuation may be compared with that given in Salmon's *Higher Plane Curves*, 2nd Edition (1873), pp. 133—137:

If the intersections of a cubic curve U_3 with any other curve V_n are divided in any manner into two systems of points, then each of these systems is said to be the residue of the other; and, in like manner, if starting with a given system of points on a cubic curve we draw through them a curve of any order, then the remaining intersections of the two curves form a system which is the residue of the given system.

If two systems of points are each of them the residue of a third system, they are residues of each other; or, what is the same thing, if two systems have a common residue, they are residues of each other.

To prove this, let A, B, C be the three systems; let $A+B$ denote the complete system of intersections of the cubic by a curve through A and B; and so $B+C$ the complete system of intersections by a curve through B and C. Then $A+B$, $B+C$ are each of them a complete intersection of the cubic by a curve; and hence $A+B+C$ is also a complete intersection of the cubic by a curve; viz. the curve through $A+B$ combined with the curve through C gives the complete system $A+B+C$; and so the curve through $B+C$ combined with the curve through A gives the complete system $B+C+A=A+B+C$; that is, $A+B+C$ is a complete intersection. Hence A is the residue of $B+C$, and C is the residue of $A+B$; that is, A and C are residues of each other.

The particular case most easily understood is when the common residue B reduces itself to a single point; the theorem is that if A and C have each of them a single point for residue, they are residues of each other; viz. if $A+P$ and $C+P'$ are each of them a complete intersection, and $P=P'$, then $A+C$ is a complete intersection.

590. Addition to Prof. Hall's Paper "On the Motion of a Particle toward an Attracting Centre at which the Force is Infinite."

[From the Messenger of Mathematics, vol. iii. (1874), pp. 149—152.]

I do not in the passage referred to expressly profess to interpret Newton's idea. After referring to his investigation I say, "The method has the advantage of explaining the paradoxical result which presents itself in the case force $\propto$ (dist.)$^{-3}$, and in some other cases where the force becomes infinite. According to theory the velocity becomes infinite at the centre, but the direction of the velocity is indeterminate. The method shows that these peculiarities arise from the use of an ideal law of force, and that they disappear when the true law is substituted for it."

The question is as to the statement (p. 34): "It follows from the same equation (3) that if the force vary as the inverse cube of the distance, or in a higher inverse ratio, the body never can arrive at the centre of force, but will continue to approach it indefinitely."

I think this must mean that the body never does arrive at the centre in any finite time, but only in an infinite time; and in fact, if the velocity were infinite and the direction determinate, the body would in a finite time go right through the centre and out on the other side.

591. A Smith's Prize Paper and Dissertation; Solutions and Remarks.

[From the Messenger of Mathematics, vol. iii. (1874), pp. 165—183, vol. iv. (1875), pp. 6—8.]

1. Find the triangular numbers which are also square.

The "mise en équation" is immediate; we have to find n, m such that

$$\tfrac{1}{2}n(n+1) = m^2;$$

or, what is the same thing,

$$(2n+1)^2 - 8m^2 = 1.$$

Observing that this is satisfied by $n = m = 1$, that is, $2n+1 = 3$, $2m = 2$, we have the general solution given by

$$2n+1+2m\surd 2 = (3+2\surd 2)^p,$$

where p is any positive integer; viz. $2n+1$, $2m$ being rational, this implies

$$2n+1-2m\surd 2 = (3-2\surd 2)^p,$$

and thence the equation in question.

The successive powers $3+2\surd 2$, $17+12\surd 2$, $99+70\surd 2$, &c., give the solutions $n, m = 1, 1$; $8, 6$; $49, 35$, &c.; viz. the square triangular numbers are $1^2 = \frac{1}{2}\cdot 1\cdot 2$; $6^2 = \frac{1}{2}\cdot 8\cdot 9$; $35^2 = \frac{1}{2}\cdot 49\cdot 50$, &c.

2. Show how to express any symmetrical function of the roots of an equation in terms of the coefficients. What objection is there to the method which employs the sums of the powers of the roots?

The ordinary method is that referred to, employing the sums of the powers of the roots; but it is a very bad one. In fact, writing

$$x^n - bx^{n-1} + cx^{n-2} - \&\text{c.} = (x-\alpha)(x-\beta)(x-\gamma)\ldots = 0,$$

leading to

$$S_1 = b, \quad S_2 = b^2 - 2c, \quad S_3 = b^3 - 3bc + 3d,$$

then if the method were employed throughout, we should have for instance to find $\Sigma\alpha^2\beta\gamma$, that is, d, from the formula

$$6\Sigma\alpha\beta\gamma = S_1^3 - 3S_1S_2 + 2S_3,$$

that is,

$$6d = b^3-3b(b^2-2c)+2(b^3-3bc+3d) = b^3-3b^3+6bc+2b^3-6bc+6d = 6d,$$

which is a mere identity, and gives no information. The objection is that the method leads to identities rather than to the actual values of the symmetric functions.

3. If A, B, C, D be the coefficients of a binary quartic, find the discriminant.

The discriminant of the binary quartic $(a, b, c, d, e \,|\, x, y)^4$ is

$$a^3e^3 + \&\text{c.},$$

a complicated expression of degree 6 in the coefficients. It may be expressed in terms of the invariants I and J; viz. the discriminant is $= I^3 - 27J^2$.

4. Find the differential equation of the curves having a common tangent with a given curve.

Taking the given curve to be $U = 0$, and considering the first and second derived equations, we obtain the required differential equation by eliminating the constants between these equations. For a curve $U = 0$ with parameters a, b, &c., the envelope of the curves obtained by varying the parameters is found from the equations $U = 0$, $\frac{dU}{da} = 0$, $\frac{dU}{db} = 0$, &c.

5. Find the locus of the vertex of a triangle of given species whose base slides along a fixed line.

If the base slide along a fixed line, the locus is a curve of the third order. The problem may be solved by considering the angles which the sides make with the base, and the ratios of the sides, as given quantities.

6. Integrate the differential equation

$$\frac{dy}{dx} + y^2 = \frac{6}{x^2}.$$

This is a particular case of Riccati's equation. Assuming $y = \frac{1}{z}\frac{dz}{dx}$, we obtain

$$\frac{d^2z}{dx^2} = \frac{6z}{x^2},$$

which is satisfied by $z = x^m$, where $m(m-1) = 6$, that is, $m = 3$ or $m = -2$.

Hence the complete integral is $z = Ax^3 + Bx^{-2}$, giving

$$y = \frac{3Ax^3 - 2Bx^{-2}}{Ax^3 + Bx^{-2}} = \frac{3Ax^5 - 2B}{Ax^5 + Bx^2}.$$

7. Find the value of

$$\int_0^{\frac{\pi}{2}} \frac{d\theta}{\sqrt{(1 - c^2 \sin^2\theta)}}.$$

This is the complete elliptic integral of the first kind, $F(c)$. It cannot be expressed in terms of elementary functions, but may be evaluated by series expansion or by Landen's transformation.

8. Find the equations of equilibrium of a rigid body under any system of forces.

The equations are:

(a) $\Sigma X = 0$, $\Sigma Y = 0$, $\Sigma Z = 0$ (resolved parts);

(b) $\Sigma L = 0$, $\Sigma M = 0$, $\Sigma N = 0$ (moments about three axes).

9. Find the moment of inertia of a triangular lamina about an axis through its centre of gravity perpendicular to its plane.

For a triangle of mass M and sides a, b, c, the moment of inertia about an axis through the centroid perpendicular to the plane is

$$I = \frac{M}{36}(a^2 + b^2 + c^2).$$

10. Find the attraction of a uniform sphere on an external particle.

The attraction is the same as if the whole mass were collected at the centre; viz. if M be the mass, r the distance of the particle from the centre, the attraction is $\frac{M}{r^2}$.

11. Find the differential equation of the orthogonal trajectories of a given family of curves.

If the given family be $f(x, y, c) = 0$, the differential equation is obtained by eliminating c between $f = 0$ and $\frac{dy}{dx} = -\frac{f_x}{f_y}$. For the orthogonal trajectories, we replace $\frac{dy}{dx}$ by $-\frac{dx}{dy}$.

12. Show that the six coordinates of a line satisfy a quadratic relation.

If the line be determined as the join of two points (x, y, z, w) and (x', y', z', w'), the six coordinates are

$$a = yz' - y'z, \quad b = zx' - z'x, \quad c = xy' - x'y,$$

$$f = xw' - x'w, \quad g = yw' - y'w, \quad h = zw' - z'w.$$

These satisfy the relation

$$af + bg + ch = 0.$$

13. Find the condition that four given lines may belong to the same hyperboloid.

The condition may be expressed by means of the six coordinates of each line. If the lines be $(a_1, b_1, c_1, f_1, g_1, h_1)$, &c., then the condition is that a certain determinant formed from these coordinates shall vanish.

14. Find the singular solution of the differential equation

$$(xp - y)^2 = a^2(1 + p^2),$$

where $p = \frac{dy}{dx}$.

The complete integral represents a family of straight lines. The singular solution is obtained by eliminating p between the equation and its derivative with respect to p. We find

$$x^2 + y^2 = a^2,$$

which represents a circle.

15. Explain the distinction between a singular solution and a particular integral.

A singular solution is a solution not obtainable from the complete integral by assigning any particular value to the arbitrary constant; it represents the envelope of the family of curves given by the complete integral. A particular integral is obtained by giving a particular value to the arbitrary constant in the complete integral.

The case of six lines is one the answer to which could not have been discovered in an examination; the relations in fact are that the six lines shall form an involution; viz. this is a system such that taking five of the lines as given, then if the sixth line belongs to the system, it is uniquely determined.

16. Find the value of the definite integral

$$\int_0^\infty \frac{\sin x}{x}\, dx.$$

The value is $\frac{\pi}{2}$. This may be proved by various methods; for instance, by considering $\int_0^\infty e^{-\alpha x} \frac{\sin x}{x}\, dx$, differentiating with respect to α, and then integrating with respect to α from $\alpha = 0$ to $\alpha = \infty$.

17. Prove that the equation $x^n - 1 = 0$ has n roots.

By De Moivre's theorem, the roots are $x = \cos \frac{2k\pi}{n} + i \sin \frac{2k\pi}{n}$, for $k = 0, 1, 2, \ldots, n-1$.

18. Find the equation of a sphere cutting four given spheres at right angles.

If the four given spheres be $S_1 = 0$, $S_2 = 0$, $S_3 = 0$, $S_4 = 0$, the required sphere is $lS_1 + mS_2 + nS_3 + pS_4 = 0$, where l, m, n, p are determined by the condition of orthogonal intersection.

19. Find the condition that the conic $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ may represent two straight lines.

The condition is

$$\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0.$$

20. Find the equation of the surface generated by a line which moves so as always to intersect three given lines.

The surface is a hyperboloid of one sheet (or a hyperbolic paraboloid if the three lines are parallel to one plane). The equation may be found by expressing that the moving line intersects each of the three given lines, and eliminating the parameters.

592. On the Mercator's-Projection of a Skew Hyperboloid of Revolution.

[From the Messenger of Mathematics, vol. iv. (1875), pp. 17—20.]

In a note "On the Mercator's-projection of a surface of revolution" read before the British Association, [555, (5)], I remarked that the surface might be, by its meridians and parallels, divided into infinitesimal squares; and that these would be on the map represented by two systems of parallel lines at right angles to each other, dividing the map into infinitesimal squares; and that, by taking the squares not infinitesimal but small, for instance, $1^\circ \times 1^\circ$, we should obtain a good approximate construction.

In the case of the skew hyperboloid of revolution, the meridians and parallels may be defined as follows. The surface is generated by a line meeting two fixed lines (the axes) at right angles. Taking any point on the surface, the meridian through it is the curve cut out by a plane through the axis of revolution; the parallel through it is the circle described by the point in its rotation about the axis.

The Mercator's-projection is obtained by representing the meridians and parallels as two systems of parallel lines at right angles to each other, the scale being such that the ratio of the distances between consecutive meridians and consecutive parallels is equal to the ratio of the corresponding arcs on the surface.

593. A Sheepshanks' Problem (1866).

[From the Messenger of Mathematics, vol. iv. (1875), pp. 34—36.]

Apply the formulæ of elliptic motion to determine the motion of a body let fall from the top of a tower at the equator.

The earth is regarded as rotating with the angular velocity ω round a fixed axis, so that the body is in fact projected from the apocentre with an angular velocity $= \omega$; and we write a for the equatorial radius, β for the height of the tower; then g denoting the force of gravity, and μ, h, n, α, e, θ, as in the theory of elliptic motion, we have

$$\mu = n^2 a^3 = g a^2, \quad h = n a^2 \omega,$$

and thence

$$n\alpha^{\frac{1}{2}} = a^{\frac{3}{2}}\omega, \quad e^2 = 1 - \frac{\omega^2}{n^2}\left(1 + \frac{\beta}{a}\right)^2.$$

The motion is thus completely determined; the body will describe an ellipse about the centre of the earth, and the time of falling to the surface may be calculated by the formulæ of elliptic motion.

594. On a Differential Equation in the Theory of Elliptic Functions.

[From the Messenger of Mathematics, vol. iv. (1875), pp. 69, 70.]

The following equation presented itself to me in connexion with the cubic transformation:

$$\frac{dQ}{dk} \cdot Q^2 - Q(k^4+1)^2 = 3(1-k^2)^2.$$

Writing as usual $k^2 = u^4$, I was aware that a solution was

$$Q = u^4 + 2uv,$$

where u, v are connected by the modular equation

$$u^4 - v^4 + 2uv(1-u^2v^2) = 0;$$

but it was no easy matter to verify that the differential equation was satisfied.

After a different solution, it occurred to me to obtain the relation between (Q, u); or, what is the same thing, to express Q as a function of u alone. The modular equation gives v as an algebraic function of u, and thence Q as an algebraic function of u.

The verification of the differential equation is effected by means of the theory of elliptic functions; the equation is found to be satisfied identically when the proper relations between the variables are established.

595. On a Senate-House Problem.

[From the Messenger of Mathematics, vol. iv. (1875), pp. 75—78.]

The following was given [5 Jan., 1874,] as a problem of elementary algebra:

"Solve the equations

$$x(2a-y)=y(2a-z)=z(2a-w)=w(2a-x)=b^2,$$

and prove that unless $b^2=2a^2$, $x=y=z=w$, but that if $b^2=2a^2$, the equations are not independent."

This is really a very remarkable theorem in regard to the intersections of a certain set of four quadric surfaces in four-dimensional space; viz. slightly altering the notation, we may write the equations in the form

$$x(2a-y)=b^2, \tag{1}$$
$$y(2a-z)=b^2, \tag{2}$$
$$z(2a-w)=b^2, \tag{3}$$
$$w(2a-x)=b^2, \tag{4}$$

From (1) and (2), we have $x=\frac{b^2}{2a-y}$, $y=\frac{b^2}{2a-z}$, whence $x=\frac{b^2}{2a-\frac{b^2}{2a-z}}$; and similarly for the other variables. Eliminating y, z, w successively, we obtain an equation for x; and similarly for each of the other variables.

The solution $x=y=z=w$ is obtained by observing that if $x=y=z=w$, then from any one of the equations $x(2a-x)=b^2$, that is, $x^2-2ax+b^2=0$, giving $x=a\pm\sqrt{(a^2-b^2)}$.

When $b^2=2a^2$, the equations are not independent; there is a one-parameter family of solutions. The condition $b^2=2a^2$ makes the four quadric surfaces have a common curve of intersection.

Writing the equations in the form

$$\frac{x}{b^2}=\frac{1}{2a-y},\quad \frac{y}{b^2}=\frac{1}{2a-z},\quad \frac{z}{b^2}=\frac{1}{2a-w},\quad \frac{w}{b^2}=\frac{1}{2a-x},$$

and multiplying, we obtain

$$\frac{xyzw}{b^8}=\frac{1}{(2a-x)(2a-y)(2a-z)(2a-w)}.$$

The complete theory involves the consideration of the four quadric surfaces in four-dimensional space, and their complete intersection.

596. Note on a Theorem of Jacobi's for the Transformation of a Double Integral.

[From the Messenger of Mathematics, vol. iv. (1875), pp. 92—94.]

Jacobi, in the Memoir "De Transformatione Integralis Duplicis..." &c., Crelle, t. viii. (1832) pp. 253—279 and 321—357, [Ges. Werke, t. iii., pp. 91—158], after establishing a theorem which includes the addition-theorem of elliptic functions, viz. this last is "the differential equation

$$\frac{d\eta}{\sqrt{(G'\cos^2\eta + G''\sin^2\eta - G)}} = \frac{d\theta}{\sqrt{(G'''\cos^2\theta + G''''\sin^2\theta - G)}}$$

has for its complete integral $G + G'\cos\eta\cos\theta + G''\sin\eta\sin\theta = 0$," (observe, as to the integral being complete, that the differential equation contains only the constant $G'''' - G' + (G'' - G''')$, whereas the integral equation contains the two constants $G' \div G$ and $G'' \div G$), obtains a corresponding theorem for double integrals; viz. this, in the corresponding special case, is as follows:

If the variables (ϕ, ψ) and (η, θ) are connected by the two equations

$$\begin{aligned}
\alpha &= \beta + \alpha'\cos\phi\cdot\cos\eta + \alpha''\sin\phi\cos\psi\cdot\sin\eta\cos\theta + \alpha'''\sin\phi\sin\psi\cdot\sin\eta\sin\theta,\\
0 &= \beta' + \beta''\cos\phi\cdot\cos\eta + \beta'''\sin\phi\cos\psi\cdot\sin\eta\cos\theta + \beta''''\sin\phi\sin\psi\cdot\sin\eta\sin\theta,
\end{aligned}$$

and if putting for shortness

$$\begin{gathered}
\alpha''\beta''' - \alpha'''\beta'' = f, \qquad \alpha'''\beta' - \alpha'\beta''' = g, \qquad \alpha'\beta'' - \alpha''\beta' = h,\\
\alpha\beta''' - \alpha'''\beta = a, \qquad \alpha''\beta - \alpha\beta'' = b, \qquad \alpha\beta'' - \alpha''\beta = c,
\end{gathered}$$

(whence $af + bg + ch = 0$);

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596.

ON JACOBI'S TRANSFORMATION OF A DOUBLE INTEGRAL.

[From the *Messenger of Mathematics*, vol. IV. (1875), pp. 89–92.]

JACOBI, in the Memoir "De Transformatione Integralis Duplicis $\iint A\,d\partial\,d\psi$ in formam $\iint B\,dt\,d\theta$ transformanda," *Crelle*, t. XVI. (1837), pp. 21–25, [*Jacobi's Ges. Werke*, t. III., pp. 91–97], considers the transformation of

$$\begin{aligned}R^2 = {} & f^2(\sin\phi\cos\psi)^2(\sin\phi\sin\psi)^2 + g^2(\sin\phi\cos\psi)^2(\cos\phi)^2 + h^2(\cos\phi)^2(\sin\phi\cos\psi) \\ & - a^2(\cos\phi)^2 - b^2(\sin\phi\cos\psi)^2 - c^2(\sin\phi\sin\psi)^2,\end{aligned}$$

$$\begin{aligned}S^2 = {} & f^2(\sin\eta\cos\theta)^2(\sin\eta\sin\theta)^2 + g^2(\sin\eta\sin\theta)^2(\cos\eta)^2 + h^2(\cos\eta)^2(\sin\eta\cos\theta)^2 \\ & - a^2(\cos\eta)^2 - b^2(\sin\eta\cos\theta)^2 - c^2(\sin\eta\sin\theta)^2,\end{aligned}$$

and obtains the theorem:

If

$$\frac{\sin\phi\, d\phi\, d\psi}{R} = \frac{\sin\eta\, d\eta\, d\theta}{S},$$

then we have

$$\begin{aligned} f^2 &= \alpha\beta''\gamma' - \alpha\beta'\gamma'', \\ g^2 &= \alpha'\beta\gamma'' - \alpha'\beta''\gamma, \\ h^2 &= \alpha''\beta'\gamma - \alpha''\beta\gamma', \end{aligned}$$

&c.

And it may be added that the integral equations are, so to speak, a complete integral of the differential relation; viz. in virtue of the identity $af + bg + ch = 0$, the differential relation contains really only four constants; the integral relations contain the six constants $\alpha : \alpha' : \alpha'' : \alpha'''$ and $\beta : \beta' : \beta'' : \beta'''$, or we have two constants introduced by the integration.

The best form of statement is, in the first theorem, to write x, y for $\cos\eta$, $\sin\eta$ $(x^2+y^2 = 1)$, ξ, η for $\cos\theta$, $\sin\theta$ $(\xi^2+\eta^2 = 1)$, and similarly in the second theorem to introduce the variables x, y, z connected by $x^2 + y^2 + z^2 = 1$, and ξ, η, ζ connected by $\xi^2 + \eta^2 + \zeta^2 = 1$; then in the first theorem $d\xi\eta$, $d\theta$ represent elements of circular arc, and in the second theorem $\sin\phi\, d\phi\, d\psi$ and $\sin\eta\, d\eta\, d\theta$ represent elements of spherical surface, and the theorems are:

I. If (x, y) are coordinates of a point on the circle $x^2 + y^2 = 1$, and (ξ, η) coordinates of a point on the circle $\xi^2 + \eta^2 = 1$, and if ds, $d\sigma$ are the corresponding circular elements, then

$$\frac{ds}{d\sigma} = \frac{\sqrt{(ax^2 + by^2 - c)}}{\sqrt{(a\xi^2 + b\eta^2 - c)}}$$

has for its complete integral $ax\xi + by\eta - c = 0$.

II. If (x, y, z) are coordinates of a point on the sphere $x^2+y^2+z^2 = 1$, and (ξ, η, ζ) coordinates of a point on the sphere $\xi^2 + \eta^2 + \zeta^2 = 1$; and if ds, $d\sigma$ are the corresponding spherical elements, and

$$\begin{aligned} \beta\gamma' - \beta'\gamma &= f, & \alpha\delta' - \alpha'\delta &= a, \\ \gamma\alpha' - \gamma'\alpha &= g, & \beta\delta' - \beta'\delta &= b, \\ \alpha\beta' - \alpha'\beta &= h, & \gamma\delta' - \gamma'\delta &= c, \end{aligned}$$

(whence $af + bg + ch = 0$); and for shortness

$$R = \sqrt{(f^2y^2z^2 + g^2z^2x^2 + h^2x^2y^2 - a^2x^2 - b^2y^2 - c^2z^2)},$$
$$S = \sqrt{(f^2\eta^2\zeta^2 + g^2\zeta^2\xi^2 + h^2\xi^2\eta^2 - a^2\xi^2 - b^2\eta^2 - c^2\zeta^2)},$$

then the differential relation

$$\frac{\sin\phi\,d\phi\,d\psi}{R} = \frac{\sin\eta\,d\eta\,d\theta}{S}$$

has for its complete integral the system

$$\alpha x\xi + \beta y\eta + \gamma z\zeta + \delta = 0,$$
$$\alpha' x\xi + \beta' y\eta + \gamma' z\zeta + \delta' = 0,$$

where by complete integral is meant a system of two equations containing two arbitrary constants.

597.

ON A DIFFERENTIAL EQUATION IN THE THEORY OF ELLIPTIC FUNCTIONS.

[From the *Messenger of Mathematics*, vol. IV. (1875), pp. 110–113.]

The differential equation

$$(2-Q^2)\frac{d^2Q}{dk^2} - 3Q\left(\frac{dQ}{dk}\right)^2 = 3(1-k^2)\frac{dQ}{dk},$$

considered ante, p. 69, [594, this volume, p. 244], belongs to a class of equations transformable into linear equations of the second order, and consequently is such that, knowing a particular solution, we can obtain the general solution.

In fact, assuming

$$z = \frac{1}{\sqrt{Q}}\exp.\left(-\frac{1}{2}\int\frac{Q}{1-k^2}\,dk\right),$$

the equation becomes

$$(2-Q^2)\frac{d^2z}{dk^2} = 3(1-k^2)\left\{3(1-k^2)\frac{dz}{dk} + 6z - 3(1-k^2)z\frac{Q'}{Q}\right\},$$

viz. omitting the terms in $\left(\dfrac{dz}{dk}\right)^2$ which destroy each other, and dividing by $3(1-k^2)$, this is

$$\frac{(2-Q^2)}{3(1-k^2)}\frac{d^2z}{dk^2} = 3(1-k^2)\frac{dz}{dk} + 6z - 3(1-k^2)z\frac{Q'}{Q},$$

or finally

$$\frac{d^2z}{dk^2} + \frac{2}{Q}\frac{dQ}{dk}\frac{dz}{dk} + \frac{z}{(1-k^2)}\frac{d^2Q}{dk^2} + \frac{3z}{(1-k^2)}\frac{dQ}{dk} = 0.$$

But knowing a particular value of Q we have

$$z = \exp.\left(-\frac{1}{2}\int\frac{Q}{1-k^2}\,dk\right),$$

a particular value of z, and thence in the ordinary manner the general value of z, giving the general value of Q.

The solution given in my former paper may be exhibited in a more simple form by introducing, instead of k, the variable α connected with it by the equation

$$k^2 = \frac{\alpha^2(2+\alpha)}{1+2\alpha}.$$

We have in fact, *Fundamenta Nova*, p. 25, [Jacobi's *Ges. Werke*, t. I., p. 76],

$$u = \frac{2+\alpha}{1+2\alpha}, \qquad v = \frac{2+\alpha}{\alpha(1+2\alpha)},$$

viz. these expressions of u, v in terms of the parameter α, are equivalent to, and replace, the modular equation

$$u^4 - v^4 + 2uv(1-u^2v^2) = 0.$$

We thence obtain

$$\frac{(1+2\alpha)^2}{\alpha^2(2+\alpha)^2} = \frac{u^2}{v^2},$$

that is,

$$uv = \frac{\alpha\sqrt{(1+2\alpha)}}{\sqrt{(2+\alpha)}}, \qquad \frac{u}{v} = \frac{(2+\alpha)}{\alpha(1+2\alpha)},$$

$$Q' - Q = \frac{1}{uv}, \qquad Q' + Q = 2uv(1+u^2v^2),$$

and the particular solution, $Q = \dfrac{u}{v} + 2uv$, becomes

$$Q = \frac{1}{\alpha}\sqrt{(2+\alpha)(1+2\alpha)}.$$

Introducing into the differential equation α in place of k, this is found to be

$$\{4\alpha^2(2+\alpha)^2 - Q^2\}\left\{\frac{d^2Q}{d\alpha^2} + \frac{3(1+2\alpha)}{\alpha(2+\alpha)}\frac{dQ}{d\alpha}\right\} - 3Q\left(\frac{dQ}{d\alpha}\right)^2 = 0.$$

But from this form it at once appears that it is convenient in place of α to introduce the new variable β, $= \alpha + \dfrac{1}{\alpha}$; the equation thus becomes

$$4Q^2\frac{d^2Q}{d\beta^2} + \frac{dQ}{d\beta}(3 + 6\beta Q^2 - Q^4) - 12 = (\beta^2-1)(\beta^2-9)Q^2,$$

satisfied by $Q = \sqrt{(5+2\beta)}$; or, what is the same thing, writing $5 + 2\beta = \gamma^2$, that is, $\beta = -\dfrac{5}{2} + \dfrac{\gamma^2}{2}$, the equation becomes

$$4Q^2 \frac{d^2Q}{d\gamma^2} + \frac{1}{\gamma}\frac{dQ}{d\gamma}(3 + 6\gamma^2 Q^2 - \gamma^4 Q^4) - 12\gamma^2 = (\gamma^2 - 1)(\gamma^2 - 9)\gamma^2 Q^2,$$

satisfied by $Q = \gamma$.

Writing here

$$Q = \frac{1}{2}(\gamma^2-1)(\gamma^2-9)z^2,$$

we have for z the equation

$$\frac{d^2z}{d\gamma^2} + \frac{3\gamma^2-14}{\gamma(\gamma^2-1)(\gamma^2-9)}\frac{dz}{d\gamma} + \frac{24\gamma^2}{\{(\gamma^2-1)(\gamma^2-9)\}^2}z = 0,$$

satisfied by

$$z = \left(\frac{\gamma^2-9}{\gamma^2-1}\right)^{\frac{1}{2}}.$$

[In fact, this value gives

$$\frac{dz}{d\gamma} = (\gamma^2-9)^{-\frac{1}{2}}(\gamma^2-1)^{-\frac{3}{2}},$$

$$\frac{d^2z}{d\gamma^2} = (-12\gamma^3+57\gamma^2+36)(\gamma^2-9)^{-\frac{3}{2}}(\gamma^2-1)^{-\frac{5}{2}},$$

which verify the equation as they should do.]

Representing for a moment the differential equation by

$$A\frac{d^2z}{d\gamma^2} + B\frac{dz}{d\gamma} + Cz = 0$$

and putting $z = z_1\int y\,d\gamma$, then assuming $z = z_1\int y\,d\gamma$, we find

$$Az_1\frac{dy}{d\gamma} + y\left(2A\frac{dz_1}{d\gamma} + Bz_1\right) = 0,$$

that is,

$$\frac{1}{y}\frac{dy}{d\gamma} + \frac{2}{z_1}\frac{dz_1}{d\gamma} + \frac{B}{A} = 0,$$

viz.

$$\frac{1}{y}\frac{dy}{d\gamma} + \frac{2}{z_1}\frac{dz_1}{d\gamma} + \frac{3\gamma^2-14}{\gamma(\gamma^2-1)(\gamma^2-9)} = 0,$$

or

$$\frac{1}{y}\frac{dy}{d\gamma} + \frac{2}{z_1}\frac{dz_1}{d\gamma} - \frac{3\gamma}{\gamma^2} - \frac{1}{2}\frac{2\gamma}{\gamma^2-1} + \frac{3}{2}\frac{2\gamma}{\gamma^2-9} = 0,$$

whence, integrating

$$\log y + 2\log z_1 - 3\log\gamma - \tfrac{1}{2}\log(\gamma^2-1) + \tfrac{3}{2}\log(\gamma^2-9) = 0,$$

that is,

$$y = \frac{1}{z_1^2}\frac{\gamma^3\sqrt{(\gamma^2-1)}}{(\gamma^2-9)^{\frac{3}{2}}} = \frac{(\gamma+1)(\gamma+3)}{(\gamma-3)^2},$$

$$\int y\,d\gamma = \gamma + \log\frac{(\gamma+1)^2(\gamma+3)^2}{(\gamma-3)^2}.$$

Hence, the general value of z is

$$z = \left(\frac{\gamma^2 - 9}{\gamma^2 - 1}\right)^{\frac{1}{2}} \left\{\gamma + K + \log\frac{(\gamma+1)^2(\gamma+3)^2}{(\gamma-3)^2}\right\},$$

the constants of integration being K and γ_0, or, what is the same the corresponding value of Q being

$$Q = \frac{1}{2}(\gamma^2 - 1)(\gamma^2 - 9)z^2,$$

which contains the single arbitrary constant K; when this vanishes, we have the foregoing particular solution $Q = \gamma$.

I recall that the expression of γ is

$$\gamma = \sqrt{(5 + 2\beta)}, = \sqrt{5 + 2\left(\alpha + \frac{1}{\alpha}\right)},$$
$$= \frac{1}{\sqrt{\alpha}}\sqrt{\{(2+\alpha)(1+2\alpha)\}},$$

where α is connected with k by the relation

$$\frac{\alpha^2(2+\alpha)}{1+2\alpha} = k^2.$$

598.

NOTE ON A PROCESS OF INTEGRATION.

[From the *Messenger of Mathematics*, vol. IV. (1875), pp. 149, 150.]

I had occasion to consider the integral

$$\int_0^R \frac{r^{q-1}\,dr}{(r^2+e^2)^{\frac{1}{2}q+\delta}},$$

where e is small in regard to R and q is negative. The integral is finite when $e=0$, and it might be imagined that it could be expanded in positive powers of e; and, assuming it to be thus expansible, that the process would simply be to expand under the integral sign in ascending powers of e, and integrate each term separately, so that the series would be in integer powers of e^2.

Take two particular cases. First, let $\delta=2$, $q=-\frac{4}{3}$; the integral is

$$\int_0^R \frac{r^{-\frac{7}{3}}\,dr}{(r^2+e^2)^{\frac{1}{3}}},$$

which, putting $r=e\tan\theta$, becomes

$$\frac{1}{e^{\frac{8}{3}}}\int_0^{\tan^{-1}\frac{R}{e}} \frac{\cos^{\frac{5}{3}}\theta\,d\theta}{\sin^{\frac{7}{3}}\theta},$$

and this, putting $\cos\theta=t^{\frac{3}{2}}$, becomes

$$\frac{3}{2e^{\frac{8}{3}}}\int \frac{t\,dt}{(1-t^3)^{\frac{5}{3}}},$$

viz. the integral is not thus obtainable: the series is right as far as it goes, but the true expansion contains a term in e^2; and the failure of the series to give the true expansion is indicated by the appearance of infinite coefficients. In fact, the indefinite integral is $\frac{3}{2}(r^2+e^2)^{\frac{1}{3}}$; taking this between the limits, it is ...

Again, let $\delta=1$, $q=-2$; the integral is

$$\int_0^R \frac{r^{-3}\,dr}{(r^2+e^2)^{\frac{1}{2}}},$$

which gives similarly

$$\int_0^R \frac{r^{-3}\,dr}{(r^2+e^2)^{\frac{1}{2}}} = \int \left(r^{-4} - \tfrac{1}{2}e^2 r^{-6} + \ldots\right) dr,$$

viz. the integral is not thus obtainable: the series is right as far as it goes, but the true expansion contains a term as $e^4 \log e$, and the failure is indicated by the infinite coefficients. In fact, the indefinite integral is

$$\left(\tfrac{1}{2}r^{-2} + \tfrac{1}{2}e^2 r^{-4}\right)\sqrt{(r^2+e^2)} + \tfrac{3}{2}e^4 \log\{r + \sqrt{(r^2+e^2)}\},$$

which between the limits is

$$\left(\tfrac{1}{2}R^{-2} + \tfrac{1}{2}e^2 R^{-4}\right)\sqrt{(R^2+e^2)} + \tfrac{3}{2}e^4 \log\frac{R+\sqrt{(R^2+e^2)}}{e} = \tfrac{1}{2}R^{-1} + \tfrac{1}{2}e^2 R^{-2} + \ldots - \tfrac{3}{2}e^4$$

In the general case, the term causing the failure is $Ke^{-\frac{2}{q}}$ when q is fractional, and $Ke^{-\frac{2}{q}} \log e$ when q is integral.

As a step towards determining the entire expansion, I notice that, writing $\omega = \dfrac{e}{r+e}$ or $r = e\omega(1-\omega)^{-1}$, the value of the integral is

$$= \frac{1}{e^{\delta}} \int_0^{\frac{R}{R+e}} \omega^{q-1}(1-\omega)^{\frac{1}{2}q-\delta}\,d\omega,$$

where ...

599.

A SMITH'S PRIZE DISSERTATION.

[From the *Messenger of Mathematics*, vol. IV. (1875), pp. 157–160.]

Write a dissertation on Bernoulli's Numbers and their use in Analysis.

The function

$$\frac{t}{e^t-1}+\frac{1}{2}t$$

is an even function of t, as appears by expressing it in the form

$$\frac{1}{2}t\frac{e^{\frac{1}{2}t}+e^{-\frac{1}{2}t}}{e^{\frac{1}{2}t}-e^{-\frac{1}{2}t}},$$

and its value for $t=0$ being obviously $=1$, we may write

$$\frac{t}{e^t-1}=1-B_1t^2+B_2t^4-B_3t^6+\&\text{c.};$$

or, what is the same thing,

$$\frac{t}{e^t-1}=1-\frac{B_1}{1.2}t^2+\frac{B_2}{1.2.3.4}t^4-\frac{B_3}{1.2.3.4.5.6}t^6+\&\text{c.},$$

where the several coefficients B_1, B_2, B_3, &c., are, as is at once seen, rational fractions, and, as it may be shown, are all of them positive. These numerical coefficients B_1, B_2, B_3, &c., are called Bernoulli's numbers.

There is no difficulty in calculating directly the first few terms; viz. we have

$$\left(1+\tfrac{1}{2}t+\tfrac{1}{6}t^2+\tfrac{1}{24}t^3+\tfrac{1}{120}t^4+\ldots\right)\left(1-\tfrac{B_1}{1.2}t^2+\tfrac{B_2}{1.2.3.4}t^4-\ldots\right)=1,$$

that is

$$1+t\left(-\tfrac{1}{2}\right)+t^2\left(-\tfrac{1}{2}\cdot\tfrac{1}{2}+\tfrac{1}{6}\right)+t^3\left(\tfrac{1}{2}\cdot\tfrac{B_1}{2}-\tfrac{1}{2}\cdot\tfrac{1}{6}+\tfrac{1}{24}\right)+\ldots=1,$$

viz.

$$B_1=\frac{1}{6},\quad 1-3B_1+B_2=0,\quad \&\text{c.},$$

which is therefore

$$B_1=\frac{1}{6},\quad B_2=\frac{1}{30},\quad B_3=\frac{1}{42},$$

and so a few more terms might have been found.

But a more convenient method is to express the numbers in terms of the differences of 0^m by means of a general formula for the expansion of a function of e^t, viz. this is

$$\phi(e^t) = \phi(1+\Delta)e^{0.t},$$

where

$$e^{0.t} = 1 + 0.t + \frac{0^2.t^2}{1.2} + \frac{0^3.t^3}{1.2.3} + \dots,$$

and the $\phi(1+\Delta)$ is to be applied to the terms 0^0, 0^1, 0^2, 0^3, &c. We have thus

$$\frac{t}{e^t-1} = \frac{t}{e^t-1}\cdot\frac{\log e^t}{\log e^t} = \frac{\log(1+\Delta)}{\Delta}e^{0.t},$$

$$\frac{t}{e^t-1} = \frac{\log(1+\Delta)}{\Delta}\left(1 + 0.t + \frac{0^2.t^2}{1.2} + \dots\right).$$

We have, as may be at once verified,

$$\frac{\log(1+\Delta)}{\Delta} = 1 - \tfrac{1}{2}\Delta + \tfrac{1}{3}\Delta^2 - \dots + \frac{2B_n}{1.2\dots 2n}\Delta^{2n},$$

and by what precedes, since the coefficient of every higher odd power of t vanishes,

$$\frac{\log(1+\Delta)}{\Delta} = 1 - \tfrac{1}{2}\Delta + \tfrac{1}{3}\Delta^2 - \dots;$$

and then, by comparing the even powers of t, that is,

$$(-)^{n-1}B_n = \frac{\log(1+\Delta)}{\Delta}0^{2n},$$

$$(-)^{n-1}B_n = \left(1 - \tfrac{1}{2}\Delta + \tfrac{1}{3}\Delta^2 - \dots + \frac{2B_n}{1.2\dots 2n}\Delta^{2n}\right)0^{2n},$$

the series for $\dfrac{\log(1+\Delta)}{\Delta}$ being stopped at this point since $\Delta^{2n+1}0^{2n} = 0$, &c.

For instance, in the case $n = 1$, we have

$$B_1 = \left(1 - \tfrac{1}{2}\Delta + \tfrac{1}{6}\Delta^2\right)0^2 = 0'^2 - \tfrac{1}{2}(1^2 - 0^2) + \tfrac{1}{6}(2^2 - 2.1^2 + 0^2) = -\tfrac{1}{2} + \tfrac{2}{3} = \tfrac{1}{6},$$

as above.

The formula shows, not only that B_n is a rational fraction but that its denominator is at most = least common multiple of the numbers 2, 3, ..., $2n+1$; the actual denominator of the fraction in its least terms is, however, much less than this, there being as to its value a theorem known as Staudt's theorem.

It does not obviously show that the Numbers are positive, or afford any indication of the rate of increase of the successive terms of the series. These last requirements are satisfied by an expression for B_n as the sum of an infinite numerical series, which expression is obtained by means of the function $\cot\theta$, as follows:

We have

$$\frac{te^{\frac{1}{2}t}}{e^t-1} = \frac{\frac{1}{2}t(e^{\frac{1}{2}t}+e^{-\frac{1}{2}t})}{e^{\frac{1}{2}t}-e^{-\frac{1}{2}t}} = 1 + \frac{B_1}{1.2}t^2 - \frac{B_2}{1.2.3.4}t^4 + \dots,$$

or, writing herein $t = 2i\theta$ ($i = \sqrt{(-1)}$ as usual), this is

$$\frac{i\theta\cos\theta}{\sin\theta} = 1 - \frac{2^2 B_1}{1.2}\theta^2 + \frac{2^4 B_2}{1.2.3.4}\theta^4 - \dots.$$

But we have

$$\theta\cot\theta = 1 - \frac{B_1}{1.2}\theta^2 - \frac{B_2}{1.2.3.4}\theta^4 - \dots;$$

$$\log\sin\theta = \log\theta + \log\left(1-\frac{\theta^2}{\pi^2}\right) + \log\left(1-\frac{\theta^2}{2^2\pi^2}\right) + \dots,$$

and thence, by differentiation,

$$\cot\theta = \frac{1}{\theta} - \frac{2\theta}{\pi^2-\theta^2} - \frac{2\theta}{4\pi^2-\theta^2} - \dots;$$

hence

$$\theta\cot\theta = 1 - 2\theta^2\left(\frac{1}{\pi^2-\theta^2} + \frac{1}{4\pi^2-\theta^2} + \dots\right),$$

that is,

$$B_n = \frac{2(1.2\dots 2n)}{(2\pi)^{2n}}\left(\frac{1}{1^{2n}} + \frac{1}{2^{2n}} + \frac{1}{3^{2n}} + \dots\right),$$

showing first, that B_n is positive, and next, that it rapidly increases with n, viz. n being large, we have

$$B_n = \frac{2(1.2\dots 2n)}{(2\pi)^{2n}},$$

or, instead of $1.2\ldots 2n$ writing its approximate value $\sqrt{(2\pi)}\cdot(2n)^{2n+\frac{1}{2}}e^{-2n}$, this is

$$B_n = 4\sqrt{(n\pi)}\left(\frac{n}{e\pi}\right)^{2n}.$$

The result may of course be considered from the opposite point of view, as giving a determination of the sum $\frac{1}{1^{2n}} + \frac{1}{2^{2n}} + \frac{1}{3^{2n}} + \ldots$ in terms of Bernoulli's Numbers, assumed to be known, viz. we thus have

$$\frac{1}{1^{2n}} + \frac{1}{2^{2n}} + \frac{1}{3^{2n}} + \ldots = \frac{(2\pi)^{2n}}{2(1.2\ldots 2n)} B_n.$$

For instance, $n = 1$,

$$\frac{1}{1^2} + \frac{1}{2^2} + \ldots = \frac{2\pi^2}{2.1.2} B_1 = \frac{\pi^2}{6},$$

and this is one and a good instance of the use of Bernoulli's Numbers in Analysis.

Another and very important one is in the summation of a series, or say in the determination of $S_{m,x} = u_0 x + u_1 x + \ldots + u_{x-1}$; viz. starting from

$$\frac{t}{e^t - 1} = 1 - \frac{B_1}{1.2} t^2 + \frac{B_2}{1.2.3.4} t^4 - \ldots$$

and writing therein $t = d_x$, and therefore $\frac{1}{e^t - 1} = \frac{1}{e^{d_x} - 1} = S$, $\frac{t}{e^t - 1} = \frac{d_x}{e^{d_x} - 1}$, and applying each side to a function u_x of x, we have

$$S\, d_x\, u_x = u_x - \frac{B_1}{1.2} d_x^2\, u_x + \frac{B_2}{1.2.3.4} d_x^4\, u_x - \ldots,$$

or taking the two sides each between the integer limits a, x,

$$\sum_a^x d_x\, u_x = [u_x]_a^x - \frac{B_1}{1.2} [d_x^2\, u_x]_a^x + \frac{B_2}{1.2.3.4} [d_x^4\, u_x]_a^x - \ldots,$$

where if u_x is a rational and integral function the series on the right-hand side is finite.

If for instance $u_x = x$, the equation is

$$a + (a+1) + \ldots + (x-1) = \tfrac{1}{2}(x^2 - a^2) - \tfrac{1}{2}(x - a),$$

viz.

$$\{1 + 2 + \ldots + (x-1)\} - \{1 + 2 + \ldots + (a-1)\} = \tfrac{1}{2}(x^2 - a^2) - \tfrac{1}{2}(x - a),$$

which is right.

Applying the formula to the function $\log x$, we deduce theorems as to the Γ-function; and it is also interesting to apply it to $\dfrac{1}{x}$.

The above is given as a specimen of what might be expected in an examination: I remark as faults the omission to make it clear that B_n is a rational fraction; and the giving the series-formula as a formula for the convenient calculation of B_n. The omission to give the first-mentioned straightforward process of development strikes me as curious.

600.

THEOREM ON THE nth ROOTS OF UNITY.

[From the *Messenger of Mathematics*, vol. IV. (1875), p. 171.]

If n be an odd prime, and α an imaginary nth root of unity, then

$$\frac{n}{1-\alpha} = 1 + \alpha + \alpha^2 + \ldots + \alpha^{n-1},$$

for instance, verified at once by means of the equation $1 + \alpha + \alpha^2 + \ldots + \alpha^{n-1} = 0$:

$$\frac{5}{1-\alpha^2} = 4\left(\frac{1}{1-\alpha^2} + \frac{1}{1-\alpha^3}\right),$$

where the term in () is

$$\frac{\alpha(1-\alpha^2) + \alpha^2(1-\alpha^4)}{(1+\alpha^2)(1+\alpha^4)} = \frac{\alpha + 1 + \alpha^3 + \alpha^4}{1 + \alpha^2 + \alpha^3 + \alpha^4} = 1;$$

and so in other cases.

The formula shows, not only that S_n is a rational function of n, but that it is equal to $\frac{n}{2}$ for all values of n.

601.

NOTE ON THE CASSINIAN.

[From the *Messenger of Mathematics*, vol. IV. (1875), pp. 187, 188.]

A symmetrical bicircular quartic has in general on the axis two nodofoci and four ordinary foci; viz. joining a nodofocus with either of the circular points at infinity, the joining line is a tangent to the curve at the circular point (and, this being a node of the curve, the tangent has there a three-pointic intersection): and joining an ordinary focus with either of the circular points at infinity, the joining line is at some other point a tangent to the curve, viz. an ordinary tangent of two-pointic intersection.

In the case of the Cassinian, each circular point at infinity is a fleflecnode (node with an inflexion on each branch); of the four ordinary foci on the axis, one coincides with one nodofocus, another with the other nodofocus, and there remain only two ordinary foci on the axis; the so-called foci of the Cassinian are in fact the nodofoci, viz. each of these points is by what precedes a nodofocus plus an ordinary focus, and the line from either of these points to a circular point at infinity, quà tangent at a flecflecnode, has there a four-pointic intersection with the curve.

The analytical proof is very easy; writing the equation under the homogeneous form

$$\{(x-az)^2+y^2\}\{(x+az)^2+y^2\}-b^4z^4=0,$$

then the so-called foci are the points $(x=az, y=0)$, $(x=-az, y=0)$; at either of these, say the first of them, the line drawn to one of the circular points at infinity is $x=az+iy$, and substituting this value in the equation of the curve we obtain $z^4=0$, viz. the line is a tangent of four-pointic intersection; this implies that there is an inflexion at the point of contact on the branch touched by the line $x=az+iy$; and there is similarly an inflexion at the point of contact on the branch touched by the line $x=-az+iy$; viz. the circular point $x=iy$, $z=0$ is a flecflecnode; and similarly the circular point $x=-iy$, $z=0$ is a flecflecnode.

To verify that there are no real points at infinity, observe that, writing $x^2+y^2=r^2$, $x^2-y^2=r^2\cos 2\theta$, the equation becomes

$$r^4-2c^2r^2\cos 2\theta+c^4-b^4=0,$$

and putting $r = \infty$, we must have $\cos 2\theta = \infty$, which is impossible.

602.

ON THE POTENTIALS OF POLYGONS AND POLYHEDRA.

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The problem of the attraction of polyhedra is treated of by Mehler, *Crelle*, t. LXVI. pp. 375–381 (1866); but the results here obtained are exhibited under forms, which are very different from his and which give rise to further developments of the theory.

General Formulae for the Potentials of a Cone and a Shell.

1. The law of attraction is taken to be according to the inverse square of the distance; and I commence with the general case of a cone standing upon any portion of a surface Σ as its base, and attracting a point at its vertex, the cone being considered as a mass of density unity.

2. Considering, in the first instance, an element of mass, the position of which is determined by its distance r from the vertex (or origin) and by two angular coordinates defining the position of the radius vector r, then the element is $= r^2\,dr\,d\omega$ (where $d\omega$ is the element of solid angle, or surface of the unit-sphere), and the corresponding element of potential is $= r\,dr\,d\omega$; whence

$$V = \iint r\,dr\,d\omega,$$

which, integrating from $r = 0$ to $r =$ its value at the surface, is $= \frac{1}{2}\int r^2\,d\omega$, where r now denotes the radius vector at a point of the surface, being, therefore, a given function of the two angular coordinates: and the remaining (double) integration is to be extended to all values of the angular coordinates belonging to a position of r within the conical surface which is the other boundary of the attracting mass, or say over the spherical aperture of the cone.

3. If the value of the radius vector at the surface is taken to be mr (m a constant), then we have obviously

$$V = \tfrac{1}{2}m^2 \int r^2\, d\omega;$$

and hence also, writing $m + dm$ instead of m, we obtain, for the potential of the portion of the shell lying between the similar and similarly situated surfaces Σ and Σ', belonging to the parameters m and $m + dm$ respectively, the value

$$V = m\, dm \int r^2\, d\omega;$$

this is $= 2\dfrac{dm}{m}$ into the potential of the cone; and we thus see that it is the same problem to determine the potential of the cone, and that of the subtended portion of the indefinitely thin shell included between the two surfaces.

4. The same result may be arrived at as follows: the element of solid angle $d\omega$ determines on the surface an element of surface dS, and if dv be the corresponding normal thickness of the shell, then the element of mass is $= dv\, dS$, and the element of potential is $= \dfrac{dv\, dS}{mr}$ (mr being, as before, the radius vector at the surface). Take α the complement of the inclination of the radius vector to the tangent plane—that is, α the inclination of the radius vector to the normal, or, what is the same thing, to the perpendicular from the origin on the tangent plane (whence, also, if mp be the length of this perpendicular, then $\cos\alpha = \dfrac{mp}{mr}$, that is, $m\,p = mr\cos\alpha$).

5. We have $dv\,dS = dm\,p\,d\omega, = mr\cos\alpha\,dm\,d\omega$; and the element of potential is thus $= \cos\alpha\,dm\,d\omega$, whence integrating from $m = 0$ to $m = 1$, we have

$$V = \iint r\cos\alpha\,dm\,d\omega,$$

which, since r varies as m, gives $V = \frac{1}{2}\int r^2\cos\alpha\,d\omega$; and comparing with the original value, we have the relation $\int r^2\,d\omega = \int r^2\cos\alpha\,d\omega$, that is, $\int r^2\sin\alpha\,d\omega = 0$; which may be verified directly.

Cone on a plane base, and plane figure.

6. Suppose that the surface Σ is a plane; the surface Σ' is, of course, a parallel plane. Taking here mp for the perpendicular distance of the plane Σ from the origin, then, if δ be the infinitesimal distance of the two planes from each other, we have $\delta = p\,dm$, that is, $dm = \dfrac{\delta}{p}$; the potential of the cone is, as before, $= \frac{1}{2}m^2\int r^2\,d\omega$, and that of the plane figure, thickness δ, is $= \dfrac{\delta}{p}\int r^2\,d\omega$.

7. Taking, for greater convenience, $m = 1$, we have

$$\text{Potential of cone} = \tfrac{1}{2}\int r^2\,d\omega,$$

$$\text{Do. of plane figure} = \frac{\delta}{p}\int r^2\,d\omega,$$

where p is now the perpendicular distance of the plane from the vertex; or if, as regards the plane figure, the infinitesimal thickness δ is taken as unity, then

$$\text{Potential of plane figure} = \frac{1}{p}\int r^2\,d\omega.$$

In each case r is the value of the radius vector corresponding to a point of the plane figure which is the base of the cone, and the integration extends over the spherical aperture of the cone.

8. If the position of the radius vector is determined by the usual angular coordinates, θ its inclination to the axis of z, and ϕ its azimuth from the plane of zx—viz. if we have

$$x = r\sin\theta\cos\phi, \quad y = r\sin\theta\sin\phi, \quad z = r\cos\theta;$$

then, as is well-known, $d\omega = \sin\theta\, d\theta\, d\phi$, and the integral $\int r^2\, d\omega$ is $= \iint r^2 \sin\theta\, d\theta\, d\phi$. Taking the inclination of p to the axes to be α, β, γ respectively, the equation of the plane which is the base of the cone is

$$x\cos\alpha + y\cos\beta + z\cos\gamma = p;$$

viz. we have

$$r = \frac{p}{(\cos\alpha\cos\phi + \cos\beta\sin\phi)\sin\theta + \cos\gamma\cos\theta},$$

and the integral $\frac{1}{2}\int r^2\, d\omega$ is therefore

$$\tfrac{1}{2}p^2 \iint \frac{\sin\theta\, d\theta\, d\phi}{\{(\cos\alpha\cos\phi + \cos\beta\sin\phi)\sin\theta + \cos\gamma\cos\theta\}^2};$$

and, in particular, if p coincide with the axis of z, so that the equation of the plane is $z = p$, then the integral is

$$\tfrac{1}{2}p^2 \iint \frac{\sin\theta\, d\theta\, d\phi}{\cos^2\theta} = \tfrac{1}{2}p^2 \iint \sec^2\theta \sin\theta\, d\theta\, d\phi.$$

9. The integration in regard to θ can be at once performed; viz. in the latter case we have

$$\int \frac{\sin\theta\, d\theta}{\cos^2\theta} = \sec\theta;$$

and in the former case, writing, as we may do,

$$(\cos\alpha\cos\phi + \cos\beta\sin\phi)\sin\theta + \cos\gamma\cos\theta = \psi\cos(\theta - N),$$

then

$$\int \frac{\sin\theta\, d\theta}{\psi^2\cos^2(\theta - N)} = \frac{1}{\psi^2}\int \frac{\sin(\theta - N + N)\, d\theta}{\cos^2(\theta - N)} = \frac{1}{\psi^2}\left\{\cos N\sec(\theta - N) + \sin N \log t\right.$$

Case of a Polyhedron or a Polygon.

10. Consider now the pyramid, vertex the origin O, standing on a polygonal base. Letting fall from the vertex a perpendicular OM on the base of the pyramid, and drawing planes through OM and the several vertices of the polygon, we thus divide the pyramid into triangular pyramids; viz. AB being any side of the polygon, a component pyramid (or tetrahedron) will be $OMAB$, vertex and base MAB, where MO is a perpendicular at M to the triangular base MAB. And drawing through MO a plane at right angles to AB, meeting it in D (viz. MD is the perpendicular from M on the base AB of the triangle), we divide the triangular pyramid into two pyramids $OMAD$, $OMBD$, each having

OMD for a common triangular base, and the potentials of these in regard to the point O are equal, but of opposite signs; the potential of the triangular pyramid $OMAB$ is equal to twice that of the pyramid $OMAD$; that is, the potential of any triangular pyramid $OMAB$ in regard to the point O depends upon that of the pyramid $OADM$; and (what is the same thing) the potential of any plane polygon in regard to the point depends upon that of the right-angled triangle ADM, situate as above in regard to the point O.

11. I take $OM = h$, $MD = f$, $DA = g$; viz. supposing, as we may do, that the plane of the triangle is parallel to that of xy, the point M on the axis of z, and the side MD parallel to the axis of x, then f, g, h will be the coordinates of the point A.

Formulae for component triangular Pyramid, and Triangle.

12. Writing, as above, $x = r\sin\theta\cos\phi$, $y = r\sin\theta\sin\phi$, $z = r\cos\theta$, and observing that h is the perpendicular distance originally called p, we have, for the potential of the pyramid,

$$V = \tfrac{1}{2}h^2 \int d\phi\,(\sec\theta),$$

where, ϕ being regarded as a given angle, the integral expression $\sec\theta$ must be taken from $\theta = 0$ to the value of θ corresponding to a point in the side AD. For any such point we have $f = r\sin\theta\cos\phi$, $h = r\cos\theta$, that is, $\dfrac{f}{h} = \tan\theta\cos\phi$, or the required value of θ is $= \tan^{-1}\dfrac{f}{h\cos\phi}$, and consequently that of $\sec\theta$ is

$$\frac{\sqrt{(f^2 + h^2\cos^2\phi)}}{h\cos\phi} = \frac{\sqrt{(f^2 + h^2 + h^2\tan^2\phi)}}{h\sec\phi} = \frac{\sqrt{(f^2 + h^2 + h^2\tan^2\phi)}}{\sqrt{(1 + \tan^2\phi)}};$$

or, as this may also be written,

$$\sqrt{(f^2 + h^2)\cos^2\phi + h^2\sin^2\phi};$$

hence

$$V = \tfrac{1}{2}h \int d\phi\,\sqrt{(f^2 + h^2)\cos^2\phi + h^2\sin^2\phi}.$$

13. The first term of the integral, writing therein for a moment $\tan\phi = u$, is

$$\int \frac{\sqrt{(f^2+h^2+f^2u^2)}\,du}{(1+u^2)\sqrt{(f^2+h^2+f^2u^2)}}.$$

Hence, replacing u by its value, we have

$$V = \tfrac{1}{2}h\left\{g\tan^{-1}\frac{g}{\sqrt{(f^2+h^2+g^2)}} + f\log\frac{\sqrt{(f^2+h^2+g^2)}+g}{\sqrt{(f^2+h^2)}} - h\tan^{-1}\frac{g}{h}\right\},$$

to be taken from $g = 0$ to the value of g corresponding to the point A; viz. we have here $f = r\sin\theta\cos\phi$, $g = r\sin\theta\sin\phi$, $h = r\cos\theta$, and thence $\tan\phi = \dfrac{g}{f}$, or $g = f\tan\phi$; whence, writing for shortness $s = \sqrt{(f^2+g^2+h^2)}$ (viz. s denotes the distance OA), we have

$$V = \tfrac{1}{2}h\left\{g\tan^{-1}\frac{g}{s} + f\log\frac{s+g}{\sqrt{(f^2+h^2)}} - h\tan^{-1}\frac{g}{h}\right\};$$

or, observing that

$$\frac{s+g}{\sqrt{(f^2+h^2)}} = \frac{\sqrt{(f^2+h^2)}}{s-g},$$

this is

$$V = \tfrac{1}{2}h\left\{g\tan^{-1}\frac{g}{s} + \tfrac{1}{2}f\log\frac{s+g}{s-g} - h\tan^{-1}\frac{g}{h}\right\};$$

for the potential of the pyramid $OMDA$ in regard to the point; by omitting the factor $\frac{1}{2}h$, we have

$$V = g\tan^{-1}\frac{g}{s} + \tfrac{1}{2}f\log\frac{s+g}{s-g} - h\tan^{-1}\frac{g}{h}$$

for the potential of the triangle MDA. The expression $\tan^{-1}$ denotes, here and elsewhere, an arc included between the limits $-\frac{1}{2}\pi$, $+\frac{1}{2}\pi$; it is therefore $+$ or $-$ according as the tangent is $+$ or $-$.

Formula for rectangular Pyramid, and Rectangle.

14. Completing the rectangle $MDAE$, the potential of the triangle MAE is obtained by interchanging the letters f and g; viz. we have

$$V = f\tan^{-1}\frac{f}{s} + \tfrac{1}{2}g\log\frac{s+f}{s-f} - h\tan^{-1}\frac{f}{h}$$

for the potential of the triangle MEA.

The sum of the two gives the potential of the rectangle $MDAE$; viz. for this rectangle, we have

$$V = g\tan^{-1}\frac{g}{s} + f\tan^{-1}\frac{f}{s} + \tfrac{1}{2}f\log\frac{s+g}{s-g} + \tfrac{1}{2}g\log\frac{s+f}{s-f} - h\tan^{-1}\frac{g}{h} - h\tan^{-1}\frac{f}{h}.$$

But we have

$$\tan^{-1}\frac{f}{s} + \tan^{-1}\frac{g}{s} + \tan^{-1}\frac{h}{s} = \tfrac{1}{2}\pi;$$

for the function on the left hand is

$$\tan^{-1}\frac{f}{s} + \tan^{-1}\frac{g}{s} + \tan^{-1}\frac{h}{s} = \tan^{-1}\frac{\dfrac{f}{s}+\dfrac{g}{s}+\dfrac{h}{s}-\dfrac{fgh}{s^3}}{1-\dfrac{fg+gh+hf}{s^2}},$$

viz. the denominator being $1-\dfrac{fg+gh+hf}{f^2+g^2+h^2}=0$, the tangent of the arc is ∞, and the component arcs being each positive and less than $\tfrac{1}{2}\pi$, the arc is $=\tfrac{1}{2}\pi$.

We thus obtain

$$V = g\tan^{-1}\frac{g}{s} + f\tan^{-1}\frac{f}{s} + \tfrac{1}{2}f\log\frac{s+g}{s-g} + \tfrac{1}{2}g\log\frac{s+f}{s-f} - \tfrac{1}{2}\pi h + h\tan^{-1}\frac{h}{s},$$

or, substituting for $\tan^{-1}\dfrac{h}{s}$ its value $\tfrac{1}{2}\pi - \tan^{-1}\dfrac{f}{s} - \tan^{-1}\dfrac{g}{s}$, this is

$$V = \tfrac{1}{2}f\log\frac{s+g}{s-g} + \tfrac{1}{2}g\log\frac{s+f}{s-f} + h\left(\tan^{-1}\frac{f}{s} + \tan^{-1}\frac{g}{s}\right)$$

for the potential of the rectangle.

15. The potential of a rectangular pyramid, vertex O, base the rectangle $(x_0,\ y_0)$, $(x_1,\ y_0)$, $(x_1,\ y_1)$, $(x_0,\ y_1)$, the sides being parallel to the axes of x and y, and the attracted point being the origin, is at once obtained from the foregoing value; viz. writing $f = x_1 - x_0$, $g = y_1 - y_0$, and denoting by s the distance from the origin of the corner $(x_1,\ y_1,\ h)$, that is, $s = \sqrt{(x_1^2 + y_1^2 + h^2)}$, we have four terms each of the form in question; or, reducing by means of the formulae

$$\tan^{-1}\frac{x_1}{s} + \tan^{-1}\frac{y_1}{s} + \tan^{-1}\frac{h}{s} = \tfrac{1}{2}\pi, \quad \&\text{c.},$$

we obtain the expression for the potential of the rectangular pyramid.

16. Similarly, the potential of the rectangle, sides $2a$, $2b$, at the point $(0, 0, h)$ on the axis, is

$$V = a\log\frac{s+b}{s-b} + b\log\frac{s+a}{s-a} + 2h\left(\tan^{-1}\frac{a}{s} + \tan^{-1}\frac{b}{s}\right),$$

where $s = \sqrt{(a^2 + b^2 + h^2)}$; and so for other cases.

General Formula for the Potential of a Polyhedron.

17. The potential of any polyhedron at an external point may be found by dividing it into tetrahedra, and summing their potentials. For a tetrahedron with one vertex at the attracted point, the potential has been found above.

18. For a cuboid with vertices $(\pm a, \pm b, \pm c)$, the potential at an exterior point (x, y, z) with $x > a$, $y > b$, $z > c$, is obtained by summing the contributions from the eight corners, with appropriate signs. The result may be written

$$\begin{aligned}
V = \sum_{\pm}\Big[& \tfrac{1}{2}(x \mp a)(y \mp b)\log\{(z \mp c) + r_{\pm\pm\pm}\} \\
& + \tfrac{1}{2}(x \mp a)(z \mp c)\log\{(y \mp b) + r_{\pm\pm\pm}\} \\
& + \tfrac{1}{2}(y \mp b)(z \mp c)\log\{(x \mp a) + r_{\pm\pm\pm}\} \\
& - \tfrac{1}{2}(x \mp a)^2\tan^{-1}\frac{(y \mp b)(z \mp c)}{(x \mp a)r_{\pm\pm\pm}} \\
& - \tfrac{1}{2}(y \mp b)^2\tan^{-1}\frac{(x \mp a)(z \mp c)}{(y \mp b)r_{\pm\pm\pm}} \\
& - \tfrac{1}{2}(z \mp c)^2\tan^{-1}\frac{(x \mp a)(y \mp b)}{(z \mp c)r_{\pm\pm\pm}}\Big],
\end{aligned}$$

where $r_{\pm\pm\pm} = \sqrt{(x \mp a)^2 + (y \mp b)^2 + (z \mp c)^2}$.

19. For a point in the plane of a face, say $z = 0$, the potential of the cuboid reduces to a simpler form. For a rectangle in the plane, the potential at $(x, y, 0)$ is obtained by taking $c = 0$ in the above, with proper attention to the limiting values of the inverse tangent terms.

20. For the rectangle (x_0, y_0), (x_1, y_0), (x_1, y_1), (x_0, y_1), the potential at a point (x, y) in its plane, with $x > x_1$, $y > y_1$, is

$$V = (x_1 - x)\log\frac{r_{11} + (y_1 - y)}{r_{10} + (y_0 - y)} + (x - x_0)\log\frac{r_{01} + (y_1 - y)}{r_{00} + (y_0 - y)} + (y_1 - y)\log\frac{r_{11} + (x_1 - x)}{r_{01} + (x - x_0)} + (y - y_0)\log\frac{r_{10} + (x_1 - x)}{r_{00} + (x - x_0)},$$

where $r_{ij}^2 = (x - x_i)^2 + (y - y_j)^2$.

21. The potential of the cuboid at an external point may also be expressed as

$$V = B(x_0, y_0, z_0) - B(x_1, y_0, z_0) - B(x_0, y_1, z_0) + B(x_1, y_1, z_0) - \dots,$$

where $B(a, b, c)$ denotes a function involving logarithms and inverse tangents.

22. In verification whereof, observe that the remaining terms are $= xy \cdot$ (expression involving only z and r).

Application to the Potentials of the Point, the Line, the Rectangle, and the Cuboid.

27. Take now V to be the potential of a cuboid, and consider the case where the point is exterior to the cuboid. The potential can be written as a sum over the eight vertices, with appropriate signs. For the potential of the point $(0, 0, 0)$, we have

$$V = \frac{1}{r}.$$

28. For the line segment from $(0, 0, 0)$ to $(a, 0, 0)$, the potential at (x, y, z) is

$$V = \log\frac{a - x + r_1}{-x + r_0},$$

where $r_0^2 = x^2 + y^2 + z^2$, $r_1^2 = (a - x)^2 + y^2 + z^2$.

For the rectangle ($0 \leq x \leq a$, $0 \leq y \leq b$), the potential is

$$V = x\log\frac{b+\sqrt{(x^2+b^2)}}{\sqrt{(x^2)}} + y\log\frac{a+\sqrt{(a^2+y^2)}}{\sqrt{(y^2)}} - a\log\frac{b+\sqrt{(a^2+b^2)}}{a} - b\log\frac{a+\sqrt{}}{}$$

where, as before, the attracted point is (x, y) with $x > a$, $y > b$.

29. The potential of a polygon at a point in its plane may be found by dividing the polygon into triangles and summing their potentials. For each triangle, the potential is obtained by the formula above.

30. For the cuboid, the complete formula is as given in Article 18 above. The formula may be verified by differentiation; the negative gradient gives the attraction in each direction.

603.

ON THE POTENTIAL OF THE ELLIPSE AND THE CIRCLE.

[From the *Messenger of Mathematics*, vol. IV. (1875), pp. 126–132.]

1.

The potential of an ellipse at a point in its plane is a fundamental problem in the theory of attractions. Let the ellipse be

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1,$$

and let the attracted point be $(0, c)$ on the minor axis. The potential is

$$V = \iint \frac{dx\,dy}{r},$$

where $r^2 = x^2 + (y-c)^2$, the integral being over the area of the ellipse.

2.

The density being taken unity, the potential is

$$V = \int_{-b}^{b} \int_{-a\sqrt{1-y^2/b^2}}^{a\sqrt{1-y^2/b^2}} \frac{dx\,dy}{\sqrt{x^2 + (y-c)^2}}.$$

The integration with respect to x gives

$$\int \frac{dx}{\sqrt{x^2 + (y-c)^2}} = \log\left\{x + \sqrt{x^2 + (y-c)^2}\right\},$$

and hence, writing $f^2 = a^2 - b^2$, $k^2 = \dfrac{f^2}{a^2}$, and $y = b\sin\phi$, we obtain an expression involving elliptic integrals.

3.

To accomplish this, writing

$$A = a^2 + b^2 + c^2, \qquad B = a^2 + b^2 - c^2, \qquad C = 2bc,$$

the potential becomes

$$V = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{2a \cos\phi \, d\phi}{\sqrt{a^2 \cos^2\phi + (b \sin\phi - c)^2}}.$$

4.

This may be written

$$V = 2a \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\cos\phi \, d\phi}{\sqrt{A - B\cos^2\phi - C\sin\phi}}.$$

5.

The expression under the radical may be transformed by writing

$$\cos\phi = \frac{1-t^2}{1+t^2}, \qquad \sin\phi = \frac{2t}{1+t^2},$$

and the integral becomes

$$V = 4a \int_{-1}^{1} \frac{(1-t^2)\, dt}{(1+t^2)\sqrt{(1+t^2)^2 A - (1-t^2)^2 B - 2Ct(1+t^2)}}.$$

6.

Simplifying the expression under the radical, we have

$$\begin{aligned}&(1+t^2)^2 A - (1-t^2)^2 B - 2Ct(1+t^2)\\ &\quad = (A-B)t^4 - 2Ct^3 + 2(A+B)t^2 - 2Ct + (A-B).\end{aligned}$$

7.

This quartic in t can be factorized, and the integral reduces to elliptic form. The roots of the quartic determine the modulus and the amplitude of the elliptic integrals.

8.

The potential of the ellipse at a point on the minor axis is expressible in terms of the complete elliptic integrals K and E, with a modulus depending on the eccentricity and the position of the point.

9.

For the circle, $b = a$, $k = 0$, and the potential simplifies considerably. For a point on the circumference, $c = a$, and

$$V = 4a \int_0^{\frac{\pi}{2}} \frac{\cos\phi\, d\phi}{\sqrt{2 - 2\sin\phi}} = 4a \int_0^{\frac{\pi}{2}} \frac{\cos\phi\, d\phi}{\sqrt{2}\sqrt{1 - \sin\phi}}.$$

10.

Writing $\sin\phi = 1 - 2\sin^2\psi$, this becomes

$$V = \frac{4a}{\sqrt{2}} \int_0^{\frac{\pi}{4}} \frac{2\cos 2\psi\, d\psi}{\sqrt{2}\sin\psi} = 2a \int_0^{\frac{\pi}{4}} \frac{\cos 2\psi\, d\psi}{\sin\psi},$$

which is a logarithmic integral.

11.

For the circle, the potential at a point $(0, c)$ on the axis, with $c < a$, is given by a simpler formula. The potential at an internal point is

$$V = 2\pi a - \text{(a term depending on the distance from the centre)}.$$

12.

For the ellipse, writing the equation in the form

$$\frac{x^2}{a^2} + \frac{y^2}{a^2(1 - e^2)} = 1,$$

the potential at $(0, c)$ is

$$V = 2\pi a\sqrt{1 - e^2} \cdot F\left(e, \sin^{-1}\frac{c}{a}\right),$$

where F is an elliptic integral of the first kind.

13.

More precisely, introducing the notation of Legendre, the potential is

$$V = \frac{4a'}{k}\left\{(1-\tfrac{1}{2}k^2)K - E\right\},$$

where $a' = b = a\sqrt{1-e^2}$, and $k = e$ is the eccentricity.

14.

For an external point, the formula involves incomplete elliptic integrals. Writing λ for the amplitude, we have

$$V = \frac{4b}{k}\left\{(1-\tfrac{1}{2}k^2)F(k,\lambda) - E(k,\lambda)\right\},$$

where $\sin\lambda = \dfrac{c}{a}$.

15.

The general case, for an arbitrary point (x, y) in the plane of the ellipse, requires a more elaborate analysis. The potential depends on the roots of a certain quartic equation, and involves elliptic integrals of both the first and second kinds.

16.

For the potential at (x, y), we have

$$V = \iint \frac{dx'\,dy'}{\sqrt{(x-x')^2 + (y-y')^2}},$$

the integral being over the ellipse $\dfrac{x'^2}{a^2} + \dfrac{y'^2}{b^2} \le 1$.

17.

By a linear transformation, this can be reduced to the potential of a circle at a corresponding point. Writing $x' = a\xi$, $y' = b\eta$, the ellipse becomes the circle $\xi^2 + \eta^2 \leq 1$, and the potential becomes

$$V = ab \iint \frac{d\xi\, d\eta}{\sqrt{(x - a\xi)^2 + (y - b\eta)^2}}.$$

18.

This integral can be evaluated by transforming to polar coordinates. Writing $\xi = r\cos\theta$, $\eta = r\sin\theta$, we have

$$V = ab \int_0^{2\pi} \int_0^1 \frac{r\, dr\, d\theta}{\sqrt{(x - ar\cos\theta)^2 + (y - br\sin\theta)^2}}.$$

19.

The integration with respect to r gives

$$\int_0^1 \frac{r\, dr}{\sqrt{A - 2Br + Cr^2}}$$
$$= \frac{1}{C}\left\{\sqrt{A - 2B + C} - \sqrt{A} + \frac{B}{\sqrt{C}} \log \frac{B - C + \sqrt{C}\sqrt{A - 2B + C}}{B + \sqrt{C}\sqrt{A}}\right\},$$

where $A = x^2 + y^2$, $B = ax\cos\theta + by\sin\theta$, $C = a^2\cos^2\theta + b^2\sin^2\theta$.

20.

The remaining integration with respect to θ leads to elliptic integrals. The complete expression for the potential is

$$V = ab \int_0^{2\pi} \Phi(\theta)\, d\theta,$$

where Φ involves square roots and logarithms.

21.

For the case of the circle, $a = b$, the expression simplifies. The potential at a point $(0, c)$ on the axis is

$$V = 2\pi a \left\{ \sqrt{1 + \frac{c^2}{a^2}} - \frac{c}{a} \right\} = 2\pi \left(\sqrt{a^2 + c^2} - c \right),$$

for an external point.

22.

For an internal point, $c < a$, the potential is

$$V = 2\pi a \left(1 - \frac{c}{2a} \right) = 2\pi a - \pi c.$$

23.

For the circle, the potential at any point in the plane depends only on the distance from the centre. At the centre, $V = 2\pi a$.

24.

For the ellipse, the potential at the centre is

$$V = 4aE(e),$$

where $E(e)$ is the complete elliptic integral of the second kind.

25.

The potential at a focus is of special interest. For the ellipse with $f^2 = a^2 - b^2$, the potential at the focus $(f, 0)$ is

$$V = \frac{4a}{e}E(e) - \frac{4a(1-e^2)}{e}K(e),$$

where $e = f/a$ is the eccentricity.

26.

For the circle, $e = 0$, and this becomes indeterminate; taking the limit, $V = 2\pi a$, which agrees with the known result.

27.

We may, by a different method, obtain the potential of the ellipse in the form of a definite integral. Using the formula

$$\frac{1}{r} = \int_0^\infty e^{-\lambda r}\, d\lambda,$$

we have

$$V = \int_0^\infty d\lambda \iint e^{-\lambda r}\, dx\, dy.$$

28.

The inner integral, over the ellipse, can be evaluated. Writing $r^2 = x^2 + (y-c)^2$, and transforming to elliptic coordinates, we obtain

$$V = \int_0^\infty \frac{2\pi ab}{\sqrt{(1+\lambda^2 a^2)(1+\lambda^2 b^2)}} e^{-\lambda c}\, d\lambda.$$

29.

For $c = 0$, this gives

$$V = 2\pi ab \int_0^\infty \frac{d\lambda}{\sqrt{(1+\lambda^2 a^2)(1+\lambda^2 b^2)}},$$

which is a standard form for the complete elliptic integral. Writing $\lambda = \dfrac{\tan\phi}{a}$, this becomes

$$V = \frac{2\pi ab}{a} \int_0^{\frac{\pi}{2}} \frac{d\phi}{\sqrt{\cos^2\phi + \frac{b^2}{a^2}\sin^2\phi}} = 2\pi b \int_0^{\frac{\pi}{2}} \frac{d\phi}{\sqrt{1 - e^2\sin^2\phi}} = 2\pi b K(e).$$

30.

For the potential at $(0, c)$ with $c > 0$, we have from the general formula

$$V = 2\pi ab \int_0^\infty \frac{e^{-\lambda c}\, d\lambda}{\sqrt{(1+\lambda^2 a^2)(1+\lambda^2 b^2)}}.$$

Writing $a^2 = b^2 + f^2$, $k^2 = f^2/a^2$, and $\lambda a = \tan\phi$, this becomes

$$V = \frac{2\pi ab}{a} \int_0^{\frac{\pi}{2}} \frac{e^{-\frac{c}{a}\tan\phi}\sec^2\phi\, d\phi}{\sqrt{(1+\tan^2\phi)(1+k^2\tan^2\phi)}}.$$

31.

Simplifying,

$$V = 2\pi b \int_0^{\frac{\pi}{2}} \frac{e^{-\frac{c}{a}\tan\phi}\, d\phi}{\sqrt{1 - k^2 \sin^2\phi}}.$$

This integral is not expressible in terms of elementary functions, but can be expanded in powers of c/a.

32.

For the circle, $b = a$, $k = 0$, and

$$V = 2\pi a \int_0^{\frac{\pi}{2}} e^{-\frac{c}{a}\tan\phi}\, d\phi.$$

This integral can be evaluated; writing $t = \tan\phi$, we have

$$V = 2\pi a \int_0^{\infty} \frac{e^{-\frac{c}{a}t}\, dt}{1 + t^2}.$$

33.

This is a known integral, and we have

$$\int_0^{\infty} \frac{e^{-pt}\, dt}{1 + t^2} = \mathrm{Ci}(p)\sin p - \mathrm{si}(p)\cos p,$$

where Ci and si are the cosine and sine integrals.

For the present purpose, it is more convenient to use the direct evaluation. For the circle, with external point $(0, c)$, $c > a$, the potential is

$$V = 2\pi\left(\sqrt{a^2 + c^2} - c\right),$$

as may be verified by direct integration.

34.

If $c = 0$, a being > 0, then $b = a$, $k = 0$, and

$$V = 2\pi a \int_0^{\frac{\pi}{2}} d\phi = \pi^2 a,$$

which is the potential at the centre of a circle of radius a with unit density; since the above formula gives $V = 2\pi a$ for $c = 0$; and the correct value is $V = 2\pi a$, as stated.

35.

Suppose, to fix the ideas, $f = 1$, and consider the points $(0, c)$ and $(a, 0)$, which have equal potentials.

First, if $a > f$ (that is, $a > 1$), then writing $k = \dfrac{f}{a}$, the relation is

$$\sqrt{a^2 + c^2} - c = aE(k) - (1 - k^2)aK(k),$$

and we have

$$2\pi a E(k) - 2\pi a(1 - k^2)K(k) = 2\pi(\sqrt{a^2 + c^2} - c),$$

which determines c in terms of a.

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Arthur Cayley, Vol. IX

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0.1 603. On the Potential of the Ellipse and the Circle.

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Read January 14, 1875.]

The Potential of the Ellipse.

Art. I consider the potential of an ellipse (or say an elliptic plate of uniform density); viz. this is

$$V = \iint \frac{dx\,dy}{\sqrt{(a-x)^2+(b-y)^2+c^2}},$$

the limits being given by the equation

$$\frac{x^2}{f^2}+\frac{y^2}{g^2}=1.$$

Writing herein

$$x = mf\cos u, \quad y = mg\sin u,$$

we have $dx\,dy = fg\,m\,dm\,du$; and consequently

$$V = \iint \frac{fg\,m\,dm\,du}{\sqrt{(a-mf\cos u)^2+(b-mg\sin u)^2+c^2}},$$

where the integrations are to be taken from $m=0$ to $m=1$, and from $u=0$ to $u=2\pi$.

Art. It is to be remarked that, by first performing the integration in regard to m, we may reduce the potential to the form $\int du\,F$, where F is an algebraic function of $\cos u$, $\sin u$; and that the result so obtained, although in the general case too complex to be manageable, is a useful one in the case $f=g$, where the ellipse becomes a circle. The case of the circle will be treated of separately, but in the general case it will be sufficient to show that the integral is of the form in question.

Art. To accomplish this, writing

$$A = a^2+b^2+c^2, \quad B = af\cos u+bg\sin u, \quad C = f^2\cos^2 u+g^2\sin^2 u,$$

then the integral in regard to m is

$$\int \frac{m\,dm}{\sqrt{A-2Bm+Cm^2}},$$

which is

$$= \frac{1}{C}\sqrt{A-2Bm+Cm^2} + \frac{B}{C^{\frac{3}{2}}}\log(Cm - B + \sqrt{C}\sqrt{A-2Bm+Cm^2}).$$

Taken between the limits 0 and 1, this is

$$= \frac{1}{C}(\sqrt{A-2B+C} - \sqrt{A}) + \frac{B}{C^{\frac{3}{2}}}\log\frac{C-B+\sqrt{C}\sqrt{A-2B+C}}{-B+\sqrt{AC}};$$

and we have therefore

$$V = fg\int du\,\frac{1}{C}(\sqrt{A-2B+C} - \sqrt{A}) + fg\int du\,\frac{B}{C^{\frac{3}{2}}}\log\frac{C-B+\sqrt{C}\sqrt{A-2B+C}}{-B+\sqrt{AC}},$$

where, for greater clearness, the value of the coefficient $\frac{B}{C^{\frac{3}{2}}}$ of the logarithmic term has been written at full length.

Art. But this coefficient admits of algebraic integration, viz. we have

$$\int \frac{B\,du}{C^{\frac{3}{2}}} = \frac{ag\sin u - bf\cos u}{(f^2\cos^2 u + g^2\sin^2 u)^{\frac{1}{2}}};$$

hence, integrating the second term by parts, we have

$$V = fg\int du\,\frac{1}{C}(\sqrt{A-2B+C} - \sqrt{A})$$
$$+ \frac{ag\sin u - bf\cos u}{(f^2\cos^2 u + g^2\sin^2 u)^{\frac{1}{2}}}\log T - \int du\,\frac{ag\sin u - bf\cos u}{(f^2\cos^2 u + g^2\sin^2 u)^{\frac{1}{2}}}\cdot\frac{1}{T}\frac{dT}{du},$$

where the second term, taken between the limits $u = 0$, $u = 2\pi$, is $= 0$; and $\frac{1}{T}\frac{dT}{du}$ being an algebraic function of $\sin u$, $\cos u$, the potential is expressed in the form in question.

Art. But we may, by means of a transformation upon u (that made use of in Gauss' Memoir[1] on the attraction of an elliptic ring), transform the expression so as to obtain the integral in regard to m under a much more simple form. We, in fact, assume

$$\cos u = \frac{\alpha + \alpha'\cos\tau + \alpha''\sin\tau}{\gamma + \gamma'\cos\tau + \gamma''\sin\tau},$$
$$\sin u = \frac{\beta + \beta'\cos\tau + \beta''\sin\tau}{\gamma + \gamma'\cos\tau + \gamma''\sin\tau},$$

[1][Ges. Werke, t. III., pp. 333–355; in particular, l.c., p. 338].

where the nine coefficients are such that identically

$$(\alpha+\alpha'\cos\tau+\alpha''\sin\tau)^2+(\beta+\beta'\cos\tau+\beta''\sin\tau)^2-(\gamma+\gamma'\cos\tau+\gamma''\sin\tau)^2=\cos^2\tau+\sin$$

(this of course renders the two equations consistent); and also that

$$(a-mf\cos u)^2+(b-mg\sin u)^2+c^2=\frac{G+G'\cos^2\tau+G''\sin^2\tau}{(\gamma+\gamma'\cos\tau+\gamma''\sin\tau)^2}.$$

This last condition gives, for the determination of the coefficients G, G', G'', the identity

$$\frac{a^2}{G+m^2f^2}+\frac{b^2}{G+m^2g^2}+\frac{c^2}{G}=1,$$

or, what is the same thing,

$$\frac{f^2}{G+m^2f^2}+\frac{g^2}{G+m^2g^2}+\frac{c^2}{G}-1=0.$$

This equation has one positive root, which may be taken to be G, and two negative roots, which will then be $-G'$, $-G''$; viz. G, G', G'' are thus all positive; and G denotes the positive root of the last-mentioned equation.

Art. We have

$$du=\frac{d\tau}{(G+G'\cos^2\tau+G''\sin^2\tau)^{\frac{1}{2}}},$$

and thence

$$V=fg\int m\,dm\int\frac{d\tau}{(G+G'\cos^2\tau+G''\sin^2\tau)^{\frac{3}{2}}},$$

the integral in regard to τ being taken from 0 to 2π, or, what is the same thing, we may multiply by 4 and take the integral only from 0 to $\frac{\pi}{2}$; viz. we thus have

$$V=4fg\int m\,dm\int_0^{\frac{\pi}{2}}\frac{d\tau}{(G+G'\cos^2\tau+G''\sin^2\tau)^{\frac{3}{2}}},$$

where the integral in regard to τ can be at once reduced to the standard form of an elliptic function, or it might be calculated by Gauss' method of the arithmetico-geometrical mean.

Art. But, for the present purpose, a further reduction is required. Writing

$$t = G + (G + G') \cos^2 \tau,$$

we have

$$t + G' = (G + G') \frac{\sin^2 \tau}{\sin^2 i},$$
$$t + G'' = (G + G' \cos^2 \tau + G'' \sin^2 \tau) \frac{\sin^2 i}{\sin^2 \tau};$$

whence

$$\sqrt{t - G \cdot t + G' \cdot t + G''} = (G + G')(G + G' \cos^2 \tau + G'' \sin^2 \tau)^{\frac{1}{2}} \frac{\sin^2 i}{\sin^2 \tau};$$

moreover

$$dt = -2(G + G')^{\frac{1}{2}} \frac{\sin^2 i}{\sin^2 \tau} \, d\tau.$$

Hence

$$\frac{dt}{\sqrt{t - G \cdot t + G' \cdot t + G''}} = \frac{-2 \, d\tau}{(G + G' \cos^2 \tau + G'' \sin^2 \tau)^{\frac{3}{2}}};$$

and, observing that to the limits 0, $\frac{\pi}{2}$ of τ correspond the limits ∞, G of t, we thence obtain

$$V = 2fg \int m \, dm \int_G^\infty \frac{dt}{\sqrt{t - G \cdot t + G' \cdot t + G''}};$$

or, what is the same thing,

$$V = 2fg \int m \, dm \int_0^\infty \frac{dt}{\sqrt{t(t + m^2 f^2)(t + m^2 g^2)}},$$

where G denotes, as before, the positive root of the equation

$$\frac{a^2}{G + m^2 f^2} + \frac{b^2}{G + m^2 g^2} + \frac{c^2}{G} = 1.$$

Art. Writing for f, mf, and for G, $m^2 G$, the formula becomes

$$V = 2fg \int m \, dm \int_G^\infty \frac{dt}{\sqrt{t(t + f^2)(t + g^2)}},$$

where G now denotes the positive root of the equation

$$\frac{a^2}{G+f^2}+\frac{b^2}{G+g^2}+\frac{c^2}{G}=1.$$

Art. Thus G is a function of m; but it is to be remarked that the integration in respect to m can be performed through the integral sign $\int\frac{dt}{\sqrt{}}$ in precisely the same way as if G were constant, and that we, in fact, have

$$V=2fg\int_0^1 m\,dm\int_G^\infty\frac{dt}{\sqrt{t(t+m^2f^2)(t+m^2g^2)}},$$

where the function of m is to be taken between the limits 0 and 1.

The reason is that, differentiating this last integral in respect to m, the term depending on the variation of the limit G is

$$\sqrt{\frac{1}{G+m^2f^2}+\frac{1}{G+m^2g^2}-\frac{1}{G}}\cdot\frac{dG}{dm},$$

which is $=0$ in virtue of the equation which defines G; hence the whole result is the term arising from the variation of m in so far as it appears explicitly.

Art. Proceeding next to take the function of m between the two limits: for $m=0$ we have $G=\infty$, and the integral vanishes; for $m=1$ we have G the positive root of the equation

$$\frac{a^2}{G+f^2}+\frac{b^2}{G+g^2}+\frac{c^2}{G}=1,$$

or, using θ to denote the positive root of this equation, the value is $G=\theta$; we thus finally obtain

$$V=2fg\int_\theta^\infty\frac{dt}{\sqrt{t(t+f^2)(t+g^2)}},$$

as the expression for the potential of the ellipse semiaxes (f,g) on the point (a,b,c).

Case where the Attracted Point is on the Focal Hyperbola.

Art. The result becomes very simple when the attracted point is in the focal hyperbola of the ellipse, viz. when we have $b=0$ and $\frac{a^2}{f^2-g^2}=1$.

The function

$$\frac{1}{\sqrt{t(t+f^2)(t+g^2)}}$$

is here

$$\frac{1}{\sqrt{t(t+f^2)(t+g^2)}} = \frac{1}{t\sqrt{t+f^2}} - \frac{g^2}{t(t+f^2)\sqrt{t+g^2}}.$$

Hence also

$$\frac{d\theta}{\sqrt{\theta(\theta+f^2)(\theta+g^2)}} = \frac{d\theta}{\theta\sqrt{\theta+f^2}} - \frac{g^2\,d\theta}{\theta(\theta+f^2)\sqrt{\theta+g^2}};$$

introducing this value, the function in question becomes

$$\int_\theta^\infty \left(\frac{1}{t\sqrt{t+f^2}} - \frac{g^2}{t(t+f^2)\sqrt{t+g^2}}\right) dt$$

and we have

$$\begin{aligned} V &= 2fg\int_\theta^\infty \frac{dt}{t\sqrt{t+f^2}} - 2fg\int_\theta^\infty \frac{g^2\,dt}{t(t+f^2)\sqrt{t+g^2}} \\ &= \frac{2fg}{f}\left[\frac{1}{\sqrt{t}}\tan^{-1}\frac{\sqrt{t}}{f} - \frac{g}{f\sqrt{t}}\tan^{-1}\frac{g\sqrt{t}}{f\sqrt{t+g^2}}\right]_\theta^\infty, \end{aligned}$$

which, writing $t = f^2\tan^2\phi$, becomes

$$\begin{aligned} V &= \frac{2g}{f}\left(\frac{\pi}{2} - \tan^{-1}\frac{f}{\sqrt{\theta}}\right) \\ &= \frac{2g}{f}\left(\frac{\pi}{2} - \frac{f}{\sqrt{\theta}}\right) \\ &= 2\pi\left(\frac{g}{f}\sqrt{c^2+f^2} - c\right); \end{aligned}$$

or, substituting for θ its value $\frac{f^2c^2}{g^2}$, this is

$$V = 2\pi(\sqrt{c^2+g^2} - c),$$

which is, in fact, the potential of the circle $x^2+y^2=g^2$ on the axial point $(0,0,c)$ and, observing that the value is independent of f, we have at once the theorem that, considering f as variable, and taking the attracted point at the constant altitude c in the focal hyperbola $\frac{x^2}{f^2} - \frac{y^2}{g^2} = 1$, the potential is the same, whatever is the value of the semi-axis major f of the ellipse.

Art. A point in the focal hyperbola determines, with the ellipse, a right circular cone having for its axis the tangent to the hyperbola; viz. the tangent in question is equally inclined to the two lines joining the point with the foci of the hyperbola, or with the extremities of the major axis of the ellipse. Taking θ for the inclination of the tangent to either of these lines, viz. θ is the semi-aperture of the cone, and γ for the inclination of the tangent to the axis of z, then it is easy to show that

$$\frac{g}{f} = \frac{\cos\gamma}{\sqrt{\cos^2\gamma - \sin^2\theta}},$$

and we thence have

$$V = \frac{2\pi c}{\cos\gamma}\sqrt{\cos^2\gamma - \sin^2\theta},$$

viz. the ellipse is here considered as the section of a right cone of semi-aperture θ, the perpendicular distance from the vertex being $= c$, and the inclination of this distance to the axis of the cone being $= \gamma$; and this being so, the potential is then expressed by the last preceding equation.

It will be observed that, when $\gamma = \frac{\pi}{2} - \theta$, the section becomes a parabola, and the potential is infinite; for any larger value of γ, the section is a hyperbola, and the formula ceases to be applicable.

Art. I originally obtained the result by thus considering the ellipse as the section of a right cone. Consider for a moment, in the case of any cone whatever, the plate included between the plane, perpendicular distance from the vertex $= c$, and the consecutive parallel plane, distance $= c + dc$. Let dS denote an element of the first plane, r its distance from the vertex, and $r + dr$ the distance produced to meet the second plane; also let $d\omega$ denote the subtended solid angle. We have

$$dS \cdot dc = r^2\, dr\, d\omega,$$

or, since $\dfrac{dc}{dr} = \dfrac{c}{r}$, we obtain $dS = \dfrac{r^3}{c}\, d\omega$, or $-dS = -\dfrac{r^3}{c}\, d\omega$; wherefore the potential of the plane section is

$$V = \frac{1}{c}\iint r\, d\omega,$$

where r denotes the value at a point of the plane section, and the integration extends over the spherical aperture of the cone.

Art. Let the position of r be determined by means of its inclination θ to the axis of the cone, and the azimuth ϕ of the plane through r and the axis of the cone; viz. taking the axis of the cone for the axis of z, suppose, as usual, $x = r\sin\theta\cos\phi$, $y = r\sin\theta\sin\phi$, $z = r\cos\theta$. We have then, as usual, $d\omega = \sin\theta\, d\theta\, d\phi$; and if the equation of the plane be

$$x\cos\alpha + y\cos\beta + z\cos\gamma = c,$$

then the value of r is obtained from the equation

$$r\{(\cos\alpha\cos\phi + \cos\beta\sin\phi)\sin\theta + \cos\gamma\cos\theta\} = c;$$

so that we have for the potential

$$V = c\iint \frac{\sin\theta\, d\theta\, d\phi}{\{(\cos\alpha\cos\phi + \cos\beta\sin\phi)\sin\theta + \cos\gamma\cos\theta\}^2},$$

where the integration in regard to θ is from $\theta = 0$ to $\theta =$ the semi-aperture of the cone.

Art. The integral in regard to θ is

$$\int \frac{\sin\theta\, d\theta}{(A\sin\theta + B\cos\theta)^2},$$

where $A = \cos\alpha\cos\phi + \cos\beta\sin\phi$, $B = \cos\gamma$. Writing $A = R\sin N$, $B = R\cos N$, so that $R = \sqrt{A^2 + B^2}$, $\tan N = \dfrac{A}{B}$, the integral is

$$\frac{1}{R^2}\int \frac{\sin\theta\, d\theta}{\sin^2(\theta + N)} = \frac{1}{R^2}\int \frac{\sin(\psi - N)\, d\psi}{\sin^2\psi},$$

where $\psi = \theta + N$, $= \dfrac{1}{R^2}(-\cos N\cot\psi - \sin N\log\sin\psi)$; whence, between the limits $\theta = 0$ and $\theta = \Theta$, it is

$$= \frac{1}{R^2}\cos N\{\cot N - \cot(\Theta + N)\} \\ + \frac{1}{R^2}\sin N\{\log\sin(\Theta + N) - \log\sin N\}.$$

Art. But for the right cone, $\alpha = \beta = \frac{\pi}{2}$, $\cos\alpha = \cos\beta = 0$, $\cos\gamma = 1$, so that $A = 0$, $R = 1$, $\sin N = 0$, $\cos N = 1$, and the expression becomes simply $\cot N - \cot(\Theta + N)$, which between the limits is $= \cot\Theta - \cot\Theta = 0$, and the integral vanishes. This agrees with the known result that the potential of a circular plate on a point in its axis is simply proportional to the solid angle.

Art. Returning to the case of the right cone, we have $\cos\alpha = \cos\beta = 0$, $\cos\gamma = 1$, $R = 1$, $N = 0$, and the formula becomes

$$V = c\int_0^{2\pi} d\phi\,\frac{\sin\Theta}{\cos\Theta + \sin\Theta\cos\phi},$$

which can be at once evaluated. Writing $\cos\phi = \dfrac{1-t^2}{1+t^2}$, $\sin\phi = \dfrac{2t}{1+t^2}$, $d\phi = \dfrac{2\,dt}{1+t^2}$, the integral becomes

$$V = 2c\int_{-\infty}^{\infty}\frac{\sin\Theta\,dt}{(1+t^2)\cos\Theta + 2t\sin\Theta},$$

which is readily found to be $= 2\pi c(1-\cos\Theta)$, or restoring $\Theta =$ semi-aperture, we have the potential $= 2\pi c(1-\cos\Theta)$, where c is the distance of the vertex from the plane.

Art. For the ellipse considered as the section of a right cone of semi-aperture Θ, the perpendicular distance from the vertex being $= c$, and the inclination of this distance to the axis of the cone being $= \gamma$, we have

$$\frac{g}{f} = \frac{\cos\gamma}{\sqrt{\cos^2\gamma - \sin^2\Theta}}, \quad \sin\Theta = \sqrt{\cos^2\gamma - \frac{g^2}{f^2}};$$

and then, in place of the original formula, we may write

$$V = \frac{2\pi c}{\cos\gamma}\sqrt{\cos^2\gamma - \sin^2\Theta},$$

as already obtained.

Art. We may also obtain the result by a direct evaluation of the integral. The potential of an elliptic plate on an external point being

$$V = 2fg\int_\theta^\infty \frac{dt}{\sqrt{t(t+f^2)(t+g^2)}},$$

where θ is the positive root of

$$\frac{a^2}{\theta+f^2} + \frac{b^2}{\theta+g^2} + \frac{c^2}{\theta} = 1,$$

let the attracted point be on the focal hyperbola, so that $b = 0$, $\dfrac{a^2}{f^2 - g^2} = 1$; then $\theta = \dfrac{f^2c^2}{g^2}$, and

$$V = 2fg \int_{\frac{f^2c^2}{g^2}}^{\infty} \frac{dt}{\sqrt{t(t+f^2)(t+g^2)}}.$$

Writing $t = f^2 \tan^2 \phi$, this becomes

$$\begin{aligned} V &= 4fg \int_{\tan^{-1}\frac{c}{g}}^{\frac{\pi}{2}} \frac{d\phi}{\sqrt{f^2 \sin^2 \phi + g^2 \cos^2 \phi}} \\ &= 4g \int_{\tan^{-1}\frac{c}{g}}^{\frac{\pi}{2}} \frac{d\phi}{\sqrt{\sin^2 \phi + \frac{g^2}{f^2} \cos^2 \phi}}. \end{aligned}$$

But when f varies and the point remains on the focal hyperbola at altitude c, we have $a^2 = f^2 - g^2$, and the value V is independent of f; taking $f = \infty$, we find

$$V = 4g \int_{\tan^{-1}\frac{c}{g}}^{\frac{\pi}{2}} \frac{d\phi}{\sin \phi} = 4g \left[-\log \cot \frac{\phi}{2} \right]_{\tan^{-1}\frac{c}{g}}^{\frac{\pi}{2}} = 2g \log \frac{\sqrt{c^2+g^2}+g}{\sqrt{c^2+g^2}-g},$$

or as before $V = 2\pi(\sqrt{c^2+g^2} - c)$.

Art. The potential of the circle may be obtained directly. Taking the circle $x^2 + y^2 = g^2$, the potential on the point $(0, 0, c)$ is

$$V = \iint \frac{dx\,dy}{\sqrt{x^2+y^2+c^2}} = \int_0^{2\pi} d\phi \int_0^g \frac{r\,dr}{\sqrt{r^2+c^2}} = 2\pi(\sqrt{g^2+c^2} - c),$$

as already found.

Art. The theorem may be stated as follows: If a series of confocal ellipses be described, and from any point of the focal hyperbola as vertex a right cone be drawn having the ellipse for its section by a plane perpendicular to the tangent at the vertex; then the solid angle of the cone, multiplied by the distance of the vertex from the plane, is constant (in fact $= 2\pi(\sqrt{c^2+g^2} - c)$).

Art. We may verify this by direct calculation of the solid angle. The element of solid angle is $d\omega = \sin\theta\, d\theta\, d\phi$, and the whole solid angle is

$$\omega = \int_0^{2\pi} d\phi \int_0^{\Theta} \sin\theta\, d\theta = 2\pi(1 - \cos\Theta),$$

where Θ is the semi-aperture. The potential is then $V = c\omega = 2\pi c(1-\cos\Theta)$, which agrees with the former result.

Art. The potential of the ellipse may also be expressed in terms of elliptic functions. Writing for shortness

$$k^2 = \frac{f^2 - g^2}{f^2}, \quad \sin^2\psi = \frac{f^2}{\theta + f^2},$$

then

$$V = \frac{4fg}{f}\int_0^{\psi} \frac{d\psi}{\sqrt{1 - k^2\sin^2\psi}} = 4gF(k,\psi),$$

where $F(k,\psi)$ is the elliptic integral of the first kind.

Art. In the case where the attracted point is in the plane of the ellipse, $c = 0$, we have $\theta = 0$ if $a^2 + b^2 < f^2$ (interior point), or θ positive if $a^2 + b^2 > f^2$ (exterior point). For an interior point, the potential is

$$V = 2fg\int_0^{\infty} \frac{dt}{\sqrt{t(t+f^2)(t+g^2)}} = 4gK(k),$$

where $K(k)$ is the complete elliptic integral of the first kind.

Art. For a point on the axis of the ellipse, $a = 0$, $b = 0$, the equation for θ becomes

$$\frac{c^2}{\theta} = 1, \quad \text{or} \quad \theta = c^2,$$

and the potential is

$$V = 2fg\int_{c^2}^{\infty} \frac{dt}{\sqrt{t(t+f^2)(t+g^2)}}.$$

If $f = g$, the ellipse becomes a circle, and we find

$$V = 2\pi(\sqrt{c^2 + f^2} - c),$$

as before.

Art. For the general case, we may express the result as follows. Let e be the eccentricity of the ellipse, so that $g^2 = f^2(1 - e^2)$; then

$$V = 4f(1-e^2)^{\frac{1}{2}}\int_0^{\psi} \frac{d\psi}{\sqrt{1 - e^2\sin^2\psi}},$$

where $\sin^2\psi = \dfrac{f^2}{\theta + f^2}$, and θ is determined by

$$\frac{a^2}{\theta + f^2} + \frac{b^2}{\theta + f^2(1 - e^2)} + \frac{c^2}{\theta} = 1.$$

Art. In particular, if the attracted point is on the normal at the end of the major axis, $a = f$, $b = 0$, then

$$\frac{f^2}{\theta + f^2} + \frac{c^2}{\theta} = 1,$$

giving $\theta^2 + f^2\theta - c^2 f^2 - c^2\theta = \theta^2 + (f^2 - c^2)\theta - c^2 f^2 = 0$, or

$$\theta = \frac{-(f^2 - c^2) + \sqrt{(f^2 - c^2)^2 + 4c^2 f^2}}{2} = \frac{-(f^2 - c^2) + \sqrt{(f^2 + c^2)^2}}{2} = \frac{-(f^2 - c^2) -}{2}$$

or $\theta = -f^2$ which is inadmissible; so $\theta = c^2$, and

$$\sin^2\psi = \frac{f^2}{c^2 + f^2}, \quad \cos^2\psi = \frac{c^2}{c^2 + f^2}.$$

Art. Suppose, to fix the ideas, $f = 1$, and consider the points $(0, c)$ and $(a, 0)$, which have equal potentials. First, if $a > f$ (that is $a > 1$), the point $(a, 0)$ is outside the ellipse, and its potential is less than that of the centre; while the point $(0, c)$, if c be properly chosen, may have any potential from 0 to ∞. The condition for equal potentials gives a relation between a and c.

Art. For the point $(0, c)$, on the minor axis, the equation for θ is

$$\frac{c^2}{\theta} + \frac{0}{\theta + f^2} + \frac{0}{\theta + g^2} = 1, \quad \text{i.e.} \quad \theta = c^2.$$

The potential is

$$V_1 = 2fg \int_{c^2}^{\infty} \frac{dt}{\sqrt{t(t + f^2)(t + g^2)}}.$$

For the point $(a, 0)$, on the major axis, the equation is

$$\frac{a^2}{\theta + f^2} = 1, \quad \theta = a^2 - f^2.$$

The potential is

$$V_2 = 2fg \int_{a^2 - f^2}^{\infty} \frac{dt}{\sqrt{t(t + f^2)(t + g^2)}}.$$

For these to be equal, we must have $c^2 = a^2 - f^2$, or $c = \sqrt{a^2 - f^2}$.

Art. This result may be verified directly. If $c = \sqrt{a^2 - f^2}$, then for the point $(0, c)$:

$$V_1 = 2fg \int_{a^2-f^2}^{\infty} \frac{dt}{\sqrt{t(t+f^2)(t+g^2)}},$$

and for the point $(a, 0)$:

$$V_2 = 2fg \int_{a^2-f^2}^{\infty} \frac{dt}{\sqrt{t(t+f^2)(t+g^2)}},$$

so that $V_1 = V_2$, as stated.

Art. The potential of the ellipse may be expressed in a different form by means of the formula

$$\int_{\theta}^{\infty} \frac{dt}{\sqrt{t(t+f^2)(t+g^2)}} = \frac{2}{\sqrt{\theta+f^2}} F\left(\sqrt{\frac{f^2-g^2}{\theta+f^2}}, \frac{\pi}{2}\right),$$

when $\theta > 0$. For an external point, the potential is thus expressed in terms of the complete elliptic integral of the first kind.

Art. The results obtained for the ellipse may be extended to the ellipsoid. The potential of an ellipsoid with semiaxes (f, g, h) on the external point (a, b, c) is

$$V = 2fgh \int_{\theta}^{\infty} \frac{dt}{\sqrt{(t+f^2)(t+g^2)(t+h^2)}},$$

where θ is the positive root of

$$\frac{a^2}{\theta+f^2} + \frac{b^2}{\theta+g^2} + \frac{c^2}{\theta+h^2} = 1.$$

This is the well-known formula of Ivory and Maclaurin.

Art. For the circle, putting $f = g$ in the formula for the ellipse, the equation for θ becomes

$$\frac{a^2+b^2}{\theta+f^2} + \frac{c^2}{\theta} = 1,$$

and the potential is

$$V = 2f^2 \int_{\theta}^{\infty} \frac{dt}{(t+f^2)\sqrt{t}} = 4f^2 \left[\frac{1}{f} \tan^{-1} \frac{\sqrt{t}}{f}\right]_{\theta}^{\infty} = 2\pi f - 4f \tan^{-1} \frac{\sqrt{\theta}}{f}.$$

Art. For the axial point $(0,0,c)$, we have $\theta = c^2$ if we write c for the distance, and then

$$V = 2\pi f - 4f\tan^{-1}\frac{c}{f} = 4f\left(\frac{\pi}{2} - \tan^{-1}\frac{c}{f}\right) = 4f\tan^{-1}\frac{f}{c},$$

or equivalently $V = 2\pi(\sqrt{c^2+f^2} - c)$, which is the same as before.

Art. The case of the circle may be treated independently by direct integration. Taking the circle $x^2+y^2=f^2$, and the attracted point $(0,0,c)$, the potential is

$$V = \int_0^{2\pi} d\phi \int_0^f \frac{r\,dr}{\sqrt{r^2+c^2}} = 2\pi\left[\sqrt{r^2+c^2}\right]_0^f = 2\pi(\sqrt{f^2+c^2} - c).$$

For an eccentric point (a,b,c), with $a^2+b^2=d^2$, the potential may be shown to be

$$V = 2\pi\sqrt{f^2+c^2+d^2-2fd\cos\psi} - 2\pi c,$$

where ψ is an auxiliary angle; or, more simply,

$$V = 2\pi(\sqrt{R^2+c^2} - c),$$

where R is the greatest distance of the point from the circumference of the circle.

0.2 604. Determination of the Attraction of an Ellipsoidal Shell on an Exterior Point.

[From the *Proceedings of the London Mathematical Society*, vol. VI. (1874–1875), pp. 58–67.
Read January 14, 1875.]

Art. The attraction of an ellipsoidal shell upon an exterior point may be determined by a method similar to that employed for the solid ellipsoid. Consider a homogeneous shell bounded by two similar and similarly situated concentric ellipsoids; say the outer ellipsoid has semiaxes (f, g, h), and the inner ellipsoid has semiaxes (f', g', h'), where $\dfrac{f'}{f} = \dfrac{g'}{g} = \dfrac{h'}{h} = m$; the thickness of the shell is variable, being proportional to the distance from the centre.

Art. Let $P = (a, b, c)$ be an exterior point, and draw through P a line cutting the ellipsoid; let this line meet the outer ellipsoid in R' and R'', and the inner ellipsoid in r' and r''. Let Q be the point where the line meets the polar plane of P with respect to the outer ellipsoid. Then the attractions of the two ellipsoids upon P are inversely proportional to the squares of their masses, and the attraction of the shell is the difference of these attractions.

Art. Taking a confocal ellipsoid through P, let its semiaxes be (F, G, H); then the potential of the outer ellipsoid at P is proportional to $\dfrac{1}{FGH}$, and the attraction is directed towards the centre. The shell may be regarded as composed of elemental shells, each bounded by similar ellipsoids, and the total attraction is the integral of the attractions of these elemental shells.

Art. Let the equation of the outer ellipsoid be

$$\frac{x^2}{f^2} + \frac{y^2}{g^2} + \frac{z^2}{h^2} = 1,$$

and let the inner ellipsoid be

$$\frac{x^2}{f'^2} + \frac{y^2}{g'^2} + \frac{z^2}{h'^2} = 1,$$

where $f' = mf$, $g' = mg$, $h' = mh$, with $m < 1$. The shell is the region between these two surfaces.

Art. Consider a ray from P meeting the outer ellipsoid at R' and R'', and the inner at r' and r''. Let ρ be the distance from P to a point on the ray, and let $\delta\rho$ be an element of length. The density being uniform, the mass of the element of the shell contained between ρ and $\rho + d\rho$ is proportional to the area of the cross-section; and the attraction of this element on P is inversely proportional to ρ^2.

Art. Let λ, μ, ν be the direction-cosines of the ray from P; then a point on the ray is $(a+\lambda\rho, b+\mu\rho, c+\nu\rho)$. This point lies on the outer ellipsoid if

$$\frac{(a+\lambda\rho)^2}{f^2} + \frac{(b+\mu\rho)^2}{g^2} + \frac{(c+\nu\rho)^2}{h^2} = 1,$$

or, writing

$$A = \frac{\lambda^2}{f^2} + \frac{\mu^2}{g^2} + \frac{\nu^2}{h^2}, \quad B = \frac{a\lambda}{f^2} + \frac{b\mu}{g^2} + \frac{c\nu}{h^2}, \quad C = \frac{a^2}{f^2} + \frac{b^2}{g^2} + \frac{c^2}{h^2} - 1,$$

we have $A\rho^2 + 2B\rho + C = 0$, giving the distances ρ_1, ρ_2 to the points R', R''.

Art. Similarly, for the inner ellipsoid, writing

$$A' = \frac{\lambda^2}{f'^2} + \frac{\mu^2}{g'^2} + \frac{\nu^2}{h'^2} = \frac{1}{m^2}A, \quad B' = \frac{1}{m^2}B, \quad C' = \frac{1}{m^2}C + \frac{1}{m^2} - 1,$$

the distances to r', r'' are given by $A'\rho^2 + 2B'\rho + C' = 0$.

Art. The attraction of the shell on P in the direction (λ, μ, ν) is found by integrating the attraction of the elemental slice. If $d\omega$ is the element of solid angle, the resolved attraction is

$$dX = \lambda\, d\omega \int \rho\, d\rho,$$

the integral being taken over the material of the shell along the ray. This gives

$$dX = \lambda\, d\omega \cdot \frac{1}{2}(R'^2 - R''^2 - r'^2 + r''^2),$$

where R', R'' are the roots for the outer surface, and r', r'' for the inner surface.

Art. The mass of the shell being

$$M = \frac{4\pi}{3} fgh(1-m^3),$$

the attraction of the shell is found to be

$$X = \frac{3M}{4\pi fgh(1-m^3)} \iint \lambda\,(R'^2 - R''^2 - r'^2 + r''^2)\, d\omega.$$

By known properties of the ellipsoid, this reduces to a simple form involving only elementary functions.

Art. The final result for the attraction of the shell on the exterior point P is as follows. Let θ be the positive root of

$$\frac{a^2}{\theta+f^2}+\frac{b^2}{\theta+g^2}+\frac{c^2}{\theta+h^2}=1,$$

and let (F, G, H) be the semiaxes of the confocal ellipsoid through P, so that $F^2=\theta+f^2$, $G^2=\theta+g^2$, $H^2=\theta+h^2$. Then the attraction of the shell is the same as that of a solid ellipsoid of semiaxes (F, G, H) and density adjusted so that the mass is equal to that of the shell; viz. the attraction is directed towards the centre, and its magnitude is

$$\frac{3M}{\sqrt{(\theta+f^2)(\theta+g^2)(\theta+h^2)}}\cdot\frac{1}{\theta^{\frac{3}{2}}}\times\text{(proper numerical factor)}.$$

Art. Analytical Expressions for the Attraction of the Shell, and for the Resolved Attractions.

Art. The attraction of the shell was shown to be expressible in terms of the confocal ellipsoid through the attracted point. If the semiaxes of this confocal be (F, G, H), then the attraction of the shell is proportional to the attraction of the solid ellipsoid with semiaxes (F, G, H); and the resolved attractions are

$$X=-\frac{3Ma}{\theta}\cdot\frac{1}{FGH},\quad Y=-\frac{3Mb}{\theta}\cdot\frac{1}{FGH},\quad Z=-\frac{3Mc}{\theta}\cdot\frac{1}{FGH},$$

where θ is determined as above.

Art. The mass of the shell is $\frac{4\pi}{3}fgh\cdot S_m$, where S_m is a factor depending on the thickness; whence

$$\text{Resolved Attraction}=\frac{M}{\text{Mass of confocal}}\times\text{Attraction of confocal},$$

and since the attraction of the confocal solid ellipsoid is known, we have the result.

Art. The direction-cosines of the attraction are proportional to $\frac{a}{\theta+f^2}$, $\frac{b}{\theta+g^2}$, $\frac{c}{\theta+h^2}$; and the resultant attraction is

$$R=\sqrt{X^2+Y^2+Z^2}=\frac{3M}{FGH}\sqrt{\frac{a^2}{(\theta+f^2)^2}+\frac{b^2}{(\theta+g^2)^2}+\frac{c^2}{(\theta+h^2)^2}}.$$

Art. For a spherical shell, $f = g = h$, the formula simplifies considerably. The confocal through P is a concentric sphere of radius $\sqrt{\theta + f^2}$, where $\theta = a^2 + b^2 + c^2 - f^2 = d^2 - f^2$, d being the distance of P from the centre. The attraction is then

$$X = -\frac{M}{d^2} \cdot \frac{a}{d}, \quad Y = -\frac{M}{d^2} \cdot \frac{b}{d}, \quad Z = -\frac{M}{d^2} \cdot \frac{c}{d},$$

which is Newton's well-known result: the spherical shell attracts as if its mass were concentrated at the centre.

0.3 605. Note on a Point in the Theory of Attraction.

[From the *Proceedings of the London Mathematical Society*, vol. VI. (1874–1875), pp. 79–81.
Read February 11, 1875.]

Art. Consider a mass of matter distributed in any manner throughout space, and let V be the potential at any point. The potential satisfies Laplace's equation $\nabla^2 V = 0$ at all points external to the attracting mass, and Poisson's equation $\nabla^2 V = -4\pi\rho$ at points where the density is ρ.

Art. It is important to observe that the potential and its first derivatives are everywhere continuous, even at the surface of the attracting body. The second derivatives, however, are discontinuous at the surface; in fact, if V_1 is the potential outside and V_2 inside, then at the surface

$$V_1 = V_2, \quad \frac{\partial V_1}{\partial n} = \frac{\partial V_2}{\partial n},$$

but

$$\frac{\partial^2 V_1}{\partial n^2} - \frac{\partial^2 V_2}{\partial n^2} = -4\pi\sigma,$$

where σ is the surface-density.

Art. For a spherical shell; the potential is a different function for external and interior points, viz. for internal points it is a constant, $= M \div$ radius; for external points it is $= M \div$ distance from the centre. But at the surface itself, the two expressions coincide, and the first derivative is continuous. The second derivative, however, has a discontinuity proportional to the surface-density.

Art. This is best shown by the annexed figure, which represents a uniform spherical shell made up of two segments, one of which is taken to be a small segment at the point of observation. The potential due to the small segment varies continuously, but its normal derivative changes abruptly by $4\pi\sigma$ on crossing the surface.

Art. The general theorem may be stated as follows: If V be the potential of any distribution of matter, and if we cross a surface at which there is a stratum of matter of surface-density σ, then the normal derivative of V changes by $-4\pi\sigma$; that is,

$$\left(\frac{\partial V}{\partial n}\right)_1 - \left(\frac{\partial V}{\partial n}\right)_2 = -4\pi\sigma,$$

where the suffixes 1 and 2 refer to the two sides of the surface. This is the well-known theorem of Green.

Art. The proof is immediate from the consideration of the elemental tube of force. Taking a small cylinder with its axis along the normal to the surface, and its two ends on opposite sides of the surface, Gauss's theorem gives at once the discontinuity of the normal force. Since the potential itself is continuous, and since the tangential derivatives are also continuous (being determined solely by the values of V on the surface), the theorem is completely established.

0.4 606. On the Expression of the Coordinates of a Point of a Quartic Curve as Functions of a Parameter.

[From the *Proceedings of the London Mathematical Society*, vol. VI. (1874–1875), pp. 81–83.
Read February 11, 1875.]

Art. The question of expressing the coordinates of a point on a plane curve as rational functions of a parameter is one of great importance in the theory of curves. For a conic, the solution is well known; for a cubic, it can be effected by means of elliptic functions. For a quartic curve, the problem is more complicated, but can be solved in certain cases by means of elliptic functions.

Art. Consider a quartic curve having a double point; such a curve may be regarded as the reciprocal of a cubic. Let the equation of the quartic be

$$U=(ax+by+c)^2(x^2+y^2+1)+2(ax+by+c)(dx^2+exy+fy^2+gx+hy+k)+\cdots=0.$$

By a suitable transformation, this may be reduced to a standard form.

Art. Let the quartic have a node at the origin; then the equation may be written

$$(ax^2+2hxy+by^2)^2+2(ax+by+c)(a'x^2+2h'xy+b'y^2)+\cdots=0.$$

Taking a line $y=tx$ through the node, the other intersections are given by a quadratic equation, whose roots determine two points on the curve. If these coincide, the line is a tangent; but in general, we obtain two points corresponding to each value of t.

Art. The condition for the line $y=tx$ to meet the quartic in two coincident points (other than the node) gives a quartic equation in t; but by introducing an auxiliary parameter, the roots of this quartic may be expressed in terms of elliptic functions. In fact, the quartic equation in t may be written

$$At^4+Bt^3+Ct^2+Dt+E=0,$$

and by the substitution $t=\tan\phi$, this may be reduced to the form

$$P\cos 2\phi+Q\sin 2\phi+R\cos 4\phi+S\sin 4\phi=T,$$

which is solvable by elliptic functions.

Art. Writing for a moment $ax + b = 2\sqrt{b^2 - ac} \cdot u$, the equation becomes

$$4u^2 - 3u + \frac{a^2d - 3abc + 2b^3}{2(b^2 - ac)\sqrt{b^2 - ac}} = 0.$$

Hence, assuming

$$\frac{a^2d - 3abc + 2b^3}{2(b^2 - ac)\sqrt{b^2 - ac}} = \cos\phi,$$

we have

$$4u^2 - 3u + \cos\phi = 0,$$

which gives u in terms of ϕ.

Art. The roots of this cubic in u are connected with the trisection of the angle; in fact, $u = \cos\frac{\phi+2k\pi}{3}$ for $k = 0, 1, 2$. The three values of u correspond to the three real intersections of the line with the cubic curve.

Art. For the quartic curve, we proceed as follows. Let the equation be

$$y^2 = (x-\alpha)(x-\beta)(x-\gamma)(x-\delta),$$

which is the standard form of a quartic with all roots real. By the substitution

$$x = \frac{\alpha(x' - \gamma) - \gamma(x' - \alpha)}{(x' - \gamma) - (x' - \alpha)} = \frac{(\alpha - \gamma)x'}{\alpha - \gamma},$$

this may be reduced to the form

$$y'^2 = x'(1 - x')(1 - k^2x'),$$

which is the elliptic normal form. The coordinates are then expressible as elliptic functions of a parameter.

Art. In general, for any plane quartic curve, the coordinates may be expressed as elliptic functions of a parameter by means of the following construction. Take two fixed points A, B on the curve, and let P be a variable point. The line AP meets the curve again in P', and the line BP' meets again in P'', and so on. The correspondence between P and P'' is a $(1,1)$ correspondence, and the parameter of P'' differs from that of P by a constant. Hence the parameter of P may be taken as an elliptic parameter.

Art. The result may be stated more precisely: A plane quartic curve is of deficiency 3; but if it have a double point, the deficiency is reduced to 2, and if it have two double points, to 1. In the latter case, the curve is unicursal, and the coordinates may be expressed rationally in terms of a parameter. If the deficiency is 2, the coordinates may be expressed in terms of elliptic functions. For the general quartic (deficiency 3), the coordinates require hyperelliptic functions of two arguments.

Art. And consequently, for the case of a single double point (deficiency 2), the coordinates (x, y) may be expressed as elliptic functions of a single parameter u. The relation between x and y is given by the equation of the curve, and either coordinate may be taken as the independent elliptic variable. The periods of the elliptic functions are determined by the moduli, which are algebraic functions of the coefficients of the quartic.

0.5 607. A Memoir on Prepotentials.

[From the *Philosophical Transactions of the Royal Society of London*, vol. CLXV. Part II. (1875), pp. 675–774.
Received April 8,—Read June 10, 1875.]

Introduction.

Art. The present memoir is concerned with the theory of "prepotentials," a term which I use to denote a class of integrals generalizing the ordinary potential of attracting matter. The potential of a body on an external point depends on the inverse distance; the prepotential involves an arbitrary index q, and reduces to the potential for particular values of q.

Art. The ordinary potential of a body of density ρ at a point (a, b, c) is

$$V = \iiint \frac{\rho\, dx\, dy\, dz}{\sqrt{(a-x)^2 + (b-y)^2 + (c-z)^2}}.$$

More generally, the prepotential is defined as

$$W = \iiint \frac{\rho\, dx\, dy\, dz}{\{(a-x)^2 + (b-y)^2 + (c-z)^2\}^{\frac{1}{2}q}},$$

where q is any positive or negative number. When $q = 1$, this reduces to the ordinary potential; when $q = 3$, it is the second polar; and so on.

Art. The prepotential may be considered in various cases according to the dimension of the attracting body:

A. The potential-point case; q arbitrary, the attracting body being a single point.

B. The potential-curve case; q arbitrary, the attracting body being a curve.

C. The potential-surface case; $q = -\frac{1}{2}$, the surface arbitrary, so that the integral is

$$\iint \frac{\rho\, dS}{\{(a-x)^2 + \cdots + (c-z)^2 + (e-w)^2\}^{\frac{1}{2}}}.$$

D. The potential-solid case; $q = -\frac{1}{2}$, the solid arbitrary.

Art. The cases of a less-than-three-dimensional volume are in the present memoir considered only incidentally. It is sufficient to remark that the line-integral and the surface-integral may be deduced as limiting cases of the volume integral, by supposing the density to become infinite in a certain manner.

Art. The potential-surface integral, provided that the point $(a, \dots, c, e)$ does not lie within the material space, satisfies Laplace's equation in the coordinates of the attracted point. I would rather say that the integral does not satisfy Laplace's equation when the attracted point is within the material space; for in this case the integral has a different form, depending on the side from which the surface is approached.

Case $q =$ Positive.

Art. I consider first the case where q is positive. The value is here

$$W = \frac{1}{2\pi^2} \frac{\Gamma(\frac{1}{2}q)}{\Gamma(\frac{1}{2}n - \frac{1}{2}q)} \int_0^\infty \frac{u^{\frac{1}{2}n - \frac{1}{2}q - 1}\, du}{(1+u)^{\frac{1}{2}n}},$$

or, since $\dfrac{1}{2\pi^2}$ is indefinitely small when n is large, we may write

$$W = \frac{\Gamma(\frac{1}{2}q)}{\pi^{\frac{1}{2}n}} \int \frac{\rho\, d\omega}{r^q},$$

where $d\omega$ is the element of volume, and r the distance.

Art. The prepotential of a plane surface of uniform density on a point at distance e from the plane is required in the case where the distance e (taken to be always positive) is indefinitely small in regard to the radius R. Writing $x = r\xi$, $y = r\eta$, $z = r\zeta$, where r is a length, the integral becomes

$$W = \int_0^R \frac{r^{n-1}\, dr}{r^q} \int \frac{\rho\, d\Omega}{(1 + \text{terms in } \xi, \eta, \dots)^{\frac{1}{2}q}},$$

where $d\Omega$ is the element of solid angle.

Art. For a disk of radius R and surface-density ρ, the prepotential at a point on the axis at distance e is

$$W = 2\pi\rho \int_0^R \frac{r\, dr}{(r^2 + e^2)^{\frac{1}{2}q}}.$$

When $q = 1$, this gives the ordinary potential; when $q > 1$, the integral converges even for $e = 0$; and when $q < 1$, the value for $e = 0$ requires special consideration.

Art. Evaluating the integral, we have

$$W = \frac{2\pi\rho}{2-q}\left[\frac{1}{(R^2+e^2)^{\frac{1}{2}q-1}} - \frac{1}{e^{q-2}}\right]$$

for $q \neq 2$; and for $q = 2$,

$$W = \pi\rho\log\frac{R^2+e^2}{e^2}.$$

When e is small compared with R, this becomes approximately

$$W = \frac{2\pi\rho}{2-q}R^{2-q} \quad (q < 2),$$

and for $q = 2$,

$$W = 2\pi\rho\log\frac{R}{e}.$$

Art. The prepotential of the remaining portion of the surface vanishes in comparison with that of the disk. But without entering into the question I assume that the prepotential of the whole surface may be obtained by integration, and that the result is finite for all values of q less than the dimension of the space.

Art. The differential equation satisfied by the prepotential may be obtained as follows. We have

$$\frac{\partial W}{\partial a} = q\iiint\frac{\rho(x-a)\,dx\,dy\,dz}{r^{q+2}},$$

and

$$\frac{\partial^2 W}{\partial a^2} = q\iiint\frac{\rho\{q(x-a)^2 - r^2\}\,dx\,dy\,dz}{r^{q+4}}.$$

Adding the similar equations for b and c, we obtain

$$\nabla^2 W = q(q-1)\iiint\frac{\rho\,dx\,dy\,dz}{r^{q+2}},$$

or

$$\nabla^2 W = q(q-1)W_{q+2},$$

where W_{q+2} denotes the prepotential with index $q+2$.

Art. This equation connects the prepotentials of different orders. In particular, for the ordinary potential ($q = 1$), we have $\nabla^2 W_1 = 0$ at external points; and for $q = 3$, $\nabla^2 W_3 = 6W_5$. These relations are useful in the solution of problems in attraction.

Art. The value of $\dfrac{\partial W}{\partial e}$, where W is given as a function of $(x, \ldots, z, w)$, requires explanation. Taking $\cos\alpha$, …, $\cos\gamma$ to be the inclinations of the normal to the coordinate axes, we have

$$\frac{\partial W}{\partial e} = \cos\alpha \frac{\partial W}{\partial x} + \cdots + \cos\gamma \frac{\partial W}{\partial z},$$

and this expression determines the rate of change of W in the direction of the normal.

Art. The prepotential of a solid sphere of radius R and uniform density ρ on an external point at distance f from the centre is

$$W = \frac{4\pi\rho}{3-q} \frac{R^3}{f^q} \cdot {}_2F_1\left(\frac{1}{2}q, \frac{3}{2}; \frac{5}{2}; \frac{R^2}{f^2}\right),$$

where ${}_2F_1$ is the hypergeometric function. For the ordinary potential ($q = 1$), this reduces to $\dfrac{M}{f}$, where $M = \dfrac{4\pi\rho R^3}{3}$ is the mass.

Art. When the attracted point is inside the sphere, the result is different. We have

$$W = \frac{4\pi\rho}{3-q}\left[\frac{R^{3-q}}{3-q} + \int_r^R \frac{r'^{2-q}\,dr'}{r'^{1-q}}\right],$$

which for $q = 1$ gives

$$W = 2\pi\rho\left(R^2 - \frac{1}{3}r^2\right),$$

the well-known formula for the internal potential of a uniform sphere.

Art. The general formula for the prepotential of a uniform solid sphere is

$$W = \frac{4\pi\rho}{(3-q)(2-q)}\left[R^{2-q} - \frac{2-q}{2}r^2 R^{-q} + \cdots\right],$$

where ρ here denotes the density at the point N; and the value of the integral is

$$W = \frac{1}{2}R^{2-q}(1 + \text{terms in } \xi, \eta, \ldots)^{-\frac{1}{2}q} \cdot \frac{\rho}{\text{etc.}}$$

Observe that this is indefinitely small, as it should be, when the point is outside the sphere.

Art. The prepotential of a surface-distribution on a point near the surface may be treated similarly. Let σ be the surface-density, and let the surface be plane near the attracted point. Then the prepotential is

$$W = \sigma \iint \frac{dS}{r^q},$$

and for a circular area of radius R,

$$W = \frac{2\pi\sigma}{2-q}\left[(R^2+e^2)^{1-\frac{1}{2}q} - e^{2-q}\right].$$

When $e = 0$, this becomes $W = \dfrac{2\pi\sigma R^{2-q}}{2-q}$ for $q < 2$; and for $q = 2$, $W = 2\pi\sigma \log R$.

Art. The discontinuity in the normal derivative of the prepotential at a surface may be found as follows. Let W_+ be the value on the positive side, and W_- on the negative side; then

$$\frac{\partial W_+}{\partial n} - \frac{\partial W_-}{\partial n} = -\frac{2\pi^{\frac{1}{2}n}\sigma}{\Gamma(\frac{1}{2}n-1)} \cdot \frac{1}{e^{q-1}},$$

where $(\cos\alpha', \dots, \cos\gamma')$ and $(\cos\alpha'', \dots, \cos\gamma'')$ are the cosine-inclinations of the normal on the two sides of the surface. For the ordinary potential ($q = 1$), this reduces to the theorem of Green.

Art. Reverting to the original form of the prepotential, we have for the solid case

$$W = \iiint \frac{\rho\, dx\, dy\, dz}{\{(a-x)^2+(b-y)^2+(c-z)^2\}^{\frac{1}{2}q}}.$$

To get rid of the constant factors, we may write this as

$$W = \frac{1}{\Gamma(\frac{1}{2}q)} \int_0^\infty \lambda^{\frac{1}{2}q-1}\, d\lambda \iiint \rho\, dx\, dy\, dz\, e^{-\lambda r^2},$$

which is a useful form for theoretical investigations.

Art. By differentiation under the integral sign, we obtain formulae for the derivatives of the prepotential. For instance,

$$\frac{\partial W}{\partial a} = \frac{q}{\Gamma(\frac{1}{2}q)} \int_0^\infty \lambda^{\frac{1}{2}q}\, d\lambda \iiint \rho(x-a)\, dx\, dy\, dz\, e^{-\lambda r^2},$$

and similarly for the higher derivatives.

Art. The prepotential may be expanded in a series of spherical harmonics when the attracting body is a sphere or an ellipsoid. For a sphere of radius R, we have

$$W = \frac{M}{f^q}\left[1 + \frac{q(q+2)}{6}\frac{R^2}{f^2} + \frac{q(q+2)(q+4)}{120}\frac{R^4}{f^4} + \cdots\right]$$

for an external point at distance $f > R$; and for an internal point at distance $r < R$,

$$W = \frac{M}{R^q}\left[1 + \frac{q(2-q)}{6}\frac{r^2}{R^2} + \frac{q(2-q)(4-q)}{120}\frac{r^4}{R^4} + \cdots\right].$$

Art. The case $q = 0$ gives simply the mass of the body; $q = -1$ gives the moment of inertia about the attracted point; and higher negative values of q give higher moments. The case $q = 2$ gives the potential of a line-distribution; $q = 3$ gives the potential of a surface-distribution; and so on.

Art. For the ellipsoid, with semiaxes (f, g, h) and uniform density ρ, the prepotential on an external point (a, b, c) is

$$W = \pi\rho fgh \int_\theta^\infty \frac{dt}{\sqrt{(t+f^2)(t+g^2)(t+h^2)}} \cdot \frac{1}{(t+\theta)^{\frac{1}{2}q}},$$

where θ is the positive root of the equation

$$\frac{a^2}{\theta+f^2} + \frac{b^2}{\theta+g^2} + \frac{c^2}{\theta+h^2} = 1.$$

This formula generalizes the well-known result of Ivory and Maclaurin.

Art. The prepotential has important applications in the theory of elasticity, in the theory of heat, and in electricity. In elasticity, the displacement due to a body-force may be expressed in terms of prepotentials; in the theory of heat, the temperature due to a variable source involves prepotential integrals; and in electricity, the potential of a variable charge-distribution is a prepotential.

Art. The memoir concludes with a table of the principal formulae for prepotentials in one, two, and three dimensions, for various values of the index q, and for the principal forms of attracting bodies: the point, the line, the disk, the sphere, and the ellipsoid. These formulae are useful in the solution of physical problems involving generalized potentials.

[Paper 607. Continued from previous pages.]

Continuation of the Memoir

Art. The remaining Annexes VIII. and IX. have no immediate reference to the theorems A, B, C, D, which are the principal objects of the memoir. The subjects to which they relate will be seen from the headings and introductory paragraphs.

Annex I. Surface and Volume of Sphere $x^2+\cdots+z^2+w^2=f^2$. Art. Nos. 51 and 52.

Art. We require in $(s+1)$-dimensional space, $\int dx\cdots dz\,dw$, the volume of the sphere $x^2+\cdots+z^2+w^2=f^2$, and $\int dS$, the surface of the same sphere.

Writing

$$x=f\xi,\ldots,z=f\zeta,\quad w=f\omega,$$

we have

$$dx\cdots dz\,dw=f^{s+1}\,d\xi\cdots d\zeta\,d\omega,$$

with the limiting condition $\xi^2+\cdots+\zeta^2+\omega^2=1$; but in order to take account as well of the negative as the positive values of $x,\ldots,z,w$, we must multiply by 2^{s+1}. The value is therefore

$$=f^{s+1}\int \xi^0\cdots\zeta^0\omega^0\,d\xi\cdots d\zeta\,d\omega,$$

extended to all positive values of $\xi,\ldots,\zeta,\omega$ such that $\xi^2+\cdots+\zeta^2+\omega^2<1$; and we obtain this by a known theorem, viz.

$$\text{Volume of }(s+1)\text{-dimensional sphere}=f^{s+1}\frac{\pi^{\frac{s+1}{2}}}{\Gamma(\frac{s+1}{2}+1)}.$$

Writing $x=f\xi,\ldots,z=f\zeta,w=f\omega$, we obtain $dS=f^s\,d\Sigma$, where $d\Sigma$ is the element of surface of the unit-sphere $\xi^2+\cdots+\zeta^2+\omega^2=1$; we have element of volume $d\xi\cdots d\zeta\,d\omega=r^s\,dr\,d\Sigma$, where r is to be taken from 0 to 1, and thence

$$\int d\xi\cdots d\zeta\,d\omega=\int_0^1 r^s\,dr\int d\Sigma=\frac{1}{s+1}\int d\Sigma,$$

that is,

$$\int d\Sigma=(s+1)\int d\xi\cdots d\zeta\,d\omega;$$

consequently $\int d\Sigma$ = surface of $(s+1)$-dimensional sphere = $\dfrac{(s+1)\pi^{\frac{s+1}{2}}}{\Gamma(\frac{s+1}{2}+1)}$.

Art. Writing $s-1$ for s, we have

$$\text{Volume of } s\text{-dimensional sphere} = f^s \frac{\pi^{\frac{s}{2}}}{\Gamma(\frac{s}{2}+1)},$$

$$\text{Surface of do.} = \frac{s\pi^{\frac{s}{2}}}{\Gamma(\frac{s}{2}+1)} f^{s-1},$$

which forms are sometimes convenient.

Writing in the first forms $s+1=3$, or in the second forms $s=3$, we find in ordinary space

$$\text{Volume of sphere} = f^3 \frac{\pi^{\frac{3}{2}}}{\Gamma(\frac{5}{2})} = f^3 \frac{4\pi}{3},$$

$$\text{and} \quad \text{Surface of sphere} = \frac{3\pi^{\frac{3}{2}}}{\Gamma(\frac{5}{2})} f^2 = 4\pi f^2,$$

as they should be.

Annex II. The Integral $\int \frac{r^{s-1}\,dr}{(r^2+e^2)^{\frac{1}{2}s+q}}$. Art. Nos. 53 to 63.

Art. The integral in question (which occurs ante, No. 2) may also be considered as arising from a prepotential integral in tridimensional space; the prepotential of an element of mass dm is taken to be $= \delta^{2q+1}$, where δ is the distance of the element from the attracted point P. Hence if the element of mass be an element of the plane $z=0$, coordinates (x,y), ρ being the density, and if the attracted point be situate in the axis of z at a distance e from the origin, the prepotential is

$$V = \iint \frac{\rho\,dx\,dy}{(x^2+y^2+e^2)^{\frac{1}{2}+q}}.$$

For convenience, it is assumed throughout that e is positive.

Suppose that the attracting body is a circular disk, radius R, having the origin for its centre (viz. that bounded by the curve $x^2 +$

$y^2 = R^2$); then writing $x = r\cos\theta$, $y = r\sin\theta$, we have

$$V = \int \frac{\rho\, r^{s-1}\, dr\, d\theta}{(r^2+e^2)^{\frac{1}{2}s+q}},$$

which, if ρ is a function of r only, is

$$= 2\pi \int_0^R \frac{\rho\, r^{s-1}\, dr}{(r^2+e^2)^{\frac{1}{2}s+q}};$$

and in particular, if $\rho = r^{2m}$, then the value is

$$2\pi \int_0^R \frac{r^{2m+s-1}\, dr}{(r^2+e^2)^{\frac{1}{2}s+q}},$$

the integral in regard to r being taken from $r = 0$ to $r = R$. It is assumed that $s-1$ is not negative, viz. it is positive or (it may be) zero. I consider the integral

$$\int_0^R \frac{r^{s-1}\, dr}{(r^2+e^2)^{\frac{1}{2}s+q}},$$

which I call the r-integral, more particularly in the case where e is small in comparison with R. It is to be observed that e not being $= 0$, and R being finite, the integral contains no infinite element, and is therefore finite, whether q is positive, negative, or zero.

Art. Writing $r = e\sqrt{v}$, the integral is

$$= \tfrac{1}{2}e^{-2q} \int_0^{\frac{R^2}{e^2}} \frac{v^{\frac{1}{2}s-1}\, dv}{(1+v)^{\frac{1}{2}s+q}},$$

the limits being 0 and $\frac{R^2}{e^2}$.

In the case where q is positive, this is

$$= \tfrac{1}{2}e^{-2q} \left\{ \int_0^\infty - \int_{\frac{R^2}{e^2}}^\infty \right\} \frac{v^{\frac{1}{2}s-1}\, dv}{(1+v)^{\frac{1}{2}s+q}},$$

viz. the first term of this is

$$\frac{1}{2}e^{-2q}\frac{\Gamma(\frac{1}{2}s)\Gamma(q)}{\Gamma(\frac{1}{2}s+q)},$$

and the second term is a term expansible in a series containing the powers $2q$, $2q+2$, &c. of the small quantity $\frac{e^2}{R^2}$, as appears by effecting therein the substitution $v = \frac{1}{u}$; viz. the value of the entire integral is by this means found to be

$$\tfrac{1}{2}e^{-2q}\frac{\Gamma(\frac{1}{2}s)\Gamma(q)}{\Gamma(\frac{1}{2}s+q)} - \tfrac{1}{2}e^{-2q}\int_0^{\frac{e^2}{R^2}} \frac{u^{q-1}\,du}{(1+u)^{\frac{1}{2}s+q}}.$$

Art. In the case where q is $= 0$, or negative, the formula fails by reason that the element $\int^\infty v^{\frac{1}{2}s-1-q}\,dv$ of the integrals $\int_0^\infty$, $\int_{\frac{R^2}{e^2}}^\infty$ becomes infinite for indefinitely large values of v. Recurring to the original form $\int_0^R \frac{r^{s-1}\,dr}{(r^2+e^2)^{\frac{1}{2}s+q}}$, it is to be observed that the integral has a finite value when $e = 0$; and it might therefore at first sight be imagined that the factor $(r^2+e^2)^{-\frac{1}{2}s-q}$ might be expanded in ascending powers of e^2, and the value of the integral consequently obtained as a series of positive powers of e^2.

But the series thus obtained is of the form $\sum_k e^{2k}\int_0^R r^{-2q-2k-1}\,dr$, where $2q$ being positive, the exponent $-2q-2k-1$ is for a sufficiently small value of k at first positive, or if negative less than -1, and the value of the integral is finite; but as k increases the exponent becomes negative, and equal or greater than -1, and the value of the integral is then infinite. The inference is that the series commences in the form $A+Be^2+Ce^4\ldots$; but that we come at last when q is fractional to a term of the form Ke^{2q}, and when q is $= 0$ or is integral, to a term of the form $Ke^{2q}\log e$; the process giving the coefficients A, B, C, ..., so long as the exponent of the corresponding term e^0, e^2, e^4, ... is less than $-2q$ (in particular $q = 0$, there is a term $\log e$, and the expansion-process does not give any term of the result), and the failure of the series after this point being indicated by the values of the subsequent coefficients coming out $= \infty$.

Art. In illustration, we may consider any of the cases in which the integral can be obtained in finite terms. For instance,

$$s=1,\quad q=-1,\quad \text{Integral is } \int_0^R (r^2+e^2)^{\frac{1}{2}}\,dr = \tfrac{1}{2}(r^2+e^2)^{\frac{1}{2}}r+\tfrac{1}{2}e^2\log(r+\sqrt{r^2+e^2}),$$

viz. expanding in ascending powers of e, this is

$$= \tfrac{1}{2}R^2 + \tfrac{1}{4}e^2 + \tfrac{1}{4}e^2\log\frac{2R}{e} + \cdots$$

or we have here a term in $e^2 \log e$. And so,

$$s = 1, \quad q = -2, \quad \text{Integral is } \int_0^R (r^2{+}e^2)^{\frac{3}{2}}\, dr = \tfrac{1}{4}(r^2{+}e^2)^{\frac{3}{2}} r + \tfrac{3}{8} e^2 (r^2{+}e^2)^{\frac{1}{2}} r + \tfrac{3}{8} e^4\, \mathrm{lo}$$

viz. expanding in ascending powers of e, this is

$$= \tfrac{1}{4} R^4 + \tfrac{3}{8} R^2 e^2 + \cdots + \tfrac{3}{8} e^4 \log \frac{2R}{e},$$

or we have here a term in $e^4 \log e$.

Art. Returning to the form

$$\tfrac{1}{2} e^{-2q} \int_0^{\frac{R^2}{e^2}} \frac{v^{\frac{1}{2}s-1}\, dv}{(1+v)^{\frac{1}{2}s+q}},$$

and writing therein $v = \frac{1-x}{x}$, or what is the same thing, $x = \frac{1}{1+v}$, and for shortness $\chi = \frac{e^2}{R^2+e^2}$, the integral is

$$= \tfrac{1}{2} e^{-2q} \int_\chi^1 x^{q-1} (1-x)^{\frac{1}{2}s-1}\, dx,$$

where observe that $q - 1$ is 0 or negative, but χ being a positive quantity less than 1, the function $x^{q-1}(1-x)^{\frac{1}{2}s-1}$ is finite for the whole extent of the integration.

Art. If $q = 0$, this is

$$\tfrac{1}{2} \int_\chi^1 x^{-1} (1-x)^{\frac{1}{2}s-1}\, dx = \tfrac{1}{2} \int_\chi^1 \frac{(1-x)^{\frac{1}{2}s-1} - 1}{x}\, dx + \tfrac{1}{2} \int_\chi^1 \frac{dx}{x},$$

where observe that, in virtue of the change made from $-(1-x)^{\frac{1}{2}s-1}$ to $-\{(1-(1-x)^{\frac{1}{2}s-1}\}$ (a function which becomes infinite, to one which does not become infinite, for $x = 0$), it has become allowable in place of $\int_0^1$ to write $\int_0^1 - \int_0^\chi$.

When e is small, the integral which is the third term of the foregoing expression is obviously a quantity of the order e^s; the first term is $\frac{1}{2}(\log \frac{2R}{e} + \text{terms in } e^2)$, which, neglecting terms in e^2, is $= \frac{1}{2} \log \frac{2R}{e}$, and hence the approximate value of the r-integral $\int_0^R \frac{r^{s-1}\, dr}{(r^2+e^2)^{\frac{1}{2}s}}$ is

$$= \log \frac{2R}{e} - \int_0^1 \frac{1 - (1-x)^{\frac{1}{2}s-1}}{x}\, dx,$$

or what is the same thing, it is

$$= \log \frac{2R}{e} - \tfrac{1}{2}\psi(\tfrac{1}{2}s),$$

where the integral in this expression is a mere numerical constant, which, when $\frac{1}{2}s - 1$ is a positive integer, has the value

$$1 + \tfrac{1}{2} + \tfrac{1}{3} + \cdots + \tfrac{1}{\frac{1}{2}s-1};$$

neglecting this in comparison with the logarithmic term, the approximate value is $= \log \dfrac{2R}{e}$.

Art. I consider also the case $q = \frac{1}{2}$; the integral is here

$$\tfrac{1}{2}e^{-1} \int_\chi^1 x^{-\frac{1}{2}} (1-x)^{\frac{1}{2}s-1}\, dx;$$

$$= \tfrac{1}{2}e^{-1} \int_\chi^1 x^{-\frac{1}{2}} \{1 - (1-x)^{\frac{1}{2}s-1}\}\, dx + \tfrac{1}{2}e^{-1} \int_\chi^1 x^{-\frac{1}{2}}\, dx;$$

$$= e^{-1}(1 - \chi^{\frac{1}{2}}) - \tfrac{1}{2}e^{-1} \int_0^\chi x^{-\frac{1}{2}} \{1 - (1-x)^{\frac{1}{2}s-1}\}\, dx;$$

Annex III. Prepotentials of Uniform Spherical Shell and Solid Sphere. Art. Nos. 64 to 92.

Art. The prepotential of the uniform spherical shell $x^2+\cdots+z^2+w^2 = f^2$, density ρ, in regard to the attracted point $(a,\dots,c,e)$, is

$$V = \int \frac{\rho\, dS}{\{(a-x)^2+\cdots+(c-z)^2+(e-w)^2\}^{\frac{1}{2}s+q}},$$

where dS is the element of surface at the point $(x,\dots,z,w)$.

We require the value of V in the two cases:

1. Where the attracted point is exterior, distance from centre $\kappa > f$;

2. Where the attracted point is interior, $\kappa < f$.

Art. The integral is at once expressible in terms of the distance κ; for, taking λ to be the cosine of inclination of the radius to the line from the centre to the attracted point, that is,

$$\lambda = \frac{ax+\cdots+cz+ew}{f\kappa},$$

then the element of surface, semiangle θ, $\lambda = \cos\theta$, is $dS = f^s \sin^{s-1}\theta\, d\theta\, d\omega_s$, where $d\omega_s$ relates to the angular coordinates other than θ; and we then have

$$(a-x)^2+\cdots+(c-z)^2+(e-w)^2 = \kappa^2 - 2f\kappa\lambda + f^2.$$

Hence

$$V = \rho f^s \int \frac{\sin^{s-1}\theta\, d\theta\, d\omega_s}{(\kappa^2-2f\kappa\lambda+f^2)^{\frac{1}{2}s+q}},$$

where $\int d\omega_s$ is the surface of the s-dimensional unit-sphere (see Annex I.), $= \dfrac{s\pi^{\frac{s}{2}}}{\Gamma(\frac{s}{2}+1)}$; and for greater convenience transforming the second factor by writing therein $\lambda = \cos\theta$, the integral is

$$V = \frac{s\pi^{\frac{s}{2}}}{\Gamma(\frac{s}{2}+1)}\rho f^s \int_0^\pi \frac{\sin^{s-1}\theta\, d\theta}{(\kappa^2-2f\kappa\cos\theta+f^2)^{\frac{1}{2}s+q}}.$$

Art. For the exterior point $\kappa > f$, writing $\kappa = f \div k$, where $k < 1$, the denominator is $= f^{s+2q}(1-2k\cos\theta+k^2)^{\frac{1}{2}s+q}$; the value of the first factor (see Annex I.) is $= \dfrac{2\pi^{\frac{s}{2}}}{\Gamma(\frac{s}{2})}$; writing y and x in place of $\cos\theta$ and k respectively, the integral is

$$V = \frac{2\pi^{\frac{s}{2}}}{\Gamma(\frac{s}{2})}\rho f^{-2q}\int_{-1}^{1}\frac{(1-y^2)^{\frac{1}{2}s-1}\,dy}{(1-2xy+x^2)^{\frac{1}{2}s+q}}.$$

Now by the formula for the zonal harmonic,

$$\frac{1}{(1-2xy+x^2)^{\frac{1}{2}}} = P_0 + P_1x + \cdots + P_nx^n + \cdots,$$

and thence

$$\frac{1}{(1-2xy+x^2)^{\frac{1}{2}s+q}} = \sum_{n=0}^{\infty} x^n \cdot f_n(y),$$

where $f_n(y)$ is a function expressible in terms of spherical harmonics; and we thence obtain

$$V = \frac{2\pi^{\frac{s}{2}}}{\Gamma(\frac{s}{2})}\rho f^{-2q}\sum_{n=0}^{\infty} x^n\int_{-1}^{1}(1-y^2)^{\frac{1}{2}s-1}f_n(y)\,dy.$$

Art. The particular case $q = -\frac{1}{2}$ gives the ordinary potential of the spherical shell; for this case the prepotential is

$$V = \frac{2\pi^{\frac{s}{2}}}{\Gamma(\frac{s}{2})}\rho f\int_{-1}^{1}\frac{(1-y^2)^{\frac{1}{2}s-1}\,dy}{(1-2xy+x^2)^{\frac{1}{2}(s-1)}}.$$

Writing $n = \frac{1}{2}s - 1$, the value is

$$V = \frac{2\pi^{n+1}}{\Gamma(n+1)}\rho f\int_{-1}^{1}\frac{(1-y^2)^n\,dy}{(1-2xy+x^2)^n}.$$

This is expressible in terms of the hypergeometric series; and for the exterior point we obtain $V = \dfrac{M}{\kappa}$, where M is the total mass $= \dfrac{2\pi^{\frac{s}{2}}}{\Gamma(\frac{s}{2})}\rho f^s \cdot f$; and for the interior point, $V = \dfrac{M}{f}$, a constant.

Art. For the solid sphere, the prepotential is

$$V = \int\frac{\rho\,dx\cdots dz\,dw}{\{(a-x)^2+\cdots+(c-z)^2+(e-w)^2\}^{\frac{1}{2}s+q}},$$

the integration extending over the volume $x^2 + \cdots + z^2 + w^2 \leq f^2$. Taking ρ to be a function of the radius $= \rho(r)$, and writing as before κ for the distance of the attracted point from the centre, we have

$$V = \frac{2\pi^{\frac{s}{2}}}{\Gamma(\frac{s}{2})} \int_0^f \rho(r)\, r^{s-1}\, dr \int_0^\pi \frac{\sin^{s-1}\theta\, d\theta}{(\kappa^2 - 2\kappa r\cos\theta + r^2)^{\frac{1}{2}s+q}}.$$

Art. Writing for shortness $R^2 = \kappa^2 - 2\kappa r\cos\theta + r^2$, the θ-integral is

$$\int_0^\pi \frac{\sin^{s-1}\theta\, d\theta}{R^{s+2q}}.$$

Writing $t = \cos\theta$, this becomes

$$\int_{-1}^1 \frac{(1-t^2)^{\frac{1}{2}s-1}\, dt}{(\kappa^2 - 2\kappa r t + r^2)^{\frac{1}{2}s+q}}.$$

We may expand the denominator in powers of $\frac{r}{\kappa}$ (exterior point) or $\frac{\kappa}{r}$ (interior point); the resulting series can be integrated term by term, and the result expressed in terms of hypergeometric functions.

Art. For the exterior point, writing $\dfrac{r}{\kappa} = k$, where $k < 1$, we have

$$\frac{1}{(\kappa^2 - 2\kappa r t + r^2)^{\frac{1}{2}s+q}} = \frac{1}{\kappa^{s+2q}(1 - 2kt + k^2)^{\frac{1}{2}s+q}}.$$

Expanding in ascending powers of k, the coefficient of k^n is a zonal harmonic function of t, which we may denote by $P_n^{(s,q)}(t)$; and thence

$$\int_{-1}^1 \frac{(1-t^2)^{\frac{1}{2}s-1} P_n^{(s,q)}(t)\, dt}{(1 - 2kt + k^2)^{\frac{1}{2}s+q}} = \frac{2^{s-1}\{\Gamma(\frac{1}{2}s)\}^2}{\Gamma(s)} \cdot k^n \cdot f_n(s,q),$$

where $f_n(s,q)$ is a rational function of n, s, q.

Art. For the uniform sphere, $\rho =$ const., the prepotential for the exterior point is

$$V = \frac{2\pi^{\frac{s}{2}}}{\Gamma(\frac{s}{2})} \frac{\rho}{\kappa^{s+2q}} \int_0^f r^{s-1}\, dr \sum_{n=0}^\infty \left(\frac{r}{\kappa}\right)^n f_n(s,q).$$

Integrating term by term,

$$V = \frac{2\pi^{\frac{s}{2}}}{\Gamma(\frac{s}{2})} \frac{\rho f^s}{\kappa^{s+2q}} \sum_{n=0}^\infty \frac{1}{s+n} \left(\frac{f}{\kappa}\right)^n f_n(s,q).$$

This series can be summed; and for the particular case $q = -\frac{1}{2}$ (ordinary potential) it gives the known result $V = \dfrac{M}{\kappa}$, where M is the total mass.

Art. For the interior point, the procedure is similar; we expand in powers of $\dfrac{\kappa}{r}$, and obtain

$$V = \frac{2\pi^{\frac{s}{2}}}{\Gamma(\frac{s}{2})}\rho \int_0^f r^{s-1}\,dr \cdot \frac{1}{r^{s+2q}} \sum_{n=0}^{\infty} \left(\frac{\kappa}{r}\right)^n f_n(s,q).$$

Here we must distinguish two cases:

1. If the attracted point is within the sphere, but not at the centre, we integrate from $r = \kappa$ to $r = f$ for the external shell, and add the integral from $r = 0$ to $r = \kappa$ for the internal shell.

2. The formula simplifies considerably when $q = -\frac{1}{2}$.

Art. For the case $q = -\frac{1}{2}$, the θ-integral becomes

$$\int_0^\pi \frac{\sin^{s-1}\theta\,d\theta}{R^{s-1}} = \frac{2^{s-1}\{\Gamma(\frac{1}{2}s)\}^2}{\Gamma(s)} \cdot \frac{1}{(\kappa r)^{s-1}} \times \begin{cases} \kappa^{s-1} & \text{if } \kappa > r, \\ r^{s-1} & \text{if } \kappa < r. \end{cases}$$

Hence for the exterior point $(\kappa > f)$,

$$V = \frac{2\pi^{\frac{s}{2}}}{\Gamma(\frac{s}{2})}\rho \int_0^f r^{s-1}\,dr \cdot \frac{1}{\kappa^{s-1}} = \frac{2\pi^{\frac{s}{2}}}{\Gamma(\frac{s}{2})}\rho\frac{f^s}{s\kappa^{s-1}} = \frac{M}{\kappa^{s-1}},$$

where $M = \dfrac{2\pi^{\frac{s}{2}}}{\Gamma(\frac{s}{2})}\dfrac{\rho f^s}{s}$ is the total mass.

Art. For the interior point $(\kappa < f)$, we divide the integration into two parts: from 0 to κ, and from κ to f. For the first part $(r < \kappa)$,

$$\int_0^\kappa r^{s-1}\,dr \cdot \frac{1}{\kappa^{s-1}} = \frac{\kappa}{s};$$

for the second part $(r > \kappa)$,

$$\int_\kappa^f r^{s-1}\,dr \cdot \frac{1}{r^{s-1}} = f - \kappa.$$

Hence

$$V = \frac{2\pi^{\frac{s}{2}}}{\Gamma(\frac{s}{2})}\rho\left(\frac{\kappa}{s} + f - \kappa\right) = \frac{2\pi^{\frac{s}{2}}}{\Gamma(\frac{s}{2})}\rho\left(f - \frac{s-1}{s}\kappa\right).$$

Art. The general function $\Pi(\alpha, \beta, \ldots, \gamma, \kappa)$ is defined as follows: let λ be the positive root of

$$\frac{\alpha^2}{\theta + f^2} + \frac{\beta^2}{\theta + g^2} + \cdots + \frac{\gamma^2}{\theta + h^2} + \frac{\kappa^2}{\theta} = 1,$$

then

$$\Pi = \int_\lambda^\infty \frac{d\theta}{\sqrt{(\theta + f^2)(\theta + g^2) \cdots (\theta + h^2)\theta}}.$$

This function reduces to the prepotential of the ellipsoid when the attracted point is exterior; and for the interior point, it takes a modified form involving the complementary integral.

Art. For the disk-integral, taking the attracted point in the axis of e at distance k from the centre, and the disk to be the region $x^2 + \cdots + z^2 \leq f^2$ in the plane $w = 0$, we have

$$V = \int \cdots \int \frac{\rho\, dx \cdots dz}{(x^2 + \cdots + z^2 + k^2)^{\frac{1}{2}s+q}}.$$

Writing $r^2 = x^2 + \cdots + z^2$, this becomes

$$V = \frac{2\pi^{\frac{s}{2}}}{\Gamma(\frac{s}{2})} \int_0^f \frac{\rho\, r^{s-1}\, dr}{(r^2 + k^2)^{\frac{1}{2}s+q}}.$$

Art. The integral just written is the same as that considered in Annex II. Hence, for q positive and k small compared to f, we have

$$V = \frac{2\pi^{\frac{s}{2}}}{\Gamma(\frac{s}{2})} \rho \left\{ \frac{\Gamma(\frac{1}{2}s)\Gamma(q)}{2\Gamma(\frac{1}{2}s + q)} k^{-2q} - \frac{f^{-2q}}{2q} + \cdots \right\}.$$

For $q = 0$,

$$V = \frac{2\pi^{\frac{s}{2}}}{\Gamma(\frac{s}{2})} \rho \left(\log \frac{2f}{k} - \tfrac{1}{2}\psi(\tfrac{1}{2}s) + \cdots \right).$$

For q negative but $-q < \frac{1}{2}s$, the formula involves a term k^{-2q}.

Art. The ring-integral arises when the attracted point is not on the axis. Taking cylindrical coordinates with the axis of the disk as axis, and letting the attracted point have coordinates (ϖ, ϕ, k), where ϖ is the distance from the axis, we have

$$V = \int \frac{\rho\, r^{s-1}\, dr\, d\omega\, d\theta}{(r^2 + \varpi^2 - 2r\varpi\cos\theta + k^2)^{\frac{1}{2}s+q}}.$$

The integration in regard to θ leads to elliptic integrals when $s = 2$ and $q = -\frac{1}{2}$; for general values of s and q, it involves hypergeometric functions.

Art. The integration in regard to θ may be effected by expanding the denominator in powers of $\dfrac{2r\varpi\cos\theta}{r^2+\varpi^2+k^2}$. Writing $R_1^2 = r^2 + \varpi^2 + k^2$, we have

$$\frac{1}{(R_1^2 - 2r\varpi\cos\theta)^{\frac{1}{2}s+q}} = \frac{1}{R_1^{s+2q}} \sum_{n=0}^{\infty} P_n^{(s,q)}(\cos\theta) \left(\frac{r\varpi}{R_1^2}\right)^n,$$

where $P_n^{(s,q)}$ are generalised Legendre coefficients; and the integration in θ then depends on the integrals $\int_0^{2\pi} P_n^{(s,q)}(\cos\theta)\, d\theta$.

Art. For the particular case where $\varpi = 0$, that is, the attracted point is on the axis, the integration simplifies; we have $R_1^2 = r^2 + k^2$, and the θ-integral gives merely 2π. Hence

$$V = \frac{4\pi^{\frac{s}{2}+1}}{\Gamma(\frac{s}{2})} \int_0^f \frac{\rho\, r^{s-1}\, dr}{(r^2+k^2)^{\frac{1}{2}s+q}},$$

which is the disk-integral already considered.

Art. We come now to the disk-integral for the case where the attracted point is indefinitely near to the disk. Taking k indefinitely small, and supposing first that q is positive, the formula of Art. 0.5 gives

$$V = \frac{\pi^{\frac{s}{2}}\Gamma(q)}{\Gamma(\frac{1}{2}s+q)}\rho k^{-2q} + \text{finite part.}$$

The prepotential thus becomes infinite as the point approaches the disk; and the infinite part is independent of the radius f, that is, it is the same as if the disk were of infinite extent.

Art. For $q = 0$, we have

$$V = \frac{2\pi^{\frac{s}{2}}}{\Gamma(\frac{s}{2})}\rho \log\frac{1}{k} + \text{finite part,}$$

where the finite part is $= \dfrac{2\pi^{\frac{s}{2}}}{\Gamma(\frac{s}{2})}\rho\left(\log 2f - \frac{1}{2}\psi(\frac{1}{2}s)\right)$.

Art. For q negative, the prepotential remains finite as $k \to 0$, provided $-q < \frac{1}{2}s$; but the derived functions in regard to k become infinite. In particular, the force perpendicular to the disk, which is proportional to $\dfrac{\partial V}{\partial k}$, behaves like k^{2q+1} for $q < -\frac{1}{2}$, like $\log k$ for $q = -\frac{1}{2}$, and like k^{-2q-1} for $-\frac{1}{2} < q < 0$.

Art. For $q = -\frac{1}{2}$ (ordinary potential of a material surface), we have $V = \text{const.} + O(k \log k)$, so that V itself is continuous, but its first derivative has a logarithmic infinity; the surface-density is $\sigma = \rho$, and the force on crossing the surface changes by $4\pi\sigma$.

Annex IV. Examples of Theorem A. Art. Nos. 93 to 112.

Art. It is remarked in the text that, in the examples which relate to the s-coordinal sphere, the equations are satisfied by functions of the form

$$V = (a^2 + \cdots + c^2 + e^2)^{-\frac{1}{2}(s-2)-q} \cdot f\left(\frac{a}{e}, \ldots, \frac{c}{e}\right),$$

or more generally by

$$V = \kappa^{-s-2q+2} \cdot f\left(\frac{a}{\kappa}, \ldots, \frac{c}{\kappa}, \frac{e}{\kappa}\right),$$

where $\kappa^2 = a^2 + \cdots + c^2 + e^2$.

Art. For the exterior point, the positive root θ of the equation

$$\frac{a^2}{\theta + f^2} + \cdots + \frac{c^2}{\theta + f^2} + \frac{e^2}{\theta} = 1$$

is such that $\theta > 0$; and the function

$$\Theta = \theta^{q-1}\{(\theta + f^2) \cdots (\theta + h^2)\}^{-\frac{1}{2}}$$

satisfies certain differential relations which connect it with the prepotential equation.

Art. Writing $f^2 = g^2 = \cdots = h^2$, the equation becomes

$$\frac{a^2 + \cdots + c^2}{\theta + f^2} + \frac{e^2}{\theta} = 1,$$

and the positive root is determined by this quadratic in θ. For the sphere, the expression for the prepotential simplifies; and we find, for the exterior point,

$$V = \frac{M}{(a^2 + \cdots + c^2 + e^2)^{\frac{1}{2}(s-1)+q}},$$

where M is a constant depending on the density and radius.

Art. The positive root θ of the equation

$$\frac{a^2 + \cdots + c^2}{\theta + f^2} + \frac{e^2}{\theta} = 1$$

has certain properties which connect themselves with the function

$$\Theta = \theta^{q-1}\{(\theta + f^2) \cdots (\theta + h^2)\}^{-\frac{1}{2}}.$$

We have, the accents denoting differentiations in regard to θ,

$$\frac{d\Theta}{da} = \frac{2a}{\theta + f^2}\frac{d\Theta}{d\theta}, \qquad \frac{d\Theta}{de} = \frac{2e}{\theta}\frac{d\Theta}{d\theta},$$

where

$$\frac{d\Theta}{d\theta} = \Theta\left\{\frac{q-1}{\theta} - \frac{1}{2}\left(\frac{1}{\theta + f^2} + \cdots + \frac{1}{\theta + h^2}\right)\right\}.$$

Art. First example: $x^2 = a^2 + \cdots + c^2$, and θ the positive root of $\dfrac{x^2}{\theta + f^2} + \dfrac{e^2}{\theta} = 1$. V is assumed $= \int_0^\infty \Theta\kappa^{2q+s-2}\,dt$, where $s+1$ is positive.

I do not work the example out; it corresponds step by step with, and is hardly more simple than, the next example, which relates to the ellipsoid. The result is

$$V = \frac{\pi^{\frac{s}{2}}\Gamma(q+1)}{\Gamma(\frac{1}{2}s + q + 1)} \cdot \frac{M}{(x^2 + e^2)^{\frac{1}{2}s+q}},$$

if $x^2 + e^2 > f^2$ (exterior point); and a different form if $x^2 + e^2 < f^2$ (interior point).

Art. Second example: the ellipsoid $\dfrac{x^2}{f^2} + \cdots + \dfrac{z^2}{h^2} \le 1$, with density $\rho = \rho_0\left(1 - \dfrac{x^2}{f^2} - \cdots - \dfrac{z^2}{h^2}\right)^m$.

The prepotential is

$$V = \int \cdots \int \frac{\rho_0\left(1 - \frac{x^2}{f^2} - \cdots - \frac{z^2}{h^2}\right)^m dx \cdots dz}{\{(a-x)^2 + \cdots + (c-z)^2 + e^2\}^{\frac{1}{2}s+q}}.$$

Art. Writing $\xi = \dfrac{x}{f}, \ldots, \zeta = \dfrac{z}{h}$, the integral becomes

$$V = \rho_0 f \cdots h \int \cdots \int \frac{(1 - \xi^2 - \cdots - \zeta^2)^m\, d\xi \cdots d\zeta}{\{(a - f\xi)^2 + \cdots + (c - h\zeta)^2 + e^2\}^{\frac{1}{2}s+q}},$$

the integration extending over the unit-sphere $\xi^2 + \cdots + \zeta^2 \le 1$.

Art. Using the formula which involves $\left(\dfrac{d^2}{de^2}\right)^{q+\frac{1}{2}}$, we find

$$\frac{dV}{de} = -2e \int \cdots \int \frac{(s+2q)\rho\, dx \cdots dz}{\{(a-x)^2 + \cdots + (c-z)^2 + e^2\}^{\frac{1}{2}s+q+1}};$$

and therefore

$$\frac{d^2V}{de^2} = \text{terms involving } \frac{\partial^2 V}{\partial a^2} + \cdots + \frac{\partial^2 V}{\partial c^2}.$$

Hence, writing $e = 0$: first, for an exterior point, or $\frac{a^2}{f^2} + \cdots + \frac{c^2}{h^2} > 1$, λ is not $= 0$, and the expression vanishes in virtue of the factor e^{2q+1}; whence also $\rho = 0$.

Art. We may in this result write $e = 0$. There are two cases, according as the attracted point is exterior or interior: if it is exterior, $\dfrac{a^2}{f^2} + \cdots + \dfrac{c^2}{h^2} > 1$, θ will denote the positive root of the equation $\dfrac{a^2}{\theta + f^2} + \cdots + \dfrac{c^2}{\theta + h^2} = 1$; if interior, $\dfrac{a^2}{f^2} + \cdots + \dfrac{c^2}{h^2} < 1$, θ will be $= 0$; and we thus have

$$\int \cdots \int \frac{(1 - \frac{x^2}{f^2} - \cdots - \frac{z^2}{h^2})^m \, dx \cdots dz}{\{(a-x)^2 + \cdots + (c-z)^2\}^{\frac{1}{2}s+q}} =$$

$$= \frac{\pi^{\frac{s}{2}} \Gamma(1+q+m)}{\Gamma(\frac{1}{2}s + q + m)} \cdot \frac{M}{\theta^{\cdots}} \quad \text{for exterior point,}$$

$$= \frac{\pi^{\frac{s}{2}} \Gamma(1+q+m)}{\Gamma(\frac{1}{2}s + q + m)} \cdot \text{(different form)} \quad \text{for interior point.}$$

Art. For the verification of the formula, we may consider the case where $m = 0$ (uniform ellipsoid). The integral then reduces to that already considered in the text; and the result may be compared with the known formulae for the attraction of an ellipsoid.

Annex V. Green's Integration of the Prepotential Equation. Art. Nos. 113 to 125.

Art. In the present Annex, I in part reproduce Green's process for the integration of the potential equation $\nabla^2 V = 0$, and in part extend it to the prepotential equation; and I determine by Green's process the corresponding prepotential integrals. I do not go into the question of the Greenian Functions of a given region, but I establish the existence of certain functions (the "Greenian Functions" of the memoir) which serve for the integration of the prepotential equation.

Art. The prepotential equation is

$$\frac{\partial^2 V}{\partial a^2} + \cdots + \frac{\partial^2 V}{\partial c^2} + \frac{\partial^2 V}{\partial e^2} + \frac{2q+1}{e}\frac{\partial V}{\partial e} = 0.$$

Assuming a solution of the form $V = \Theta\Phi$, where Θ is a function of θ only, and Φ a function of $\alpha, \beta, \ldots, \gamma$ (without θ), such that κ being a function of θ only, we have

$$\nabla^2(\Theta\Phi) - \kappa\Theta\Phi = 0.$$

Art. The equation to be satisfied by Θ and Φ is then

$$\Phi\frac{d^2\Theta}{d\theta^2} + \frac{2q+1}{\theta}\frac{d\Theta}{d\theta}\Phi + \Theta\nabla^2\Phi - \kappa\Theta\Phi = 0.$$

Dividing by $\Theta\Phi$ and separating variables, we obtain

$$\frac{1}{\Theta}\frac{d^2\Theta}{d\theta^2} + \frac{2q+1}{\theta}\frac{1}{\Theta}\frac{d\Theta}{d\theta} - \kappa = -\frac{\nabla^2\Phi}{\Phi} = \text{const}.$$

Art. Solutions of the form in question are

$$\begin{aligned}
&\Phi = 1, \quad \kappa = 0,\\
&\Phi = \alpha, \quad \kappa = \frac{1}{\theta + f^2}\cdot\frac{2q+2}{\theta},\\
&\Phi = \beta, \quad \kappa = \frac{1}{\theta + g^2}\cdot\frac{2q+2}{\theta},
\end{aligned}$$

and so on for the other variables.

Art. And it can be shown next that there is a solution of the form

$$\Phi = A\alpha^2 + B\beta^2 + \cdots + C\gamma^2 + D,$$

where A, B, ..., C, D are constants to be determined.

Assuming that this satisfies $\nabla^2\Phi - \kappa\Phi = 0$, we must have identically

$$\nabla^2\Phi - \kappa(A\alpha^2 + B\beta^2 + \cdots + C\gamma^2 + D) = 0;$$

so that, from the term in α^2, we have

$$2A\left(\frac{1}{\theta + f^2}\right)^2 - \frac{\kappa A\alpha^2}{\theta + f^2} = 0,$$

or what is the same thing,

$$\frac{2A}{\theta + f^2} - \kappa A = 0;$$

with the like equations from β^2, ..., γ^2; and from the constant term we have

$$-\kappa D = 0.$$

Art. Multiplying this last by f^2, and adding it to the first, we obtain

$$A(2q + 2 + n + \tfrac{1}{2}\kappa f^2) + \kappa f^2 D = 0;$$

viz. putting for shortness $n = \frac{1}{\theta+f^2} + \cdots + \frac{1}{\theta+h^2}$, and similarly

$$B(2q + 2 + n + \tfrac{1}{2}\kappa g^2) + \kappa g^2 D = 0,$$

and similarly

$$C(2q + 2 + n + \tfrac{1}{2}\kappa h^2) + \kappa h^2 D = 0.$$

Art. To these we join the foregoing equation

$$\frac{A}{\theta + f^2} + \frac{B}{\theta + g^2} + \cdots + \frac{C}{\theta + h^2} = 0.$$

Eliminating A, B, ..., C, D, we have an equation which determines κ as a function of θ; and the equations then determine the ratios of A, B, ..., C, D, so that these quantities will be given as determinate multiples of an arbitrary quantity M. The equation for κ is in fact

$$\frac{1}{\theta + f^2}\frac{1}{2q + 2 + n + \frac{1}{2}\kappa f^2} + \cdots + \frac{1}{\theta + h^2}\frac{1}{2q + 2 + n + \frac{1}{2}\kappa h^2} + 1 = 0;$$

and the values of A, B, $\ldots$, C, D are then

$$A = \frac{Mf^2}{2q+2+n+\frac{1}{2}\kappa f^2}, \quad B = \frac{Mg^2}{2q+2+n+\frac{1}{2}\kappa g^2}, \quad \ldots$$

values which seem to be dependent on θ: if they were so, it would be fatal to the success of the process; but they are really independent of θ.

Art. That they are independent of θ depends on the theorems; that we have

$$\kappa_0 = -\frac{2(2q+2+n)}{f^2},$$

where κ_0 is a quantity independent of θ; and then multiplying out and reducing, we obtain

$$\kappa_0(2q+2+n) - (2q+2)n + \tfrac{1}{2}\kappa\kappa_0 f^2 = 0,$$

viz. the equation divides out by the factor $g^2 - f^2$, thereby becoming

$$\kappa_0(2q+2+n) - (2q+2)n + \tfrac{1}{4}\kappa\kappa_0 d = 0,$$

that is, it gives for κ the foregoing value: hence clearly, κ having this value, we obtain by symmetry

$$2q+2+n+\tfrac{1}{2}\kappa(f^2+\theta), \quad 2q+2+n+\tfrac{1}{2}\kappa(g^2+\theta), \ldots$$

proportional to

$$2q+2+\tfrac{1}{2}\kappa_0 f^2, \quad 2q+2+\tfrac{1}{2}\kappa_0 g^2, \ldots;$$

viz. the ratios, not only of $A : B$, but of $A : B : \cdots : C$ will be independent of θ.

Art. To complete the transformation, starting with the foregoing value of κ, we have

$$2q+2+n+\tfrac{1}{2}\kappa(f^2+\theta) = (2q+2+n)\frac{\theta+f^2}{f^2}, \text{ \&c.};$$

so that we have

$$A(2q+2+\tfrac{1}{2}\kappa_0 f^2) + \kappa_0 f^2 D = 0,$$

$$B(2q+2+\tfrac{1}{2}\kappa_0 g^2) + \kappa_0 g^2 D = 0,$$

$$C(2q+2+\tfrac{1}{2}\kappa_0 h^2)+\kappa_0 h^2 D=0,$$

and

$$\frac{A}{\theta+f^2}+\frac{B}{\theta+g^2}+\cdots+\frac{C}{\theta+h^2}=0.$$

Art. The equation for κ_0 is of the order s; there are consequently s functions of the form in question, and each of the terms α^2, β^2, …, γ^2 can be expressed as a linear function of these. It thus appears that any quadric function of α, β, …, γ can be expressed as a sum of Greenian functions; viz. the form is

$$A'\frac{\alpha^2}{2q+2+\frac{1}{2}\kappa_0' f^2}+B'\frac{\beta^2}{2q+2+\frac{1}{2}\kappa_0' g^2}+\cdots+C'\frac{\gamma^2}{2q+2+\frac{1}{2}\kappa_0' h^2}+D'\left(1-\frac{\alpha^2}{f^2}-\cdots-\right.$$

(s lines), viz. the terms multiplied by D', D'', &c. respectively are those answering to the roots κ_0', κ_0'', … of the equation in κ_0.

The general conclusion is that any rational and integral function of α, β, …, γ can be expressed as a sum of Greenian functions.

Art. We have next to integrate the equation

$$\frac{d^2\Theta}{d\theta^2}+\frac{2q+1}{\theta}\frac{d\Theta}{d\theta}=\kappa\Theta.$$

Suppose $\kappa=0$, a particular solution is $\Theta=$ const.; and the general solution is $\Theta=A+B\theta^{-2q}$.

Art. And therefore

$$\frac{d\Theta}{d\theta}=-2qB\theta^{-2q-1},$$

and

$$\frac{d^2\Theta}{d\theta^2}=2q(2q+1)B\theta^{-2q-2}.$$

The equation to be verified is

$$\frac{1}{\sqrt{(\theta+f^2)(\theta+g^2)\cdots(\theta+h^2)}}\left\{\frac{d^2}{d\theta^2}+\frac{2q+1}{\theta}\frac{d}{d\theta}\right\}\Theta=0;$$

or putting for shortness

$$\Omega=\sqrt{(\theta+f^2)(\theta+g^2)\cdots(\theta+h^2)},$$

$$\frac{1}{\Omega}\frac{d}{d\theta}\left(\Omega\frac{d\Theta}{d\theta}\right)+\frac{2q+1}{\theta}\frac{d\Theta}{d\theta}=0,$$

which is true. And, generally, the particular solution is deduced from the value of Φ by writing therein

$$\frac{\sqrt{\theta + f^2}}{\sqrt{p + f^2 + \cdots + h^2}}, \quad \frac{\sqrt{\theta + g^2}}{\sqrt{p + g^2 + \cdots + h^2}}, \ldots$$

in place of α, β, $\ldots$, γ respectively: say the value thus obtained is $\Phi = H$, where H is what Φ becomes by the above substitution.

Art. Represent for a moment the equation in κ_0 by

$$F(\kappa_0) = 0,$$

and assume

$$\Theta = \int \frac{H\, d\kappa_0}{F'(\kappa_0)},$$

the limits being such that H vanishes at each limit; then this value of Θ, substituted in the partial differential equation, satisfies it; and we thus obtain the general integral of the prepotential equation in a form analogous to that given by Green for the potential equation.

[End of transcription for pages 351–400. Continued in subsequent pages.]

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(Concluded from p. 400.)

Art. CXVIII. (continued.)
and therefore

$$8\theta' + 2PH^{\frac{1}{2}}z + 4\frac{d\theta'}{de} = 0;$$

viz. multiplying by $\frac{\theta'}{H^{\frac{1}{2}}}$, this is

$$4\frac{d\theta'^2}{de} + P\theta'^2 = 0;$$

or viz. substituting for P its value, this is

$$\frac{d\theta'^2}{de} = \frac{\theta'^2}{2}\left(\frac{1}{e+p} + \frac{1}{e+g} + \ldots + \frac{1}{e+h}\right).$$

Hence, integrating,

$$\theta'^2 = \frac{C}{(e+p)(e+g)\ldots(e+h)},$$

C an arbitrary constant; and

$$W = C\int_e^\infty \frac{H^{\frac{1}{2}}\,de}{(e+p)(e+g)\ldots(e+h)},$$

Y arbitrary,

$$\chi H^{\frac{1}{2}}\int_e^\infty \frac{de}{(e+p)(e+g)\ldots(e+h)},$$

where the constants of integration are C, Y; or, what is the same thing, taking T the same function of t that H is of e (viz. T is what Φ becomes on writing therein

$$\frac{1}{\sqrt{(f^2+t)(g^2+t)\ldots(h^2+t)}} \text{ in place of } \frac{1}{\sqrt{(f^2+e)(g^2+e)\ldots(h^2+e)}},$$

in place of α, β, ..., γ respectively), then

$$e = \frac{1}{2} - CH\int \frac{T^{\frac{1}{2}}\,dt}{(f^2+t)(g^2+t)\ldots(h^2+t)},$$

where χ is taken $= \infty$: we thus have

$$V = m = -CH \int_e^\infty \frac{T^{\frac{1}{2}}\,dt}{(f^2+t)(g^2+t)\dots(h^2+t)}.$$

Recollecting that

$$\frac{1}{(e+p)(e+g)\dots(e+h)} = \frac{1}{e^{s+1}\Phi e},$$

so that for $e = \infty$ we have $\alpha^2+\beta^2+\ldots+\gamma^2+\epsilon^2 = \chi^2$, the assumption $\chi = \infty$ comes to making V vanish for infinite values of $(a, b, \ldots, c, e)$.

125. We have to find the value of ρ corresponding to the foregoing value of V; viz. W being the value of V, on writing therein $(x, y, \ldots, z)$ in place of $(a, b, \ldots, c)$, then (theorem A)

$$\rho = \frac{-1}{2(\pi i)^s\Gamma(q+1)}\frac{dW}{de}.$$

C. IX. 51

Take X the same function of $(x, y, \ldots, z, e)$ that ϵ is of $(a, b, \ldots, c, e)$: viz. take X the positive root of

$$\frac{x^2}{f^2+X} + \ldots + \frac{z^2}{h^2+X} - \frac{1}{X} = 1,$$

and let $(\xi, \eta, \ldots, \zeta, \tau)$ correspond to $(\alpha, \beta, \ldots, \gamma, \epsilon)$, viz.

$$\xi = \frac{x}{\sqrt{f^2+X}}, \quad \eta = \frac{y}{\sqrt{g^2+X}}, \quad \ldots, \quad \zeta = \frac{z}{\sqrt{h^2+X}}, \quad \tau = \sqrt{\frac{1}{f^2+X} + \ldots + \frac{\cdots}{h^2 \cdots}}$$

so that W is the same function of $(\xi, \eta, \ldots, \tau)$ that V is of $(\alpha, \beta, \ldots, \epsilon)$: say this is

$$W = C\int_0^X \frac{T^{\frac{1}{2}}\,dt}{(f^2+t)(g^2+t)\ldots(h^2+t)},$$

then we have for ρ the value

$$\rho = \frac{-C}{2(\pi i)^s \Gamma(q+1)} \left\{ \frac{T^{\frac{1}{2}}}{(f^2+X)(g^2+X)\ldots(h^2+X)} \frac{dX}{de} \right\}_{e=0},$$

where e is to be put $= 0$.

126. Suppose e is $= 0$; then, if $\frac{x^2}{f^2} + \frac{y^2}{g^2} + \ldots + \frac{z^2}{h^2} > 1$, X is not $= 0$ but is the positive root of $\frac{x^2}{f^2+X} + \ldots + \frac{z^2}{h^2+X} = 1$; $\epsilon = 0$: and we have $\rho = 0$, viz. ρ is $= 0$ for all points outside the ellipsoid $\frac{x^2}{f^2} + \frac{y^2}{g^2} + \ldots + \frac{z^2}{h^2} = 1$.

But if $\frac{x^2}{f^2} + \frac{y^2}{g^2} + \ldots + \frac{z^2}{h^2} < 1$, then, on writing $e = 0$, we have $X = 0$, $\tau^2 = \frac{1}{f^2} + \frac{1}{g^2} + \ldots + \frac{1}{h^2}$,

$$\frac{dX}{de} = \frac{-X}{\frac{x^2}{(f^2+X)^2} + \ldots + \frac{z^2}{(h^2+X)^2} + \frac{1}{X^2}},$$

$$\rho = \frac{-C\Gamma(\frac{1}{2}s+q)}{2(\pi i)^s\Gamma(q+1)} \cdot \frac{1}{A_0 fg\ldots hX^{\frac{1}{2}s+q}},$$

where the term in () is $= -2\Theta$.

Hence

$$\rho = \frac{-\Gamma(\frac{1}{2}s+q)}{2\pi^{\frac{1}{2}s}\Gamma(q+1)} \cdot \frac{C}{A_0 fg\ldots h}\left(1 - \frac{x^2}{f^2} - \ldots - \frac{z^2}{h^2}\right)^{q-1},$$

where ψ_0, A_0 are what ψ, A become on writing therein $X = 0$.

It will be remembered that A is what H becomes on changing therein e into X; hence A_0 is what H becomes on writing therein $e = 0$. Moreover ψ is what Φ becomes on changing therein α, β, ..., γ into ξ, η, ..., ζ.

Writing $X = 0$, we have $\xi = \frac{x}{f}$, $\eta = \frac{y}{g}$, $\ldots$, $\zeta = \frac{z}{h}$; hence ψ_0 is what Φ becomes on changing therein α, β, $\ldots$, γ into $\frac{x}{f}$, $\frac{y}{g}$, $\ldots$, $\frac{z}{h}$.

And it is proper in Φ to restore the original

For the case last considered

$$V = \frac{\Gamma(\frac{1}{2}s+q)}{\pi^{\frac{1}{2}s}\Gamma(q+1)} \cdot \frac{C}{A_0 fg\ldots h}\left(1 - \frac{a^2}{f^2} - \ldots - \frac{c^2}{h^2}\right)^{q-1},$$

$$S = \pm \frac{\Gamma(\frac{1}{2}s+q)}{\pi^{\frac{1}{2}s}\Gamma(q+1)} \cdot \frac{C}{A_0 fg\ldots h}\int \frac{dx\ldots dz}{(1 - \frac{x^2}{f^2} - \ldots - \frac{z^2}{h^2})^{1-q}},$$

or $S =$ the same function with t for q,

$$A = \frac{\Gamma(\frac{1}{2}s+q)}{\pi^{\frac{1}{2}s}\Gamma(q+1)} \cdot \frac{C}{A_0 fg\ldots h} \cdot \frac{\pi^{\frac{1}{2}s}\Gamma(1-q)}{\Gamma(\frac{1}{2}s-q+1)}(fg\ldots h),$$

where ω is the root of the equation

$$\frac{a^2}{f^2+\omega} + \ldots + \frac{c^2}{h^2+\omega} = 1.$$

Annex VI.

Examples of Theorem C.

Art. Nos. 129 to 132.

129. First example: relating to the $(s+1)$-coordinal sphere $a^2 + \ldots + z^2 + w^2 = f^2$.

Assume

$$V = \frac{M}{(a^2 + \ldots + c^2 + e^2)^{\frac{1}{2}s-q}}, \quad V' = \frac{M}{f^{s-2q}}, \quad V'' = M' \quad (M, M' \text{ constants});$$

these at the surface give

$$\frac{M}{f^{s-2q}} = \frac{M'}{f^{s-2q}}, \quad \text{viz. } M = M'.$$

which at the surface is f^{2q}.

Hence

$$\frac{dV}{dn} = \frac{-2(\frac{1}{2}s-q)M}{f^{s-2q+1}}, \quad \frac{dV'}{dn} = \frac{2(\frac{1}{2}s-q)M}{f^{s-2q+1}}, \quad \frac{dV''}{dn} = 0,$$

$$P = \frac{-4(\frac{1}{2}s-q)M}{f^{s-2q+1}} = \frac{M}{2(\pi i)^s\Gamma(q+1)} \quad \text{(constant)}.$$

130. Writing for convenience $M = \rho f^{s-2q-2} S f^{s-2q}$ ($S f^{s-2q}$ a constant which may be put $= 1$), also $a^2 + \ldots + c^2 + e^2 = \varpi^2$, we have $\rho = S f^{s-2q}$, and consequently

$$\int \frac{S\, f\, dS}{\{(a-x)^2 + \ldots + (c-z)^2 + (e-w)^2\}^{\frac{1}{2}s-q}}$$

$$= \frac{\pi^{\frac{1}{2}s}\Gamma(\frac{1}{2}s-q)}{\Gamma(\frac{1}{2}s)} \cdot \frac{1}{\varpi^{s-2q}}, \text{ for exterior point } \varpi > f,$$

$$= \frac{\pi^{\frac{1}{2}s}\Gamma(\frac{1}{2}s-q)}{\Gamma(\frac{1}{2}s)} \cdot \frac{1}{f^{s-2q}}, \text{ for interior point } \varpi < f.$$

By making $a, \ldots, c, e$ all indefinitely large, we find

$$\int \frac{f\, dS}{\{(a-x)^2 + \ldots + (c-z)^2 + (e-w)^2\}^{\frac{1}{2}s-q}} = \frac{\pi^{\frac{1}{2}s}\Gamma(\frac{1}{2}s-q)}{\Gamma(\frac{1}{2}s)} \cdot \frac{1}{(a^2 + \ldots + c^2 + e^2)^{\frac{1}{2}}}$$

viz. the expression on the right-hand side

V is not infinite at any point outside the cylinder; and it vanishes at infinity, except indeed when only the coordinate e is infinite, and its form at infinity is not $= M \div (a^2 + \ldots + c^2 + e^2)^{\frac{1}{2}s-q}$.

V'' is not infinite for any point within the cylinder; and at the surface we have $V' = V''$.

We have

$$\frac{dV'}{dn} = \frac{dV''}{dn} = P,$$

where $\frac{dV''}{dn} = 0$, and therefore

$$P = \frac{-4(\frac{1}{2}s - q)M}{4(fg\ldots h)^{\frac{1}{2}s-q+1}} = \frac{-M}{(fg\ldots h)^{\frac{1}{2}s-q}} \quad \text{(constant)};$$

or, what is the same thing, writing $M = \rho(fg\ldots h)^{\frac{1}{2}s-q+1}$, To prove it, write $w = A\tan\theta$, then the integral is in the first place converted into

$$\frac{-1}{A^{s-2}}\int \cos^{s-2}\theta\, d\theta,$$

which, putting $\cos\theta = \sqrt{x}$ and therefore $\sin\theta = \sqrt{1-x}$, becomes

$$= \frac{1}{2A^{s-2}}\int x^{\frac{1}{2}s-2}(1-x)^{-\frac{1}{2}}\, dx,$$

which has the value in question.

Hence, replacing A by its value, we have

$$\int \frac{S\, f\, dS}{\{(a-x)^2+\ldots+(c-z)^2+w^2\}^{\frac{1}{2}s-q}} = \frac{\pi^{\frac{1}{2}(s-1)}\Gamma(\frac{1}{2}s-q)}{\Gamma(\frac{1}{2}s)(s-2)(fg\ldots h)^{\frac{1}{2}s-q-1}}\cdot\frac{}{\{(a^2+\ldots -}$$

that is,

$$\int \frac{f\, dS}{\{(a-x)^2+\ldots+(c-z)^2\}^{\frac{1}{2}s-q}} = \frac{\pi^{\frac{1}{2}(s-1)}\Gamma(\frac{1}{2}s-q)}{(s-2)\Gamma(\frac{1}{2}s)(fg\ldots h)^{\frac{1}{2}s-q-1}}.$$

the value of the interior potential must be

$$M' = \frac{\Gamma(\frac{1}{2}s)\Gamma(\frac{1}{2}s-q)}{\pi^{\frac{1}{2}s}\Gamma(\frac{1}{2}s)}\left(\frac{a^2+\ldots+c^2+e^2}{f^2}\right)^{q-1}.$$

The corresponding values of W are of course

$$\frac{\Gamma(\frac{1}{2}s-q)}{\pi^{\frac{1}{2}s}\Gamma(\frac{1}{2}s)}\cdot\frac{(x^2+\ldots+z^2+w^2)^{q-1}}{f^{2q-2}} \text{ for } \varpi > f,$$

$$\frac{\Gamma(\frac{1}{2}s-q)}{\pi^{\frac{1}{2}s}\Gamma(\frac{1}{2}s)}\cdot\frac{1}{f^{s-2}} \text{ for } \varpi < f,$$

and we thence find

$$\rho = 0, \quad \frac{1}{2}(x^2+\ldots+z^2+w^2)^{q-1},$$

$$\frac{\Gamma(\frac{1}{2}s-q)}{\pi^{\frac{1}{2}s}\Gamma(q+1)}\cdot\frac{\Gamma(\frac{1}{2}s+q)}{\Gamma(\frac{1}{2}s)}\cdot\frac{1}{f^{s+2q}},$$

if $x^2+\ldots+z^2+w^2=\varpi^2$.

Assuming for M the value $\frac{\pi^{\frac{1}{2}s}\Gamma(q+1)}{\Gamma(\frac{1}{2}s+q)}f^{s+2q}$, the last value becomes $\rho = 1$; writing for shortness $a^2+\ldots+$

coordinates $(a, \ldots, c)$, the $s - s'$ coordinates $(d, \ldots, e)$, and the $(s+1)$th coordinate u: and the integration is extended over the $(s-s')$-dimensional figure $w = -\infty$ to $+\infty$, $\ldots$, $t = -\infty$ to $+\infty$.

And it is also assumed that q is positive.

It is at once clear that we may reduce the integral to

$$\int \frac{dw \ldots dt}{\{(a-x)^2 + \ldots + (c-z)^2 + u^2 + v^2 + \ldots + t^2\}^{\frac{1}{2}s+q}},$$

say for shortness

$$\int \frac{dw \ldots dt}{U^{\frac{1}{2}s+q}},$$

where $K^2 = (a-x)^2 + \ldots + (c-z)^2 + u^2$, is a constant as regards

Annex IX.

The Gauss–Jacobi Theory of Epiperic Integrals.

Art. No. 138.

138. The formula obtained (Annex IV. No. 110) is proved only for positive values of m; but writing therein $q = 0$, $m = -q$, it becomes

$$\int \frac{dx \ldots dz}{\{(a-x)^2 + \ldots + (c-z)^2 + e^2\}^{\frac{1}{2}s}}$$

$$= \frac{\pi^{\frac{1}{2}s}\Gamma(-q)}{\Gamma(\frac{1}{2}s - q)} \cdot \frac{1}{e^{s+2q}} \left(1 - \frac{a^2}{f^2} - \ldots - \frac{c^2}{h^2}\right)^{-q},$$

a formula which is obtainable as a particular case of the more general formula

$$\int \frac{dx \ldots dz\, dw}{\{(\ast\!\!\!\$\, x, \ldots, z, w, 1)\}^{\frac{1}{2}s}}$$

$$= \frac{\pi^{\frac{1}{2}s}\Gamma(-q)}{\Gamma(\frac{1}{2}s - q)} \cdot \frac{1}{e^{s+2q}} \left(1 - \frac{a^2}{f^2} - \ldots - \frac{c^2}{h^2}\right)^{-q}.$$

C. IX. 55

140. The determination of the integral depends upon formulae for the transformation of the spherical element dS, and of the quadric function $(x, y, \ldots, z, w, 1)^2$.

First, as regards the spherical element dS; let the $s+1$ variables $x, y, \ldots, z, w$ which satisfy $x^2 + y^2 + \ldots + z^2 + w^2 = 1$ be regarded as functions of the s independent variables θ, ϕ, $\ldots$, ψ; then we have

$$dS = \begin{vmatrix} \frac{dx}{d\theta} & \frac{dy}{d\theta} & \cdots & \frac{dz}{d\theta} & \frac{dw}{d\theta} \\ \frac{dx}{d\phi} & \frac{dy}{d\phi} & \cdots & \frac{dz}{d\phi} & \frac{dw}{d\phi} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \frac{dx}{d\psi} & \frac{dy}{d\psi} & \cdots & \frac{dz}{d\psi} & \frac{dw}{d\psi} \\ x & y & \cdots & z & w \end{vmatrix} d\theta\, d\phi \ldots d\psi.$$

let $d\Sigma$ represent the spherical element belonging to the coordinates ξ, η, ..., ζ, ω.

Considering these as functions of the foregoing s independent variables θ, ϕ, ..., ψ, we have

$$d\Sigma = \begin{vmatrix} \frac{d\xi}{d\theta} & \frac{d\eta}{d\theta} & \cdots & \frac{d\zeta}{d\theta} & \frac{d\omega}{d\theta} \\ \frac{d\xi}{d\phi} & \frac{d\eta}{d\phi} & \cdots & \frac{d\zeta}{d\phi} & \frac{d\omega}{d\phi} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \frac{d\xi}{d\psi} & \frac{d\eta}{d\psi} & \cdots & \frac{d\zeta}{d\psi} & \frac{d\omega}{d\psi} \\ \xi & \eta & \cdots & \zeta & \omega \end{vmatrix} d\theta\, d\phi \ldots d\psi.$$

and observing that

$$(\$ \, x, y, \ldots, z, w, 1)^2 = 4(\$ \, X, Y, \ldots, Z, W, T)^2,$$

the integral becomes

$$\int \frac{d\Sigma}{\{(\$ \, X, Y, \ldots, Z, W, T)\}^{\frac{1}{2}s+q}},$$

where $X, Y, \ldots, Z, W, T$ denote given linear functions (with constant coefficients) of the $s+1$ variables ξ, η, $\ldots$, ζ, ω, or, what is the same thing, given linear functions of the $s+2$ quantities ξ, η, $\ldots$, ζ, ω, 1, such that identically

$$X^2 + Y^2 + \ldots + Z^2 + W^2 - T^2 = \xi^2 + \eta^2 + \ldots + \zeta^2 + \omega^2 - 1.$$

We have then $\xi^2 + \eta^2 + \ldots + \zeta^2 + \omega^2 = 1$, and $d\Sigma$ a where

$$\frac{d\Sigma}{dS} = \frac{(ab\ldots ce)^{\frac{1}{2}}}{\{(\$\, X, \ldots, Z, W, T)\}^{\frac{1}{2}s+q}},$$

and consequently

$$\frac{d\Sigma}{dS} = \frac{(ab\ldots ce)^{\frac{1}{2}}}{\{(\$\, x, \ldots, z, w, 1)\}^{\frac{1}{2}s+q}}.$$

Hence, writing dS to denote the spherical element corresponding to the point $(x, y, \ldots, z, w)$, we have, by a former formula,

$$\frac{d\Sigma}{dS} = \frac{(ab\ldots ce)^{\frac{1}{2}}}{\{(\$\, x, \ldots, z, w, 1)\}^{\frac{1}{2}s+q}}.$$

Hence, integrating each side, and observing that $\int dS$, taken over the whole spherical surface $x^2 + y^2 + \ldots + z^2 + w^2 = 1$, is $= 2(\pi)^{\frac{1}{2}s+1} \div \Gamma(\frac{1}{2}s+1)$, we have

$$\int \frac{d\Sigma}{\{(\$\, X, \ldots, Z, W, T)\}^{\frac{1}{2}s+q}} = \frac{2(\pi)^{\frac{1}{2}s+1}}{\Gamma(\frac{1}{2}s+1)} \cdot \frac{(ab\ldots ce)^{\frac{1}{2}}}{\{(\$\, a, \ldots, c, e, 1)\}^{\frac{1}{2}s+q}}.$$

or, if for $a, \ldots, c, e$ we restore the values $A-L, \ldots, C-L, E-L$, then

$$= \frac{2\pi^{\frac{1}{2}s+1}}{\Gamma(\frac{1}{2}s+1)} \int dt\,\{(t+A)\ldots(t+C)(t+E)(t+L)\}^{-\frac{1}{2}};$$

viz. we thus have

$$\int \frac{d\Sigma}{U^{\frac{1}{2}s+q}},$$

where $(t+A)\ldots(t+C)(t+E)(t+L)$ is in fact a given rational and integral function of t; viz. it is $= -\,\text{Disct.}\,[(\$\, X \ldots Z, W, T)^2 + t(X^2 + \ldots + Z^2 + W^2 - T^2)]$.

147. Consider, in particular, the integral

$$\int \frac{dS}{\{(a-fx)^2 + \ldots + (c-hz)^2 + (e-kw)^2 + l\}^{\frac{1}{2}s+q}},$$

here

$$\{(a-fx)^2 + \ldots + (c-hz)^2 + (e-kw)^2 + l\}^{\frac{1}{2}}$$
$$= \sqrt{2}\{(aT-fX)^2 + \ldots + (cT-hZ)^2 + (eT-kW)^2 + lT^2\}^{\frac{1}{2}}.$$

provisional assumption. Or, writing for greater convenience b to denote the positive quantity and the formula thus is

$$\int \frac{dS}{\{(a-x)^2+\ldots+(c-z)^2+e^2\}^{\frac{1}{2}s+q}} = \frac{2\pi^{\frac{1}{2}s}\Gamma(-q)}{\Gamma(\frac{1}{2}s-q)} \cdot \frac{1}{e^{s+2q}} \left(1-\frac{a^2}{f^2}-\ldots-\frac{c^2}{h^2}\right)^{-q}$$

150. We have thus, for the prepotential of the solid ellipsoid, the theorem

$$\int \frac{dx \ldots dz\, dw}{\{(a-x)^2+\ldots+(c-z)^2+(e-w)^2\}^{\frac{1}{2}s+q}} = \frac{\pi^{\frac{1}{2}s}\Gamma(-q)}{\Gamma(\frac{1}{2}s-q)} \cdot \frac{1}{e^{s+2q}} \left(1-\frac{a^2}{f^2}-\ldots-\frac{c}{h}\right.$$

which agrees with the result obtained for the potential of the solid ellipsoid in the particular case $q=-\frac{1}{2}$.

The proof applies without change to the prepotential of the shell of the ellipsoid, or shell included between two similar and similarly situated ellipsoids; and it applies also to the prepotential of the ellipsoidal plate or lamina included between two non-similar ellipsoids.

In particular, for the potential $(q=-\frac{1}{2})$ of the solid ellipsoid, we have

$$\int \frac{dx \ldots dz\, dw}{\{(a-x)^2+\ldots+(c-z)^2+(e-w)^2\}^{\frac{1}{2}(s-1)}} = \frac{\pi^{\frac{1}{2}s}\Gamma(\frac{1}{2})}{\Gamma(\frac{1}{2}s+\frac{1}{2})} \cdot \frac{1}{e^{s-1}} \left(1-\frac{a^2}{f^2}-\ldots-\right.$$

The formula may be presented in a different form by introducing in place of q a new parameter $p = -q$; and we then have

$$\int \frac{dx \ldots dz\, dw}{\{(a-x)^2 + \ldots + (c-z)^2 + (e-w)^2\}^{\frac{1}{2}s-p}} = \frac{\pi^{\frac{1}{2}s}\Gamma(p)}{\Gamma(\frac{1}{2}s+p)} \cdot \frac{1}{e^{s-2p}} \left(1 - \frac{a^2}{f^2} - \ldots - \frac{c}{h}\right.$$

151. The foregoing results may be generalized by considering the integral

$$\int \frac{dx \ldots dz\, dw}{\{(a-x)^2 + \ldots + (c-z)^2 + (e-w)^2\}^{\frac{1}{2}s-p}} \cdot \frac{1}{(1 - \frac{x^2}{f^2} - \ldots - \frac{z^2}{h^2} - \frac{w^2}{e^2})^{1-p}},$$

which, for the case $p = \frac{1}{2}$, reduces to the potential of the solid ellipsoid.

152. The integral

$$\int \frac{dx \ldots dz\, dw}{\{(a-x)^2 + \ldots + (c-z)^2 + (e-w)^2\}^{\frac{1}{2}s-p}}$$

taken over the ellipsoid

$$\frac{x^2}{f^2} + \frac{y^2}{g^2} + \ldots + \frac{z^2}{h^2} + \frac{w^2}{e^2} = 1,$$

may be written as a double integral in the form

$$V = \int_0^1 \int_0^1 \frac{\phi(u)\, du\, dt}{\{(a-x)^2 + \ldots + (c-z)^2 + (e-w)^2\}^{\frac{1}{2}s-p}},$$

where $\phi(u)$ is a function depending on the form of the integral.

function under the integral sign is to be replaced by zero whenever the values u, t are such that u is less than $\frac{x^2}{f^2}+\ldots+\frac{z^2}{h^2}+\frac{w^2}{e^2}$, viz. when the values belong to a point in the shaded portion of the strip; the integral is therefore to be extended only over the unshaded portion of the strip; viz. the value is

$$V=\frac{\Gamma(-q)\Gamma(\frac{1}{2}s)}{\pi^{\frac{1}{2}s}\Gamma(\frac{1}{2}s-q)}\iint\frac{\phi(u)\,du\,dt}{\{(a-x)^2+\ldots+(c-z)^2+(e-w)^2\}^{\frac{1}{2}s-p}},$$

the double integral being taken over the unshaded portion of the strip; or, what is the same thing, the integral in regard to u is to be taken from

$$u=\frac{x^2}{f^2+t}+\ldots+\frac{z^2}{h^2+t}+\frac{w^2}{e^2+t}$$

(say from $u=\sigma$) to $u=1$, and then the integral in regard to t is to be taken from $t=0$ to $t=\infty$, where, as before, θ is the positive root of the equation $\sigma=1$, that is of

$$\frac{a^2}{f^2+\theta}+\ldots+\frac{c^2}{h^2+\theta}+\frac{e^2}{e^2+\theta}=1.$$

162. Write $u=\sigma+(1-\sigma)x$, and therefore $u-\sigma=(1-\sigma)x$, $1-u=(1-\sigma)(1-x)$ and $du=(1-\sigma)\,dx$; then the limits $(1,0)$ of x correspond to the limits $(1,\sigma)$ of u, and the formula becomes

$$V=\frac{\Gamma(-q)\Gamma(\frac{1}{2}s)}{\pi^{\frac{1}{2}s}\Gamma(\frac{1}{2}s-q)}\int_0^\infty\frac{(1-\sigma)^{-q}\,dt}{\{(a-x)^2+\ldots+(c-z)^2+(e-w)^2\}^{\frac{1}{2}s-p}},$$

where σ is retained in place of its value $\frac{a^2}{f^2+t}+\ldots+\frac{c^2}{h^2+t}+\frac{e^2}{e^2+t}$.

This is, in fact, a form (deduced from Boole's result in the memoir of 1846) given by me, *Cambridge and Dublin Mathematical Journal*, vol. II. (1847), p. 219, [44].

If in particular $\phi(u)=(1-u)^{\frac{1}{2}s+q}$, then $\phi\{\sigma+(1-\sigma)x\}=(1-\sigma)^{\frac{1}{2}s+q}(1-x)^{\frac{1}{2}s+q}$, and thence

$$\int_0^1 x^{-q-1}(1-x)^{\frac{1}{2}s+q}\,dx=\frac{\Gamma(-q)\Gamma(\frac{1}{2}s+q+1)}{\Gamma(\frac{1}{2}s+1)},$$

and then, restoring for σ its value, we have as the value of the integral

$$\int\frac{dx\ldots dz\,dw}{\{(a-x)^2+\ldots+(c-z)^2+(e-w)^2\}^{\frac{1}{2}s-p}}$$

taken over the ellipsoid $\frac{x^2}{f^2} + \ldots + \frac{w^2}{e^2} = 1$.

This is, in fact, the theorem of Annex IV. No. 110 in its general form; but the proof assumes that q is positive. C. IX. 58

The formula may be presented in a different form by introducing in place of q a new parameter $p = -q$; and we then have

$$\int \frac{dx \dots dz\, dw}{\{(a-x)^2 + \dots + (c-z)^2 + (e-w)^2\}^{\frac{1}{2}s-p}} = \frac{\pi^{\frac{1}{2}s}\Gamma(p)}{\Gamma(\frac{1}{2}s+p)} \cdot \frac{1}{e^{s-2p}} \left(1 - \frac{a^2}{f^2} - \dots - \frac{c}{h}\right.$$

163. The foregoing results may be generalized by considering the integral

$$\int \frac{dx \dots dz\, dw}{\{(a-x)^2 + \dots + (c-z)^2 + (e-w)^2\}^{\frac{1}{2}s-p}} \cdot \frac{1}{(1 - \frac{x^2}{f^2} - \dots - \frac{z^2}{h^2} - \frac{w^2}{e^2})^{1-p}},$$

which, for the case $p = \frac{1}{2}$, reduces to the potential of the solid ellipsoid.

The proof here given is subject to the same limitations as the proof in Annex IV.; and it would seem that the formula is not generally true, but only for such values of p as satisfy certain conditions.

164. The theorem of Annex IV. No. 110 may be presented under a more general form by writing therein q for $-\frac{1}{2}s$, and $-q$ for p; and we then have

$$\int \frac{dx \dots dz\, dw}{\{(a-x)^2 + \dots + (c-z)^2 + (e-w)^2\}^{\frac{1}{2}s+q}} = \frac{\pi^{\frac{1}{2}s}\Gamma(-q)}{\Gamma(\frac{1}{2}s-q)} \cdot \frac{1}{e^{s+2q}} \left(1 - \frac{a^2}{f^2} - \dots - \frac{c}{h}\right.$$

which is the general theorem for the prepotential of the solid ellipsoid.

165. The formula obtained for the prepotential of the solid ellipsoid may be written in the form

$$V = \frac{\pi^{\frac{1}{2}s}\Gamma(-q)}{\Gamma(\frac{1}{2}s-q)} \int_0^\infty \frac{dt}{\{(t+f^2)\dots(t+h^2)(t+e^2)\}^{\frac{1}{2}}} \cdot \frac{1}{(t+\theta)^{-q}},$$

where θ is the positive root of the equation

$$\frac{a^2}{f^2+\theta} + \dots + \frac{c^2}{h^2+\theta} + \frac{e^2}{e^2+\theta} = 1.$$

And this form puts in evidence the fact that V is a solution of the partial differential equation

$$\frac{d^2V}{da^2} + \dots + \frac{d^2V}{dc^2} + \frac{d^2V}{de^2} = \frac{1}{e^{2q+2}} \cdot V.$$

166. The theorem of No. 138 may be generalized as follows: writing therein $-q$ for q, and multiplying both sides by e^{s+2q}, we have

$$\int \frac{dx\dots dz\,dw}{\{(a-x)^2+\dots+(c-z)^2+(e-w)^2\}^{\frac{1}{2}s+q}} = \frac{\pi^{\frac{1}{2}s}\Gamma(-q)}{\Gamma(\frac{1}{2}s-q)} \cdot \frac{1}{e^{s+2q}} \left(1 - \frac{a^2}{f^2} - \dots - \frac{c}{h}\right.$$

which, when $q=0$, reduces to the theorem for the potential of the solid ellipsoid.

167. The foregoing formulae may be applied to the determination of the potential of an ellipsoidal shell, and to the investigation of the attractions of ellipsoids.

168. The theorem of No. 138 may also be extended to the case of the prepotential of a spherical shell; and the results obtained for the potential of the ellipsoid may be extended to the case of the prepotential of the ellipsoid.

169. The theorem of Annex IV. No. 110 may be presented under a more general form. Writing therein q for $-\frac{1}{2}s$, and $-q$ for p, we have

$$\int \frac{dx \ldots dz\, dw}{\{(a-x)^2 + \ldots + (c-z)^2 + (e-w)^2\}^{\frac{1}{2}s+q}} = \frac{\pi^{\frac{1}{2}s}\Gamma(-q)}{\Gamma(\frac{1}{2}s-q)} \cdot \frac{1}{e^{s+2q}} \left(1 - \frac{a^2}{f^2} - \ldots - \frac{c}{h}\right.$$

which is the general theorem for the prepotential of the solid ellipsoid.

The formula may also be written in the form

$$\int \frac{dx \ldots dz\, dw}{\{(a-x)^2 + \ldots + (c-z)^2 + (e-w)^2\}^{\frac{1}{2}s+q}} = \frac{\pi^{\frac{1}{2}s}\Gamma(-q)}{\Gamma(\frac{1}{2}s-q)} \cdot \frac{1}{\{(a^2 + \ldots + c^2 + e^2)\}^{\frac{1}{2}s-}}$$

which is valid for all values of q.

170. The foregoing investigations relate to the prepotential of the solid ellipsoid; but similar results may be obtained for the prepotential of the ellipsoidal shell and of the ellipsoidal plate.

171. The theorem of No. 138 gives the prepotential of the solid ellipsoid in the form of a definite integral; and it may be shown that this integral may be evaluated in finite terms when s is an even integer.

172. The results obtained in the present memoir may be applied to the investigation of the potential of the ellipsoid and of the attractions of ellipsoids; and they furnish a direct proof of the theorems established by Ivory and Maclaurin.

173. The theorem of Annex IV. No. 110 may be extended to the case where the coordinates are connected by any number of linear relations; and the results may be applied to the determination of the potential of a body of any form.

174. The foregoing investigations complete the theory of prepotentials as far as it has been developed in the present memoir; and it is hoped that they may serve as a foundation for further researches in this interesting branch of mathematical physics. C. IX. 61

function under the integral sign is to be replaced by zero whenever the values u, t are such that u is less than $\frac{x^2}{f^2+t} + \ldots + \frac{z^2}{h^2+t} + \frac{w^2}{e^2+t}$, viz. when the values belong to a point in the shaded portion of the strip; the integral is therefore to be extended only over the unshaded portion of the strip; viz. the value is

$$V = \frac{\Gamma(-q)\Gamma(\frac{1}{2}s)}{\pi^{\frac{1}{2}s}\Gamma(\frac{1}{2}s-q)} \iint \frac{\phi(\sigma + (1-\sigma)x)\,(1-\sigma)\,dx\,dt}{\{(a-x)^2 + \ldots + (c-z)^2 + (e-w)^2\}^{\frac{1}{2}s-p}},$$

the double integral being taken over the unshaded portion of the strip; or, what is the same thing, the integral in regard to u is to be taken from

$$u = \frac{x^2}{f^2+t} + \ldots + \frac{z^2}{h^2+t} + \frac{w^2}{e^2+t}$$

(say from $u = \sigma$) to $u = 1$, and then the integral in regard to t is to be taken from $t = \theta$ to $t = \infty$, where, as before, θ is the positive root of the equation $\sigma = 1$, that is of

$$\frac{a^2}{f^2+\theta} + \ldots + \frac{c^2}{h^2+\theta} + \frac{e^2}{e^2+\theta} = 1.$$

162. Write $u = \sigma + (1-\sigma)x$, and therefore $u - \sigma = (1-\sigma)x$, $1-u = (1-\sigma)(1-x)$ and $du = (1-\sigma)\,dx$; then the limits $(1, 0)$ of x correspond to the limits $(1, \sigma)$ of u, and the formula becomes

$$\int_0^1 x^{-q-1}\phi\{\sigma + (1-\sigma)x\}\,dx = (1-\sigma)^{-q}\int_0^1 x^{-q-1}(1-x)^{\frac{1}{2}s+q}\,dx,$$

where σ is retained in place of its value $\frac{a^2}{f^2+t} + \ldots + \frac{c^2}{h^2+t} + \frac{e^2}{e^2+t}$.

This is, in fact, a form (deduced from Boole's result in the memoir of 1846) given by me, *Cambridge and Dublin Mathematical Journal*, vol. II. (1847), p. 219, [44].

If in particular $\phi(u) = (1-u)^{\frac{1}{2}s+q}$, then $\phi\{\sigma + (1-\sigma)x\} = (1-\sigma)^{\frac{1}{2}s+q}(1-x)^{\frac{1}{2}s+q}$, and thence

$$\int_0^1 x^{-q-1}(1-x)^{\frac{1}{2}s+q}\,dx = \frac{\Gamma(-q)\Gamma(\frac{1}{2}s+q+1)}{\Gamma(\frac{1}{2}s+1)},$$

and then, restoring for σ its value, we have as the value of the integral

$$\int \frac{dx \ldots dz\,dw}{\{(a-x)^2 + \ldots + (c-z)^2 + (e-w)^2\}^{\frac{1}{2}s-p}}$$

taken over the ellipsoid $\frac{x^2}{f^2} + \ldots + \frac{w^2}{e^2} = 1$.

This is, in fact, the theorem of Annex IV. No. 110 in its general form; but the proof assumes that q is positive. C. IX. 62

608. [Extract from a] Report on Mathematical Tables,[From the *Report of the British Association for the Advancement of Science*, (1873), pp. 3, 4.]

It was necessary as a preliminary to form a classification of mathematical (numerical) tables; and the following classification was drawn up by Prof. Cayley and adopted by the Committee.

A. Auxiliary for non-logarithmic computations.

1. Multiplication.
2. Quarter-squares.
3. Squares, cubes, and higher powers, and reciprocals.

B. Logarithmic and circular.

4. Logarithms (Briggian) and antilogarithms (do.); addition and subtraction logarithms, &c.
5. Circular functions (sines, cosines, &c.), natural, and lengths of circular arcs.
6. Circular functions (sines, cosines, &c.), logarithmic.

C. Exponential.

7. Hyperbolic logarithms.
8. Do. antilogarithms (e^x) and $\frac{1}{2}\log\tan(45^\circ + \frac{1}{2}\phi)$, and hyperbolic sines, cosines, &c., natural and logarithmic.

D. Algebraic constants.

9. Accurate integer or fractional values. Bernoulli's Numbers, $A^{(n)}0^n$, &c. Binomial coefficients.
10. Decimal values auxiliary to the calculation of series.

E. Transcendental constants.

11. Transcendental constants, e, π, γ, &c., and their powers and functions.

F. Arithmological.

12. Divisors and prime numbers. Prime roots. The Canon arithmeticus, &c.

13. The Pellian equation.
14. Partitions.
15. Quadratic forms $a^2 + b^2$, &c., and partition of numbers into squares, cubes, and biquadrates.
16. Binary, ternary, &c. quadratic, and higher forms.
17. Complex theories.

G. Transcendental functions.

18. Elliptic.
19. Gamma.
20. Sine-integral, cosine-integral, and exponential-integral.
21. Bessel's and allied functions.
22. Planetary coefficients for given $\frac{m}{a}$.
23. Logarithmic transcendental.
24. Miscellaneous.

Several of these classes need some little explanation. Thus D 9 and 10 are intended to include the same class of constants, the only difference being that in 9 accurate values are given, while in 10 they are only approximate; thus, for example, the accurate Bernoulli's numbers as vulgar fractions, and the decimal values of the same to (say) ten places are placed in different classes, as the former are of theoretical interest, while the latter are only of use in calculation.

It is not necessary to enter into further detail with respect to the classification, as in point of fact it is only very partially followed in the Report. C. IX. 63–64

609. On the Analytical Forms called Factions,[From the *Report of the British Association for the Advancement of Science*, (1875), p. 10.]

A FACTION is a product of differences such that each letter occurs the same number of times; thus we have a *quadrifaction* where each letter occurs twice, a *cubifaction* where each letter occurs three times, and so on.

A *broken faction* is one which is a product of factions having no common letter; thus

$$(a-b)^2(c-d)(d-e)(e-c)$$

is a broken quadrifaction, the product of the quadrifactions $(a-b)^2$ and $(c-d)(d-e)(e-c)$.

We have, in regard to quadrifactions, the theorem that every quadrifaction is a sum of broken quadrifactions such that each component quadrifaction contains two or else three letters.

Thus we have the identity

$$2(a-b)(b-c)(c-d)(d-a) = (b-c)^2\cdot(a-d)^2-(c-a)^2\cdot(b-d)^2+(a-b)^2\cdot(c-d)^2,$$

which verifies the theorem in the case of a quadrifaction of four letters; but the verification even in the next following case of a quadrifaction of five letters is a matter of some difficulty.

The theory is connected with that of the invariants of a system of binary quantics. C. IX. 64

610. On the Analytical Forms called Trees, with Application to the Theory of Chemical Combinations,[From the *Report of the British Association for the Advancement of Science*, (1875), pp. 257–305.]

I HAVE in two papers "On the Analytical forms called Trees," *Phil. Mag.* vol. XIII. (1857), pp. 172–176, [203], and ditto, vol. XX. (1859), pp. 374–378, [247], considered this theory; and in a paper "On the Mathematical Theory of Isomers," ditto, vol. XLVII. (1874), p. 444, [586], pointed out its connexion with modern chemical theory.

In particular, as regards the paraffins C_nH_{2n+2}, we have n atoms of carbon connected by $n-1$ bands, under the restriction that from each carbon-atom there proceed at most 4 bands (or, in the language of the papers first referred to, we have n knots connected by $n-1$ branches), in the form of a tree; for instance; $n=5$, such forms (and the only such forms) are

[scale=0.8] (a) at (0,0) ; (b) at (1,0) ; (c) at (2,0) ; (d) at (3,0) ; (e) at (4,0) ; (a)–(b)–(c)–(d)–(e); (a) circle (2pt) node[below] 1; (b) circle (2pt) node[below] 2; (c) circle (2pt) node[below] 2; (d) circle (2pt) node[below] 2; (e) circle (2pt) node[below] 3;

And if, under the foregoing restriction of only 4 bands from a carbon-atom, we connect with each carbon-atom the greatest possible number of hydrogen-atoms, as shown in the diagrams by the affixed numerals, we see that the number of hydrogen-atoms is 12 $(= 2.5 + 2)$; and we have thus the representations of three different paraffins, C_5H_{12}.

It should be observed that the tree-symbol of the paraffin is obtained from the carbon-tree by the mere affixture to each knot of a number of single branches equal to the number of hydrogen-atoms belonging to the corresponding carbon-atom; or say that the carbon-tree is the *root* of the paraffin-tree.

The theory of the analytical forms called trees was established by me in the two papers above referred to; and I have recently, in the paper "On the Mathematical Theory of Isomers," [586], given a much more simple investigation, and have also shown the connexion of the theory with the chemical theory of isomerism.

In the present paper I propose to give a complete investigation of the theory of trees, and to apply it to the determination of the number of isomers of the paraffins and other hydrocarbons.

1. I commence with the definition and fundamental properties of trees.

A *tree* is a figure consisting of a number of points, called *knots*, connected by lines, called *branches*, in such a way that there is no closed polygon; or, what is the same thing, a tree is a connected figure in which the number of branches is one less than the number of knots.

The number of branches which proceed from any knot is called the *order* of the knot; and a knot of order 1 is called a *terminal knot*.

The *distance* of any two knots is the number of branches in the shortest path from one to the other.

The *altitude* of a tree is the greatest distance of any terminal knot from a certain knot or pair of knots, called the *centre* or *bicentre* of the tree.

2. It may be shown that every tree has a unique centre or bicentre. For, starting from any terminal knot and proceeding along the tree, we must at length arrive at a knot from which all the remaining knots are at distances not greater than the distance already traversed; and this knot is the centre, or else there is a pair of knots each at the same distance from it, which pair is the bicentre.

3. A *root-tree* is a tree in which one knot is distinguished as the *root*; and a *centre-tree* is a tree in which the centre is regarded as the root.

4. The problem of finding the number of distinct trees with n knots is thus reduced to the problem of finding the number of centre-trees and bicentre-trees with n knots; and this again depends on the problem of finding the number of root-trees with n knots. C. IX. 65

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5. Two trees are considered as identical when they can be derived from each other by a continuous deformation without passing any branch through a knot; or, what is the same thing, two trees are identical when there is a one-to-one correspondence between their knots such that the distances of corresponding knots are equal.

6. The number of root-trees with n knots may be determined by means of generating functions. Let $F(x)$ be the generating function for the number of root-trees, so that the coefficient of x^n in $F(x)$ is the number of root-trees with n knots; and let $f(x)$ be the generating function for the number of centre-trees.

Then it may be shown that

$$F(x) = x \ \exp\left\{F(x) + \tfrac{1}{2}F(x^2) + \tfrac{1}{3}F(x^3) + \dots\right\},$$

and that

$$f(x) = x \ \exp\left\{F(x) + \tfrac{1}{2}F(x^2) + \tfrac{1}{3}F(x^3) + \dots\right\} - \tfrac{1}{2}\left\{F(x)^2 - F(x^2)\right\}.$$

7. The number of bicentre-trees is given by the coefficient of x^n in $\frac{1}{2}\{F(x)^2 + F(x^2)\}$; and the total number of trees is the sum of the numbers of centre-trees and bicentre-trees.

8. The foregoing formulae may be applied to the determination of the number of paraffins C_nH_{2n+2}. The number of paraffins is equal to the number of carbon centre-trees and bicentre-trees with n knots, subject to the condition that no knot has more than 4 branches.

9. To adapt the formulae to this case, we introduce a second variable t, and consider the generating function

$$\Phi(x,t) = x\left\{t + t\Phi(x,t) + t\frac{\Phi(x^2,t^2)}{2} + \dots\right\},$$

where the coefficient of $t^\alpha x^n$ is the number of carbon root-trees with n knots and α terminal knots.

10. The generating function for the carbon centre-trees is then

$$\Psi(x,t) = \Phi(x,t) - \tfrac{1}{2}\{\Phi(x,t)^2 - \Phi(x^2,t^2)\},$$

and the generating function for the carbon bicentre-trees is

$$\tfrac{1}{2}\{\Phi(x,t)^2 + \Phi(x^2,t^2)\}.$$

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11. By developing these generating functions, we obtain the numbers of the several classes of trees. The results are exhibited in the following tables.

Table I.—General Root-Trees.

Index X, or num.	Index t, or num.	Altitude or number of column											
		1	2	3	4	5	6	7	8	9	10	11	12
1	1	1											
2	1	1											
	2	1											
3	1	1											
	2	1	1										
	3	1											
4	1	1											
	2	1	2	1									
	3	1	1										
	4	1											
5	1	1											
	2	1	3	3	1								
	3	1	2	1									
	4	1	1										
	5	1											
6	1	1											
	2	1	4	6	4	1							
	3	1	3	3	1								
	4	1	2	1									
	5	1	1										
	6	1											

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12. The generating functions for the successive columns of Table I are obtained as follows. Denoting by GF, column r the generating function for the trees in column r, we have

for column 0: x,
for column 1: $tx^2 + t^2x^3 + t^3x^4 + \ldots = tx^2(1-tx)^{-1}$,
for column 2: $tx^3 + t^2(x^4 + x^5) + t^3(x^5 + x^6 + x^7) + \ldots$,
and so on indefinitely.

The successive generating functions are connected by the formula

$$\text{GF, col. } (r{+}1) = tx{\cdot}\text{GF, col. } r{+}tx^2{\cdot}\text{GF, col. } (r{-}1){+}\ldots{+}tx^{r+1}{\cdot}\text{GF, col. } 0,$$

which serves to calculate them in succession.

13. The numbers in Table I may be verified by means of the following formula for the total number of root-trees with n knots:

$$R_n = \frac{1}{n-1} \sum_{d|n-1} \phi(d) \cdot R_{\frac{n-1}{d}},$$

where $\phi(d)$ is Euler's totient function.

14. *Table II.—General Centre- and Bicentre-Trees.*

The table of centre-trees is obtained from that of root-trees by omitting the root-trees which have a bicentre; and the table of bicentre-trees is obtained by a simple formula.

15. The generating function for the bicentre-trees is

$$\tfrac{1}{2}\{F(x)^2 + F(x^2)\},$$

and the coefficients of this generating function give the numbers of bicentre-trees. C. IX. 67

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16. The numbers of centre-trees and bicentre-trees for the first few values of n are as follows:

n	Centre-trees	Bicentre-trees	Total
1	1	0	1
2	0	1	1
3	1	0	1
4	1	1	2
5	2	0	2
6	3	1	4
7	6	0	6
8	11	1	12
9	23	0	23
10	47	2	49

17. *Application to the Paraffins* C_nH_{2n+2}.

The case of carbon-trees, where the number of branches from a knot is at most 4, is that which presents the chief chemical interest, as giving the paraffins C_nH_{2n+2}.

I call to mind here that the theory of the carbon-root-trees is established as an analytical result for its own sake and as the foundation for the other cases, but that it is the number of the carbon centre- and bicentre-trees which is the number of the paraffins.

The theory extends to the case where the number of branches from a knot is at most $= 5$, or $=$ any larger number; but I have not developed the formulae.

18. I pass now to the analytical theory: considering first the case of general root-trees, we endeavour to find for a given altitude N the number of trees of a given number of knots n and main branches a, or say the generating function

$$\sum a_n x^n t^a,$$

where the coefficient gives the number of the trees in question.

And we assume that the problem is solved for the cases of the several inferior altitudes 0, 1, 2, 3, $\ldots$, $N-1$.

This being so, observe that a tree of altitude N can be built up as shown in the figure, which I call the *edification diagram*, by combining one or more trees of altitude $N-1$ with a single tree of altitude not exceeding $N-1$; viz. in the figure, $N=3$, we have the two trees a, b, each of altitude 2, combined, as shown by the dotted lines, with the tree c of altitude 1: the whole number of knots in the resulting tree being the sum of the numbers of knots in the component trees. C. IX. 68

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19. The edification diagram gives at once the formula for the generating function. If $\Phi_N(x,t)$ denote the generating function for trees of altitude N, then

$$\Phi_N(x,t) = x \sum \frac{t^m}{m!} \left(\Phi_{N-1} + \Phi_{N-2} + \ldots + \Phi_0\right)^m,$$

where the sum extends over all values of m from 1 to ∞, and the expansion of the power is to be interpreted as giving all combinations of the generating functions Φ_{N-1}, Φ_{N-2}, ..., Φ_0.

20. The formula may be written in the more convenient form

$$\Phi_N(x,t) = x \left\{ \exp\left(t \sum_{r=0}^{N-1} \Phi_r \right) - 1 \right\},$$

and from this the generating functions may be calculated in succession.

21. The generating function for the centre-trees is obtained from that for the root-trees by a simple transformation. If $F(x,t)$ denote the generating function for root-trees, then the generating function for centre-trees is

$$\Psi(x,t) = F(x,t) - \tfrac{1}{2}\{F(x,t)^2 - F(x^2,t^2)\}.$$

22. For the carbon-trees (paraffins), we have the additional condition that no knot has more than 4 branches. The generating function is then obtained by truncating the general generating function after the terms involving t^4.

23. The number of paraffins C_nH_{2n+2} for the first few values of n is given in the following table:

n	Number of paraffins
1	1
2	1
3	1
4	2
5	3
6	5
7	9
8	18
9	35
10	75
11	159
12	355
13	802
14	1858
15	4347
16	10359
17	24894
18	60523
19	148284
20	366319

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24. The theory of trees may be extended to the case of *di-trees*, or trees with two roots, and more generally to *multi-trees* with any number of roots. The generating functions for these cases may be obtained by obvious modifications of the formulae for single trees.

25. The theory may also be applied to the determination of the number of isomers of other classes of hydrocarbons, such as the olefines C_nH_{2n}, the acetylenes C_nH_{2n-2}, and the aromatic compounds.

26. In the case of the olefines, we have n carbon-atoms connected by n bands, forming a figure with one closed polygon; and the number of isomers is equal to the number of distinct figures with n knots and n branches.

27. The theory of trees is thus seen to have important applications to chemistry; and it is hoped that it may also be found useful in other branches of science.

28. I conclude this paper with a brief summary of the principal results obtained.

29. The number of root-trees with n knots is given by the coefficient of x^n in the generating function

$$F(x) = x\exp\{F(x) + \tfrac{1}{2}F(x^2) + \tfrac{1}{3}F(x^3) + \ldots\}.$$

30. The number of centre-trees with n knots is given by the coefficient of x^n in

$$F(x) - \tfrac{1}{2}\{F(x)^2 - F(x^2)\}.$$

31. The number of bicentre-trees with n knots is given by the coefficient of x^n in

$$\tfrac{1}{2}\{F(x)^2 + F(x^2)\}.$$

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32. The number of paraffins C_nH_{2n+2} is equal to the number of carbon centre- and bicentre-trees with n knots, subject to the condition that no knot has more than 4 branches.

33. The theory may be extended to the case where the number of branches from a knot is at most = 5, or = any larger number; and it may also be applied to the determination of the number of isomers of other classes of hydrocarbons.

34. The results obtained in this paper complete the theory of trees as far as it has been developed; and it is hoped that they may serve as a foundation for further researches in this interesting branch of analysis.

Tables.

Table I.—General Root-Trees.

X	t	1	2	3	4	5	6	7	8	9	10	11	12	Total
1	1	1												1
2	1	1												1
	2	1												1
	Total	2												2
3	1	1												1
	2	1	1											2
	3	1												1
	Total	3	1											4
4	1	1												1
	2	1	2	1										4
	3	1	1											2
	4	1												1
	Total	4	3	1										8
5	1	1												1
	2	1	3	3	1									8
	3	1	2	1										4
	4	1	1											2
	5	1												1
	Total	6	8	4	1									20

6	1	1												1
	2	1	4	6	4	1								16
	3	1	3	3	1									8
	4	1	2	1										4
	5	1	1											2
	6	1												1
	Total	10	18	13	5	1								48
7	1	1												1
	2	1	6	10	10	5	1							33
	3	1	3	5	3	1								13
	4	1	3	3	1									8
	5	1	2	1										4
	6	1	1											2
	7	1												1
	Total	14	38	36	19	6	1							115

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Table II.—General Centre- and Bicentre-Trees.

X	t	1	2	3	4	5	6	7	8	9	10	11	12	Total
1	1	1												1
2	1		1											1
	Total		1											1
3	1	1												1
	2													0
	Total	1												1
4	1	1												1
	2	1												1
	Total	2												2
5	1	1	1											2
	2	1												1
	3	1												1
	Total	3	1											4
6	1	1	2	1										4
	2	1	1											2
	3	1	1											2
	Total	3	4	1										8
7	1	2	4	3	1									10
	2	1	2	1										4
	3	1	2	1										4
	Total	4	8	5	1									18
8	1	2	8	8	4	1								23
	2	1	4	3	1									9
	3	1	3	2	1									7
	4	1	1											2
	Total	5	16	13	6	1								41

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Subsidiary Table for GF's of Tables I. and II.

Index of x	1	2	3	4	5	6	7	8	9	10	11	12	13
GF, column 0	x												
GF, column 1													
First factor	1												
Second factor	1	1											
GF, column 2													
First factor	1	1	1										
Second factor	1	1	1	1									
GF, column 3													
First factor	1	1	1	1	1								
Second factor	1	1	1	1	1	1							
GF, column 4													
First factor	1	1	1	1	1	1	1						
Second factor	1	1	1	1	1	1	1	1					

The first factors are the generating functions for the columns of Table I, and the second factors are the generating functions for the columns of Table II.

38. The subsidiary table shows that the generating functions for the successive columns are connected by simple multiplicative relations. Thus, for column r, the generating function is the product of the first factor and the second factor for that column.

39. And so also it shows that, as regards Table IV. (centre part), the GF's of the successive columns are

for column: x, $\quad 1 : t^2x^3{\cdot}X, \quad 2 : t^2x^4{\cdot}X, \quad 3 : t^2x^5{\cdot}X, \quad 4 : t^2x^6{\cdot}X, \quad 5 : t^2x^7{\cdot}X,$

viz. that the successive GF's are

$$x,\ t^2x^3,\ t^2x^4,\ t^2x^5,\ t^2x^6,\ t^2x^7,\ \ldots$$

agreeing in fact with Table IV. C. IX. 73

40. The theory of the carbon-trees, or trees in which the order of each knot is at most 4, is obtained from the general theory by simply omitting from the generating functions all terms corresponding to knots of order greater than 4.

41. The generating function for the carbon root-trees is thus

$$\Phi(x, t) = x\left\{t + t\Phi_1 + t\frac{\Phi_2}{2} + t\frac{\Phi_3}{6} + t\frac{\Phi_4}{24}\right\},$$

where Φ_r denotes the generating function for trees in which the root has order r.

42. By developing this generating function, we obtain the numbers of carbon root-trees. The results are exhibited in Table III.

Table III.—Oxygen Root-Trees.

X	t	1	2	3	4	5	6	7	8	9	10	11
1	1	1										
2	1	1										
	2	1										
3	1	1										
	2	1	1									
	3	1										
4	1	1										
	2	1	2	1								
	3	1	1									
	4	1										
5	1	1										
	2	1	3	3	1							
	3	1	2	1								
	4	1	1									
	5	1										
6	1	1										
	2	1	4	6	4	1						
	3	1	3	3	1							
	4	1	2	1								
	5	1	1									
	6	1										

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43. The generating function for the oxygen centre-trees is obtained from that for the oxygen root-trees by the same formula as in the general case; and the numbers are exhibited in Table IV.

Table IV.—Oxygen Centre- and Bicentre-Trees.

X	t	1	2	3	4	5	6	7	8	9	10	11
1	1	1										
2	1		1									
3	1	1										
	2											
4	1	1										
	2	1										
5	1	1	1									
	2	1										
	3	1										
6	1	1	2	1								
	2	1	1									
	3	1	1									

44. The number of paraffins for the first 20 values of n has been calculated by means of these formulae, and the results are given in the following table:

n	Number of paraffins
1	1
2	1
3	1
4	2
5	3
6	5
7	9
8	18
9	35
10	75
11	159
12	355
13	802
14	1858
15	4347
16	10359
17	24894
18	60523
19	148284
20	366319

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Subsidiary Table for GF's of Tables III. and IV.

Index of x	1	2	3	4	5	6	7	8	9	10	11	12	13
GF, column 0	x												
GF, column 1													
First factor	1	1											
Second factor	1												
GF, column 2													
First factor	1	1	1										
Second factor	1	1											
GF, column 3													
First factor	1	1	1	1									
Second factor	1	1	1										
GF, column 4													
First factor	1	1	1	1	1								
Second factor	1	1	1	1									
GF, column 5													
First factor	1	1	1	1	1	1							
Second factor	1	1	1	1	1								

45. The table shows that the generating functions for the oxygen trees are obtained from those for the general trees by simply omitting the terms corresponding to orders greater than 4.

46. And so also it shows that, as regards Table IV. (centre part), the GF's of the successive columns are

for column: x, $1 : t^2x^3{\cdot}X$, $2 : t^2x^4{\cdot}X$, $3 : t^2x^5{\cdot}X$, $4 : t^2x^6{\cdot}X$, $5 : t^2x^7{\cdot}X$,

viz. that the successive GF's are

$$x,\ t^2x^3,\ t^2x^4,\ t^2x^5,\ t^2x^6,\ t^2x^7,\ \ldots$$

agreeing in fact with Table IV. C. IX. 76

Subsidiary Table for GF's of Tables I. and II. (continued).

H_4 Index of x	1	2	3	4	5	6	7	8	9	10	11	12	13
GF, column 4													
First factor	1	2	3	(1)									
Second factor	1	1	2	3	4	6	6	7	8	9	10	11	12
GF, column 5													
First factor	1	2	5	8	(1)								
Second factor	1	1	2	3	4	6	6	7	8	9	10	11	12

47. The subsidiary table for Tables III. and IV. is constructed in the same manner as that for Tables I. and II.; and it serves to verify the generating functions for the oxygen trees.

48. The generating functions for the carbon-trees are obtained from those for the oxygen-trees by omitting the terms corresponding to knots of order greater than 4; or, what is the same thing, by truncating the generating functions after the terms involving t^4.

49. The numbers of carbon root-trees, carbon centre-trees, and carbon bicentre-trees are exhibited in Tables V., VI., and VII. respectively.

50. The generating functions for the carbon-trees may be written in the form

$$\Phi(x,t) = x\sum_{m=1}^{4} \frac{t^m}{m!}\left(\sum_{r=0}^{N-1} \Phi_r\right)^m,$$

where the sum is limited to $m = 4$, corresponding to the restriction that no knot has more than 4 branches.

51. The coefficients of this generating function give the numbers of carbon root-trees; and from these the numbers of carbon centre-trees and bicentre-trees are obtained by the formulae already given.

52. The results of the calculation are exhibited in the following tables.

Table V.—Boron Root-Trees.

X	t	1	2	3	4	5	6	7	8	9	10	11	12	Total
1	1	1												1
2	1	1												1
	2	1												1
	Total	2												2
3	1	1												1
	2	1	1											2
	3	1												1
	Total	3	1											4
4	1	1												1
	2	1	2	1										4
	3	1	1											2
	4	1												1
	Total	4	3	1										8
5	1	1												1
	2	1	3	3	1									8
	3	1	2	1										4
	4	1	1											2
	5	1												1
	Total	6	8	4	1									20

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Table VI.—Boron Centre- and Bicentre-Trees.

X	t	1	2	3	4	5	6	7	8	9	10	11	12	Total
1	1	1												1
2	1		1											1
3	1	1												1
	2													0
4	1	1												1
	2	1												1
5	1	1	1											2
	2	1												1
	3	1												1
6	1	1	2	1										4
	2	1	1											2
	3	1	1											2
7	1	2	4	3	1									10
	2	1	2	1										4
	3	1	2	1										4

53. The numbers in the tables may be verified by means of the following formula for the total number of boron root-trees with n knots:

$$B_n = \sum \frac{1}{m!} \binom{m}{m_1, m_2, \ldots} B_{i_1}^{m_1} B_{i_2}^{m_2} \ldots,$$

where the sum extends over all partitions of $n-1$ into parts $i_1, i_2, \ldots$, and $m = m_1 + m_2 + \ldots$. C. IX. 79

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54. The theory of the boron-trees, in which the order of each knot is at most 3, is similar to that of the carbon-trees; and the numbers may be obtained by the same methods.

55. The generating function for the boron root-trees is

$$\Phi(x,t) = x\left\{t + t\Phi_1 + t\frac{\Phi_2}{2}\right\},$$

where the sum is limited to terms involving t and t^2, corresponding to the restriction that no knot has more than 3 branches.

56. The numbers of boron centre-trees and bicentre-trees are obtained from the generating function by the same formulae as in the general case.

57. The theory may be extended to the case of trees in which the order of each knot is at most k, for any value of k; and the generating function is then

$$\Phi(x,t) = x\sum_{m=1}^{k}\frac{t^m}{m!}\left(\sum_{r=0}^{N-1}\Phi_r\right)^m.$$

58. The number of trees with n knots, in which the order of each knot is at most k, may thus be determined for any values of n and k.

59. The theory of trees has important applications to the theory of chemical combinations, and in particular to the determination of the number of isomers of the paraffins and other hydrocarbons.

60. I have given in this paper a complete investigation of the theory of trees, and have shown how it may be applied to the determination of the number of isomers of the paraffins and other hydrocarbons.

61. The results obtained may be summarized as follows:

1. The number of root-trees with n knots is given by the coefficient of x^n in the generating function $F(x) = x\exp\{F(x) + \frac{1}{2}F(x^2) + \frac{1}{3}F(x^3) + \ldots\}$.

2. The number of centre-trees with n knots is given by the coefficient of x^n in $F(x) - \frac{1}{2}\{F(x)^2 - F(x^2)\}$.

3. The number of bicentre-trees with n knots is given by the coefficient of x^n in $\frac{1}{2}\{F(x)^2 + F(x^2)\}$.

4. The number of paraffins C_nH_{2n+2} is equal to the number of carbon centre- and bicentre-trees with n knots.

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62. The theory may be extended to the case of trees in which the order of each knot is at most = 5, or = any larger number; but I have not developed the formulae.

63. The theory may also be applied to the determination of the number of isomers of other classes of hydrocarbons, such as the olefines C_nH_{2n}, the acetylenes C_nH_{2n-2}, and the aromatic compounds.

64. The theory of trees is thus seen to have important applications to chemistry; and it is hoped that it may also be found useful in other branches of science.

65. The results obtained in this paper complete the theory of trees as far as it has been developed; and it is hoped that they may serve as a foundation for further researches in this interesting branch of analysis.

---

66. I conclude with a remark on the connexion of the theory of trees with the theory of invariants. The generating function for the number of root-trees is closely connected with the generating function for the number of invariants of a system of binary quantics; and the theory of trees may be regarded as a particular case of the theory of invariants.

67. This connexion will be developed in a future paper; and it is hoped that it may lead to important results in both theories. C. IX. 81

68. The theory of trees may be extended to the case of *multi-trees*, or trees with any number of roots. The generating function for multi-trees is obtained by a simple modification of the formula for single trees.

69. The generating function for r-trees (trees with r roots) is

$$\Phi_r(x) = \frac{1}{r!}\left\{\Phi(x)\right\}^r + \text{terms involving } \Phi(x^2), \Phi(x^3), \ldots$$

70. The number of multi-trees with n knots and r roots is given by the coefficient of x^n in $\Phi_r(x)$.

71. The theory of multi-trees has applications to the theory of chemical combinations, in the case of compounds with multiple bonds.

72. I have recently received from Prof. Schröder a memoir on the same subject, in which he obtains some of the same results by a different method. His memoir will appear in the forthcoming volume of the *Zeitschrift für Combinatorik.*

73. The methods employed in the present paper are entirely different from those of Prof. Schröder; and it is hoped that the comparison of the two methods may lead to further advances in the theory.

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74. The theory of trees may also be applied to the determination of the number of isomers of the aromatic compounds. In these compounds, the carbon-atoms are arranged in a closed ring; and the number of isomers is equal to the number of distinct arrangements of n carbon-atoms in a ring, subject to the condition that each carbon-atom is connected with at most 3 other carbon-atoms.

75. The generating function for the number of aromatic isomers is obtained from that for the number of trees by a simple modification; and the coefficients of the generating function give the numbers of isomers.

76. The numbers of aromatic isomers for the first few values of n are as follows:

n	Number of aromatic isomers
6	1
7	1
8	2
9	3
10	6
11	12
12	29
13	73
14	199
15	567

77. The theory may also be applied to the case of polycyclic compounds, in which the carbon-atoms are arranged in two or more closed rings. C. IX.

78. The generating function for the number of polycyclic isomers is obtained by combining the generating functions for the several rings; and the coefficients of the resulting generating function give the numbers of isomers.

79. The theory of trees is thus seen to have wide applications to chemistry; and it is hoped that it may be extended to other branches of science.

80. I conclude this paper with a brief account of the history of the theory of trees.

81. The theory was originated by me in the two papers "On the Analytical forms called Trees," published in the *Philosophical Magazine* in 1857 and 1859. In these papers I established the fundamental principles of the theory, and gave the first applications to chemistry.

82. The theory was afterwards developed by me in several papers, and was applied to the determination of the number of isomers of various classes of hydrocarbons.

83. In the paper "On the Mathematical Theory of Isomers," published in the *Philosophical Magazine* in 1874, I pointed out the connexion of the theory with the chemical theory of isomerism, and gave a more simple investigation of the fundamental formulae.

84. The present paper contains a complete investigation of the theory, and applies it to the determination of the number of isomers of the paraffins and other hydrocarbons.

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85. and so on indefinitely; viz. observing that the first factors, as shown by the Table, are $(1-tx)^{-1}$ $[t^0=1]$, $(1-tx^2)^{-1}$ $[t^1=x]$, &c., the Table in fact shows that as regards Table III. the GF's are

$$\text{for column: } x, \quad 1: tx^2+t^2x^3\cdot X, \quad 2: tx^3+t^2x^4\cdot X+t^3x^5, \quad 3: tx^4+t^2(x^5+x^6)\cdot X+t^3$$

$$4: tx^5+t^2(x^6+x^7+x^8)\cdot X+t^4x^9, \quad 5: tx^6+t^2(x^7+x^8+x^9+x^{10})\cdot X+t^5x^{11};$$

viz. developing as far as t^5, that the successive GF's are

$$\text{column: } x, \quad 1: tx^2+t^2x^3, \quad 2: tx^3+t^2(x^4+x^5), \quad 3: tx^4+t^2(x^5+x^6+x^7),$$

$$4: tx^5+t^2(x^6+x^7+x^8+x^9), \quad 5: tx^6+t^2(x^7+x^8+x^9+x^{10}+x^{11})$$

&c., agreeing with Table III.

And so also it shows that, as regards Table IV. (centre part), the GF's of the successive columns are

$$\text{for column: } x, \quad 1: t^2x^3\cdot X, \quad 2: t^2x^4\cdot X, \quad 3: t^2x^5\cdot X, \quad 4: t^2x^6\cdot X, \quad 5: t^2x^7\cdot X;$$

viz. that the successive GF's are

$$x,\ t^2x^3,\ t^2x^4,\ t^2x^5,\ t^2x^6,\ t^2x^7,\ \ldots$$

agreeing in fact with Table IV. C. IX. 85

Table V.—Boron Root-Trees.

X	t	1	2	3	4	5	6	7	8	9	10	11	12	Total
1	1	1												1
2	1	1												1
	2	1												1
	Total	2												2
3	1	1												1
	2	1	1											2
	3	1												1
	Total	3	1											4
4	1	1												1
	2	1	2	1										4
	3	1	1											2
	4	1												1
	Total	4	3	1										8
5	1	1												1
	2	1	3	3	1									8
	3	1	2	1										4
	4	1	1											2
	5	1												1
	Total	6	8	4	1									20
6	1	1												1
	2	1	4	6	4	1								16
	3	1	3	3	1									8
	4	1	2	1										4
	5	1	1											2
	6	1												1
	Total	10	18	13	5	1								48
7	1	1												1
	2	1	6	10	10	5	1							33
	3	1	3	5	3	1								13
	4	1	3	3	1									8
	5	1	2	1										4
	6	1	1											2
	7	1												1
	Total	14	38	36	19	6	1							115
8	1	1												1
	2	1	8	18	18	10	4	1						60
	3	1	4	9	4	1								19
	4	1	3	3	1									8
	5	1	2	1										4

	6	1	1											2
	7	1	1											2
	8	1												1
	Total	20	40	37	21	7	1							127
9	1	1												1
	2	1	10	28	30	20	9	3	1					102
	3	1	5	15	8	3	1							33
	4	1	4	6	3	1								15
	5	1	3	3	1									8
	6	1	2	1										4
	7	1	1											2
	8	1	1											2
	9	1												1
	Total	25	71	82	59	28	8	1						275
10	1	1												1
	2	1	14	46	60	48	26	10	4	1				210
	3	1	7	28	19	9	3	1						68
	4	1	5	11	6	3	1							27
	5	1	4	6	3	1								15
	6	1	3	3	1									8
	7	1	2	1										4
	8	1	1											2
	9	1	1											2
	10	1												1
	Total	29	119	170	147	88	36	9	1					598

C. IX. 86

Arthur Cayley

Collected Mathematical Papers

Volume IX, Pages 451–500

0.6 Paper 610: On the Analytical Forms Called Trees, with Application to the Theory of Chemical Combinations

[From the *Philosophical Magazine*, vol. XIII. (1874).]

[Page 451]

Table VI.—Boron Centre- and Bicentre-Trees.

Centre-Trees. *Altitude or number of column.*

x	t	0	1	2	3	4	5	Centre	Grand Tot.
1	0	1	1					1	1
	T	1	1						
2	—							0	1
3	2		1	1				1	1
	T		1	1					
4	2		1	1				1	2
	T		1	1					
5	2		1	1				1	2
	T		1	1					
6	2		1	1				2	4
	3		1	1					
	T		2	2					
7	2		1	1				4	6
	3		2	2					
	T		3	1	4				
8	2		2	2				5	11
	3		2	1	3				
	T		2	3	5				
9	2		1	5	1	6		10	18
	3		1	3	4				
	T		1	8	1	10			
10	2		1	5	3	8		19	37
	3		1	9	1	11			
	T		1	14	4	19			
11	2		6	11	1	18		36	66
	3		14	4		18			
	T		20	15	1	36			
12	2		4	22	4	30		68	135
	3		21	16	1	38			
	T		25	38	5	68			
13	2		3	44	19	1	67	138	265
	3		24	42	5		71		
	T		27	86	24	1	138		

Bicentre-Trees. *Altitude.*

x	0	1	2	3	4	5
1						
2		1				
3						
4		1				
5		1				
6		1	1			
7			2			
8			5	1		
9			5	3		
10			6	11	1	
11			4	22	4	
12			3	44	19	1
13		1	68	53	5	

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Subsidiary Table for *GF*'s of Tables V. and VI.

Index of x.

t	0	1	2	3	4	5	6	7	8	9	10	11	12	13	
*								−1	−6	−20	−53	−125	−274	−571	*GF, c.7*
0	1							1	6	20	53	125	274		1st f.
0		1	1	1	2	3	6	11	22	39	70	118	200	324	2nd f.
1			1	1	1	3	5	12	23	51	101	207	398	773	
2											1	4	19		
*									−1	−7	−27	−81	−213	−516	*GF, c.8*
0	1								1	7	34	129	432		1st f.
0		1	1	1	2	4	8	17	39	88	203	464	1056	2381	2nd f.
1			1	1	1	3	6	15	33	82	193	473	1135	2743	
2											1		9		
*										−1	−8	−35	−117	−338	*GF, c.9*
0	1									1	8	43	180		1st f.
0		1	1	1	2	4	8	17	39	89	210	498	1185	2813	2nd f.
1			1	1	1	3	6	15	33	82	194	481	1178	2924	
*											−1	−9	−44	−162	*GF, c.10*
0	1										1	9	44		1st f.
0		1	1	1	2	4	8	17	39	89	211	506	1228	2993	2nd f.
1			1	1	1	3	6	15	33	82	194	482	1187	2977	
*												−1	−10	−54	*GF, c.11*
0	1											1	10		1st f.
0		1	1	1	2	4	8	17	39	89	211	507	1237	3048	2nd f.
1			1	1	1	3	6	15	33	82	194	482	1188	2987	
*													−1	−11	*GF, c.12*
0	1												1		1st f.
0		1	1	1	2	4	8	17	39	89	211	507	1238	3055	2nd f.
1			1	1	1	3	6	15	33	82	194	482	1188	2988	
*														−12	*GF, c.13*
0	1														1st f.
0		1	1	1	2	4	8	17	39	89	211	507	1238	3056	2nd f.
1			1	1	1	3	6	15	33	82	194	482	1188	2988	

The subsidiary table continues on page 453 with columns 0–6 in the same format, showing the generating functions, first factors, and second factors for each column.

[Page 453]

Subsidiary Table (*continued*). Columns 0–6:

Column 0. GF: 1. First factor: 1. Second factor: 1.

Column 1. GF: -1. First factor: 1. Second factor: 1.

Column 2. GF: $-1, -1$. First factor: $1, 1$. Second factors: $(1, 1, 1)$ and (1).

Column 3. GF: $-1, -2, -2, -1, -1$. First factor: $1, 2, 2, 1, 1$. Second factors: $(1, 1, 1, 2, 2, 1, 1)$ and $(1, 1, 1, 2, 2, 1, 1)$.

Column 4. GF: $-1, -3, -5, -7, -8, -9, -7, -7, -4, -3$. First factor: $1, 3, 5, 7, 8, 9, 7, 7, 4$. Second factors: $(1, 1, 1, 2, 3, 4, 4, 5, 4, 4, 3, 2, 1)$ and $(1, 1, 1, 3, 5, 10, 14, 23, 29, 40, 46, 55, 128)$.

Column 5. GF: $-1, -4, -9, -17, -29, -45, -66, -89, -118$. First factor: $1, 4, 9, 17, 29, 45, 66, 89$. Second factors: $(1, 1, 1, 2, 3, 5, 6, 8, 8, 9, 7, 7, 4)$ and $(1, 1, 1, 3, 6, 14, 27, 56, 103, 194, 343, 605, 117)$.

Column 6. GF: $-1, -5, -14, -32, -66, -127, -231, -405$. First factor: $1, 5, 14, 32, 66, 127, 231$. Second factors: $(1, 1, 1, 2, 3, 6, 10, 17, 25, 38, 52, 73, 93)$ and $(1, 1, 1, 3, 6, 15, 32, 75, 160, 350, 732, 1534, 1093)$.

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Table VII.—Carbon Root-trees.

Altitude or number of column.

x	t	0	1	2	3	4	5	6	7	8	9	10	11	12
1	0	1	1											
	T	1	1											
2	1		1	1										
	T		1	1										
3	1		1	1	1									
	2		1	1										
	T		1	1	2									
4	1			1	1	2								
	2			1		1								
	3		1			1								
	T		1	2	1	4								
5	1			1	2	1	4							
	2			2	1		3							
	3			1			1							
	4		1				1							
	T		1	4	3	1	9							
6	1				2	4	3	1	8					
	2			2	3	1		6						
	3			2	1			3						
	4			1				1						
	T			5	8	4	1	18						
7	1				4	8	4	1	17					
	2			2	8	4	1		15					
	3			3	3	1			7					
	4			2	1				3					
	T			7	16	13	5	1	42					

Table VII continued on pages 455–456 with values for $x = 8$ to 13.

[Pages 455–456]

Table VII (*continued*). For $x = 8$ to 13 the values continue similarly. For $x = 8$: $t = 1$: (5, 15, 13, 5, 1, 39); $t = 2$: (1, 13, 13, 5, 1, 33); $t = 3$: (3, 9, 4, 1, 17); $t = 4$: (3, 3, 1, 7); Total: (7, 30, 33, 19, 6, 1, 96). For $x = 9$: Total (8, 52, 78, 57, 26, 7, 1, 229). For $x = 10$: Total (7, 85, 169, 156, 89, 34, 8, 1, 549). For $x = 11$: Total (7, 135, 355, 394, 272, 130, 43, 9, 1, 1346). For $x = 12$: Total (5, 202, 712, 953, 770, 439, 181, 53, 10, 1, 3326). For $x = 13$: Total (5, 300, 1399, 2222, 2065, 1355, 665, 243, 63, 11, 1, 8329).

Table VIII.—Carbon Centre- and Bicentre-Trees.

Centre-Trees.

x	t	0	1	2	3	4	5	Centre	Grand Tot.
1	0	1	1					1	1
2	—							0	1
3	2		1	1				1	1
4	2		1	1				1	2
5	2		1	1	2			2	3
6	2		1	1				2	5
	3		1	1					
	T		2	2					
7	2		2	1	3			6	9
	3		2	2	1				
	4		1						
	T		5	1	6				
8	2		1	2	3			9	18
	3		3	1	4				
	4		2	2					
	T		6	3	9				
9	2		1	7	1	9		20	35
	3		3	3	6				
	4		4	1	5				
	T		8	11	1	20			
10	2		3	12	3	15		37	75
	3		3	11	1	15			
	4		4	3	7				
	T		7	26	4	37			
11	2		2	23	14	1	38	86	159
	3		2	24	4		30		
	4		5	12	1		18		
	T		7	59	19	1	86		
12	2		1	30	39	4	73	183	357
	3		1	54	19	1	75		
	4		4	27	4		35		
	T		5	111	62	5	183		
13	2		1	42	108	23	1	419	799
	3		1	88	63	5		157	
	4		4	63	20	1		88	
	T		5	193	191	29	1	419	

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Subsidiary Table for GF's of Tables VII. and VIII.

The table gives, for columns 0–13, the generating function coefficients, first factors, and second factors. Column 0: $GF = 1$; column 1: $GF = -1$; column 2: $GF = -1, -1, -1$; column 3: $GF = -1, -2, -4, -4, -5, -4, -4, -3, -2, -1, -1$; column 4: $GF = -1, -3, -8, -15, -27, -43, -67, -97, -136, -183$; column 5: $GF = -1, -4, -13, -32, -74, -155, -316, -612, -1160$; column 6: $GF = -1, -5, -19, -56, -151, -374, -889, -2032$; column 7: $GF = -1, -6, -26, -88, -267, -743, -1968$; columns 8–13 continue similarly.

[Pages 458–459]

Subsidiary Table (*continued*). The table continues with the same format for the remaining columns, showing first factors (with entries marked (1) where applicable) and second factors with their full coefficient sequences.

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I annex the following two Tables of (centre- and bicentre-) trees as far as I have completed them.

Table A:

Valency not greater than

Knots	0	1	2 Oxygen	3 Boron	4 Carbon	5	6	7	8	Gen.
1	1	1	1	1	1	1	1	1	1	1
2		1	1	1	1	1	1	1	1	1
3			1	1	1	1	1	1	1	1
4			1	2	2	2	2	2	2	2
5			1	2	3	3	3	3	3	3
6			1	4	5	6	6	6	6	6
7			1	6	9	10	11	11	11	11
8			1	11	18	21	22	23	23	23
9			1	18	35	42	45	46	47	47
10			1	37	75					106
11			1	66	159					235
12			1	135	357					551
13			1	265	799					1301

Table B:

Actual Valency.

Knots	0	1	2	3	4	5	6	7	8
1	1	1							
2		1							
3			1						
4			1	1					
5			1	1	1				
6			1	3	1	1			
7			1	5	3	1	1		
8			1	10	7	3	1	1	
9			1	17	17	7	3	1	1
10			1	36	38				
11			1	65	93				
12			1	134	222				
13			1	264	534				

In A, the columns 2, 3, 4, and the last column are the totals given by the Tables IV., VI., VIII., and II., and the remaining numbers of columns 5, 6, 7, 8 have been found by trial; and, in B, the several columns are the differences of the columns of A. The signification is obvious; for instance, if the number of knots is $= 9$, then Table A, if the valency, or the maximum number of branches from a knot, is

is	$= 2$,	3,	4,	5,	6,	7,	8 or any greater number,
No. of trees	$= 1$,	18,	35,	42,	45,	46,	47;

viz. with 9 knots the tree can have at most 8 branches from a knot, so that the number of trees having at most 8 branches from a knot is $= 47$, the whole number of trees with 9 knots; and so the number of knots being as before $= 9$, Table B shows that the number of 47 is made up of the numbers

$$1,\quad 17,\quad 17,\quad 7,\quad 3,\quad 1,\quad 1;$$

viz. 1 is the No. of trees, at most 2 branches from a knot,

17	"	3	"	at least one 3-branch knot.		
17	"	4	"		" 4	"
7	"	5	"		" 5	"
3	"	6	"		" 6	"
1	"	7	"		" 7	"
1	"	8	"		" 8	"

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I annex also a plate showing the figures of the $1+1+2+3+6+11+23+47$ trees of $1,\ 2,\ 3,\ \ldots,\ 9$ knots, classified according to their altitudes and number of main branches; and as to the bicentre-trees, according to the number of main branches from *each* point of the bicentre. The affixed numbers show in each case the greatest number of branches from a knot; so that when this is (2), the knots may be oxygen-, boron-, carbon-, &c., atoms; when (3), boron-, carbon-, &c., atoms; when (4), carbon-, &c., atoms; and so on.

0.7 Paper 611: Report of the Committee on Mathematical Tables

[From the *Report of the British Association for the Advancement of Science* (1875), pp. 305–336.]

[Page 461]

611.

REPORT OF THE COMMITTEE ON MATHEMATICAL TABLES:

CONSISTING OF PROFESSOR CAYLEY, F.R.S., PROFESSOR STOKES, F.R.S.,
PROFESSOR SIR W. THOMSON, F.R.S., PROFESSOR H. J. S. SMITH, F.R.S.,
AND J. W. L. GLAISHER, F.R.S.

THE present Report (say Report 1875) is in continuation of that by Mr Glaisher, published in the volume for 1873, and here cited as Report 1873.

Report 1873 extends to all those tables which are at p. 3 (l.c.) included under the headings:

- A, auxiliary for non-logarithmic calculation, 1, 2, 3;
- B, logarithmic and circular, 4, 5, 6;
- C, exponential, 7, 8 (but only partially to C. 8), other than those tables of C referred to as "h.l $\tan(45^\circ + \frac{1}{2}\phi)$"; and also partially (see Art. 24, pp. 81–83) to the tables included under the heading "E. 11, transcendental constants ϵ, π, γ, &c., and their powers and functions."

A future Report will comprise the tables, or further tables, included under the headings:

- C. 8. Hyperbolic antilogarithms (e^x) and h.l $\tan(45^\circ + \frac{1}{2}\phi)$, and hyperbolic sines, cosines, &c.
- D. Algebraic constants.
    9. Accurate integer or fractional values. Bernoulli's Numbers, $\Delta^n 0^m$, &c. Binomial coefficients.
    10. Decimal values auxiliary to the calculation of series.
- E. 11. Transcendental constants ϵ, π, γ, &c., and their powers and functions.

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11. A catalogue of 61 pairs of numbers is given by

Euler, *Op. Arith. Coll.* t. i. pp. 144—145; it occupies about one page. The paper, "De Numeris Amicabilibus," pp. 102—145, contains an elaborate investigation of the theory, by means whereof all but two of the pairs of numbers are obtained. The first pair is the above-mentioned one, $2^2.5.11$ and $2^2.71$ (= 220 and 284); and the fifty-ninth pair is the high numbers

$$3^5.7^2.13.19.53.6959 \quad \text{and} \quad 3^5.7^2.13.19.179.2087.$$

The last two pairs are referred to as "formae diversae a praecedentibus;" viz. these are

$$\begin{cases} 2^2.19.41 \\ 2^2.199 \end{cases} \quad \text{and} \quad \begin{cases} 2^3.41.467 \\ 2^2.19.233. \end{cases}$$

12. A Table of the Frequency of Primes is given by

Gauss, *Tafel der Frequenz der Primzahlen, Werke*, t. II. pp. 436—443; viz. this extends to 3,000,000.

The first part, extending to 1,000,000, = 1000 thousand, shows how many primes there are in each thousand: a specimen is

1,, 168;
2,, 135;
3,, 127;
4,, 120;
5,, 119;
&c.;

viz. in the first thousand there are 168 primes, in the second thousand 135 primes, and so on.

For the second and third millions the frequency is given for each ten thousand: a specimen is

1,000,000 to 1,100,000.

	0	1	2	3	4	5	6	7	8	9	Sum
1	1										1
2		1					1			1	4
3			4	2	2	3	1	2	3	3	21
4	2	8	5	4	3	6	9	4	5	8	54
5	11	10	8	18	12	10	10	12	15	8	114
6	14	14	18	21	16	22	19	15	17	15	171
7	26	17	23	23	24	24	17	22	20	21	217
8	19	19	21	7	14	15	20	17	15	17	164
9	11	13	9	13	14	14	12	13	11	16	126
10	8	6	8	5	9	5	5	9	7	9	71
11	6	6	4	6	3	1	3	1	4	5	39
12	1	1	2	1	1	1		2	2	1	12
13	1	1			1		1	1	1		6
	752	719	732	700	731	698	713	722	706	737	7210,

$$\int \frac{dx}{\log x} = 7212.99;$$

viz. in the interval 1,000,000 to 1,010,000, 100 hundreds, there is 1 hundred containing 1 prime, there are 2 hundreds each containing 4 primes, 11 hundreds each containing 5 primes,..., 1 hundred containing 13 primes, so that, as

$$\begin{array}{rrcr} 1\times & 1 & = & 1, \\ 4\times & 2 & = & 8, \\ 5\times & 11 & = & 55, \\ & & \vdots & \\ 13\times & 1 & = & 13, \\ \hline 100 & & & 752, \end{array}$$

the whole 10,000 contains 752 primes; the next 10,000 contains 719 primes, and so on; the whole 100,000 thus containing 752+719+&c. = 7210 primes, which number is at the foot compared with the theoretic approximate value

$$\int \frac{dx}{\log x} \text{ (limits 1,000,000 to 1,010,000)} = 7212.99.$$

The integral in question is represented by the notation $\operatorname{Li} x$ or $\operatorname{li} x$.

p. 443. We have the like tables 1,000,000 to 2,000,000 and 2,000,000 to 3,000,000, showing for each 100,000 how many hundreds there are containing 0 prime, 1 prime, 2 primes, up to (the largest number) 17 primes.

13. It is noticed that

the 26,379th hundred contains no prime,
the 27,050th hundred contains 17 primes.

It may be observed that, if $N = 2.3.5\ldots p$, the product of all the primes up to p, then each of the numbers $N+1$ and $N+q$ (if q be the prime next succeeding p) is or is not a prime; but the intermediate numbers $N+2$, $N+3, \ldots$, $N+q-1$ are certainly composite; viz. we thus have at least $q-2$ consecutive composites. To obtain in this manner 99 consecutive composites, the value of N would be $=$ $2.3.5\ldots 97$, viz. this is a number far exceeding 2,637,900; but, in fact, the hundred numbers 2,637,901 to 2,638,000 are all composite.

Legendre, in his *Essai sur la Théorie des Nombres* (1st edit., 1798; 2nd edit., 1808, supplement, 1816: references to this edition), gives for the number of primes inferior to a given limit x the approximate formula

$$\frac{x}{\log x - 1.08366};$$

and p. 394, and supplement, p. 62, he compares for each 10,000 up to 100,000, and for each 100,000 up to 1,000,000, the values as computed by this formula with the actual numbers of primes exhibited by the tables of Vega and Chernac. Thus for $x = 1{,}000{,}000$, the computed value is 78,543, the actual value 78,493.

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14. A table of frequency is given by

Glaisher, J. W. L., *British Association Report* for 1872, p. 20. This gives for the second and the ninth millions, respectively divided into intervals of 50,000, the actual number of primes in each interval, as compared with the theoretic value $\mathrm{li}\,x' - \mathrm{li}\,x$; and also deduced therefrom, by the formula $\log \frac{1}{2}(x' + x)$, a table of the average interval between two consecutive primes; this average interval increases very slowly: at the beginning and the end of the second million the values are 13·76 and 14·58 (theoretic values 13·84 and 14·50); at the beginning and the end of the ninth million 16·02 and 15·95 (theoretic values 15·90 and 16·01).

15. Coming under the head of Divisor Tables, some tables by Reuschle and Gauss may be here referred to. These are:—

Reuschle, *Mathematische Abhandlung, zahlentheoretische Tabellen sammt einer dieselben treffenden Correspondenz mit der verewigten C. G. J. Jacobi*, 4°, pp. 1—61* (1856). The tables belonging to the present subject are

A. Tafeln zur Zerlegung von $a^n - 1$ (pp. 18—22).

I. Table of the prime factors of $10^n - 1$, viz.

(a. pp. 18—19.) Complete decomposition of $10^n - 1$, $n = 1$ to 42: and $10^n + 1$, $n = 1$ to 21. Some values of n are omitted.

A specimen is

$$10^{13} - 1 = 3^2.53.79.265371653,$$
$$10^{13} + 1 = 11.189.1058313049.$$

(b. p. 19.) List of the specific prime factors f of $10^n - 1$, or the prime factors of the residue after separation of the analytical factors, for those values of n for which the complete decomposition is unknown, and omitting those values for which no factor is known, $n = 25$ to 243.

A specimen is

n	f
25	21401.

The meaning seems to be, residue of $10^{25} - 1$ is $1 + 10^5 + 10^{10} + 10^{15} + 10^{20}$, and this contains the prime factor 21401; but it is not clear why this is the "specific prime factor."

II. Prime factors of $a^n - 1$ for different values of a and n.

(a. p. 20) gives for 41 values of a (2, 3, &c. at intervals to 100) and for the following values of n the decompositions of the residues or specific factors of $a^n - 1$; viz. these are

$$\begin{array}{lll} n = & 1, & a-1; \\ " & 2, & a+1; \\ " & 3, & a^2+a+1; \\ " & 6, & a^2-a+1; \\ " & 4, & a^2+1; \\ " & 5, & a^4+a^3+a^2+a+1; \\ " & 10, & a^4-a^3+a^2-a+1; \\ " & 8, & a^4+1; \\ " & 12, & a^4-a^2+1. \end{array}$$

A specimen is

a	$a-1$	a^2-1	a^3-1	a^6-1	a^4-1	a^5-1	$a^{10}-1$	a^8-1	$a^{12}-1$
10	3^2	11	$3^3.37$	7.13	101	41.271	9091	73.137	9901

(b. p. 21.) Specific prime factors for the numbers 2, 3, 5, 6, 7, 10, (the powers 4, 8, 9 being omitted as coming under 2 and 3), for the exponents 1 to 42.

A specimen is

n	2^n-1	3^n-1	5^n-1	6^n-1	7^n-1	10^n-1
19	524287	1597.363889	$191.x$	$191.x$	$419.x$	

where the x denotes that the other factor is not known to be prime. And so, where no number is given, as in $10^{19}-1$, it is not known whether the number $(= 1 + 10^1 + 10^2 + \ldots + 10^{18})$ is or is not prime.

Addition, p. 22. For $a = 2$, the complete decomposition of the prime factor of $2^n - 1$ is given for values of $n, = 44, 45, \ldots$ at intervals to 156.

A specimen is

n	f
44	397.2113,

viz.

$$2^{20} - 2^{18} + 2^{16} - \ldots - 2^2 + 1, = 838861 = 397.2113.$$

$n = 31$, Fermat's prime. $n = 37$, the first case for which the decomposition is not given completely. $n = 41$, the first case for which no factor is known.

[Pages 467–468]

16. Gauss, Tafel zur Cyclotechnie, *Werke*, t. II. pp. 478—495, shows, for 2452 numbers of the several forms a^2+1, a^2+4, a^2+9, ..., a^2+81, the values of a such that the number in question is a product of prime factors no one of which exceeds 200, and exhibits all the odd prime factors of each such number. The table is in nine parts, zerlegbare a^2+1, zerlegbare a^2+4, &c., with to each part a subsidiary table, as presently mentioned. Thus a specimen is

zerlegbare a^2+9

1	5
2	13
4	5.5
5	17
7	29
8	73
⋮	⋮
1411168679	5.5.13.17.17.89.113.157.173.197.197;

viz. 1^2+9, odd prime factor is 5, 2^2+9, ", ", 13, 4^2+9, ", ", factors are 5, 5, and so on.

And the subsidiary table is

5	1, 4, 79
13	2, 11, 41
17	5, 29, 46, 379, 1042
⋮	⋮

showing that the numbers a for which the largest factor is 5 are 1, 4, 79; those for which it is 13 are 2, 11, 41; and so on.

The object of the table is explained in the *Bemerkungen*, (l. c., p. 523), by Schering, the editor of the volume, viz. it is to facilitate the calculation of the circular arcs the cotangents of which are rational numbers. To take a simple example, it appears to be by means of it that Gauss obtained, among other formulae, the following:

$$\frac{\pi}{4} = 12\arctan\tfrac{1}{18} + 8\arctan\tfrac{1}{57} - 5\arctan\tfrac{1}{239},$$

and

$$= 12\arctan\tfrac{1}{38} + 20\arctan\tfrac{1}{57} + 7\arctan\tfrac{1}{239} + 24\arctan\tfrac{1}{268}.$$

[Pages 468–469]

17. *Prime Roots.*—Let p be a prime number; then there exist $\varpi(p-1)$ inferior integers g, such that all the numbers 1, 2, ..., $p-1$ are, to the modulus p,

$$\equiv 1,\ g,\ g^2,\ \ldots,\ g^{p-2}\ (g^{p-1} \text{ is of course } \equiv 1).$$

This being so, g is said to be a prime root of p; and moreover the several numbers g^a, where a is any number whatever less than and prime to $p-1$, constitute the series of the $\varpi(p-1)$ prime roots of p.

18. A small table of prime roots, $p = 3$ to 37, is given by **Euler**, *Op. Arith. Coll.* t. i. pp. 525—526.

19. A table, p and p^m, 3 to 97, is given by **Gauss**, "Disquisitiones Arithmeticae," 1801, (*Werke*, t. I. p. 468).

20. Jacobi's *Canon Arithmeticus*, the complete title of which is *Canon Arithmeticus sive tabula quibus exhibentur pro singulis numeris primis vel primorum potestatibus infra 1000 numeri ad datos indices et indices ad datos numeros pertinentes. Edidit C. G. J. Jacobi. Berolini, 1839.* 4°.

The contents are as follows:—

	Pages
Introductio	i to xl
Tabulae numerorum ad indices datos pertinentium	1—221
Tabulae residuorum et indicum sibi mutuo respondentium	222—238
Hujus tabula ea pars quae pertinet ad modulos formae 2^n	239—240.

[Pages 470–472]

21. Jacobi, *Crelle*, t. xxx. (1846), pp. 181, 182, a table of m' for the argument m, such that $1+g^m \equiv g^{m'}$ (mod. p), $p = 7$ to 103, and $m = 0$ to 102.

22. Reuschle's Memoir, the following relating to prime roots:—

a. Table of the Hauptexponenten of the six roots 10, 5, 2, 6, 3, 7 for all prime numbers of the first 1000, together with the least primitive root of each of these numbers (pp. 42—46).

A specimen is as follows:—

		10		5		2		6		3		7		w
p	$p-1$	e	n	e	n	e	n	e	n	e	n	e	n	
53	$2^2.13$	13	4	52	1	52	1	26	2	52	1	26	2	2

where e is the Hauptexponent or indicatrix of the root, $n = \frac{p-1}{e}$, w the least primitive root.

b. The like table for the roots 10 and 2 for all prime numbers from 1000 to 5000 (pp. 47—53).

c. The like table for the root 10 for all prime numbers between 5000 and 15000 (pp. 53—61).

23. For a composite number n, if $N = \varpi(n)$ be the number of integers less than n and prime to it, then if x be any number less than n and prime to it, we have $x^N \equiv 1$ (mod. n). But we have in this case no analogue of a prime root. The maximum indicator I is defined. For instance, $n = 12$, $N = \varpi(n) = 4$; the maximum indicator I is $= 2$.

[Page 472]

A table of the maximum indicator $n = 1$ to 1000 is given by **Cauchy**, *Exer. d'Analyse &c.*, t. II. (1841), pp. 36—40.

24. Cayley, Specimen Table $M \equiv a^\alpha b^\beta$ (mod. N) for any prime or composite modulus; *Quart. Math. Journ.* vol. ix. (1868), pp. 95, 96, and folding sheet, [397].

[F. 12. Divisors, &c.] *continued.* Prime Roots. The Canon Arithmeticus, Quadratic residues. Art. II.

25. *Quadratic Residues.*—In regard to a given prime number p, a number N is or is not a quadratic residue according as the index of N is even or odd.

A small table, $p = 3$ to 97 and $N = -1$ and (prime values) 3 to 97, is given by **Gauss**, "Disquisitiones Arithmeticae," 1801; Table II. (*Werke*, t. I. p. 469).

The same table carried from $p = 3$ to 503, and prime values $N = 3$ to 997, is given by **Gauss**, *Werke*, t. II. pp. 400—409.

26. Table III. for the conversion into decimals of a vulgar fraction, denominator p or p^μ, not exceeding 100; and more extensively in *Werke*, t. II. pp. 412—434, denominators primes or powers of primes, p^μ up to 997.

[Pages 473–474]

The required mantissae, denoted in the table by (0), (1), (2), (3), (4), are those of $\frac{10}{11}$, $\frac{10.2}{11}$, $\frac{10.2^2}{11}$, $\frac{10.2^3}{11}$, $\frac{10.2^4}{11}$; viz. these fractions are respectively $=$ ·9090..., 1·8181..., 3·6363..., 7·2727..., 14·5454...; or their mantissae are 90, 81, 63, 27, 54.

And we accordingly have as a specimen

$$11 \mid (1)\ldots 81,\quad (2)\ldots 63,\quad (3)\ldots 27,\quad (4)\ldots 54,\quad (0)\ldots 90.$$

Or again, as another specimen, $r = 2$:—

$$27 \mid (1)\ldots 740,\quad (2)\ldots 481,\quad (3)\ldots 962,\quad (4)\ldots 925,\quad (5)\ldots 851,\quad (0)\ldots 370.$$

The table in this form extends to $p^\mu = 463$.

In the latter part of the table $p^\mu = 467$ to 997, we have only the mantissae of $\frac{100}{p^\mu}$.

[F. 13. The Pellian Equation.] Art. III.

27. The Pellian equation is $y^2 = ax^2 + 1$, a being a given integer number, which is not a square (or rather, if it be, the only solution is $y = 1$, $x = 0$), and x, y being numbers to be determined: what is required is the least values of x, y, since these, being known, all other values can be found. A small table $a = 2$ to 68 is given by **Euler**, *Op. Arith. Coll.* t. I. p. 8. The Memoir is "Solutio problematum Diophanteorum per numeros integros," pp. 4—10, 1732—33.

The form of the table is

a	$x\ (= p)$	$y\ (= q)$
2	2	3
3	1	2
5	4	9
⋮	⋮	⋮
68	4	33.

Even here, for some of the values of a, the values of x, y are extremely large; thus $a = 61$, $x = 226{,}153{,}980$, $y = 1{,}766{,}399{,}049$.

And probably tables of a like extent may be found elsewhere; in particular, a table of the solution of $y^2 = ax^2 \pm 1$ ($-$ when the value of a is such that there is a solution of $y^2 = ax^2 - 1$, and $+$ for other values of a), $a = 2$ to 135, is given by **Legendre**, *Théorie des*

Nombres, 2nd ed. 1808, in the Table X. (one page) at the end of the work.

For the before-mentioned number 61, the equation is $y^2 = 61x^2 - 1$, and the values are $x = 3805$, $y = 29718$; much smaller than Euler's values for the equation $y^2 = 61x^2 + 1$.

[Pages 475–478]

28. The most extensive table, however, is given by **Degen**, *Canon Pellianus, sive Tabula simplicissimam equationis celebratissimae*: $y^2 = ax^2+1$, *solutionem, pro singulis numeri dati valoribus ab 1 usque ad 1000 in numeris rationalibus, iisdemque integris exhibens.* Auctore Carolo Ferdinando Degen. Hafn (Copenhagen) apud Gerhardum Bonnarum, 1817. 8vo. pp. iv to xxiv and 1 to 112.

The first table (pp. 3—106), is entitled as "Tabula I. Solutionem Equationis $y^2 - ax^2 - 1 = 0$ exhibens." It, in fact, also gives the expression of $\sqrt{a}$ as a continued fraction; thus a specimen is

$$\begin{array}{r|ccccc}
209 & 14 & 2 & 5 & 3 & (2) \\
 & 1 & 13 & 5 & 8 & 11 \\
 & 3220 & & & & \\
 & 46551 & & & &
\end{array}$$

Here the first line gives the continued fraction, viz.

$$\sqrt{209} = 14 + \cfrac{1}{2 + \cfrac{1}{5 + \cfrac{1}{3 + \cfrac{1}{2 + \cfrac{1}{3 + \cfrac{1}{5 + \cfrac{1}{2 + \cfrac{1}{28 + \cfrac{1}{2 + \&c.}}}}}}}}}$$

the period being $(2, 5, 3, 2, 3, 5, 2)$ indicated by $2, 5, 3$ (2). [The number of terms in the period is here odd, but it may be even; for instance, the period $(1, 1, 5, 5, 1, 1)$ is indicated by $1, 1$ $(5, 5)$.]

The second line contains auxiliary numbers presenting themselves in the process; thus, if $R^2 = 239$, we have $R = 14 + \frac{1}{\alpha}$,

$$\alpha = \frac{1}{R-14} = \frac{1(R+14)}{209-14^2} = \frac{R+14}{13} = 2 + \frac{1}{\beta},$$
$$\beta = \frac{13}{R-12} = \frac{13(R+12)}{209-12^2} = \frac{R+12}{5} = 5 + \frac{1}{\gamma},$$
$$\gamma = \frac{5}{R-13} = \frac{5(R+13)}{209-13^2} = \frac{R+13}{8} = 3 + \frac{1}{\delta},$$

&&c.,

where the second line 1, 13, 5,. . . shows the numerical factors of the third column. The value of this second line as a result is not very obvious.

The third line gives x, and the fourth line y.

29. The second table, pp. 109—112, is entitled "Tabula II. Solutionem equationis $y^2 - ax^2 + 1 = 0$, quotiescunque valor ipsius a talem admiserat, exhibens"; viz. it is remarked that this is only possible (but see *infra*) for those values of a which in Table I. correspond to a period of an even number of terms, as shown by two equal numbers in brackets; thus $a = 13$, the period of $\sqrt{13}$ given in Table I. is $(1,1,1,1)$ as shown by the top line $3, 1\ (1,1)$, and accordingly 13 is one of the numbers in Table II.; and we have there $13 \begin{cases} 5 \\ 18. \end{cases}$

Or take another specimen, $241 \begin{cases} 4574225 \\ 71011068; \end{cases}$ viz. the first line gives the value of x, and the second line the value of y (least values), for which $y^2 - ax^2 = -1$.

It is to be noticed that $a = 2$ and $a = 5$, for which we have obviously the solutions $(x = 1,\ y = 1)$ and $(x = 1,\ y = 2)$ respectively, are exceptional numbers not satisfying the test above referred to; and (apparently for this reason) the values in question, 2 and 5, are omitted from the table.

[Page 479]

30. Cayley, "Table des plus petites solutions impaires de l'équation $x^2 - Dy^2 = \pm 4$, $D \equiv 5$ (mod. 8)." *Crelle*, t. LIII. (1857), page 371 (one page), [231].

As regards the theory of quadratic forms, it is important to know whether for a given value of D ($\equiv 5$, mod. 8) there does or does not exist a solution, in odd numbers, of the equation $x^2 - Dy^2 = 4$. As remarked in the paper, "Note sur l'équation $x^2 - Dy^2 = \pm 4$, $D \equiv 5$ (mod. 8)," pp. 369—371, [231], this can be determined for values of D of the form in question up to $D = 997$ by means of Degen's Table; and the solutions, when they exist, of the equation $x^2 - Dy^2 = 4$, as also of the equation $x^2 - Dy^2 = -4$, can be obtained up to the same value of D.

Observe that when the equation $x^2 - Dy^2 = -4$ is possible, the equation $x^2 - Dy^2 = 4$ is also possible, and that its least solution is obtained very readily from that of the other equation; it is therefore sufficient to tabulate the solution of $x^2 - Dy^2 = \pm 4$, the sign being $-$ when the corresponding equation is possible, and being in other cases $+$. Hence the form of the Table: viz. as a specimen we have

D	$\pm$	x	y
757	imposs.		
765	$+$	83	3
773	$-$	139	5
781	imposs.		
$\vdots$			

that is, if $D = 757$ or 781, there is no solution of either $x^2 - Dy^2 = +4$ or $= -4$; if $D = 765$, there is a solution $x = 83$, $y = 3$ of $x^2 - Dy^2 = +4$, but none of $x^2 - Dy^2 = -4$; if $D = 773$, there is a solution $x = 139$, $y = 5$ of $x^2 - Dy^2 = -4$, and therefore also a solution of $x^2 - Dy^2 = +4$; and so in other cases.

[F. 14. Partitions.] Art. IV.

31. The problem of Partitions is closely connected with that of Derivations. Thus if it be asked in how many ways can the number n be expressed as a sum of three parts, the parts being 0, 1, 2, 3, and each part being repeatable an indefinite number of times, it is clear that n is at most $= 9$, and that for the values of $m = 0, 1, \ldots, 9$ shown by the top line of the annexed table, the number of partitions has the values shown by the bottom line thereof:—

0	1	2	3	4	5	6	7	8	9
a^3	a^2b	a^2c	a^2d	abd	acd	ad^2	bd^2	cd^2	d^3
		ab^2	abc	ac^2	b^2d	bcd	c^2d		
			b^3	b^2c	bc^2	c^3			
1	1	2	3	3	3	3	2	1	1

But taking a, b, c, d to stand for 0, 1, 2, 3 respectively, the actual partitions of the required form are exhibited by the literal terms of the table (these being obtained, each column from the preceding one, by the method of derivations, or say by the rule of the last and last but one), and the numbers of the bottom line are simply the number of terms in the several columns respectively.

32. A set of such literal tables, say of tables $\begin{pmatrix} a, b, c, \ldots, k \\ = 0, 1, 2, \ldots, m \end{pmatrix}^n$, for different values of n and m (where the number of letters is $= m+1$), would be extremely interesting and valuable. The tables for a given value of m and for different values of n are, it is clear, the proper foundation of the theory of the binary quantic $(a, b, c, \ldots, k \between x, 1)^m$, which corresponds to such value of m.

33. But the question at present is as to the *number* of terms in a column, that is, as to the number of the partitions of a given form: the analytical theory has been investigated by Euler and others. The expression for the number of partitions is usually obtained as equal to the coefficient of x^n in the development, in ascending powers of x, of a given rational function of x: for instance, if there is no limitation as to the number of the parts, but if the parts are 1, 2, 3, m (viz. a part may have any value not exceeding m), each part being repeatable an indefinite number of times, then

$$\text{Number of partitions of } n = \text{coefficient of } x^n \text{ in } \frac{1}{(1-x)(1-x^2)(1-x^3)\ldots(1-x^m)}$$

and we can, by actual development, obtain for any given values of m, n the number of partitions.

These have been tabulated $m = 1, 2, \ldots, 20$, and $m = \infty$ (viz. there is in this case no limit as to the largest part), and $n = 1$ to 59, by **Euler**, *Op. Arith. Coll.* t. I. pp. 97—101, given in the paper "De Partitione Numerorum," pp. 73—101, (1750).

34. The generating function for any given value of m is, it is clear, $= \frac{1}{1-x^m}$ multiplied by that for the next preceding value of m, and it thus appears how each line of the table is calculated from that which precedes it. The auxiliary numbers are *printed*; thus a specimen is

Valores numeri n.

m	0	1	2	3	4	5	6	7	8	9	10
					1	1	2	3	5	6	9
4	1	1	2	3	5	6	9	11	15	18	23
						1	1	2	3	5	7
5	1	1	2	3	5	7	10	13	18	23	30

viz. suppose the numbers in the second 4-line known: then simply moving these each five steps onward we have the (auxiliary) numbers of the first 5-line; and thence by a mere addition the required series of numbers shown by the second 5-line.

35. More extensive tables are contained in the memoir by **Marsano**, *Sulle leggi delle derivate generali delle funzioni di funzioni et sulla teoria delle forme di partizione dei numeri interi*, (4$^{\text{o}}$. Genova, 1870), pp. 1—281.

[Pages 482–483]

Table I. (16 pages) is, in fact, Euler's table, showing in how many ways the number n can be made up with the parts 1, 2, 3, $\ldots$, m; but the extent is greater, viz. n is from 1 to 103, and m from 1 to 102.

Table II. (6 pages). The successive lines give the coefficients in the expansions of

$$S,\ \frac{S}{1-x},\ \frac{S}{1-x.1-x^2},\ \ldots,\ \frac{S}{1-x.1-x^2\ldots 1-x^{35}},$$

where $S = \frac{1}{(1-x)(1-x^2)(1-x^3)}\ldots$ ad inf., each expanded as far as x^{53}, and further continued as regards the first ten lines.

Table III. (2 pages). The successive lines give the coefficients in the expansions of

$$S^2,\ \frac{S^2}{1-x},\ \frac{S^2}{1-x.1-x^2},\ \ldots,\ \frac{S^2}{1-x.1-x^2\ldots 1-x^{9}},$$

each expanded as far as x^{55}.

36. As regards Tables II. and III., the analytical explanations have been given in the first instance; but it is easy to see that the tables give numbers of partitions. Thus, in Table II., the second line gives the coefficients in the development of

$$\frac{1}{(1-x)^2(1-x^2)(1-x^3)\ldots};$$

viz. these are 1, 2, 4, 7, 12, 19, 30,$\ldots$, being the number of ways in which the numbers 0, 1, 2, 3, 4, &c. respectively can be made up with the parts 1, $1'$, 2, 3, 4, &c.;

Similarly, the third line shows the number of ways in which these numbers respectively can be made up with the parts 1, $1'$, 2, $2'$, 3, 4, 5, &c.

[F. 15. Quadratic forms $a^2 + b^2$, &c., and Partitions of Numbers into squares, cubes, and biquadrates.] Art. V.

37. The forms here referred to present themselves in the various complex theories. Thus $N = a^2 + b^2, = (a + bi)(a - bi)$; this means that, in the theory of the complex numbers $a + bi$ (a and b integers), N is not a prime but a composite number.

It is well known that an ordinary prime number $= 3$, mod. 4, is not expressible as a sum $a^2 + b^2$, being, in fact, a prime in the complex theory as well as in the ordinary one: but that an ordinary prime number $= 1$, mod. 4, is (in one way only) $= a^2 + b^2$; so that it is in the complex theory a composite number.

38. Tables presenting themselves in the several complex theories are:—

(a) Tables of the representation of a number as a sum of two squares.

(b) Tables of the representation of a number as a sum of four squares.

(c) Tables of the complex divisors of a number in the theory of $a + bi$.

(d) Tables relating to the theory of the forms $a^2 + b^2$, $a^2 + 2b^2$, $a^2 + 3b^2$, &c.

39. The question of the partition of a number into squares has been considered by **Legendre**, *Théorie des Nombres*, t. I. pp. 304—315, and by **Jacobi**, *Crelle*, t. II. pp. 55—56.

The question of the partition of a number into cubes or biquadrates is considered by **Euler**, in various memoirs.

[F. 16. Binary, ternary, &c., quadratic and higher forms.] Art. VI.

40. The theory of binary quadratic forms, as is well known, was established by **Gauss**, "Disquisitiones Arithmeticae," and the work contains a large number of tables.

The theory relates to the form $ax^2+2bxy+cy^2$, or as it is written (a, b, c), where the discriminant is $b^2-ac = D$, a given number positive or negative.

The tables in the "Disquisitiones" relate chiefly to the following subjects:

- The number of classes for a given discriminant.
- The characters of the several classes.
- The composition of classes.
- The ambiguous classes.
- The periods of reduced forms.

41. For negative discriminants, Gauss gives (Table I. pp. 552—566) the number of classes and the number of genera for each discriminant $-D$, $D = 1$ to 305, and for certain larger values to $D = 997$.

42. For positive discriminants, he gives (Table II. pp. 566—573) the number of classes and the number of genera for each discriminant D, $D = 1$ to 199, and for certain larger values to $D = 997$.

43. For the ternary quadratic forms, Gauss gives (Art. 288) the classification of the positive forms according to their determinants from $D = 1$ to $D = 100$.

[F. 17. Complex theories.] Art. VII.

44. Under this head are included tables relating to complex numbers of various forms.

(a) Complex numbers $a + bi$, $i = \sqrt{-1}$. The theory is given by **Gauss**, "Theoria residuorum biquadraticorum," Commentationes recentiores, t. VII. (Gottingae, 1832).

Tables relating to this theory are given by **Jacobi** and **Reuschle** (see No. 15 above).

(b) Complex numbers $a + b\rho$, where ρ is a primitive cube root of unity. The theory is given by **Jacobi**, "De compositione numerorum e cubis integris," *Crelle*, t. II. (1827).

(c) Complex numbers $a+b\sqrt{-2}$, $a+b\sqrt{-3}$, $a+b\sqrt{-5}$, &c. The theories of these forms are given by **Dirichlet**, in various memoirs in *Crelle*.

(d) Complex numbers $a+b\sqrt{2}$, $a+b\sqrt{3}$, &c. Theories of these and similar forms are given by **Zolotareff**, **Markoff**, and others.

45. The tables relating to these several complex theories are of great variety. They include:

- Tables of prime numbers in the complex theory.
- Tables of the complex factors of rational numbers.
- Tables of the units of the complex theory.
- Tables of quadratic and higher residues in the complex theory.
- Tables relating to the theory of ideals.

46. A very important collection of tables relating to the theory of complex numbers is given by **Reuschle**, *Mathematische Abhandlung* (already cited, No. 15).

The tables relating to complex theories are:

V. (a, b, c) Tafeln der komplexen Primzahlen.

These give tables of prime numbers in the several complex theories:

(a) $a + bi$, $a^2 + b^2 < 10000$ (pp. 23—34).

(b) $a + b\rho$, $a^2 - ab + b^2 < 10000$ (pp. 34—44).

(c) $a + b\sqrt{-2}$, $a^2 + 2b^2 < 10000$ (pp. 44—52).

A specimen of table (a) is

Norm	Primes
5	$1+2i$, $2+i$
13	$2+3i$, $3+2i$
17	$1+4i$, $4+i$
$\vdots$	$\vdots$

47. It is proper to notice that the several theories here referred to are closely connected with the theory of binary quadratic forms, and with the theory of the decomposition of numbers into sums of squares.

[Pages 497–499]

The following is a list of the more important tables relating to the theory of complex numbers, not included in the preceding enumeration:

1. **Dirichlet**, Tables relating to complex numbers of the form $a + b\sqrt{-5}$, *Abh. d. K. Akad. d. Wiss. zu Berlin*, 1841.

2. **Eisenstein**, Tables of cubic residues and reciprocity, *Crelle*, t. XXVII. (1844).

3. **Kummer**, Tables relating to the theory of ideal numbers, *Berliner Abh.* 1857, 1859, 1860.

4. **Kronecker**, Tables relating to the theory of complex multiplication, *Berliner Abh.* 1861, 1862, 1863.

5. **Weber**, Tables relating to the theory of elliptic functions and complex multiplication, *Elliptische Functionen und algebraische Zahlen*, 1891.

48. In connexion with the foregoing theories, there are tables relating to the *congruence of numbers*, and to the *theory of Galois imaginaries.*

A congruence-table for a prime modulus p gives, for every number a, the least positive residue of a to the modulus p. Such tables are given by **Gauss**, *Werke*, t. II. pp. 478—495, for the moduli $p = 2, 3, 5, 7, 11, 13$.

49. Tables relating to the theory of Galois imaginaries, that is, to the theory of functions of a variable which satisfies an irreducible congruence of a given degree to a prime modulus, have been given by **Galois**, "Sur la théorie des nombres," *Bulletin de Ferussac*, t. XIII. (1830), and by **Serret**, *Cours d'Algèbre supérieure.*

50. We have hitherto been concerned with tables which are purely arithmetical, that is, tables in which the entries are numbers. But there are also tables of a mixed character, in which the entries are analytical expressions, or formulae, involving letters which denote numbers. Such tables occur in the theory of elliptic functions, in the theory of hyperelliptic functions, and in the theory of Abelian functions generally.

The most important tables of this kind are those relating to the *modular equations* and the *singular moduli.* These have been given by **Jacobi**, *Fundamenta nova theoriae functionum ellipticarum*, 1829; **Soethero**, *Modulargleichungen*, 1878; **Kiepert**, in various memoirs in *Math. Ann.*; **Weber**, *Lehrbuch der Algebra*, t. III., 1908.

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51. The Committee have now completed their survey of the tables which are included under the headings referred to at the commencement of the present Report.

They would wish to express their sense of the great value of the work which has been done by Mr Glaisher in the preparation of the Reports for 1872 and 1873, and to record their opinion that the publication of these Reports has been of great service to mathematics.

They would also express their hope that the present Report may be found useful, and that it may tend to promote the preparation and publication of mathematical tables, the importance of which, for all purposes of research and calculation, can hardly be over-estimated.

A. CAYLEY, G. G. STOKES, W. THOMSON, H. J. S. SMITH, J. W. L. GLAISHER.

October, 1875.

0.8 Paper 612: Note sur une formule d'intégration indéfinie

[From the *Comptes Rendus de l'Académie des Sciences*, vol. LXXX. (1875), pp. 735—736.]

[Page 500]

La formule dont il s'agit est celle qui donne l'intégrale indéfinie

$$\int \frac{dx}{(1-x^2)^{3/2}(1-k^2x^2)^{1/2}}.$$

On sait que les intégrales indéfinies des fonctions elliptiques de la première espèce et de la seconde espèce s'expriment par des fonctions définies (les fonctions F, E de Legendre); mais pour les intégrales de la troisième espèce, on a en général besoin d'une nouvelle transcendant. Il y a toutefois des cas particuliers où l'intégrale indéfinie de la troisième espèce peut s'exprimer au moyen des seules fonctions F et E; et c'est un cas de cette espèce que je considère ici.

Posons, avec Legendre,

$$F(k,\phi) = \int_0^\phi \frac{d\phi}{\Delta(k,\phi)},$$
$$E(k,\phi) = \int_0^\phi \Delta(k,\phi)\,d\phi,$$

où $\Delta(k,\phi) = \sqrt{1-k^2\sin^2\phi}$.

On a la formule

$$\int \frac{d\phi}{\Delta^3} = \frac{1}{1-k^2}E - \frac{k^2}{1-k^2}\frac{\sin\phi\cos\phi}{\Delta}.$$

Pour le vérifier, on différencie le second membre; on a

$$dE = \Delta\,d\phi,$$
$$d\left(\frac{\sin\phi\cos\phi}{\Delta}\right) = \frac{\cos^2\phi-\sin^2\phi}{\Delta}\,d\phi + \frac{k^2\sin^2\phi\cos^2\phi}{\Delta^3}\,d\phi;$$

d'où, en observant que $\cos^2\phi-\sin^2\phi = 1-2\sin^2\phi = 2\cos^2\phi-1$, et que $\Delta^2 = 1-k^2\sin^2\phi$, on trouve

$$d\left(\frac{1}{1-k^2}E - \frac{k^2}{1-k^2}\frac{\sin\phi\cos\phi}{\Delta}\right) = \frac{d\phi}{\Delta^3}.$$

La formule peut s'écrire sous la forme plus symétrique

$$\int \frac{d\phi}{\Delta^3} = \frac{E - k^2 \sin\phi \cos\phi/\Delta}{1 - k^2}.$$

On a ainsi l'intégrale indéfinie cherchée, exprimée au moyen des fonctions F et E seulement.

Cambridge, 26 avril 1875.

0.9 612. Note sur une formule d'intégration indéfinie[2]

Pour démontrer la formule, j'écris

$$u = x^{-m}(A + Bx + Cx^2 + \ldots + Kx^{\theta-1}),$$

et aussi pour abréger

$$X = (x+p)^{m+n-\theta+1}(x+p+q)^{-n},$$

ce qui donne

$$\frac{X}{(x+p)(x+p+q)} = (x+p)^{m+n-\theta}(x+p+q)^{-n-1}.$$

L'équation à vérifier est donc

$$Xu = \int \frac{X(x+q)^\theta\,dx}{x^{m+1}(x+p)(x+p+q)},$$

ou, en différentiant et divisant par X,

$$\frac{X'}{X}u + u' = \frac{(x+q)^\theta}{x^{m+1}(x+p)(x+p+q)},$$

ou enfin

$$[(m+n-\theta+1)(x+p+q) - n(x+p)]u + (x+p)(x+p+q)u' = \frac{(x+q)^\theta}{x^{m+1}},$$

où u' dénote $\dfrac{du}{dx}$. Il ne s'agit donc que d'exprimer que cette équation ait une intégrale

$$u = x^{-m}(A + Bx + Cx^2 + \ldots + Kx^{\theta-1}).$$

En supposant que cela soit ainsi, et en effectuant la substitution, les termes en $x^{-m+\theta}$ se détruisent, et l'on obtient une équation qui contient des termes en x^{-m-1}, x^{-m}, ..., $x^{-m+\theta-1}$, savoir $(\theta+1)$ termes. On a ainsi, entre les θ coefficients A, B, C, ..., K un système de $(\theta+1)$ équations linéaires, ce qui implique une condition entre les

[2][From the *Comptes Rendus de l'Académie des Sciences*, vol. LXXX (1875), pp. 857–859.]

constantes m, n, p, q; mais, cette condition satisfaite, les équations se réduisent à θ équations indépendantes, et les coefficients seront ainsi déterminés.

Par exemple, soit $\theta = 2$; l'équation différentielle est

$$[m-1\,p+m+n-1\,q+m-1\,x]u+[p^2+pq+x(2p+q)+x^2]u' = x^{-m-1}(q+x)^2,$$

laquelle doit être satisfaite par $u = Ax^{-m} + Bx^{-m+1}$. Cela donne

x^{-m-1}	x^{-m}	x^{-m+1}	
	$(m-1\,p+m+n-1\,q)A,$	$(m-1\,p+m+n-1\,q)B,$	$(m-1$
$-m(p^2+pq)A,$	$-(m-1)(p^2+pq)B,$		
$n-m(2p+q)A,$	$-(m-1)(2p+q)B,$		
$n-mA,$	$-(m-1)B$		
$n-q^2,$	$-2q,$	-1	

c'est-à-dire

$$\begin{aligned}
&1+[(m-1)\,p-nq]B+[(m+1)\,p-(n-1)\,q]A=0,\\
&2q+(m-1)(p^2+pq)B+[(m-1)\,p-nq]A=0,\\
&q^2+m(p^2+pq)A=0,
\end{aligned}$$

ce qui donne une condition, déterminant $= 0$, et l'on a une condition de cette même forme pour une valeur quelconque de θ.

En formant, puis en réduisant les expressions de ces déterminants, on obtient pour toutes les valeurs $\theta = 1, 2, 3, 4, \ldots$ respectively les suites d'équations que voici:

$$0=\begin{vmatrix} 1, & mp-nq \\ q, & m(p^2+pq) \end{vmatrix} = \begin{vmatrix} [m]^1\,p\,(p+q) \\ q\,([m]\,p-[n]\,q)^1 \end{vmatrix} = ([m]\,p^2+[n]\,q^2)^1,$$

$$0=\begin{vmatrix} 1, & \overline{m-1}\,p-nq, & 1 \\ 2q, & m(p^2+pq), & \overline{m-1}(p^2+pq) \\ q^2, & \cdot, & m(p^2+pq) \end{vmatrix} = \begin{vmatrix} [m]^2\,p^2(p+q)^2 \\ q^2\,([m]\,p-[n]\,q)^2 \end{vmatrix}$$

$$= -[m]^2\,p^2(p+q)\,q\,([m-1]\,p-[n]\,q)^1 = ([m]\,p^2+[n]\,q^2)^2,$$

$$0=\begin{vmatrix} 1, & \overline{m-2}\,p-nq, & 1, & 2 \\ 3q, & m(p^2+pq), & \overline{m-1}(p^2+pq), & \overline{m-2}(p^2+pq) \\ 3q^2, & \cdot, & m(p^2+pq), & (m-1)(2p+q) \\ q^3, & \cdot, & \cdot, & m(p^2+pq) \end{vmatrix}$$

$$= -[m]^3\, p^3(p+q)\, q^2\, ([m-2]\, p-[n]\, q)^2+3\, [m]^3\, p^2(p+q)^2\, q^2\, ([m-2]\, p-[n]\, q)^1 = ([m]\, p^2+$$

$$0 = \begin{vmatrix} 1, & \overline{m-3}\, p - nq, & 1, & 2, & 3 \\ 4q, & m(p^2+pq), & \overline{m-1}(p^2+pq), & \overline{m-2}(p^2+pq), & \overline{m-3}(p^2+pq) \\ 6q^2, & \cdot, & m(p^2+pq), & (m-1)(2p+q), & (m-2)(2p+q) \\ 4q^3, & \cdot, & \cdot, & m(p^2+pq), & (m-1)(2p+q) \\ q^4, & \cdot, & \cdot, & \cdot, & m(p^2+pq) \end{vmatrix}$$

$$= -[m]^4\, p^4(p+q)\, q^3\, ([m-3]\, p-[n]\, q)^3+6\, [m]^4\, p^3(p+q)^2\, q^3\, ([m-3]\, p-[n]\, q)^2-4\, [m]^4\, p^2($$

et ainsi de suite. Les notations $([m]\, p-[n]\, q)^1$, $([m]\, p-[n]\, q)^2, \ldots$ ont des significations semblables à celles de $([m]\, p^2+[n]\, q^2)^1$, $([m]\, p^2+[n]\, q^2)^2, \ldots$, auparavant expliquées. On a, par exemple,

$$([m]\, p-[n]\, q)^2 = [m]^2\, p^2 - 2\, [m]^1\, [n]^1\, pq + [n]^2\, q^2.$$

Considérons, par exemple, le deuxième déterminant: ceci contient trois termes en 1, $2q$, q^2 respectivement; le premier terme est

$$1 \cdot (m-1)(p^2+pq) \cdot m(p^2+pq),$$

c'est-à-dire

$$[m]^2\, p^2(p+q)^2;$$

le deuxième terme est

$$2q \cdot -m(p^2+pq)[(m-1)\, p - nq],$$

c'est-à-dire

$$-2\, [m]^1\, p\, (p+q)\, q\, ([m-1]\, p-[n]\, q)^1;$$

le troisième terme est

$$q^2[(\overline{m-1}\, p - nq)(\overline{m+1}\, p - \overline{n-1}\, q) - (m-1)(p^2+pq)],$$

c'est-à-dire

$$q^2[(m^2-m)\, p^2 - 2mnpq + (n^2-n)\, q^2] = q^2([m]\, p-[n]\, q)^2.$$

Et de même le troisième déterminant est composé de quatre termes en 1, $3q$, $3q^2$, q^3 respectivement, lesquels sont les quatre termes de la première expression transformée; et ainsi pour le quatrième déterminant, etc. Au moyen de ces premières transformées, on obtient sans peine les expressions finales $([m]\,p^2+[n]\,q^2)^1$, $([m]\,p^2+[n]\,q^2)^2$, $\ldots$

En écrivant $z-\frac{1}{2}(p+q)$ au lieu de x, et puis $\frac{1}{2}(p+q)=\alpha$, $\frac{1}{2}(p-q)=a$, la formule devient

$$\int \frac{(z-a)^\theta (z+a)^{m+n-\theta}\,dz}{(z-\alpha)^{m+1}(z+\alpha)^{n+1}},$$

et la valeur algébrique

$$=(z+\alpha)^{m+n-\theta-1}(z-\alpha)^{-m}(z+\alpha)^{-n}(A'+B'z+\ldots+K'z^{\theta-1}),$$

pourvu qu'on ait entre les quantités m, n, a, α la relation

$$\{[m](\alpha+a)^2+[n](\alpha-a)^2\}^\theta=0.$$

En écrivant $\theta=m$, on a la formule de MM. Serret et Liouville, laquelle, en y écrivant $\dfrac{(\alpha+a)^2}{4a\alpha}=\zeta$ et $\dfrac{(\alpha-a)^2}{4a\alpha}=\zeta-1$, peut s'écrire sous la forme $\{[m]\,\zeta+[n]\,(\zeta-1)\}^\theta=0$. Je remarque que l'équation en ζ ne donne pas *toujours* pour ζ des valeurs réelles, positives et plus grandes que l'unité: par exemple, pour $\theta=1$, on a $\zeta=\dfrac{n}{m+n}$, valeur qui ne peut pas satisfaire à ces conditions. Je n'ai pas cherché dans quel cas ces conditions (qui ont rapport à l'application des formules à la représentation des fonctions elliptiques) subsistent.

0.10 613. On the group of points G^1_4 on a sextic curve with five double points[3]

The present note relates to a special group of points considered incidentally by MM. Brill and Nöther in their paper "Ueber die algebraischen Functionen und ihre Anwendung in der Geometrie," *Math. Annalen*, t. VII. pp. 268–310 (1874).

I recall some of the fundamental notions. We have a basis-curve which to fix the ideas may be taken to be of the order $n, = p+1$, with $\frac{1}{2}p\,(p-3)$ dps, and therefore of the "Geschlecht" or deficiency p; any curve of the order $n-3, = p-2$ passing through the $\frac{1}{2}p\,(p-3)$ dps is said to be an adjoint curve. We may have, on the basis-curve, a special group G^q_Q of Q points ($Q \not< 2p-2$); viz. this is the case when the Q points are such that every adjoint curve through $Q-q$ of them—that is, every curve of the order $p-2$ through $\frac{1}{2}p\,(p-3)$ dps and the $Q-q$ points—passes through the remaining q points of the group: the number q may be termed the "speciality" of the group: if $q=0$, the group is an ordinary one.

It may be observed that a special group G^q_Q is chiefly noteworthy in the case where $Q-q$ is so small that the adjoint curve is not completely determined: thus if $p=5$, viz. if the basis-curve be a sextic with 5 dps, then we may have a special group G^2_6, but there is nothing remarkable in this; the 6 points are intersections with the sextic of an arbitrary cubic through the 5 dps—the cubic of course intersects the sextic in the 5 dps counting as 10 points, and in 8 other points—and such cubic is completely determined by means of the 5 dps and any 4 of the 6 points. But contrariwise, there is something remarkable in the group G^1_4 about to be considered: viz. we have here on the sextic 4 points, such that every cubic through the 5 dps and through 3 of the 4 points (through 8 points in all) passes through the remaining one of the 4 points.

The whole number of intersections of the basis-curve with an adjoint, exclusive of the dps counting as $p\,(p-3)$ points, is of course $= 2p-2$: hence an adjoint through the Q points of a group G^q_Q meets the basis-curve besides in $R, = 2p-2-Q$, points; we have then the "Riemann-Roch" theorem that these R points form a special group G^r_R, where

$$Q+R=2p-2,$$

[3][From the *Mathematische Annalen*, vol. VIII. (1875), pp. 359–362.]

as just mentioned, and

$$Q - R = 2q - 2r;$$

viz. dividing in any manner the $2p-2$ intersections of the basis-curve by an adjoint into groups of Q and R points respectively, these will be special groups, or at least one of them will be a special group, G_Q^q, G_R^r, such that their specialities q, r are connected by the foregoing relation $Q - R = 2q - 2r$.

The Authors give (*l.c.*, p. 293) a Table showing for a given basis-curve, or given value of p, and for a given value of r, the least value of R and the corresponding values of q, Q: this table is conveniently expressed in the following form.

The least value of

$$R = p - \frac{p}{r+1} + r;$$

and then

$$q = \frac{p}{r+1} - 1, \qquad Q = p + \frac{p}{r+1} - r - 2,$$

where $\dfrac{p}{r+1}$ denotes the integer equal to or next less than the fraction.

It is, I think, worth while to present the table in the more developed form:

n	p	Dps	$r=$					
			1	2	3	4	5	6
4	3	0	G_3^1 G_1^0	G_4^2 G_0^0	. .	. .	. .	. .
5	4	2	G_3^1 G_3^1	G_5^2 G_1^0	G_6^3 G_0^0	. .	. .	. .
6	5	5	G_4^1 G_4^1	G_6^2 G_2^0	G_7^3 G_1^0	G_8^4 G_0^0	. .	. .
7	6	9	G_4^1 G_6^2	G_6^2 G_4^1	G_8^3 G_2^0	G_9^4 G_1^0	G_{10}^5 G_0^0	. .
8	7	14	G_5^1 G_7^2	G_7^2 G_5^1	G_9^3 G_3^0	G_{10}^4 G_2^0	G_{11}^5 G_1^0	G_{12}^6 G_0^0
⋮								

where the table shows the values of $\dfrac{G_R^r}{G_Q^q}$ for any given values of p, r.

I recur to the case $p = 5$ and the group G_4^1, which is the subject of the present note: viz. we have here a sextic curve with 5 dps, and on it a group of 4 points G_4^1, such that every cubic through the 5 dps and through 3 points of the group, 8 points in all, passes through the remaining 1 point.

MM. Brill and Nöther show (by consideration of a rational transformation of the whole figure) that, given 2 points of the group, it is possible, and possible in 5 different ways, to determine the remaining 2 points of the group.

I remark that the 5 dps and the 4 points of the group form "an ennead" or system of the nine intersections of two cubic curves: and that the question is, given the 5 dps and 2 points on the sextic, to show how to determine on the sextic a pair of points forming with the 7 points an ennead: and to show that the number of solutions is $= 5$.

We have the following "Geiser-Cotterill" theorem:

If seven of the points of an ennead are fixed, and the eighth point describes a curve of the order n passing $\alpha_1, \alpha_2, \ldots, \alpha_7$ times through the seven points respectively, then will the ninth point describe a curve of the order ν passing $\alpha_1, \alpha_2, \ldots, \alpha_7$ times through the seven points respectively: where

$$\begin{aligned} \nu &= 8n - 3\Sigma\alpha, \\ \alpha_1 &= 3n - \alpha_1 - \Sigma\alpha, \\ &\vdots\;\vdots \\ \alpha_7 &= 3n - \alpha_7 - \Sigma\alpha, \end{aligned}$$

and conversely

$$\begin{aligned} n &= 8\nu - 3\Sigma\alpha, \\ \alpha_1 &= 3\nu - \alpha_1 - \Sigma\alpha, \\ &\vdots\;\vdots \\ \alpha_7 &= 3\nu - \alpha_7 - \Sigma\alpha, \end{aligned}$$

(Geiser, *Crelle-Borchardt*, t. LXVII. (1867), pp. 78–90; the complete form, as just stated, and which was obtained by Mr Cotterill, has not

I believe been published): and also Geiser's theorem "the locus of the coincident eighth and ninth points is a sextic passing twice through each of the seven points."

The sextic and the curve n intersect in $6n$ points, among which are included the seven points counting as $2\Sigma\alpha$ points: the number of the remaining points is $= 6n - 2\Sigma\alpha$. Similarly, the sextic and the curve ν intersect in 6ν points, among which are included the seven points counting as $2\Sigma\alpha$ points: the number of the remaining points is $6\nu - 2\Sigma\alpha$ $(= 6n - 2\Sigma\alpha)$. The points in question are, it is clear, common intersections of the sextic, and the curves n, ν: viz. of the intersections of the curves n, ν, a number $6n - 2\Sigma\alpha$, $= 6\nu - 2\Sigma\alpha$, $= 3n + 3\nu - \Sigma\alpha - \Sigma\alpha$ lie on the sextic.

The curves n, ν intersect in $n\nu$ points, among which are included the seven points counting $\Sigma\alpha\alpha$ times: the number of the remaining intersections is therefore $n\nu - \Sigma\alpha\alpha$, but among these are included the $3n + 3\nu - \Sigma\alpha - \Sigma\alpha$ points on the sextic; omitting these, there remain $n\nu - 3(n+\nu) - \Sigma\alpha\alpha + \Sigma\alpha + \Sigma\alpha$ points, or, what is the same thing, $(n-3)(\nu-3) - \Sigma(\alpha-1)(\alpha-1) - 2$ points: it is clear that these must form pairs such that, the eighth point being either point of a pair, the ninth point will be the remaining point of the pair: the number of pairs is of course

$$\tfrac{1}{2}[(n-3)(\nu-3) - \Sigma(\alpha-1)(\alpha-1) - 2],$$

and we have thus the solution of the question, given the seven points to determine the number of pairs of points on the curve n (or on the curve ν) such that each pair may form with the seven points an ennead.

In particular, if $n = 6$; $\alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5, \alpha_6, \alpha_7 = 2, 2, 2, 2, 2, 1, 1$ respectively, viz. if the curve be a sextic having 5 of the points for dps, and the remaining two for simple points, then we find $\nu = 12$; $\alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5, \alpha_6, \alpha_7 = 4, 4, 4, 4, 4, 5, 5$ respectively, and the number of pairs is

$$= \tfrac{1}{2}[3\cdot 9 - 5(2-1)(4-1) - 2], = \tfrac{1}{2}(27 - 15 - 2), = 5,$$

viz. starting with the 5 dps and any 2 points of the group G_4^1 we can, in 5 different ways, determine the remaining 2 points of the group.

In reference to the number $3p - 3$ of parameters in the curves belonging to a given value of p, it may be remarked as follows. Such a curve is rationally transformable into a curve of the order $p+1$ with $\frac{1}{2}p\,(p-3)$ dps, and therefore containing $\frac{1}{2}(p+1)(p+4)$, $-\frac{1}{2}p\,(p-3)$, $= 4p+2$ parameters. Employing an arbitrary homographic transformation to establish any assumed relations between the parameters, the number is diminished to $4p+2-8$, $= 4p-6$; and again employing a rational transformation by means of adjoint curves of the order $p-2$ drawn through the dps and $p-3$ points of the curve—thereby transforming the curve into one of the same order $p+1$ and deficiency p—then, assuming that the $p-3$ parameters (or constants on which depend the positions of the $p-3$ points) can be disposed of so as to establish $p-3$ relations between the parameters and so further diminish the number by $p-3$, the required number of parameters will finally be $4p-6-(p-3) = 3p-3$.

Cambridge, 26th October, 1874.

0.11 614. On a problem of projection[4]

I measure off on three rectangular axes the distances $\Omega X = \Omega Y = \Omega Z, = \theta$; and then, in a plane through Ω drawing in arbitrary directions the three lines ΩA, ΩB, ΩC, $= a$, b, c respectively, I assume that A, B, C (fig. 1) are the parallel projections of X, Y, Z respectively; viz. taking ΩO as the direction of the projecting lines, then ΩA, ΩB, ΩC being given in position and magnitude, we have to find θ, and the position of the line ΩO.

[Figure 1: Spherical diagram showing points A, B, C, X, Y, Z, O on a sphere with center Ω, with great circle arcs connecting them]

This is in fact a case of a more general problem solved by Prof. Pohlke in 1853, (see the paper by Schwarz, "Elementarer Beweis des Pohlke'schen Fundamentalsatzes der Axonometrie," *Crelle*, t. LXIII. (1864), pp. 309–314), viz. the three lines ΩX, ΩY, ΩZ may be any three axes given in magnitude and direction, and their parallel projection is to be similar to the three lines ΩA, ΩB, ΩC. Schwarz obtains a very elegant construction, which I will first reproduce. We may imagine through Ω a plane cutting at right angles the projecting lines, say in the points X', Y', Z'; we have then in plano a triad of lines $\Omega X'$, $\Omega Y'$, $\Omega Z'$ which are an orthogonal projection of ΩX, ΩY, ΩZ; and are also an orthogonal projection of a plane triad similar to ΩA, ΩB, ΩC; quà such last-mentioned projection, the triangles $\Omega Y'Z'$, $\Omega Z'X'$, $\Omega X'Y'$, must be proportional to the triangles ΩBC, ΩCA, ΩAB; that is, we have to find an orthogonal projection of ΩX, ΩY, ΩZ, such that the triangles $\Omega Y'Z'$, $\Omega Z'X'$, $\Omega X'Y'$, which are the projections of ΩYZ, ΩZX, ΩXY respectively, shall be in given ratios. There is no difficulty in the solution of this problem; referring everything to a sphere centre Ω, let the normals to the planes ΩYZ, ΩZX, ΩXY, meet the sphere in the points X'', Y'', Z'' respectively, and the projecting line through Ω meet the sphere in the point O, then the projection of ΩYZ is to ΩYZ as $\cos OX'' : 1$; and the like as to the projections of ΩZX and ΩXY; that is, in the given spherical triangle $X''Y''Z''$, we have to find a point O, such that the cosines of the distances OX'', OY'', OZ'' are in given ratios: we have at once,

[4] [From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XIII. (1875), pp. 19–29.]

through X'', Y'', Z'' respectively, three arcs meeting in the required point O.

The projecting lines being thus obtained, say these are the three parallel lines X', Y', Z', we have next to draw through Ω a plane meeting these in the points A', B', C' such that the triangle $A'B'C'$ is similar to the given triangle ABC; for this being so, the triangles $\Omega B'C'$, $\Omega C'A'$, $\Omega A'B'$ being the projections of, and therefore proportional to $\Omega Y'Z'$, $\Omega Z'X'$, $\Omega X'Y'$, that is, proportional to ΩBC, ΩCA, ΩAB, will, it is clear, be similar to these triangles respectively; that is, we have the triad $\Omega A'$, $\Omega B'$, $\Omega C'$, a projection of ΩX, ΩY, ΩZ, and similar to the triad ΩA, ΩB, ΩC, which is what was required.

It remains only to show how the given three parallel lines X', Y', Z', not in the same plane, can be cut by a plane in a triangle similar to a given triangle ABC.

[Figure 2: Planar diagram showing points A'', B'', C'', D, E, K, X, Y, Z with lines forming triangles]

Imagine the three lines at right angles to the plane of the paper, meeting the plane of the paper in the given points X, Y, Z (fig. 2) respectively. On the base YZ describe a triangle $A''YZ$ similar to the given triangle ABC; and through A'', X with centre on the line YZ, describe a circle meeting this line in the points D and K. Then in the plane, through YZ at right angles to the plane of the paper, we may draw a line meeting the lines Y, Z in the points B'', C'' respectively, such that joining XB'', XC'' we obtain a triangle $XB''C''$ similar to $A''YZ$, that is, to the given triangle ABC.

Taking K the centre of the circle, suppose that its radius is $= l$, and that we have $KY = \beta$, $KZ = \gamma$; also $YX = \sigma$, $ZX = \tau$; $YA'' = \sigma''$, $ZA'' = \tau''$. If for a moment x, y denote the coordinates of X, then

$$\sigma^2 = (x-\beta)^2 + y^2, = l^2 + \beta^2 - 2\beta x,$$
$$\tau^2 = (x-\gamma)^2 + y^2, = l^2 + \gamma^2 - 2\gamma x,$$

and thence

$$\gamma\sigma^2 - \beta\tau^2 = \gamma(l^2+\beta^2) - \beta(l^2+\gamma^2),$$

that is,

$$\gamma\sigma^2 - \beta\tau^2 = (\gamma-\beta)(l^2-\beta\gamma);$$

viz. this is the equation of the circle in terms of the vectors σ, τ; we have therefore in like manner

$$\gamma\sigma''^2 - \beta\tau''^2 = (\gamma-\beta)(l^2-\beta\gamma).$$

We may determine θ so as to satisfy the two equations

$$\sigma''^2 = \sigma^2\cos^2\theta + (l+\beta)^2\sin^2\theta,$$
$$\tau''^2 = \tau^2\cos^2\theta + (l+\gamma)^2\sin^2\theta;$$

in fact, these equations give

$$\gamma\sigma''^2 - \beta\tau''^2 = (\gamma\sigma^2-\beta\tau^2)\cos^2\theta + \{\gamma(l^2+\beta^2) - \beta(l^2+\gamma^2)\}\sin^2\theta,$$

which, the left-hand side and the coefficients of $\cos^2\theta$, and $\sin^2\theta$ on the right-hand side being each $= (\gamma-\beta)(l^2-\beta\gamma)$, is, in fact, an identity.

But in the figure, if θ, determined as above, denote the angle at D, then

$$(XB'')^2 = XY^2 + YB''^2 = \sigma^2 + (l+\beta)^2\tan^2\theta,$$
$$(ZC'')^2 = XZ^2 + ZC''^2 = \tau^2 + (l+\gamma)^2\tan^2\theta,$$

that is,

$$XB'' = \sigma''\sec\theta, \qquad ZC'' = \tau''\sec\theta,$$

or, since $B''C'' = YZ\sec\theta$ [$= (\gamma-\beta)\sec\theta$], the triangle $XB''C''$ is, as mentioned, similar to the triangle $A''YZ$.

I was not acquainted with the foregoing construction when my paper was written; but the analytical investigation of the particular case is nevertheless interesting, and I proceed to consider it.

Taking (fig. 1) Ω as the centre of a sphere and projecting on this sphere, we have A, B, C given points on a great circle; and we have to find the point O, such that there may be a trirectangular triangle XYZ, the vertices of which lie in OA, OB, OC respectively, and for which

$$\frac{\sin OX}{\sin OA} = \frac{a}{\theta}, \quad \frac{\sin OY}{\sin OB} = \frac{b}{\theta}, \quad \frac{\sin OZ}{\sin OC} = \frac{c}{\theta}.$$

I take the arcs BC, CA, $AB = \alpha$, β, γ respectively, $\alpha+\beta+\gamma = 2\pi$; and the required arcs OA, OB, OC are taken to be ξ, η, ζ respectively; these are connected by the relation

$$\sin\alpha\cos\xi + \sin\beta\cos\eta + \sin\gamma\cos\zeta = 0,$$

to obtain which, observe that from the triangles OAB, OAC, we have

$$\cos A = \frac{\cos\eta - \cos\xi\cos\gamma}{\sin\xi\sin\gamma} = -\frac{\cos\zeta - \cos\xi\cos\beta}{\sin\xi\sin\beta},$$

that is,

$$\sin\beta(\cos\eta - \cos\xi\cos\gamma) + \sin\gamma(\cos\zeta - \cos\xi\cos\beta) = 0,$$

which, with $\sin\alpha = -\sin(\beta+\gamma)$, gives the required relation. We have

$$\sin OX = \frac{a}{\theta}\sin\xi, \quad \sin OY = \frac{b}{\theta}\sin\eta, \quad \sin OZ = \frac{c}{\theta}\sin\zeta;$$

and then from the triangles OBC, OCA, OAB, and the quadrantal triangles OYZ, OZX, OXY, we have

$$\cos BOC = \frac{\cos\alpha - \cos\eta\cos\zeta}{\sin\eta\sin\zeta} = -\frac{\sqrt{\left(1 - \frac{b^2}{\theta^2}\sin^2\eta\right)}\sqrt{\left(1 - \frac{c^2}{\theta^2}\sin^2\zeta\right)}}{\frac{bc}{\theta^2}\sin\eta\sin\zeta}, \text{ \&c.};$$

that is,

$$\begin{aligned}
bc(\cos\alpha - \cos\eta\cos\zeta) &= -\sqrt{(\theta^2 - b^2\sin^2\eta)}\sqrt{(\theta^2 - c^2\sin^2\zeta)},\\
ca(\cos\beta - \cos\zeta\cos\xi) &= -\sqrt{(\theta^2 - c^2\sin^2\zeta)}\sqrt{(\theta^2 - a^2\sin^2\xi)},\\
ab(\cos\gamma - \cos\xi\cos\eta) &= -\sqrt{(\theta^2 - a^2\sin^2\xi)}\sqrt{(\theta^2 - b^2\sin^2\eta)},
\end{aligned}$$

which, when rationalized, are quadric equations in $\cos\xi$, $\cos\eta$, $\cos\zeta$. The first equation, in fact, gives

$$b^2c^2(\cos\alpha - \cos\eta\cos\zeta)^2 = (\theta^2 - b^2 + b^2\cos^2\eta)(\theta^2 - c^2 + c^2\cos^2\zeta),$$

that is,

$$\begin{aligned}
(\theta^2 - b^2)(\theta^2 - c^2) - b^2c^2\cos^2\alpha + (\theta^2 - b^2)c^2\cos^2\zeta + (\theta^2 - c^2)b^2\cos^2\eta&\\
+ 2b^2c^2\cos\alpha\cos\eta\cos\zeta = 0,& \quad (1)
\end{aligned}$$

or, what is the same thing,

$$\begin{aligned}
-\left(1 - \frac{b^2c^2\cos^2\alpha}{(b^2 - \theta^2)(c^2 - \theta^2)}\right) + \frac{c^2}{c^2 - \theta^2}\cos^2\zeta + \frac{b^2}{b^2 - \theta^2}\cos^2\eta&\\
- \frac{2b^2c^2}{(b^2 - \theta^2)(c^2 - \theta^2)}\cos\alpha\cos\eta\cos\zeta = 0.& \quad (2)
\end{aligned}$$

Completing the system, we have

$$-\left(1-\frac{c^2a^2\cos^2\beta}{(c^2-\theta^2)(a^2-\theta^2)}\right)+\frac{a^2}{a^2-\theta^2}\cos^2\xi+\frac{c^2}{c^2-\theta^2}\cos^2\zeta \\ -\frac{2c^2a^2}{(c^2-\theta^2)(a^2-\theta^2)}\cos\beta\cos\zeta\cos\xi=0, \quad (3)$$

$$-\left(1-\frac{a^2b^2\cos^2\gamma}{(a^2-\theta^2)(b^2-\theta^2)}\right)+\frac{b^2}{b^2-\theta^2}\cos^2\eta+\frac{a^2}{a^2-\theta^2}\cos^2\xi \\ -\frac{2a^2b^2}{(a^2-\theta^2)(b^2-\theta^2)}\cos\gamma\cos\xi\cos\eta=0, \quad (4)$$

and, as above,

$$\sin\alpha\cos\xi+\sin\beta\cos\eta+\sin\gamma\cos\zeta=0.$$

It seems difficult from these equations to eliminate ξ, η, ζ, so as to obtain an equation in θ; but I employ some geometrical considerations.

Taking Π as the pole of the circle ABC, and drawing ΠX, ΠY, ΠZ to meet the circle in p, q, r respectively, then, if α'', β'', γ'' are the cosine-inclinations of O to X, Y, Z respectively, we have

$$\sin Xp,\quad \sin Yq,\quad \sin Zr=\alpha'',\ \beta'',\ \gamma''.$$

From the right parallel triangles BYq and CZr, we have

$$\sin Yq=\sin BY\sin B,$$
$$\sin Zr=\sin CZ\sin C,$$

and, thence,

$$\frac{\sin Yq}{\sin Zr}=\frac{\sin BY}{\sin CZ}\cdot\frac{\sin OC}{\sin OB},$$

or, since

$$BY=OB-OY,\quad CZ=OC-OZ,$$

and thence

$$\sin BY=\frac{\sin\eta}{\theta}\{\surd(\theta^2-b^2\sin^2\eta)-b\cos\eta\},$$
$$\sin CZ=\frac{\sin\zeta}{\theta}\{\surd(\theta^2-c^2\sin^2\zeta)-c\cos\zeta\},$$

we obtain

$$\frac{\beta''}{\gamma''} = \frac{\sqrt{(\theta^2 - b^2\sin^2\eta)} - b\cos\eta}{\sqrt{(\theta^2 - c^2\sin^2\zeta)} - c\cos\zeta}.$$

We have thence

$$\beta''\sqrt{(\theta^2 - c^2\sin^2\zeta)} - \gamma''\sqrt{(\theta^2 - b^2\sin^2\eta)} = \beta'' c\cos\zeta - \gamma'' b\cos\eta,$$

or, squaring and reducing

$$\beta''^2(\theta^2 - c^2) + \gamma''^2(\theta^2 - b^2) + 2\beta''\gamma''\{-\sqrt{(\theta^2 - c^2\sin^2\zeta)}\,\sqrt{(\theta^2 - b^2\sin^2\eta)} + bc\cos\eta\cos\zeta\} =$$

that is,

$$\beta''^2(\theta^2 - c^2) + \gamma''^2(\theta^2 - b^2) + 2\beta''\gamma''\cdot bc\cos\alpha = 0;$$

and, similarly,

$$\gamma''^2(\theta^2 - a^2) + \alpha''^2(\theta^2 - c^2) + 2\gamma''\alpha''\cdot ca\cos\beta = 0,$$
$$\alpha''^2(\theta^2 - b^2) + \beta''^2(\theta^2 - a^2) + 2\alpha''\beta''\cdot ab\cos\gamma = 0,$$

or, what is the same thing,

$$\frac{\beta''^2}{b^2 - \theta^2} + \frac{\gamma''^2}{c^2 - \theta^2} - \frac{2bc\cos\alpha}{b^2 - \theta^2\cdot c^2 - \theta^2}\beta''\gamma'' = 0,$$
$$\frac{\gamma''^2}{c^2 - \theta^2} + \frac{\alpha''^2}{a^2 - \theta^2} - \frac{2ca\cos\beta}{c^2 - \theta^2\cdot a^2 - \theta^2}\gamma''\alpha'' = 0,$$
$$\frac{\alpha''^2}{a^2 - \theta^2} + \frac{\beta''^2}{b^2 - \theta^2} - \frac{2ab\cos\gamma}{a^2 - \theta^2\cdot b^2 - \theta^2}\alpha''\beta'' = 0;$$

writing

$$\alpha'',\ \beta'',\ \gamma'' = X\sqrt{(a^2 - \theta^2)},\quad Y\sqrt{(b^2 - \theta^2)},\quad Z\sqrt{(c^2 - \theta^2)},$$

and

$$\frac{bc\cos\alpha}{\sqrt{(b^2 - \theta^2\cdot c^2 - \theta^2)}},\ \frac{ca\cos\beta}{\sqrt{(c^2 - \theta^2\cdot a^2 - \theta^2)}},\ \frac{ab\cos\gamma}{\sqrt{(a^2 - \theta^2\cdot b^2 - \theta^2)}} = f,\ g,\ h,$$

the equations are

$$Y^2 + Z^2 - 2fYZ - a = 0,$$
$$Z^2 + X^2 - 2gZX - b = 0,$$
$$X^2 + Y^2 - 2hXY - c = 0.$$

Writing the last two under the form

$$X^2 - 2gZX + Z^2 = 0,$$
$$X^2 - 2hYX + Y^2 = 0,$$

and eliminating X', we have

$$-4(1-g^2)(1-h^2)\,Y'^2 Z'^2 + (Y'^2 + Z'^2 - 2gh\,Y'Z')^2 = 0,$$

which, in virtue of the first equation, is

$$-4(1-g^2)(1-h^2)\,Y'^2 Z'^2 + 4(gh-f)^2\,Y'^2 Z'^2 = 0,$$

that is,

$$(1-g^2)(1-h^2) - (gh-f)^2 = 0;$$

or, what is the same thing,

$$1 - f^2 - g^2 - h^2 + 2fgh = 0.$$

I remark that we may write

$$gh - f = \surd(1-g^2)\,\surd(1-h^2),$$
$$hf - g = \surd(1-h^2)\,\surd(1-f^2),$$
$$fg - h = \surd(1-f^2)\,\surd(1-g^2),$$

the signs on the right-hand side being either all +, or else one + and two −, so that the product is +. In fact, multiplying the assumed equations, we have

$$\begin{aligned} f^2g^2h^2 - fgh(f^2+g^2+h^2) + g^2h^2 + h^2f^2 + f^2g^2 - fgh \\ = 1 - f^2 - g^2 - h^2 + g^2h^2 + h^2f^2 + f^2g^2 - f^2g^2h^2, \end{aligned} \tag{5}$$

that is,

$$1 - f^2 - g^2 - h^2 + fgh(1 + f^2 + g^2 + h^2) - 2f^2g^2h^2 = 0,$$

or,

$$(1 - f^2 - g^2 - h^2 + 2fgh)(1 - fgh) = 0,$$

which is right; but with a different combination of signs the result would not have been obtained.

Substituting for f, g, h their values, we have

$$(a^2-\theta^2)(b^2-\theta^2)(c^2-\theta^2)-b^2c^2(a^2-\theta^2)\cos^2\alpha-c^2a^2(b^2-\theta^2)\cos^2\beta \\ -a^2b^2(c^2-\theta^2)\cos^2\gamma+2a^2b^2c^2\cos\alpha\cos\beta\cos\gamma=0, \quad (6)$$

The three given equations are

$$\begin{aligned} Y^2+Z^2-2fYZ-a&=0,\\ Z^2+X^2-2gZX-b&=0,\\ X^2+Y^2-2hXY-c&=0, \end{aligned}$$

say these are $U=0$, $V=0$, $W=0$; it is to be shown that $cV-bW$, $aW-cU$, $bU-aV$, each contain the linear factor in question. We have

$$cV-bW=(c-b)X^2-bY^2+cZ^2-2cgZX+2bhXY;$$

or, what is the same thing,

$$a(cV-bW)=a(c-b)X^2-h^2Y^2+g^2Z^2-2gg^2ZX+2hh^2XY.$$

Assuming this

$$=(aX+hY+gZ)(\lambda X-hY+gZ),$$

we have

$$\begin{aligned} a\lambda&=a(c-b),\\ g(a+\lambda)&=-2gg^2,\\ h(-a+\lambda)&=2hh^2, \end{aligned}$$

that is,

$$\lambda=c-b,\quad a+\lambda=-2gg,\quad -a+\lambda=2hh:$$

but $\lambda=c-b,=-h^2+g^2$, and the other two equations are $a+c-b+2gg=0$, $a+b-c+2hh=0$, which are identically true.

The values of X, Y, Z are thus determined as the coordinates of the intersection of the conic with the plane $AX+BY+CZ=0$; or, what is the same thing, of the line

$$\begin{aligned} AX+BY+CZ&=0,\\ aX+hY+gZ&=0, \end{aligned}$$

with any one of the three cylinders.

We may, however, complete the analytical solution in a different manner as follows:

Assuming as above $\surd(bc) = f$, $\surd(ca) = g$, $\surd(ab) = h$, and hence $h\surd(c) - g\surd(b) = 0$, we obtain from the second and the third equations

$$Y = hX + \surd(c)\,\surd(1 - X^2), \quad Z = gX - \surd(b)\,\surd(1 - X^2),$$

(the signs are one $+$ the other $-$, in order that this may consist with the equation $aX + hY + gZ = 0$). Substituting in $AX + BY + CZ = 0$, we have

$$(A + Bh + Cg)X + \{B\surd(c) - C\surd(b)\}\surd(1 - X^2) = 0,$$

that is,

$$(A + Bh + Cg)^2X^2 - (B^2c + C^2b - 2BCf)(1 - X^2) = 0,$$

which is

$$= -\theta^2\left(\frac{\sin^2\alpha}{a^2 - \theta^2} + \frac{\sin^2\beta}{b^2 - \theta^2} + \frac{\sin^2\gamma}{c^2 - \theta^2}\right),$$

so that we have

$$\Lambda X^2 = \left(\frac{\sin^2\alpha}{a^2 - \theta^2} + \frac{\sin^2\beta}{b^2 - \theta^2} + \frac{\sin^2\gamma}{c^2 - \theta^2}\right).$$

Similarly,

$$\Lambda Y^2 = \left(\frac{\sin^2\alpha}{a^2 - \theta^2} + \frac{\sin^2\beta}{b^2 - \theta^2} + \frac{\sin^2\gamma}{c^2 - \theta^2}\right),$$

$$\Lambda Z^2 = \left(\frac{\sin^2\alpha}{a^2} + \frac{\sin^2\beta}{b^2} + \frac{\sin^2\gamma}{c^2 - \theta^2}\right);$$

and hence also

$$\Lambda(1-X^2) = \frac{-\theta^2\sin^2\alpha}{a^2(a^2 - \theta^2)}, \quad \Lambda(1-Y^2) = \frac{-\theta^2\sin^2\beta}{b^2(b^2 - \theta^2)}, \quad \Lambda(1-Z^2) = \frac{-\theta^2\sin^2\gamma}{c^2(c^2 - \theta^2)},$$

where

$$\Lambda = \frac{\sin^2\alpha}{a^2} + \frac{\sin^2\beta}{b^2} + \frac{\sin^2\gamma}{c^2}.$$

The equation in X is

$$\Lambda\left(1-\frac{a^2\cos^2\xi}{a^2-\theta^2}\right)=\frac{-\theta^2\sin^2\alpha}{a^2(a^2-\theta^2)},$$

that is,

$$\Lambda(a^2\sin^2\xi-\theta^2)=-\theta^2\frac{\sin^2\alpha}{a^2},$$

or

$$a^2\sin^2\xi=\theta^2\left(1-\frac{\sin^2\alpha}{a^2\Lambda}\right),$$

and the like for η, ζ. Writing for greater convenience $\dfrac{\sin^2\alpha}{a^2}$, $\dfrac{\sin^2\beta}{b^2}$, $\dfrac{\sin^2\gamma}{c^2}=p,\ q,\ r$, then $\Lambda=p+q+r$, and we have

$$\sin^2\xi=\frac{\theta^2}{a^2}\frac{q+r}{p+q+r},\quad \sin^2\eta=\frac{\theta^2}{b^2}\frac{r+p}{p+q+r},\quad \sin^2\zeta=\frac{\theta^2}{c^2}\frac{p+q}{p+q+r},$$

(whence also $a^2\sin^2\xi+b^2\sin^2\eta+c^2\sin^2\zeta=2\theta^2$: as a simple verification, observe that, if the projection is rectangular, the axes being all equally inclined to the plane of projection, then $\xi=\eta=\zeta=90^\circ$, $a=b=c=\theta\sin s$, and the equation is $3\sin^2 s=2$; s, s are here the sides of an isosceles quadrantal triangle, the included angle being 120°, that is, we have $\cos 120^\circ\ (=-\tfrac{1}{2})=-\cot^2 s$, that is, $\cot^2 s=\tfrac{1}{2}$, or $\sin^2 s=\tfrac{2}{3}$, which is right).

I remark, that a geometrical solution may be obtained upon very different principles. We have on a sphere the trirectangular triangle XYZ, which by parallel lines is projected into ABC. Every great circle of the sphere is projected into an ellipse having double contact at the extremities of a diameter with the ellipse which is the apparent contour of the sphere. Moreover, if the arc of great circle XY is a quadrant, then the radius through X and the tangent at Y are parallel to each other, whence, if Ω be the projection of the centre, and AB the projection of the arc XY, then in the projection the line ΩA and the tangent at B are parallel to each other. It is now easy to derive a construction: with centre Ω, and conjugate semi-axes $(\Omega B,\ \Omega C)$, $(\Omega C,\ \Omega A)$, $(\Omega A,\ \Omega B)$ respectively, describe three ellipses; and find a concentric ellipse having double contact with each of these (there are in fact two such ellipses, one touching the three ellipses internally, and giving an imaginary solution; the other touching them externally, which is the ellipse intended). Drawing then through the ellipse a right cylinder (there are two such cylinders, but only one of them is real), and inscribing in it a sphere, and projecting on to the surface of the sphere by lines parallel to the axis of the cylinder, the three ellipses are projected into three great circles cutting at right angles, or, say, the elliptic arcs BC, CA, AB are projected into the trirectangular triangle XYZ.

0.12 615. On the conic torus[5]

The equation $p + \surd(qr) + \surd(st) = 0$, where p, q, r, s, t are linear functions of the coordinates (x, y, z, w), and as such are connected by a linear relation, belongs to a quartic surface having a nodal conic $(p = 0,\ qr - st = 0)$; and four nodes (conical points), viz. these are the intersections of the line $q = 0$, $r = 0$ with the quadric surface $p^2 - qr - st = 0$, and of the line $r = 0$, $s = 0$ with the same surface. The quartic surface has also four tropes (planes which touch the surface along a conic); viz. these are the planes $q = 0$, $r = 0$, $s = 0$, $t = 0$, the conic of contact or tropal conic in each plane being the intersection of the plane with the before-mentioned quadric surface $p^2 - qr - st = 0$. The planes $q = 0$, $r = 0$, and also the conics in these planes pass through two of the nodes, say A, C; and the planes $s = 0$, $t = 0$, and also the conics in these planes pass through the remaining two nodes, say B, D; so that the relations of the surface are as is shown in fig. 1. It is to be added that AB, BC, CD, DA (but not AC or BD) are lines on the surface.

The planes $q = 0$, $r = 0$, which contain the tropal conics through A, C, are in general distinct from the planes ABC, ADC which contain the line-pairs BA, BC and DA, DC respectively: and so also the planes $s = 0$, $t = 0$, which contain the tropal conics through B, D, are in general distinct from the planes ABD, CBD which contain the line-pairs AB, AD and CB, CD respectively.

If, however, the identical linear relation contain only p, s, t, then the planes $q = 0$, $r = 0$ will be the planes ABC, ADC respectively: and the tropal conics in these planes will consequently be the line-pairs BA, BC, and DA, DC respectively. But the planes $s = 0$, $t = 0$ will continue to be distinct from the planes ABD, CBD: and the tropal conics in the planes $s = 0$, $t = 0$ will remain proper conics.

A surface of the last-mentioned form is

$$mz + \surd(xy) + \surd(w^2 - z^2) = 0,$$

viz. this has the nodal conic $z = 0$, $xy - w^2 = 0$, the nodes $\{x = 0,\ y = 0,\ (m^2 + 1)z^2 - w^2 = 0\}$, and $(z = 0,\ w = 0,\ x = 0)$, $(z = 0,\ w = 0,\ y = 0)$, and the tropes $x = 0$, $y = 0$, $z + w = 0$, $z - w = 0$; but the planes $z + w = 0$ and $z - w = 0$ are ordinary tropal planes

[5][From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XIII. (1875), pp. 127–129.]

each touching the surface in a proper conic; the planes $x=0$, $y=0$ special planes each touching along a line-pair.

[Figure 1: Tetrahedral diagram showing nodes A, B, C, D with planes $q=0$, $r=0$, $s=0$, $t=0$ indicated]

The equation in question, writing therein $w=1$ and $x+iy$, $x-iy$ in place of (x,y) respectively, is

$$\{\surd(x^2+y^2)+mz\}^2=1-z^2,$$

which is derived from

$$(x+mz)^2=1-z^2,$$

by the change of x into $\surd(x^2+y^2)$; and the surface is consequently the torus generated by the rotation of the conic $(x+mz)^2=1-z^2$ about its diameter. Or, what is the same thing, the surface

$$mz+\surd(xy)+\surd(w^2-z^2)=0,$$

regarding therein (x,y) as circular coordinates and w as being $=1$, is a torus. The rational equation is $U=0$, where we have

$$\begin{aligned}U&=\{(m^2+1)z^2-w^2+xy\}^2-4m^2z^2xy\\&=\{xy+(1-m^2)z^2-w^2\}^2+4m^2z^2(z^2-w^2)\\&=x^2y^2+(m^2+1)^2z^4+w^4+(2-2m^2)z^2xy-(2+2m^2)z^2w^2-2xyw^2.\end{aligned}$$

I find that the Hessian H of this function U contains the factor $xy+(1-m^2)z^2-w^2$, viz. that we have

$$H=\{xy+(1-m^2)z^2-w^2\}\,H',$$

where H' is a sextic function; and it thus appears that the before-mentioned quadric surface $p^2-qr-st=0$, that is $xy+(1-m^2)z^2-w^2=0$, is a node-couple quadric surface of the torus.

0.13 616. A geometrical illustration of the cubic transformation in elliptic functions[6]

Consider the cubic curve

$$x^3 + y^3 + z^3 + 6lxyz = 0.$$

If through one of the inflexions $z = 0$, $x + y = 0$, we draw an arbitrary line $z = u(x + y)$, we have at the other intersections of this line with the curve

$$u\{u^3(x + y)^3 + 6lxy\} + x^2 - xy + y^2 = 0;$$

that is,

$$(u^3 + 1)(x^2 + y^2) + 2xy(u^3 + 3lu - \tfrac{1}{2}) = 0;$$

and from this equation it appears that the ratio $x : y$ is given as a function involving the square root of

$$(u^3 + 3lu - \tfrac{1}{2})^2 - (u^3 + 1)^2,$$

which, rejecting a factor 3, is

$$= (2u^3 + 3lu + \tfrac{1}{2})(lu - \tfrac{1}{2}).$$

It may be noticed that $lu - \frac{1}{2} = 0$ gives the value of u, which in the equation $z = u(x+y)$ belongs to the tangent at the inflexion; and $2u^3 + 3lu + \frac{1}{2} = 0$ gives the values which belong to the three tangents from the inflexion.

It thus appears that the coordinates x, y, z of any point of the curve can be expressed as proportional to functions of u involving the radical

$$\sqrt{\{(lu - \tfrac{1}{2})(2u^3 + 3lu + \tfrac{1}{2})\}},$$

and the theory of the curve is connected with that of a quasi-elliptic integral depending on this radical.

Taking m an imaginary cube root of unity, and writing for a moment X, Y, Z instead of x, y, z, the equation of the cubic curve may be written

$$XYZ = \text{const.},$$

[6] [From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XIII. (1875), pp. 211–216.]

where

$$\begin{aligned}X &= x + y + z,\\ Y &= x + my + m^2 z,\\ Z &= x + m^2 y + mz.\end{aligned}$$

The cubic transformation may be represented geometrically as follows: taking ξ, η, ζ as the coordinates of a point on a cubic curve, we construct a point (x, y, z) such that

$$\frac{x}{\xi^2} = \frac{y}{\eta^2} = \frac{z}{\zeta^2},$$

or, what is the same thing, we take x, y, z proportional to ξ^2, η^2, ζ^2; then (x, y, z) will be a point on a sextic curve; and conversely, from a point on the sextic we derive a point on the cubic. This is the cubic transformation in its geometrical form.

0.14 617. On the scalene transformation of a plane curve[7]

The transformation by reciprocal radius vectors can be effected mechanically by Sylvester's Peaucellier-cell. But, employing a more general cell (considered incidentally by him) which may be called the scalene-cell, we have the scalene transformation in question[8]; viz. if, in two curves, r, r' are radius vectors belonging to the same angle (or say opposite angles) θ, then the relation between r, r' is

$$rr'(r+r') + (m^2 - l^2)\, r + (m^2 - n^2)\, r' = 0;$$

or, as this may also be written,

$$r^2 + \left(r' - \frac{l^2 - m^2}{r'}\right) r + m^2 - n^2 = 0.$$

The transformation is, it will be seen, an interesting one for its own sake, independently of the remarkably simple mechanical construction, viz. the scalene cell is simply a system of 3 pairs of equal rods PA, QA; PB, QB; PC, QC (fig. 1, p. 528), jointed together at and capable of rotating about the points P, Q, A, B, C; the three lengths PA, PB, PC (say these are $= l$, m, n) are all of them unequal: in the case of any two of them equal, we have Peaucellier or isosceles cell. The effect of the arrangement is that the points A, B, C are retained in a right line, the distances BA, $= r'$, and BC, $= r$, being connected by the above-mentioned equation; so that taking B as a fixed point, if the point A describe any given curve, the point C will describe the corresponding or transformed curve.

In the case where the given curve is a right line or a circle, we may through B draw at right angles to the curve the axis $x'Bx$: viz. in the case of the circle,

[Figure 1: Mechanical diagram showing the scalene cell with points P, Q, A, B, C, D and rod lengths l, m, n]

[7][From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XIII. (1875), pp. 321–328.]

[8]The transformation itself, and doubtless many of the results obtained by means of it, are familiar to Prof. Sylvester; and I abandon all claim to priority.

The equation of the Cartesian is $r^2 + (A\cos\theta + N)\,r + B = 0$, viz. this is $x^2 + y^2 + Ax + B = -N\sqrt{(x^2 + y^2)}$, or writing $N^2 = a$, it is

$$(x^2 + y^2 + Ax + B)^2 = a(x^2 + y^2),$$

which is the form considered in the preceding paper. It may be further observed in regard to it that, starting from the focal equation $r = ls + m$, where r, s are the distances of a point (x, y) of the Cartesian from any two of its three foci, this equation gives $r^2 - l^2 s^2 + m^2 = 2mr$, or writing $r^2 = x^2 + y^2$, $s^2 = (x - \alpha)^2 + y^2$, the function on the left-hand is of the form $(1 - l^2)(x^2 + y^2 + Ax + B)$, whence, assuming $\dfrac{2m}{1 - l^2} = \sqrt{(a)}$, the equation becomes as above

$$(x^2 + y^2 + Ax + B)^2 = a\,(x^2 + y^2).$$

Taking the distance $r = \sqrt{(x^2 + y^2)}$ to be measured from a given focus, it is easy to see that, no matter which of the other two foci we associate with it, we obtain the *same* equation $(x^2 + y^2 + Ax + B)^2 = a\,(x^2 + y^2)$; viz. starting with any one focus, we connect with it a determinate circle $x^2 + y^2 + Ax + B = 0$, and a determinate coefficient a, such that taking this focus as the origin, the equation of the curve is

$$(x^2 + y^2 + Ax + B)^2 = a\,(x^2 + y^2);$$

but there are for the given curve three such forms of equation, according as the origin is taken at one or other of the three foci.

0.15 618. On the mechanical description of a Cartesian[9]

If in the scalene cell we take C for a fixed point and taking $BA = r$, we have $AC = \dfrac{m^2 - l^2}{r}$, whence $BC = r + \dfrac{m^2 - l^2}{r}$, or D being a point at the distance CD, $= \alpha$, from the point C, and denoting BD by r', we have $r' = r + \dfrac{m^2 - l^2}{r} + \alpha$, which is an equation of the required form; whence, if the point D describe a circle passing through B, then the point A will describe a Cartesian.

[Figure 1: Mechanical diagram showing the scalene cell with points P, Q, A, B, C, D and rod lengths l, m, n]

The equation of the Cartesian is $r^2 + (A\cos\theta + N)\,r + B = 0$, viz. this is $x^2 + y^2 + Ax + B = -N\sqrt{(x^2 + y^2)}$, or writing $N^2 = a$, it is

$$(x^2 + y^2 + Ax + B)^2 = ax^2 + ay^2,$$

which is the form considered in the preceding paper. It may be further observed in regard to it that, starting from the focal equation $r = ls + m$, where r, s are the distances of a point (x, y) of the Cartesian from any two of its three foci, this equation gives $r^2 - l^2 s^2 + m^2 = 2mr$, or writing $r^2 = x^2 + y^2$, $s^2 = (x - \alpha)^2 + y^2$, the function on the left-hand is of the form $(1 - l^2)(x^2 + y^2 + Ax + B)$, whence, assuming $\dfrac{2m}{1 - l^2} = \sqrt{(a)}$, the equation becomes as above

$$(x^2 + y^2 + Ax + B)^2 = a\,(x^2 + y^2).$$

Taking the distance $r = \sqrt{(x^2 + y^2)}$ to be measured from a given focus, it is easy to see that, no matter which of the other two foci we associate with it, we obtain the *same* equation $(x^2 + y^2 + Ax + B)^2 = a\,(x^2 + y^2)$; viz. starting with any one focus, we connect with it a determinate circle $x^2 + y^2 + Ax + B = 0$, and a determinate coefficient a, such that taking this focus as the origin, the equation of the curve is

$$(x^2 + y^2 + Ax + B)^2 = a\,(x^2 + y^2);$$

[9] [From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XIII. (1875), pp. 329–330.]

but there are for the given curve three such forms of equation, according as the origin is taken at one or other of the three foci.

(Addition, Feb. 1875.) It is obviously the same thing, but I find that it is mechanically more convenient to derive the Cartesian from the Limaçon $r' = -N - A\cos\theta$, by the transformation $r' = r + \dfrac{B}{r}$: I have on this principle constructed an apparatus whereby the Cartesian is described on a rotating board by a pencil moving in a fixed line.

0.16 619. On an algebraical operation[10]

I consider

$$\Omega F(a,\, x),$$

an operation Ω performed upon $F(a,\, x)$ a rational function of $(a,\, x)$; viz. F being first expanded or regarded as expanded in ascending powers of a, the coefficients of the several powers are then to be expanded or regarded as expanded in ascending powers of x, and the operation consists in the rejection of all negative powers of x.

In the cases intended to be considered, F contains only positive powers of a: but this restriction is not necessary to the theory.

The investigation has reference to the functions $A(x)$ of my "Ninth Memoir on Quantics," *Phil. Trans.*, t. CLXI. (1871), pp. 17–50, [462]; for instance, we there have as regards the covariants of a quadric

$$A(x) - \frac{1}{x^2} A\left(\frac{1}{x}\right) = \frac{1 - x^{-2}}{1 - ax^2 \cdot 1 - a \cdot 1 - ax^{-2}},$$

and consequently, in the present notation,

$$A(x) = \Omega\, \frac{1 - x^{-2}}{1 - ax^2 \cdot 1 - a \cdot 1 - ax^{-2}};$$

by a process of development and summation, the value of this expression was found to be

$$= \frac{1}{1 - ax^2 \cdot 1 - a^2},$$

and the like for the other forms.

[10][From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XIII. (1875), pp. 369–375.]

0.17 620. Correction of two numerical errors in Sohnke's paper respecting modular equations[11]

In Sohnke's paper "Aequationes modulares pro transformatione functionum ellipticarum," *Crelle*, t. XVI. (1837), there is, on p. 113, an obvious error in the expression of u^6, viz. the term q^{18} is given with the same numerical coefficient as it had in u^6: this remark was made to me by Mr W. Barrett Davis, who finds that the term of u^6 should be

$$+13569463\,q^{18}.$$

In the expression of u^{18} (*l.c.*, p. 115), I had remarked that, in the coefficient of q^{18}, a figure must have dropped out. Mr Davis has verified this, and finds that the figure omitted is a 1 in the unit's place, and thus that the correct value[12] is

$$+80177033781\,q^{18}.$$

Cambridge, 26 October, 1875.

[11] [From the *Journal für die reine und angewandte Mathematik* (Crelle), t. LXXXI. (1876), p. 229.]

[12] [The former of these numbers should replace the number 15063859 in the Table, p. 128 of this volume; the latter has been introduced in the Table, p. 129.]

0.18 621. On the number of the univalent radicals C_nH_{2n+1}[13]

I have just remarked that the determination is contained in my paper "On the Analytical Forms called Trees, &c.," *British Association Report*, 1875, [610]; in fact, in the form C_nH_{2n+1}, there is one carbon atom distinguished from the others by its being combined with (instead of 4, only) 3 other atoms; viz. these are 3 carbon atoms, 2 carbon atoms and 1 hydrogen atom, or else 1 carbon atom and 2 hydrogen atoms (CH_3, methyl, is an exception; but here the number is = 1). The number of carbon atoms thus combined with the first-mentioned atom is the number of main branches, which is thus = 3, 2, or 1; hence we have, number of radicals C_nH_{2n+1} is =

No. of carbon root-trees C_n with one main branch,
+ No. of with two main branches,
+ No. of with three main branches;

and the three terms for the values $n = 1$ to 13 are given in Table VII. (pp. 454, 455 of this volume) of the paper referred to.

Thus, if $n = 5$, an extract from the Table (p. 454 of this volume), is

Index x, or number of knots	Index t, or number of main branches	Altitude 0	1	2	3	4	
5	1			1	2	1	4
	2			2	1		3
	3			1			1
	4		1				1
	Total ...		1	4	3	1	9

[13][From the *Philosophical Magazine*, series 5, vol. III. (1877), pp. 34, 35.]

0.19 622. On a system of equations connected with Malfatti's problem[14]

I consider the equations

$$\begin{aligned} X &= by^2 + cz^2 - 2fyz - a\,(bc - f^2), = 0, \\ Y &= cz^2 + ax^2 - 2gzx - b\,(ca - g^2), = 0, \\ Z &= ax^2 + by^2 - 2hxy - c\,(ab - h^2), = 0, \end{aligned}$$

where the constants $(a,\, b,\, c,\, f,\, g,\, h)$ are such that

$$K = abc - af^2 - bg^2 - ch^2 + 2fgh, = 0.$$

Hence, writing as usual $(A,\, B,\, C,\, F,\, G,\, H)$ to denote the inverse coefficients

$$(bc - f^2,\ ca - g^2,\ ab - h^2,\ gh - af,\ hf - bg,\ fg - ch),$$

we have $(A,\, B,\, C,\, F,\, G,\, H \between x,\, y,\, z)^2 =$ the square of a linear function, $= (\alpha x + \beta y + \gamma z)^2$ suppose; that is,

$$(A,\, B,\, C,\, F,\, G,\, H) = (\alpha^2,\ \beta^2,\ \gamma^2,\ \beta\gamma,\ \gamma\alpha,\ \alpha\beta).$$

It is to be shown that the three quadric surfaces $X = 0$, $Y = 0$, $Z = 0$ intersect in a conic Θ lying in the plane $a\alpha x + b\beta y + c\gamma z = 0$, and in two points I, J; or more completely, that

$$\begin{array}{lr} \text{the surfaces } Y,\, Z \text{ meet in the conic } \Theta \text{ and a conic } & P, \\ \qquad\quad Z,\, X & Q, \\ \qquad\quad X,\, Y & R, \end{array}$$

where the conics P, Q, R intersect in the two points I, J; and moreover that the pairs of planes

$$\begin{aligned} Y = 0,\ Z = 0 &\text{ each contain a line through } I \text{ (or } J), \\ Z = 0,\ X = 0 &\text{ each contain a line through } I \text{ (or } J), \\ X = 0,\ Y = 0 &\text{ each contain a line through } I \text{ (or } J). \end{aligned}$$

The equation of the surface $X = 0$ may be written

$$X = (\sqrt{a}\,x + i\sqrt{b}\,y + i\sqrt{c}\,z)(\sqrt{a}\,x - i\sqrt{b}\,y - i\sqrt{c}\,z) - 2\sqrt{bc}\,yz - 2fyz - a\,(bc - f^2),$$

[14][From the *Proceedings of the London Mathematical Society*, vol. VII. (1875–1876), pp. 38–42. Read December 9, 1875.]

and the tangent plane at a point of the conic Θ meets the surface in two generators, one in each sheet, which are the intersections of the surface by the planes

$$\sqrt{a}\,x \pm (i\sqrt{b}\,y + i\sqrt{c}\,z) = 0.$$

The points I, J are determined as follows: taking the signs all positive, we find for $(x^2, y^2, z^2, yz, zx, xy)$ the values (A, B, C, F', G', H'), giving two points of intersection

$$\left(\sqrt{A},\ \frac{H'}{\sqrt{A}},\ \frac{G'}{\sqrt{A}}\right) \text{ and } \left(-\sqrt{A},\ -\frac{H'}{\sqrt{A}},\ -\frac{G'}{\sqrt{A}}\right).$$

Taking the signs one positive and the other two negative, say

$$\sqrt{BC} = F', \quad \sqrt{CA} = -G', \quad \sqrt{AB} = -H',$$

we find for $(x^2, y^2, z^2, yz, zx, xy)$ the values

$$\left(0,\ \frac{Cc}{b},\ \frac{Bb}{c},\ F',\ 0,\ 0\right),$$

viz. we have thus two intersections

$$\left(0,\ \sqrt{\frac{Cc}{b}},\ F'\sqrt{\frac{b}{Cc}}\right), \quad \left(0,\ -G'\sqrt{\frac{Cc}{b}},\ -F'\sqrt{\frac{b}{Cc}}\right);$$

and the other combinations of signs give the remaining two pairs of intersections

$$\left(G'\sqrt{\frac{c}{Aa}},\ 0,\ \sqrt{\frac{Aa}{c}}\right), \quad \left(-G'\sqrt{\frac{c}{Aa}},\ 0,\ -\sqrt{\frac{Aa}{c}}\right),$$

and

$$\left(\sqrt{\frac{Bb}{a}},\ H'\sqrt{\frac{a}{Bb}},\ 0\right), \quad \left(-\sqrt{\frac{Bb}{a}},\ -H'\sqrt{\frac{a}{Bb}},\ 0\right).$$

But the most convenient statement of the result is that the values of $(ax^2, by^2, cz^2, yz, zx, xy)$, for the four pairs of points respectively, are

$$\begin{array}{l}
(aA, bB, cC, F', G', H'),\\
(\ \ 0, cC, bB, F', \ \ 0, 0),\\
(cC, 0, \ \ aA, 0, \ \ G', 0),\\
(bB, aA, \ \ 0, 0, \ \ \ 0, H');
\end{array}$$

there is no difficulty in substituting these values in the original equations, and in verifying that the equations are in each case satisfied.

End of Transcription

This completes the transcription of pages 501–550 from Volume IX of Arthur Cayley's *Collected Mathematical Papers.*

Papers included:

The Collected Mathematical Papers of Arthur Cayley, F.R.S.

Volume IX — Pages 551 to 600

623. On Three-Bar Motion.

[From the *Proceedings of the London Mathematical Society*, vol. vii. (1875–1876), pp. 136–166.
Read March 10, 1876.]

ON THREE-BAR MOTION. [From the Proceedings of the London Mathematical Society, vol. vii. (187.5—1876), pp. 186—166. Read March 10, 1876.] The discovery by Mr Roberts of the triple generation of a Three-Bar Curve, throws a new light on the whole theory, and is a copious source of further develop- ments*. The present paper gives in its most simple form the theorem of the triple generation ; it also establishes the relation between the nodes and foci ; and it con- tains other researches. I have made on the subject a further investigation, which I give in a separate paper, "On the Bicursal Sextic," [624]; but the two papers are intimately related and should be read in connection. The Three-Bar Curve is derived from the motion of a system of three bars of given lengths pivoted to each other, and to two fixed points, so as to form the three sides of a quadrilateral, the fourth side of which is the line joining the two fixed points ; the curve is described by a point rigidly connected with the middle bar ; or, what is more convenient, we take the middle bar to be a triangle pivoted at the extremities of the ba.se to the other two bars (say, the radial bars), and having its vertex for the describing point. Including the constants of position and magnitude, the Three-Bar Curve thus depends on nine parameters ; viz. these are the coordinates of the two fixed points, the lengths of the connecting bars, and the three sides of the triangle. It is known that the curve is a tricircular trinodal sextic, and the equation of such a curve contains 27—6 — 6—3, =12 constants. Imposing on the curve the condition that the three nodes lie upon a given curve, the number of constants is reduced to 12—3, =9: and it is in this way that the Three-Bar Curve is distinguished from the general tricircular * See his paper "On Three-Bar Motion in Plane Space," I.e., vol. vii., pp. 15—23, which contains more than I had supposed of the results here arrived at. There is no question as to Mr Roberts' priority in all his results.

ON THREE-BAR MOTION. trinodal sextic ; viz. in the Three-Bar Curve the two fixed points are foci, and they determine a third focus * ; and the condition is that the nodes are situate on the circle thiough the 3 foci. The nodes are two of them arbitrary points on the circle ; and |the third of them is a point such that, measuring the distances along the circle from any fixed point of the circumference, the sum of the distances of the nodes is equal to the sum of the distances of the foci. Considering the two fixed points as given, the curve depends upon five parameters, viz. the lengths of the connecting bars and the sides of the triangle. Taking the form of the triangle as given, there are then only three parameters, say the lengths of the connecting bars and the base of the triangle; in this case the third focus is determined, and therefore the circle through the three foci ; we may then take two of the nodes as given points on this circle, and thereby establish two relations between the three parameters, in fact, we thereby determine the differences of the squares of the lengths in question : but the third node is then an absolutely determined point on the circle, and we cannot make use of it for com- pleting the determination of the parameters ; viz. one parameter remains arbitrary. Or, what is the same thing, given the three foci and also the three nodes, consistently with the foregoing conditions, viz. the nodes lie in the centre through the three foci, the sum of the distances of the nodes being equal to the sum of the distances of the foci : we have a singly infinite series of three-bar curves. In reference to the notation proper for the theorem of the triple generation, I shall, when only a single node of generation is attended to, take the curve to be generated as shown in the annexed Figure 1 ; viz. is the generating point, 0(7i5, the triangle, (7, B the fixed points, CC^ and BB^^ the radial bars. The sides of the Fig. 1. triangle are a„ 6„ Ci ; its angles are 0, = A, B^, = B, C, , =C: the bars CC, and BBi are =0^ and Oj respectively, and the distance CB is = a. The sides «i, 6„ Ci may be put =Ai(sin.d, sin 2?, sin C), and the Hues a„ Oj, 03= (A-„ k.j, A:s)sin4, viz. the original data Ui, 6„ c„ Oj, a.„ a^, may be replaced by the angles A, B, C {A+B + C = tr) and the lines A„ kj, k,. And it is convenient to mention at once that the third focus A is then a point such that ABC is a triangle similar and congruent to OB^Ci. * A focus is a point, given as the intersection of a tangent to the curve from one circular point at infinity with a tangent from the other circular point at infinity ; if the circular points are simple or multiple points on the curve, then the tangent or tangents at a circular point alvould be excluded from the tangents from the point ; and the intersection of two such tangents at the two circular points respectively is not an ordinary focus; but, as the points in question are the only kind of foci occurring in the present paper, I have in the text called them foci

* ON THREE-BAR, MOTION. It may be remarked that, producing 00^ and BBi to meet in a point a, this is the centre of instantaneous rotation of the triangle, and therefore aO is the normal to the curve at 0, I proceed to show that the three nodes F, G, H are in the circle circumscribed about ABC, and that their positions are such that (the distances being measured along the circle as before) we have the property, Sum of the distances of F, 6, H is equal to the Sum of the distances of J., B, C. Supposing to be at a node F, we have then the two equal triangles FB^G^, FBiCi, such that (7„ C/ are equidistant from G, and B^, B{ equidistant from B. Hence the angles B^FB^, G^FCi are equal; consequently the halves of these angles GFC^ and Fig. 2. BFBi are equal; whence the angle GFB is equal to the angle G/FB^', that is, to the angle A ; ot F lies on a circle through B, C such that the segment upon BG contains the angle A, that is, upon the circle through A, B, G. To complete the investigation of the nodes, suppose GF = t, BF = a : then the condition Z GFC^ = Z BFBi gives fci' + T'-c' ^ Ci' + g'-a,' 26,T 2c,a that is, c,<r (6,» + T» - a,») - 6,T (c,= + o-' -«,») = ; and the condition that F is on the circle gives cr' + t' — 2o-T cos A = a". These equations give six values of (a, t) corresponding in pairs to each other; viz. if (o-„ T,) is a solution, then (— cr„ — t,) is also a solution; and to each pair of solutions corresponds a single point on the circle, viz. we have thus the three nodes F, G, H. Writing the foregoing equation in the form {c, (6,' - a,") o- - 6, (ci" - a,') t} (o-» + t' - 2<rT cos A) + a" (c,o-t» - 6,aV) = 0, and putting the left-hand side =M (a- — Pit) (a- — p.iT) {a — p^r) ; then, if a, /3, 7 denote 008.4 +1 sin .4, cos J5 + 1 sin 5, cosG+iainG respectively, putting first o- = ot and next <r = - , and dividing one of these results by the other, we find C — b,a _ g -/),. a-pi. a-p, Cia — bi~l-pi'x.l-pia.l-p3a' c. IX. 70

ON THREE-BAR MOTION. The left-hand side is here _ sin (7 — a sin 5 _ sin 0— sin 5 (cos A +i sin A) ~ a sin 0— sin B sin (7 (cos A +i sin 4) — sin B _ sin A (cos B — i sin B) _ y ~ —sinA (cos C — i8mC)~ /8 or the equation is a-p^.a-p^.a-p, ^ y l-pia.l-p^a.l-psa /3' Also, \{}vriting / for the angle FOB, we have cr = . , a f\{} t, viz. the values of sin (A. +j sin/ sin g sin A „, , „ , Pi, Pt, P> are . . /, -. , . , .\{}—r , . ,.„. . We thence find ^ ^ -^ sm (^ -t-/) sin (A+g) sm (il -I- h) a -jpi _ sin (A +f) (cos .4 + z sin A) — sin/_ cos (A +f) + i sin (A +/) 1 — op, sin (A +f) — (cos ^ + 1 sin j4) sin/~ cos/— i sin/ = cos (4 -t- 2/") + 1 sin (^ + 2/) with the like values for the other two values. Hence, writing also - I = - cos((7-£)- isin(C-£) = cos (tt + C-B) + isin (tt + C-B), the equation becomes cos(3il + 2/+ 25r-H2A)-|-isin(3-4 -I- 2/+ 2g + 2h) = cos(w + C- B) + i ain {-n- + C - B), that is, SA + 2/+2g + 2h = -TT + C-B, or, what is the same thing, 2f+2g + 2h = '7r + C-B-SA. Fig. 3. 73 Reckoning the angles round the centre from a point on the circumference, if A', B', C, F, G', H' are the angles belonging to the points A, B, G, F, G, H respectively, then A' = \{} + 2C, F' = X+2/, B' = \{}, G' = \{} + 2g, C' = X + 2C + 2B, H' = \{} + 2h; and therefore A' + B' + C' = SX + 4,C + 2B, F'+G' + H' = S\{} + 2f+2g-^2h, =3\{} + 'ir + C-B -SA;

ON THREE-BAR MOTION. that is, A' + B'+C'-F' -G'-H' = –ir + 3 (A +B+G); or, omitting an angle 2-n; this is A' + B" + C' = F' + G' + H', the equation which determines the relation between the three nodes on the circle ABC Reverting to the equation CiO-(6,^ + t^ — Oa") — iir (cr + a-^ — a,^) = 0, which belongs to a node : if we consider the form of the triangle as given, and write bi, c, = ki sin B, ki sin G, this becomes a sin C (bi' - a^) - t sin 5 (Ci' - af) + ar (t smC-a sin 5) = ; viz. considering the node as given, then the values of tr, t are given, and the equation establishes a relation between the values of bi — a^ and Ci — a^*. If a second node be given, we have a second relation between these same quantities, and the two equations give the values of the two quantities, viz. the values of k^&hx^B — k^sm^ A, k' k' k^ k^ ki^sin^C — kj'airi'A, or, what is the same thing, the value of -r-^— i . f „ , .-l~. — • ! n- ° am'A Bin'B smM sm^B It thus appears that, if Z„ L, I, are any values of k^, k^, k, belonging to a given system of three nodes, the general values of ki, k„ ks belonging to the same system of three nodes are ki'^li' + u sin- A, Atj" = ij» + M sin= £, iV = 's" + m sin" (7, where u is an arbitrary constant. It may be added that there will be a node at B, if the equation is satisfied by k k« T = 0, 0- = a, for the condition is b^ —a^=0: that is, if -^-^. = . '„ : similarly, there will smil sm.S •' k k, be a node at C, if <>.^—a^ = 0, that is, if ^—i-j = ^—=; and a node at A, if sm ^ sm jB -r-*7i = -.—Si . If two of these equations are satisfied, the third equation is also Bin C smB ^ ^ k k k satisfied, viz. we then have ^-^.^ = . - „ = • ^^ ; and the three nodes coincide with the svaA smij sinO three foci respectively. If, in Figure 2 (p. 553), the points C„ Gi coincide on the line GF, and therefore also the points fi„ 5,' coincide on the line BF, then, instead of a node at F, we have a cusp. We have in this case a triangle the sides of which are a^+b^, Os + Ci, a, and the included angle between the first two sides is = J. : we have, therefore, the relation a" = (oj + bif + (tts + Ci)' - 2 (a, + b^) (a, + <h) cos A. Substituting herein for a, Oj, bi, &c., the values A; sin .4, kiSmA, kiSinB, &c., the equation is l(?Bixi^A = (Jcie,inB->t-hsmAf + {kiSia G+kiSiuAf — 2 (i, sin B + kisinA) {ki sin C + Ajj sin .4) cos A. * Considering, in the equation, Oj and a^ aa the distances of a variable point P from the points G and B respectively, the equation represents a circle having its centre on the line GB. Similarly, when a second node is given, the corresponding equation represents another circle, having its centre on the line GB, and the intersections of the two circles determine a, and a„ the lengths of the radial bars, in order that the curve may have the given nodes. 70—2

ON THREE-BAR MOTION. Expanding the right-hand side and reducing by means of A + B + C=ir, the whole becomes divisible by sin' J, and we have jt» = jfc,» + jfcj» -f- ^' — 2^^, cos A + 2ktk, cosB + 2kiki cos C viz. considering A, B, C, k^, k^, k^ as given, this equation determines k so that the curve may have a cusp. The equation is one of the system of four equations A« = jfc,» -f Ar," -I- ki^ — 2kiki cos A + ^k^k^ cos £ + 2^^, cos G, l(?z=k^ + kt' + kt* + 2kikt cos A - 2k,ki cos B + 2kiki cos G, k^=zki^ + k^ + k^ + 2i2^'8 cos A + 2^,^;, cos B — 2kiki cos C, k' = ki* + A,' + k,^ — 2^2 ij cos A — 2kiki cos JS — 2A, A-, cos G, which belong to the different arrangements CG^F or GFCi, BB^F or BFB^, of the three points on the lines BF and GF ; if k has any of these four values, the curve will have a cusp. If two of the equations subsist together, we have a curve with two cusps. Taking ki, k^, k^, and also cos 4, coaB, cos C, as positive, viz. assuming that the triangle is acute-angled, the fourth equation cannot subsist with any one of the others: but two of the others may subsist together, for instance, the first and second will do so, k ki if k,ktCosA=k,kiCoaB, that is, if —^-= —=, and then }<? = ki' + k^^ + k,^ + 2kiki cos G coaA coaB the curve has then two cusps. Similarly, the three equations may subsist together, viz. we must then have n?! iCj K^ i» = ii' + k^ + k^ + 2ktkt cos A ; cos A cos B cos G ' writing herein ki, k^, Ai, = \{}cos4, \{}co8 5, XcosO, we find A» = X." (cos' A + cos' B + cos' G -|- 2 cos J. cos 5 cos (7) = X» viz. if ki,k.„k, are respectively = A; cos J., kcoaB, kcosG, the curve has then three cusps. It will be recollected that, if ki : ki : k3 = sin A : sin B : sin G, the nodes coincide with the foci ; the two sets of conditions subsist together, if A— B = G = 60°; ki = kt

= k, = ik, viz. we have then a curve with three cusps coinciding with the three foci respectively. Before going further, I will establish the theorem for the triple generation of the curve. The theorem which gives the triple generation may be stated as follows. See Figures 4, 5, 6*. Imagine a triangle ABG and a point 0, through which point are drawn lines parallel to the sides dividing the triangle into three triangles OBfi^, OGtA^, OAiB„ * Figure 6 (substantially the same as Fig. 5) belongs to the same curve as Figures 1 and 2, and it exhibits the triple generation of this carve : the generating point being taken at a node (the same node as in Figure 2), and the two positions OBjCj and OBi'Ci of one of the triangles being shown in the figure.

ON THREE-BAR MOTION. similar inter se and to the original triangle, and into three parallelograms OA^AA„ OBjBB^, OCiCCj. Then, considering the three triangles as pivoted together at the point 0, and replacing the exterior sides of the parallelograms by pairs of bars AtAAt, B^BBi, GfiCi pivoted together at A, B, C, and to the triangles at A^, A^, B^, B^, C„ C„ the figure thus consisting of the three triangles and the six bars; let the Fig. 4. Fig. 5. Fig. 6. 4' three triangles be turned at pleasure about the point 0, so as to displace in any manner the points A, B, C: we have the theorem that the triangle ABG will remain always similar to the original triangle ABC, that is, to each of the three triangles OBiCi, OG^Ai, OAiBt'. and further, that, starting from any given positions of the three triangles, we may so move them as not to alter the triangle ABG in magnitude whence, conversely, fixing the three points A, B, G, the point will be moveable in a curve. Assuming this, it is clear that the locus of the point is simultaneously the locus given by The triangle OBiGi , connected by bars BiB and (7,(7 to fixed points B, G, „ OGfAi, „ G^G „ A2A „ G, A, OA,B„ „ A,A „ B,B „ A,B; or, that we have a triple generation of the same three-bar curve. It may be remarked that the intersection of the lines BB^ and CCi is the axis of the instantaneous rotation of the triangle OB^Gi, so that, joining this intersection with the point 0, we have the normal at to the locus; and similarly for the other two triangles. It of course follows that the intersections of BB^ and (7(7„ of (7Cj and AA^, and of AAi and BB^, lie on a line through 0, viz. this line is the normal at 0. The result depends on the following theorem : viz. starting with the similar triangles OB^Gu AjOG^, A^BtO, say, the angles of these are A, B, G, so that the sides are fc,(sin.4, sin 5, sin C), kj{sinA, sin 5, sin (7), kt(eixiA, sin 5, sin (7); then it follows that the sides of the triangle ABG are k (sin A, sin B, sin G), the value of k being given by the equation ifc« = jfc,« + A^» + A;,' -f- 2kjc, cos{X-A) + 2kA coa (Y-B)+ 2kA cos {Z- G),

ON THREE-BAR MOTION. where X, F, Z denote the angles A^OA„ BfiBi, CiOC^ respectively: whence, since A +B + C=7r, we have also X+Y + Z=7r. If therefore the angles X, Y, Z vary in any manner subject to this last relation and to the equation 4" = const., the triangle ABC will be constant in magnitude. There is no difficulty in proving the theorem. Writing 00 =j, and OB = a, also /L COCi = -^, and Z BOB^ = 4>, we have . , , , , , o, „ sin i/r sin Z , 6, + a, cos Z T* = Oi* + a,* + zoiaa cos Z, = , cos -Jr = — = , . , , , , - „ sin <f> sin F , Cj + a, cos F <r» = Ci» + o,* + 2cia3 cos F, —- , cosd»=-= = ; and then a^=r"-\{}-(j^- 2x0- cos {A -V -^ \{}- (f) = t" + o"' — 2to- cos A cos 4> cos •</r + 2to- cos A sin <^ sin i/r + 2to- sin A sin t^ cos <^ + 2t<7 sin A cos ^ sin ^ = 6i» + Ci= + ttj' + a^ + 26iaj cos Z + 2cia, cos F — 2 cos ^ (6j + a^ cos ^) (ci + a^ cos F) + 2 cos ^ . ajOj sin F sin Z + 2 sin ^ sin Z . a^ (ci + a, cos F) + 2 sin ^ sin F. 03(61 + 03 cos Z) = bi" + Ci" - 26,c, cos A + a.j» + O3' + 2ojja3 [— cos A (— sin Fsin Z+ cos Fcos Z) + sin A (sin Fcos Z + cos Fsin Z)] + 203 [(C] — 6, cos A) cos F + 61 sin J. sin F] + 2aj [(6j — Ci cos J.) cos ^ + Ci sin A sin Z]. We have here 61° + Cj' — 2i»iCi cos ^4 = Oi" : the second line is = -2aj05cos(ul + F+Z) which, by virtue of F+^=7r — X, is = 20303 cos (X — ^): and in the third and fourth lines Ci — 61 cos il = Oi cos 5, 61 sin ^ = o, sin B, bi — Ci cos A = ai cos C, Ci sin 4 = a, sin C whence these lines are 20301 cos (F — J5), 2ai02 cos (Z — 0) : the equation therefore is a» = Oi' + a,' + o,» + 20,03 cos (X -A) + 2a^ cos (F- 5) + 2o,as cos {Z - 0), which, putting therein for a„ Oj, a, the values A;isin J, k^sin A, Ar,8in4, and assuming OR above A:* = Ai" + Jks» + ifc,» + 2A;A cos (Z - .4) + 2^-3^ cos (F- J?) + 2A;iA;, cos (Z - (7), becomes a" = jfc'sin'il, or say a = fc8in^; and similarly h = k8mB, c = ksaxC, that is, {a, h, c) = A(8in^, sinJB, sinC), the required theorem.

ON THREE-BAR MOTION. Before proceeding to find the equation of the curve, I insert, by way of lemma, the following investigation: Three triads (A, B, G), {F, G, H), {I, J, K) of points in a line, or of lines through a point, may be in cubic involution ; viz. representing A, B, &c. by the equations x — ay = 0, x — hy = 0, &c., then this is the case when the cubic functions {x-ay){x-hy){x-cy), (x -fy) (x - gy) (x - hy), (x-iy){x-jy)(x-ky), are connected by a linear equation. Regarding I, J, K as given, the condition establishes between {A, B,

C) and (F, G, H) two relations: viz. these are (i_a)(i-6)(i-c) : (j-a)(j-b){j-c) : (k- a)ik-b)(k-c) = (i -/) (^ - 9) (i -h):U -f) (j - g) (j - h) : (k -f) {k -g){k- h). But, if if be regarded as indeterminate, then the condition establishes only the single relation (i - a) (i - b) (i - c) : (j - a) (j - b) (j - c) = (»• -/) (i -g){i-h): {j -f) (j - g) U - h), which relation, if i = 0, j = <x> , takes the form abc=fgh. When K is thus indeter- minate, we may say that the triads {A, B, 0), {F, G, H) are in cubic involution with the duad /, J. li A, B, &c. are points on a conic, then, considering the pencils obtained by joining these points with a point on the conic, if the cubic involution exists for any particular position of 0, it will exist for every position whatever of ; hence, considering triads of points on a conic, we may have a cubic involution between three triads, or between two triads and a duad, as above. Taking x = 0, y=0 for the equations of the tangents at the points I, J respectively, and z = for the equation of the line joining these two points, the equation of the conic may be taken to be xy—!? = 0, and consequently the coordinates of any point A on the conic may be taken to be x : y : z = a : - : \{}. It is then readily shown that a, y9, 7, f, g, h referring to the points A, B, G, F, G, H respectively, the condition for the cubic involution of (^1, B, C), {F, G, H) with the duad (/, J) is a0'y=fgh. And we thence at once prove the theorem, that there exists a cubic curve J^IgI(;FGH, viz. a cubic curve passing through /, and having there the tangent JA, having at / a node with the tangents IB, IG to the two branches respectively, and pa.ssing through the points F, G, H ; viz. that the triads {A, B, G), (F, G, H) being in cubic involution with (/, J) as above, there exists a cubic curve satisfying these 2 + 5+3, =10 conditions. In fact, the equation of the cubic curve is JJJcFGH; [y-^^{x-^z){x-'iz) 9 + — {(«- oir) {x - ^z) {x - riz) - {x -fz) {x - gz) {x - hz)] = 0,

ON THREE-BAR MOTION. where observe that second term is an integral function - 2I' (— Mx + Nz), if, for shortness, d M=a + ^ + y-f-g-h, N = ^y + ya + a^ - gh- h/-/g. In fact, the equations of the lines JA, IB, IC are y— = 0, x — ^z = 0, x — yz = 0, respectively, and we at once see that these lines are tangents at the points /, J respectively; moreover, at the point F, we have x, y, z—f, 51 !• Substituting these values, the equation becomes (j-^)(/-)8)(/-7) + |.(/-a)(/-y3)(/-7) = 0, viz. the equation is satisfied identically, or the curve passes through F; and similarly the curve passes through and H. In precisely the same manner there exists a cubic curve Ij^J^JcFGH; viz. this is IJ^JoFQH; {X -az)(y- 1) (y - J)

.f|(.-9(.-|)(.-5)-(,-|)(.-i)(.4))=». where the second term is an integral function, az' (— M'y + N'z) ; if, for shortness, a P y f g h a^y ' PI 7a «/3 gh hf fg a^y in virtue of the relation a^y=fgk; so that the second term is in fact =-^(— Gx+ Bz). py Writing for shortness J^, I^ to denote these two cubics respectively, we have four other like cubics, Jb{=J^IcIaFGH), Ib{= IbJcJaFGH), Jc{=JcIa^bJ^GH), and !(,{= IcJaJbFGH); the equations being Ja; (y-^)(^-yS^)(^-7^) + f {-Mx+Nz) = o, Jb\{} (y-|)(«-7^)(*–a2)+| {-Mx+Nz) = 0, Jc\{} {y–Mx-az){x-fiz)-\{}– {-Mx + Nz) = 0, \{} y/ 7 /.; ix-az)(y-'^{y-'^ + ^^{-Ny + Mz)^0, h\{} (^-7^)(y-~)(y-|)+^(-iVy + ii/^) = o.

ON THREE-BAR MOTION. We require the differences of the products I^Jaj IrJb, IqJc- We find IsJB = {»=-oiz)(.<e- ^z)(j^-l^)[y-^-^[y - -^{y -^^ + ^;^i-Mx + Nz){- Ny + Mz) ^^<r-Ny + M^) {y-1) (« - 7^) (« - o^) 7« z' , J(- i»f^ + Nz) {x - ^z) (y _ J) (y - i) ; let fl denote the sum of the two expressions in the first line. Similarly, we have 7eJc = n + ^ (-% + -A^^) (y - -) (a; - 0^) (a; - /3z) + ^(^-Mx^Nz){x-rfz)(y-'-){^y-'-^). We have thence IsJb - IcJc = Z-' j^ (»,- o^) (- Ny + ilfz) - (y - ^) (- Mx + iV^)J the factors in { } are respectively fl 1 BO that we have = (i - i) {,xy - z^) and {m - ^) {xy - z% IsJh - IcJc = [-^-^^i^-M)z^xy- z^f. The constant factor M IB U 7] (a = (S-f)-(|-f)'—- if P„ P„ P, denote respectively the functions N_ M N M N M /S*/ a ' 7a /9 ' a/8 7 Attending to the equation 0/87 ^fgh, it appears that we have D 1/ a ^ LX /I 1 1 1 1 1\{} with like values for P, and P,. C. IX. 71

ON THREE-BAR MOTION. We have thus I^J„ - I,.Ja = - (A - Ps) z' {xy - z^r. and similarly I„J„ -I^J^=- (P, - A) z^ {xy - z% IaJa - IbJh = - (-P. - P.) ^ (^ - zj. Any function IaJa + ^' (xy — z^f, where \{} is arbitrary, can of course be expressed in the form IaJa + (^+ Pi)z^ix!/ — z")-, where 6 is arbitrary, and therefore in the three equivalent forms lAJA+(0 + Pi)z^(xy-2j, lBJB+(S + Pd^(xy-z'y, IaJc+(0 + Ps)z^xy-zJ. We have z = 0, the line IJ : and xi/ — z^ = 0, the conic IJABCFOH. The equation I^J^+X2i'{an/ — z"y=0 may thus be written in the more complete form IA J„JcFOH . JA InlcFOH + \{} (IJy {IJABCFOHf = 0, and we hence see that it is the equation of a sextic curve, having a triple point at /, the tangents there being IA, IB, IC; having a triple point at J, the tangents there being JA, JB, JC ; and having a node (double point) at each of the points F, G, H. There are thus in all (6 + 3)4-(6 + 3) + 3 + 3 + 3, =27 conditions, and these would in general be sufficient to determine the sextic. The data are, however, related in a special manner; viz.

regarding the points /, J, F, 0, H as arbitrary, the lines IA, IB, IC, JA, JB, JC are not arbitrary, but satisfy the conditions that A, B are arbitrary points, and C a determinate point, on the conic IJABC. And the foregoing result shows that, this being so, there exists a sextic satisfying the foregoing conditions, but containing in its equation an arbitrary constant \{} or 0, and that the equation may be presented under the three forms IaJbJcPOH . JA IbIcFGH + {6 + P,) {IJf {IJABCFGHy = 0, &c., corresponding to the partitions A, BC; B, GA; C, AB o{ the three points A, B, C. In the case where /, J are the circular points at infinity, the conic IJABCFGH is a circle passing through the six points A, B, C, F, G, H; and the condition of the cubic involution of the triads {A, B, C) and {F, G, H) with the points (/, /) is easily seen to be equivalent to the following relation, viz. the sum of the distances (measured along the circle from any fixed point of the circumference) of the three points A, B, C is equal to the sum of the distances of the three points F, G, H. The sextic is a tricircular sextic having the three points A, B, C for foci, and having three nodes F, G, H, on the circle ABC, two of them being arbitrary points, and the third of them a determinate point on this circle. And it appears that there exists a sextic satisfying the foregoing conditions, and containing in its equation an arbitrary parameter.

ON THREE-BAR MOTION. I proceed to find the equation of the curve. Consider the curve (see Fig. 1, p. 552) as generated by the point 0, the vertex of the triangle OC\{}Bi, connected by the bars Gfi and BiB with the fixed points C and B respectively; and suppose, as before, CB = a, 0^0 = a^, BiB = a3, BiCi = ai, OCi = hi, OBi = Ci ; and draw as in the figure the parallelograms 0^00^0 and BiBBfi ; then may be considered as the intersection of a circle, centre G^ and radius C.2O, with a circle, centre B^ and radius B^O. Take zGfiB=d, /.B,BG = <^: the lines GG,_, BB^ are parallel to 00„ OjB, respectively, and consequently + <f>=-7r — A, a relation between the two variable angles 6, <f>. Taking the origin at G and the axis of x along the line GB, that of y being at right angles to it : the cooi-dinates of G« are (61 cos d, 61 sin 6), and those of iJ, are (a - Ci cos if>, c, sin (f>) ; the equations of the circles thus are (x — 6] cos 6y +(y — bi sin tif = a^, (x — a + Ci cos ^y + (2/ — Ci sin <^)^ = a^ ; whence + 26ja- COS 9 + 2b^ sin 6= a? + y"->r b,' - Oa', — 2ci {x — a) cos <^ + 2ciy sin tf>=(x — ay + y' + c{- — a^, which equations, writing therein for 6 its value ='ir — A — <f> and eliminating the single parameter <f>, give the equation of the curve. We in fact have - 26,a; cos (A +<f>) + 2% sin (-4 + <^) = a;' + y' + 61' - (h\{} — 2ci{x — a)cos<f) +2cjysm<p ={x-af + y*-\{}-c^-a^; or say these are P cos <^ + Q sin </) = R, P' cos </) + Q' sin ^ = R, where P = - 26,a; cos A + 26;jr sin A, P' = - 2ci (a; - a), Q = 26,a; sin 4 + 26iy cos j1, Q' = 2ciy, R= a? + ^^ + 6,^ - ((2-, i2' = (a:-a)» + y= + c,^-a,l The equations give therefore cos(^ : sine/) : -l = QR-Q'R : RF - R'P : FQ' - P'Q, whence (Qii' - e'ii)= + (RF - R'Py = (PQ' - FQy ; and it hence follows that the nodes are the common intersections of the three curves QR' - Q'R = 0, RF-RP = 0, PQ' - FQ = 0. 71—2

ON THREE-BAR MOTION. We have, retaining R and R' to denote their values, QR -Q'R^-i [(Re, - R\{} cosA)y- R% sin A . x], RP' -R'P=-2 [(Rci - R\{} cos A) {x-a)-\{}- R% sin ^ (y - a cot ^1)], PQ' - P'Q = - 4 6,Ci [x{x - a) -iry {y- a cot A)]. Observing that iZ = 0, iJ' = are circles ; the equation QR' — QR = is a circular cubic through the point x = Q, y = ; the equation RP" — R'P = 0, a circular cubic through the point x=a, y = a cot A ; and the equation PQ' — PQ = 0, a circle through these two points (and also the points a; = 0, y = a cot A; x = a, y = 0). Hence the first and third curves intersect in the point (« = 0, y = 0), in the circular points at infinity, and in three other points which are the nodes; viz. the curve has three nodes, say these are F, 0, H. The second and third curves intersect in the point (x = 0, y = acot A), in the circular points at infinity, and in the three nodes. As regards the first and second curves, it is readily shown that these touch at the circular points at infinity ; viz. they intersect in these points each twice, in the two finite intersections of the circles jR = 0, R' = 0, and in the three nodes. The three nodes F, G, H thus lie in the circle x(x — a) + y(y — a cot A) = 0, which passes through the points (x = 0, y = 0) and (x = a, y = 0), that is, the points G and B. Assuming 6= r „-, the circle also passes through the point x = bcosG, y = 6 sin C, that is, the point A of the figure. Thus the three nodes F, 0, H lie in the circle circumscribed about the triangle ABC. Writing, for greater convenience, R = a?+y^-e!', R' = a? + y^ -2ax-f\{} the nodes F, 0, H lie on the two curves Ciy(a^+2/"-e=)-6,sin A {x -\{}-y cot A)(a? + y" -2ax-p)=0, af + y^=a(x + y cot A). The first of these is [c,3/ - 6, sin A(x + y cot A)]{a^ + y^) + [6j sin A(x + y cot A)p - c,e=y] + 2a6, s\{}nA(x + ycotA)x = 0. We may combine these equations so as to obtain the equation of the triad of lines OF, CO, CH ; viz. multiplying the second and the third terms of the first equation (X^ -I. •J/3\{}* /»>'- 4. 7y2 by

„t/^,.„,^t A\{}a ^^^ ~l— ; . A\{} (each = 1 in virtue of the second equation), the a' (x + y cot Af a{x-\{}-ycotA) ^ " equation becomes divisible by a? + y": and, throwing this out, the equation is c^y — 6, sin A{x-\{}-y cot .4) + [6, sin 4 (. + j,cot ^)/' - c,e'<,]^.^^ + 26,a; sin A = Q,

ON THREE-BAR MOTION. where the first and the third terms together are = (ci — b^ cos A) y + b^x sin A, viz. this is =aisin B{x + y cot B). Hence, writing also in the second term Oj sin B for &i sin A , the equation is 1 Ce^ 1 {x + ycotAy(x + ycotB) + -^(x + ycotA)p-^^^^y'^{aP + f) = 0; or say this is (x sinA+y cos Ay (x aia B + y cos B) sin A sin B -^^^ i^{x sin A+y cos A)/' -^y'^iaf + f)=0; viz. there is a term in af + y-, and another term {x sin A + y cos A)' (x sin B + y cos 5). Suppose for a moment that the angles FBC, GBC, HBG are called F, G, H ; then the function on the left hand must be = ilf (a; sin ^— y cos F) (x sin G — ycos G) (x siuH -y cos H). Writing in the identity x = iy, we have (cos A +iBinAy (cos £ + j sin J?) = — ilf (cos F — i sin F) (cos G — i sin G) (cos H — i sin iT) and similarly, writing x = — iy, we have the like equation with — i instead of +i; whence, dividing the two equations and taking the logarithms, 4.A+2B = 2?«7r -F-G-H, which leads as before to the relation A' +B' + C = F' + G' + H'. Ip completion of the investigation, observe that M is determinately +1 or — 1 and that sin AainB \{}. . , . . -, c, ,) Ux sin A+y cos ^)/' - ^ ^I/[is the linear factor of M {xainF—y cos F) (x sin G—y cos G)(x8inH—y cos H) — (xsin A +y cos Ay{xsinB + y cos j5), which remains after throwing out the factor of + y". Calling this linear factor px + qy, we have . °'P. p =/'sin^, . ":\{} j, =/^cosA-'^e', Bin Asm B •' siaAmnB •' 6, or, as this last equation may be written, a'q ,. . sin (7 , . . —D =/" cos 4 - ^ f^e\{} sin il sm ij •' sm B Hence, writing a = A;8in.4, we have f^=-=-^n^ e^=-. ^(pcosA—qsinA): •' sin 5 smC^^ ^ '

ON THREE-BAR MOTION. substituting for/* and e* their values, we have - /t,> sin^C + kt^Bin" A =^^ + ^sinM, sin 5 - A,' sin' B + h^ sin' A = -.—~ (p cos A— q sin A), smC ^ ^ or, what is the same thing. ^-,—f + –rn = i A •—p • .n ip + S"l' A Sin B) sin' A sm' C sin' A sin B sm' (7 ^^ A– sin ^sin G sin ii/ sin A sin C . sin A sin ^ sin ' r-T—. + .-vn = > .— . , „ -, (» COS 4 — o sin ^ sm' A sin' B sm- A sin- 5 sm G ^^ ^ ' _ ^–^ sin (A - F) sin (^ - G) sin (^ - H) sin ^ sin if . sin A sin JS sin C which are the relations connecting k^, k.„ k^, when the foci and nodes are given. It is to be remarked that if, for instance, F = and G = A, then i, : k. : ks =

sin .4 : sin £ : sin C ; the nodes in this case coincide with the foci. A simple example is when A=B = G; the three triangles are here equal equilateral triangles. The general equations show that, if l^, l^, I3 are values of ki, k„ k, belonging to a given set of nodes and foci, then the values ki’ = li- + usiQ^ A, kfl’ = 1^- +usm^ B, k3^=ls- + usui-C (where u is arbitrary) will belong to the same set of nodes and foci. I write the equation of the curve in the form {{QR -Q:R) + i {RP’ - B’F)} [QE -QR-i {RF - R’F)\{} - (PQ’ - P’Qf = 0, where iQR’ - QR) + i (RP’ - R’P) = (A;, - Rlh cos A)i{x- a- iy) - R% sin A{x-i(y-a cot A)}. Calling /, J the circular points (« , a; + ii/ = 0) and (oc , x — iy=0), this is a nodal circular cubic having / for an ordinary point, but J for a node. Moreover, one of the tangents at J is the line x — iy = 0, that is, the line JB ; in fact, writing as before R = x- + y’- — e-, R’ = x- +y" — 2ax —/-, then, when x— iy = 0, we have R = -e-, R’ = —2ax—f’, and the equation becomes {-ce^ + b, cos A (2cuc +/’)] (- ia) + 6, sin .4 {2ax -f-/’) (ia cot -4) = ; viz. the term in x here disappears, or the three intersections are at infinity. The other tangent at J is the line x — a — iy = 0, that is, the line JG ; in fact, when a — a-iy = 0, that is, y = — i{x — a), we have R = 2aa; - a’ - e’, R’ = — a^ —/-, and the equation becomes {C (2cw; - a« - e’) -I- 6, cos .4 (a» +/’)} . -f 6, sin .4 (a’ +/’) . a (1 -f t cot .4) = 0,

ON THREE-BAR MOTION. viz. the three intersections are here at infinity. The tangent at / is the line x — b cos C + i(y — bsmC) = 0, that is, the line lA ; in fact, writing this in the form y = ix — ib (cos C + i sin O) = ix — iby, (if for a moment cos C + ismC= y, and similarly cos A + isinA = a, cos B + iainB= 0); then, y having this value, we find R = 2bxy - by - e-, R’=2(by-a)x- by -f% and the equation becomes = _|%_6y_/=; Ci{2byx-b"-y- -e’)^ 2c i (2a; — a — by) — bicoa A (— r, "" — b-’f — e’) — btsmA (— -^x— by — ^](2x+ ia cot A — by) = 0. The coefficient of x’ is here * 2.-(fc„.^.!). or, since &iC = &c„ this is = 2ibc.{y + ^), =0, in virtue of the relation A + jB + C = tt, giving a^Sy = — 1 : hence there is only one finite intersection, or the line IA is a tangent. The cubic in question QR’ -QR + i {RF - R’P) = . is thus a nodal circular cubic which it is convenient to represent in the form (IjJ„JrFGH) = 0; viz. this is a cubic, through / with the tangent IA, having / as a node with the tangents JB, JC, and through the points F, G, H. Observe that, if F, 0, H were arbitrary, this would be 2 + 5 + 3, = 10 conditions. The before-mentioned relation is, in fact, the condition in order to the existence of the cubic. Similarly the cubic QR’ - Q’R - i

(RP' - R'P) = is the cubic {JJ„I,FGH)^0. The circle PQ'-P'Q^O is the conic through /, J, A, B, C, F, G, H; or it may in like manner be written (IJABCFGH) = ; and we may write (/./) = 0, as the equation of the line infinity. The functions denoted as above contain implicitly

ON THREE-BAR MOTION. constant multipliers which give, in the equation of the three-bar curve, one arbitrary parameter—and the equation thus is {I^JsJcFQH) (JJJoFOH) - e (Ijy (IJABCFOHy = 0, a form which puts in evidence that /, / are triple points having the tangents lA, IB, IC, and JA, JB, JC respectively (whence also A, B, C are foci), and that F, 0, H are nodes ; viz. the result is as follows : Taking A, B, C, F, 0, H points in a circle, such that, Sum of the distances (being the angular distances from a fixed point in the circumference) oi A, B,G \{}s equal to the sum of the distances of F, G, H : then there exist the cubics {IaJb Jc^OH) = 0, {J^IglcFGH) = 0, and the sextic is as above. Writing for shortness il^J^JaFGH) = IA , {JJJcFGH) = JA then the above form is clearly one of three equivalent forms = laJo - e,£i\{} This implies an identical linear relation between the functions IaJa> ^b/b. IcJc\{} whence also U and Q? are each of them a linear function of any two of these quantities. I originally obtained the equation of the curve in a form which, though far less valuable than the preceding one, is nevertheless worth preserving; viz. the equation {QR - qny + (rp' - Rpy = {pq - rqy may be written (if - i^ - (?) (E" - F' - Q'-') - {RR - PP' - QQJ = 0, which equation, substituting therein for P, Q, R, P', Q, R their values, gives the form in question. Proceeding to the reduction, we have ^ ={!i? + ff-2 (6,' + a,') (a? + y') + (6,^ - a,J = (x' + y'-bi + <h){a? + y–b,-a^); R'^-P'»-Q'* = (x-a + y"+ c' - (h'Y - ici'ix-a +f) = {x-a + y')= – 2 (c» + a,») (x - a + y*) + (c," - a,>)« {x-a +i/'-Ci + (h){x-a +y^-Ci-at)-

ON THREE-BAR MOTION. But the reduction of RR' — PP'— QQ' is somewhat longer. We have RR' - PF -Q(^ = (*•= + y' + b,^ - ai) (x -a' + y'+ c/-' - a,') — 46iCi {xx — a + y^) cos A — 46,Ci ay sin A : and here 26iCi cos A = b^- + c,- — ft]-, 2x (x — a) + 2y'- = afl + y^ + {x — a)- + y- — a-, also h,CiSm A = OiPi, if jw, be the perpendicular distance of from the base Bfii. Hence the second line is — (bi- + c,- — Ui^) {x' + y^ + x — a +y- — a?') — ^aa^piy, and the whole is whence, finallj', we have f* RR' - PP' - QQ' = (jf + y' + a,– a/ - c,») (x-a +y' + a,- - a^' 6,-) + (a= + a,= - a.f - a^") (6,^ + c{' - Oi^) - 4>aa^p^y. Hence the equation of the curve is (ar + y"- -bi + ai)(af + y^-bi-a,)(x- a + y- - Ci + a,) (x -a + y–Ci- a,) - {(x' +y'- + a,' - «./ - cr) (x-a +f + a," - a,= - 6,-) + (a» + ai' - ai - ct,=) (6,-^ + c,' - a.^) _ ^aa^p.yY = 0, where pi is given in terms of the constants a„ 6„ Ci by the equation There are in the equation two terms, (ar' + y')', {x-a +y"y, which destroy each other, and the remaining terms are of the order 6 at most. Hence the curve is a sextic and it is, moreover, readily seen that the curve is tricircular. Assuming this, it appears at once that the lines x+iy = 0, x — iy=0 are tangents to the cui-ve at the two circular points at infinity. In fact, assuming either of these equations, we have r= + y= = 0, and the equation becomes (6,' - a,») (— -Zax + a:' — Cj + Os) (— 2ax + a' — Ci - Ua - {(oi" - a^' - ci') (- 2ax + a= + a,'-' - a,' - bi') + (a' + ar - a,' - a-/) {b,' + c,= - 0,'} - '^aa.p.yY = 0, a quadric equation. Hence there are on each of the two lines only two finite inter- sections, or the number of intersections at infinity is = 4 ; viz. the line is a tangent c. IX. 72

ON THREE-BAR MOTION. to one of the branches at the triple point. Similarly, the lines a; — a + ty = 0, x — a — iy = are tangents. Thus the points C and B are foci. It might with some- what more difficulty be shown from the equation that the point a; = 6 cos C, y = 6 sin (7 (where, as before, 6= .^p), viz. the point A of the figure, is a focus; but I have not verified this directly. It clearly follows, if we generate the curve by means of the triangle OAfit and the fixed points G, A. Hence A, B, G are a triad of foci, and the theorem as to the nodes is that these lie on the circle drawn through the three foci A, B, G. I prove in a somewhat different manner, for the sake of the further theory which arises, the theorem of the triple genei-ation ; for this purpose, constructing the foregoing Figure 2 (p. 553) by means of the three triangles OBfi^, OG^A^, OA^B:„ but without assuming anything as to the form or position of the triangle ABG, I draw through a line Ox, the position of which is in the first instance arbitrary, say its inclination to OO, is =v; and drawing Oy at right angles to Ox, I proceed, in regaixl to these axes, to find the cooixiinates of the points G, B. We have, for G, X — (u cos v + bj cos (V + Z), y = a„ sin y + 6, sin {v +Z); for B, a; = Ci cos (w + .4 + ^) + a, cos {v + A + Z + Y), y = Ci&\{}n{v + A + Z) + a-iSm {v\{}- A + Z Jf Y); or, writing for Y+Z the value tt — X, so that v + A+Z+Y=ir + v + A-X, the coordinates of B are x = Ci cos (v + A + Z) — a3Cos(v + A — X), y = Cisin (v + A + Z) - a-jsin (v + A — X). Taking the two values of y equal to each other, the equation to determine v is aj sin 1/ + 6i sin (v + Z) — c, sin (v + A + Z) + Ossin (v + A - X) = 0. We make the line Ox parallel to BG, so that, writing X — lu cos V +6, cos {v + Z), X — a = Ci cos (v + A + Z) — a3 cos (v + A — X), we have a - rtj cos u + 6, cos (v + Z) — c,cos{v+A+Z) + a3Cos{v + A - X), which determines the distance BG, = a. And moreover, writing y = O2 sin w + 61 sin (v + Z), = c, sin (v + yl + Z) — a, sin {v-\{}- A — X),

ON THREE-BAR MOTION. we have y as the perpendicular distance of from BG, and * and (a — a;) as the two parts into which BG is divided by the foot of this perpendicular. In the reduction of the formulse we assume that the three triangles are similar; viz. we write («!, k, Ci), {CU, b^, Co), (tta, 63, c) = ^-, (sin.4, sin 5, sinC), A;j(sin^, sin 5, sinC), &s(sin^, sin 5, sin (7); and we use when required the relation A -{ B + G= ir. The equation for v becomes ki sin (u - G + Z) + L sin v + k; sin {v + A - X) = 0, which may be written Z sin w — ilf cos u = 0, where L =L + i-, cos (Z -G) + ^'3 cos (X — A), M= -k,8m{Z-G)-\{}-k,8m(X-A); hence, putting k^ = fc,- + /;/ + k,^ + 2kjc^ cos (Z - .4) + 2^-^ cos (F - 5) + 2^1^:3 cos {Z - G), we have D + M^ = k', that is, •J'D-^M- = k, and therefore ^ sin V = Af, k cos v = Z, which gives the value of v; and then, after all reductions, kx = k^%m B cos G + k.^ sm A +k^.O \{}-kjC3»va. A COS {X — A) + ^/i [— sin B cos (1' f A)] + kjci [.sin (5 - A) cos {Z+ A) + 2 sin .4 sin 5 sin (.2^ 4- .4)], k(a — x)= ki' sin C cos B + k^- . + A;/ sin A + Lk^ sin A cos (AT — .4 + k^; [sin (C- ^) cos(r + ^) + 2 sin A sin C sin (F+ A)] + ^,io [- sin Gcoa (Z+A)], and Ay = A;,' sin B nin C+k^Jc, sin .4 sin (X - .4) + k^ki sin B sin (1'+ .4) + k^k^ sin Csin (Z + 4). The first and second equations give ka^k'sin A, that is, a = A'sin.4; and, similarly, b = ksm B, c = A; sin (7 ; viz. we have (a, b, c) = A;(sin .4, sin B, sin 0), or the triangle ABG is similar to the other three triangles, its magnitude being given by the foregoing equation for k'. These are the properties which give the triple generation. 72—2

ON THREE-BAR MOTION. Changing the notation of the coordinates, a!id writing (x, y, z) for the perpend- icular distances from on the sides of the triangle ABC, we have, as above, kx = ir,' sin BsinC + kjc^ sin A sin {X — A) + kjct sin 5 sin (F+ ^) + A;,AjSin CsiniZ + A), and therefore ky = k/ sin 5 sin C + A-^ sin A?.m{X + B} + k^kj sin B sin {Y-B) + A-,A-j sin G sin (Z + B), kz = k^ sin Csm A + kjc^ sin A sin {X + 0) + kjc^ sin B sin (F + C) + A:,^;(TM)sin C sin (^ — (7), values which give, as they should do, X sin A-\{}-ysinB + zsmC=k? sin A sin £ sin C. Taking {x, y, z) as simply proportional (instead of equal) to the perpendicular distances, then {x, y, z) will be a system of trilinear coordinates in which the equation of the line infinity is a; sin ^4. + y sin 5 + 2: sin (7 = ; and considering (x, y, z) as proportional to the foregoing values, and in these X, Y, Z as connected by the equation X +Y+ Z^ir and by the equation which determines Ar", the coordinates {x, y, z) are given as proportional to functions of a single parameter, so that the equations in effect determine the curve which is the locus of 0. But to determine the order, &c., the trigonometrical functions must be expressed algebraically ; and this is done most readily by introducing instead of X, Y, Z the functions cosX + isinZ, cosF + isinF, cos^ + isin^, =f, 7;, ?'; and we may at the same time, in place of ^, B, G, introduce the functions cos A +ism A, cos B + i sin B, cos (7 + i sin 0, = a, /3, 7. The relation X +Y + Z =Tr gives fj7^=-l; and similarly A + B+G = tt gives a/37 = -l. We have cos(Z-4) = j(| + |), isin(Z-^)=j(^-|), &c.; the equation k' = A;i" + &c. becomes A:^ = A;,^ + A-,^ + AV + kJc, (| + |) + kk Q + f) + ^^^' (7 + ?) or, as this may be written, (_ k^ + k,^ + 1^ + k,') + t,k, (I -«'??)+ kA (I - /Sr?) + A^ k, (^ - y^v) = 0. Also the value of x is proportional to

ON THREE-BAR MOTION. or, what is the same thing, to with the like expressions as to the values of y and z. Introducing for homogeneity a quantity as, viz. writing - , - , - in place of f , rj, ^, we have the parameters {?. '7. ?. «*>) connected by the homogeoeous equations (- /c' + k,- + L' + k,') «» + k, A.-3 (^^ - a„?) + tA (^^ - ;Sr|) + k, k, (^- - y^rj^ = 0, and the ratios of the coordinates are »^ + hk. (^ - ^) («,a, + f) + A-,L (7 - ^) (<« + ^^) + A..A.. (/3 - J) (r;« + f) + ^'.A;, (7 - ^) (f^ + 7?,) . Suppose, for shortness, these are x : y : z = P : Q : R. Observe that the form of the equations is fj;5'+(a' = 0, 11 = 0, and x : y : z-F : Q : B, where fi and F, Q, R are each of them a quadric function of the form ((o\{} a>^, cori, wif, Tjf, ff, f ??), the terms in ^, T)^, ^ being wanting. Treating (^, 77, f, ta) as the coordinates of a point in space, the equation ^rj^+m^^O is a cubic surface having a binode at each of the points (? = 0, o) = 0), (77 = 0, ft)=0), (f=0, a) = 0), and the second equation is that of a quadric surface passing through these three points; hence the two equations together represent a sextic in space, or say a skew sextic, having a node at each of these three points. The equations X : y : z = F : Q : R establish a (1, 1) correspondence between the locus of and this skew sextic. To find the degree of the locus we intersect it by the arbitrary line ax + by + cz = 0; viz. we intersect the skew sextic by the quadric surface aF +bQ + cR = 0. This is a surface passing through the three nodes of the skew sextic, and it there- fore besides iotersects the skew sextic in 12 — 2.3, =6 points. Hence the locus is (as it should be) a sextic.

ON THREE-BAR MOTION. I consider the point 17 = 0, 5' = 0, a> = 0, or say the point (1, 0, 0, 0), of the skew sextic. This is a node, and for the consecutive point on one branch we have 7] : f : 0) = me : le' : ne, where e is infinitesimal. The equation of the cubic surface gives Ivi + n" = 0, and the equation of the quadric surface gives kjc, . kjc^fq = 0, that -is, k3a> = ayki7], which, in fact, determines the ratio I : m; but it will be convenient to retain the equation in this form. For the corresponding values of (x, y, z) we have a=:y:z= ^•,(a-^) 1 + ^,(^,-1)5 ' .k,(a-^'ia>+h{y-^^7„ which, writing for k^a its value = ay^ji;, become a- : y : 2= (a–) y+(y–]-= 07-'^ + - ^ \{} aj \{} yj a ' a. a. ya. (a–)a/37+(7-l)^ :_„ + i_ay: + « (« - -j ay' + (7 - -] 7 : *V - 7' + 7' - 1. the last set of values being obtained by aid of the relation 0/87 = — 1 ; viz. we thus have that is, 1 , ^ 7 which are, in fact, the values belonging to one of the circular points at infinity. For the consecutive point on the other branch we should obtain in like manner X : y : z = y : —\{} : - , which are the values belonging to the other circular point at infinity; viz. the node (1, 0, 0, 0) of the skew sextic corresponds to the circular points at infinity. But, in like manner, the other two nodes (0, 1, 0, 0) and (0, 0, 1, 0) each correspond to the circular points at infinity, or say we have in the skew sextic the three nodes each corresponding to one circular point at infiuity, and the same three nodes each coiresponding to the other circular point at infinity; viz. we thus prove that each of the circular points at infinity is a triple point on the locus of 0. In order not to interrupt the demonstration, I have assumed the formulae which, in the system of coordinates defined by taking x, y, z proportional to the perpendiculai-s on the sides of a triangle ABC, or where the equation of the line infinity is xsinA+yamB + zsinC=0,

ON THREE-BAR MOTION. give the cii-cular points at infinity ; viz. writing cos A +ismA, cos B + i sin B, cos C + i sin C = a, /3, 7, the coordinates for the two points respectively are X : y : z = — I : 7-o ^^^^ x : y : z = — \{} : -:/8 = -:— l:a =7:— 1:- 7 a = y3:^:-l. =!:«:- 1, a ^ the three vahies for each point being equivalent in virtue of the relation a/S^ = — 1. This is, in fact, under a different i'orni, the theorem given in my Smith's Prize paper for 1875; viz. the theorem was: If \{}, /t, v are the inclinations to a fixed line of the perpendiculars let fall from an interior point on the sides of the fundamental triangle ABC, then, in the system of trilinear coordinates in which the co<jrdinates of a point P are proportional to the triangles PBC, PGA, PAB (or where the equation of the line infinity is x + y-\{}-z = 0), the coordinates of the circular points at infinity are proportional, those of the one point to e'^ sin {fi — v), e*'* sin {v — \{}), e'" sin (X, — /*), and those of the other point to e~'^ sin (/a — i^), e"'" sin (y — X), e"*" sin (\{} — /i). In the plane curve, the lines drawn from A, B, C to the circular points at infinity are To the one point. To the other point. From A, ay + z = 0, y + az = 0; „ B. ^z + a;=0, z+^x=0; „ C, yx + y = 0, a;+'yy = 0. Each of these lines, quot, tangent at a triple point, meets the curve in the circular point at infinity counted four times, and in two other points. The corresponding points on the skew sextic .should be a node counted twice, the two other nodes counted each once, and two other points. The proof that this is so would show that the points A, B, C are a triad of foci. There is also the question of the determination of the values of (^, rj, f, «) which correspond to the nodes of the plane curve. But I have not further pursued the theory. Addition.—Since writing the foregoing paper, I have found that the relation between the nodes and foci (sum of angular distances of the foci = sum of angular distances of the nodes) may be expressed in a different form ; viz. the triangle of the foci and the triangle of the nodes are circumscribed to a parabola (having its focus on the circle); and I have made in relation to the question the following further investigations: Considering a circle : and a parabola having its focus at K, a point of the circle then if, as usual, /, ./ are the circular points at infinity, we have UK a triangle inscribed in the circle and circumscribed to the parabola ; hence there exists a

ON THREE-BAR MOTION. singly-infinite series of in- and circumscribed triangles, so that, drawing from a point A of the circle tangents to the paiabola again meeting the circle in the points B and C respectively, BC will be a tangent to the parabola; or, what is the same thing, starting with the triangle ABC inscribed in the circle, we can, with the arbitrary point K on the circle as focus, describe a parabola touching the three sides of the triangle ABC; viz. the parabola described to touch two of the sides of the triangle will touch the third side. Taking, then, a circle radius ^k, and upon it the three points A, B, C determined by the angles 2a, 2/8, 27 respectively (viz. the coordinates of A are x, y = ^k cos 2o, ^ A sin 2a, &c.), and a point K determined by the angle 2ie (suppose for a moment the origin is at K), the equation of a parabola having K for its focus will be xr + y" = {x cos 26 + 1/ sin 20 — p)", or, what is the same thing, {x sin 2^-7/ cos 2^)- + 2p (x cos 20 + y sin 26) -f = 0, where 0, p are in the first instance arbitrary ; and the condition in order that (x + riy + f = may be a tangent is easily found to be P (?' + ¥) + 2f cos 20 + 27j sin 26 = 0. It is to be shown that p, 6 can be determined so that the parabola shall touch each of the lines BC, CA, AB. , Taking the origin at the centre, the equation of BC is X cos (/3 -f 7) + 1/ sin (/3 + 7) — i i' cos (f3—y) = 0, as is at once verified by showing that this equation is satisfied by the values x,y — ^k cos 20, JA; sin 2/9, and =^4 cos 27, ^ A sin 27. Hence, transforming to the point K as origin, the equation is [x + ^k cos 2k] cos (j8 + 7) -I- [y + ^i sin 2«] sin (/8 -I- 7) - ^k cos (/3 — 7) = ; viz. this is a; cos ()9 -(- 7) -H y sin (/8 -f 7) - P [cos (/9 - 7) - cos (/S + 7 - 2k)] = ; or, finally, it is a; cos (y9 -I- 7) -f y sin (/8 + 7) — k sin (« — 0) sin (« — 7) = 0. Hence the condition of contact with the line BC is p = 2ksm(K — 0) sin (k - 7) cos (20 — y9 — 7) and, similarly, the condition of contact with the line CA is p = 2k sin (k — 7) sin (k — a) cos (20 — 7 — a)

» ON THREE-BAR MOTION. viz. these conditions determine the unknown quantities p, 0. It is at once seen that we have 2^-/3- 7 = ^77 -(«-«), that is, 26 = ^Tr - k + ci + ^ + y; and then p = 2k sin (k — a) sin (k — /3) sin (k — y); from symmetry, we see that the parabola touches also the side AB. Suppose, next, F, G are points on the circle determined by the angles 2/", 2g; retaining p and to denote their values, 2) = 2k sin (k — a) sin (« - yS) sin (« — 7), and 26 = ^tt — «+a+/3 + 7, the condition, in order that FG may be a tangent, is p = 2k sin (« -/) sin (« - ^r) cos (2^ -f-g); viz. determining A by the equation a + + y=f+ff + h, this is p = 2t sin (« —/) sin (« — 51) sin (k — h), or, what is the same thing, sin (« — a) sin (/c — yS) sin (« — 7) = sin (« —y) sin (« — g) sin (« — A) viz, this equation, considering therein h as standing for a + yS + 7 —f—g, is the relation which must subsist between f and g, in order that the line FG may be a tangent to the parabola. And then, h being determined as above, and the point H on the circle being determined by the angle 2h, it is clear that the lines GH, HF will also be tangents to the parabola; viz. FGH will be an in- and cii-cumscribed triangle, provided only f, g, h satisfy the above-mentioned two equations. The latter of these, if / g, h satisfy only the relation a + ^ + y=f+g +

h, serves to determine k; and then, 6 and k denoting as above, the equation of the parabola is a~' + y^ = {x cos 26 + y sin 26 - pf and it thus appears that the condition in question, a + ^+y=f+g + h, is equivalent to the condition that the triangles ABC, FGH shall be circumscribed to the same parabola. It is to be remarked that the distances KA, KB, &c. are equal to A; sin (k - a), k sin (« — /3), &c. ; hence the condition sin (k — a) sin (/c — B) sin (« — 7) = sin (k —f) sin {k — g) sin {k - h) becomes KA.KB.KC = KF.KG.KH; viz. the focus if is a point on the circle such that the product of its (linear) flistances from the foci A, B, G is equal to the product of its (linear) distances from the nodes F, G, H. c. IX, 73

ON THREE-BAR MOTION. It is to be remarked that the foregoing equation in k determines a single position of the point K ; viz. it determines tan k, and therefore sin 2/c and cos 2/c, linejirly. The equation is, in fiwit, a cubic equation in tan «, satisfied identically by tanK = i and tan k= —i, and therefore reducible to a linear equation. Write for a moment tan /c = w, and (tan AC — tan a) (tan k - tan y9) (tan k — tan 7) = w' — pw* + 50) — r, (tan K — tan/) (tan « — tan g) (tan k - tan h) = to' —p'a>- + 9'a) — r ; also M = cos/cos ^f cos h -T- cos a cos ^ cos 7. Then we have «' — pto' + g'ft) — r = ilf (o)' — ^w' + 5'a) - r'), where ^ = r + 3/(r'-p), g=l + if (g^'-l). Substituting these values, the equation becomes o,3 _ ru)" + a —r = M {(o"— r'co^ + w - r'), viz. dividing by m^ + 1, this is co — r = M(<o — r'); or substituting for r, r', M their values, (cos a cos yS cos 7 — cos/ cos ^r cos h) tan /c = (sin a sin /3 sin 7 - sin/sin ^r sin h), which is the value of tan*, and then . „ 2 tan K -. 1 — tan' k Sm ZK = :;——- —- , COS '2k = :; ^ ., . 1 + tan- K 1 -f tan" It may be further noticed that, if the parabola intersect the circle in a point L, and the tangent at L to the parabola again meet the circle in M, then, if 2i, 2m are the angles for the points L, M, we have I, m, m for values of / g, h, whence I, m are determined by the equations I + 2m = a + /8 + 7, sin (« — Z) sin- {k — m) = sin (k — a) sin (k — /3) sin {k — 7) but as the circle intersects the parabola not only in two real points, but in two other imaginary points, there is no simple formula for the determination of I and m. To determine the linkage when the nodes are given, suppose that, in the generation by 0, the vertex of the triangle OBiCi, we have at the node F: then, if t, a are the distances of G, B from the node in question, we have, as in the memoir, (6,= -I- T» - Oj") c,<r = (ci" -t- <T- - 0,=) 6,T,

ON THREE-BAR MOTION. that is, (61" - a,') CO- - (c,^ - a/)

61T + <TT (CjT - 6,0-) = 0, or, what is the same thing, (61= — a^-) ca — (Ci" — tta") br + ar (ct — ba) = 0. Suppose, as in the figure, that F is between B and A; then, if AF=p, we have CT = 60- + ap, and the equation becomes (6,« - aa^) CO- - (c,'^ - 03') 6t + a/wrr = 0. Similarly, if, as in the figure, G is on the other side of A, that is, between A and C, and if /»', <r', t' be the distances A G, BG, CG, then ba' = ct' + ap', that is, ct' — ba = — ap', and the corresponding equation is We hence find But we have that is, ahio, and (6,* - a./) ca' - (Ci» - a^) br' - apaW = 0. {b{- - a,") c (<7t' - (t't) + aW {pa + pV) = 0, (ci'-' — a,^) 6 (o-t' — ct't) + aaa (pr + /j't') = 0. BG.CF=BF.GG + BC.FG, a'r = ar' -f a . FG, or ai' — o't = — a . /^G pa + pV = .4/'. £i' + ^(? . BG, = i^'G . C//, = FG . t", pr+p'r' =AF.GF+AG.CG, = FG . BH, =FG.a", as may be shown without difficulty, p", a", t" being the distances AH, BH, GH. Hence the equations become c (6,' - a^') - tt't" = 0, h (c' - a/) - aa'a" = 0, showing that, the foci being as in the figure, b^^ — ai and Ci° — a^ are each of them positive; viz. that, in the generation by the triangle OC^Bi, the radial bars a^, a^ are shorter than the sides 6„ c, respectively. Substituting for 6„ &c. the values k^sinB, &c.; also, instead of A^, k^, k„ introducing the quantities Xj, X^, \{}.j, where ki, k«, ki = \{}i&\{}nA, XiSmB, A^sinO, these equations become c (\{},^ - V) sin' A sin" i^ = tt't", b (X,» - \{}^) sin' ^ sin= C = aa'a" ; or, as these may be written, putting for shortness Jlf = sin .4 sin B sin G, M'k' (X,» -W) = c tt't", M'k'i\{}^-\{}') = baa'a". 73—2

ON THREE-BAR MOTION. All the quantities have so far been regarded as positive, and the formulae are applicable to the particular figure ; but, to present them in a form applicable to any order of the nodes and foci, we have only to write the equations in the forms M' (X,» - V) = *' sin (o - /9) sin (/- 7) sin (g - 7) sin (h - 7), M^ (X,» - V) = Ic' sin (a - 7) sin (/- /3) sin (g - p) sin {h - ff) And these may be replaced by the system M-' (V - V) = ^ sin (/3 - 7) sin (/- o) sin (g - a) siu {h - a), M' (V - V) = i^ sin (7 - a) sin (/- ff) sin (g - ;8) sin {h - /8), M' (V - V) = it" sin (a - /3) sin (/ - 7) sin (jr - 7) sin (h - 7), since the first of these equations is implied in the other two ; and then, reverting to the original form, we may write M^l<f(K?-\{}:?) = BC.FA.OA.HA, M^t- (V - Xr) ^CA.FB.GB. HE, M^h? (\{},' - V) = AB.FC.GC . HC. it being understood that the distances BC, FA, &a, which enter into these equations, are not all positive, but that they stand for isin(/3 — 7), k8m(f—a), &c., and that their signs are

to be taken accordingly. Or, again, these may be written BC {c^ - b./) = FA.GA.HA, CA (a,' - cr) =FB.GB. HB, AB{h('-a.?) = FC .GC .HC, where the signs are as just mentioned. We may say that ±(02'- — 63') is the modulus for the focus A ; and the formula then shows that this modulus, taken positively, is equal to the product of the distances FA, GA, HA of A from the three nodes respectively, divided by BC, the distance of the other two foci from each other.

624] 624. ON THE BICURSAL SEXTIC. [From the Proceedings of the London Mathematical Society, vol. vii. (1875—1876), pp. 166—172. Read March 10, 1876.] In the paper " On the mechanical description of certain sextic curves," Proceeditigs of the London Matlienuiticul Society, vol. IV. (1872), pp. 105—111, [504], I obtained the bicursal sextic as a rational transformation of a binodal quartic. The theory was in effect as follow.s: taking fi, P, Q, R, each of them a function of \{}, fi of the form (*5l, X.)"(l, /*)*, and considering (\{}, m) -^ connected by the equation fl = 0, (viz. X, /x. being coordinate.s, this represents a t)inodal quartic), then, if we assume x : y : z = P : Q : R, the locus of the point {x, y, z) is a curve rationally connected with the binodal quartic, viz. the points of the two curves have with each other a (1, 1) correspondence; whence the locus in question, say the curve U=0, is bicursal. The degree is obtained as the number of the intersections of the curve by an arbitrary line, or, what is the same thing, the number of the variable intersections of the corresponding \{}/*-curves n=0, aP + 0Q + yR=O, viz. each of these being a quartic curve having the .same two nodes, the nodes each count as 4 intersections, and the number of the remaining intersections is 4.4 — 2.4, =8, and thus the curve 6'=0 is in general of the order 8. But if the curves fl = 0, P = 0, Q = 0, R = have (besides the nodes) k common intersections, then these are also fixed intersections of the two curves fl = 0, aP + 0Q + yR = 0, and the number of variable intersections is reduced to H — k: we have thus H — k as the order of the curve U= 0. In particular, if k = 2, then the curve is a bicursal sextic. The theory assumes a different and more simple form if, in the several functions n, P, Q, R, we suppose that the terms in X-, /a* are wanting. The curves H = 0, P = 0, Q = 0, R = are here cubics having two common points; the curve 17=0, qua

624. On the Bicursal Sextic.

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(1875–1876), pp. 166–172.
Read March 10, 1876.]

ON THE BICURSAL SEXTIC. rational transformation of the cubic ft = 0, is still a bicursal curve; but its order is given as the number of the variable intersections of the cubics n = 0, aP + 0Q+'YR = O, viz. this is =3.3-2, =7. But if the curves n = 0, P = 0, Q = 0, ^ = have (besides the before-mentioned two common points) k other common points, then the number of the variable intersections is =7 — k: and this is therefore the order of the curve U=0. In particular, if k=l, then the curve is a bicursal sextic. And, in the present paper, I consider the binodal sextic as thus obtained, viz. as given by the equations (1 = 0, X : y : z = P : Q : R, where fl = 0, P = 0, Q = 0, R = are cubics, having (in all) three common points. The bicursal sextic has in general 9 nodes ; but 3 of these may unite together into a triple point: this will be the case if, in the series of cui-ves aP +^Q + yR = 0, there are any two curves which have 3 common intersections with the curve n = 0. (Observe that we throughout disregard the 3 common points of the curves 12 = 0, P = 0, Q = 0, jR = 0, and attend only to the 6 variable points of intersection of the curves 12 = and aP + ^Q + yR=0,— the meaning is, that there are two curves of the series such that, attending only to the 6 variable intersections of each of them with the curve f2 = 0, there are three common intersections.) For, supposing the two curves to be aP + ^Q + yR = and a'P + ^Q + y'R = 0, then any curve whatever aP + 0Q + yR+e(a'P + 0Q + y'R) = O has the same three intersections with the curve 12 = 0, say these are the points Ai, A3, Af, the coordinates of which are independent of 0. Hence the line {ax + 0y+yz) + (a'x + 0y + y'z) = intersects the curve U^=0 in six points, three of which, as corresponding to the points Ai, At, A3, are independent of 6, viz. they are the same three points for any line whatever of the series; and this means that the curve U = has at the point (ax + ^y + yz = 0, a'x + ^'y + y'z = 0) a triple point; and that to this triple point correspond the three points Ai, A^, A3. We may, in the series of lines ax + ^y + yz+6 {a'x+ fi'y + y'z) = 0, rationally determine 80 that one of the three variable points of intersection shall correspond to .4„ A.;„ or A.i ; viz. must be such that the curve aP + ^Q + yR + {a'P

+ /3'Q + y'R) = shall touch the curve 12 = at one of the points A^, A.j, A3. The three lines thus determined are the three tangents to the curve at the triple point : and the three branches may be considered as corresponding to the three points Ai, A„ A3, respectively. There is no loss of generality in assuming that the triple point is the point (x=0, ^ = 0); the condition then simply is that the curves P = 0, R = shall have three cximmon intersections with the curve f2 = ; and the tangents at the triple point are x+ 0z = O, being so determined that one of the three variable points of inter- section shall correspond to one of the three points .4„ A^, A3: in particular, if this ifl the case for the line x = 0, then this line will be one of the tangents at the triple point.

ON THE BICURSAL SEXTIC. The bicursal sextic may have a second triple point, viz. three other nodes may unite together into a triple point. The theory is precisely the same : we must have two other curves aP + /SQ + 'yR = 0, a'F + ^'Q + y'E = 0, having with the curve ft = three common interaections Bi, B.„ B^: there is then a second triple point {ax + fiy + <y2= 0, a'x + ^'y + 7'^ = 0) and, to find the tangents at this point, we must determine 6 so that one of the variable points of intersection of the line ax+^y + yz+6 {a'x + fi'y + y'z) = with the sextic shall correspond with £„ B«, or B^; viz. must be such that the curve al^-i- ^Q + yR+e (a'P + yS'Q + y'R) = shall touch the curve fi = at one of the points Bi, Bj, B3. In particular, if, as before, the curves P = 0, R = have three common intersections with the curve = 0, and if, moreover, the curves Q = 0, R = Q have three common intersections with the curve 12 = 0, then the bicursal sextic will have the two triple points (a; = 0, ^ = 0) and (y=0, 2 = 0); and it may further happen that the line x = is a tangent at the first triple point, and the line y =0 a, tangent at the second triple point. The sextic may in like manner have a third triple point, but this is a special case which I do not at present consider. I write for greater convenience - , - in place of \{}, u., so as to make il, P, Q, R V V each of them a homogeneous cubic function of (X„ fi, v); and I give to these functions, not the most general values belonging to a bicursal sextic with two triple points, but the values in the form obtained fijr them, as appearing further on, in the problem of three-bar motion ; viz. the equations £1 = 0, x : y : z = P : Q : R are respectively taken to be . V (hv\{} +/\{}») + 11(^1^ + evK + gK?) + /t' (/v + h\{}) = 0, X : y : z=\{}fi {a\{} + bfi) : 1^ (c\{} + d/x.) : X/j-v. The four curves n = 0, P = 0, Q = 0, R = have thus

the three common intersections (^4 = 0, i'=0), (i' = 0, X, = ()), (\{} = 0, fj, = 0), represented in the figure by the points A, B, C ; the curve drawn in the figure is the curve n = 0, and the points F, G, U are the third points of intersection of the cubic with the lines BG, CA, AB respectively. The equation P + 6R = is here \{}/t (a\{} + V + ^^) = 0, which intersects £1 = in the points C, A, G, G, B, F, and the three intersections by the line a\{} + bfi + 6i> = 0;

ON THE BICUR8AL 8EXTIC. viz. excluding the fixed points A, B, C, the six intersections are C, F, G, and the three intersections by the line. Hence, of the six intei-sections, we have C, F, G independent of 0, or we have (a; = 0, z = 0) a triple point, say /, corresponding to the three points C, F, G, viz. these are the points (X, /i. v) = {0. 0. 1), (0, g, -/), (h, 0. -/), (C, F, G). The equation Q + 6R = is i' {cXv + dfiv + 6\{}fi) = ; viz. the line v = meets fl = in the three points A, B, H, and the conic c\{}v + dfiv + 6\{}ix = meets fi=0 in the points A, B, C, and three other points: hence, rejecting the points A, B, C, the six points of intersection are the points A, B, H, and the three variable points of inter- section by the conic ; or we have {y = 0, ^ = 0) a triple point, say J, corresponding to the three points A, B, H, viz. these are the points {\{}, ^, v) = {\{}, 0, 0), (0, 1, 0), {h, -g, 0), {A,B, H). To find the tangents at the triple point 7, these are x+ 6z = Q, where ^ is to be successively determined by the conditions that the line aX + bfi+ 6v = shall pass through the points 0, F, G* ; viz. we thus have ^ = 0, x = 0, the tangent conesponding to the point C, (0, 0, 1), 0= + ^j, fx+bgz=0, „ „ „ „ F,iO,g,-f), 0^ + j, fx + ahz = 0, „ „ „ „ G,(h,0,-f). And similarly, at the triple point J, the tangents are y + 0z = O, wheie is to be successively determined by the conditions that the conic cvX + dv/j. + 0\{}fjL = shall pass through the point H, and shall touch the cubic at the points A, B; viz. we thus have ^ = 0, y = 0, the tangent corresponding to the point H, {h, — g, 0), S = j, fy + cgz = o, „ „ „ „ ^,(1,0,0), e=j, fy+dhz^a, „ „ „ „ B.{o,i,o). The two last values of are obtained by the consideration that the equations of the tangents to fl = at the points A, B respectively, are gfi+/v = 0, h\{}+fp = 0, where X, fi, v are current coordinates of a point on the tangent : it may be added that the equation of the tjingent at the point C is h\{} + gfj, = 0. * Obuerve the somewhat altered form of the condition : 9 is to be determined so that the cubic \{}n{a\{} + bn + 6ii) = shall touch the cubic fi = at one of the points C, F, G : but, as the first-mentioned cubic breaks up, and the component curve a\{} + bfi + 0y = O does not pass through any one of these points, this can only mean that shall be so determined as that the line shall pass through one of these points, viz. that there shall be at the point, not a proper contact, but a double intersection, arising from a node of the cubic X/t(aX + fc^ + ei') = 0. And the like case happens for the other triple point; viz. there the cubic y{cy\{} + dvfii + 6\{}ii) = Q is to touch the cubic fi=0 at one of the points ^1, B, H ; the component conic e»\{}+diiii, + e\{}iJ.=Q passes through the points A and B but not through H; hence the conditions for 6 are, that the conic shall touch the cnbio at A or B, or that it shall pass through H.

ON THE BICURSAL SEXTIC. The three-bar curve may be represented by means of a system of equations of the last-mentioned form, viz. x : y : z=\{}fi, {aX + bfj.) : v' (c\{} -t- dfi) : Xfiv, where \{}, fi, v are connected as above ; or, taking X, Y as ordinary rectangular coordinates, x, y, and z are here the circular coordinates - = X + iY, - = X—iY, and z=l; and the parameters - , - denote like functions cos + i sin 6, cos <f> + i sin <b of angles which are the V V r CD inclinations of two bars to a fixed line. Using, for convenience, Figure 2 of my paper on Three-bar Motion, (p. 553 of this volume), the curve is considered as the locus of the vertex of the triangle OC\{}Bi, connected by the bars 0,(7 and BjB with the fixed points B and C respectively; and we have CGi = a.2, Of, =6„ OBi=Ci, (7,5, = a„ B^B^a^. Also, to avoid confusion with the foregoing notation of the present paper, instead of call- ing it a, I take BC^a^: the angle OGi-Bi is = C„ and cosC, + isinCi is taken = y. Hence, taking the origin at G, the axis of X coinciding with GB and that of Y being at right angles to it : taking also 6, <f), i/r for the inclinations of CO„ C,£„ and BiB to GB, we have a., cos d + a, cos <f> — a = — u^ cos \{}}r, o^ sin ^ + a, sin <f> = Og sin i^ ; viz. writing cos ^ -)- i sin ^ = \{}, cos <f> + i Bin <f> = fi, these give a.j\{} + Uifi — tto = — Ha (cos y(r — i sin •^), "i c + «i «» = — flj (cos y + I sm yjr), that is, («„\{} + ff,/t — a) f ftj - -h a, «o j — a-j- = ; VIZ. (a," + a,' -1- a./ - a,=) + «,aj (^ + "j - "oOs i^ + yj- o^o^i (/* + -) = 0, for the relation between the parameters \{}, /i. And then X = 03 cos + bi cos {<f> + Gi), Y = Hj sin -1- 6, sin (<^ + 0,) viz. if X, y = X +iY, X —lY, then 1 , 1 y = flj - -t- 62— which equations determine the coordinates (a;, y) in terms of the parameters \{}, fi con- nected by the foregoing relation. C. IX. 74

ON THE BICUR8AL 8EXTIC. Writing for homogeneity - , - in place of \{}, ft, and - , - in place of x, y, the equations become (a,'

+ a,' + «/ - a^) \{}fiv + a, cu (V + /t») v - a,a,/i (i/^ + X-') - rt„a,X (/*' + 1^=) = 0, and X : y : z = (ai\{} + b.,y,ti)XiJi : f ~ \{} + a.,/i) >/- : Xfiv. Comparing with the foregoing equations e \{}fiu +/(X' + ti^)v + g(i^ + X') fi+ h(ijL^ + V-) \{} = 0, and X : y : z = (a\{}+ hfi) X/i : (cX -t- dfi) v- : Xfiv, the equations agree together, and we have /= + Oitt-j, 9 = -Woaa, /t = -ttoCt„ rt = fl2, 6 = ti7.. c = 6. 7i The tangents at the triple points thus are x=0, aiX — aobiyiZ=0, X — a„z = 0, a,y z = 0, 7i ?/ - «,0 = ; viz. restoring the rectangular coordinates, and for y substituting the value cosC+ism 0, for a« writing a, and taking b= ' , we have i: + tF=0, X-iF=0, Z + iF= 6(cos C + isinO), X-iY^ b (cos C-isin C), X + iF= a„, JV — jF = a„ viz. the first two intersect in the point (0, 0), the second two in the point (6 cos (7, 6 sin (7), the third two in the point (a, 0): the first and third of these are the points B and C, the second of them i.s the point A of the figure ; viz. the formuhe give the point A, forming, with B and C, a triad of foci.

625] 625. ON THE CONDITION FOR THE EXISTENCE OF A SURFACE CUTTING AT RIGHT ANGLES A GIVEN SET OF LINES. [From the Proceedings of the London Mathematical Society, vol. Vlll. (1876—1877), pp. 53-r57. Read December 14, 1876.] In a congruency or doubly infinite system of right lines, the direction-cosines a, /8, 7 of the line through any given point («, y, z), are expressible as functions of X, y, z; and it was shown by Sir W. R. Hamilton in a very elegant manner that, in order to the existence of a surface (or, what is the same thing, a set of parallel surfaces) cutting the lines at right angles, adx + ^dy + '^dz must be an exact differential : when this is so, writing V = I {adx + ^dy + '^dz), we have V = c, the equation of the system of parallel surfaces each cutting the given lines at right angles. The proof is as follows :—If the surface exists, its differential equation is adx + ^dy + <^dz = 0, and this equation must therefore be integrable by a factor. Now the functions a, ^, 7 are such that a' + /3^ + 7"= 1, and they besides satisfy a system of partial differential equations which Hamilton deduces from the geometrical notion of a congruency ; viz. passing from the point {x, y, z) to the consecutive point on the line, that is, to the point whose coordinates are x + pa, y + p/S, z+ py (p infinitesimal), the line belonging to this point is the original line; and conse- quently a,)8, 7, considered as functions of x, y, z, must remain unaltered when these variables are changed into x + pa, y + p^, z + py, respectively. We thus obtain the equations da doL da _ dtc dy dz ' dx dy dz ' dx dy dz 74—2

625. On the Condition for the Existence of a Surface Cutting at Right Angles a Given Set of Lines.

[From the *Proceedings of the London Mathematical Society*, vol. viii. (1876–1877), pp. 53–57.
Read December 14, 1876.]

ON THE CONDITION FOR THE EXISTENCE OF A Combining herewith the equations obtained by differentiation of a-' + yS"- + 7= = 1 , viz, — 4-S—+ ^ =0 dx dx dx ~ ' da ndB dy „ <^d,^^d,- 'ydl-" da . ^rf/8 dfy . and subtracting the corresponding equations, we obtain three equations which may be written . o . _ <^/3 dy d^ da da rfy3 dz dy ' dx dz' dy dx' or, what is the same thing, d^ _dy dy _da da ^dff _, j.o 7. dz dy' dx dz' dy dx ' ' ' " and, multiplying by a, /S, 7, and adding, k - (^ - ^^ a (^ - ^"^ (— - ^^\{} \{}dz dy) \{}dx dz) \{}dy dx)' We thus see that, if the function on the right-hand vanish, then k = 0, and conse- quently also d^ _dr/ d/y '^^ da d^ 1, _ n dz dy' dx dz' dy dx ~ ' viz. if the equation adx + 0dy + ydz = be integrable, then adx + 0dy + ydz is an exact differential ; which is the theorem in question. But it is interesting to obtain the first mentioned set of differential equations from the analytical equations of a congruency, viz. these are x = viz + p, y = nz + q, where m, n, p, q are functions of two arbitrary parametei-s, or, what is the ssxme thing, p, q are given functions of vi, n ; and therefore, from the three equations, m, n are given functions of x, y, z. And it is also interesting to express in terms of these quantities m, n, considered as functions of x, y, z, the condition for the existence of the set of surfaces. We have MRU ^ m n 1 , n r, i—— a, yS, 7= p, p, p, where ii = VI -|- w* + «'; and thence without difficulty / d , _ d , d\{} 1 f/i ^s f dm dm , dm\{} (dn dn , dn\{}\{} ('£-4/^S-i[- (.) - (.)].

SURFACE CUTTING AT RIGHT ANGLES A GIVEN SET OF LINES. SO that the required equations in a, /8, 7 will be stitisfied if only dm dm dm wt j^ + « -7- + -y = 0, dx ay dz dn , dn dn . m-r- +n-j- +-T- =0, oar dy dz and it is to be shown that these equations hold good. Writing for shortness dp = Adm + Bdn, dq = Cdm + Bdn, the equations of the line give _ dm . dm „ dn dx dx dx' dm , dm „ dn = z J- + A , +B ^ , dy dy dy dm . dm r, dn — m = z ,~+A J + a -jf , dz dz dz dn ^ dm , „ d)i = z J +C J- +B -J- , (Ix ax ax , _ dn „ dm „ dn dy dy dy' dn „ dm . r. d>i n = z J- + C 'j~ + D~j-;

dz dz dz or, writing so that identically dm dn dm dn dm dn dm dn dm dn dm dn ' ^' ~ dy dz dz dy ' dz dx dx dz ' dx dy dy dx' dm dm dm _ - dx dy dz ' ^ dn dn dn „ X , + M J +Vj- =0, dx dy dz then in each set, multiplying by X, fi, v and adding, so as to eliminate A, B, G, D, we find X — mv = 0, fi — nv = 0. Substituting these values of X, /a in the last preceding equations, v divides out, and we have the two equations in question. The foregoing equations give further .„„„ 1 dn 1 dm A, B, C, D = -z + - J-, J-, V dy V dy 1 dm V dx -3 + 1 dm V dx Taking for a, /9, 7 the before-mentioned values, we find da dfi _1 (dm dn\{} m f dm dn\{} _]}_ (dm dn\{} dy~ dx~ R\{}dbj~dx)~W r' rf7 "^ " dy) R" V' dx'^"' dx) 1 (., „, dm „ , ,.dn , fdm dn\{}[and similarly, but using the equations dm dm dm . dn , dn . dn dx dy dz dx dy dz

ON THE CONDITION FOR THE EXISTENCE OF A to eliminate the coefficients ^ , j- which in the first instance present themselves, we find dy _da _ n \{} ^,) di'dz'R'X " " " J- whence, multiplying by 7, a, /3, and adding, fd0 dy\{} ^/dY_da\{} (doc _d^\{} "\{}dz~dy)^^\{}dx dzJ^'^Kdy da;) 1 {.. , „.dni „ , .,^dn , /dm dn\{} 1 + m' + n- (^ ' dy ^ ' dx \{}dx dy] ' or we have „ „, dm „ .,v dn /dnt d?i\{} as the condition for the existence of the set of surfaces. It is clear that the condition is satisfied when the lines are the normals of a given surface: seeking the surfaces which cut the lines at right angles, we obtain the parallel surfaces; and we are led to the theorem that any parallel surface is the locus of the extremity of a line of constant length measured off from each point of the surface along the normal—or, what is equivalent thereto, the parallel surface is the envelope of a sphere of constant radius having its centre on the surface. I will verify the theorem for the case of the ellipsoid. Taking X, Y, Z as the X- r» Z^ coordinates of a point on the ellipsoid —^ +-Tr +^;; = 1. and x, y, z as current co- ordinates, the equations of the normal are a' 6° c° •^ (a; - Z) = ^(2/ - F) = ^(^ - Z), (= A, suppose). We have therefore and thence ' ' a'-i-X' 6'-t-\{}' c= + \{}' gV fry c"^° _ 1 an equation which determines X as a function of x, y, z. X Y Z The direction-cosines a, fi, y of the normal are proportional to —„ ji, -^, that

SURFACE CUTTING AT RIGHT ANGLES A GIVEN SET OF LINES. is, to , ,—^— , —— , and the equation a-+/8-+7-=l then determines their absohite magnitudes: the equation adx + fidy + •ydz = dV thus is xdx ydy zdz +#^ + ^/, a» + \{} i^ + X c» + \{} „, ^^ J^r- = dV, a? v" z^ ..+ ' -" {a? + Xf (hfi + Xf {(f + \{})- viz.

the left-hand side, considering therein \{} as a given function of V, is an exact differential. We verify this by finding the value of V, viz. writing down the two equations ar' y- z- V'^ rt=+x "^ PTx "^ c=Tx X ~ ' these are equivalent in virtue of the equation that determines X: and it is to be shown that, regai-ding F as given by either of them, say by the second equation, we have for dV its foregoing value. In fact, differentiating the second equation, the term in rfX disappears by virtue of the first equation, and the result is xdx ydy zdz V dV «= + X 6= + X C-' + X X V in which substituting for — its value from the first equation, we have for dV the value in question. Regarding F as a given constant, the two equations give, by elimination of X, an equation (^ (x, y, z, V) = 0, which is, in fact, the surface parallel to the ellipsoid an<l at a constant normal distance = F from it.

[626 626. ON THE GENERAL DIFFERENTIAL EQUATION '%+%=0, VA Vi WHERE X, Y ARE THE SAME QUARTIC FUNCTIONS OF X, y RESPECTIVELY. [From the Proceedings of tlie Londmi Mathematical Society, vol. Vlll. (1876—1877), pp. 184—199. Read February 8, 1877.] Write (d=a+b6 + c0' + d0' + ed*, the general quartic function of 6 ; and let it be required to integrate by Abel's theorem the differential equation We have a particular integral of da; ,dy_ X-, X, 1, iJX =0, y% y, 1. VF Z-, z, 1, ^'Z W-, lu, 1, VTF dx dy_ dz dw _ and consequently the above equation, taking therein z, w as constants, is the general integral of viz. the two constants z, w must enter in such wise that the equation contains only a single constant; whence also, attributing to w any special value, we have the general integral with z as the arbitrary constant.

^ dx dii 626] ON THE GENERAL DIFFERENTIAL EQUATION 71?+ 'JY~ Take w = 00 ; the equation becomes a?, X, 1, tJX = 0, y^ 2/, 1. ^Y Z-, z, 1, v^ 1 , 0, 0, Ve a relation between x, y, z which may be otherwise expressed by means of the identity eiO' + ^0 + 'if-{ee* + de^ + 00' + he + a) = {2^e-d){d -x)(d - y){d - z), or, what is the same thing, e (27 + /3^) - c = - {2Be -d){x + y + z), e 2/S7 -6= {l^e - d) iyz + zx -\{}- ooy), 67- —a = — (2/3e — d) xyz, where /3, 7 are indeterminate coefficients which are to be eliminated. •I Write then we have giving /9a; + 7 + P = 0, /32/ + 7 + Q = 0; /8 : 7 : 1=Q-P : Py-Qx : x-y. Substituting these values in the first of the preceding three equations, we have that is. ^2(Py-Q.)(.-y) + (Q-P)»_^^_{2(Q-f)e_^|^^^^^^^_ {x-yf y or, reducing by j^,-P^)^(Q-P. 2JQ-P)) ^^^ (x-y {x-yY x-y) + ^); Qy — Px = y' — a? + xsJX-yJY this is Q-P-v'-^^"^. -,.-^+to-.,* if J/ = ^4^^r gj_v_v ^_v '_j^2xy+ 1(x + y) -r - iix + y) z + Iz -T { »Je{x-y) " e ^ ^' >je ^ ' y> ^ ^^ = c + d(x + y+z) + e(x + yy. We have Euler's solution in the far more simple form M"=C + d(x + y) + e(x + yy, C. IX. 75 I

626. On the General Differential Equation $dx/\sqrt{X} + dy/\sqrt{Y} = 0$.

[From the *Proceedings of the London Mathematical Society*, vol. viii. (1876–1877), pp. 57–62.
Read December 14, 1876.]

dx dy ^ r^nr. ON THE GENERAL DIFFERENTIAL EQUATION /y + lY~^- L^Sfi where G is the arbitrary constant. It is to be observed that, in the particular case where e = 0, the first equation becomes M" = c + d {x •{ y + z); and the two results for this case agi-ee on putting G = c-\{}-dz. But it is required to identify the two solutions in the general case where e is not =0. I remark that I have, in my Treatise on Elliptic Fuivctions, Chap, xiv., further developed the theory of Euler's solution, and have shown that, regarding G as variable, and writing g = ad- + h'e - 2bcd + C [- 4ae + 6d + (C - c)=], then the given equation between the vai-iables x, y, G corresponds to the differential equation dx dy dG ^ a result which will be useful for effecting the identification. The Abelian solution may be written e^J^^.^0^-a?-f + ~-2{x + y)~\{}-c-d{x + y) = z\{}d + 2e(x + y)-2M^e]-, and substituting for M its value, and multiplying by {x — yY, the equation becomes 2^e{x-y){x^/X-y^/Y)-eia^ + f)(x-yy + WX-^y)^' -2(x'-f)WX-^/Y)^e-c(x-yf-d{x+y)(x-yy = z(x-y)ldix-y) + 2e{af-f)-2WX-^/Y)'Je}. On the left-hand side, the rational part is X + Y+c(- af + 2xy - f) + d{- X-" + x^y + xy" - f) + e (- x* + 2a?y - 2a?y" + 2a;i/' - y*\{} which, substituting therein for X, Y their values, becomes = 2a + 6 (a; + y) + c . 2xy + dxy{x + y) + e. 2xy {a? — xy+y^); and the irrational part is at once found to be = 2^/e{x-y){x^/Y-y^/X)-2^TY. The equation thus is 2a -I- 6 (a; + y) + c . 2xy + dxy(x + y) + e. 2xy (a?-xy + y^M + 2^e(x-y) (x^Y-y^X)-2 VZ F z = {x - y) [d (x-y) + 2e (a? - y=) - 2(^X- JY) /e] which equation is thus a form of the general integral of -j^+tv-^^' ^^^ *'® ^ ^ particular integral of "Tr +^ + 7^= ^-

-, dx dii ON THE GENERAL DIFFERENTIAL EQUATION y= + -^ = 0. Multiplying the numerator and the denominator by d{x-y)^2e{x^-f)-{-2{^/X-'JY)^e, the denominator becomes = {x-yr \{}\{}d + 2e{x + y)Y-4.e (^^^Z^Yl \{} x — y J which, introducing herein the C of Euler's equation, is = (a;-y)»(d=-4eC). We have therefore z{x — yf (ci' — 4e(7) = {2a + b(x + y) +c. 2xy + dxy{x + y) + e. 2xy {a? — xy->r- y-) ^2^e{x-y)(x^JY-y>JX)-2^XY] x {d(x-y) + 2e(ay'-y') + 2^eWX - ^Y)}. Using S to denote the same value as before, the function on the right-hand is, in fact, = (x - yy {2be -cd + dC + 2'^e VS} and, this being so, the required relation between z, C is z (d» - 4e(7) ={2be-cd + dC+2>^e VS). To prove this, we have first, from the equation to express 6 as a function of x, y. This equation, regarding therein (7 as a variable, gives and we have therefore dx dy dC _ .JX'^7fY'^^~ ' dx ay viz. V-^ J will bs * symmetrical function of x, y. Putting, as before x-y we have C=M^-d{x + y)-e{x-ir yf, and thence We have dM^ 1 X' yz-VF dx ^ x — y 2»JX i^ — yY ' 75—2

(id* cIai ON THE GENERAL DIFFERENTIAL EQUATION - 7-v' + -7v^=0' and hence ^/g(a;-y)•: ^X {x-yy\{}2M~ - d-2e(a; + j/)^ = -(x-y)X'{^/X-^Y) + 2iX+V-2'^TV)^X + (d + 2ex + y)(x-yy^/X = [(x -y)X' + 2X + 2Y + (d+2eiTy) {x - yf] ^X + [(x-y)X'-4X]^Y. We obtain at once the coefficient of \{}/Y, and with little more difficulty that of it/X; and the result is VS (x - y)' = - [4a + Sbx + 2caf + dx' + y(b + 2cx + 3d^ + 4eaf)] -JY + [4a + nby + 2cy- -{ df + x{b + 2cy + My- + 4ev')] i/X. We have also C(x-yy = WX-^Yr-d(x + y)(x-yy-e{x + yy(x-yy = X -irY -d{a? - x-'y -xxf + y') - e{af -23?f-\{}-y*) -2 -JT? = 2a + 6 (a; + y) + c(a!= + y») + da;i/ (« + y) + 2ea?y^ - 2 \{}/XY, or, say C(x-yy = 2a{x-y) + b{af'-y-) + c(x'- a?y + xy"-if) ->r d xy {a? - f) + 2ear'«/» (x-y)-2(x-y) ^/XY. We can hence form the expression of (x - yy {2be -cd + dC+2>Je v'S), viz. this is = {ibe -cd)(x-yy + 2ad {x-y)+ bd (of - y^) + cd {a? - a?y -if xy' -f) + d^ xy {x? - y^) + 2de xY (x-y)- 2d (x - y) vT? + 2 Ve {[- (4a + 36a; + 2c«» + daf) -y(b + 2cx + Sdof + iex")] V Y + [(4o + 3% + 2cy' + df) + x{b + 2cy + My- + 4ey^)] ^/X], and this should be = {2a + 6 (« + y) + c . 2xy -\{}-dxy{x + y) + e. 2xy (af — xy-i- if) -\{}-2^/eix-y)(x^Y-y^/X)-2s/XY} x {d(x- y) + 2e{af- y^) + 2 y/eWX - >s/Y)\{}. The function on the right-hand is, in fact, = {2a + b(x + y) + c.2xy + dxy{x-\{}-y) + e. 2xy (of - xy + y^ - 2 -s/XY] X [d{x-y)+2e(x'-f)\{} +ieix-y)WX - s/Y)(x^Y-y ^/X) + 2 Ve WX - i^Y) {2a + b {x + y) + c .2xy + d xy{x + y) + e .2xy {x' - xy + y^) - 2 >JXY\{} ^2^Je{x-y){x^Y-y ^X) [d{x - y)-^2e(a?-y%

dx dy ON THE GENERAL DIFFERENTIAL EQUATION Ty"^"/!^" viz. this is = {2a+h{x + y) + c .ixy + dxy {x + y) + e. 2xy {a? - xy + y^)] X {d(x-y) + 2e{a^-y')] + 4e(x -y){-xY- yX) - 2 ^XY [d(x - y) + 2e(afi - y'-)} + 4.eix - y)(x + 7/) \{}/XY + 2>/e(<s/X ['2a + b{x + y) + c.2xy + dxy(x + y) + e.2xy(af-xy

+ y^y + 2Y-(x-y)y[d{x-y) + 2e(a^-f)]} — <JY {2a + b{x + y) + c. 2xy + dxy{x + y) + e. 2xy {x:- — xy + y-) + 2X-ix-y)x[d(x-y) + 2e{af-y')]] which is, ia fact, equal to the expression on the left-hand side. To complete the theory, we require to express \{}/Z as a function of x, y. It would be impracticable to effect this by direct substitution of the foregoing value EURi 1* fi'ii fi P' of z; but, observing that the value in question is a solution of -7^ + "7w-+ ~rw = 0. or, what is the same thing, that —ps- + -j^ j~—^> 'ly + Ty ;/" ~ ^' ^^® "^"^ iro\{}x\{}. either of these equations, considering therein ^ as a given function of x, y, calculate '^Z. Writing for shortness J-2^/ey{x-y)l^X + 2^/ex(x-y)^Y-2^JXY where R-2'Je{x-y)s/X + 2'Je{x-tj)>JY R=={x-yy\{}d + 2e{x-\{}-y)], J=2a + b{x + y) + 2cxy + dxy{x + y) + 2exy(a? — xy + y^) ; N or, if for a moment 8 = 7^, then that is, dx IJP\{} dx dx) s/X' ,_ ^Xf„dD j.dN\{} il dX dR dJ dx ' dx' dx' or, writing for shortness X', R', J to denote the derived functions dY i Y' is afterwards written to denote -5— , but as the final formulae contain only dy ^ X', =-j-, and Y', =-=- , this does not occasion any defect of symmetry), we find n= N[R'^X-2^JeX-^e{x-y)X' + 2^Je^XY] -D{J'^/X-2s/eyX-^/e(x-y)yX' + 2^e(2x-y)^XY-X'^Y};

dx cly ON THE GENERAL DIFFERENTIAL EQUATION , t> + -;>, = 0. and substituting herein for N, D their values, and arranging the terms, we find where ^— J\{}2X + {x-y)X'] -2{x-y)yR'X ^Ry{2X+{x-y)X'] + 2{x-y)XJ' + 2(x-y)X'Y, 6 = — 4ey (x — y) X -2e(x-y)x[2X + {.T-7/)X'} -2R'X + RX' + 2e{x-y)y{2X + ix-y)X'] {4ie(x-y)(2x-y)X, S)= JR' + 2e{x-y)y{2X + (.T-y)X'} + ^x{x — y) Y -RJ' -2e{x-y)y[2X+{x-y)X'\{} -4ie(x-y)(2x-y) Y, 2)= 2J + 2{x-y)xR' + 2{2X+{x-y)X'] -2{2x-y)R -2(x-y)X' -2{x-y)J\{} where the terms have been written down as they immediately present themselves ; but, collecting and arranging, we have 21 = 2X {-J + Ry -2Y) + {x- y) [2XJ' + 2X'Y- X'J - 2yR'X + yRX'], 33= JR'-J'R-^e{x-yyY, 6 = - -IXR' + X'R + 4e (*• - yy X-2e{x- yf X', <!>= 2J+4>X-2Rx + 2{x-y)(xR'-R-J'). To reduce these expressions, writing M = d + 2e{x + y), A =c + d(x + y) + e{x' + y"), we have R = (x — yf M, and therefore R = 2{x — y) M + 2e {x - yf ; also J = X+Y-{x-yyA; also, from the original form, J' = b + 2cy + d {2xy -^y") + e(6x'^y -4!xy'+ 2^). The final values are ?l = -X^-QXY- P + (x-ijy {A"-+(-b + day)M+xyM% 8= {x-y)M{^Y-¥{,x-y)Y'\{}+2e{x-yyY', (^ =.-(x-y)M {iX -(x-y) X'} -2eix -yy X'. D= ^(X+Y) + 4,e(x-yy, which, once obtained, may be verified without difficulty.

^ dx chi ^„„ ON THE GENERAL DIFFERENTIAL EQUATION np + Jy — ^' ^^^ Verification of 21.—The equation is - Z^ - 6ZF- Y^ + {x- yy | A-" + (- 6 + dxy) M+xyM"} = 2X(-J + Ri/-2Y) + (x- y) {2XJ' + 2X'Y- X'J- 2yRX + yRX'} ; or, putting tor shortness A- + (-b + (h;y)M + xyM-== V, this is (x-yy'7= Z-^ + 6ZF+F= + 2X{-X-3Y+{x-yyA+(x-yfyM] + (x-y}{ 2XJ' + 2X'Y-X'J- 2yR'X + yRX'} = -X-'+ Y' +2(x- yy XA + 2(x- yy yXM + (x-y) \{}2XJ' + 2Z' F- X'J - 2yR'X + yRX'} ; we have -X" { Y^= -{X - Y){X +Y), where X-Y divides by x-y, =(x-y)il suppose ; hence, throwing out the factor x — y, the equation becomes (x-yfS^ =-n{k+Y) + 2(x-y)XA + 2(x-y)yXM + 2XJ' + 2Z'F - Z' {Z + F- (« - yy A} -2yX[2(x-y)M + 2(x- yy e} + {x - yy yMX', = -n(Z+F) + 2ZJ'-Z'(Z-F) + 2{x-y)XA- 2(x-y) yXM + {x-yy X'A - 4 (a; - yy eyX + {x- yy xjMX'. We have 2ZJ" = J'(Z+ Y) + J'{X-Y), and hence the first line is = (-n + J')(^ + F) + J"'(Z-F); — H + J', as will be shown, divides by x — y, or say it is ={x — y) ^, and, as before, Z — F is = (a; — y) n ; hence, throwing out the factor x — y, the equation becomes (a.— y)» V = * (Z + F) + ft (J' - Z') + 2ZA - 2yXM + {x-y) [X'A - 4eyZ + yMX'}. We have il=^h + c{x -^^ y) + d{a? + xy +f)Jt eia? \{}- x^'y + xy" + _»/'), and thence - ft + ./' = c (- a; + y) + d (- a.'^ + a^) + e (- j? + oai?y - bxy"" + if) ; or, dividing this by (x — y), we find ^ = -c — dx — e(x'— ixy + y"), or, as this may be wiitten, 4> = — A + rfy + ^xy.

dx dy ^ r ON THE GENERAL DIFFERENTIAL EQUATION -y= + -y-^=0. We find, moreover, J'-X' = 2c(-a;+y) + d(-3ic' + 2xy + f) + e(-^ + Gofiy - ixy* + 2y»), which divides by {x — y), the quotient being - 2c - d (3a; + y) - e {^ -2xy+ 2y-), viz. this is = -2A- {x-y)(d + 2ex). Hence the equation now is (a; - y)= V = (Z + F) [- A + dy + 4>eay\{} + 2ZA - 2yXM + {x-y) n {- 2A - (a;-y){d+ 2ex)} + {x-y) { X'A-'^yX+yMX']. The first line is {X + Y) [-K + yM-\{}-2{x-y)ye] + 2ZA - 2yXM, which is = {A-yM){X-Y) + 2{x-y)ey{X+Y); hence, throwing out the factor x — y, the equation becomes {x-y)V ={A-yM)il + 2eyiX + Y)-2An + X'A-4>eyX + yMX' -{x-y)il(d + 2ex) = iA+yM){- n + X') - 2ey {X - ¥)- (x -y) n(d + 2ex). We have - n + X' = c(x- y) + d{2x' - xy - y-) + e (Sx^ -x'y-xy– f), which is ={x—y){A-\{}-xM): also (Z — F) = (« — y) li, as before; whence, throwing out the factor x — y, the equation is V =(A + xM){A + yM) - 2eyD. -{d + 2ex) 12, that is, V = (A + xM) (A + yM) - Mil viz. substituting for V its value, reducing, and throwing out the factor M, the equation becomes — b + dxy = (a; + y) A — 12, which is right. Verification of 33.—The equation is J [2{x-y)M -^-^eix-yYl-J' {x-yfM - ^{x-y)' Y = 4:{x-y)MY +ix- yf MY' + 2e (a; - yf Y', which, throvnng out the factor x-y, is 0=2M(-J+2Y) + {x-y)M(,J'+Y') + 2e{x-y){-J-v2Y) + 2e{x-yf Y'.

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On Three-Bar Motion. [Continued]

that is,

$$(b_1{}^2 - a_2{}^2)\, c\sigma - (c_1{}^2 - a_3{}^2)\, b\tau + \sigma\tau\,(c_1\tau - b_1\sigma) = 0,$$

or, what is the same thing,

$$(b_1{}^2 - a_2{}^2)\, c\sigma - (c_1{}^2 - a_3{}^2)\, b\tau + \sigma\tau\,(c\tau - b\sigma) = 0.$$

Suppose, as in the figure, that F is between B and A; then, if $AF = \rho$, we have $c\tau = b\sigma + a\rho$, and the equation becomes

$$(b_1{}^2 - a_2{}^2)\, c\sigma - (c_1{}^2 - a_3{}^2)\, b\tau + a\rho\sigma\tau = 0.$$

Similarly, if, as in the figure, G is on the other side of A, that is, between A and C, and if ρ', σ', τ' be the distances AG, BG, CG, then $b\sigma' = c\tau' + a\rho'$, that is, $c\tau' - b\sigma' = -a\rho'$, and the corresponding equation is

$$(b_1{}^2 - a_2{}^2)\, c\sigma' - (c_1{}^2 - a_3{}^2)\, b\tau' - a\rho'\sigma'\tau' = 0.$$

We hence find

$$\begin{aligned}
(b_1{}^2 - a_2{}^2)\, c\,(\sigma\tau' - \sigma'\tau) + a\tau\tau'\,(\rho\sigma + \rho'\sigma') &= 0,\\
(c_1{}^2 - a_3{}^2)\, b\,(\sigma\tau' - \sigma'\tau) + a\sigma\sigma'\,(\rho\tau + \rho'\tau') &= 0.
\end{aligned}$$

But we have

$$BG \cdot CF = BF \cdot CG + BC \cdot FG,$$

that is,

$$\sigma'\tau = \sigma\tau' + a \cdot FG, \quad \text{or } \sigma\tau' - \sigma'\tau = -a \cdot FG;$$

also,

$$\rho\sigma + \rho'\sigma' = AF \cdot BF + AG \cdot BG, \; = FG \cdot CH, \; = FG \cdot \tau'',$$

and

$$\rho\tau + \rho'\tau' = AF \cdot CF + AG \cdot CG, \; = FG \cdot BH, \; = FG \cdot \sigma'',$$

as may be shown without difficulty, ρ'', σ'', τ'' being the distances AH, BH, CH. Hence the equations become

$$c\,(b_1{}^2 - a_2{}^2) - \tau\tau'\tau'' = 0,$$
$$b\,(c_1{}^2 - a_3{}^2) - \sigma\sigma'\sigma'' = 0,$$

showing that, the foci being as in the figure, $b_1{}^2 - a_2{}^2$ and $c_1{}^2 - a_3{}^2$ are each of them positive; viz. that, in the generation by the triangle OC_1B_1, the radial bars a_2, a_3 are shorter than the sides b_1, c_1 respectively. Substituting for b_1, &c. the values $k_1 \sin B$, &c.; also, instead of k_1, k_2, k_3, introducing the quantities λ_1, λ_2, λ_3, where

$$k_1,\ k_2,\ k_3 = \lambda_1 \sin A,\ \lambda_2 \sin B,\ \lambda_3 \sin C,$$

these equations become

$$c\,(\lambda_1{}^2 - \lambda_2{}^2) \sin^2 A\, \sin^2 B = \tau\tau'\tau'',$$
$$b\,(\lambda_1{}^2 - \lambda_3{}^2) \sin^2 A\, \sin^2 C = \sigma\sigma'\sigma'';$$

or, as these may be written, putting for shortness $M = \sin A \sin B \sin C$,

$$M^2 k^2\,(\lambda_1{}^2 - \lambda_2{}^2) = c\,\tau\tau'\tau'',$$
$$M^2 k^2\,(\lambda_1{}^2 - \lambda_3{}^2) = b\,\sigma\sigma'\sigma''.$$

All the quantities have so far been regarded as positive, and the formulae are applicable to the particular figure; but, to present them in a form applicable to any order of the nodes and foci, we have only to write the equations in the forms

$$M^2\,({\lambda_1}^2 - {\lambda_2}^2) = k^2 \sin(\alpha - \beta) \sin(f - \gamma) \sin(g - \gamma) \sin(h - \gamma),$$
$$M^2\,({\lambda_1}^2 - {\lambda_3}^2) = k^2 \sin(\alpha - \gamma) \sin(f - \beta) \sin(g - \beta) \sin(h - \beta);$$

and these may be replaced by the system

$$M^2\,({\lambda_2}^2 - {\lambda_3}^2) = k^2 \sin(\beta - \gamma) \sin(f - \alpha) \sin(g - \alpha) \sin(h - \alpha),$$
$$M^2\,({\lambda_3}^2 - {\lambda_1}^2) = k^2 \sin(\gamma - \alpha) \sin(f - \beta) \sin(g - \beta) \sin(h - \beta),$$
$$M^2\,({\lambda_1}^2 - {\lambda_2}^2) = k^2 \sin(\alpha - \beta) \sin(f - \gamma) \sin(g - \gamma) \sin(h - \gamma),$$

since the first of these equations is implied in the other two; and then, reverting to the original form, we may write

$$M^2k^2\,({\lambda_2}^2 - {\lambda_3}^2) = BC \cdot FA \cdot GA \cdot HA,$$
$$M^2k^2\,({\lambda_3}^2 - {\lambda_1}^2) = CA \cdot FB \cdot GB \cdot HB,$$
$$M^2k^2\,({\lambda_1}^2 - {\lambda_2}^2) = AB \cdot FC \cdot GC \cdot HC,$$

it being understood that the distances BC, FA, &c., which enter into these equations, are not all positive, but that they stand for $k\sin(\beta - \gamma)$, $k\sin(f - \alpha)$, &c., and that their signs are to be taken accordingly. Or, again, these may be written

$$BC\,({c_2}^2 - {b_3}^2) = FA \cdot GA \cdot HA,$$
$$CA\,({a_3}^2 - {c_1}^2) = FB \cdot GB \cdot HB,$$
$$AB\,({b_1}^2 - {a_2}^2) = FC \cdot GC \cdot HC,$$

where the signs are as just mentioned. We may say that $\pm\,({c_2}^2 - {b_3}^2)$ is the modulus for the focus A; and the formula then shows that this modulus, taken positively, is equal to the product of the distances FA, GA, HA of A from the three nodes respectively, divided by BC, the distance of the other two foci from each other.

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On the Bicursal Sextic.

[From the *Proceedings of the London Mathematical Society*, vol. VII. (1875—1876),
pp. 166—172. Read March 10, 1876.]

In the paper "On the mechanical description of certain sextic curves," *Proceedings of the London Mathematical Society*, vol. IV. (1872), pp. 105—111, [504], I obtained the bicursal sextic as a rational transformation of a binodal quartic. The theory was in effect as follows: taking Ω, P, Q, R, each of them a function of λ, μ of the form $(^*\backslash\!\backslash 1, \lambda)^2(1, \mu)^2$, and considering (λ, μ) as connected by the equation $\Omega = 0$, (viz. λ, μ being coordinates, this represents a binodal quartic), then, if we assume $x : y : z = P : Q : R$, the locus of the point (x, y, z) is a curve rationally connected with the binodal quartic, viz. the points of the two curves have with each other a (1, 1) correspondence; whence the locus in question, say the curve $U = 0$, is bicursal. The degree is obtained as the number of the intersections of the curve by an arbitrary line, or, what is the same thing, the number of the variable intersections of the corresponding $\lambda\mu$-curves

$$\Omega = 0, \quad \alpha P + \beta Q + \gamma R = 0,$$

viz. each of these being a quartic curve having the same two nodes, the nodes each count as 4 intersections, and the number of the remaining intersections is $4.4 - 2.4 = 8$, and thus the curve $U = 0$ is in general of the order 8. But if the curves $\Omega = 0$, $P = 0$, $Q = 0$, $R = 0$ have (besides the nodes) k common intersections, then these are also fixed intersections of the two curves $\Omega = 0$, $\alpha P + \beta Q + \gamma R = 0$, and the number of variable intersections is reduced to $8 - k$; we have thus $8 - k$ as the order of the curve $U = 0$. In particular, if $k = 2$, then the curve is a bicursal sextic.

The theory assumes a different and more simple form if, in the several functions Ω, P, Q, R, we suppose that the terms in λ^2, μ^2 are wanting. The curves $\Omega = 0$, $P = 0$, $Q = 0$, $R = 0$ are here cubics having two common points; the curve $U = 0$, *qui* rational transformation of the cubic $\Omega = 0$, is still a bicursal curve; but its order is given as the number of the variable intersections of the cubics

$$\Omega = 0, \quad \alpha P + \beta Q + \gamma R = 0,$$

viz. this is $= 3.3 - 2 = 7$. But if the curves $\Omega = 0$, $P = 0$, $Q = 0$, $R = 0$ have (besides the before-mentioned two common points) k other common points, then the number of the variable intersections is $= 7 - k$: and this is therefore the order of the curve $U = 0$. In particular, if $k = 1$, then the curve is a bicursal sextic. And, in the present paper, I consider the binodal sextic as thus obtained, viz. as given by the equations $\Omega = 0$, $x : y : z = P : Q : R$, where $\Omega = 0$, $P = 0$, $Q = 0$, $R = 0$ are cubics, having (in all) three common points.

The bicursal sextic has in general 9 nodes; but 3 of these may unite together into a triple point: this will be the case if, in the series of curves $\alpha P + \beta Q + \gamma R = 0$, there are any two curves which have 3 common intersections with the curve $\Omega = 0$. (Observe that we throughout disregard the 3 common points of the curves $\Omega = 0$, $P = 0$, $Q = 0$, $R = 0$, and attend only to the 6 variable points of intersection of the curves $\Omega = 0$ and $\alpha P + \beta Q + \gamma R = 0$,—the meaning is, that there are two curves of the series such that, attending only to the 6 variable intersections of each of them with the curve $\Omega = 0$, there are three common intersections.) For, supposing the two curves to be $\alpha P + \beta Q + \gamma R = 0$ and $\alpha' P + \beta' Q + \gamma' R = 0$, then any curve whatever

$$\alpha P + \beta Q + \gamma R + \theta\,(\alpha' P + \beta' Q + \gamma' R) = 0$$

has the same three intersections with the curve $\Omega = 0$, say these are the points A_1, A_2, A_3, the coordinates of which are independent of θ. Hence the line

$$(\alpha x + \beta y + \gamma z) + \theta\,(\alpha' x + \beta' y + \gamma' z) = 0$$

intersects the curve $U = 0$ in six points, three of which, as corresponding to the points A_1, A_2, A_3, are independent of θ, viz. they are the same three points for any line whatever of the series; and this means that the curve $U = 0$ has at the point

$$(\alpha x + \beta y + \gamma z = 0, \quad \alpha' x + \beta' y + \gamma' z = 0)$$

a triple point; and that to this triple point correspond the three points A_1, A_2, A_3.

We may, in the series of lines $\alpha x + \beta y + \gamma z + \theta(\alpha' x + \beta' y + \gamma' z) = 0$, rationally determine θ so that one of the three variable points of intersection shall correspond to A_1, A_2, or A_3; viz. θ must be such that the curve $\alpha P + \beta Q + \gamma R + \theta(\alpha' P + \beta' Q + \gamma' R) = 0$ shall touch the curve $\Omega = 0$ at one of the points A_1, A_2, A_3. The three lines thus determined are the three tangents to the curve at the triple point: and the three branches may be considered as corresponding to the three points A_1, A_2, A_3, respectively.

There is no loss of generality in assuming that the triple point is the point $(x = 0,\ z = 0)$; the condition then simply is that the curves $P = 0$, $R = 0$ shall have three common intersections with the curve $\Omega = 0$; and the tangents at the triple point are $x + \theta z = 0$, θ being so determined that one of the three variable points of intersection shall correspond to one of the three points A_1, A_2, A_3: in particular, if this is the case for the line $x = 0$, then this line will be one of the tangents at the triple point.

The bicursal sextic may have a second triple point, viz. three other nodes may unite together into a triple point. The theory is precisely the same: we must have two other curves $\alpha P + \beta Q + \gamma R = 0$, $\alpha' P + \beta' Q + \gamma' R = 0$, having with the curve $\Omega = 0$ three common intersections B_1, B_2, B_3: there is then a second triple point

$$(\alpha x + \beta y + \gamma z = 0, \quad \alpha' x + \beta' y + \gamma' z = 0);$$

and, to find the tangents at this point, we must determine θ so that one of the variable points of intersection of the line

$$\alpha x + \beta y + \gamma z + \theta\,(\alpha' x + \beta' y + \gamma' z) = 0$$

with the sextic shall correspond with B_1, B_2, or B_3; viz. θ must be such that the curve $\alpha P + \beta Q + \gamma R + \theta(\alpha' P + \beta' Q + \gamma' R) = 0$ shall touch the curve $\Omega = 0$ at one of the points B_1, B_2, B_3. In particular, if, as before, the curves $P = 0$, $R = 0$ have three common intersections with the curve $\Omega = 0$, and if, moreover, the curves $Q = 0$, $R = 0$ have three common intersections with the curve $\Omega = 0$, then the bicursal sextic will have the two triple points $(x = 0,\ z = 0)$ and $(y = 0,\ z = 0)$; and it may further happen that the line $x = 0$ is a tangent at the first triple point, and the line $y = 0$ a tangent at the second triple point. The sextic may in like manner have a third triple point, but this is a special case which I do not at present consider.

I write for greater convenience $\dfrac{\lambda}{\nu}$, $\dfrac{\mu}{\nu}$ in place of λ, μ, so as to make Ω, P, Q, R each of them a homogeneous cubic function of (λ, μ, ν); and I give to these functions, not the most general values belonging to a bicursal sextic with two triple points, but the values in the form obtained for them, as appearing further on, in the problem of three-bar motion; viz. the equations $\Omega = 0$, $x : y : z = P : Q : R$ are respectively taken to be

$$\nu\,(h\nu\lambda + f\lambda^2) + \mu\,(g\nu^2 + e\nu\lambda + g\lambda^2) + \mu^2\,(f\nu + h\lambda) = 0,$$
$$x : y : z = \lambda\mu\,(a\lambda + b\mu) : \nu^2(c\lambda + d\mu) : \lambda\mu\nu.$$

The four curves $\Omega = 0$, $P = 0$, $Q = 0$, $R = 0$ have thus the three common intersections

$$(\mu = 0,\ \nu = 0), \quad (\nu = 0,\ \lambda = 0), \quad (\lambda = 0,\ \mu = 0),$$

represented in the figure by the points A, B, C; the curve drawn in the figure is the curve $\Omega = 0$, and the points F, G, H are the third points of intersection of the cubic with the lines BC, CA, AB respectively.

[Figure: Triangle ABC with cubic curve $\Omega = 0$ passing through A, B, C and points F on BC, G on CA, H on AB.]

The equation $P + \theta R = 0$ is here $\lambda\mu\,(a\lambda + b\mu + \theta\nu) = 0$, which intersects $\Omega = 0$ in the points C, A, G, C, B, F, and the three intersections by the line $a\lambda + b\mu + \theta\nu = 0$;

viz. excluding the fixed points A, B, C, the six intersections are C, F, G, and the three intersections by the line. Hence, of the six intersections, we have C, F, G independent of θ, or we have $(x=0,\ z=0)$ a triple point, say I, corresponding to the three points C, F, G, viz. these are the points

$$(\lambda,\, \mu,\, \nu) = (0,\, 0,\, 1), \quad (0,\, g,\, -f), \quad (h,\, 0,\, -f), \qquad (C,\, F,\, G).$$

The equation $Q + \theta R = 0$ is $\nu\,(c\lambda\nu + d\mu\nu + \theta\lambda\mu) = 0$; viz. the line $\nu = 0$ meets $\Omega = 0$ in the three points A, B, H, and the conic $c\lambda\nu + d\mu\nu + \theta\lambda\mu = 0$ meets $\Omega = 0$ in the points A, B, C, and three other points: hence, rejecting the points A, B, C, the six points of intersection are the points A, B, H, and the three variable points of intersection by the conic; or we have $(y=0,\ z=0)$ a triple point, say J, corresponding to the three points A, B, H, viz. these are the points

$$(\lambda,\, \mu,\, \nu) = (1,\, 0,\, 0), \quad (0,\, 1,\, 0), \quad (h,\, -g,\, 0), \qquad (A,\, B,\, H).$$

To find the tangents at the triple point I, these are $x + \theta z = 0$, where θ is to be successively determined by the conditions that the line $a\lambda + b\mu + \theta\nu = 0$ shall pass through the points C, F, G*; viz. we thus have

$\theta = 0,$	$x = 0,$	the tangent corresponding to the point	$C,\ (0,\, 0,\, 1),$
$\theta = +\dfrac{bg}{f},$	$fx + bgz = 0,$	"	$F,\ (0,\, g,$
$\theta = +\dfrac{ah}{f},$	$fx + ahz = 0,$	"	$G,\ (h,\, 0,$

And similarly, at the triple point J, the tangents are $y + \theta z = 0$, where θ is to be successively determined by the conditions that the conic $c\nu\lambda + d\nu\mu + \theta\lambda\mu = 0$ shall pass through the point H, and shall touch the cubic at the points A, B; viz. we thus have

$\theta = 0,$	$y = 0,$	the tangent corresponding to the point	$H,\ (h,\, -g,\, 0),$
$\theta = \dfrac{cg}{f},$	$fy + cgz = 0,$	"	$A,\ (1,\, 0,\, 0)$
$\theta = \dfrac{dh}{f},$	$fy + dhz = 0,$	"	$B,\ (0,\, 1,\, 0)$

The two last values of θ are obtained by the consideration that the equations of the tangents to $\Omega = 0$ at the points A, B respectively, are $g\mu + f\nu = 0$, $h\lambda + f\nu = 0$, where λ, μ, ν are current coordinates of a point on the tangent: it may be added that the equation of the tangent at the point C is $h\lambda + g\mu = 0$.

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* Observe the somewhat altered form of the condition: θ is to be determined so that the cubic $\lambda\mu\,(a\lambda + b\mu + \theta\nu) = 0$ shall touch the cubic $\Omega = 0$ at one of the points C, F, G: but, as the first-mentioned cubic breaks up, and the component curve $a\lambda + b\mu + \theta\nu = 0$ does not pass through any one of these points, this can only mean that θ shall be so determined as that the line shall pass through one of these points, viz. that there shall be at the point, not a proper contact, but a double intersection, arising from a node of the cubic $\lambda\mu\,(a\lambda + b\mu + \theta\nu) = 0$. And the like case happens for the other triple point; viz. there the cubic $\nu\,(c\nu\lambda + d\nu\mu + \theta\lambda\mu) = 0$ is to touch the cubic $\Omega = 0$ at one of the points A, B, H; the component conic $c\nu\lambda + d\nu\mu + \theta\lambda\mu = 0$ passes through the points A and B but not through H; hence the conditions for θ are, that the conic shall touch the cubic at A or B, or that it shall pass through H.

The three-bar curve may be represented by means of a system of equations of the last-mentioned form, viz. $x:y:z=\lambda\mu\,(a\lambda+b\mu):\nu^2(c\lambda+d\mu):\lambda\mu\nu$, where λ, μ, ν are connected as above; or, taking X, Y as ordinary rectangular coordinates, x, y, and z are here the circular coordinates $\dfrac{x}{z}=X+iY$, $\dfrac{y}{z}=X-iY$, and $z=1$; and the parameters $\dfrac{\lambda}{\nu}$, $\dfrac{\mu}{\nu}$ denote like functions $\cos\theta+i\sin\theta$, $\cos\phi+i\sin\phi$ of angles which are the inclinations of two bars to a fixed line. Using, for convenience, Figure 2 of my paper on Three-bar Motion, (p. 553 of this volume), the curve is considered as the locus of the vertex O of the triangle OC_1B_1, connected by the bars C_1C and B_1B with the fixed points B and C respectively; and we have $CC_1=a_2$, $OC_1=b_1$, $OB_1=c_1$, $C_1B_1=a_1$, $B_1B=a_3$. Also, to avoid confusion with the foregoing notation of the present paper, instead of calling it a, I take $BC=a_0$: the angle OC_1B_1 is $=C_1$, and $\cos C_1+i\sin C_1$ is taken $=\gamma$.

Hence, taking the origin at C, the axis of X coinciding with CB and that of Y being at right angles to it: taking also θ, ϕ, ψ for the inclinations of CC_1, C_1B_1, and B_1B to CB, we have

$$\begin{aligned} a_2\cos\theta+a_1\cos\phi-a&=-a_3\cos\psi,\\ a_2\sin\theta+a_1\sin\phi\quad&=\quad a_3\sin\psi; \end{aligned}$$

viz. writing $\cos\theta+i\sin\theta=\lambda$, $\cos\phi+i\sin\phi=\mu$, these give

$$\begin{aligned} a_2\lambda+a_1\mu-a_0&=-a_3\,(\cos\psi-i\sin\psi),\\ a_2\frac{1}{\lambda}+a_1\frac{1}{\mu}-a_0&=-a_3\,(\cos\psi+i\sin\psi), \end{aligned}$$

that is,

$$(a_2\lambda+a_1\mu-a)\left(a_2\frac{1}{\lambda}+a_1\frac{1}{\mu}-a_0\right)-{a_3}^2=0;$$

viz.

$$({a_0}^2+{a_1}^2+{a_2}^2-{a_3}^2)+a_1a_2\left(\frac{\mu}{\lambda}+\frac{\lambda}{\mu}\right)-a_0a_2\left(\lambda+\frac{1}{\lambda}\right)-a_0a_1\left(\mu+\frac{1}{\mu}\right)=0,$$

for the relation between the parameters λ, μ. And then

$$\begin{aligned} X&=a_2\cos\theta+b_1\cos(\phi+C_1),\\ Y&=a_2\sin\theta+b_1\sin(\phi+C_1); \end{aligned}$$

viz. if x, $y = X + iY$, $X - iY$, then

$$\begin{cases} x = a_2\lambda + b_2\gamma_1\mu, \\ y = a_2\dfrac{1}{\lambda} + b_2\dfrac{1}{\gamma_1\mu}, \end{cases}$$

which equations determine the coordinates (x, y) in terms of the parameters λ, μ connected by the foregoing relation.

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Writing for homogeneity $\dfrac{\lambda}{\nu}$, $\dfrac{\mu}{\nu}$ in place of λ, μ, and $\dfrac{x}{z}$, $\dfrac{y}{z}$ in place of x, y, the equations become

$$({a_0}^2+{a_1}^2+{a_2}^2-{a_3}^2)\,\lambda\mu\nu+a_1a_2\,(\lambda^2+\mu^2)\,\nu-a_0a_2\mu\,(\nu^2+\lambda^2)-a_0a_1\lambda\,(\mu^2+\nu^2)=0,$$

and

$$x:y:z=(a_2\lambda+b_2\gamma_1\mu)\,\lambda\mu:\left(\frac{b_2}{\gamma_1}\lambda+a_2\mu\right)\nu^2:\lambda\mu\nu.$$

Comparing with the foregoing equations

$$e\,\lambda\mu\nu+f(\lambda^2+\mu^2)\,\nu+g(\nu^2+\lambda^2)\,\mu+h(\mu^2+\nu^2)\,\lambda=0,$$

and

$$x:y:z=(a\lambda+b\mu)\,\lambda\mu:(c\lambda+d\mu)\,\nu^2:\lambda\mu\nu,$$

the equations agree together, and we have

$$\begin{cases} e={a_0}^2+{a_1}^2+{a_2}^2-{a_3}^2,\\ f=+a_1a_2,\\ g=-a_0a_2,\\ h=-a_0a_1,\\ a=\quad a_2,\\ b=\quad b_1\gamma_1,\\ c=\quad \dfrac{b_1}{\gamma_1},\\ d=\quad a_2.\end{cases}$$

The tangents at the triple points thus are

$$\begin{array}{rcc} x=0, & ext & y=0,\\ a_1x-a_0b_1\gamma_1z=0, & ext & a_1y-\dfrac{a_0b_1}{\gamma_1}z=0,\\ x-a_0z=0, & ext & y-a_0z=0; \end{array}$$

viz. restoring the rectangular coordinates, and for γ substituting the value $\cos C+i\sin C$, for a_0 writing a, and taking $b=\dfrac{ab_1}{a_1}$, we have

$$\begin{array}{ccc} X+iY=0, & ext & X-iY=0,\\ X+iY=b\,(\cos C+i\sin C), & ext & X-iY=b\,(\cos C-i\sin C),\\ X+iY=a_0, & ext & X-iY=a_0; \end{array}$$

viz. the first two intersect in the point (0, 0), the second two in the point $(b\cos C,\ b\sin C)$, the third two in the point $(a,\ 0)$: the first and third of these are the points B and C, the second of them is the point A of the figure; viz. the formulae give the point A, forming, with B and C, a triad of foci.

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On the Condition for the Existence of a Surface Cutting at Right Angles a Given Set of Lines.

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In a congruency or doubly infinite system of right lines, the direction-cosines α, β, γ of the line through any given point (x, y, z), are expressible as functions of x, y, z; and it was shown by Sir W. R. Hamilton in a very elegant manner that, in order to the existence of a surface (or, what is the same thing, a set of parallel surfaces) cutting the lines at right angles, $\alpha dx+\beta dy+\gamma dz$ must be an exact differential: when this is so, writing $V = \int(\alpha dx + \beta dy + \gamma dz)$, we have $V = c$, the equation of the system of parallel surfaces each cutting the given lines at right angles.

The proof is as follows:—If the surface exists, its differential equation is $\alpha dx + \beta dy + \gamma dz = 0$, and this equation must therefore be integrable by a factor. Now the functions α, β, γ are such that $\alpha^2+\beta^2+\gamma^2 = 1$, and they besides satisfy a system of partial differential equations which Hamilton deduces from the geometrical notion of a congruency; viz. passing from the point (x, y, z) to the consecutive point on the line, that is, to the point whose coordinates are $x + \rho\alpha$, $y+\rho\beta$, $z+\rho\gamma$ (ρ infinitesimal), the line belonging to this point is the original line; and consequently α, β, γ, considered as functions of x, y, z, must remain unaltered when these variables are changed into $x + \rho\alpha$, $y + \rho\beta$, $z + \rho\gamma$, respectively. We thus obtain the equations

$$\alpha\frac{d\alpha}{dx}+\beta\frac{d\alpha}{dy}+\gamma\frac{d\alpha}{dz}=0,$$
$$\alpha\frac{d\beta}{dx}+\beta\frac{d\beta}{dy}+\gamma\frac{d\beta}{dz}=0,$$
$$\alpha\frac{d\gamma}{dx}+\beta\frac{d\gamma}{dy}+\gamma\frac{d\gamma}{dz}=0.$$

Combining herewith the equations obtained by differentiation of

$\alpha^2 + \beta^2 + \gamma^2 = 1$, viz.

$$\alpha\,\frac{d\alpha}{dx} + \beta\,\frac{d\beta}{dx} + \gamma\,\frac{d\gamma}{dx} = 0,$$
$$\alpha\,\frac{d\alpha}{dy} + \beta\,\frac{d\beta}{dy} + \gamma\,\frac{d\gamma}{dy} = 0,$$
$$\alpha\,\frac{d\alpha}{dz} + \beta\,\frac{d\beta}{dz} + \gamma\,\frac{d\gamma}{dz} = 0,$$

and subtracting the corresponding equations, we obtain three equations which may be written

$$\alpha : \beta : \gamma = \frac{d\beta}{dz} - \frac{d\gamma}{dy} : \frac{d\gamma}{dx} - \frac{d\alpha}{dz} : \frac{d\alpha}{dy} - \frac{d\beta}{dx},$$

or, what is the same thing,

$$\frac{d\beta}{dz} - \frac{d\gamma}{dy},\ \frac{d\gamma}{dx} - \frac{d\alpha}{dz},\ \frac{d\alpha}{dy} - \frac{d\beta}{dx} = k\alpha,\ k\beta,\ k\gamma,$$

and, multiplying by α, β, γ, and adding,

$$k = \alpha\Big(\frac{d\beta}{dz} - \frac{d\gamma}{dy}\Big) + \beta\Big(\frac{d\gamma}{dx} - \frac{d\alpha}{dz}\Big) + \gamma\Big(\frac{d\alpha}{dy} - \frac{d\beta}{dx}\Big).$$

We thus see that, if the function on the right-hand vanish, then $k = 0$, and consequently also

$$\frac{d\beta}{dz} - \frac{d\gamma}{dy},\ \frac{d\gamma}{dx} - \frac{d\alpha}{dz},\ \frac{d\alpha}{dy} - \frac{d\beta}{dx} \text{ each } = 0;$$

viz. if the equation $\alpha dx + \beta dy + \gamma dz = 0$ be integrable, then $\alpha dx + \beta dy + \gamma dz$ is an exact differential; which is the theorem in question.

But it is interesting to obtain the first mentioned set of differential equations from the analytical equations of a congruency, viz. these are $x = mz + p$, $y = nz + q$, where m, n, p, q are functions of two arbitrary parameters, or, what is the same thing, p, q are given functions of m, n; and therefore, from the three equations, m, n are given functions of x, y, z. And it is also interesting to express in terms of these quantities m, n, considered as functions of x, y, z, the condition for the existence of the set of surfaces.

We have

$$\alpha,\ \beta,\ \gamma = \frac{m}{R},\ \frac{n}{R},\ \frac{1}{R},\quad \text{where } R = \sqrt{1 + m^2 + n^2};$$

and hence without difficulty

$$\begin{aligned}
\Big(\alpha\frac{d}{dx}+\beta\frac{d}{dy}+\gamma\frac{d}{dz}\Big)\alpha &= \frac{1}{R^4}\Big[(1+n^2)\Big(m\frac{dm}{dx}+n\frac{dm}{dy}+\frac{dm}{dz}\Big)-mn\Big(m\frac{dn}{dx}+n\frac{dn}{dy}\\
\Big(\alpha\frac{d}{dx}+\beta\frac{d}{dy}+\gamma\frac{d}{dz}\Big)\beta &= \frac{1}{R^4}\Big[-mn\Big(\qquad\qquad\qquad\qquad\Big)+(1+m^2)\Big(\qquad\qquad\qquad\qquad\Big)\Big],\\
\Big(\alpha\frac{d}{dx}+\beta\frac{d}{dy}+\gamma\frac{d}{dz}\Big)\gamma &= \frac{1}{R^4}\Big[-(\qquad\qquad\qquad\qquad)\quad-(\qquad\qquad\qquad\qquad)\Big],
\end{aligned}$$

so that the required equations in α, β, γ will be satisfied if only

$$\begin{aligned}
m\frac{dm}{dx}+n\frac{dm}{dy}+\frac{dm}{dz}&=0,\\
m\frac{dn}{dx}+n\frac{dn}{dy}+\frac{dn}{dz}&=0,
\end{aligned}$$

and it is to be shown that these equations hold good.

Writing for shortness $dp = A\,dm + B\,dn$, $dq = C\,dm + D\,dn$, the equations of the line give

$$\left.\begin{aligned}
1 &= z\frac{dm}{dx}+A\frac{dm}{dx}+B\frac{dn}{dx},\\
0 &= z\frac{dm}{dy}+A\frac{dm}{dy}+B\frac{dn}{dy},\\
-m &= z\frac{dm}{dz}+A\frac{dm}{dz}+B\frac{dn}{dz},
\end{aligned}\right|\quad
\begin{aligned}
0 &= z\frac{dn}{dx}+C\frac{dm}{dx}+D\frac{dn}{dx},\\
1 &= z\frac{dn}{dy}+C\frac{dm}{dy}+D\frac{dn}{dy},\\
-n &= z\frac{dn}{dz}+C\frac{dm}{dz}+D\frac{dn}{dz};
\end{aligned}$$

or, writing

$$\lambda,\ \mu,\ \nu = \frac{dm}{dy}\frac{dn}{dz}-\frac{dm}{dz}\frac{dn}{dy},\ \frac{dm}{dz}\frac{dn}{dx}-\frac{dm}{dx}\frac{dn}{dz},\ \frac{dm}{dx}\frac{dn}{dy}-\frac{dm}{dy}\frac{dn}{dx},$$

so that identically

$$\begin{aligned}
\lambda\frac{dm}{dx}+\mu\frac{dm}{dy}+\nu\frac{dm}{dz}&=0,\\
\lambda\frac{dn}{dx}+\mu\frac{dn}{dy}+\nu\frac{dn}{dz}&=0,
\end{aligned}$$

then in each set, multiplying by λ, μ, ν and adding, so as to eliminate A, B, C, D, we find

$$\lambda - m\nu = 0,\qquad \mu - n\nu = 0.$$

Substituting these values of λ, μ in the last preceding equations, ν divides out, and we have the two equations in question.

The foregoing equations give further

$$A,\ B,\ C,\ D = -z + \frac{1}{\nu}\frac{dn}{dy},\ -\frac{1}{\nu}\frac{dm}{dy},\ -\frac{1}{\nu}\frac{dn}{dx},\ -z + \frac{1}{\nu}\frac{dm}{dx}.$$

Taking for α, β, γ the before-mentioned values, we find

$$\begin{aligned}\frac{d\alpha}{dy} - \frac{d\beta}{dx} &= \frac{1}{R}\Big(\frac{dm}{dy} - \frac{dn}{dx}\Big) - \frac{m}{R^3}\Big(m\,\frac{dm}{dy} + n\,\frac{dn}{dy}\Big) - \frac{n}{R^3}\Big(m\,\frac{dm}{dx} + n\,\frac{dn}{dx}\Big)\\ &= \frac{1}{R^3}\Big\{(1+n^2)\,\frac{dm}{dy} - (1+m^2)\,\frac{dn}{dx} + mn\Big(\frac{dm}{dx} - \frac{dn}{dy}\Big)\Big\};\end{aligned}$$

and similarly, but using the equations

$$m\,\frac{dm}{dx} + n\,\frac{dm}{dy} + \frac{dm}{dz} = 0, \quad m\,\frac{dn}{dx} + n\,\frac{dn}{dy} + \frac{dn}{dz} = 0,$$

to eliminate the coefficients $\dfrac{dm}{dz}$, $\dfrac{dn}{dz}$ which in the first instance present themselves, we find

$$\begin{aligned}\frac{d\beta}{dz} - \frac{d\gamma}{dy} &= \frac{m}{R^3}\Big\{(1+n^2)\,\frac{dm}{dy} - (1+m^2)\,\frac{dn}{dx} + mn\Big(\frac{dm}{dx} - \frac{dn}{dy}\Big)\Big\},\\ \frac{d\gamma}{dx} - \frac{d\alpha}{dz} &= \frac{n}{R^3}\Big\{ \qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad \Big\};\end{aligned}$$

whence, multiplying by γ, α, β, and adding,

$$\begin{aligned}&\alpha\Big(\frac{d\beta}{dz} - \frac{d\gamma}{dy}\Big) + \beta\Big(\frac{d\gamma}{dx} - \frac{d\alpha}{dz}\Big) + \gamma\Big(\frac{d\alpha}{dy} - \frac{d\beta}{dx}\Big)\\ &= \frac{1}{1+m^2+n^2}\Big\{(1+n^2)\,\frac{dm}{dy} - (1+m^2)\,\frac{dn}{dx} + mn\Big(\frac{dm}{dx} - \frac{dn}{dy}\Big)\Big\};\end{aligned}$$

or we have

$$(1+n^2)\,\frac{dm}{dy} - (1+m^2)\,\frac{dn}{dx} + mn\Big(\frac{dm}{dx} - \frac{dn}{dy}\Big) = 0$$

as the condition for the existence of the set of surfaces.

It is clear that the condition is satisfied when the lines are the normals of a given surface: seeking the surfaces which cut the lines at right angles, we obtain the parallel surfaces; and we are led to the theorem that any parallel surface is the locus of the extremity of a line of constant length measured off from each point of the surface along the normal—or, what is equivalent thereto, the parallel surface is the envelope of a sphere of constant radius having its centre on the surface. I will verify the theorem for the case of the ellipsoid. Taking X, Y, Z as the coordinates of a point on the ellipsoid $\dfrac{X^2}{a^2}+\dfrac{Y^2}{b^2}+\dfrac{Z^2}{c^2}=1$, and x, y, z as current coordinates, the equations of the normal are

$$\frac{a^2}{X}(x-X)=\frac{b^2}{Y}(y-Y)=\frac{c^2}{Z}(z-Z),\ (=\lambda \text{ suppose}).$$

We have therefore

$$X,\ Y,\ Z=\frac{a^2x}{a^2+\lambda},\ \frac{b^2y}{b^2+\lambda},\ \frac{c^2z}{c^2+\lambda},$$

and thence

$$\frac{a^2x^2}{(a^2+\lambda)^2}+\frac{b^2y^2}{(b^2+\lambda)^2}+\frac{c^2z^2}{(c^2+\lambda)^2}=1,$$

an equation which determines λ as a function of x, y, z.

The direction-cosines α, β, γ of the normal are proportional to $\dfrac{X}{a^2}, \dfrac{Y}{b^2}, \dfrac{Z}{c^2}$, that is, to $\dfrac{x}{a^2+\lambda}, \dfrac{y}{b^2+\lambda}, \dfrac{z}{c^2+\lambda}$, and the equation $\alpha^2+\beta^2+\gamma^2=1$ then determines their absolute magnitudes: the equation $\alpha dx+\beta dy+\gamma dz=dV$ thus is

$$\frac{\dfrac{x\,dx}{a^2+\lambda}+\dfrac{y\,dy}{b^2+\lambda}+\dfrac{z\,dz}{c^2+\lambda}}{\sqrt{\dfrac{x^2}{(a^2+\lambda)^2}+\dfrac{y^2}{(b^2+\lambda)^2}+\dfrac{z^2}{(c^2+\lambda)^2}}}=dV,$$

viz. the left-hand side, considering therein λ as a given function of V, is an exact differential. We verify this by finding the value of V, viz. writing down the two equations

$$\frac{x^2}{(a^2+\lambda)^2}+\frac{y^2}{(b^2+\lambda)^2}+\frac{z^2}{(c^2+\lambda)^2}-\frac{V^2}{\lambda^2}=0,$$

$$\frac{x^2}{a^2+\lambda}+\frac{y^2}{b^2+\lambda}+\frac{z^2}{c^2+\lambda}-\frac{V^2}{\lambda}=1,$$

these are equivalent in virtue of the equation that determines λ; and it is to be shown that, regarding V as given by either of them, say by the second equation, we have for dV its foregoing value. In fact, differentiating the second equation, the term in $d\lambda$ disappears by virtue of the first equation, and the result is

$$\frac{x\,dx}{a^2+\lambda}+\frac{y\,dy}{b^2+\lambda}+\frac{z\,dz}{c^2+\lambda}-\frac{V\,dV}{\lambda}=0,$$

in which substituting for $\dfrac{V}{\lambda}$ its value from the first equation, we have for dV the value in question. Regarding V as a given constant, the two equations give, by elimination of λ, an equation $\phi\,(x,\,y,\,z,\,V)=0$, which is, in fact, the surface parallel to the ellipsoid and at a constant normal distance $=V$ from it.

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On the General Differential Equation $\dfrac{dx}{\sqrt{X}} + \dfrac{dy}{\sqrt{Y}} = 0$, where X, Y are the Same Quartic Functions of x, y respectively.

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Write $\Theta = a + b\theta + c\theta^2 + d\theta^3 + e\theta^4$, the general quartic function of θ; and let it be required to integrate by Abel's theorem the differential equation

$$\frac{dx}{\sqrt{X}} + \frac{dy}{\sqrt{Y}} = 0.$$

We have

$$\begin{vmatrix} x^2, & x, & 1, & \sqrt{X} \\ y^2, & y, & 1, & \sqrt{Y} \\ z^2, & z, & 1, & \sqrt{Z} \\ w^2, & w, & 1, & \sqrt{W} \end{vmatrix} = 0,$$

a particular integral of

$$\frac{dx}{\sqrt{X}} + \frac{dy}{\sqrt{Y}} + \frac{dz}{\sqrt{Z}} + \frac{dw}{\sqrt{W}} = 0;$$

and consequently the above equation, taking therein z, w as constants, is the general integral of

$$\frac{dx}{\sqrt{X}} + \frac{dy}{\sqrt{Y}} = 0,$$

viz. the two constants z, w must enter in such wise that the equation contains only a single constant; whence also, attributing to w any special value, we have the general integral with z as the arbitrary constant.

Take $w = \infty$; the equation becomes

$$\begin{vmatrix} x^2, & x, & 1, & \sqrt{X} \\ y^2, & y, & 1, & \sqrt{Y} \\ z^2, & z, & 1, & \sqrt{Z} \\ 1, & 0, & 0, & \sqrt{e} \end{vmatrix} = 0,$$

a relation between x, y, z which may be otherwise expressed by means of the identity

$$e\,(\theta^2+\beta\theta+\gamma)^2-(e\theta^4+d\theta^3+c\theta^2+b\theta+a) = (2\beta e-d)\,(\theta-x)(\theta-y)(\theta-z),$$

or, what is the same thing,

$$\begin{aligned} e\,(2\gamma+\beta^2) - c &= -\,(2\beta e-d)\,(x+y+z), \\ e\cdot 2\beta\gamma \quad - b &= \quad (2\beta e-d)\,(yz+zx+xy), \\ e\cdot\gamma^2 \quad - a &= -\,(2\beta e-d)\,xyz, \end{aligned}$$

where β, γ are indeterminate coefficients which are to be eliminated.

Write

$$x^2 - \frac{\sqrt{X}}{\sqrt{e}} = P, \quad y^2 - \frac{\sqrt{Y}}{\sqrt{e}} = Q;$$

then we have

$$\beta x + \gamma + P = 0, \quad \beta y + \gamma + Q = 0;$$

giving

$$\beta : \gamma : 1 = Q - P : Py - Qx : x - y.$$

Substituting these values in the first of the preceding three equations, we have

$$e\,\frac{2\,(Py-Qx)\,(x-y)+(Q-P)^2}{(x-y)^2} - c = -\left\{\frac{2\,(Q-P)\,e}{x-y} - d\right\}(x+y+z),$$

that is,

$$e\left\{\frac{2\,(Qy-Px)}{x-y} + \frac{(Q-P)^2}{(x-y)^2} + \frac{2\,(Q-P)}{x-y}\,z\right\} = c + d\,(x+y+z);$$

or, reducing by

$$Qy - Px = y^3 - x^3 + \frac{x\sqrt{X} - y\sqrt{Y}}{\sqrt{e}},$$

$$Q-P = y^2-x^2+\frac{\sqrt{X}-\sqrt{Y}}{\sqrt{e}}, \quad = y^2-x^2+(y-x)\,\frac{M}{\sqrt{e}}, \text{ if } M = \frac{\sqrt{X}-\sqrt{Y}}{x-y},$$

this is

$$e\left\{\frac{2\,(x\sqrt{X}-y\sqrt{Y})}{\sqrt{e}\,(x-y)}+2xy+\frac{M^2}{e}-2\,(x+y)\,\frac{M}{\sqrt{e}}-2\,(x+y)\,z+2z\,\frac{M}{\sqrt{e}}\right\}$$
$$= c+d\,(x+y+z)+e\,(x+y)^2.$$

We have Euler's solution in the far more simple form

$$M^2 = C+d\,(x+y)+e\,(x+y)^2,$$

where C is the arbitrary constant. It is to be observed that, in the particular case where $e=0$, the first equation becomes

$$M^2 = c+d\,(x+y+z);$$

and the two results for this case agree on putting $C = c+dz$.

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But it is required to identify the two solutions in the general case where e is not $= 0$. I remark that I have, in my *Treatise on Elliptic Functions*, Chap. xiv., further developed the theory of Euler's solution, and have shown that, regarding C as variable, and writing

$$\mathfrak{C} = ad^2 + b^2e - 2bcd + C\left[-4ae + bd + (C-c)^2\right],$$

then the given equation between the variables x, y, C corresponds to the differential equation

$$\frac{dx}{\sqrt{X}} + \frac{dy}{\sqrt{Y}} + \frac{dC}{\sqrt{\mathfrak{C}}} = 0,$$

a result which will be useful for effecting the identification. The Abelian solution may be written

$$e\left\{\frac{2\,(x\sqrt{X} - y\sqrt{Y})}{\sqrt{e}\,(x-y)} - x^2 - y^2 + \frac{M^2}{e} - 2\,(x+y)\,\frac{M}{\sqrt{e}}\right\} - c - d\,(x+y)$$
$$= z\left\{d + 2e\,(x+y) - 2M\sqrt{e}\right\};$$

and substituting for M its value, and multiplying by $(x-y)^2$, the equation becomes

$$\begin{aligned}&2\sqrt{e}\,(x-y)\,(x\sqrt{X} - y\sqrt{Y}) - e\,(x^2+y^2)\,(x-y)^2 + (\sqrt{X} - \sqrt{Y})^2\\ &\quad - 2\,(x^2-y^2)\,(\sqrt{X}-\sqrt{Y})\,\sqrt{e} - c\,(x-y)^2 - d\,(x+y)\,(x-y)^2\\ &\qquad = z\,(x-y)\left\{d\,(x-y) + 2e\,(x^2-y^2) - 2\,(\sqrt{X}-\sqrt{Y})\,\sqrt{e}\right\}.\end{aligned}$$

On the left-hand side, the rational part is

$$X + Y + c\,(-x^2+2xy-y^2) + d\,(-x^3+x^2y+xy^2-y^3) + e\,(-x^4+2x^3y-2x^2y^2+2xy^3-y^4),$$

which, substituting therein for X, Y their values, becomes

$$= 2a + b\,(x+y) + c\,2xy + d\,xy\,(x+y) + e\,2xy\,(x^2-xy+y^2);$$

and the irrational part is at once found to be

$$= 2\sqrt{e}\,(x-y)\,(x\sqrt{Y} - y\sqrt{X}) - 2\sqrt{X}\sqrt{Y}.$$

The equation thus is

$$z = \frac{\left\{\begin{aligned}&2a + b\,(x+y) + c\,2xy + d\,xy\,(x+y) + e\,2xy\,(x^2-xy+y^2)\\ &\qquad\qquad + 2\sqrt{e}\,(x-y)\,(x\sqrt{Y} - y\sqrt{X}) - 2\sqrt{X}\sqrt{Y}\end{aligned}\right\}}{(x-y)\left\{d\,(x-y) + 2e\,(x^2-y^2) - 2\,(\sqrt{X}-\sqrt{Y})\,\sqrt{e}\right\}},$$

which equation is thus a form of the general integral of $\dfrac{dx}{\sqrt{X}}+\dfrac{dy}{\sqrt{Y}}=0$, and also a particular integral of $\dfrac{dx}{\sqrt{X}}+\dfrac{dy}{\sqrt{Y}}+\dfrac{dz}{\sqrt{Z}}=0$.

Multiplying the numerator and the denominator by

$$d\,(x-y)+2e\,(x^2-y^2)+2\,(\sqrt{X}-\sqrt{Y})\,\sqrt{e},$$

the denominator becomes

$$=(x-y)^2\left[\{d+2e\,(x+y)\}^2-4e\left(\frac{\sqrt{X}-\sqrt{Y}}{x-y}\right)^2\right],$$

which, introducing herein the C of Euler's equation, is

$$=(x-y)^2\,(d^2-4eC).$$

We have therefore

$$\begin{aligned} z\,(x-y)^2\,(d^2-4eC) = \{&2a+b\,(x+y)+c\,2xy+d\,xy\,(x+y)+e\,2xy\,(x^2-xy+y^2)\\ &+2\sqrt{e}\,(x-y)\,(x\sqrt{Y}-y\sqrt{X})-2\sqrt{X}\sqrt{Y}\}\\ &\times\ \{d\,(x-y)+2e\,(x^2-y^2)+2\sqrt{e}\,(\sqrt{X}-\sqrt{Y})\}.\end{aligned}$$

Using $\mathfrak{C}$ to denote the same value as before, the function on the right-hand is, in fact,

$$=(x-y)^2\,\{2be-cd+dC+2\sqrt{e}\,\sqrt{\mathfrak{C}}\};$$

and, this being so, the required relation between z, C is

$$z\,(d^2-4eC)=\{2be-cd+dC+2\sqrt{e}\,\sqrt{\mathfrak{C}}\}.$$

To prove this, we have first, from the equation

$$\left(\frac{\sqrt{X}-\sqrt{Y}}{x-y}\right)^2 = C + d\,(x+y) + e\,(x+y)^2,$$

to express $\mathfrak{C}$ as a function of x, y. This equation, regarding therein C as a variable, gives

$$\frac{dx}{\sqrt{X}} + \frac{dy}{\sqrt{Y}} + \frac{dC}{\sqrt{\mathfrak{C}}} = 0;$$

and we have therefore

$$-\sqrt{\mathfrak{C}} = \sqrt{X}\,\frac{dC}{dx} = \sqrt{Y}\,\frac{dC}{dy},$$

viz. $\sqrt{X}\,\dfrac{dC}{dx}$ will be a symmetrical function of x, y. Putting, as before,

$$M = \frac{\sqrt{X}-\sqrt{Y}}{x-y},$$

we have

$$C = M^2 - d\,(x+y) - e\,(x+y)^2,$$

and thence

$$\frac{dC}{dx} = 2M\,\frac{dM}{dx} - d - 2e\,(x+y).$$

We have

$$\frac{dM}{dx} = \frac{1}{x-y}\,\frac{X'}{2\sqrt{X}} - \frac{\sqrt{X}-\sqrt{Y}}{(x-y)^2},$$

and hence

$$\begin{aligned}
\sqrt{\mathfrak{C}}\,(x-y)^2 &= -\sqrt{X}\,(x-y)^2\left\{2M\,\frac{dM}{dx} - d - 2e\,(x+y)\right\}\\
&= -(x-y)\,X'\,(\sqrt{X}-\sqrt{Y}) + 2\,(X+Y-2\sqrt{X}\sqrt{Y})\,\sqrt{X}\\
&\qquad\qquad + (d+2e\,\overline{x+y})\,(x-y)^2\,\sqrt{X}\\
&= [(x-y)\,X' + 2X + 2Y + (d+2e\,\overline{x+y})\,(x-y)^2]\,\sqrt{X}\\
&\quad + [(x-y)\,X' - 4X]\,\sqrt{Y}.
\end{aligned}$$

We obtain at once the coefficient of $\sqrt{Y}$, and with little more difficulty that of $\sqrt{X}$; and the result is

$$\begin{aligned}
\sqrt{\mathfrak{C}}\,(x-y)^2 &= -[4a+3bx+2cx^2+dx^3+y\,(b+2cx+3dx^2+4ex^3)]\,\sqrt{Y}\\
&\quad + [4a+3by+2cy^2+dy^3+x\,(b+2cy+3dy^2+4ey^3)]\,\sqrt{X}.
\end{aligned}$$

We have also

$$\begin{aligned}C\,(x-y)^2 &= (\sqrt{X}-\sqrt{Y})^2 - d\,(x+y)\,(x-y)^2 - e\,(x+y)^2\,(x-y)^2\\ &= X+Y-d\,(x^3-x^2y-xy^2+y^3)-e\,(x^4-2x^2y^2+y^4)-2\sqrt{X}\sqrt{Y}\\ &= 2a+b\,(x+y)+c\,(x^2+y^2)+d\,xy\,(x+y)+2e\,x^2y^2-2\sqrt{X}\sqrt{Y},\end{aligned}$$

or, say

$$\begin{aligned}C\,(x-y)^2 = 2a\,(x-y)+b\,(x^2-y^2)+c\,(x^3-x^2y+xy^2-y^3)+d\,xy\,(x^2-y^2)\\ +\,2e\,x^2y^2\,(x-y)-2\,(x-y)\,\sqrt{X}\sqrt{Y}.\end{aligned}$$

We can hence form the expression of

$$(x-y)^2\,\{2be-cd+dC+2\sqrt{e}\,\sqrt{\mathfrak{C}}\},$$

viz. this is

$$\begin{aligned}&= (2be-cd)\,(x-y)^2+2ad\,(x-y)+bd\,(x^2-y^2)+cd\,(x^3-x^2y+xy^2-y^3)+d^2\,xy\,(x^2-y^2\\ &\qquad +\,2de\,x^2y^2\,(x-y)-2d\,(x-y)\,\sqrt{X}\sqrt{Y}\\ &+2\sqrt{e}\,\{[-(4a+3bx+2cx^2+dx^3)-y\,(b+2cx+3dx^2+4ex^3)]\,\sqrt{Y}\\ &\quad +[(4a+3by+2cy^2+dy^3)+x\,(b+2cy+3dy^2+4ey^3)]\,\sqrt{X}\},\end{aligned}$$

and this should be

$$\begin{aligned}&= \{2a+b\,(x+y)+c\,2xy+d\,xy\,(x+y)+e\,2xy\,(x^2-xy+y^2)\\ &+2\sqrt{e}\,(x-y)\,(x\sqrt{Y}-y\sqrt{X})-2\sqrt{X}\sqrt{Y}\}\times\{d\,(x-y)+2e\,(x^2-y^2)+2\sqrt{e}\,(\sqrt{X}-\sqrt{Y})\}.\end{aligned}$$

To complete the theory, we require to express $\sqrt{Z}$ as a function of x, y. It would be impracticable to effect this by direct substitution of the foregoing value of z; but, observing that the value in question is a solution of $\dfrac{dx}{\sqrt{X}}+\dfrac{dy}{\sqrt{Y}}+\dfrac{dz}{\sqrt{Z}}=0$, or, what is the same thing, that $\dfrac{1}{\sqrt{X}}+\dfrac{1}{\sqrt{Z}}\dfrac{dz}{dx}=0$, $\dfrac{1}{\sqrt{Y}}+\dfrac{1}{\sqrt{Z}}\dfrac{dz}{dy}=0$, we can from either of these equations, considering therein z as a given function of x, y, calculate $\sqrt{Z}$.

Writing for shortness

$$z = \frac{J - 2\sqrt{e}\,y\,(x-y)\,\sqrt{X} + 2\sqrt{e}\,x\,(x-y)\,\sqrt{Y} - 2\sqrt{X}\sqrt{Y}}{R - 2\sqrt{e}\,(x-y)\,\sqrt{X} + 2\sqrt{e}\,(x-y)\,\sqrt{Y}},$$

where

$$R = (x-y)^2\,\{d + 2e\,(x+y)\},$$

$$J = 2a + b\,(x+y) + 2c\,xy + d\,xy\,(x+y) + 2e\,xy\,(x^2 - xy + y^2);$$

or, if for a moment $z = \dfrac{N}{D}$, then

$$\frac{dz}{dx} = \frac{1}{D^2}\Big(D\,\frac{dN}{dx} - N\,\frac{dD}{dx}\Big) = -\frac{\sqrt{Z}}{\sqrt{X}},$$

that is,

$$\sqrt{Z} = \frac{\sqrt{X}}{D^2}\Big(N\,\frac{dD}{dx} - D\,\frac{dN}{dx}\Big), \;= \frac{\Omega}{D^2}\text{ suppose;}$$

or, writing for shortness X', R', J to denote the derived functions $\dfrac{dX}{dx}$, $\dfrac{dR}{dx}$, $\dfrac{dJ}{dx}$, we find

$$\begin{aligned}\Omega &= N\,\{R'\,\sqrt{X} - 2\sqrt{e}\,X - \sqrt{e}\,(x-y)\,X' + 2\sqrt{e}\,\sqrt{X}\sqrt{Y}\}\\ &-D\,\{J'\,\sqrt{X} - 2\sqrt{e}\,yX - \sqrt{e}\,(x-y)\,yX' + 2\sqrt{e}\,(2x-y)\,\sqrt{X}\sqrt{Y} - X'\,\sqrt{Y}\};\end{aligned}$$

and, after the reductions which Cayley carries out in detail (pp. 599–602), we obtain the final result for $\sqrt{Z}$. The verification of the several subsidiary equations, marked $\mathfrak{A}$, $\mathfrak{B}$, $\mathfrak{C}$, $\mathfrak{D}$, proceeds through successive eliminations of the factors $(x-y)$, establishing at each step the correctness of the algebraic reductions.

Verification of $\mathfrak{A}$.—The equation is

$$\begin{aligned}&-X^2 - 6XY - Y^2 + (x-y)^4\,\{\Lambda^2 + (-b + dxy)\,M + xyM^2\}\\ &= 2X\,(-J + Ry - 2Y) + (x-y)\,\{2XJ' + 2X'Y - X'J - 2yR'X + yRX'\};\end{aligned}$$

or, putting for shortness

$$\Lambda^2 + (-b + dxy)\,M + xyM^2 = \nabla,$$

the successive reductions yield

$$\begin{aligned}(x-y)^2\,\nabla = \Phi\,(X+Y) + \Omega\,(J' - X') + 2X\Lambda - 2yXM&\\ + (x-y)\,\{X'\Lambda - 4eyX + yMX'\}&,\end{aligned}$$

and finally, throwing out the factor $x-y$, the equation reduces to the identity

$$-b + dxy = (x+y)\,\Lambda - \Omega,$$

which is right.

Verification of $\mathfrak{B}$.—Similarly, the equation

$$\begin{aligned} J\,\{2\,(x-y)\,M + 2e\,(x-y)^2\} - J'\,(x-y)^2\,M - 4e\,(x-y)^2\,Y \\ = 4\,(x-y)\,MY + (x-y)^2\,MY' + 2e\,(x-y)^3\,Y', \end{aligned}$$

leads, after throwing out the factor $x-y$, to

$$0 = 2M\,(-J+2Y) + (x-y)\,M\,(J'+Y') + 2e\,(x-y)\,(-J+2Y) + 2e\,(x-y)^2\,Y',$$

and thence, after further reductions, to the identity $0=0$.

Verification of $\mathfrak{C}$.—We have

$$\begin{aligned} -2X\,\{2\,(x-y)\,M + 2e\,(x-y)^2\} + (x-y)^2\,X'M + 4e\,(x-y)^2\,X - 2e\,(x-y)^3\,X' \\ = -(x-y)\,M\,\{4X - (x-y)\,X'\} - 2e\,(x-y)^2\,X', \end{aligned}$$

which is, in fact, an identity.

Verification of $\mathfrak{D}$.—The equation may be written

$$\begin{aligned} 4X + 4Y + 4e\,(x-y)^4 = \quad & 2X + 2Y - 2\,(x-y)^2\,\Lambda \\ & + 4X - 2x\,(x-y)^2\,M \\ & + 2\,(x-y)\,\{2\,(x-y)\,xM + 2ex\,(x-y)^2 - M\,(x-y)^2 - J' \end{aligned}$$

viz. this is

$$\begin{aligned} 0 = 2X - 2Y - 4e\,(x-y)^4 - 2\,(x-y)^2\,\Lambda + 2x\,(x-y)^2\,M \\ + 4ex\,(x-y)^3 - 2M\,(x-y)^3 - 2\,(x-y)\,J'. \end{aligned}$$

The first term $2\,(X-Y)$ is divisible by $2\,(x-y)$; throwing this factor out, the equation becomes

$$\begin{aligned} 0 = b + c\,(x+y) + d\,(x^2+xy+y^2) + e\,(x^3+x^2y+xy^2+y^3) - J' \\ - 2e\,(x-y)^2 - (x-y)\,\Lambda + x\,(x-y)\,M + 2ex\,(x-y)^2 - M\,(x-y)^2. \end{aligned}$$

Substituting for J' its value, the first line becomes

$$c\,(x-y) + d\,(x^2-xy) + e\,(x^3-5x^2y+5xy^2-y^3),$$

which is divisible by $(x-y)$; hence, throwing out this factor, the equation is

$$0 = c+dx+e\,(x^2-4xy+y^2)-\Lambda+xM-2e\,(x-y)^2+2ex\,(x-y)-M\,(x-y),$$

where the sum of all the terms but the last is $= d\,(x-y)+e\,(2x^2-2xy)$: hence, again throwing out the factor $x-y$, the equation becomes

$$0 = d + 2ex - 2e\,(x-y) + 2ex - M,$$

which is right.

Recapitulating, we have for the general integral of $\dfrac{dx}{\sqrt{X}}+\dfrac{dy}{\sqrt{Y}}=0$, or for a particular integral of $\dfrac{dx}{\sqrt{X}}+\dfrac{dy}{\sqrt{Y}}+\dfrac{dz}{\sqrt{Z}}=0$,

$$z = \frac{J - 2\sqrt{e}\,(x-y)\,y\,\sqrt{X} + 2\sqrt{e}\,(x-y)\,x\,\sqrt{Y} - 2\sqrt{X}\sqrt{Y}}{(x-y)^2\,M - 2\sqrt{e}\,(x-y)\,\sqrt{X} + 2\sqrt{e}\,(x-y)\,\sqrt{Y}},$$

the corresponding value of $\sqrt{Z}$ being

$$\sqrt{Z} = \frac{\left\{\begin{array}{l} \sqrt{e}\,[-X^2-6XY-Y^2+(x-y)^4\,\{\Lambda^2+(-b+dxy)\,M+xyM^2\}] \\ +\,[\{4Y+(x-y)\,Y'\}\,M+2e\,(x-y)^2\,Y']\,(x-y)\,\sqrt{X} \\ -\,[\{4X-(x-y)\,X'\}\,M+2e\,(x-y)^2\,X']\,(x-y)\,\sqrt{Y} \\ +\,[4\,(X+Y)+4e\,(x-y)^4]\,\sqrt{X}\sqrt{Y} \end{array}\right\}}{\{(x-y)^2\,M-2\sqrt{e}\,(x-y)\,\sqrt{X}+2\sqrt{e}\,(x-y)\,\sqrt{Y}\}^2},$$

where, as before,

$$\begin{aligned} M &= d + 2e\,(x+y), \\ \Lambda &= c + d\,(x+y) + e\,(x^2+y^2), \\ J &= 2a + b\,(x+y) + 2cxy + dxy\,(x+y) + exy\,(x^2-xy+y^2); \end{aligned}$$

also X is the general quartic function $a+bx+cx^2+dx^3+ex^4$, and Y, Z are the same functions of y, z respectively.

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Geometrical Illustration of a Theorem relating to an Irrational Function of an Imaginary Variable.

[From the *Proceedings of the London Mathematical Society*, vol. VIII. (1876—1877),
pp. 212—214. Read May 11, 1876.]

If we have v, a function of u, determined by an equation $f(u, v) = 0$, then to any given imaginary value $x + iy$ of u there belong two or more values, in general imaginary, $x' + iy'$ of v; and for the complete understanding of the relation between the two imaginary variables, we require to know the series of values $x' + iy'$ which correspond to a given series of values $x + iy$, of u, respectively. We must for this purpose take x, y as the coordinates of a point P in a plane Π, and x', y' as the coordinates of a corresponding point P' in another plane Π'. The series of values $x + iy$ of u is then represented by means of a curve in the first plane, and the series of values $x' + iy'$ of v by means of a corresponding curve in the second plane. The correspondence between the two points P and P' is of course established by the two equations into which the given equation $f(x + iy,\ x' + iy') = 0$ breaks up, on the assumption that x, y, x', y' are all of them real. If we assume that the coefficients in the equation are real, then the two equations are

$$f(x + iy,\ x' + iy') + f(x - iy,\ x' - iy') = 0,$$
$$f(x + iy,\ x' + iy') - f(x - iy,\ x' - iy') = 0;$$

viz. if in these equations we regard either set of coordinates, say (x, y), as constants, then the other set (x', y') are the coordinates of any real point of intersection of the curves represented by these equations respectively.

I consider the particular case where the equation between u, v is $u^2 + v^2 = a^2$: we have here $(x + iy)^2 + (x' + iy')^2 = a^2$: so that, to a given point P in the first plane, there correspond in general two points P_1', P_2' in the second plane; but to each of the points A and B, coordinates $(a, 0)$ and $(-a, 0)$, there corresponds only a single point in the second plane.

We have here a particular case of a well-known theorem: viz. if from a given point P we pass by a closed curve, not containing within it either of the points A or B, back to the initial point P, we pass in the other plane from P_1' by a closed curve back to P_1'; and similarly from P_2' by a closed curve back to P_2': but if the closed curve described by P contain within it A or B, then, in the other plane, we pass continuously from P_1' to P_2'; and also continuously from P_2' to P_1'.

The relations between (x, y), (x', y') are

$$\begin{aligned} x'^2 - y'^2 &= a^2 - (x^2 - y^2), \\ x'y' &= -xy, \end{aligned}$$

whence also

$$(x'^2 + y'^2)^2 = a^4 - 2a^2(x^2 - y^2) + (x^2 + y^2)^2.$$

And if the point (x, y) describe a curve $x^2 + y^2 = \phi\,(x^2 - y^2)$, then will the point (x', y') describe a curve $x'^2 + y'^2 = \psi\,(x'^2 - y'^2)$, obtained by the elimination of $x^2 - y^2$ from the two equations

$$\begin{aligned} x'^2 - y'^2 &= \quad a^2 - \qquad (x^2 - y^2), \\ (x'^2 + y'^2)^2 &= a^4 - 2a^2(x^2 - y^2) + \phi\,(x^2 - y^2); \end{aligned}$$

viz. this is

$$(x'^2 + y'^2)^2 = -a^4 + 2a^2(x'^2 - y'^2) + \phi\,\{a^2 - (x'^2 - y'^2)\}.$$

In particular, if the one curve be $(x^2 + y^2)^2 = \alpha + \beta\,(x^2 - y^2)$; then the other curve is

$$(x'^2 + y'^2)^2 = -a^4 + 2a^2(x'^2 - y'^2) + \alpha + \beta\,\{a^2 - (x'^2 - y'^2)\},$$

that is,

$$(x'^2 + y'^2)^2 = \alpha' + \beta'\,(x'^2 - y'^2),$$

where

$$\alpha' = -a^4 + \beta a^2 + \alpha, \quad \beta' = 2a^2 - \beta.$$

Writing for greater simplicity $a = 1$, then $\alpha' = -1 + \alpha + \beta$, $\beta' = 2 - \beta$; in particular, if $\alpha = 0$, then $\alpha' = -1 + \beta$, $\beta' = 2 - \beta$.

Supposing successively $\beta < 1$, $\beta = 1$, and $\beta > 1$, then in each case P describes a closed curve or half figure-of-eight, as shown in the annexed P-figure; but in the first case the point A is inside the curve, in the second case on it, and in the third case outside it, as shown by the letters A, A, A of the figure; and, corresponding to the three cases respectively, we have the three P'-figures, the curve in the first of them consisting of two ovals, in the second of them being a figure of eight, and in the third a twice-indented or pinched oval: the small figures 1, 2, 3, 4 in the P-figure, and 1', 2', 3', 4' and 1", 2", 3", 4" in the P'-figures serve to show the corresponding positions of the points P and P_1', P_2' respectively; and the courses are further indicated by the arrows. And we thus see how the two separate closed curves described by P_1' and P_2', as in figure 1, change into the single closed curve described one half of it by P_1' and the other half of it by P_2', as in figure 3.

[Figures: P-Figure and three P'-Figures showing the transformation of curve shapes as the point A moves from inside to outside the curve.]

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On the Circular Relation of Möbius.

[From the *Proceedings of the London Mathematical Society*, vol. VIII. (1876—1877), pp. 217—222. Read May 10, 1877.]

I consider two systems of points on a right line and on a circle respectively, the points of each system being determined each by a value of a variable parameter u; and in regard thereto, the circular relation, as defined by Möbius. A system of points on a right line is a range; such a system on a circle is a series. The term homography (or projective relation) is applicable to two ranges or to two series, but not to a range and a series. The term circular relation applies to a range and a series; or, more exactly, given a range of points A, B, C, D, &c., and a series of points A', B', C', D', &c., then, when the two systems are or may be so determined that the anharmonic ratios of any four points of the one system are equal to the anharmonic ratios of the corresponding four points of the other system, the range and the series are said to have a circular relation to each other.

In particular, if the two systems be two ranges A, B, C, D and A', B', C', D', and if we determine them by means of distances from an origin O on the line, so that the coordinates of the two points respectively are ω and ω' (coordinates measured in the same direction), then, if the relation between ω and ω' be given by the equation

$$\omega' = \frac{a\omega + b}{c\omega + d},$$

the relation is homographic; and we have

$$\omega - a\cdot\omega' - a' = \omega - b\cdot\omega' - b' = \omega - c\cdot\omega' - c' = \omega - u\cdot\omega' - u' = \Delta \text{ (suppose)},$$

viz. we have $\omega - u \cdot \omega' - u' =$ a given value; which is the most simple form of the relation between u, u'.

Interpreting everything in the first instance in regard to the homographic ranges, the equations show that there is in the first range a point O, and in the second range a point O', such that OA, &c. denoting distances, we have

$$OA \cdot O'A' = OB \cdot O'B' = OC \cdot O'C' = OU \cdot O'U' \, (= \Delta);$$

or, what is the same thing, if in the line of the first range we construct A_1, B_1, C_1, U_1 by the formulæ

$$OA \cdot OA_1 = OB \cdot OB_1 = OC \cdot OC_1 = OU \cdot OU_1 = \Delta,$$

that is, invert the first range in regard to the centre O and squared radius Δ, then we have a range O, A_1, B_1, C_1, U_1 *equal* to the range O', A', B', C', U'; viz. the distances of corresponding points are equal in the two cases: or say a range O, A_1, B_1, C_1, U_1 impossible upon O', A', B', C', U'.

The like result holds for the circular relation, but the interpretation must be explained more in detail. And, first, it is to be remarked that O in the first figure is the point corresponding to any point whatever at infinity in the second figure; viz. writing $u' = \xi' + i\eta', = \infty$, then, whatever value we give to the ratio of the two infinite quantities ξ', η', we obtain the same complex value of ω, that is, the same coordinates for the point O. And, similarly, O' in the second figure is the point corresponding to any point whatever at infinity in the first figure.

To determine O, we have the equation

$$\frac{\omega - a}{\omega - b} = \frac{b' - c'}{c' - a'} \cdot \frac{c - a}{a - b}.$$

Any such equation gives at once the geometrical construction, viz. $\omega - a = OA \cdot e^{i\,OAx}$, where OA is the distance of the points O, A regarded as positive, and OAx is the inclination of the line OA regarded as drawn from A to O to the line Ax, such angle being measured in the sense Ax to Ay; where Ax, Ay are the lines drawn from A in the senses x positive and y positive respectively: and so in other cases.

The equation is therefore equivalent to the two equations

$$\frac{OA}{OB} = \frac{B'C'}{C'A'} \cdot \frac{CA}{AB},$$

and

$$\angle OAx - \angle OBx = \angle B'C'x - \angle C'A'x + \angle CAx - \angle ABx.$$

The former of these expresses that O is in a certain circle which, having its centre on the line AB, cuts AB and AB produced in the one or the other sense; the latter that it is in the segment described on a determinate side of AB and containing a given angle: hence O, as the intersection of the segment with the first-mentioned circle, is a uniquely determined point. Similarly O' is a uniquely determined point.

It is not obvious how to construct Δ, from its original value as given above (but, ω being known, we can without difficulty construct it from the value

$$-\Delta \cdot \omega - a = b - c \cdot c' - a' \cdot a' - b'),$$

nor consequently Δ from its expression in terms of Λ: but, ω and ω' being known, we can construct Δ from the expression $\omega - a \cdot \omega' - a' = \Delta$; supposing it thus constructed, $= ke^{i\theta}$ suppose, then if, with centre O and squared radius k, we invert the first figure, thereby obtaining the points A_1, B_1, C_1, U_1 such that

$$OA \cdot OA_1 = OB \cdot OB_1 = OC \cdot OC_1 = OU \cdot OU_1 = k,$$

(the points A_1, B_1, C_1, U_1 being on the lines OA, OB, OC, OU respectively,) then the equations

$$\omega - a \cdot \omega' - a' = \omega - b \cdot \omega' - b' = \omega - c \cdot \omega' - c' = \omega - u \cdot \omega' - u' = ke^{i\theta}$$

give

$$\omega - a \cdot \omega' - a' = OA \cdot OA_1 \cdot e^{i\theta},$$

that is,

$$OA \cdot O'A' = OA \cdot OA_1, \text{ or, simply, } O'A' = OA_1,$$

and

$$\angle AOx + \angle AO'x' = \theta,$$

or, what is the same thing,

$$\angle A_1Ox + \angle A'O'x' = \theta,$$

and so for the other letters, viz. we have

$$O'A',\ O'B',\ O'C',\ O'U' = OA_1,\ OB_1,\ OC_1,\ OU_1,$$

respectively; and further

$$\angle\text{s } A_1Ox,\ B_1Ox,\ C_1Ox,\ U_1Ox = \theta - A'O'x',\ \theta - B'O'x',\ \theta - C'O'x',\ \theta - U'O'x',$$

respectively; that is, the series O', A', B', C', U' is superposable on the range O, A_1, B_1, C_1, U_1; and the superposition is in fact that which would be obtained by bringing the point O' to coincide with O, and the line $O'A'$ to make with the line OA the angle θ.

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On the Linear Transformation of the Integral $\int \frac{du}{\sqrt{U}}$.

[From the *Proceedings of the London Mathematical Society*, vol. VIII. (1876—1877),
pp. 223—228. Read June 14, 1877.]

I remark that the general linear transformation depends on the theory of the circular relation of Möbius, as explained in the foregoing paper [628]. But I consider the general question of the linear transformation. If a', b', c', d' correspond homographically to a, b, c, d, then to represent these values a', b', c', d' we have the points A', B', C', D', connected with A, B, C, D according to the circular relation of Möbius; and then, making u', a', b', c', d' correspond homographically to u, a, b, c, d, and representing in like manner the variables u, u' by the points U, U' respectively, we have the circular relation between the two systems U, A, B, C, D and U', A', B', C', D'.

Before going further I remark that the distinction of the convex and reentrant cases is not an invariable one; the figures are transformable the one into the other. Thus, taking C on the line BD (that is, between B and D, not on the line produced), there is not this relation between B', C', D', and the figure $A'B'C'D'$ is convex or reentrant as the case may be. Giving to C an infinitesimal displacement to the one side or the other of the line BD, we have in the one case a convex figure, in the other case a reentrant figure $ABCD$; but the corresponding displacement of C' being infinitesimal, the figure $A'B'C'D'$ remains for either displacement, convex or reentrant, as it originally was; that is, we have a convex figure $ABCD$ and a reentrant figure $ABCD$, each corresponding to the figure $A'B'C'D'$ (which is convex, or else reentrant, as the case may be).

Writing for convenience

$$a,\ b,\ c,\ f,\ g,\ h = b-c,\ c-a,\ a-b,\ a-d,\ b-d,\ c-d,$$
$$a',\ b',\ c',\ f',\ g',\ h' = b'-c',\ c'-a',\ a'-b',\ a'-d',\ b'-d',\ c'-d',$$

so that identically

$$af+bg+ch=0, \quad a'f'+b'g'+c'h'=0,$$

then the homographic relation between $(a,\, b,\, c,\, d)$, $(a',\, b',\, c',\, d')$ may be written in the forms

$$af : bg : ch = a'f' : b'g' : c'h',$$

or, what is the same thing, there exists a quantity N such that

$$\frac{a'f'}{af} = \frac{b'g'}{bg} = \frac{c'h'}{ch} = N^2.$$

The relation between u, u' may be written in the forms

$$\frac{u'-a'}{u'-d'} = P\,\frac{u-a}{u-d},\quad \frac{u'-b'}{u'-d'} = Q\,\frac{u-b}{u-d},\quad \frac{u'-c'}{u'-d'} = R\,\frac{u-c}{u-d};$$

and then, writing for u, u' their corresponding values, we find

$$P = \frac{b'h}{bh'} = \frac{e'g}{eg'},\quad Q = \frac{e'f}{cf'} = \frac{a'h}{ah'},\quad R = \frac{a'g}{ag'} = \frac{b'f}{bf'},$$

giving

$$f^2PN^2 = f'^2QR,\quad g^2QN^2 = g'^2RP,\quad h^2RN^2 = h'^2PQ,\quad \sqrt{PQR} = \frac{fgh}{f'g'h'}\,N^3.$$

Differentiating any one of the equations in (u, u'), for instance the first of them, we find

$$\frac{f'\,du'}{(u'-d')^2} = \frac{fP\,du}{(u-d)^2};$$

then, forming the equation

$$\frac{\sqrt{e\cdot u'-a'\cdot u'-b'\cdot u'-c'\cdot u'-d'}}{(u'-d')^2} = \pm\,\frac{\sqrt{PQR}\,\sqrt{e\cdot u-a\cdot u-b\cdot u-c\cdot u-d}}{(u-d)^2},$$

and attending to the relation $f^2PN^2 = f'^2QR$, we obtain

$$\pm\,\frac{N\,du'}{\sqrt{U'}} = \frac{du}{\sqrt{U}},$$

which is the differential relation between u, u'.

We have in connection with A, B, C, D the point O, and in connection with A', B', C', D' the point O'. As U describes the right line AB, U' describes the arc not containing O' of the circle $A'B'O'$; for observe that O' corresponds in the second figure to the point at infinity on the line AB, viz. as U passes from A to B, not passing through the point at infinity, U' must pass from A' to B', not passing through the point O', that is, it must describe, not the arc $\widehat{A'O'B'}$, but the remaining arc $2\pi - \widehat{A'O'B'}$, say this is the arc $\widetilde{A'B'}$. The integral in regard to u' is thus not the rectilinear integral $(A'B')$, but the integral along the just-mentioned circular arc, say this is denoted by $(\widetilde{A'B'})$; and we thus have

$$(AB) = \pm\,N\,(\widetilde{A'B'}).$$

But we have $(\widetilde{A'B'})$ = or not = $(A'B')$, according as the chord $A'B'$ and the arc $\widetilde{A'B'}$ do not include between them either of the points C', D', or include between them one or both of these points; and in the same cases respectively

$$(AB) = \text{ or not } = \pm N\,(A'B').$$

Of course we may in any way interchange the letters, and write under the like circumstances

$$(AC) = \text{ or not } = \pm N\,(A'C'), \text{ \&c.}$$

To prove in the case of a convex quadrilateral that (AC) is not $= \pm\,(BD)$, it is sufficient to consider the integral $\displaystyle\int \frac{du}{\sqrt{u^4-1}}$, where A, B, C, D are the points (1, 0), (0, 1), (−1, 0), and (0, −1) respectively, and where, writing $v = iu$, it at once appears that we have

$$\int_{-1}^{1} \frac{du}{\sqrt{u^4-1}} = \pm\, i \int_{-i}^{i} \frac{du}{\sqrt{u^4-1}},$$

that is,

$$(AC) = \pm\, i\,(BD), \text{ not } (AC) = \pm\,(BD).$$

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